

*An Essay
Concerning
Human
Understanding*

by John Locke

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AN ESSAY CONCERNING HUMAN UNDERSTANDING (1690)

by

John Locke

**TO THE RIGHT HONOURABLE
LORD THOMAS, EARL OF PEMBROKE AND MONT-
GOMERY,
BARRON HERBERT OF CARDIFF,
LORD ROSS, OF KENDAL, PAR, FITZHUGH,
MARMION, ST. QUINTIN,
AND SHURLAND; LORD PRESIDENT OF HIS
MAJESTY'S MOST
HONOURABLE PRIVY COUNCIL; AND LORD LIEU-
TENANT OF
THE COUNTY OF WILTS, AND OF SOUTH WALES.**

MY LORD,

THIS TREATISE, which is grown up under your lordship's eye, and has ventured into the world by your order, does now, by a natural kind of right, come to your lordship for that protection which you several years since promised it. It is not that I think any name, how great soever, set at the beginning of a book, will be able to cover the faults that are to be found in it. Things in print must stand and fall by their own worth, or the reader's fancy. But there being nothing more to be desired for truth than a fair unprejudiced hearing, nobody is more likely to procure me that than your lordship, who are allowed to have got so intimate an acquaintance with her, in her more retired recesses. Your lordship is known to have so far advanced your speculations in the most abstract and general knowledge of things, beyond the ordinary reach or common methods, that your allowance and approbation of the design of this Treatise will at least preserve it from being condemned without reading, and will prevail to have those parts a little weighted, which might otherwise perhaps be

thought to deserve no consideration, for being somewhat out of the common road. The imputation of Novelty is a terrible charge amongst those who judge of men's heads, as they do of their perukes, by the fashion, and can allow none to be right but the received doctrines. Truth scarce ever yet carried it by vote anywhere at its first appearance: new opinions are always suspected, and usually opposed, without any other reason but because they are not already common. But truth, like gold, is not the less so for being newly brought out of the mine. It is trial and examination must give it price, and not any antique fashion; and though it be not yet current by the public stamp, yet it may, for all that, be as old as nature, and is certainly not the less genuine. Your lordship can give great and convincing instances of this, whenever you please to oblige the public with some of those large and comprehensive discoveries you have made of truths hitherto unknown, unless to some few, from whom your lordship has been pleased not wholly to conceal them. This alone were a sufficient reason, were there no other, why I should dedicate this

Essay to your lordship; and its having some little correspondence with some parts of that nobler and vast system of the sciences your lordship has made so new, exact, and instructive a draught of, I think it glory enough, if your lordship permit me to boast, that here and there I have fallen into some thoughts not wholly different from yours. If your lordship think fit that, by your encouragement, this should appear in the world, I hope it may be a reason, some time or other, to lead your lordship further; and you will allow me to say, that you here give the world an earnest of something that, if they can bear with this, will be truly worth their expectation. This, my lord, shows what a present I here make to your lordship; just such as the poor man does to his rich and great neighbour, by whom the basket of flowers or fruit is not ill taken, though he has more plenty of his own growth, and in much greater perfection. Worthless things receive a value when they are made the offerings of respect, esteem, and gratitude: these you have given me so mighty and peculiar reasons to have, in the highest degree, for your lordship, that if

Human Understanding

they can add a price to what they go along with, proportionable to their own greatness, I can with confidence brag, I here make your lordship the richest present you ever received. This I am sure, I am under the greatest obligations to seek all occasions to acknowledge a long train of favours I have received from your lordship; favours, though great and important in themselves, yet made much more so by the forwardness, concern, and kindness, and other obliging circumstances, that never failed to accompany them. To all this you are pleased to add that which gives yet more weight and relish to all the rest: you vouchsafe to continue me in some degrees of your esteem, and allow me a place in your good thoughts, I had almost said friendship. This, my lord, your words and actions so constantly show on all occasions, even to others when I am absent, that it is not vanity in me to mention what everybody knows: but it would be want of good manners not to acknowledge what so many are witnesses of, and every day tell me I am indebted to your lordship for. I wish they could as easily assist my gratitude, as they convince me of the

great and growing engagements it has to your lordship. This I am sure, I should write of the Understanding without having any, if I were not extremely sensible of them, and did not lay hold on this opportunity to testify to the world how much I am obliged to be, and how much I am,

MY LORD,

Your Lordship's most humble and most obedient
servant,

JOHN LOCKE

Dorset Court,
24th of May, 1689

EPISTLE TO THE READER

I HAVE PUT INTO THY HANDS what has been the diversion of some of my idle and heavy hours. If it has the good luck to prove so of any of thine, and thou hast but half so much pleasure in reading as I had in writing it, thou wilt as little think thy money, as I do my pains, ill bestowed. Mistake not this for a commendation of my work; nor conclude, because I was pleased with the doing of it, that therefore I am fondly taken with it now it is done. He that hawks at larks and sparrows has no less sport, though a much less considerable quarry, than he that flies at nobler game: and he is little acquainted with the subject of this treatise—the *understanding*—who does not know that, as it is the most elevated faculty of the soul, so it is employed with a greater and more constant delight than any of the other. Its searches after truth are a sort of hawking and hunting, wherein the very pursuit makes a great part of the pleasure. Every step the mind takes in its progress towards Knowledge makes some discovery, which is not only new, but the best too, for the time at least.

For the understanding, like the eye, judging of objects only by its own sight, cannot but be pleased with what it discovers, having less regret for what has escaped it, because it is unknown. Thus he who has raised himself above the alms-basket, and, not content to live lazily on scraps of begged opinions, sets his own thoughts on work, to find and follow truth, will (whatever he lights on) not miss the hunter's satisfaction; every moment of his pursuit will reward his pains with some delight; and he will have reason to think his time not ill spent, even when he cannot much boast of any great acquisition.

This, Reader, is the entertainment of those who let loose their own thoughts, and follow them in writing; which thou oughtest not to envy them, since they afford thee an opportunity of the like diversion, if thou wilt make use of thy own thoughts in reading. It is to them, if they are thy own, that I refer myself: but if they are taken upon trust from others, it is no great matter what they are; they are not following truth, but some meaner consideration; and it is not worth while to be concerned what he says or thinks, who says or thinks

only as he is directed by another. If thou judgest for thyself I know thou wilt judge candidly, and then I shall not be harmed or offended, whatever be thy censure. For though it be certain that there is nothing in this Treatise of the truth whereof I am not fully persuaded, yet I consider myself as liable to mistakes as I can think thee, and know that this book must stand or fall with thee, not by any opinion I have of it, but thy own. If thou findest little in it new or instructive to thee, thou art not to blame me for it. It was not meant for those that had already mastered this subject, and made a thorough acquaintance with their own understandings; but for my own information, and the satisfaction of a few friends, who acknowledged themselves not to have sufficiently considered it.

Were it fit to trouble thee with the history of this Essay, I should tell thee, that five or six friends meeting at my chamber, and discoursing on a subject very remote from this, found themselves quickly at a stand, by the difficulties that rose on every side. After we had awhile puzzled ourselves, without coming any nearer a

resolution of those doubts which perplexed us, it came into my thoughts that we took a wrong course; and that before we set ourselves upon inquiries of that nature, it was necessary to examine our own abilities, and see what objects our understandings were, or were not, fitted to deal with. This I proposed to the company, who all readily assented; and thereupon it was agreed that this should be our first inquiry. Some hasty and undigested thoughts, on a subject I had never before considered, which I set down against our next meeting, gave the first entrance into this Discourse; which having been thus begun by chance, was continued by intreaty; written by incoherent parcels; and after long intervals of neglect, resumed again, as my humour or occasions permitted; and at last, in a retirement where an attendance on my health gave me leisure, it was brought into that order thou now seest it.

This discontinued way of writing may have occasioned, besides others, two contrary faults, viz., that too little and too much may be said in it. If thou findest anything wanting, I shall be glad that what I have written gives

thee any desire that I should have gone further. If it seems too much to thee, thou must blame the subject; for when I put pen to paper, I thought all I should have to say on this matter would have been contained in one sheet of paper; but the further I went the larger prospect I had; new discoveries led me still on, and so it grew insensibly to the bulk it now appears in. I will not deny, but possibly it might be reduced to a narrower compass than it is, and that some parts of it might be contracted, the way it has been writ in, by catches, and many long intervals of interruption, being apt to cause some repetitions. But to confess the truth, I am now too lazy, or too busy, to make it shorter.

I am not ignorant how little I herein consult my own reputation, when I knowingly let it go with a fault, so apt to disgust the most judicious, who are always the nicest readers. But they who know sloth is apt to content itself with any excuse, will pardon me if mine has prevailed on me, where I think I have a very good one. I will not therefore allege in my defence, that the same notion, having different respects, may be convenient or

necessary to prove or illustrate several parts of the same discourse, and that so it has happened in many parts of this: but waiving that, I shall frankly avow that I have sometimes dwelt long upon the same argument, and expressed it different ways, with a quite different design. I pretend not to publish this Essay for the information of men of large thoughts and quick apprehensions; to such masters of knowledge I profess myself a scholar, and therefore warn them beforehand not to expect anything here, but what, being spun out of my own coarse thoughts, is fitted to men of my own size, to whom, perhaps, it will not be unacceptable that I have taken some pains to make plain and familiar to their thoughts some truths which established prejudice, or the abstractedness of the ideas themselves, might render difficult. Some objects had need be turned on every side; and when the notion is new, as I confess some of these are to me; or out of the ordinary road, as I suspect they will appear to others, it is not one simple view of it that will gain it admittance into every understanding, or fix it there with a clear and lasting impression.

There are few, I believe, who have not observed in themselves or others, that what in one way of proposing was very obscure, another way of expressing it has made very clear and intelligible; though afterwards the mind found little difference in the phrases, and wondered why one failed to be understood more than the other. But everything does not hit alike upon every man's imagination. We have our understandings no less different than our palates; and he that thinks the same truth shall be equally relished by every one in the same dress, may as well hope to feast every one with the same sort of cookery: the meat may be the same, and the nourishment good, yet every one not be able to receive it with that seasoning; and it must be dressed another way, if you will have it go down with some, even of strong constitutions. The truth is, those who advised me to publish it, advised me, for this reason, to publish it as it is: and since I have been brought to let it go abroad, I desire it should be understood by whoever gives himself the pains to read it. I have so little affection to be in print, that if I were not flattered this Essay might be of

some use to others, as I think it has been to me, I should have confined it to the view of some friends, who gave the first occasion to it. My appearing therefore in print being on purpose to be as useful as I may, I think it necessary to make what I have to say as easy and intelligible to all sorts of readers as I can. And I had much rather the speculative and quick-sighted should complain of my being in some parts tedious, than that any one, not accustomed to abstract speculations, or prepossessed with different notions, should mistake or not comprehend my meaning.

It will possibly be censured as a great piece of vanity or insolence in me, to pretend to instruct this our knowing age; it amounting to little less, when I own, that I publish this Essay with hopes it may be useful to others. But, if it may be permitted to speak freely of those who with a feigned modesty condemn as useless what they themselves write, methinks it savours much more of vanity or insolence to publish a book for any other end; and he fails very much of that respect he owes the public, who prints, and consequently expects men should

read, that wherein he intends not they should meet with anything of use to themselves or others: and should nothing else be found allowable in this Treatise, yet my design will not cease to be so; and the goodness of my intention ought to be some excuse for the worthlessness of my present. It is that chiefly which secures me from the fear of censure, which I expect not to escape more than better writers. Men's principles, notions, and relishes are so different, that it is hard to find a book which pleases or displeases all men. I acknowledge the age we live in is not the least knowing, and therefore not the most easy to be satisfied. If I have not the good luck to please, yet nobody ought to be offended with me. I plainly tell all my readers, except half a dozen, this Treatise was not at first intended for them; and therefore they need not be at the trouble to be of that number. But yet if any one thinks fit to be angry and rail at it, he may do it securely, for I shall find some better way of spending my time than in such kind of conversation. I shall always have the satisfaction to have aimed sincerely at truth and usefulness, though in one of the

meanest ways. The commonwealth of learning is not at this time without master-builders, whose mighty designs, in advancing the sciences, will leave lasting monuments to the admiration of posterity: but every one must not hope to be a Boyle or a Sydenham; and in an age that produces such masters as the great Huygenius and the incomparable Mr. Newton, with some others of that strain, it is ambition enough to be employed as an under-labourer in clearing the ground a little, and removing some of the rubbish that lies in the way to knowledge;—which certainly had been very much more advanced in the world, if the endeavours of ingenious and industrious men had not been much cumbered with the learned but frivolous use of uncouth, affected, or unintelligible terms, introduced into the sciences, and there made an art of, to that degree that Philosophy, which is nothing but the true knowledge of things, was thought unfit or incapable to be brought into well-bred company and polite conversation. Vague and insignificant forms of speech, and abuse of language, have so long passed for mysteries of science; and hard and misapplied

words, with little or no meaning, have, by prescription, such a right to be mistaken for deep learning and height of speculation, that it will not be easy to persuade either those who speak or those who hear them, that they are but the covers of ignorance, and hindrance of true knowledge. To break in upon the sanctuary of vanity and ignorance will be, I suppose, some service to human understanding; though so few are apt to think they deceive or are deceived in the use of words; or that the language of the sect they are of has any faults in it which ought to be examined or corrected, that I hope I shall be pardoned if I have in the Third Book dwelt long on this subject, and endeavoured to make it so plain, that neither the inveterateness of the mischief, nor the prevalency of the fashion, shall be any excuse for those who will not take care about the meaning of their own words, and will not suffer the significancy of their expressions to be inquired into.

I have been told that a short Epitome of this Treatise, which was printed in 1688, was by some condemned without reading, because innate ideas were denied in it;

they too hastily concluding, that if innate ideas were not supposed, there would be little left either of the notion or proof of spirits. If any one take the like offence at the entrance of this Treatise, I shall desire him to read it through; and then I hope he will be convinced, that the taking away false foundations is not to the prejudice but advantage of truth, which is never injured or endangered so much as when mixed with, or built on, falsehood.

In the Second Edition I added as followeth:—

The bookseller will not forgive me if I say nothing of this New Edition, which he has promised, by the correctness of it, shall make amends for the many faults committed in the former. He desires too, that it should be known that it has one whole new chapter concerning Identity, and many additions and amendments in other places. These I must inform my reader are not all new matter, but most of them either further confirmation of what I had said, or explications, to prevent others being mistaken in the sense of what was formerly printed, and not any variation in me from it.

I must only except the alterations I have made in Book II. chap. xxi.

What I had there written concerning Liberty and the Will, I thought deserved as accurate a view as I am capable of; those subjects having in all ages exercised the learned part of the world with questions and difficulties, that have not a little perplexed morality and divinity, those parts of knowledge that men are most concerned to be clear in. Upon a closer inspection into the working of men's minds, and a stricter examination of those motives and views they are turned by, I have found reason somewhat to alter the thoughts I formerly had concerning that which gives the last determination to the Will in all voluntary actions. This I cannot forbear to acknowledge to the world with as much freedom and readiness as I at first published what then seemed to me to be right; thinking myself more concerned to quit and renounce any opinion of my own, than oppose that of another, when truth appears against it. For it is truth alone I seek, and that will always be welcome to me, when or from whencesoever it comes.

But what forwardness soever I have to resign any opinion I have, or to recede from anything I have writ, upon the first evidence of any error in it; yet this I must own, that I have not had the good luck to receive any light from those exceptions I have met with in print against any part of my book, nor have, from anything that has been urged against it, found reason to alter my sense in any of the points that have been questioned. Whether the subject I have in hand requires often more thought and attention than cursory readers, at least such as are prepossessed, are willing to allow; or whether any obscurity in my expressions casts a cloud over it, and these notions are made difficult to others' apprehensions in my way of treating them; so it is, that my meaning, I find, is often mistaken, and I have not the good luck to be everywhere rightly understood.

Of this the ingenious author of the Discourse Concerning the Nature of Man has given me a late instance, to mention no other. For the civility of his expressions, and the candour that belongs to his order, forbid me to think that he would have closed his Preface with an

insinuation, as if in what I had said, Book II. ch. xxvii, concerning the third rule which men refer their actions to, I went about to make virtue vice and vice virtue unless he had mistaken my meaning; which he could not have done if he had given himself the trouble to consider what the argument was I was then upon, and what was the chief design of that chapter, plainly enough set down in the fourth section and those following. For I was there not laying down moral rules, but showing the original and nature of moral ideas, and enumerating the rules men make use of in moral relations, whether these rules were true or false: and pursuant thereto I tell what is everywhere called virtue and vice; which “alters not the nature of things,” though men generally do judge of and denominate their actions according to the esteem and fashion of the place and sect they are of.

If he had been at the pains to reflect on what I had said, Bk. I. ch. ii. sect. 18, and Bk. II. ch. xxviii. sects. 13, 14, 15 and 20, he would have known what I think of the eternal and unalterable nature of right and wrong, and what I call virtue and vice. And if he had observed

that in the place he quotes I only report as a matter of fact what others call virtue and vice, he would not have found it liable to any great exception. For I think I am not much out in saying that one of the rules made use of in the world for a ground or measure of a moral relation is—that esteem and reputation which several sorts of actions find variously in the several societies of men, according to which they are there called virtues or vices. And whatever authority the learned Mr. Lowde places in his Old English Dictionary, I daresay it nowhere tells him (if I should appeal to it) that the same action is not in credit, called and counted a virtue, in one place, which, being in disrepute, passes for and under the name of vice in another. The taking notice that men bestow the names of “virtue” and “vice” according to this rule of Reputation is all I have done, or can be laid to my charge to have done, towards the making vice virtue or virtue vice. But the good man does well, and as becomes his calling, to be watchful in such points, and to take the alarm even at expressions, which, standing alone by themselves, might sound ill and be suspected.

'Tis to this zeal, allowable in his function, that I forgive his citing as he does these words of mine (ch. xxviii. sect. II): "Even the exhortations of inspired teachers have not feared to appeal to common repute, Philip. iv. 8"; without taking notice of those immediately preceding, which introduce them, and run thus: "Whereby even in the corruption of manners, the true boundaries of the law of nature, which ought to be the rule of virtue and vice, were pretty well preserved. So that even the exhortations of inspired teachers," &c. By which words, and the rest of that section, it is plain that I brought that passage of St. Paul, not to prove that the general measure of what men called virtue and vice throughout the world was, the reputation and fashion of each particular society within itself; but to show that, though it were so, yet, for reasons I there give, men, in that way of denominating their actions, did not for the most part much stray from the Law of Nature; which is that standing and unalterable rule by which they ought to judge of the moral rectitude and gravity of their actions, and accordingly denominate them virtues or vices.

Had Mr. Lowde considered this, he would have found it little to his purpose to have quoted this passage in a sense I used it not; and would I imagine have spared the application he subjoins to it, as not very necessary. But I hope this Second Edition will give him satisfaction on the point, and that this matter is now so expressed as to show him there was no cause for scruple.

Though I am forced to differ from him in these apprehensions he has expressed, in the latter end of his preface, concerning what I had said about virtue and vice, yet we are better agreed than he thinks in what he says in his third chapter (p. 78) concerning "natural inscription and innate notions." I shall not deny him the privilege he claims (p. 52), to state the question as he pleases, especially when he states it so as to leave nothing in it contrary to what I have said. For, according to him, "innate notions, being conditional things, depending upon the concurrence of several other circumstances in order to the soul's exerting them," all that he says for "innate, imprinted, impressed notions" (for of innate ideas he says nothing at all), amounts at last only to

this—that there are certain propositions which, though the soul from the beginning, or when a man is born, does not know, yet “by assistance from the outward senses, and the help of some previous cultivation,” it may afterwards come certainly to know the truth of; which is no more than what I have affirmed in my First Book. For I suppose by the “soul’s exerting them,” he means its beginning to know them; or else the soul’s “exerting of notions” will be to me a very unintelligible expression; and I think at best is a very unfit one in this, it misleading men’s thoughts by an insinuation, as if these notions were in the mind before the “soul exerts them,” i.e. before they are known;—whereas truly before they are known, there is nothing of them in the mind but a capacity to know them, when the “concurrency of those circumstances,” which this ingenious author thinks necessary “in order to the soul’s exerting them,” brings them into our knowledge.

P. 52 I find him express it thus: “These natural notions are not so imprinted upon the soul as that they naturally and necessarily exert themselves (even in chil-

dren and idiots) without any assistance from the outward senses, or without the help of some previous cultivation.” Here, he says, they exert themselves, as p. 78, that the “soul exerts them.” When he has explained to himself or others what he means by “the soul’s exerting innate notions,” or their “exerting themselves”; and what that “previous cultivation and circumstances” in order to their being exerted are—he will I suppose find there is so little of controversy between him and me on the point, bating that he calls that “exerting of notions” which I in a more vulgar style call “knowing,” that I have reason to think he brought in my name on this occasion only out of the pleasure he has to speak civilly of me; which I must gratefully acknowledge he has done everywhere he mentions me, not without conferring on me, as some others have done, a title I have no right to.

There are so many instances of this, that I think it justice to my reader and myself to conclude, that either my book is plainly enough written to be rightly understood by those who peruse it with that attention and

indifferency, which every one who will give himself the pains to read ought to employ in reading; or else that I have written mine so obscurely that it is in vain to go about to mend it. Whichever of these be the truth, it is myself only am affected thereby; and therefore I shall be far from troubling my reader with what I think might be said in answer to those several objections I have met with, to passages here and there of my book; since I persuade myself that he who thinks them of moment enough to be concerned whether they are true or false, will be able to see that what is said is either not well founded, or else not contrary to my doctrine, when I and my opposer come both to be well understood.

If any other authors, careful that none of their good thoughts should be lost, have published their censures of my Essay, with this honour done to it, that they will not suffer it to be an essay, I leave it to the public to value the obligation they have to their critical pens, and shall not waste my reader's time in so idle or ill-natured an employment of mine, as to lessen the satisfaction any one has in himself, or gives to others, in so

hasty a confutation of what I have written.

The booksellers preparing for the Fourth Edition of my Essay, gave me notice of it, that I might, if I had leisure, make any additions or alterations I should think fit. Whereupon I thought it convenient to advertise the reader, that besides several corrections I had made here and there, there was one alteration which it was necessary to mention, because it ran through the whole book, and is of consequence to be rightly understood. What I thereupon said was this:—

Clear and distinct ideas are terms which, though familiar and frequent in men's mouths, I have reason to think every one who uses does not perfectly understand. And possibly 'tis but here and there one who gives himself the trouble to consider them so far as to know what he himself or others precisely mean by them. I have therefore in most places chose to put determinate or determined, instead of clear and distinct, as more likely to direct men's thoughts to my meaning in this matter. By those denominations, I mean some object in the mind, and consequently determined, i.e. such as it

is there seen and perceived to be. This, I think, may fitly be called a determinate or determined idea, when such as it is at any time objectively in the mind, and so determined there, it is annexed, and without variation determined, to a name or articulate sound, which is to be steadily the sign of that very same object of the mind, or determinate idea.

To explain this a little more particularly. By determinate, when applied to a simple idea, I mean that simple appearance which the mind has in its view, or perceives in itself, when that idea is said to be in it: by determined, when applied to a complex idea, I mean such an one as consists of a determinate number of certain simple or less complex ideas, joined in such a proportion and situation as the mind has before its view, and sees in itself, when that idea is present in it, or should be present in it, when a man gives a name to it. I say should be, because it is not every one, nor perhaps any one, who is so careful of his language as to use no word till he views in his mind the precise determined idea which he resolves to make it the sign of. The want of this is the

cause of no small obscurity and confusion in men's thoughts and discourses.

I know there are not words enough in any language to answer all the variety of ideas that enter into men's discourses and reasonings. But this hinders not but that when any one uses any term, he may have in his mind a determined idea, which he makes it the sign of, and to which he should keep it steadily annexed during that present discourse. Where he does not, or cannot do this, he in vain pretends to clear or distinct ideas: it is plain his are not so; and therefore there can be expected nothing but obscurity and confusion, where such terms are made use of which have not such a precise determination.

Upon this ground I have thought determined ideas a way of speaking less liable to mistakes, than clear and distinct: and where men have got such determined ideas of all that they reason, inquire, or argue about, they will find a great part of their doubts and disputes at an end; the greatest part of the questions and controversies that perplex mankind depending on the doubtful

and uncertain use of words, or (which is the same) indetermined ideas, which they are made to stand for. I have made choice of these terms to signify, (1) Some immediate object of the mind, which it perceives and has before it, distinct from the sound it uses as a sign of it. (2) That this idea, thus determined, i.e. which the mind has in itself, and knows, and sees there, be determined without any change to that name, and that name determined to that precise idea. If men had such determined ideas in their inquiries and discourses, they would both discern how far their own inquiries and discourses went, and avoid the greatest part of the disputes and wranglings they have with others.

Besides this, the bookseller will think it necessary I should advertise the reader that there is an addition of two chapters wholly new; the one of the Association of Ideas, the other of Enthusiasm. These, with some other larger additions never before printed, he has engaged to print by themselves, after the same manner, and for the same purpose, as was done when this Essay had the second impression.

In the Sixth Edition there is very little added or altered. The greatest part of what is new is contained in the twenty-first chapter of the second book, which any one, if he thinks it worth while, may, with a very little labour, transcribe into the margin of the former edition.

INTRODUCTION:

AN ESSAY

CONCERNING HUMAN UNDERSTANDING

As thou knowest not what is the way of the Spirit, nor how the bones do grow in the womb of her that is with child: even so thou knowest not the works of God, who maketh all things.—Eccles. 11. 5.

Quam bellum est velle confiteri potius nescire quod nescias, quam ista effutientem nauseare, atque ipsum sibi displicere.—Cicero, *de Natur. Deor.* l. i.

INTRODUCTION

1. An Inquiry into the understanding, pleasant and useful. Since it is the understanding that sets man above the rest of sensible beings, and gives him all the advantage and dominion which he has over them; it is certainly a subject, even for its nobleness, worth our labour to inquire into. The understanding, like the eye, whilst

it makes us see and perceive all other things, takes no notice of itself; and it requires art and pains to set it at a distance and make it its own object. But whatever be the difficulties that lie in the way of this inquiry; whatever it be that keeps us so much in the dark to ourselves; sure I am that all the light we can let in upon our minds, all the acquaintance we can make with our own understandings, will not only be very pleasant, but bring us great advantage, in directing our thoughts in the search of other things.

2. Design. This, therefore, being my purpose—to inquire into the original, certainty, and extent of human knowledge, together with the grounds and degrees of belief, opinion, and assent;—I shall not at present meddle with the physical consideration of the mind; or trouble myself to examine wherein its essence consists; or by what motions of our spirits or alterations of our bodies we come to have any sensation by our organs, or any ideas in our understandings; and whether those ideas do in their formation, any or all of them, depend on matter or not. These are speculations which, however

curious and entertaining, I shall decline, as lying out of my way in the design I am now upon. It shall suffice to my present purpose, to consider the discerning faculties of a man, as they are employed about the objects which they have to do with. And I shall imagine I have not wholly misemployed myself in the thoughts I shall have on this occasion, if, in this historical, plain method, I can give any account of the ways whereby our understandings come to attain those notions of things we have; and can set down any measures of the certainty of our knowledge; or the grounds of those persuasions which are to be found amongst men, so various, different, and wholly contradictory; and yet asserted somewhere or other with such assurance and confidence, that he that shall take a view of the opinions of mankind, observe their opposition, and at the same time consider the fondness and devotion wherewith they are embraced, the resolution and eagerness wherewith they are maintained, may perhaps have reason to suspect, that either there is no such thing as truth at all, or that mankind hath no sufficient means to attain a certain knowledge of it.

3. Method. It is therefore worth while to search out the bounds between opinion and knowledge; and examine by what measures, in things whereof we have no certain knowledge, we ought to regulate our assent and moderate our persuasion. In order whereunto I shall pursue this following method:—

First, I shall inquire into the original of those ideas, notions, or whatever else you please to call them, which a man observes, and is conscious to himself he has in his mind; and the ways whereby the understanding comes to be furnished with them.

Secondly, I shall endeavour to show what knowledge the understanding hath by those ideas; and the certainty, evidence, and extent of it. Thirdly, I shall make some inquiry into the nature and grounds of faith or opinion: whereby I mean that assent which we give to any proposition as true, of whose truth yet we have no certain knowledge. And here we shall have occasion to examine the reasons and degrees of assent.

4. Useful to know the extent of our comprehension. If by this inquiry into the nature of the understanding, I

can discover the powers thereof; how far they reach; to what things they are in any degree proportionate; and where they fail us, I suppose it may be of use to prevail with the busy mind of man to be more cautious in meddling with things exceeding its comprehension; to stop when it is at the utmost extent of its tether; and to sit down in a quiet ignorance of those things which, upon examination, are found to be beyond the reach of our capacities. We should not then perhaps be so forward, out of an affectation of an universal knowledge, to raise questions, and perplex ourselves and others with disputes about things to which our understandings are not suited; and of which we cannot frame in our minds any clear or distinct perceptions, or whereof (as it has perhaps too often happened) we have not any notions at all. If we can find out how far the understanding can extend its view; how far it has faculties to attain certainty; and in what cases it can only judge and guess, we may learn to content ourselves with what is attainable by us in this state.

5. Our capacity suited to our state and concerns. For

though the comprehension of our understandings comes exceeding short of the vast extent of things, yet we shall have cause enough to magnify the bountiful Author of our being, for that proportion and degree of knowledge he has bestowed on us, so far above all the rest of the inhabitants of this our mansion. Men have reason to be well satisfied with what God hath thought fit for them, since he hath given them (as St. Peter says) *pana pros zoen kaieusebeian*, whatsoever is necessary for the conveniences of life and information of virtue; and has put within the reach of their discovery, the comfortable provision for this life, and the way that leads to a better. How short soever their knowledge may come of an universal or perfect comprehension of whatsoever is, it yet secures their great concerns, that they have light enough to lead them to the knowledge of their Maker, and the sight of their own duties. Men may find matter sufficient to busy their heads, and employ their hands with variety, delight, and satisfaction, if they will not boldly quarrel with their own constitution, and throw away the blessings their hands are filled

with, because they are not big enough to grasp everything. We shall not have much reason to complain of the narrowness of our minds, if we will but employ them about what may be of use to us; for of that they are very capable. And it will be an unpardonable, as well as childish peevishness, if we undervalue the advantages of our knowledge, and neglect to improve it to the ends for which it was given us, because there are some things that are set out of the reach of it. It will be no excuse to an idle and untoward servant, who would not attend his business by candle light, to plead that he had not broad sunshine. The Candle that is set up in us shines bright enough for all our purposes. The discoveries we can make with this ought to satisfy us; and we shall then use our understandings right, when we entertain all objects in that way and proportion that they are suited to our faculties, and upon those grounds they are capable of being proposed to us; and not peremptorily or intemperately require demonstration, and demand certainty, where probability only is to be had, and which is sufficient to govern all our concernments.

If we will disbelieve everything, because we cannot certainly know all things, we shall do much what as wisely as he who would not use his legs, but sit still and perish, because he had no wings to fly.

6. Knowledge of our capacity a cure of scepticism and idleness. When we know our own strength, we shall the better know what to undertake with hopes of success; and when we have well surveyed the powers of our own minds, and made some estimate what we may expect from them, we shall not be inclined either to sit still, and not set our thoughts on work at all, in despair of knowing anything; nor on the other side, question everything, and disclaim all knowledge, because some things are not to be understood. It is of great use to the sailor to know the length of his line, though he cannot with it fathom all the depths of the ocean. It is well he knows that it is long enough to reach the bottom, at such places as are necessary to direct his voyage, and caution him against running upon shoals that may ruin him. Our business here is not to know all things, but those which concern our conduct. If we can find out those

measures, whereby a rational creature, put in that state in which man is in this world, may and ought to govern his opinions, and actions depending thereon, we need not to be troubled that some other things escape our knowledge.

7. Occasion of this essay. This was that which gave the first rise to this Essay concerning the understanding. For I thought that the first step towards satisfying several inquiries the mind of man was very apt to run into, was, to take a survey of our own understandings, examine our own powers, and see to what things they were adapted. Till that was done I suspected we began at the wrong end, and in vain sought for satisfaction in a quiet and sure possession of truths that most concerned us, whilst we let loose our thoughts into the vast ocean of Being; as if all that boundless extent were the natural and undoubted possession of our understandings, wherein there was nothing exempt from its decisions, or that escaped its comprehension. Thus men, extending their inquiries beyond their capacities, and letting their thoughts wander into those depths where they

can find no sure footing, it is no wonder that they raise questions and multiply disputes, which, never coming to any clear resolution, are proper only to continue and increase their doubts, and to confirm them at last in perfect scepticism. Whereas, were the capacities of our understandings well considered, the extent of our knowledge once discovered, and the horizon found which sets the bounds between the enlightened and dark parts of things; between what is and what is not comprehensible by us, men would perhaps with less scruple acquiesce in the avowed ignorance of the one, and employ their thoughts and discourse with more advantage and satisfaction in the other.

8. What "Idea" stands for. Thus much I thought necessary to say concerning the occasion of this Inquiry into human Understanding. But, before I proceed on to what I have thought on this subject, I must here in the entrance beg pardon of my reader for the frequent use of the word idea, which he will find in the following treatise. It being that term which, I think, serves best to stand for whatsoever is the object of the understanding

when a man thinks, I have used it to express whatever is meant by phantasm, notion, species, or whatever it is which the mind can be employed about in thinking; and I could not avoid frequently using it.

I presume it will be easily granted me, that there are such ideas in men's minds: every one is conscious of them in himself; and men's words and actions will satisfy him that they are in others.

Our first inquiry then shall be,—how they come into the mind.

BOOK I

Neither Principles nor Ideas Are Innate

Chapter I

No Innate Speculative Principles

1. The way shown how we come by any knowledge, sufficient to prove it not innate. It is an established opinion amongst some men, that there are in the understanding certain innate principles; some primary notions, *koinai ennoiai*, characters, as it were stamped upon the mind of man; which the soul receives in its very first being, and brings into the world with it. It would be sufficient to convince unprejudiced readers of the falseness of this supposition, if I should only show (as I hope I shall in the following parts of this Discourse) how men, barely by the use of their natural faculties, may attain to all the knowledge they have, without the help of any innate impressions; and may arrive at certainty, without any such original notions or principles. For I imagine any one will easily grant that it would be

impertinent to suppose the ideas of colours innate in a creature to whom God hath given sight, and a power to receive them by the eyes from external objects: and no less unreasonable would it be to attribute several truths to the impressions of nature, and innate characters, when we may observe in ourselves faculties fit to attain as easy and certain knowledge of them as if they were originally imprinted on the mind.

But because a man is not permitted without censure to follow his own thoughts in the search of truth, when they lead him ever so little out of the common road, I shall set down the reasons that made me doubt of the truth of that opinion, as an excuse for my mistake, if I be in one; which I leave to be considered by those who, with me, dispose themselves to embrace truth wherever they find it.

2. General assent the great argument. There is nothing more commonly taken for granted than that there are certain principles, both speculative and practical, (for they speak of both), universally agreed upon by all mankind: which therefore, they argue, must needs be the

constant impressions which the souls of men receive in their first beings, and which they bring into the world with them, as necessarily and really as they do any of their inherent faculties.

3. Universal consent proves nothing innate. This argument, drawn from universal consent, has this misfortune in it, that if it were true in matter of fact, that there were certain truths wherein all mankind agreed, it would not prove them innate, if there can be any other way shown how men may come to that universal agreement, in the things they do consent in, which I presume may be done.

4. "What is, is," and "It is impossible for the same thing to be and not to be," not universally assented to. But, which is worse, this argument of universal consent, which is made use of to prove innate principles, seems to me a demonstration that there are none such: because there are none to which all mankind give an universal assent. I shall begin with the speculative, and instance in those magnified principles of demonstration, "Whatsoever is, is," and "It is impossible for the same

thing to be and not to be"; which, of all others, I think have the most allowed title to innate. These have so settled a reputation of maxims universally received, that it will no doubt be thought strange if any one should seem to question it. But yet I take liberty to say, that these propositions are so far from having an universal assent, that there are a great part of mankind to whom they are not so much as known.

5. Not on the mind naturally imprinted, because not known to children, idiots, &c. For, first, it is evident, that all children and idiots have not the least apprehension or thought of them. And the want of that is enough to destroy that universal assent which must needs be the necessary concomitant of all innate truths: it seeming to me near a contradiction to say, that there are truths imprinted on the soul, which it perceives or understands not: imprinting, if it signify anything, being nothing else but the making certain truths to be perceived. For to imprint anything on the mind without the mind's perceiving it, seems to me hardly intelligible. If therefore children and idiots have souls, have minds,

with those impressions upon them, they must unavoidably perceive them, and necessarily know and assent to these truths; which since they do not, it is evident that there are no such impressions. For if they are not notions naturally imprinted, how can they be innate? and if they are notions imprinted, how can they be unknown? To say a notion is imprinted on the mind, and yet at the same time to say, that the mind is ignorant of it, and never yet took notice of it, is to make this impression nothing. No proposition can be said to be in the mind which it never yet knew, which it was never yet conscious of. For if any one may, then, by the same reason, all propositions that are true, and the mind is capable ever of assenting to, may be said to be in the mind, and to be imprinted: since, if any one can be said to be in the mind, which it never yet knew, it must be only because it is capable of knowing it; and so the mind is of all truths it ever shall know. Nay, thus truths may be imprinted on the mind which it never did, nor ever shall know; for a man may live long, and die at last in ignorance of many truths which his mind was capable of

knowing, and that with certainty. So that if the capacity of knowing be the natural impression contended for, all the truths a man ever comes to know will, by this account, be every one of them innate; and this great point will amount to no more, but only to a very improper way of speaking; which, whilst it pretends to assert the contrary, says nothing different from those who deny innate principles. For nobody, I think, ever denied that the mind was capable of knowing several truths. The capacity, they say, is innate; the knowledge acquired. But then to what end such contest for certain innate maxims? If truths can be imprinted on the understanding without being perceived, I can see no difference there can be between any truths the mind is capable of knowing in respect of their original: they must all be innate or all adventitious: in vain shall a man go about to distinguish them. He therefore that talks of innate notions in the understanding, cannot (if he intend thereby any distinct sort of truths) mean such truths to be in the understanding as it never perceived, and is yet wholly ignorant of. For if these words “to be

in the understanding” have any propriety, they signify to be understood. So that to be in the understanding, and not to be understood; to be in the mind and never to be perceived, is all one as to say anything is and is not in the mind or understanding. If therefore these two propositions, “Whatsoever is, is,” and “It is impossible for the same thing to be and not to be,” are by nature imprinted, children cannot be ignorant of them: infants, and all that have souls, must necessarily have them in their understandings, know the truth of them, and assent to it.

6. That men know them when they come to the use of reason, answered. To avoid this, it is usually answered, that all men know and assent to them, when they come to the use of reason; and this is enough to prove them innate. I answer:

7. Doubtful expressions, that have scarce any significance, go for clear reasons to those who, being preposessed, take not the pains to examine even what they themselves say. For, to apply this answer with any tolerable sense to our present purpose, it must signify one

of these two things: either that as soon as men come to the use of reason these supposed native inscriptions come to be known and observed by them; or else, that the use and exercise of men's reason, assists them in the discovery of these principles, and certainly makes them known to them.

8. If reason discovered them, that would not prove them innate. If they mean, that by the use of reason men may discover these principles, and that this is sufficient to prove them innate; their way of arguing will stand thus, viz. that whatever truths reason can certainly discover to us, and make us firmly assent to, those are all naturally imprinted on the mind; since that universal assent, which is made the mark of them, amounts to no more but this,—that by the use of reason we are capable to come to a certain knowledge of and assent to them; and, by this means, there will be no difference between the maxims of the mathematicians, and theorems they deduce from them: all must be equally allowed innate; they being all discoveries made by the use of reason, and truths that a rational creature may cer-

tainty come to know, if he apply his thoughts rightly that way.

9. It is false that reason discovers them. But how can these men think the use of reason necessary to discover principles that are supposed innate, when reason (if we may believe them) is nothing else but the faculty of deducing unknown truths from principles or propositions that are already known? That certainly can never be thought innate which we have need of reason to discover; unless, as I have said, we will have all the certain truths that reason ever teaches us, to be innate. We may as well think the use of reason necessary to make our eyes discover visible objects, as that there should be need of reason, or the exercise thereof, to make the understanding see what is originally engraven on it, and cannot be in the understanding before it be perceived by it. So that to make reason discover those truths thus imprinted, is to say, that the use of reason discovers to a man what he knew before: and if men have those innate impressed truths originally, and before the use of reason, and yet are always ignorant of

them till they come to the use of reason, it is in effect to say, that men know and know them not at the same time.

10. No use made of reasoning in the discovery of these two maxims. It will here perhaps be said that mathematical demonstrations, and other truths that are not innate, are not assented to as soon as proposed, wherein they are distinguished from these maxims and other innate truths. I shall have occasion to speak of assent upon the first proposing, more particularly by and by. I shall here only, and that very readily, allow, that these maxims and mathematical demonstrations are in this different: that the one have need of reason, using of proofs, to make them out and to gain our assent; but the other, as soon as understood, are, without any the least reasoning, embraced and assented to. But I withal beg leave to observe, that it lays open the weakness of this subterfuge, which requires the use of reason for the discovery of these general truths: since it must be confessed that in their discovery there is no use made of reasoning at all. And I think those who give this answer

will not be forward to affirm that the knowledge of this maxim, "That it is impossible for the same thing to be and not to be," is a deduction of our reason. For this would be to destroy that bounty of nature they seem so fond of, whilst they make the knowledge of those principles to depend on the labour of our thoughts. For all reasoning is search, and casting about, and requires pains and application. And how can it with any tolerable sense be supposed, that what was imprinted by nature, as the foundation and guide of our reason, should need the use of reason to discover it?

11. And if there were, this would prove them not innate. Those who will take the pains to reflect with a little attention on the operations of the understanding, will find that this ready assent of the mind to some truths, depends not, either on native inscription, or the use of reason, but on a faculty of the mind quite distinct from both of them, as we shall see hereafter. Reason, therefore, having nothing to do in procuring our assent to these maxims, if by saying, that "men know and assent to them, when they come to the use of

reason," be meant, that the use of reason assists us in the knowledge of these maxims, it is utterly false; and were it true, would prove them not to be innate.

12. The coming to the use of reason not the time we come to know these maxims. If by knowing and assenting to them "when we come to the use of reason," be meant, that this is the time when they come to be taken notice of by the mind; and that as soon as children come to the use of reason, they come also to know and assent to these maxims; this also is false and frivolous. First, it is false; because it is evident these maxims are not in the mind so early as the use of reason; and therefore the coming to the use of reason is falsely assigned as the time of their discovery. How many instances of the use of reason may we observe in children, a long time before they have any knowledge of this maxim, "That it is impossible for the same thing to be and not to be?" And a great part of illiterate people and savages pass many years, even of their rational age, without ever thinking on this and the like general propositions. I grant, men come not to the knowledge of these gen-

eral and more abstract truths, which are thought innate, till they come to the use of reason; and I add, nor then neither. Which is so, because, till after they come to the use of reason, those general abstract ideas are not framed in the mind, about which those general maxims are, which are mistaken for innate principles, but are indeed discoveries made and verities introduced and brought into the mind by the same way, and discovered by the same steps, as several other propositions, which nobody was ever so extravagant as to suppose innate. This I hope to make plain in the sequel of this Discourse. I allow therefore, a necessity that men should come to the use of reason before they get the knowledge of those general truths; but deny that men's coming to the use of reason is the time of their discovery.

13. By this they are not distinguished from other knowable truths. In the mean time it is observable, that this saying, that men know and assent to these maxims "when they come to the use of reason," amounts in reality of fact to no more but this,—that they are never known

nor taken notice of before the use of reason, but may possibly be assented to some time after, during a man's life; but when is uncertain. And so may all other knowable truths, as well as these; which therefore have no advantage nor distinction from others by this note of being known when we come to the use of reason; nor are thereby proved to be innate, but quite the contrary.

14. If coming to the use of reason were the time of their discovery it would not prove them innate. But, secondly, were it true that the precise time of their being known and assented to were, when men come to the use of reason; neither would that prove them innate. This way of arguing is as frivolous as the supposition itself is false. For, by what kind of logic will it appear that any notion is originally by nature imprinted in the mind in its first constitution, because it comes first to be observed and assented to when a faculty of the mind, which has quite a distinct province, begins to exert itself? And therefore the coming to the use of speech, if it were supposed the time that these maxims are first assented to, (which it may be with as much

truth as the time when men come to the use of reason,) would be as good a proof that they were innate, as to say they are innate because men assent to them when they come to the use of reason. I agree then with these men of innate principles, that there is no knowledge of these general and self-evident maxims in the mind, till it comes to the exercise of reason: but I deny that the coming to the use of reason is the precise time when they are first taken notice of, and if that were the precise time, I deny that it would prove them innate. All that can with any truth be meant by this proposition, that men "assent to them when they come to the use of reason," is no more but this,—that the making of general abstract ideas, and the understanding of general names, being a concomitant of the rational faculty, and growing up with it, children commonly get not those general ideas, nor learn the names that stand for them, till, having for a good while exercised their reason about familiar and more particular ideas, they are, by their ordinary discourse and actions with others, acknowledged to be capable of rational conversation. If assent-

ing to these maxims, when men come to the use of reason, can be true in any other sense, I desire it may be shown; or at least, how in this, or any other sense, it proves them innate.

15. The steps by which the mind attains several truths. The senses at first let in particular ideas, and furnish the yet empty cabinet, and the mind by degrees growing familiar with some of them, they are lodged in the memory, and names got to them. Afterwards, the mind proceeding further, abstracts them, and by degrees learns the use of general names. In this manner the mind comes to be furnished with ideas and language, the materials about which to exercise its discursive faculty. And the use of reason becomes daily more visible, as these materials that give it employment increase. But though the having of general ideas and the use of general words and reason usually grow together, yet I see not how this any way proves them innate. The knowledge of some truths, I confess, is very early in the mind but in a way that shows them not to be innate. For, if we will observe, we shall find it still to be about ideas, not innate, but ac-

quired; it being about those first which are imprinted by external things, with which infants have earliest to do, which make the most frequent impressions on their senses. In ideas thus got, the mind discovers that some agree and others differ, probably as soon as it has any use of memory; as soon as it is able to retain and perceive distinct ideas. But whether it be then or no, this is certain, it does so long before it has the use of words; or comes to that which we commonly call "the use of reason." For a child knows as certainly before it can speak the difference between the ideas of sweet and bitter (i.e. that sweet is not bitter), as it knows afterwards (when it comes to speak) that wormwood and sugar-plums are not the same thing.

16. Assent to supposed innate truths depends on having clear and distinct ideas of what their terms mean, and not on their innateness. A child knows not that three and four are equal to seven, till he comes to be able to count seven, and has got the name and idea of equality; and then, upon explaining those words, he presently assents to, or rather perceives the truth of

that proposition. But neither does he then readily assent because it is an innate truth, nor was his assent wanting till then because he wanted the use of reason; but the truth of it appears to him as soon as he has settled in his mind the clear and distinct ideas that these names stand for. And then he knows the truth of that proposition upon the same grounds and by the same means, that he knew before that a rod and a cherry are not the same thing; and upon the same grounds also that he may come to know afterwards "That it is impossible for the same thing to be and not to be," as shall be more fully shown hereafter. So that the later it is before any one comes to have those general ideas about which those maxims are; or to know the signification of those general terms that stand for them; or to put together in his mind the ideas they stand for; the later also will it be before he comes to assent to those maxims;—whose terms, with the ideas they stand for, being no more innate than those of a cat or a weasel, he must stay till time and observation have acquainted him with them; and then he will be in a capacity to know the truth of

these maxims, upon the first occasion that shall make him put together those ideas in his mind, and observe whether they agree or disagree, according as is expressed in those propositions. And therefore it is that a man knows that eighteen and nineteen are equal to thirty-seven, by the same self-evidence that he knows one and two to be equal to three: yet a child knows this not so soon as the other; not for want of the use of reason, but because the ideas the words eighteen, nineteen, and thirty-seven stand for, are not so soon got, as those which are signified by one, two, and three.

17. Assenting as soon as proposed and understood, proves them not innate. This evasion therefore of general assent when men come to the use of reason, failing as it does, and leaving no difference between those suppose innate and other truths that are afterwards acquired and learnt, men have endeavoured to secure an universal assent to those they call maxims, by saying, they are generally assented to as soon as proposed, and the terms they are proposed in understood: seeing all men, even children, as soon as they hear and under-

stand the terms, assent to these propositions, they think it is sufficient to prove them innate. For since men never fail after they have once understood the words, to acknowledge them for undoubted truths, they would infer, that certainly these propositions were first lodged in the understanding, which, without any teaching, the mind, at the very first proposal immediately closes with and assents to, and after that never doubts again.

18. If such an assent be a mark of innate, then "that one and two are equal to three, that sweetness is not bitterness," and a thousand the like, must be innate. In answer to this, I demand whether ready assent given to a proposition, upon first hearing and understanding the terms, be a certain mark of an innate principle? If it be not, such a general assent is in vain urged as a proof of them: if it be said that it is a mark of innate, they must then allow all such propositions to be innate which are generally assented to as soon as heard, whereby they will find themselves plentifully stored with innate principles. For upon the same ground, viz. of assent at first hearing and understanding the terms, that men would

have those maxims pass for innate, they must also admit several propositions about numbers to be innate; and thus, that one and two are equal to three, that two and two are equal to four, and a multitude of other the like propositions in numbers, that everybody assents to at first hearing and understanding the terms, must have a place amongst these innate axioms. Nor is this the prerogative of numbers alone, and propositions made about several of them; but even natural philosophy, and all the other sciences, afford propositions which are sure to meet with assent as soon as they are understood. That "two bodies cannot be in the same place" is a truth that nobody any more sticks at than at these maxims, that "it is impossible for the same thing to be and not to be," that "white is not black," that "a square is not a circle," that "bitterness is not sweetness." These and a million of such other propositions, as many at least as we have distinct ideas of, every man in his wits, at first hearing, and knowing what the names stand for, must necessarily assent to. If these men will be true to their own rule, and have assent at first hearing and under-

standing the terms to be a mark of innate, they must allow not only as many innate propositions as men have distinct ideas, but as many as men can make propositions wherein different ideas are denied one of another. Since every proposition wherein one different idea is denied of another, will as certainly find assent at first hearing and understanding the terms as this general one, "It is impossible for the same thing to be and not to be," or that which is the foundation of it, and is the easier understood of the two, "The same is not different"; by which account they will have legions of innate propositions of this one sort, without mentioning any other. But, since no proposition can be innate unless the ideas about which it is be innate, this will be to suppose all our ideas of colours, sounds, tastes, figure, &c., innate, than which there cannot be anything more opposite to reason and experience. Universal and ready assent upon hearing and understanding the terms is, I grant, a mark of self-evidence; but self-evidence, depending not on innate impressions, but on something else, (as we shall show hereafter,) belongs to several

propositions which nobody was yet so extravagant as to pretend to be innate.

19. Such less general propositions known before these universal maxims. Nor let it be said, that those more particular self-evident propositions, which are assented to at first hearing, as that "one and two are equal to three," that "green is not red," &c., are received as the consequences of those more universal propositions which are looked on as innate principles; since any one, who will but take the pains to observe what passes in the understanding, will certainly find that these, and the like less general propositions, are certainly known, and firmly assented to by those who are utterly ignorant of those more general maxims; and so, being earlier in the mind than those (as they are called) first principles, cannot owe to them the assent wherewith they are received at first hearing.

20. "One and one equal to Two, &c., not general nor useful," answered. If it be said, that these propositions, viz. "two and two are equal to four," "red is not blue," &c., are not general maxims, nor of any great use, I

answer, that makes nothing to the argument of universal assent upon hearing and understanding. For, if that be the certain mark of innate, whatever proposition can be found that receives general assent as soon as heard and understood, that must be admitted for an innate proposition, as well as this maxim, "That it is impossible for the same thing to be and not to be," they being upon this ground equal. And as to the difference of being more general, that makes this maxim more remote from being innate; those general and abstract ideas being more strangers to our first apprehensions than those of more particular self-evident propositions; and therefore it is longer before they are admitted and assented to by the growing understanding. And as to the usefulness of these magnified maxims, that perhaps will not be found so great as is generally conceived, when it comes in its due place to be more fully considered.

21. These maxims not being known sometimes till proposed, proves them not innate. But we have not yet done with "assenting to propositions at first hearing and understanding their terms." It is fit we first take

notice that this, instead of being a mark that they are innate, is a proof of the contrary; since it supposes that several, who understand and know other things, are ignorant of these principles till they are proposed to them; and that one may be unacquainted with these truths till he hears them from others. For, if they were innate, what need they be proposed in order to gaining assent, when, by being in the understanding, by a natural and original impression, (if there were any such,) they could not but be known before? Or doth the proposing them print them clearer in the mind than nature did? If so, then the consequence will be, that a man knows them better after he has been thus taught them than he did before. Whence it will follow that these principles may be made more evident to us by others' teaching than nature has made them by impression: which will ill agree with the opinion of innate principles, and give but little authority to them; but, on the contrary, makes them unfit to be the foundations of all our other knowledge; as they are pretended to be. This cannot be denied, that men grow first acquainted with many of these

self-evident truths upon their being proposed: but it is clear that whosoever does so, finds in himself that he then begins to know a proposition, which he knew not before, and which from thenceforth he never questions; not because it was innate, but because the consideration of the nature of the things contained in those words would not suffer him to think otherwise, how, or whensoever he is brought to reflect on them. And if whatever is assented to at first hearing and understanding the terms must pass for an innate principle, every well-grounded observation, drawn from particulars into a general rule, must be innate. When yet it is certain that not all, but only sagacious heads, light at first on these observations, and reduce them into general propositions: not innate, but collected from a preceding acquaintance and reflection on particular instances. These, when observing men have made them, unobserving men, when they are proposed to them, cannot refuse their assent to.

22. Implicitly known before proposing, signifies that the mind is capable of understanding them, or else sig-

nifies nothing. If it be said, the understanding hath an implicit knowledge of these principles, but not an explicit, before this first hearing (as they must who will say "that they are in the understanding before they are known,") it will be hard to conceive what is meant by a principle imprinted on the understanding implicitly, unless it be this,—that the mind is capable of understanding and assenting firmly to such propositions. And thus all mathematical demonstrations, as well as first principles, must be received as native impressions on the mind; which I fear they will scarce allow them to be, who find it harder to demonstrate a proposition than assent to it when demonstrated. And few mathematicians will be forward to believe, that all the diagrams they have drawn were but copies of those innate characters which nature had engraven upon their minds.

23. The argument of assenting on first hearing, is upon a false supposition of no precedent teaching. There is, I fear, this further weakness in the foregoing argument, which would persuade us that therefore those maxims are to be thought innate, which men admit at first hear-

ing; because they assent to propositions which they are not taught, nor do receive from the force of any argument or demonstration, but a bare explication or understanding of the terms. Under which there seems to me to lie this fallacy, that men are supposed not to be taught nor to learn anything *de novo*; when, in truth, they are taught, and do learn something they were ignorant of before. For, first, it is evident that they have learned the terms, and their signification; neither of which was born with them. But this is not all the acquired knowledge in the case: the ideas themselves, about which the proposition is, are not born with them, no more than their names, but got afterwards. So that in all propositions that are assented to at first hearing, the terms of the proposition, their standing for such ideas, and the ideas themselves that they stand for, being neither of them innate, I would fain know what there is remaining in such propositions that is innate. For I would gladly have any one name that proposition whose terms or ideas were either of them innate. We by degrees get ideas and names, and learn their appropriated connexion

one with another; and then to propositions made in such terms, whose signification we have learnt, and wherein the agreement or disagreement we can perceive in our ideas when put together is expressed, we at first hearing assent; though to other propositions, in themselves as certain and evident, but which are concerning ideas not so soon or so easily got, we are at the same time no way capable of assenting. For, though a child quickly assents to this proposition, "That an apple is not fire," when by familiar acquaintance he has got the ideas of those two different things distinctly imprinted on his mind, and has learnt that the names apple and fire stand for them; yet it will be some years after, perhaps, before the same child will assent to this proposition, "That it is impossible for the same thing to be and not to be"; because that, though perhaps the words are as easy to be learnt, yet the signification of them being more large, comprehensive, and abstract than of the names annexed to those sensible things the child hath to do with, it is longer before he learns their precise meaning, and it requires more time plainly to form in

his mind those general ideas they stand for. Till that be done, you will in vain endeavour to make any child assent to a proposition made up of such general terms; but as soon as ever he has got those ideas, and learned their names, he forwardly closes with the one as well as the other of the forementioned propositions: and with both for the same reason; viz. because he finds the ideas he has in his mind to agree or disagree, according as the words standing for them are affirmed or denied one of another in the proposition. But if propositions be brought to him in words which stand for ideas he has not yet in his mind, to such propositions, however evidently true or false in themselves, he affords neither assent nor dissent, but is ignorant. For words being but empty sounds, any further than they are signs of our ideas, we cannot but assent to them as they correspond to those ideas we have, but no further than that. But the showing by what steps and ways knowledge comes into our minds; and the grounds of several degrees of assent, being the business of the following Discourse, it may suffice to have only touched on it here, as one reason that made

me doubt of those innate principles.

24. Not innate, because not universally assented to. To conclude this argument of universal consent, I agree with these defenders of innate principles,—that if they are innate, they must needs have universal assent. For that a truth should be innate and yet not assented to, is to me as unintelligible as for a man to know a truth and be ignorant of it at the same time. But then, by these men's own confession, they cannot be innate; since they are not assented to by those who understand not the terms; nor by a great part of those who do understand them, but have yet never heard nor thought of those propositions; which, I think, is at least one half of mankind. But were the number far less, it would be enough to destroy universal assent, and thereby show these propositions not to be innate, if children alone were ignorant of them.

25. These maxims not the first known. But that I may not be accused to argue from the thoughts of infants, which are unknown to us, and to conclude from what passes in their understandings before they express it; I

say next, that these two general propositions are not the truths that first possess the minds of children, nor are antecedent to all acquired and adventitious notions: which, if they were innate, they must needs be. Whether we can determine it or no, it matters not, there is certainly a time when children begin to think, and their words and actions do assure us that they do so. When therefore they are capable of thought, of knowledge, of assent, can it rationally be supposed they can be ignorant of those notions that nature has imprinted, were there any such? Can it be imagined, with any appearance of reason, that they perceive the impressions from things without, and be at the same time ignorant of those characters which nature itself has taken care to stamp within? Can they receive and assent to adventitious notions, and be ignorant of those which are supposed woven into the very principles of their being, and imprinted there in indelible characters, to be the foundation and guide of all their acquired knowledge and future reasonings? This would be to make nature take pains to no purpose; or at least to write very ill; since its characters could not be read by those eyes which saw other things

very well: and those are very ill supposed the clearest parts of truth, and the foundations of all our knowledge, which are not first known, and without which the undoubted knowledge of several other things may be had. The child certainly knows, that the nurse that feeds it is neither the cat it plays with, nor the blackmoor it is afraid of: that the wormseed or mustard it refuses, is not the apple or sugar it cries for: this it is certainly and undoubtedly assured of: but will any one say, it is by virtue of this principle, "That it is impossible for the same thing to be and not to be," that it so firmly assents to these and other parts of its knowledge? Or that the child has any notion or apprehension of that proposition at an age, wherein yet, it is plain, it knows a great many other truths? He that will say, children join in these general abstract speculations with their sucking-bottles and their rattles, may perhaps, with justice, be thought to have more passion and zeal for his opinion, but less sincerity and truth, than one of that age.

26. And so not innate. Though therefore there be several general propositions that meet with constant and ready assent, as soon as proposed to men grown up,

who have attained the use of more general and abstract ideas, and names standing for them; yet they not being to be found in those of tender years, who nevertheless know other things, they cannot pretend to universal assent of intelligent persons, and so by no means can be supposed innate;—it being impossible that any truth which is innate (if there were any such) should be unknown, at least to any one who knows anything else. Since, if they are innate truths, they must be innate thoughts: there being nothing a truth in the mind that it has never thought on. Whereby it is evident, if there by any innate truths, they must necessarily be the first of any thought on; the first that appear.

27. Not innate, because they appear least where what is innate shows itself clearest. That the general maxims we are discoursing of are not known to children, idiots, and a great part of mankind, we have already sufficiently proved: whereby it is evident they have not an universal assent, nor are general impressions. But there is this further argument in it against their being innate: that these characters, if they were native and original impressions,

should appear fairest and clearest in those persons in whom yet we find no footsteps of them; and it is, in my opinion, a strong presumption that they are not innate, since they are least known to those in whom, if they were innate, they must needs exert themselves with most force and vigour. For children, idiots, savages, and illiterate people, being of all others the least corrupted by custom, or borrowed opinions; learning and education having not cast their native thoughts into new moulds; nor by super-inducing foreign and studied doctrines, confounded those fair characters nature had written there; one might reasonably imagine that in their minds these innate notions should lie open fairly to every one's view, as it is certain the thoughts of children do. It might very well be expected that these principles should be perfectly known to naturals; which being stamped immediately on the soul, (as these men suppose,) can have no dependence on the constitution or organs of the body, the only confessed difference between them and others. One would think, according to these men's principles, that all these native beams of light (were there any such) should, in those

who have no reserves, no arts of concealment, shine out in their full lustre, and leave us in no more doubt of their being there, than we are of their love of pleasure and abhorrence of pain. But alas, amongst children, idiots, savages, and the grossly illiterate, what general maxims are to be found? What universal principles of knowledge? Their notions are few and narrow, borrowed only from those objects they have had most to do with, and which have made upon their senses the frequentest and strongest impressions. A child knows his nurse and his cradle, and by degrees the playthings of a little more advanced age; and a young savage has, perhaps, his head filled with love and hunting, according to the fashion of his tribe. But he that from a child untaught, or a wild inhabitant of the woods, will expect these abstract maxims and reputed principles of science, will, I fear, find himself mistaken. Such kind of general propositions are seldom mentioned in the huts of Indians: much less are they to be found in the thoughts of children, or any impressions of them on the minds of naturals. They are the language and business of the schools and academies of learned na-

tions, accustomed to that sort of conversation or learning, where disputes are frequent; these maxims being suited to artificial argumentation and useful for conviction, but not much conducing to the discovery of truth or advancement of knowledge. But of their small use for the improvement of knowledge I shall have occasion to speak more at large, 1. 4, c. 7.

28. Recapitulation. I know not how absurd this may seem to the masters of demonstration. And probably it will hardly go down with anybody at first hearing. I must therefore beg a little truce with prejudice, and the forbearance of censure, till I have been heard out in the sequel of this Discourse, being very willing to submit to better judgments. And since I impartially search after truth, I shall not be sorry to be convinced, that I have been too fond of my own notions; which I confess we are all apt to be, when application and study have warmed our heads with them.

Upon the whole matter, I cannot see any ground to think these two speculative Maxims innate: since they are not universally assented to; and the assent they so

generally find is no other than what several propositions, not allowed to be innate, equally partake in with them: and since the assent that is given them is produced another way, and comes not from natural inscription, as I doubt not but to make appear in the following Discourse. And if these “first principles” of knowledge and science are found not to be innate, no other speculative maxims can (I suppose), with better right pretend to be so.

Chapter II

No Innate Practical Principles

1. No moral principles so clear and so generally received as the forementioned speculative maxims. If those speculative Maxims, whereof we discoursed in the foregoing chapter, have not an actual universal assent from all mankind, as we there proved, it is much more visible concerning practical Principles, that they come short of an universal reception: and I think it will be hard to instance any one moral rule which can pretend to so

general and ready an assent as, “What is, is”; or to be so manifest a truth as this, that “It is impossible for the same thing to be and not to be.” Whereby it is evident that they are further removed from a title to be innate; and the doubt of their being native impressions on the mind is stronger against those moral principles than the other. Not that it brings their truth at all in question. They are equally true, though not equally evident. Those speculative maxims carry their own evidence with them: but moral principles require reasoning and discourse, and some exercise of the mind, to discover the certainty of their truth. They lie not open as natural characters engraven on the mind; which, if any such were, they must needs be visible by themselves, and by their own light be certain and known to everybody. But this is no derogation to their truth and certainty; no more than it is to the truth or certainty of the three angles of a triangle being equal to two right ones: because it is not so evident as “the whole is bigger than a part,” nor so apt to be assented to at first hearing. It may suffice that these moral rules are capable of demonstration: and there-

fore it is our own faults if we come not to a certain knowledge of them. But the ignorance wherein many men are of them, and the slowness of assent wherewith others receive them, are manifest proofs that they are not innate, and such as offer themselves to their view without searching.

2. Faith and justice not owned as principles by all men. Whether there be any such moral principles, wherein all men do agree, I appeal to any who have been but moderately conversant in the history of mankind, and looked abroad beyond the smoke of their own chimneys. Where is that practical truth that is universally received, without doubt or question, as it must be if innate? Justice, and keeping of contracts, is that which most men seem to agree in. This is a principle which is thought to extend itself to the dens of thieves, and the confederacies of the greatest villains; and they who have gone furthest towards the putting off of humanity itself, keep faith and rules of justice one with another. I grant that outlaws themselves do this one amongst another: but it is without receiving these as the innate laws of nature.

They practise them as rules of convenience within their own communities: but it is impossible to conceive that he embraces justice as a practical principle, who acts fairly with his fellow-highwayman, and at the same time plunders or kills the next honest man he meets with. Justice and truth are the common ties of society; and therefore even outlaws and robbers, who break with all the world besides, must keep faith and rules of equity amongst themselves; or else they cannot hold together. But will any one say, that those that live by fraud or rapine have innate principles of truth and justice which they allow and assent to?

3. Objection: "though men deny them in their practice, yet they admit them in their thoughts," answered. Perhaps it will be urged, that the tacit assent of their minds agrees to what their practice contradicts. I answer, first, I have always thought the actions of men the best interpreters of their thoughts. But, since it is certain that most men's practices, and some men's open professions, have either questioned or denied these principles, it is impossible to establish an universal consent, (though

we should look for it only amongst grown men,) without which it is impossible to conclude them innate. Secondly, it is very strange and unreasonable to suppose innate practical principles, that terminate only in contemplation. Practical principles, derived from nature, are there for operation, and must produce conformity of action, not barely speculative assent to their truth, or else they are in vain distinguished from speculative maxims. Nature, I confess, has put into man a desire of happiness and an aversion to misery: these indeed are innate practical principles which (as practical principles ought) do continue constantly to operate and influence all our actions without ceasing: these may be observed in all persons and all ages, steady and universal; but these are inclinations of the appetite to good, not impressions of truth on the understanding. I deny not that there are natural tendencies imprinted on the minds of men; and that from the very first instances of sense and perception, there are some things that are grateful and others unwelcome to them; some things that they incline to and others that they fly: but this makes noth-

ing for innate characters on the mind, which are to be the principles of knowledge regulating our practice. Such natural impressions on the understanding are so far from being confirmed hereby, that this is an argument against them; since, if there were certain characters imprinted by nature on the understanding, as the principles of knowledge, we could not but perceive them constantly operate in us and influence our knowledge, as we do those others on the will and appetite; which never cease to be the constant springs and motives of all our actions, to which we perpetually feel them strongly impelling us.

4. Moral rules need a proof, ergo not innate. Another reason that makes me doubt of any innate practical principles is, that I think there cannot any one moral rule be proposed whereof a man may not justly demand a reason: which would be perfectly ridiculous and absurd if they were innate; or so much as self-evident, which every innate principle must needs be, and not need any proof to ascertain its truth, nor want any reason to gain it approbation. He would be thought void of com-

mon sense who asked on the one side, or on the other side went to give a reason why "it is impossible for the same thing to be and not to be." It carries its own light and evidence with it, and needs no other proof: he that understands the terms assents to it for its own sake or else nothing will ever be able to prevail with him to do it. But should that most unshaken rule of morality and foundation of all social virtue, "That one should do as he would be done unto," be proposed to one who never heard of it before, but yet is of capacity to understand its meaning; might he not without any absurdity ask a reason why? And were not he that proposed it bound to make out the truth and reasonableness of it to him? Which plainly shows it not to be innate; for if it were it could neither want nor receive any proof; but must needs (at least as soon as heard and understood) be received and assented to as an unquestionable truth, which a man can by no means doubt of. So that the truth of all these moral rules plainly depends upon some other antecedent to them, and from which they must be deduced; which could not be if either they were innate or

so much as self-evident.

5. Instance in keeping compacts. That men should keep their compacts is certainly a great and undeniable rule in morality. But yet, if a Christian, who has the view of happiness and misery in another life, be asked why a man must keep his word, he will give this as a reason:- Because God, who has the power of eternal life and death, requires it of us. But if a Hobbist be asked why? he will answer:—Because the public requires it, and the Levathan will punish you if you do not. And if one of the old philosophers had been asked, he would have answered:—Because it was dishonest, below the dignity of a man, and opposite to virtue, the highest perfection of human nature, to do otherwise.

6. Virtue generally approved, not because innate, but because profitable. Hence naturally flows the great variety of opinions concerning moral rules which are to be found among men, according to the different sorts of happiness they have a prospect of, or propose to themselves; which could not be if practical principles were innate, and imprinted in our minds immediately by the

hand of God. I grant the existence of God is so many ways manifest, and the obedience we owe him so congruous to the light of reason, that a great part of mankind give testimony to the law of nature: but yet I think it must be allowed that several moral rules may receive from mankind a very general approbation, without either knowing or admitting the true ground of morality; which can only be the will and law of a God, who sees men in the dark, has in his hand rewards and punishments and power enough to call to account the proudest offender. For, God having, by an inseparable connexion, joined virtue and public happiness together, and made the practice thereof necessary to the preservation of society, and visibly beneficial to all with whom the virtuous man has to do; it is no wonder that every one should not only allow, but recommend and magnify those rules to others, from whose observance of them he is sure to reap advantage to himself. He may, out of interest as well as conviction, cry up that for sacred, which, if once trampled on and profaned, he himself cannot be safe nor secure. This, though it takes noth-

ing from the moral and eternal obligation which these rules evidently have, yet it shows that the outward acknowledgment men pay to them in their words proves not that they are innate principles: nay, it proves not so much as that men assent to them inwardly in their own minds, as the inviolable rules of their own practice; since we find that self-interest, and the conveniences of this life, make many men own an outward profession and approbation of them, whose actions sufficiently prove that they very little consider the Lawgiver that prescribed these rules; nor the hell that he has ordained for the punishment of those that transgress them.

7. Men's actions convince us that the rule of virtue is not their internal principle. For, if we will not in civility allow too much sincerity to the professions of most men, but think their actions to be the interpreters of their thoughts, we shall find that they have no such internal veneration for these rules, nor so full a persuasion of their certainty and obligation. The great principle of morality, "To do as one would be done to," is more commended than practised. But the breach of this rule can-

not be a greater vice, than to teach others, that it is no moral rule, nor obligatory, would be thought madness, and contrary to that interest men sacrifice to, when they break it themselves. Perhaps conscience will be urged as checking us for such breaches, and so the internal obligation and establishment of the rule be preserved.

8. Conscience no proof of any innate moral rule. To which I answer, that I doubt not but, without being written on their hearts, many men may, by the same way that they come to the knowledge of other things, come to assent to several moral rules, and be convinced of their obligation. Others also may come to be of the same mind, from their education, company, and customs of their country; which persuasion, however got, will serve to set conscience on work; which is nothing else but our own opinion or judgment of the moral rectitude or pravity of our own actions; and if conscience be a proof of innate principles, contraries may be innate principles; since some men with the same bent of conscience prosecute what others avoid.

9. Instances of enormities practised without remorse.

But I cannot see how any men should ever transgress those moral rules, with confidence and serenity, were they innate, and stamped upon their minds. View but an army at the sacking of a town, and see what observation or sense of moral principles, or what touch of conscience for all the outrages they do. Robberies, murders, rapes, are the sports of men set at liberty from punishment and censure. Have there not been whole nations, and those of the most civilized people, amongst whom the exposing their children, and leaving them in the fields to perish by want or wild beasts has been the practice; as little condemned or scrupled as the begetting them? Do they not still, in some countries, put them into the same graves with their mothers, if they die in childbirth; or despatch them, if a pretended astrologer declares them to have unhappy stars? And are there not places where, at a certain age, they kill or expose their parents, without any remorse at all? In a part of Asia, the sick, when their case comes to be thought desperate, are carried out and laid on the earth before they are dead; and left there, exposed to wind

and weather, to perish without assistance or pity. It is familiar among the Mingrelians, a people professing Christianity, to bury their children alive without scruple. There are places where they eat their own children. The Caribbees were wont to geld their children, on purpose to fat and eat them. And Garcilasso de la Vega tells us of a people in Peru which were wont to fat and eat the children they got on their female captives, whom they kept as concubines for that purpose, and when they were past breeding, the mothers themselves were killed too and eaten. The virtues whereby the Tououpinambos believed they merited paradise, were revenge, and eating abundance of their enemies. They have not so much as a name for God, and have no religion, no worship. The saints who are canonized amongst the Turks, lead lives which one cannot with modesty relate. A remarkable passage to this purpose, out of the voyage of Baumgarten, which is a book not every day to be met with, I shall set down at large, in the language it is published in. Ibi (sc. prope Belbes in Ægypto) *vidimus sanctum unum Saracenicum inter arenarum cumulos,*

ita ut ex utero matris prodiit nudum sedentem. Mos-est, ut didicimus, Mahometistis, ut eos, qui amentes et sine ratione sunt, prosanctis colant et venerentur. Insuper et eos, qui cum diu vitam egerint inquinatissimam, voluntariam demum poenitentiam et paupertatem, sanctitate venerandos deputant. Ejusmodi vero genus hominum libertatem quandam effrenem habent, domos quos volunt intrandi, edendi, bibendi, et quod majus est, concumbendi; ex quo concubitu, si proles secuta fuerit, sancta similiter habetur. His ergo hominibus dum vivunt, magnos exhibent honores; mortuis vero vel templa vel monumenta extruunt amplissima, eosque contingere ac sepelire maximae fortunae ducunt loco. Audivimus haec dicta et dicenda per interpretem a Mucrelo nostro. Insuper sanctum illum, quem eo loco vidimus, publicitus apprime commendari, eum esse hominem sanctum, divinum ac integritate praecipuum; eo quod, nec foeminarum unquam esset, nec puerorum, sed tantummodo asellarum concubitor atque mularum. (Peregr. Baumgarten, 1. ii. c. I. p. 73.) More of the same kind concerning these precious saints amongst the Turks

may be seen in Pietro della Valle, in his letter of the 25th of January, 1616.

Where then are those innate principles of justice, piety, gratitude, equity, chastity? Or where is that universal consent that assures us there are such inbred rules? Murders in duels, when fashion has made them honourable, are committed without remorse of conscience: nay, in many places innocence in this case is the greatest ignominy. And if we look abroad to take a view of men as they are, we shall find that they have remorse, in one place, for doing or omitting that which others, in another place, think they merit by.

10. Men have contrary practical principles. He that will carefully peruse the history of mankind, and look abroad into the several tribes of men, and with indifferency survey their actions, will be able to satisfy himself, that there is scarce that principle of morality to be named, or rule of virtue to be thought on, (those only excepted that are absolutely necessary to hold society together, which commonly too are neglected betwixt distinct societies,) which is not, somewhere or other, slighted and

condemned by the general fashion of whole societies of men, governed by practical opinions and rules of living quite opposite to others.

11. Whole nations reject several moral rules. Here perhaps it will be objected, that it is no argument that the rule is not known, because it is broken. I grant the objection good where men, though they transgress, yet disown not the law; where fear of shame, censure, or punishment carries the mark of some awe it has upon them. But it is impossible to conceive that a whole nation of men should all publicly reject and renounce what every one of them certainly and infallibly knew to be a law; for so they must who have it naturally imprinted on their minds. It is possible men may sometimes own rules of morality which in their private thoughts they do not believe to be true, only to keep themselves in reputation and esteem amongst those who are persuaded of their obligation. But it is not to be imagined that a whole society of men should publicly and professedly disown and cast off a rule which they could not in their own minds but be infallibly certain was a law; nor be

ignorant that all men they should have to do with knew it to be such: and therefore must every one of them apprehend from others all the contempt and abhorrence due to one who professes himself void of humanity: and one who, confounding the known and natural measures of right and wrong, cannot but be looked on as the professed enemy of their peace and happiness. Whatever practical principle is innate, cannot but be known to every one to be just and good. It is therefore little less than a contradiction to suppose, that whole nations of men should, both in their professions and practice, unanimously and universally give the lie to what, by the most invincible evidence, every one of them knew to be true, right, and good. This is enough to satisfy us that no practical rule which is anywhere universally, and with public approbation or allowance, transgressed, can be supposed innate.—But I have something further to add in answer to this objection.

12. The generally allowed breach of a rule, proof that it is not innate. The breaking of a rule, say you, is no argument that it is unknown. I grant it: but the gener-

ally allowed breach of it anywhere, I say, is a proof that it is not innate. For example: let us take any of these rules, which, being the most obvious deductions of human reason, and conformable to the natural inclination of the greatest part of men, fewest people have had the impudence to deny or inconsideration to doubt of. If any can be thought to be naturally imprinted, none, I think, can have a fairer pretence to be innate than this: “Parents, preserve and cherish your children.” When, therefore, you say that this is an innate rule, what do you mean? Either that it is an innate principle which upon all occasions excites and directs the actions of all men; or else, that it is a truth which all men have imprinted on their minds, and which therefore they know and assent to. But in neither of these senses is it innate. First, that it is not a principle which influences all men’s actions, is what I have proved by the examples before cited: nor need we seek so far as Mingrelia or Peru to find instances of such as neglect, abuse, nay, and destroy their children; or look on it only as the more than brutality of some savage and barbarous nations, when

we remember that it was a familiar and uncondemned practice amongst the Greeks and Romans to expose, without pity or remorse, their innocent infants. Secondly, that it is an innate truth, known to all men, is also false. For, "Parents preserve your children," is so far from an innate truth, that it is no truth at all: it being a command, and not a proposition, and so not capable of truth or falsehood. To make it capable of being assented to as true, it must be reduced to some such proposition as this: "It is the duty of parents to preserve their children." But what duty is, cannot be understood without a law; nor a law be known or supposed without a lawmaker, or without reward and punishment; so that it is impossible that this, or any other, practical principle should be innate, i.e. be imprinted on the mind as a duty, without supposing the ideas of God, of law, of obligation, of punishment, of a life after this, innate: for that punishment follows not in this life the breach of this rule, and consequently that it has not the force of a law in countries where the generally allowed practice runs counter to it, is in itself evident. But these

ideas (which must be all of them innate, if anything as a duty be so) are so far from being innate, that it is not every studious or thinking man, much less every one that is born, in whom they are to be found clear and distinct; and that one of them, which of all others seems most likely to be innate, is not so, (I mean the idea of God,) I think, in the next chapter, will appear very evident to any considering man.

13. If men can be ignorant of what is innate, certainty is not described by innate principles. From what has been said, I think we may safely conclude, that whatever practical rule is in any place generally and with allowance broken, cannot be supposed innate; it being impossible that men should, without shame or fear, confidently and serenely, break a rule which they could not but evidently know that God had set up, and would certainly punish the breach of, (which they must, if it were innate,) to a degree to make it a very ill bargain to the transgressor. Without such a knowledge as this, a man can never be certain that anything is his duty. Ignorance or doubt of the law, hopes to escape the knowl-

edge or power of the law-maker, or the like, may make men give way to a present appetite; but let any one see the fault, and the rod by it, and with the transgression, a fire ready to punish it; a pleasure tempting, and the hand of the Almighty visibly held up and prepared to take vengeance, (for this must be the case where any duty is imprinted on the mind,) and then tell me whether it be possible for people with such a prospect, such a certain knowledge as this, wantonly, and without scruple, to offend against a law which they carry about them in indelible characters, and that stares them in the face whilst they are breaking it? Whether men, at the same time that they feel in themselves the imprinted edicts of an Omnipotent Law-maker, can, with assurance and gaiety, slight and trample underfoot his most sacred injunctions? And lastly, whether it be possible that whilst a man thus openly bids defiance to this innate law and supreme Lawgiver, all the bystanders, yea, even the governors and rulers of the people, full of the same sense both of the law and Law-maker, should silently connive, without testifying their dislike or laying the least blame

on it? Principles of actions indeed there are lodged in men's appetites; but these are so far from being innate moral principles, that if they were left to their full swing they would carry men to the overturning of all morality. Moral laws are set as a curb and restraint to these exorbitant desires, which they cannot be but by rewards and punishments that will overbalance the satisfaction any one shall propose to himself in the breach of the law. If, therefore, anything be imprinted on the minds of all men as a law, all men must have a certain and unavoidable knowledge that certain and unavoidable punishment will attend the breach of it. For if men can be ignorant or doubtful of what is innate, innate principles are insisted on, and urged to no purpose; truth and certainty (the things pretended) are not at all secured by them; but men are in the same uncertain floating estate with as without them. An evident indubitable knowledge of unavoidable punishment, great enough to make the transgression very unelidable, must accompany an innate law; unless with an innate law they can suppose an innate Gospel too. I would not

here be mistaken, as if, because I deny an innate law, I thought there were none but positive laws. There is a great deal of difference between an innate law, and a law of nature; between something imprinted on our minds in their very original, and something that we, being ignorant of, may attain to the knowledge of, by the use and due application of our natural faculties. And I think they equally forsake the truth who, running into contrary extremes, either affirm an innate law, or deny that there is a law knowable by the light of nature, i.e. without the help of positive revelation.

14. Those who maintain innate practical principles tell us not what they are. The difference there is amongst men in their practical principles is so evident that I think I need say no more to evince, that it will be impossible to find any innate moral rules by this mark of general assent; and it is enough to make one suspect that the supposition of such innate principles is but an opinion taken up at pleasure; since those who talk so confidently of them are so sparing to tell us which they are. This might with justice be expected from those men

who lay stress upon this opinion; and it gives occasion to distrust either their knowledge or charity, who, declaring that God has imprinted on the minds of men the foundations of knowledge and the rules of living, are yet so little favourable to the information of their neighbours, or the quiet of mankind, as not to point out to them which they are, in the variety men are distracted with. But, in truth, were there any such innate principles there would be no need to teach them. Did men find such innate propositions stamped on their minds, they would easily be able to distinguish them from other truths that they afterwards learned and deduced from them; and there would be nothing more easy than to know what, and how many, they were. There could be no more doubt about their number than there is about the number of our fingers; and it is like then every system would be ready to give them us by tale. But since nobody, that I know, has ventured yet to give a catalogue of them, they cannot blame those who doubt of these innate principles; since even they who require men to believe that there are such innate propo-

sitions, do not tell us what they are. It is easy to foresee, that if different men of different sects should go about to give us a list of those innate practical principles, they would set down only such as suited their distinct hypotheses, and were fit to support the doctrines of their particular schools or churches; a plain evidence that there are no such innate truths. Nay, a great part of men are so far from finding any such innate moral principles in themselves, that, by denying freedom to mankind, and thereby making men no other than bare machines, they take away not only innate, but all moral rules whatsoever, and leave not a possibility to believe any such, to those who cannot conceive how anything can be capable of a law that is not a free agent. And upon that ground they must necessarily reject all principles of virtue, who cannot put morality and mechanism together, which are not very easy to be reconciled or made consistent.

15. Lord Herbert's innate principles examined. When I had written this, being informed that my Lord Herbert had, in his book *De Veritate*, assigned these innate prin-

ciples, I presently consulted him, hoping to find in a man of so great parts, something that might satisfy me in this point, and put an end to my inquiry. In his chapter *De Instinctu Naturali*, p. 72, ed. 1656, I met with these six marks of his *Notitiae*, *Communes*:—1. *Prioritas*. 2. *Independentia*. 3. *Universalitas*. 4. *Certitudo*. 5. *Necessitas*, i.e. as he explains it, *faciunt ad hominis conservationem*. 6. *Modus conformationis*, i.e. *Assensus nulla interposita mora*. And at the latter end of his little treatise *De Religione Laici*, he says this of these innate principles: *Adeo ut non uniuscujusvis religionis confinio arctentur quae ubique vigent veritates. Sunt enim in ipsa mente caelitus descriptae, nullisque traditionibus, sive scriptis, sive non scriptis, obnoxiae, p. 3. And Veritates nostrae catholicae, quae tanquam indubia Dei emata in foro interiori descriptae.*

Thus, having given the marks of the innate principles or common notions, and asserted their being imprinted on the minds of men by the hand of God, he proceeds to set them down, and they are these: 1. *Esse aliquod supremum numen*. 2. *Numen illud coli debere*. 3. *Virtutem*

cum pietate conjunctam optimam esse rationem cultus divini. 4. Resipiscendum esse a peccatis. 5. Dari praemium vel paenam post hanc vitam transactam. Though I allow these to be clear truths, and such as, if rightly explained, a rational creature can hardly avoid giving his assent to, yet I think he is far from proving them innate impressions in foro interiori descriptae. For I must take leave to observe:—

16. These five either not all, or more than all, if there are any. First, that these five propositions are either not all, or more than all, those common notions written on our minds by the finger of God; if it were reasonable to believe any at all to be so written. Since there are other propositions which, even by his own rules, have as just a pretence to such an original, and may be as well admitted for innate principles, as at least some of these five he enumerates, viz. “Do as thou wouldst be done unto.” And perhaps some hundreds of others, when well considered.

17. The supposed marks wanting. Secondly, that all his marks are not to be found in each of his five propositions, viz. his first, second, and third marks agree per-

fectly to neither of them; and the first, second, third, fourth, and sixth marks agree but ill to his third, fourth, and fifth propositions. For, besides that we are assured from history of many men, nay whole nations, who doubt or disbelieve some or all of them, I cannot see how the third, viz. “That virtue joined with piety is the best worship of God,” can be an innate principle, when the name or sound virtue, is so hard to be understood; liable to so much uncertainty in its signification; and the thing it stands for so much contended about and difficult to be known. And therefore this cannot be but a very uncertain rule of human practice, and serve but very little to the conduct of our lives, and is therefore very unfit to be assigned as an innate practical principle.

18. Of little use if they were innate. For let us consider this proposition as to its meaning, (for it is the sense, and not sound, that is and must be the principle or common notion,) viz. “Virtue is the best worship of God,” i.e. is most acceptable to him; which, if virtue be taken, as most commonly it is, for those actions which,

according to the different opinions of several countries, are accounted laudable, will be a proposition so far from being certain, that it will not be true. If virtue be taken for actions conformable to God's will, or to the rule prescribed by God—which is the true and only measure of virtue when virtue is used to signify what is in its own nature right and good—then this proposition, "That virtue is the best worship of God," will be most true and certain, but of very little use in human life: since it will amount to no more but this, viz. "That God is pleased with the doing of what he commands;"—which a man may certainly know to be true, without knowing what it is that God doth command; and so be as far from any rule or principle of his actions as he was before. And I think very few will take a proposition which amounts to no more than this, viz. "That God is pleased with the doing of what he himself commands," for an innate moral principle written on the minds of all men, (however true and certain it may be,) since it teaches so little. Whosoever does so will have reason to think hundreds of propositions innate principles; since there are many

which have as good a title as this to be received for such, which nobody yet ever put into that rank of innate principles.

19. Scarce possible that God should engrave principles in words of uncertain meaning. Nor is the fourth proposition (viz. "Men must repent of their sins") much more instructive, till what those actions are that are meant by sins be set down. For the word peccata, or sins, being put, as it usually is, to signify in general ill actions that will draw punishment upon the doers, what great principle of morality can that be to tell us we should be sorry, and cease to do that which will bring mischief upon us; without knowing what those particular actions are that will do so? Indeed this is a very true proposition, and fit to be incated on and received by those who are supposed to have been taught what actions in all kinds are sins: but neither this nor the former can be imagined to be innate principles; nor to be of any use if they were innate, unless the particular measures and bounds of all virtues and vices were engraven in men's minds, and were innate principles also, which I

think is very much to be doubted. And, therefore, I imagine, it will scarcely seem possible that God should engrave principles in men's minds, in words of uncertain signification, such as virtues and sins, which amongst different men stand for different things: nay, it cannot be supposed to be in words at all, which, being in most of these principles very general, names, cannot be understood but by knowing the particulars comprehended under them. And in the practical instances, the measures must be taken from the knowledge of the actions themselves, and the rules of them,—abstracted from words, and antecedent to the knowledge of names; which rules a man must know, what language soever he chance to learn, whether English or Japan, or if he should learn no language at all, or never should understand the use of words, as happens in the case of dumb and deaf men. When it shall be made out that men ignorant of words, or untaught by the laws and customs of their country, know that it is part of the worship of God, not to kill another man; not to know more women than one; not to procure abortion; not to

expose their children; not to take from another what is his, though we want it ourselves, but on the contrary, relieve and supply his wants; and whenever we have done the contrary we ought to repent, be sorry, and resolve to do so no more;—when I say, all men shall be proved actually to know and allow all these and a thousand other such rules, all of which come under these two general words made use of above, viz. *virtutes et peccata*, virtues and sins, there will be more reason for admitting these and the like, for common notions and practical principles. Yet, after all, universal consent (were there any in moral principles) to truths, the knowledge whereof may be attained otherwise, would scarce prove them to be innate; which is all I contend for.

20. Objection, “innate principles may be corrupted,” answered. Nor will it be of much moment here to offer that very ready but not very material answer, viz. that the innate principles of morality may, by education, and custom, and the general opinion of those amongst whom we converse, be darkened, and at last quite worn out of the minds of men. Which assertion of theirs, if true,

quite takes away the argument of universal consent, by which this opinion of innate principles is endeavoured to be proved; unless those men will think it reasonable that their private persuasions, or that of their party, should pass for universal consent;—a thing not unfrequently done, when men, presuming themselves to be the only masters of right reason, cast by the votes and opinions of the rest of mankind as not worthy the reckoning. And then their argument stands thus:—"The principles which all mankind allow for true, are innate; those that men of right reason admit, are the principles allowed by all mankind; we, and those of our mind, are men of reason; therefore, we agreeing, our principles are innate;"—which is a very pretty way of arguing, and a short cut to infallibility. For otherwise it will be very hard to understand how there be some principles which all men do acknowledge and agree in; and yet there are none of those principles which are not, by depraved custom and ill education, blotted out of the minds of many men: which is to say, that all men admit, but yet many men do deny and dissent from them. And

indeed the supposition of such first principles will serve us to very little purpose; and we shall be as much at a loss with as without them, if they may, by any human power—such as the will of our teachers, or opinions of our companions—be altered or lost in us: and notwithstanding all this boast of first principles and innate light, we shall be as much in the dark and uncertainty as if there were no such thing at all: it being all one to have no rule, and one that will warp any way; or amongst various and contrary rules, not to know which is the right. But concerning innate principles, I desire these men to say, whether they can or cannot, by education and custom, be blurred and blotted out; if they cannot, we must find them in all mankind alike, and they must be clear in everybody; and if they may suffer variation from adventitious notions, we must then find them clearest and most perspicuous nearest the fountain, in children and illiterate people, who have received least impression from foreign opinions. Let them take which side they please, they will certainly find it inconsistent with visible matter of fact and daily observation.

21. Contrary principles in the world. I easily grant that there are great numbers of opinions which, by men of different countries, educations, and tempers, are received and embraced as first and unquestionable principles; many whereof, both for their absurdity as well as oppositions to one another, it is impossible should be true. But yet all those propositions, how remote soever from reason, are so sacred somewhere or other, that men even of good understanding in other matters, will sooner part with their lives, and whatever is dearest to them, than suffer themselves to doubt, or others to question, the truth of them.

22. How men commonly come by their principles. This, however strange it may seem, is that which every day's experience confirms; and will not, perhaps, appear so wonderful, if we consider the ways and steps by which it is brought about; and how really it may come to pass, that doctrines that have been derived from no better original than the superstition of a nurse, or the authority of an old woman, may, by length of time and consent of neighbours, grow up to the dignity of principles

in religion or morality. For such, who are careful (as they call it) to principle children well, (and few there be who have not a set of those principles for them, which they believe in,) instil into the unwary, and as yet unprejudiced, understanding, (for white paper receives any characters,) those doctrines they would have them retain and profess. These being taught them as soon as they have any apprehension; and still as they grow up confirmed to them, either by the open profession or tacit consent of all they have to do with; or at least by those of whose wisdom, knowledge, and piety they have an opinion, who never suffer those propositions to be otherwise mentioned but as the basis and foundation on which they build their religion and manners, come, by these means, to have the reputation of unquestionable, self-evident, and innate truths.

23. Principles supposed innate because we do not remember when we began to hold them. To which we may add, that when men so instructed are grown up, and reflect on their own minds, they cannot find anything more ancient there than those opinions, which were

taught them before their memory began to keep a register of their actions, or date the time when any new thing appeared to them; and therefore make no scruple to conclude, that those propositions of whose knowledge they can find in themselves no original, were certainly the impress of God and nature upon their minds, and not taught them by any one else. These they entertain and submit to, as many do to their parents with veneration; not because it is natural; nor do children do it where they are not so taught; but because, having been always so educated, and having no remembrance of the beginning of this respect, they think it is natural.

24. How such principles come to be held. This will appear very likely, and almost unavoidable to come to pass, if we consider the nature of mankind and the constitution of human affairs; wherein most men cannot live without employing their time in the daily labours of their callings; nor be at quiet in their minds without some foundation or principle to rest their thoughts on. There is scarcely any one so floating and superficial in

his understanding, who hath not some revered propositions, which are to him the principles on which he bottoms his reasonings, and by which he judgeth of truth and falsehood, right and wrong; which some, wanting skill and leisure, and others the inclination, and some being taught that they ought not to examine, there are few to be found who are not exposed by their ignorance, laziness, education, or precipitancy, to take them upon trust.

25. Further explained. This is evidently the case of all children and young folk; and custom, a greater power than nature, seldom failing to make them worship for divine what she hath inured them to bow their minds and submit their understandings to, it is no wonder that grown men, either perplexed in the necessary affairs of life, or hot in the pursuit of pleasures, should not seriously sit down to examine their own tenets; especially when one of their principles is, that principles ought not to be questioned. And had men leisure, parts, and will, who is there almost that dare shake the foundations of all his past thoughts and actions, and endure

to bring upon himself the shame of having been a long time wholly in mistake and error? Who is there hardy enough to contend with the reproach which is everywhere prepared for those who dare venture to dissent from the received opinions of their country or party? And where is the man to be found that can patiently prepare himself to bear the name of whimsical, sceptical, or atheist; which he is sure to meet with, who does in the least scruple any of the common opinions? And he will be much more afraid to question those principles, when he shall think them, as most men do, the standards set up by God in his mind, to be the rule and touchstone of all other opinions. And what can hinder him from thinking them sacred, when he finds them the earliest of all his own thoughts, and the most revered by others?

26. A worship of idols. It is easy to imagine how, by these means, it comes to pass that men worship the idols that have been set up in their minds; grow fond of the notions they have been long acquainted with there; and stamp the characters of divinity upon absurdities

and errors; become zealous votaries to bulls and mon-keys, and contend too, fight, and die in defence of their opinions. *Dum solos credit habendos esse deos, quos ipse colit.* For, since the reasoning faculties of the soul, which are almost constantly, though not always warily nor wisely employed, would not know how to move, for want of a foundation and footing, in most men, who through laziness or avocation do not, or for want of time, or true helps, or for other causes, cannot penetrate into the principles of knowledge, and trace truth to its fountain and original, it is natural for them, and almost unavoidable, to take up with some borrowed principles; which being reputed and presumed to be the evident proofs of other things, are thought not to need any other proof themselves. Whoever shall receive any of these into his mind, and entertain them there with the reverence usually paid to principles, never venturing to examine them, but accustoming himself to believe them, because they are to be believed, may take up, from his education and the fashions of his country, any absurdity for innate principles; and by long poring on the

same objects, so dim his sight as to take monsters lodged in his own brain for the images of the Deity, and the workmanship of his hands.

27. Principles must be examined. By this progress, how many there are who arrive at principles which they believe innate may be easily observed, in the variety of opposite principles held and contended for by all sorts and degrees of men. And he that shall deny this to be the method wherein most men proceed to the assurance they have of the truth and evidence of their principles, will perhaps find it a hard matter any other way to account for the contrary tenets, which are firmly believed, confidently asserted, and which great numbers are ready at any time to seal with their blood. And, indeed, if it be the privilege of innate principles to be received upon their own authority, without examination, I know not what may not be believed, or how any one's principles can be questioned. If they may and ought to be examined and tried, I desire to know how first and innate principles can be tried; or at least it is reasonable to demand the marks and characters whereby the genu-

ine innate principles may be distinguished from others: that so, amidst the great variety of pretenders, I may be kept from mistakes in so material a point as this. When this is done, I shall be ready to embrace such welcome and useful propositions; and till then I may with modesty doubt; since I fear universal consent, which is the only one produced, will scarcely prove a sufficient mark to direct my choice, and assure me of any innate principles.

From what has been said, I think it past doubt, that there are no practical principles wherein all men agree; and therefore none innate.

Chapter III

**Other considerations concerning Innate Principles,
both Speculative and Practical**

1. Principles not innate, unless their ideas be innate. Had those who would persuade us that there are innate principles not taken them together in gross, but considered separately the parts out of which those propositions are made, they would not, perhaps, have been so forward to believe they were innate. Since, if the ideas which made up those truths were not, it was impossible that the propositions made up of them should be innate, or our knowledge of them be born with us. For, if the ideas be not innate, there was a time when the mind was without those principles; and then they will not be innate, but be derived from some other original. For, where the ideas themselves are not, there can be no knowledge, no assent, no mental or verbal propositions about them.

2. Ideas, especially those belonging to principles, not born with children. If we will attentively consider new-

born children, we shall have little reason to think that they bring many ideas into the world with them. For, bating perhaps some faint ideas of hunger, and thirst, and warmth, and some pains, which they may have felt in the womb, there is not the least appearance of any settled ideas at all in them; especially of ideas answering the terms which make up those universal propositions that are esteemed innate principles. One may perceive how, by degrees, afterwards, ideas come into their minds; and that they get no more, nor other, than what experience, and the observation of things that come in their way, furnish them with; which might be enough to satisfy us that they are not original characters stamped on the mind.

3. "Impossibility" and "identity" not innate ideas. "It is impossible for the same thing to be, and not to be," is certainly (if there be any such) an innate principle. But can any one think, or will any one say, that "impossibility" and "identity" are two innate ideas? Are they such as all mankind have, and bring into the world with them? And are they those which are the first in children, and

antecedent to all acquired ones? If they are innate, they must needs be so. Hath a child an idea of impossibility and identity, before it has of white or black, sweet or bitter? And is it from the knowledge of this principle that it concludes, that wormwood rubbed on the nipple hath not the same taste that it used to receive from thence? Is it the actual knowledge of impossible *est idem esse, et non esse*, that makes a child distinguish between its mother and a stranger; or that makes it fond of the one and flee the other? Or does the mind regulate itself and its assent by ideas that it never yet had? Or the understanding draw conclusions from principles which it never yet knew or understood? The names impossibility and identity stand for two ideas, so far from being innate, or born with us, that I think it requires great care and attention to form them right in our understandings. They are so far from being brought into the world with us, so remote from the thoughts of infancy and childhood, that I believe, upon examination it will be found that many grown men want them.

4. "Identity," an idea not innate. If identity (to in-

stance that alone) be a native impression, and consequently so clear and obvious to us that we must needs know it even from our cradles, I would gladly be resolved by any one of seven, or seventy years old, whether a man, being a creature consisting of soul and body, be the same man when his body is changed? Whether Euphorbus and Pythagoras, having had the same soul, were the same men, though they lived several ages asunder? Nay, whether the cock too, which had the same soul, were not the same with both of them? Whereby, perhaps, it will appear that our idea of sameness is not so settled and clear as to deserve to be thought innate in us. For if those innate ideas are not clear and distinct, so as to be universally known and naturally agreed on, they cannot be subjects of universal and undoubted truths, but will be the unavoidable occasion of perpetual uncertainty. For, I suppose every one's idea of identity will not be the same that Pythagoras and thousands of his followers have. And which then shall be true? Which innate? Or are there two different ideas of identity, both innate?

5. What makes the same man? Nor let any one think that the questions I have here proposed about the identity of man are bare empty speculations; which, if they were, would be enough to show, that there was in the understandings of men no innate idea of identity. He that shall with a little attention reflect on the resurrection, and consider that divine justice will bring to judgment, at the last day, the very same persons, to be happy or miserable in the other, who did well or ill in this life, will find it perhaps not easy to resolve with himself, what makes the same man, or wherein identity consists; and will not be forward to think he, and every one, even children themselves, have naturally a clear idea of it.

6. Whole and part, not innate ideas. Let us examine that principle of mathematics, viz. that the whole is bigger than a part. This, I take it, is reckoned amongst innate principles. I am sure it has as good a title as any to be thought so; which yet nobody can think it to be, when he considers [that] the ideas it comprehends in it, whole and part, are perfectly relative; but the positive ideas to which they properly and immediately belong

are extension and number, of which alone whole and part are relations. So that if whole and part are innate ideas, extension and number must be so too; it being impossible to have an idea of a relation, without having any at all of the thing to which it belongs, and in which it is founded. Now, whether the minds of men have naturally imprinted on them the ideas of extension and number, I leave to be considered by these who are the patrons of innate principles.

7. Idea of worship not innate. That God is to be worshipped, is, without doubt, as great a truth as any that can enter into the mind of man, and deserves the first place amongst all practical principles. But yet it can by no means be thought innate, unless the ideas of God and worship are innate. That the idea the term worship stands for is not in the understanding of children, and a character stamped on the mind in its first original, I think will be easily granted, by any one that considers how few there be amongst grown men who have a clear and distinct notion of it. And, I suppose, there cannot be anything more ridiculous than to say, that children

have this practical principle innate, "That God is to be worshipped," and yet that they know not what that worship of God is, which is their duty. But to pass by this.

8. Idea of God not innate. If any idea can be imagined innate, the idea of God may, of all others, for many reasons, be thought so; since it is hard to conceive how there should be innate moral principles, without an innate idea of a Deity. Without a notion of a law-maker, it is impossible to have a notion of a law, and an obligation to observe it. Besides the atheists taken notice of amongst the ancients, and left branded upon the records of history, hath not navigation discovered, in these later ages, whole nations, at the bay of Soldania, in Brazil, [in Boranday,] and in the Caribbee islands, &c., amongst whom there was to be found no notion of a God, no religion? Nicholaus del Techo, in *Literis ex Paraquaria, de Caiguarum Conversione*, has these words: *Reperi eam gentem nullum nomen habere quod Deum, et hominis animam significet; nulla sacra habet, nulla idola*. These are instances of nations where uncultivated nature has

been left to itself, without the help of letters and discipline, and the improvements of arts and sciences. But there are others to be found who have enjoyed these in a very great measure, who yet, for want of a due application of their thoughts this way, want the idea and knowledge of God. It will, I doubt not, be a surprise to others, as it was to me, to find the Siamites of this number. But for this, let them consult the King of France's late envoy thither, who gives no better account of the Chinese themselves. And if we will not believe La Loubere, the missionaries of China, even the Jesuits themselves, the great encomiasts of the Chinese, do all to a man agree, and will convince us, that the sect of the literari, or learned, keeping to the old religion of China, and the ruling party there, are all of them atheists. Vid. Navarette, in the *Collection of Voyages*, vol. i., and *Historia Cultus Sinensium*. And perhaps, if we should with attention mind the lives and discourses of people not so far off, we should have too much reason to fear, that many, in more civilized countries, have no very strong and clear impressions of a Deity upon their minds,

and that the complaints of atheism made from the pulpit are not without reason. And though only some profigate wretches own it too barefacedly now; yet perhaps we should hear more than we do of it from others, did not the fear of the magistrate's sword, or their neighbour's censure, tie up people's tongues; which, were the apprehensions of punishment or shame taken away, would as openly proclaim their atheism as their lives do.

9. The name of God not universal or obscure in meaning. But had all mankind everywhere a notion of a God, (whereof yet history tells us the contrary,) it would not from thence follow, that the idea of him was innate. For, though no nation were to be found without a name, and some few dark notions of him, yet that would not prove them to be natural impressions on the mind; no more than the names of fire, or the sun, heat, or number, do prove the ideas they stand for to be innate; because the names of those things, and the ideas of them, are so universally received and known amongst mankind. Nor, on the contrary, is the want of such a name, or the absence of such a notion out of men's

minds, any argument against the being of a God; any more than it would be a proof that there was no loadstone in the world, because a great part of mankind had neither a notion of any such thing nor a name for it; or be any show of argument to prove that there are no distinct and various species of angels, or intelligent beings above us, because we have no ideas of such distinct species, or names for them. For, men being furnished with words, by the common language of their own countries, can scarce avoid having some kind of ideas of those things whose names those they converse with have occasion frequently to mention to them. And if they carry with it the notion of excellency, greatness, or something extraordinary; if apprehension and concernment accompany it; if the fear of absolute and irresistible power set it on upon the mind,—the idea is likely to sink the deeper, and spread the further; especially if it be such an idea as is agreeable to the common light of reason, and naturally deducible from every part of our knowledge, as that of a God is. For the visible marks of extraordinary wisdom and power appear so plainly in all

the works of the creation, that a rational creature, who will but seriously reflect on them, cannot miss the discovery of a Deity. And the influence that the discovery of such a Being must necessarily have on the minds of all that have but once heard of it is so great, and carries such a weight of thought and communication with it, that it seems stranger to me that a whole nation of men should be anywhere found so brutish as to want the notion of a God, than that they should be without any notion of numbers, or fire.

10. Ideas of God and idea of fire. The name of God being once mentioned in any part of the world, to express a superior, powerful, wise, invisible Being, the suitableness of such a notion to the principles of common reason, and the interest men will always have to mention it often, must necessarily spread it far and wide; and continue it down to all generations: though yet the general reception of this name, and some imperfect and unsteady notions conveyed thereby to the unthinking part of mankind, prove not the idea to be innate; but only that they who made the discovery had made a right

use of their reason, thought maturely of the causes of things, and traced them to their original; from whom other less considering people having once received so important a notion, it could not easily be lost again.

11. Idea of God not innate. This is all could be inferred from the notion of a God, were it to be found universally in all the tribes of mankind, and generally acknowledged, by men grown to maturity in all countries. For the generality of the acknowledging of a God, as I imagine, is extended no further than that; which, if it be sufficient to prove the idea of God innate, will as well prove the idea of fire innate; since I think it may be truly said, that there is not a person in the world who has a notion of a God, who has not also the idea of fire. I doubt not but if a colony of young children should be placed in an island where no fire was, they would certainly neither have any notion of such a thing, nor name for it, how generally soever it were received and known in all the world besides; and perhaps too their apprehensions would be as far removed from any name, or notion, of a God, till some one amongst them had

employed his thoughts to inquire into the constitution and causes of things, which would easily lead him to the notion of a God; which having once taught to others, reason, and the natural propensity of their own thoughts, would afterwards propagate, and continue amongst them.

12. Suitable to God's goodness, that all men should have an idea of Him, therefore naturally imprinted by Him, answered. Indeed it is urged, that it is suitable to the goodness of God, to imprint upon the minds of men characters and notions of himself, and not to leave them in the dark and doubt in so grand a concernment; and also, by that means, to secure to himself the homage and veneration due from so intelligent a creature as man; and therefore he has done it. This argument, if it be of any force, will prove much more than those who use it in this case expect from it. For, if we may conclude that God hath done for men all that men shall judge is best for them, because it is suitable to his goodness so to do, it will prove, not only that God has imprinted on the minds of men an idea of himself, but that he hath plainly stamped there, in fair characters, all

that men ought to know or believe of him; all that they ought to do in obedience to his will; and that he hath given them a will and affections conformable to it. This, no doubt, every one will think better for men, than that they should, in the dark, grope after knowledge, as St. Paul tells us all nations did after God (Acts 17. 27); than that their wills should clash with their understandings, and their appetites cross their duty. The Romanists say it is best for men, and so suitable to the goodness of God, that there should be an infallible judge of controversies on earth; and therefore there is one. And I, by the same reason, say it is better for men that every man himself should be infallible. I leave them to consider, whether, by the force of this argument, they shall think that every man is so. I think it a very good argument to say,—the infinitely wise God hath made it so; and therefore it is best. But it seems to me a little too much confidence of our own wisdom to say,—“I think it best; and therefore God hath made it so.” And in the matter in hand, it will be in vain to argue from such a topic, that God hath done so, when certain experience shows

us that he hath not. But the goodness of God hath not been wanting to men, without such original impressions of knowledge or ideas stamped on the mind; since he hath furnished man with those faculties which will serve for the sufficient discovery of all things requisite to the end of such a being; and I doubt not but to show, that a man, by the right use of his natural abilities, may, without any innate principles, attain a knowledge of a God, and other things that concern him. God having endued man with those faculties of knowledge which he hath, was no more obliged by his goodness to plant those innate notions in his mind, than that, having given him reason, hands, and materials, he should build him bridges or houses,—which some people in the world, however of good parts, do either totally want, or are but ill provided of, as well as others are wholly without ideas of God and principles of morality, or at least have but very ill ones; the reason in both cases, being, that they never employed their parts, faculties, and powers industriously that way, but contented themselves with the opinions, fashions, and things of their country, as

they found them, without looking any further. Had you or I been born at the Bay of Soldania, possibly our thoughts and notions had not exceeded those brutish ones of the Hottentots that inhabit there. And had the Virginia king Apochancana been educated in England, he had been perhaps as knowing a divine, and as good a mathematician as any in it; the difference between him and a more improved Englishman lying barely in this, that the exercise of his faculties was bounded within the ways, modes, and notions of his own country, and never directed to any other or further inquiries. And if he had not any idea of a God, it was only because he pursued not those thoughts that would have led him to it.

13. Ideas of God various in different men. I grant that if there were any ideas to be found imprinted on the minds of men, we have reason to expect it should be the notion of his Maker, as a mark God set on his own workmanship, to mind man of his dependence and duty; and that herein should appear the first instances of human knowledge. But how late is it before any such notion is

discoverable in children? And when we find it there, how much more does it resemble the opinion and notion of the teacher, than represent the true God? He that shall observe in children the progress whereby their minds attain the knowledge they have, will think that the objects they do first and most familiarly converse with are those that make the first impressions on their understandings; nor will he find the least footsteps of any other. It is easy to take notice how their thoughts enlarge themselves, only as they come to be acquainted with a greater variety of sensible objects; to retain the ideas of them in their memories; and to get the skill to compound and enlarge them, and several ways put them together. How, by these means, they come to frame in their minds an idea men have of a Deity, I shall hereafter show.

14. Contrary and inconsistent ideas of God under the same name. Can it be thought that the ideas men have of God are the characters and marks of himself, engraven in their minds by his own finger, when we see that, in the same country, under one and the same name, men

have far different, nay often contrary and inconsistent ideas and conceptions of him? Their agreeing in a name, or sound, will scarce prove an innate notion of him.

15. Gross ideas of God. What true or tolerable notion of a Deity could they have, who acknowledged and worshipped hundreds? Every deity that they owned above one was an infallible evidence of their ignorance of Him, and a proof that they had no true notion of God, where unity, infinity, and eternity were excluded. To which, if we add their gross conceptions of corporeity, expressed in their images and representations of their deities; the amours, marriages, copulations, lusts, quarrels, and other mean qualities attributed by them to their gods; we shall have little reason to think that the heathen world, i.e. the greatest part of mankind, had such ideas of God in their minds as he himself, out of care that they should not be mistaken about him, was author of. And this universality of consent, so much argued, if it prove any native impressions, it will be only this:—that God imprinted on the minds of all men speaking the same language, a name for himself, but not any idea; since those

people who agreed in the name, had, at the same time, far different apprehensions about the thing signified. If they say that the variety of deities worshipped by the heathen world were but figurative ways of expressing the several attributes of that incomprehensible Being, or several parts of his providence, I answer: what they might be in the original I will not here inquire; but that they were so in the thoughts of the vulgar I think nobody will affirm. And he that will consult the voyage of the Bishop of Beryte, c. 13, (not to mention other testimonies,) will find that the theology of the Siamites professedly owns a plurality of gods: or, as the Abbe de Choisy more judiciously remarks in his *Journal du Voyage de Siam*, 107/177, it consists properly in acknowledging no God at all.

16. Idea of God not innate although wise men of all nations come to have it. If it be said, that wise men of all nations came to have true conceptions of the unity and infinity of the Deity, I grant it. But then this, First, excludes universality of consent in anything but the name; for those wise men being very few, perhaps one of a thousand, this universality is very narrow.

Secondly, it seems to me plainly to prove, that the truest and best notions men have of God were not imprinted, but acquired by thought and meditation, and a right use of their faculties: since the wise and considerate men of the world, by a right and careful employment of their thoughts and reason, attained true notions in this as well as other things; whilst the lazy and inconsiderate part of men, making far the greater number, took up their notions by chance, from common tradition and vulgar conceptions, without much beating their heads about them. And if it be a reason to think the notion of God innate, because all wise men had it, virtue too must be thought innate; for that also wise men have always had.

17. Odd, low, and pitiful ideas of God common among men. This was evidently the case of all Gentilism. Nor hath even amongst Jews, Christians, and Mahometans, who acknowledged but one God, this doctrine, and the care taken in those nations to teach men to have true notions of a God, prevailed so far as to make men to have the same and the true ideas of him. How many

even amongst us, will be found upon inquiry to fancy him in the shape of a man sitting in heaven; and to have many other absurd and unfit conceptions of him? Christians as well as Turks have had whole sects owning and contending earnestly for it,—that the Deity was corporeal, and of human shape: and though we find few now amongst us who profess themselves Anthropomorphites, (though some I have met with that own it,) yet I believe he that will make it his business may find amongst the ignorant and uninstructed Christians many of that opinion. Talk but with country people, almost of any age, or young people almost of any condition, and you shall find that, though the name of God be frequently in their mouths, yet the notions they apply this name to are so odd, low, and pitiful, that nobody can imagine they were taught by a rational man; much less that they were characters written by the finger of God himself. Nor do I see how it derogates more from the goodness of God, that he has given us minds unfurnished with these ideas of himself, than that he hath sent us into the world with bodies unclothed; and

that there is no art or skill born with us. For, being fitted with faculties to attain these, it is want of industry and consideration in us, and not of bounty in him, if we have them not. It is as certain that there is a God, as that the opposite angles made by the intersection of two straight lines are equal. There was never any rational creature that set himself sincerely to examine the truth of these propositions that could fail to assent to them; though yet it be past doubt that there are many men, who, having not applied their thoughts that way, are ignorant both of the one and the other. If any one think fit to call this (which is the utmost of its extent) universal consent, such an one I easily allow; but such an universal consent as this proves not the idea of God, any more than it does the idea of such angles, innate.

18. If the idea of God be not innate, no other can be supposed innate. Since then though the knowledge of a God be the most natural discovery of human reason, yet the idea of him is not innate, as I think is evident from what has been said; I imagine there will be scarce any other idea found that can pretend to it. Since if God

hath set any impression, any character, on the understanding of men, it is most reasonable to expect it should have been some clear and uniform idea of Himself; as far as our weak capacities were capable to receive so incomprehensible and infinite an object. But our minds being at first void of that idea which we are most concerned to have, it is a strong presumption against all other innate characters. I must own, as far as I can observe, I can find none, and would be glad to be informed by any other.

19. Idea of substance not innate. I confess there is another idea which would be of general use for mankind to have, as it is of general talk as if they had it; and that is the idea of substance; which we neither have nor can have by sensation or reflection. If nature took care to provide us any ideas, we might well expect they should be such as by our own faculties we cannot procure to ourselves; but we see, on the contrary, that since, by those ways whereby other ideas are brought into our minds, this is not, we have no such clear idea at all; and therefore signify nothing by the word substance but

only an uncertain supposition of we know not what, i.e. of something whereof we have no [particular distinct positive] idea, which we take to be the substratum, or support, of those ideas we do know.

20. No propositions can be innate, since no ideas are innate. Whatever then we talk of innate, either speculative or practical, principles, it may with as much probability be said, that a man hath £100 sterling in his pocket, and yet denied that he hath there either penny, shilling, crown, or other coin out of which the sum is to be made up; as to think that certain propositions are innate when the ideas about which they are can by no means be supposed to be so. The general reception and assent that is given doth not at all prove, that the ideas expressed in them are innate; for in many cases, however the ideas came there, the assent to words expressing the agreement or disagreement of such ideas, will necessarily follow. Every one that hath a true idea of God and worship, will assent to this proposition, "That God is to be worshipped," when expressed in a language he understands; and every rational man that hath not

thought on it to-day, may be ready to assent to this proposition to-morrow; and yet millions of men may be well supposed to want one or both those ideas to-day. For, if we will allow savages, and most country people, to have ideas of God and worship, (which conversation with them will not make one forward to believe,) yet I think few children can be supposed to have those ideas, which therefore they must begin to have some time or other; and then they will also begin to assent to that proposition, and make very little question of it ever after. But such an assent upon hearing, no more proves the ideas to be innate, than it does that one born blind (with cataracts which will be couched to-morrow) had the innate ideas of the sun, or light, or saffron, or yellow; because, when his sight is cleared, he will certainly assent to this proposition, "That the sun is lucid, or that saffron is yellow." And therefore, if such an assent upon hearing cannot prove the ideas innate, it can much less the propositions made up of those ideas. If they have any innate ideas, I would be glad to be told what, and how many, they are.

21. No innate ideas in the memory. To which let me add: if there be any innate ideas, any ideas in the mind which the mind does not actually think on, they must be lodged in the memory; and from thence must be brought into view by remembrance; i.e. must be known, when they are remembered, to have been perceptions in the mind before; unless remembrance can be without remembrance. For, to remember is to perceive anything with memory, or with a consciousness that it was perceived or known before. Without this, whatever idea comes into the mind is new, and not remembered; this consciousness of its having been in the mind before, being that which distinguishes remembering from all other ways of thinking. Whatever idea was never perceived by the mind was never in the mind. Whatever idea is in the mind, is, either an actual perception, or else, having been an actual perception, is so in the mind that, by the memory, it can be made an actual perception again. Whenever there is the actual perception of any idea without memory, the idea appears perfectly new and unknown before to the understanding. When-

ever the memory brings any idea into actual view, it is with a consciousness that it had been there before, and was not wholly a stranger to the mind. Whether this be not so, I appeal to every one's observation. And then I desire an instance of an idea, pretended to be innate, which (before any impression of it by ways hereafter to be mentioned) any one could revive and remember, as an idea he had formerly known; without which consciousness of a former perception there is no remembrance; and whatever idea comes into the mind without that consciousness is not remembered, or comes not out of the memory, nor can be said to be in the mind before that appearance. For what is not either actually in view or in the memory, is in the mind no way at all, and is all one as if it had never been there. Suppose a child had the use of his eyes till he knows and distinguishes colours; but then cataracts shut the windows, and he is forty or fifty years perfectly in the dark; and in that time perfectly loses all memory of the ideas of colours he once had. This was the case of a blind man I once talked with, who lost his sight by the small-pox

when he was a child, and had no more notion of colours than one born blind. I ask whether any one can say this man had then any ideas of colours in his mind, any more than one born blind? And I think nobody will say that either of them had in his mind any ideas of colours at all. His cataracts are couched, and then he has the ideas (which he remembers not) of colours, *de novo*, by his restored sight, conveyed to his mind, and that without any consciousness of a former acquaintance. And these now he can revive and call to mind in the dark. In this case all these ideas of colours, which, when out of view, can be revived with a consciousness of a former acquaintance, being thus in the memory, are said to be in the mind. The use I make of this is,—that whatever idea, being not actually in view, is in the mind, is there only by being in the memory; and if it be not in the memory, it is not in the mind; and if it be in the memory, it cannot by the memory be brought into actual view without a perception that it comes out of the memory; which is this, that it had been known before, and is now remembered. If therefore there be any innate ideas,

they must be in the memory, or else nowhere in the mind; and if they be in the memory, they can be revived without any impression from without; and whenever they are brought into the mind they are remembered, i.e. they bring with them a perception of their not being wholly new to it. This being a constant and distinguishing difference between what is, and what is not in the memory, or in the mind;—that what is not in the memory, whenever it appears there, appears perfectly new and unknown before; and what is in the memory, or in the mind, whenever it is suggested by the memory, appears not to be new, but the mind finds it in itself, and knows it was there before. By this it may be tried whether there be any innate ideas in the mind before impression from sensation or reflection. I would fain meet with the man who, when he came to the use of reason, or at any other time, remembered any of them; and to whom, after he was born, they were never new. If any one will say, there are ideas in the mind that are not in the memory, I desire him to explain himself, and make what he says intelligible.

22. Principles not innate, because of little use or little certainty. Besides what I have already said, there is another reason why I doubt that neither these nor any other principles are innate. I that am fully persuaded that the infinitely wise God made all things in perfect wisdom, cannot satisfy myself why he should be supposed to print upon the minds of men some universal principles; whereof those that are pretended innate, and concern speculation, are of no great use; and those that concern practice, not self-evident; and neither of them distinguishable from some other truths not allowed to be innate. For, to what purpose should characters be graven on the mind by the finger of God, which are not clearer there than those which are afterwards introduced, or cannot be distinguished from them? If any one thinks there are such innate ideas and propositions, which by their clearness and usefulness are distinguishable from all that is adventitious in the mind and acquired, it will not be a hard matter for him to tell us which they are; and then every one will be a fit judge whether they be so or no. Since if there be such innate

ideas and impressions, plainly different from all other perceptions and knowledge, every one will find it true in himself of the evidence of these supposed innate maxims, I have spoken already: of their usefulness I shall have occasion to speak more hereafter.

23. Difference of men's discoveries depends upon the different application of their faculties. To conclude: some ideas forwardly offer themselves to all men's understanding; and some sorts of truths result from any ideas, as soon as the mind puts them into propositions: other truths require a train of ideas placed in order, a due comparing of them, and deductions made with attention, before they can be discovered and assented to. Some of the first sort, because of their general and easy reception, have been mistaken for innate: but the truth is, ideas and notions are no more born with us than arts and sciences; though some of them indeed offer themselves to our faculties more readily than others; and therefore are more generally received: though that too be according as the organs of our bodies and powers of our minds happen to be employed; God having fitted

men with faculties and means to discover, receive, and retain truths, according as they are employed. The great difference that is to be found in the notions of mankind is, from the different use they put their faculties to. Whilst some (and those the most) taking things upon trust, misemploy their power of assent, by lazily enslaving their minds to the dictates and dominion of others, in doctrines which it is their duty carefully to examine, and not blindly, with an implicit faith, to swallow; others, employing their thoughts only about some few things, grow acquainted sufficiently with them, attain great degrees of knowledge in them, and are ignorant of all other, having never let their thoughts loose in the search of other inquiries. Thus, that the three angles of a triangle are quite equal to two right ones is a truth as certain as anything can be, and I think more evident than many of those propositions that go for principles; and yet there are millions, however expert in other things, who know not this at all, because they never set their thoughts on work about such angles. And he that certainly knows this proposition may yet be utterly igno-

rant of the truth of other propositions, in mathematics itself, which are as clear and evident as this; because, in his search of those mathematical truths, he stopped his thoughts short and went not so far. The same may happen concerning the notions we have of the being of a Deity. For, though there be no truth which a man may more evidently make out to himself than the existence of a God, yet he that shall content himself with things as he finds them in this world, as they minister to his pleasures and passions, and not make inquiry a little further into their causes, ends, and admirable contrivances, and pursue the thoughts thereof with diligence and attention, may live long without any notion of such a Being. And if any person hath by talk put such a notion into his head, he may perhaps believe it; but if he hath never examined it, his knowledge of it will be no perfecter than his, who having been told, that the three angles of a triangle are equal to two right ones, takes it upon trust, without examining the demonstration; and may yield his assent as a probable opinion, but hath no knowledge of the truth of it; which yet his faculties, if carefully employed,

were able to make clear and evident to him. But this only, by the by, to show how much our knowledge depends upon the right use of those powers nature hath bestowed upon us, and how little upon such innate principles as are in vain supposed to be in all mankind for their direction; which all men could not but know if they were there, or else they would be there to no purpose. And which since all men do not know, nor can distinguish from other adventitious truths, we may well conclude there are no such.

24. Men must think and know for themselves. What censure doubting thus of innate principles may deserve from men, who will be apt to call it pulling up the old foundations of knowledge and certainty, I cannot tell;— I persuade myself at least that the way I have pursued, being conformable to truth, lays those foundations surer. This I am certain, I have not made it my business either to quit or follow any authority in the ensuing Discourse. Truth has been my only aim; and wherever that has appeared to lead, my thoughts have impartially followed, without minding whether the footsteps of any other lay

that way or not. Not that I want a due respect to other men's opinions; but, after all, the greatest reverence is due to truth: and I hope it will not be thought arrogance to say, that perhaps we should make greater progress in the discovery of rational and contemplative knowledge, if we sought it in the fountain, in the consideration of things themselves; and made use rather of our own thoughts than other men's to find it. For I think we may as rationally hope to see with other men's eyes, as to know by other men's understandings. So much as we ourselves consider and comprehend of truth and reason, so much we possess of real and true knowledge. The floating of other men's opinions in our brains, makes us not one jot the more knowing, though they happen to be true. What in them was science, is in us but opiniatrety; whilst we give up our assent only to reverend names, and do not, as they did, employ our own reason to understand those truths which gave them reputation. Aristotle was certainly a knowing man, but nobody ever thought him so because he blindly embraced, and confidently vented the opinions of another.

And if the taking up of another's principles, without examining them, made not him a philosopher, I suppose it will hardly make anybody else so. In the sciences, every one has so much as he really knows and comprehends. What he believes only, and takes upon trust, are but shreds; which, however well in the whole piece, make no considerable addition to his stock who gathers them. Such borrowed wealth, like fairy money, though it were gold in the hand from which he received it, will be but leaves and dust when it comes to use.

25. Whence the opinion of innate principles. When men have found some general propositions that could not be doubted of as soon as understood, it was, I know, a short and easy way to conclude them innate. This being once received, it eased the lazy from the pains of search, and stopped the inquiry of the doubtful concerning all that was once styled innate. And it was of no small advantage to those who affected to be masters and teachers, to make this the principle of principles,—that principles must not be questioned. For, having once established this tenet,—that there are innate principles, it

put their followers upon a necessity of receiving some doctrines as such; which was to take them off from the use of their own reason and judgment, and put them on believing and taking them upon trust without further examination: in which posture of blind credulity, they might be more easily governed by, and made useful to some sort of men, who had the skill and office to principle and guide them. Nor is it a small power it gives one man over another, to have the authority to be the dictator of principles, and teacher of unquestionable truths; and to make a man swallow that for an innate principle which may serve to his purpose who teacheth them. Whereas had they examined the ways whereby men came to the knowledge of many universal truths, they would have found them to result in the minds of men from the being of things themselves, when duly considered; and that they were discovered by the application of those faculties that were fitted by nature to receive and judge of them, when duly employed about them.

26. Conclusion. To show how the understanding proceeds herein is the design of the following Discourse;

which I shall proceed to when I have first premised, that hitherto,—to clear my way to those foundations which I conceive are the only true ones, whereon to establish those notions we can have of our own knowledge,—it hath been necessary for me to give an account of the reasons I had to doubt of innate principles. And since the arguments which are against them do, some of them, rise from common received opinions, I have been forced to take several things for granted; which is hardly avoidable to any one, whose task is to show the falsehood or improbability of any tenet;—it happening in controversial discourses as it does in assaulting of towns; where, if the ground be but firm whereon the batteries are erected, there is no further inquiry of whom it is borrowed, nor whom it belongs to, so it affords but a fit rise for the present purpose. But in the future part of this Discourse, designing to raise an edifice uniform and consistent with itself, as far as my own experience and observation will assist me, I hope to erect it on such a basis that I shall not need to shore it up with props and buttresses, leaning on borrowed or begged

foundations: or at least, if mine prove a castle in the air, I will endeavour it shall be all of a piece and hang together. Wherein I warn the reader not to expect undeniable cogent demonstrations, unless I may be allowed the privilege, not seldom assumed by others, to take my principles for granted; and then, I doubt not, but I can demonstrate too. All that I shall say for the principles I proceed on is, that I can only appeal to men's own unprejudiced experience and observation whether they be true or not; and this is enough for a man who professes no more than to lay down candidly and freely his own conjectures, concerning a subject lying somewhat in the dark, without any other design than an unbiased inquiry after truth.

BOOK II

Of Ideas

Chapter I

Of Ideas in general, and their Original

1. Idea is the object of thinking. Every man being conscious to himself that he thinks; and that which his mind is applied about whilst thinking being the ideas that are there, it is past doubt that men have in their minds several ideas,—such as are those expressed by the words whiteness, hardness, sweetness, thinking, motion, man, elephant, army, drunkenness, and others: it is in the first place then to be inquired, How he comes by them?

I know it is a received doctrine, that men have native ideas, and original characters, stamped upon their minds in their very first being. This opinion I have at large examined already; and, I suppose what I have said in the foregoing Book will be much more easily admitted, when I have shown whence the understanding may get

all the ideas it has; and by what ways and degrees they may come into the mind;—for which I shall appeal to every one's own observation and experience.

2. All ideas come from sensation or reflection. Let us then suppose the mind to be, as we say, white paper, void of all characters, without any ideas:—How comes it to be furnished? Whence comes it by that vast store which the busy and boundless fancy of man has painted on it with an almost endless variety? Whence has it all the materials of reason and knowledge? To this I answer, in one word, from *experience*. In that all our knowledge is founded; and from that it ultimately derives itself. Our observation employed either, about external sensible objects, or about the internal operations of our minds perceived and reflected on by ourselves, is that which supplies our understandings with all the materials of thinking. These two are the fountains of knowledge, from whence all the ideas we have, or can naturally have, do spring.

3. The objects of sensation one source of ideas. First, our Senses, conversant about particular sensible objects,

do convey into the mind several distinct perceptions of things, according to those various ways wherein those objects do affect them. And thus we come by those ideas we have of yellow, white, heat, cold, soft, hard, bitter, sweet, and all those which we call sensible qualities; which when I say the senses convey into the mind, I mean, they from external objects convey into the mind what produces there those perceptions. This great source of most of the ideas we have, depending wholly upon our senses, and derived by them to the understanding, I call *sensation*.

4. The operations of our minds, the other source of them. Secondly, the other fountain from which experience furnisheth the understanding with ideas is,—the perception of the operations of our own mind within us, as it is employed about the ideas it has got;—which operations, when the soul comes to reflect on and consider, do furnish the understanding with another set of ideas, which could not be had from things without. And such are perception, thinking, doubting, believing, reasoning, knowing, willing, and all the different actings

of our own minds;—which we being conscious of, and observing in ourselves, do from these receive into our understandings as distinct ideas as we do from bodies affecting our senses. This source of ideas every man has wholly in himself; and though it be not sense, as having nothing to do with external objects, yet it is very like it, and might properly enough be called internal sense. But as I call the other *sensation*, so I call this *reflection*, the ideas it affords being such only as the mind gets by reflecting on its own operations within itself. By reflection then, in the following part of this discourse, I would be understood to mean, that notice which the mind takes of its own operations, and the manner of them, by reason whereof there come to be ideas of these operations in the understanding. These two, I say, viz. external material things, as the objects of *sensation*, and the operations of our own minds within, as the objects of *reflection*, are to me the only originals from whence all our ideas take their beginnings. The term operations here I use in a large sense, as comprehending not barely the actions of the mind about its ideas, but some sort of

passions arising sometimes from them, such as is the satisfaction or uneasiness arising from any thought.

5. All our ideas are of the one or the other of these. The understanding seems to me not to have the least glimmering of any ideas which it doth not receive from one of these two. External objects furnish the mind with the ideas of sensible qualities, which are all those different perceptions they produce in us; and the mind furnishes the understanding with ideas of its own operations. These, when we have taken a full survey of them, and their several modes, combinations, and relations, we shall find to contain all our whole stock of ideas; and that we have nothing in our minds which did not come in one of these two ways. Let any one examine his own thoughts, and thoroughly search into his understanding; and then let him tell me, whether all the original ideas he has there, are any other than of the objects of his senses, or of the operations of his mind, considered as objects of his reflection. And how great a mass of knowledge soever he imagines to be lodged there, he will, upon taking a strict view, see that he has

not any idea in his mind but what one of these two have imprinted;—though perhaps, with infinite variety compounded and enlarged by the understanding, as we shall see hereafter.

6. Observable in children. He that attentively considers the state of a child, at his first coming into the world, will have little reason to think him stored with plenty of ideas, that are to be the matter of his future knowledge. It is by degrees he comes to be furnished with them. And though the ideas of obvious and familiar qualities imprint themselves before the memory begins to keep a register of time or order, yet it is often so late before some unusual qualities come in the way, that there are few men that cannot recollect the beginning of their acquaintance with them. And if it were worth while, no doubt a child might be so ordered as to have but a very few, even of the ordinary ideas, till he were grown up to a man. But all that are born into the world, being surrounded with bodies that perpetually and diversely affect them, variety of ideas, whether care be taken of it or not, are imprinted on the minds of children. Light

and colours are busy at hand everywhere, when the eye is but open; sounds and some tangible qualities fail not to solicit their proper senses, and force an entrance to the mind;—but yet, I think, it will be granted easily, that if a child were kept in a place where he never saw any other but black and white till he were a man, he would have no more ideas of scarlet or green, than he that from his childhood never tasted an oyster, or a pine-apple, has of those particular relishes.

7. Men are differently furnished with these, according to the different objects they converse with. Men then come to be furnished with fewer or more simple ideas from without, according as the objects they converse with afford greater or less variety; and from the operations of their minds within, according as they more or less reflect on them. For, though he that contemplates the operations of his mind, cannot but have plain and clear ideas of them; yet, unless he turn his thoughts that way, and considers them attentively, he will no more have clear and distinct ideas of all the operations of his mind, and all that may be observed therein, than

he will have all the particular ideas of any landscape, or of the parts and motions of a clock, who will not turn his eyes to it, and with attention heed all the parts of it. The picture, or clock may be so placed, that they may come in his way every day; but yet he will have but a confused idea of all the parts they are made up of, till he applies himself with attention, to consider them each in particular.

8. Ideas of reflection later, because they need attention. And hence we see the reason why it is pretty late before most children get ideas of the operations of their own minds; and some have not any very clear or perfect ideas of the greatest part of them all their lives. Because, though they pass there continually, yet, like floating visions, they make not deep impressions enough to leave in their mind clear, distinct, lasting ideas, till the understanding turns inward upon itself, reflects on its own operations, and makes them the objects of its own contemplation. Children when they come first into it, are surrounded with a world of new things, which, by a constant solicitation of their senses, draw the mind con-

stantly to them; forward to take notice of new, and apt to be delighted with the variety of changing objects. Thus the first years are usually employed and diverted in looking abroad. Men's business in them is to acquaint themselves with what is to be found without; and so growing up in a constant attention to outward sensations, seldom make any considerable reflection on what passes within them, till they come to be of riper years; and some scarce ever at all.

9. The soul begins to have ideas when it begins to perceive. To ask, at what time a man has first any ideas, is to ask, when he begins to perceive;—having ideas, and perception, being the same thing. I know it is an opinion, that the soul always thinks, and that it has the actual perception of ideas in itself constantly, as long as it exists; and that actual thinking is as inseparable from the soul as actual extension is from the body; which if true, to inquire after the beginning of a man's ideas is the same as to inquire after the beginning of his soul. For, by this account, soul and its ideas, as body and its extension, will begin to exist both at the same time.

10. The soul thinks not always; for this wants proofs. But whether the soul be supposed to exist antecedent to, or coeval with, or some time after the first rudiments of organization, or the beginnings of life in the body, I leave to be disputed by those who have better thought of that matter. I confess myself to have one of those dull souls, that doth not perceive itself always to contemplate ideas; nor can conceive it any more necessary for the soul always to think, than for the body always to move: the perception of ideas being (as I conceive) to the soul, what motion is to the body; not its essence, but one of its operations. And therefore, though thinking be supposed never so much the proper action of the soul, yet it is not necessary to suppose that it should be always thinking, always in action. That, perhaps, is the privilege of the infinite Author and Preserver of all things, who “never slumbers nor sleeps;” but is not competent to any finite being, at least not to the soul of man. We know certainly, by experience, that we sometimes think; and thence draw this infallible consequence,—that there is something in us that has a

power to think. But whether that substance perpetually thinks or no, we can be no further assured than experience informs us. For, to say that actual thinking is essential to the soul, and inseparable from it, is to beg what is in question, and not to prove it by reason;—which is necessary to be done, if it be not a self-evident proposition. But whether this, “That the soul always thinks,” be a self-evident proposition, that everybody assents to at first hearing, I appeal to mankind. It is doubted whether I thought at all last night or no. The question being about a matter of fact, it is begging it to bring, as a proof for it, an hypothesis, which is the very thing in dispute: by which way one may prove anything, and it is but supposing that all watches, whilst the balance beats, think, and it is sufficiently proved, and past doubt, that my watch thought all last night. But he that would not deceive himself, ought to build his hypothesis on matter of fact, and make it out by sensible experience, and not presume on matter of fact, because of his hypothesis, that is, because he supposes it to be so; which way of proving amounts to this, that

I must necessarily think all last night, because another supposes I always think, though I myself cannot perceive that I always do so.

But men in love with their opinions may not only suppose what is in question, but allege wrong matter of fact. How else could any one make it an inference of mine, that a thing is not, because we are not sensible of it in our sleep? I do not say there is no soul in a man, because he is not sensible of it in his sleep; but I do say, he cannot think at any time, waking or sleeping: without being sensible of it. Our being sensible of it is not necessary to anything but to our thoughts; and to them it is; and to them it always will be necessary, till we can think without being conscious of it.

11. It is not always conscious of it. I grant that the soul, in a waking man, is never without thought, because it is the condition of being awake. But whether sleeping without dreaming be not an affection of the whole man, mind as well as body, may be worth a waking man's consideration; it being hard to conceive that anything should think and not be conscious of it. If the

soul doth think in a sleeping man without being conscious of it, I ask whether, during such thinking, it has any pleasure or pain, or be capable of happiness or misery? I am sure the man is not; no more than the bed or earth he lies on. For to be happy or miserable without being conscious of it, seems to me utterly inconsistent and impossible. Or if it be possible that the soul can, whilst the body is sleeping, have its thinking, enjoyments, and concerns, its pleasures or pain, apart, which the man is not conscious of nor partakes in,—it is certain that Socrates asleep and Socrates awake is not the same person; but his soul when he sleeps, and Socrates the man, consisting of body and soul, when he is waking, are two persons: since waking Socrates has no knowledge of, or concernment for that happiness or misery of his soul, which it enjoys alone by itself whilst he sleeps, without perceiving anything of it; no more than he has for the happiness or misery of a man in the Indies, whom he knows not. For, if we take wholly away all consciousness of our actions and sensations, especially of pleasure and pain, and the concernment that accompanies it, it

will be hard to know wherein to place personal identity.

12. If a sleeping man thinks without knowing it, the sleeping and waking man are two persons. The soul, during sound sleep, thinks, say these men. Whilst it thinks and perceives, it is capable certainly of those of delight or trouble, as well as any other perceptions; and it must necessarily be conscious of its own perceptions. But it has all this apart: the sleeping man, it is plain, is conscious of nothing of all this. Let us suppose, then, the soul of Castor, while he is sleeping, retired from his body; which is no impossible supposition for the men I have here to do with, who so liberally allow life, without a thinking soul, to all other animals. These men cannot then judge it impossible, or a contradiction, that the body should live without the soul; nor that the soul should subsist and think, or have perception, even perception of happiness or misery, without the body. Let us then, I say, suppose the soul of Castor separated during his sleep from his body, to think apart. Let us suppose, too, that it chooses for its scene of thinking the body of another man, v.g. Pollux, who is sleeping

without a soul. For, if Castor's soul can think, whilst Castor is asleep, what Castor is never conscious of, it is no matter what place it chooses to think in. We have here, then, the bodies of two men with only one soul between them, which we will suppose to sleep and wake by turns; and the soul still thinking in the waking man, whereof the sleeping man is never conscious, has never the least perception. I ask, then, whether Castor and Pollux, thus with only one soul between them, which thinks and perceives in one what the other is never conscious of, nor is concerned for, are not two as distinct persons as Castor and Hercules, or as Socrates and Plato were? And whether one of them might not be very happy, and the other very miserable? Just by the same reason, they make the soul and the man two persons, who make the soul think apart what the man is not conscious of. For, I suppose nobody will make identity of persons to consist in the soul's being united to the very same numerical particles of matter. For if that be necessary to identity, it will be impossible, in that constant flux of the particles of our bodies, that any

man should be the same person two days, or two moments, together.

13. Impossible to convince those that sleep without dreaming, that they think. Thus, methinks, every drowsy nod shakes their doctrine, who teach that the soul is always thinking. Those, at least, who do at any time sleep without dreaming, can never be convinced that their thoughts are sometimes for four hours busy without their knowing of it; and if they are taken in the very act, waked in the middle of that sleeping contemplation, can give no manner of account of it.

14. That men dream without remembering it, in vain urged. It will perhaps be said,—That the soul thinks even in the soundest sleep, but the memory retains it not. That the soul in a sleeping man should be this moment busy a thinking, and the next moment in a waking man not remember nor be able to recollect one jot of all those thoughts, is very hard to be conceived, and would need some better proof than bare assertion to make it be believed. For who can without any more ado, but being barely told so, imagine that the greatest

part of men do, during all their lives, for several hours every day, think of something, which if they were asked, even in the middle of these thoughts, they could remember nothing at all of? Most men, I think, pass a great part of their sleep without dreaming. I once knew a man that was bred a scholar, and had no bad memory, who told me he had never dreamed in his life, till he had that fever he was then newly recovered of, which was about the five or six and twentieth year of his age. I suppose the world affords more such instances: at least every one's acquaintance will furnish him with examples enough of such as pass most of their nights without dreaming.

15. Upon this hypothesis, the thoughts of a sleeping man ought to be most rational. To think often, and never to retain it so much as one moment, is a very useless sort of thinking; and the soul, in such a state of thinking, does very little, if at all, excel that of a looking-glass, which constantly receives variety of images, or ideas, but retains none; they disappear and vanish, and there remain no footsteps of them; the looking-

glass is never the better for such ideas, nor the soul for such thoughts. Perhaps it will be said, that in a waking man the materials of the body are employed, and made use of, in thinking; and that the memory of thoughts is retained by the impressions that are made on the brain, and the traces there left after such thinking; but that in the thinking of the soul, which is not perceived in a sleeping man, there the soul thinks apart, and making no use of the organs of the body, leaves no impressions on it, and consequently no memory of such thoughts. Not to mention again the absurdity of two distinct persons, which follows from this supposition, I answer, further,—That whatever ideas the mind can receive and contemplate without the help of the body, it is reasonable to conclude it can retain without the help of the body too; or else the soul, or any separate spirit, will have but little advantage by thinking. If it has no memory of its own thoughts; if it cannot lay them up for its own use, and be able to recall them upon occasion; if it cannot reflect upon what is past, and make use of its former experiences, reasonings, and contemplations, to what purpose

does it think? They who make the soul a thinking thing, at this rate, will not make it a much more noble being than those do whom they condemn, for allowing it to be nothing but the subtilist parts of matter. Characters drawn on dust, that the first breath of wind effaces; or impressions made on a heap of atoms, or animal spirits, are altogether as useful, and render the subject as noble, as the thoughts of a soul that perish in thinking; that, once out of sight, are gone forever, and leave no memory of themselves behind them. Nature never makes excellent things for mean or no uses: and it is hardly to be conceived that our infinitely wise Creator should make so admirable a faculty which comes nearest the excellency of his own incomprehensible being, to be so idly and uselessly employed, at least a fourth part of its time here, as to think constantly, without remembering any of those thoughts, without doing any good to itself or others, or being any way useful to any other part of the creation, If we will examine it, we shall not find, I suppose, the motion of dull and senseless matter, any where in the universe, made so little use of and so wholly thrown away.

16. On this hypothesis, the soul must have ideas not derived from sensation or reflection, of which there is no appearance. It is true, we have sometimes instances of perception whilst we are asleep, and retain the memory of those thoughts: but how extravagant and incoherent for the most part they are; how little conformable to the perfection and order of a rational being, those who are acquainted with dreams need not be told. This I would willingly be satisfied in,—whether the soul, when it thinks thus apart, and as it were separate from the body, acts less rationally than when conjointly with it, or no. If its separate thoughts be less rational, then these men must say, that the soul owes the perfection of rational thinking to the body: if it does not, it is a wonder that our dreams should be, for the most part, so frivolous and irrational; and that the soul should retain none of its more rational soliloquies and meditations.

17. If I think when I know it not, nobody else can know it. Those who so confidently tell us that the soul always actually thinks, I would they would also tell us, what those ideas are that are in the soul of a child,

before or just at the union with the body, before it hath received any by sensation. The dreams of sleeping men are, as I take it, all made up of the waking man's ideas; though for the most part oddly put together. It is strange, if the soul has ideas of its own that it derived not from sensation or reflection, (as it must have, if it thought before it received any impressions from the body,) that it should never, in its private thinking, (so private, that the man himself perceives it not,) retain any of them the very moment it wakes out of them, and then make the man glad with new discoveries. Who can find it reason that the soul should, in its retirement during sleep, have so many hours' thoughts, and yet never light on any of those ideas it borrowed not from sensation or reflection; or at least preserve the memory of none but such, which, being occasioned from the body, must needs be less natural to a spirit? It is strange the soul should never once in a man's whole life recall over any of its pure native thoughts, and those ideas it had before it borrowed anything from the body; never bring into the waking man's view any other ideas but what have a

tang of the cask, and manifestly derive their original from that union. If it always thinks, and so had ideas before it was united, or before it received any from the body, it is not to be supposed but that during sleep it recollects its native ideas; and during that retirement from communicating with the body, whilst it thinks by itself, the ideas it is busied about should be, sometimes at least, those more natural and congenial ones which it had in itself, underived from the body, or its own operations about them: which, since the waking man never remembers, we must from this hypothesis conclude either that the soul remembers something that the man does not; or else that memory belongs only to such ideas as are derived from the body, or the mind's operations about them.

18. How knows any one that the soul always thinks? For if it be not a self-evident proposition, it needs proof. I would be glad also to learn from these men who so confidently pronounce that the human soul, or, which is all one, that a man always thinks, how they come to know it; nay, how they come to know that they them-

selves think when they themselves do not perceive it. This, I am afraid, is to be sure without proofs, and to know without perceiving. It is, I suspect, a confused notion, taken up to serve an hypothesis; and none of those clear truths, that either their own evidence forces us to admit, or common experience makes it impudence to deny. For the most that can be said of it is, that it is possible the soul may always think, but not always retain it in memory. And I say, it is as possible that the soul may not always think; and much more probable that it should sometimes not think, than that it should often think, and that a long while together, and not be conscious to itself, the next moment after, that it had thought.

19. "That a man should be busy in thinking, and yet not retain it the next moment," very improbable. To suppose the soul to think, and the man not to perceive it, is, as has been said, to make two persons in one man. And if one considers well these men's way of speaking, one should be led into a suspicion that they do so. For they who tell us that the soul always thinks, do never,

that I remember, say that a man always thinks. Can the soul think, and not the man? Or a man think, and not be conscious of it? This, perhaps, would be suspected of jargon in others. If they say the man thinks always, but is not always conscious of it, they may as well say his body is extended without having parts. For it is altogether as intelligible to say that a body is extended without parts, as that anything thinks without being conscious of it, or perceiving that it does so. They who talk thus may, with as much reason, if it be necessary to their hypothesis, say that a man is always hungry, but that he does not always feel it; whereas hunger consists in that very sensation, as thinking consists in being conscious that one thinks. If they say that a man is always conscious to himself of thinking, I ask, How they know it? Consciousness is the perception of what passes in a man's own mind. Can another man perceive that I am conscious of anything, when I perceive it not myself? No man's knowledge here can go beyond his experience. Wake a man out of a sound sleep, and ask him what he was that moment thinking of. If he himself be

conscious of nothing he then thought on, he must be a notable diviner of thoughts that can assure him that he was thinking. May he not, with more reason, assure him he was not asleep? This is something beyond philosophy; and it cannot be less than revelation, that discovers to another thoughts in my mind, when I can find none there myself. And they must needs have a penetrating sight who can certainly see that I think, when I cannot perceive it myself, and when I declare that I do not; and yet can see that dogs or elephants do not think, when they give all the demonstration of it imaginable, except only telling us that they do so. This some may suspect to be a step beyond the Rosicrucians; it seeming easier to make one's self invisible to others, than to make another's thoughts visible to me, which are not visible to himself. But it is but defining the soul to be "a substance that always thinks," and the business is done. If such definition be of any authority, I know not what it can serve for but to make many men suspect that they have no souls at all; since they find a good part of their lives pass away without thinking. For no defini-

tions that I know, no suppositions of any sect, are of force enough to destroy constant experience; and perhaps it is the affectation of knowing beyond what we perceive, that makes so much useless dispute and noise in the world.

20. No ideas but from sensation and reflection, evident, if we observe children. I see no reason, therefore, to believe that the soul thinks before the senses have furnished it with ideas to think on; and as those are increased and retained, so it comes, by exercise, to improve its faculty of thinking in the several parts of it; as well as, afterwards, by compounding those ideas, and reflecting on its own operations, it increases its stock, as well as facility in remembering, imagining, reasoning, and other modes of thinking.

21. State of a child in the mother's womb. He that will suffer himself to be informed by observation and experience, and not make his own hypothesis the rule of nature, will find few signs of a soul accustomed to much thinking in a new-born child, and much fewer of any reasoning at all. And yet it is hard to imagine that the

rational soul should think so much, and not reason at all. And he that will consider that infants newly come into the world spend the greatest part of their time in sleep, and are seldom awake but when either hunger calls for the teat, or some pain (the most importunate of all sensations), or some other violent impression on the body, forces the mind to perceive and attend to it;—he, I say, who considers this, will perhaps find reason to imagine that a foetus in the mother's womb differs not much from the state of a vegetable, but passes the greatest part of its time without perception or thought; doing very little but sleep in a place where it needs not seek for food, and is surrounded with liquor, always equally soft, and near of the same temper; where the eyes have no light, and the ears so shut up are not very susceptible of sounds; and where there is little or no variety, or change of objects, to move the senses.

22. The mind thinks in proportion to the matter it gets from experience to think about. Follow a child from its birth, and observe the alterations that time makes, and you shall find, as the mind by the senses comes more

and more to be furnished with ideas, it comes to be more and more awake; thinks more, the more it has matter to think on. After some time it begins to know the objects which, being most familiar with it, have made lasting impressions. Thus it comes by degrees to know the persons it daily converses with, and distinguishes them from strangers; which are instances and effects of its coming to retain and distinguish the ideas the senses convey to it. And so we may observe how the mind, by degrees, improves in these; and advances to the exercise of those other faculties of enlarging, compounding, and abstracting its ideas, and of reasoning about them, and reflecting upon all these; of which I shall have occasion to speak more hereafter.

23. A man begins to have ideas when he first has sensation. What sensation is. If it shall be demanded then, when a man begins to have any ideas, I think the true answer is,—when he first has any sensation. For, since there appear not to be any ideas in the mind before the senses have conveyed any in, I conceive that ideas in the understanding are coeval with sensation; which is

such an impression or motion made in some part of the body, as produces some perception in the understanding. It is about these impressions made on our senses by outward objects that the mind seems first to employ itself, in such operations as we call perception, remembering, consideration, reasoning, &c.

24. The original of all our knowledge. In time the mind comes to reflect on its own operations about the ideas got by sensation, and thereby stores itself with a new set of ideas, which I call ideas of reflection. These are the impressions that are made on our senses by outward objects that are extrinsical to the mind; and its own operations, proceeding from powers intrinsical and proper to itself, which, when reflected on by itself, become also objects of its contemplation—are, as I have said, the original of all knowledge. Thus the first capacity of human intellect is,—that the mind is fitted to receive the impressions made on it; either through the senses by outward objects, or by its own operations when it reflects on them. This is the first step a man makes towards the discovery of anything, and the groundwork

whereon to build all those notions which ever he shall have naturally in this world. All those sublime thoughts which tower above the clouds, and reach as high as heaven itself, take their rise and footing here: in all that great extent wherein the mind wanders, in those remote speculations it may seem to be elevated with, it stirs not one jot beyond those ideas which sense or reflection have offered for its contemplation.

25. In the reception of simple ideas, the understanding is for the most part passive. In this part the understanding is merely passive; and whether or no it will have these beginnings, and as it were materials of knowledge, is not in its own power. For the objects of our senses do, many of them, obtrude their particular ideas upon our minds whether we will or not; and the operations of our minds will not let us be without, at least, some obscure notions of them. No man can be wholly ignorant of what he does when he thinks. These simple ideas, when offered to the mind, the understanding can no more refuse to have, nor alter when they are imprinted, nor blot them out and make new ones itself,

than a mirror can refuse, alter, or obliterate the images or ideas which the objects set before it do therein produce. As the bodies that surround us do diversely affect our organs, the mind is forced to receive the impressions; and cannot avoid the perception of those ideas that are annexed to them.

Chapter II Of Simple Ideas

1. Uncompounded appearances. The better to understand the nature, manner, and extent of our knowledge, one thing is carefully to be observed concerning the ideas we have; and that is, that some of them are simple and some complex.

Though the qualities that affect our senses are, in the things themselves, so united and blended, that there is no separation, no distance between them; yet it is plain, the ideas they produce in the mind enter by the senses simple and unmixed. For, though the sight and touch often take in from the same object, at the same time,

different ideas;—as a man sees at once motion and colour; the hand feels softness and warmth in the same piece of wax: yet the simple ideas thus united in the same subject, are as perfectly distinct as those that come in by different senses. The coldness and hardness which a man feels in a piece of ice being as distinct ideas in the mind as the smell and whiteness of a lily; or as the taste of sugar, and smell of a rose. And there is nothing can be plainer to a man than the clear and distinct perception he has of those simple ideas; which, being each in itself uncompounded, contains in it nothing but one uniform appearance, or conception in the mind, and is not distinguishable into different ideas.

2. The mind can neither make nor destroy them. These simple ideas, the materials of all our knowledge, are suggested and furnished to the mind only by those two ways above mentioned, viz. sensation and reflection. When the understanding is once stored with these simple ideas, it has the power to repeat, compare, and unite them, even to an almost infinite variety, and so can make at pleasure new complex ideas. But it is not in the

power of the most exalted wit, or enlarged understanding, by any quickness or variety of thought, to invent or frame one new simple idea in the mind, not taken in by the ways before mentioned: nor can any force of the understanding destroy those that are there. The dominion of man, in this little world of his own understanding being muchwhat the same as it is in the great world of visible things; wherein his power, however managed by art and skill, reaches no farther than to compound and divide the materials that are made to his hand; but can do nothing towards the making the least particle of new matter, or destroying one atom of what is already in being. The same inability will every one find in himself, who shall go about to fashion in his understanding one simple idea, not received in by his senses from external objects, or by reflection from the operations of his own mind about them. I would have any one try to fancy any taste which had never affected his palate; or frame the idea of a scent he had never smelt: and when he can do this, I will also conclude that a blind man hath ideas of colours, and a deaf man true distinct notions of sounds.

3. Only the qualities that affect the senses are imaginable. This is the reason why—though we cannot believe it impossible to God to make a creature with other organs, and more ways to convey into the understanding the notice of corporeal things than those five, as they are usually counted, which he has given to man—yet I think it is not possible for any man to imagine any other qualities in bodies, howsoever constituted, whereby they can be taken notice of, besides sounds, tastes, smells, visible and tangible qualities. And had mankind been made but with four senses, the qualities then which are the objects of the fifth sense had been as far from our notice, imagination, and conception, as now any belonging to a sixth, seventh, or eighth sense can possibly be;—which, whether yet some other creatures, in some other parts of this vast and stupendous universe, may not have, will be a great presumption to deny. He that will not set himself proudly at the top of all things, but will consider the immensity of this fabric, and the great variety that is to be found in this little and inconsiderable part of it which he has to do with, may be apt to think that, in other mansions of it,

there may be other and different intelligent beings, of whose faculties he has as little knowledge or apprehension as a worm shut up in one drawer of a cabinet hath of the senses or understanding of a man; such variety and excellency being suitable to the wisdom and power of the Maker. I have here followed the common opinion of man's having but five senses; though, perhaps, there may be justly counted more;—but either supposition serves equally to my present purpose.

Chapter III

Of Simple Ideas of Sense

1. Division of simple ideas. The better to conceive the ideas we receive from sensation, it may not be amiss for us to consider them, in reference to the different ways whereby they make their approaches to our minds, and make themselves perceivable by us. First, then, There are some which come into our minds by one sense only.

Secondly, There are others that convey themselves into the mind by more senses than one.

Thirdly, Others that are had from reflection only.

Fourthly, There are some that make themselves way, and are suggested to the mind by all the ways of sensation and reflection.

We shall consider them apart under these several heads.

Ideas of one sense. There are some ideas which have admittance only through one sense, which is peculiarly adapted to receive them. Thus light and colours, as white, red, yellow, blue; with their several degrees or shades and mixtures, as green, scarlet, purple, sea-green, and the rest, come in only by the eyes. All kinds of noises, sounds, and tones, only by the ears. The several tastes and smells, by the nose and palate. And if these organs, or the nerves which are the conduits to convey them from without to their audience in the brain,—the mind's presence-room (as I may so call it)—are any of them so disordered as not to perform their functions, they have no postern to be admitted by; no other way to bring themselves into view, and be perceived by the understanding.

The most considerable of those belonging to the touch, are heat and cold, and solidity: all the rest, consisting

almost wholly in the sensible configuration, as smooth and rough; or else, more or less firm adhesion of the parts, as hard and soft, tough and brittle, are obvious enough.

2. Few simple ideas have names. I think it will be needless to enumerate all the particular simple ideas belonging to each sense. Nor indeed is it possible if we would; there being a great many more of them belonging to most of the senses than we have names for. The variety of smells, which are as many almost, if not more, than species of bodies in the world, do most of them want names. Sweet and stinking commonly serve our turn for these ideas, which in effect is little more than to call them pleasing or displeasing; though the smell of a rose and violet, both sweet, are certainly very distinct ideas. Nor are the different tastes, that by our palates we receive ideas of, much better provided with names. Sweet, bitter, sour, harsh, and salt are almost all the epithets we have to denominate that numberless variety of relishes, which are to be found distinct, not only in almost every sort of creatures, but in the different

parts of the same plant, fruit, or animal. The same may be said of colours and sounds. I shall, therefore, in the account of simple ideas I am here giving, content myself to set down only such as are most material to our present purpose, or are in themselves less apt to be taken notice of though they are very frequently the ingredients of our complex ideas; amongst which, I think, I may well account solidity, which therefore I shall treat of in the next chapter.

Chapter IV

Idea of Solidity

1. We receive this idea from touch. The idea of solidity we receive by our touch: and it arises from the resistance which we find in body to the entrance of any other body into the place it possesses, till it has left it. There is no idea which we receive more constantly from sensation than solidity. Whether we move or rest, in what posture soever we are, we always feel something under us that support us, and hinders our further sink-

ing downwards; and the bodies which we daily handle make us perceive that, whilst they remain between them, they do, by an insurmountable force, hinder the approach of the parts of our hands that press them. That which thus hinders the approach of two bodies, when they are moved one towards another, I call solidity. I will not dispute whether this acceptance of the word solid be nearer to its original signification than that which mathematicians use it in. It suffices that I think the common notion of solidity will allow, if not justify, this use of it; but if any one think it better to call it impenetrability, he has my consent. Only I have thought the term solidity the more proper to express this idea, not only because of its vulgar use in that sense, but also because it carries something more of positive in it than impenetrability; which is negative, and is perhaps more a consequence of solidity, than solidity itself. This, of all other, seems the idea most intimately connected with, and essential to body; so as nowhere else to be found or imagined, but only in matter. And though our senses take no notice of it, but in masses of matter, of a bulk

sufficient to cause a sensation in us: yet the mind, having once got this idea from such grosser sensible bodies, traces it further, and considers it, as well as figure, in the minutest particle of matter that can exist; and finds it inseparably inherent in body, wherever or however modified.

2. Solidity fills space. This is the idea which belongs to body, whereby we conceive it to fill space. The idea of which filling of space is,—that where we imagine any space taken up by a solid substance, we conceive it so to possess it, that it excludes all other solid substances; and will for ever hinder any other two bodies, that move towards one another in a straight line, from coming to touch one another, unless it removes from between them in a line not parallel to that which they move in. This idea of it, the bodies which we ordinarily handle sufficiently furnish us with.

3. Distinct from space. This resistance, whereby it keeps other bodies out of the space which it possesses, is so great, that no force, how great soever, can surmount it. All the bodies in the world, pressing a drop of water on

all sides, will never be able to overcome the resistance which it will make, soft as it is, to their approaching one another, till it be removed out of their way: whereby our idea of solidity is distinguished both from pure space, which is capable neither of resistance nor motion; and from the ordinary idea of hardness. For a man may conceive two bodies at a distance, so as they may approach one another, without touching or displacing any solid thing, till their superficies come to meet; whereby, I think, we have the clear idea of space without solidity. For (not to go so far as annihilation of any particular body) I ask, whether a man cannot have the idea of the motion of one single body alone, without any other succeeding immediately into its place? I think it is evident he can: the idea of motion in one body no more including the idea of motion in another, than the idea of a square figure in one body includes the idea of a square figure in another. I do not ask, whether bodies do so exist, that the motion of one body cannot really be without the motion of another. To determine this either way, is to beg the question for or against a vacuum. But

my question is,—whether one cannot have the idea of one body moved, whilst others are at rest? And I think this no one will deny. If so, then the place it deserted gives us the idea of pure space without solidity; whereinto any other body may enter, without either resistance or protrusion of anything. When the sucker in a pump is drawn, the space it filled in the tube is certainly the same whether any other body follows the motion of the sucker or not: nor does it imply a contradiction that, upon the motion of one body, another that is only contiguous to it should not follow it. The necessity of such a motion is built only on the supposition that the world is full; but not on the distinct ideas of space and solidity, which are as different as resistance and not resistance, protrusion and not protrusion. And that men have ideas of space without a body, their very disputes about a vacuum plainly demonstrate, as is shown in another place.

4. From hardness. Solidity is hereby also differenced from hardness, in that solidity consists in repletion, and so an utter exclusion of other bodies out of the space it

possesses: but hardness, in a firm cohesion of the parts of matter, making up masses of a sensible bulk, so that the whole does not easily change its figure. And indeed, hard and soft are names that we give to things only in relation to the constitutions of our own bodies; that being generally called hard by us, which will put us to pain sooner than change figure by the pressure of any part of our bodies; and that, on the contrary, soft, which changes the situation of its parts upon an easy and unpainful touch.

But this difficulty of changing the situation of the sensible parts amongst themselves, or of the figure of the whole, gives no more solidity to the hardest body in the world than to the softest; nor is an adamant one jot more solid than water. For, though the two flat sides of two pieces of marble will more easily approach each other, between which there is nothing but water or air, than if there be a diamond between them; yet it is not that the parts of the diamond are more solid than those of water, or resist more; but because the parts of water, being more easily separable from each other, they will, by a

side motion, be more easily removed, and give way to the approach of the two pieces of marble. But if they could be kept from making place by that side motion, they would eternally hinder the approach of these two pieces of marble, as much as the diamond; and it would be as impossible by any force to surmount their resistance, as to surmount the resistance of the parts of a diamond. The softest body in the world will as invincibly resist the coming together of any other two bodies, if it be not put out of the way, but remain between them, as the hardest that can be found or imagined. He that shall fill a yielding soft body well with air or water, will quickly find its resistance. And he that thinks that nothing but bodies that are hard can keep his hands from approaching one another, may be pleased to make a trial, with the air inclosed in a football. The experiment, I have been told, was made at Florence, with a hollow globe of gold filled with water, and exactly closed; which further shows the solidity of so soft a body as water. For the golden globe thus filled, being put into a press, which was driven by the extreme force of screws,

the water made itself way through the pores of that very close metal, and finding no room for a nearer approach of its particles within, got to the outside, where it rose like a dew, and so fell in drops, before the sides of the globe could be made to yield to the violent compression of the engine that squeezed it.

5. On solidity depend impulse, resistance, and protrusion. By this idea of solidity is the extension of body distinguished from the extension of space:—the extension of body being nothing but the cohesion or continuity of solid, separable, movable parts; and the extension of space, the continuity of unsolid, inseparable, and immovable parts. Upon the solidity of bodies also depend their mutual impulse, resistance, and protrusion. Of pure space then, and solidity, there are several (amongst which I confess myself one) who persuade themselves they have clear and distinct ideas; and that they can think on space, without anything in it that resists or is protruded by body. This is the idea of pure space, which they think they have as clear as any idea they can have of the extension of body: the idea of the distance between the oppo-

site parts of a concave superficies being equally as clear without as with the idea of any solid parts between: and on the other side, they persuade themselves that they have, distinct from that of pure space, the idea of something that fills space, that can be protruded by the impulse of other bodies, or resist their motion. If there be others that have not these two ideas distinct, but confound them, and make but one of them, I know not how men, who have the same idea under different names, or different ideas under the same name, can in that case talk with one another; any more than a man who, not being blind or deaf, has distinct ideas of the colour of scarlet and the sound of a trumpet, could discourse concerning scarlet colour with the blind man I mentioned in another place, who fancied that the idea of scarlet was like the sound of a trumpet.

6. What solidity is. If any one ask me, What this solidity is, I send him to his senses to inform him. Let him put a flint or a football between his hands, and then endeavour to join them, and he will know. If he thinks this not a sufficient explication of solidity, what it is,

and wherein it consists; I promise to tell him what it is, and wherein it consists, when he tells me what thinking is, or wherein it consists; or explains to me what extension or motion is, which perhaps seems much easier. The simple ideas we have, are such as experience teaches them us; but if, beyond that, we endeavour by words to make them clearer in the mind, we shall succeed no better than if we went about to clear up the darkness of a blind man's mind by talking; and to discourse into him the ideas of light and colours. The reason of this I shall show in another place.

Chapter V Of Simple Ideas of Divers Senses

Ideas received both by seeing and touching. The ideas we get by more than one sense are, of space or extension, figure, rest, and motion. For these make perceivable impressions, both on the eyes and touch; and we can receive and convey into our minds the ideas of the extension, figure, motion, and rest of bodies, both by seeing

and feeling. But having occasion to speak more at large of these in another place, I here only enumerate them.

Chapter VI

Of Simple Ideas of Reflection

1. Simple ideas are the operations of mind about its other ideas. The mind receiving the ideas mentioned in the foregoing chapters from without, when it turns its view inward upon itself, and observes its own actions about those ideas it has, takes from thence other ideas, which are as capable to be the objects of its contemplation as any of those it received from foreign things.

2. The idea of perception, and idea of willing, we have from reflection. The two great and principal actions of the mind, which are most frequently considered, and which are so frequent that every one that pleases may take notice of them in himself, are these two:—

Perception, or Thinking; and
Volition, or Willing.

The power of thinking is called the Understanding, and the power of volition is called the Will; and these two powers or abilities in the mind are denominated faculties.

Of some of the modes of these simple ideas of reflection, such as are remembrance, discerning, reasoning, judging, knowledge, faith, &c., I shall have occasion to speak hereafter.

Chapter VII

Of Simple Ideas of both Sensation and Reflection

1. Ideas of pleasure and pain. There be other simple ideas which convey themselves into the mind by all the ways of sensation and reflection, viz. pleasure or delight, and its opposite, pain, or uneasiness; power; existence; unity.

2. Mix with almost all our other ideas. Delight or uneasiness, one or other of them, join themselves to almost all our ideas both of sensation and reflection: and there is scarce any affection of our senses from with-

out, any retired thought of our mind within, which is not able to produce in us pleasure or pain. By pleasure and pain, I would be understood to signify, whatsoever delights or molests us; whether it arises from the thoughts of our minds, or anything operating on our bodies. For, whether we call it satisfaction, delight, pleasure, happiness, &c., on the one side, or uneasiness, trouble, pain, torment, anguish, misery, &c., on the other, they are still but different degrees of the same thing, and belong to the ideas of pleasure and pain, delight or uneasiness; which are the names I shall most commonly use for those two sorts of ideas.

3. As motives of our actions. The infinite wise Author of our being, having given us the power over several parts of our bodies, to move or keep them at rest as we think fit; and also. by the motion of them, to move ourselves and other contiguous bodies, in which consist all the actions of our body: having also given a power to our minds, in several instances, to choose, amongst its ideas, which it will think on, and to pursue the inquiry of this or that subject with consideration and attention, to ex-

cite us to these actions of thinking and motion that we are capable of,—has been pleased to join to several thoughts, and several sensations a perception of delight. If this were wholly separated from all our outward sensations, and inward thoughts, we should have no reason to prefer one thought or action to another; negligence to attention, or motion to rest. And so we should neither stir our bodies, nor employ our minds, but let our thoughts (if I may so call it) run adrift, without any direction or design, and suffer the ideas of our minds, like unregarded shadows, to make their appearances there, as it happened, without attending to them. In which state man, however furnished with the faculties of understanding and will, would be a very idle, inactive creature, and pass his time only in a lazy, lethargic dream. It has therefore pleased our wise Creator to annex to several objects, and the ideas which we receive from them, as also to several of our thoughts, a concomitant pleasure, and that in several objects, to several degrees, that those faculties which he had endowed us with might not remain wholly idle and unemployed by us.

4. An end and use of pain. Pain has the same efficacy and use to set us on work that pleasure has, we being as ready to employ our faculties to avoid that, as to pursue this: only this is worth our consideration, that pain is often produced by the same objects and ideas that produce pleasure in us. This their near conjunction, which makes us often feel pain in the sensations where we expected pleasure, gives us new occasion of admiring the wisdom and goodness of our Maker, who, designing the preservation of our being, has annexed pain to the application of many things to our bodies, to warn us of the harm that they will do, and as advices to withdraw from them. But he, not designing our preservation barely, but the preservation of every part and organ in its perfection, hath in many cases annexed pain to those very ideas which delight us. Thus heat, that is very agreeable to us in one degree, by a little greater increase of it proves no ordinary torment: and the most pleasant of all sensible objects, light itself, if there be too much of it, if increased beyond a due proportion to our eyes, causes a very painful sensation. Which is wisely and favourably so ordered

by nature, that when any object does, by the vehemency of its operation, disorder the instruments of sensation, whose structures cannot but be very nice and delicate, we might, by the pain, be warned to withdraw, before the organ be quite put out of order, and so be unfitted for its proper function for the future. The consideration of those objects that produce it may well persuade us, that this is the end or use of pain. For, though great light be insufferable to our eyes, yet the highest degree of darkness does not at all disease them: because that, causing no disorderly motion in it, leaves that curious organ unharmed in its natural state. But yet excess of cold as well as heat pains us: because it is equally destructive to that temper which is necessary to the preservation of life, and the exercise of the several functions of the body, and which consists in a moderate degree of warmth; or, if you please, a motion of the insensible parts of our bodies, confined within certain bounds.

5. Another end. Beyond all this, we may find another reason why God hath scattered up and down several degrees of pleasure and pain, in all the things that environ

and affect us; and blended them together in almost all that our thoughts and senses have to do with;—that we, finding imperfection, dissatisfaction, and want of complete happiness, in all the enjoyments which the creatures can afford us, might be led to seek it in the enjoyment of Him with whom there is fullness of joy, and at whose right hand are pleasures for evermore.

6. Goodness of God in annexing pleasure and pain to our other ideas. Though what I have here said may not, perhaps, make the ideas of pleasure and pain clearer to us than our own experience does, which is the only way that we are capable of having them; yet the consideration of the reason why they are annexed to so many other ideas, serving to give us due sentiments of the wisdom and goodness of the Sovereign Disposer of all things, may not be unsuitable to the main end of these inquiries: the knowledge and veneration of him being the chief end of all our thoughts, and the proper business of all understandings.

7. Ideas of existence and unity. Existence and Unity are two other ideas that are suggested to the understand-

ing by every object without, and every idea within. When ideas are in our minds, we consider them as being actually there, as well as we consider things to be actually without us;—which is, that they exist, or have existence. And whatever we can consider as one thing, whether a real being or idea, suggests to the understanding the idea of unity.

8. Idea of power. Power also is another of those simple ideas which we receive from sensation and reflection. For, observing in ourselves that we do and can think, and that we can at pleasure move several parts of our bodies which were at rest; the effects, also, that natural bodies are able to produce in one another, occurring every moment to our senses,—we both these ways get the idea of power.

9. Idea of succession. Besides these there is another idea, which, though suggested by our senses, yet is more constantly offered to us by what passes in our minds; and that is the idea of succession. For if we look immediately into ourselves, and reflect on what is observable there, we shall find our ideas always, whilst we

are awake, or have any thought, passing in train, one going and another coming, without intermission.

10. Simple ideas the materials of all our knowledge. These, if they are not all, are at least (as I think) the most considerable of those simple ideas which the mind has, and out of which is made all its other knowledge; all which it receives only by the two forementioned ways of sensation and reflection.

Nor let any one think these too narrow bounds for the capacious mind of man to expatiate in, which takes its flight further than the stars, and cannot be confined by the limits of the world; that extends its thoughts often even beyond the utmost expansion of Matter, and makes excursions into that incomprehensible Inane. I grant all this, but desire any one to assign any simple idea which is not received from one of those inlets before mentioned, or any complex idea not made out of those simple ones. Nor will it be so strange to think these few simple ideas sufficient to employ the quickest thought, or largest capacity; and to furnish the materials of all that various knowledge, and more various fan-

cies and opinions of all mankind, if we consider how many words may be made out of the various composition of twenty-four letters; or if, going one step further, we will but reflect on the variety of combinations that may be made with barely one of the above-mentioned ideas, viz. number, whose stock is inexhaustible and truly infinite: and what a large and immense field doth extension alone afford the mathematicians?

Chapter VIII

Some further considerations concerning our Simple Ideas of Sensation

1. Positive ideas from privative causes. Concerning the simple ideas of Sensation, it is to be considered,—that whatsoever is so constituted in nature as to be able, by affecting our senses, to cause any perception in the mind, doth thereby produce in the understanding a simple idea; which, whatever be the external cause of it, when it comes to be taken notice of by our discerning faculty, it is by the mind looked on and considered there to be a

real positive idea in the understanding, as much as any other whatsoever; though, perhaps, the cause of it be but a privation of the subject.

2. Ideas in the mind distinguished from that in things which gives rise to them. Thus the ideas of heat and cold, light and darkness, white and black, motion and rest, are equally clear and positive ideas in the mind; though, perhaps, some of the causes which produce them are barely privations, in those subjects from whence our senses derive those ideas. These the understanding, in its view of them, considers all as distinct positive ideas, without taking notice of the causes that produce them: which is an inquiry not belonging to the idea, as it is in the understanding, but to the nature of the things existing without us. These are two very different things, and carefully to be distinguished; it being one thing to perceive and know the idea of white or black, and quite another to examine what kind of particles they must be, and how ranged in the superficies, to make any object appear white or black.

3. We may have the ideas when we are ignorant of their physical causes. A painter or dyer who never inquired

into their causes hath the ideas of white and black, and other colours, as clearly, perfectly, and distinctly in his understanding, and perhaps more distinctly, than the philosopher who hath busied himself in considering their natures, and thinks he knows how far either of them is, in its cause, positive or privative; and the idea of black is no less positive in his mind than that of white, however the cause of that colour in the external object may be only a privation.

4. Why a privative cause in nature may occasion a positive idea. If it were the design of my present undertaking to inquire into the natural causes and manner of perception, I should offer this as a reason why a privative cause might, in some cases at least, produce a positive idea; viz. that all sensation being produced in us only by different degrees and modes of motion in our animal spirits, variously agitated by external objects, the abatement of any former motion must as necessarily produce a new sensation as the variation or increase of it; and so introduce a new idea, which depends only on a different motion of the animal spirits in that organ.

5. Negative names need not be meaningless. But whether this be so or not I will not here determine, but appeal to every one's own experience, whether the shadow of a man, though it consists of nothing but the absence of light (and the more the absence of light is, the more discernible is the shadow) does not, when a man looks on it, cause as clear and positive idea in his mind as a man himself, though covered over with clear sunshine? And the picture of a shadow is a positive thing. Indeed, we have negative names, which stand not directly for positive ideas, but for their absence, such as insipid, silence, nihil, &c.; which words denote positive ideas, v.g. taste, sound, being, with a signification of their absence.

6. Whether any ideas are due to causes really privative. And thus one may truly be said to see darkness. For, supposing a hole perfectly dark, from whence no light is reflected, it is certain one may see the figure of it, or it may be painted; or whether the ink I write with makes any other idea, is a question. The privative causes I have here assigned of positive ideas are according to the

common opinion; but, in truth, it will be hard to determine whether there be really any ideas from a privative cause, till it be determined, whether rest be any more a privation than motion.

7. Ideas in the mind, qualities in bodies. To discover the nature of our ideas the better, and to discourse of them intelligibly, it will be convenient to distinguish them as they are ideas or perceptions in our minds; and as they are modifications of matter in the bodies that cause such perceptions in us: that so we may not think (as perhaps usually is done) that they are exactly the images and resemblances of something inherent in the subject; most of those of sensation being in the mind no more the likeness of something existing without us, than the names that stand for them are the likeness of our ideas, which yet upon hearing they are apt to excite in us.

8. Our ideas and the qualities of bodies. Whatsoever the mind perceives in itself, or is the immediate object of perception, thought, or understanding, that I call idea; and the power to produce any idea in our mind, I call

quality of the subject wherein that power is. Thus a snowball having the power to produce in us the ideas of white, cold, and round,—the power to produce those ideas in us, as they are in the snowball, I call qualities; and as they are sensations or perceptions in our understandings, I call them ideas; which ideas, if I speak of sometimes as in the things themselves, I would be understood to mean those qualities in the objects which produce them in us.

9. Primary qualities of bodies. Qualities thus considered in bodies are,

First, such as are utterly inseparable from the body, in what state soever it be; and such as in all the alterations and changes it suffers, all the force can be used upon it, it constantly keeps; and such as sense constantly finds in every particle of matter which has bulk enough to be perceived; and the mind finds inseparable from every particle of matter, though less than to make itself singly be perceived by our senses: v.g. Take a grain of wheat, divide it into two parts; each part has still solidity, extension, figure, and mobility: divide it again,

and it retains still the same qualities; and so divide it on, till the parts become insensible; they must retain still each of them all those qualities. For division (which is all that a mill, or pestle, or any other body, does upon another, in reducing it to insensible parts) can never take away either solidity, extension, figure, or mobility from any body, but only makes two or more distinct separate masses of matter, of that which was but one before; all which distinct masses, reckoned as so many distinct bodies, after division, make a certain number. These I call original or primary qualities of body, which I think we may observe to produce simple ideas in us, viz. solidity, extension, figure, motion or rest, and number.

10. Secondary qualities of bodies. Secondly, such qualities which in truth are nothing in the objects themselves but power to produce various sensations in us by their primary qualities, i.e. by the bulk, figure, texture, and motion of their insensible parts, as colours, sounds, tastes, &c. These I call secondary qualities. To these might be added a third sort, which are allowed to be barely

powers; though they are as much real qualities in the subject as those which I, to comply with the common way of speaking, call qualities, but for distinction, secondary qualities. For the power in fire to produce a new colour, or consistency, in wax or clay,—by its primary qualities, is as much a quality in fire, as the power it has to produce in me a new idea or sensation of warmth or burning, which I felt not before,—by the same primary qualities, viz. the bulk, texture, and motion of its insensible parts.

11. How bodies produce ideas in us. The next thing to be considered is, how bodies produce ideas in us; and that is manifestly by impulse, the only way which we can conceive bodies to operate in.

12. By motions, external, and in our organism. If then external objects be not united to our minds when they produce ideas therein; and yet we perceive these original qualities in such of them as singly fall under our senses, it is evident that some motion must be thence continued by our nerves, or animal spirits, by some parts of our bodies, to the brains or the seat of sensation,

there to produce in our minds the particular ideas we have of them. And since the extension, figure, number, and motion of bodies of an observable bigness, may be perceived at a distance by the sight, it is evident some singly imperceptible bodies must come from them to the eyes, and thereby convey to the brain some motion; which produces these ideas which we have of them in us.

13. How secondary qualities produce their ideas. After the same manner, that the ideas of these original qualities are produced in us, we may conceive that the ideas of secondary qualities are also produced, viz. by the operation of insensible particles on our senses. For, it being manifest that there are bodies and good store of bodies, each whereof are so small, that we cannot by any of our senses discover either their bulk, figure, or motion,—as is evident in the particles of the air and water, and others extremely smaller than those; perhaps as much smaller than the particles of air and water, as the particles of air and water are smaller than peas or hail-stones;—let us suppose at present that the differ-

ent motions and figures, bulk and number, of such particles, affecting the several organs of our senses, produce in us those different sensations which we have from the colours and smells of bodies; v.g. that a violet, by the impulse of such insensible particles of matter, of peculiar figures and bulks, and in different degrees and modifications of their motions, causes the ideas of the blue colour, and sweet scent of that flower to be produced in our minds. It being no more impossible to conceive that God should annex such ideas to such motions, with which they have no similitude, than that he should annex the idea of pain to the motion of a piece of steel dividing our flesh, with which that idea hath no resemblance.

14. They depend on the primary qualities. What I have said concerning colours and smells may be understood also of tastes and sounds, and other the like sensible qualities; which, whatever reality we by mistake attribute to them, are in truth nothing in the objects themselves, but powers to produce various sensations in us; and depend on those primary qualities, viz. bulk, figure,

texture, and motion of parts as I have said.

15. Ideas of primary qualities are resemblances; of secondary, not. From whence I think it easy to draw this observation,—that the ideas of primary qualities of bodies are resemblances of them, and their patterns do really exist in the bodies themselves, but the ideas produced in us by these secondary qualities have no resemblance of them at all. There is nothing like our ideas, existing in the bodies themselves. They are, in the bodies we denominate from them, only a power to produce those sensations in us: and what is sweet, blue, or warm in idea, is but the certain bulk, figure, and motion of the insensible parts, in the bodies themselves, which we call so.

16. Examples. Flame is denominated hot and light; snow, white and cold; and manna, white and sweet, from the ideas they produce in us. Which qualities are commonly thought to be the same in those bodies that those ideas are in us, the one the perfect resemblance of the other, as they are in a mirror, and it would by most men be judged very extravagant if one should say otherwise.

And yet he that will consider that the same fire that, at one distance produces in us the sensation of warmth, does, at a nearer approach, produce in us the far different sensation of pain, ought to bethink himself what reason he has to say—that this idea of warmth, which was produced in him by the fire, is actually in the fire; and his idea of pain, which the same fire produced in him the same way, is not in the fire. Why are whiteness and coldness in snow, and pain not, when it produces the one and the other idea in us; and can do neither, but by the bulk, figure, number, and motion of its solid parts?

17. The ideas of the primary alone really exist. The particular bulk, number, figure, and motion of the parts of fire or snow are really in them,—whether any one's senses perceive them or no: and therefore they may be called real qualities, because they really exist in those bodies. But light, heat, whiteness, or coldness, are no more really in them than sickness or pain is in manna. Take away the sensation of them; let not the eyes see light or colours, nor the ears hear sounds; let the palate

not taste, nor the nose smell, and all colours, tastes, odours, and sounds, as they are such particular ideas, vanish and cease, and are reduced to their causes, i.e. bulk, figure, and motion of parts.

18. The secondary exist in things only as modes of the primary. A piece of manna of a sensible bulk is able to produce in us the idea of a round or square figure; and by being removed from one place to another, the idea of motion. This idea of motion represents it as it really is in manna moving: a circle or square are the same, whether in idea or existence, in the mind or in the manna. And this, both motion and figure, are really in the manna, whether we take notice of them or no: this everybody is ready to agree to. Besides, manna, by its bulk, figure, texture, and motion of its parts, has a power to produce the sensations of sickness, and sometimes of acute pains or gripings in us. That these ideas of sickness and pain are not in the manna, but effects of its operations on us, and are nowhere when we feel them not; this also every one readily agrees to. And yet men are hardly to be brought to think that sweetness and whiteness are

not really in manna; which are but the effects of the operations of manna, by the motion, size, and figure of its particles, on the eyes and palate: as the pain and sickness caused by manna are confessedly nothing but the effects of its operations on the stomach and guts, by the size, motion, and figure of its insensible parts, (for by nothing else can a body operate, as has been proved): as if it could not operate on the eyes and palate, and thereby produce in the mind particular distinct ideas, which in itself it has not, as well as we allow it can operate on the guts and stomach, and thereby produce distinct ideas, which in itself it has not. These ideas, being all effects of the operations of manna on several parts of our bodies, by the size, figure number, and motion of its parts;—why those produced by the eyes and palate should rather be thought to be really in the manna, than those produced by the stomach and guts; or why the pain and sickness, ideas that are the effect of manna, should be thought to be nowhere when they are not felt; and yet the sweetness and whiteness, effects of the same manna on other parts of the body,

by ways equally as unknown, should be thought to exist in the manna, when they are not seen or tasted, would need some reason to explain.

19. Examples. Let us consider the red and white colours in porphyry. Hinder light from striking on it, and its colours vanish; it no longer produces any such ideas in us: upon the return of light it produces these appearances on us again. Can any one think any real alterations are made in the porphyry by the presence or absence of light; and that those ideas of whiteness and redness are really in porphyry in the light, when it is plain it has no colour in the dark? It has, indeed, such a configuration of particles, both night and day, as are apt, by the rays of light rebounding from some parts of that hard stone, to produce in us the idea of redness, and from others the idea of whiteness; but whiteness or redness are not in it at any time, but such a texture that hath the power to produce such a sensation in us.

20. Pound an almond, and the clear white colour will be altered into a dirty one, and the sweet taste into an oily one. What real alteration can the beating of the

pestle make in any body, but an alteration of the texture of it?

21. Explains how water felt as cold by one hand may be warm to the other. Ideas being thus distinguished and understood, we may be able to give an account how the same water, at the same time, may produce the idea of cold by one hand and of heat by the other: whereas it is impossible that the same water, if those ideas were really in it, should at the same time be both hot and cold. For, if we imagine warmth, as it is in our hands, to be nothing but a certain sort and degree of motion in the minute particles of our nerves or animal spirits, we may understand how it is possible that the same water may, at the same time, produce the sensations of heat in one hand and cold in the other; which yet figure never does, that never producing—the idea of a square by one hand which has produced the idea of a globe by another. But if the sensation of heat and cold be nothing but the increase or diminution of the motion of the minute parts of our bodies, caused by the corpuscles of any other body, it is easy to be understood, that if that motion be

greater in one hand than in the other; if a body be applied to the two hands, which has in its minute particles a greater motion than in those of one of the hands, and a less than in those of the other, it will increase the motion of the one hand and lessen it in the other; and so cause the different sensations of heat and cold that depend thereon.

22. An excursion into natural philosophy. I have in what just goes before been engaged in physical inquiries a little further than perhaps I intended. But, it being necessary to make the nature of sensation a little understood; and to make the difference between the qualities in bodies, and the ideas produced by them in the mind, to be distinctly conceived, without which it were impossible to discourse intelligibly of them;—I hope I shall be pardoned this little excursion into natural philosophy; it being necessary in our present inquiry to distinguish the primary and real qualities of bodies, which are always in them (viz. solidity, extension, figure, number, and motion, or rest, and are sometimes perceived by us, viz. when the bodies they are in are big enough

singly to be discerned), from those secondary and imputed qualities, which are but the powers of several combinations of those primary ones, when they operate without being distinctly discerned;—whereby we may also come to know what ideas are, and what are not, resemblances of something really existing in the bodies we denominate from them.

23. Three sorts of qualities in bodies. The qualities, then, that are in bodies, rightly considered, are of three sorts:—

First, The bulk, figure, number, situation, and motion or rest of their solid parts. Those are in them, whether we perceive them or not; and when they are of that size that we can discover them, we have by these an idea of the thing as it is in itself; as is plain in artificial things. These I call primary qualities.

Secondly, The power that is in any body, by reason of its insensible primary qualities, to operate after a peculiar manner on any of our senses, and thereby produce in us the different ideas of several colours, sounds, smells, tastes, &c. These are usually called sensible qualities.

Thirdly, The power that is in any body, by reason of the particular constitution of its primary qualities, to make such a change in the bulk, figure, texture, and motion of another body, as to make it operate on our senses differently from what it did before. Thus the sun has a power to make wax white, and fire to make lead fluid. These are usually called powers.

The first of these, as has been said, I think may be properly called real, original, or primary qualities; because they are in the things themselves, whether they are perceived or not: and upon their different modifications it is that the secondary qualities depend.

The other two are only powers to act differently upon other things: which powers result from the different modifications of those primary qualities.

24. The first are resemblances; the second thought to be resemblances, but are not; the third neither are nor are thought so. But, though the two latter sorts of qualities are powers barely, and nothing but powers, relating to several other bodies, and resulting from the different modifications of the original qualities, yet they

are generally otherwise thought of. For the second sort, viz, the powers to produce several ideas in us, by our senses, are looked upon as real qualities in the things thus affecting us: but the third sort are called and esteemed barely powers. v.g. The idea of heat or light, which we receive by our eyes, or touch, from the sun, are commonly thought real qualities existing in the sun, and something more than mere powers in it. But when we consider the sun in reference to wax, which it melts or blanches, we look on the whiteness and softness produced in the wax, not as qualities in the sun, but effects produced by powers in it. Whereas, if rightly considered, these qualities of light and warmth, which are perceptions in me when I am warmed or enlightened by the sun, are no otherwise in the sun, than the changes made in the wax, when it is blanched or melted, are in the sun. They are all of them equally powers in the sun, depending on its primary qualities; whereby it is able, in the one case, so to alter the bulk, figure, texture, or motion of some of the insensible parts of my eyes or hands, as thereby to produce in me the idea of light or

heat; and in the other, it is able so to alter the bulk, figure, texture, or motion of the insensible parts of the wax, as to make them fit to produce in me the distinct ideas of white and fluid.

25. Why the secondary are ordinarily taken for real qualities, and not for bare powers. The reason why the one are ordinarily taken for real qualities, and the other only for bare powers, seems to be, because the ideas we have of distinct colours, sounds, &c., containing nothing at all in them of bulk, figure, or motion, we are not apt to think them the effects of these primary qualities; which appear not, to our senses, to operate in their production, and with which they have not any apparent congruity or conceivable connexion. Hence it is that we are so forward to imagine, that those ideas are the resemblances of something really existing in the objects themselves: since sensation discovers nothing of bulk, figure, or motion of parts in their production; nor can reason show how bodies, by their bulk, figure, and motion, should produce in the mind the ideas of blue or yellow, &c. But, in the other case, in the operations of

bodies changing the qualities one of another, we plainly discover that the quality produced hath commonly no resemblance with anything in the thing producing it; wherefore we look on it as a bare effect of power. For, through receiving the idea of heat or light from the sun, we are apt to think it is a perception and resemblance of such a quality in the sun; yet when we see wax, or a fair face, receive change of colour from the sun, we cannot imagine that to be the reception or resemblance of anything in the sun, because we find not those different colours in the sun itself. For, our senses being able to observe a likeness or unlikeness of sensible qualities in two different external objects, we forwardly enough conclude the production of any sensible quality in any subject to be an effect of bare power, and not the communication of any quality which was really in the efficient, when we find no such sensible quality in the thing that produced it. But our senses, not being able to discover any unlikeness between the idea produced in us, and the quality of the object producing it, we are apt to imagine that our ideas are re-

semblances of something in the objects, and not the effects of certain powers placed in the modification of their primary qualities, with which primary qualities the ideas produced in us have no resemblance.

26. Secondary qualities twofold; first, immediately perceivable; secondly, mediately perceivable. To conclude. Besides those before-mentioned primary qualities in bodies, viz. bulk, figure, extension, number, and motion of their solid parts; all the rest, whereby we take notice of bodies, and distinguish them one from another, are nothing else but several powers in them, depending on those primary qualities; whereby they are fitted, either by immediately operating on our bodies to produce several different ideas in us; or else, by operating on other bodies, so to change their primary qualities as to render them capable of producing ideas in us different from what before they did. The former of these, I think, may be called secondary qualities immediately perceivable: the latter, secondary qualities, mediately perceivable.

Chapter IX Of Perception

1. Perception the first simple idea of reflection. *Perception*, as it is the first faculty of the mind exercised about our ideas; so it is the first and simplest idea we have from reflection, and is by some called thinking in general. Though thinking, in the propriety of the English tongue, signifies that sort of operation in the mind about its ideas, wherein the mind is active; where it, with some degree of voluntary attention, considers anything. For in bare naked perception, the mind is, for the most part, only passive; and what it perceives, it cannot avoid perceiving.

2. Reflection alone can give us the idea of what perception is. What perception is, every one will know better by reflecting on what he does himself, when he sees, hears, feels, &c., or thinks, than by any discourse of mine. Whoever reflects on what passes in his own mind cannot miss it. And if he does not reflect, all the words in the world cannot make him have any notion of it.

3. Arises in sensation only when the mind notices the organic impression. This is certain, that whatever alterations are made in the body, if they reach not the mind; whatever impressions are made on the outward parts, if they are not taken notice of within, there is no perception. Fire may burn our bodies with no other effect than it does a billet, unless the motion be continued to the brain, and there the sense of heat, or idea of pain, be produced in the mind; wherein consists actual perception.

4. Impulse on the organ insufficient. How often may a man observe in himself, that whilst his mind is intently employed in the contemplation of some objects, and curiously surveying some ideas that are there, it takes no notice of impressions of sounding bodies made upon the organ of hearing, with the same alteration that uses to be for the producing the idea of sound? A sufficient impulse there may be on the organ; but it not reaching the observation of the mind, there follows no perception: and though the motion that uses to produce the idea of sound be made in the ear, yet no sound is heard. Want of sensation, in this case, is not through any de-

fect in the organ, or that the man's ears are less affected than at other times when he does hear: but that which uses to produce the idea, though conveyed in by the usual organ, not being taken notice of in the understanding, and so imprinting no idea in the mind, there follows no sensation. So that wherever there is sense or perception, there some idea is actually produced, and present in the understanding.

5. Children, though they may have ideas in the womb, have none innate. Therefore I doubt not but children, by the exercise of their senses about objects that affect them in the womb, receive some few ideas before they are born, as the unavoidable effects, either of the bodies that environ them, or else of those wants or diseases they suffer; amongst which (if one may conjecture concerning things not very capable of examination) I think the ideas of hunger and warmth are two: which probably are some of the first that children have, and which they scarce ever part with again.

6. The effects of sensation in the womb. But though it be reasonable to imagine that children receive some ideas

before they come into the world, yet these simple ideas are far from those innate principles which some contend for, and we, above, have rejected. These here mentioned, being the effects of sensation, are only from some affections of the body, which happen to them there, and so depend on something exterior to the mind; no otherwise differing in their manner of production from other ideas derived from sense, but only in the precedence of time. Whereas those innate principles are supposed to be quite of another nature; not coming into the mind by any accidental alterations in, or operations on the body; but, as it were, original characters impressed upon it, in the very first moment of its being and constitution.

7. Which ideas appear first, is not evident, nor important. As there are some ideas which we may reasonably suppose may be introduced into the minds of children in the womb, subservient to the necessities of their life and being there: so, after they are born, those ideas are the earliest imprinted which happen to be the sensible qualities which first occur to them; amongst which light

is not the least considerable, nor of the weakest efficacy. And how covetous the mind is to be furnished with all such ideas as have no pain accompanying them, may be a little guessed by what is observable in children new-born; who always turn their eyes to that part from whence the light comes, lay them how you please. But the ideas that are most familiar at first, being various according to the divers circumstances of children's first entertainment in the world, the order wherein the several ideas come at first into the mind is very various, and uncertain also; neither is it much material to know it.

8. Sensations often changed by the judgment. We are further to consider concerning perception, that the ideas we receive by sensation are often, in grown people, altered by the judgment, without our taking notice of it. When we set before our eyes a round globe of any uniform colour, v.g. gold, alabaster, or jet, it is certain that the idea thereby imprinted on our mind is of a flat circle, variously shadowed, with several degrees of light and brightness coming to our eyes. But we having, by use,

been accustomed to perceive what kind of appearance convex bodies are wont to make in us; what alterations are made in the reflections of light by the difference of the sensible figures of bodies;—the judgment presently, by an habitual custom, alters the appearances into their causes. So that from that which is truly variety of shadow or colour, collecting the figure, it makes it pass for a mark of figure, and frames to itself the perception of a convex figure and an uniform colour; when the idea we receive from thence is only a plane variously coloured, as is evident in painting. To which purpose I shall here insert a problem of that very ingenious and studious promoter of real knowledge, the learned and worthy Mr. Molyneux, which he was pleased to send me in a letter some months since; and it is this:—"Suppose a man born blind, and now adult, and taught by his touch to distinguish between a cube and a sphere of the same metal, and nighly of the same bigness, so as to tell, when he felt one and the other, which is the cube, which the sphere. Suppose then the cube and sphere placed on a table, and the blind man be made to see:

quaere, whether by his sight, before he touched them, he could now distinguish and tell which is the globe, which the cube?" To which the acute and judicious proposer answers, "Not. For, though he has obtained the experience of how a globe, how a cube affects his touch, yet he has not yet obtained the experience, that what affects his touch so or so, must affect his sight so or so; or that a protuberant angle in the cube, that pressed his hand unequally, shall appear to his eye as it does in the cube."—I agree with this thinking gentleman, whom I am proud to call my friend, in his answer to this problem; and am of opinion that the blind man, at first sight, would not be able with certainty to say which was the globe, which the cube, whilst he only saw them; though he could unerringly name them by his touch, and certainly distinguish them by the difference of their figures felt. This I have set down, and leave with my reader, as an occasion for him to consider how much he may be beholden to experience, improvement, and acquired notions, where he thinks he had not the least use of, or help from them. And the rather,

because this observing gentleman further adds, that "having, upon the occasion of my book, proposed this to divers very ingenious men, he hardly ever met with one that at first gave the answer to it which he thinks true, till by hearing his reasons they were convinced."

9. This judgment apt to be mistaken for direct perception. But this is not, I think, usual in any of our ideas, but those received by sight. Because sight, the most comprehensive of all our senses, conveying to our minds the ideas of light and colours, which are peculiar only to that sense; and also the far different ideas of space, figure, and motion, the several varieties whereof change the appearances of its proper object, viz. light and colours; we bring ourselves by use to judge of the one by the other. This, in many cases by a settled habit,—in things whereof we have frequent experience, is performed so constantly and so quick, that we take that for the perception of our sensation which is an idea formed by our judgment; so that one, viz. that of sensation, serves only to excite the other, and is scarce taken notice of itself;—as a man who reads or hears with attention and

understanding, takes little notice of the characters or sounds, but of the ideas that are excited in him by them.

10. How, by habit, ideas of sensation are unconsciously changed into ideas of judgment. Nor need we wonder that this is done with so little notice, if we consider how quick the actions of the mind are performed. For, as itself is thought to take up no space, to have no extension; so its actions seem to require no time, but many of them seem to be crowded into an instant. I speak this in comparison to the actions of the body. Any one may easily observe this in his own thoughts, who will take the pains to reflect on them. How, as it were in an instant, do our minds, with one glance, see all the parts of a demonstration, which may very well be called a long one, if we consider the time it will require to put it into words, and step by step show it another? Secondly, we shall not be so much surprised that this is done in us with so little notice, if we consider how the facility which we get of doing things, by a custom of doing, makes them often pass in us without our notice. Habits, especially such as are begun very early, come at

last to produce actions in us, which often escape our observation. How frequently do we, in a day, cover our eyes with our eyelids, without perceiving that we are at all in the dark! Men that, by custom, have got the use of a by-word, do almost in every sentence pronounce sounds which, though taken notice of by others, they themselves neither hear nor observe. And therefore it is not so strange, that our mind should often change the idea of its sensation into that of its judgment, and make one serve only to excite the other, without our taking notice of it.

11. Perception puts the difference between animals and vegetables. This faculty of perception seems to me to be, that which puts the distinction betwixt the animal kingdom and the inferior parts of nature. For, however vegetables have, many of them, some degrees of motion, and upon the different application of other bodies to them, do very briskly alter their figures and motions, and so have obtained the name of sensitive plants, from a motion which has some resemblance to that which in animals follows upon sensation: yet I suppose it is all

bare mechanism; and no otherwise produced than the turning of a wild oat-beard, by the insinuation of the particles of moisture, or the shortening of a rope, by the affusion of water. All which is done without any sensation in the subject, or the having or receiving any ideas.

12. Perception in all animals. Perception, I believe, is, in some degree, in all sorts of animals; though in some possibly the avenues provided by nature for the reception of sensations are so few, and the perception they are received with so obscure and dull, that it comes extremely short of the quickness and variety of sensation which is in other animals; but yet it is sufficient for, and wisely adapted to, the state and condition of that sort of animals who are thus made. So that the wisdom and goodness of the Maker plainly appear in all the parts of this stupendous fabric, and all the several degrees and ranks of creatures in it.

13. According to their condition. We may, I think, from the make of an oyster or cockle, reasonably conclude that it has not so many, nor so quick senses as a man,

or several other animals; nor if it had, would it, in that state and incapacity of transferring itself from one place to another, be bettered by them. What good would sight and hearing do to a creature that cannot move itself to or from the objects wherein at a distance it perceives good or evil? And would not quickness of sensation be an inconvenience to an animal that must lie still where chance has once placed it, and there receive the afflux of colder or warmer, clean or foul water, as it happens to come to it?

14. Decay of perception in old age. But yet I cannot but think there is some small dull perception, whereby they are distinguished from perfect insensibility. And that this may be so, we have plain instances, even in mankind itself. Take one in whom decrepit old age has blotted out the memory of his past knowledge, and clearly wiped out the ideas his mind was formerly stored with, and has, by destroying his sight, hearing, and smell quite, and his taste to a great degree, stopped up almost all the passages for new ones to enter; or if there be some of the inlets yet half open, the impressions made are

scarcely perceived, or not at all retained. How far such an one (notwithstanding all that is boasted of innate principles) is in his knowledge and intellectual faculties above the condition of a cockle or an oyster, I leave to be considered. And if a man had passed sixty years in such a state, as it is possible he might, as well as three days, I wonder what difference there would be, in any intellectual perfections, between him and the lowest degree of animals.

15. Perception the inlet of all materials of knowledge. Perception then being the first step and degree towards knowledge, and the inlet of all the materials of it; the fewer senses any man, as well as any other creature, hath; and the fewer and duller the impressions are that are made by them, and the duller the faculties are that are employed about them,—the more remote are they from that knowledge which is to be found in some men. But this being in great variety of degrees (as may be perceived amongst men) cannot certainly be discovered in the several species of animals, much less in their particular individuals. It suffices me only to have remarked

here,—that perception is the first operation of all our intellectual faculties, and the inlet of all knowledge in our minds. And I am apt too to imagine, that it is perception, in the lowest degree of it, which puts the boundaries between animals and the inferior ranks of creatures. But this I mention only as my conjecture by the by; it being indifferent to the matter in hand which way the learned shall determine of it.

Chapter X Of Retention

1. Contemplation. The next faculty of the mind, whereby it makes a further progress towards knowledge, is that which I call retention; or the keeping of those simple ideas which from sensation or reflection it hath received. This is done two ways.

First, by keeping the idea which is brought into it, for some time actually in view, which is called contemplation.

2. Memory. The other way of retention is, the power to revive again in our minds those ideas which, after im-

printing, have disappeared, or have been as it were laid aside out of sight. And thus we do, when we conceive heat or light, yellow or sweet,—the object being removed. This is memory, which is as it were the storehouse of our ideas. For, the narrow mind of man not being capable of having many ideas under view and consideration at once, it was necessary to have a repository, to lay up those ideas which, at another time, it might have use of. But, our ideas being nothing but actual perceptions in the mind, which cease to be anything when there is no perception of them; this laying up of our ideas in the repository of the memory signifies no more but this,—that the mind has a power in many cases to revive perceptions which it has once had, with this additional perception annexed to them, that it has had them before. And in this sense it is that our ideas are said to be in our memories, when indeed they are actually nowhere;—but only there is an ability in the mind when it will to revive them again, and as it were paint them anew on itself, though some with more, some with less difficulty; some more lively, and others more

obscurely. And thus it is, by the assistance of this faculty, that we are said to have all those ideas in our understandings which, though we do not actually contemplate, yet we can bring in sight, and make appear again, and be the objects of our thoughts, without the help of those sensible qualities which first imprinted them there.

3. Attention, repetition, pleasure and pain, fix ideas. Attention and repetition help much to the fixing any ideas in the memory. But those which naturally at first make the deepest and most lasting impressions, are those which are accompanied with pleasure or pain. The great business of the senses being, to make us take notice of what hurts or advantages the body, it is wisely ordered by nature, as has been shown, that pain should accompany the reception of several ideas; which, supplying the place of consideration and reasoning in children, and acting quicker than consideration in grown men, makes both the old and young avoid painful objects with that haste which is necessary for their preservation; and in both settles in the memory a caution for the future.

4. Ideas fade in the memory. Concerning the several degrees of lasting, wherewith ideas are imprinted on the memory, we may observe,—that some of them have been produced in the understanding by an object affecting the senses once only, and no more than once; others, that have more than once offered themselves to the senses, have yet been little taken notice of: the mind, either heedless, as in children, or otherwise employed, as in men intent only on one thing; not setting the stamp deep into itself. And in some, where they are set on with care and repeated impressions, either through the temper of the body, or some other fault, the memory is very weak. In all these cases, ideas in the mind quickly fade, and often vanish quite out of the understanding, leaving no more footsteps or remaining characters of themselves than shadows do flying over fields of corn, and the mind is as void of them as if they had never been there.

5. Causes of oblivion. Thus many of those ideas which were produced in the minds of children, in the beginning of their sensation, (some of which perhaps, as of

some pleasures and pains, were before they were born, and others in their infancy,) if the future course of their lives they are not repeated again, are quite lost, without the least glimpse remaining of them. This may be observed in those who by some mischance have lost their sight when they were very young; in whom the ideas of colours having been but slightly taken notice of, and ceasing to be repeated, do quite wear out; so that some years after, there is no more notion nor memory of colours left in their minds, than in those of people born blind. The memory of some men, it is true, is very tenacious, even to a miracle. But yet there seems to be a constant decay of all our ideas, even of those which are struck deepest, and in minds the most retentive; so that if they be not sometimes renewed, by repeated exercise of the senses, or reflection on those kinds of objects which at first occasioned them, the print wears out, and at last there remains nothing to be seen. Thus the ideas, as well as children, of our youth, often die before us: and our minds represent to us those tombs to which we are approaching; where, though the brass and

marble remain, yet the inscriptions are effaced by time, and the imagery moulders away. The pictures drawn in our minds are laid in fading colours; and if not sometimes refreshed, vanish and disappear. How much the constitution of our bodies and the make of our animal spirits are concerned in this; and whether the temper of the brain makes this difference, that in some it retains the characters drawn on it like marble, in others like freestone, and in others little better than sand, I shall not here inquire; though it may seem probable that the constitution of the body does sometimes influence the memory, since we oftentimes find a disease quite strip the mind of all its ideas, and the flames of a fever in a few days calcine all those images to dust and confusion, which seemed to be as lasting as if graved in marble.

6. Constantly repeated ideas can scarce be lost. But concerning the ideas themselves, it is easy to remark, that those that are oftenest refreshed (amongst which are those that are conveyed into the mind by more ways than one) by a frequent return of the objects or actions that produce them, fix themselves best in the memory,

and remain clearest and longest there; and therefore those which are of the original qualities of bodies, vis. solidity, extension, figure, motion, and rest; and those that almost constantly affect our bodies, as heat and cold; and those which are the affections of all kinds of beings, as existence, duration, and number, which almost every object that affects our senses, every thought which employs our minds, bring along with them;—these, I say, and the like ideas, are seldom quite lost, whilst the mind retains any ideas at all.

7. In remembering, the mind is often active. In this secondary perception, as I may so call it, or viewing again the ideas that are lodged in the memory, the mind is oftentimes more than barely passive; the appearance of those dormant pictures depending sometimes on the will. The mind very often sets itself on work in search of some hidden idea, and turns as it were the eye of the soul upon it; though sometimes too they start up in our minds of their own accord, and offer themselves to the understanding; and very often are roused and tumbled out of their dark cells into open daylight, by

turbulent and tempestuous passions; our affections bringing ideas to our memory, which had otherwise lain quiet and unregarded. This further is to be observed, concerning ideas lodged in the memory, and upon occasion revived by the mind, that they are not only (as the word revive imports) none of them new ones, but also that the mind takes notice of them as of a former impression, and renews its acquaintance with them, as with ideas it had known before. So that though ideas formerly imprinted are not all constantly in view, yet in remembrance they are constantly known to be such as have been formerly imprinted; i.e. in view, and taken notice of before, by the understanding.

8. Two defects in the memory, oblivion and slowness. Memory, in an intellectual creature, is necessary in the next degree to perception. It is of so great moment, that, where it is wanting, all the rest of our faculties are in a great measure useless. And we in our thoughts, reasonings, and knowledge, could not proceed beyond present objects, were it not for the assistance of our memories; wherein there may be two defects:—

First, That it loses the idea quite, and so far it produces perfect ignorance. For, since we can know nothing further than we have the idea of it, when that is gone, we are in perfect ignorance.

Secondly, That it moves slowly, and retrieves not the ideas that it has, and are laid up in store, quick enough to serve the mind upon occasion. This, if it be to a great degree, is stupidity; and he who, through this default in his memory, has not the ideas that are really preserved there, ready at hand when need and occasion calls for them, were almost as good be without them quite, since they serve him to little purpose. The dull man, who loses the opportunity, whilst he is seeking in his mind for those ideas that should serve his turn, is not much more happy in his knowledge than one that is perfectly ignorant. It is the business therefore of the memory to furnish to the mind those dormant ideas which it has present occasion for; in the having them ready at hand on all occasions, consists that which we call invention, fancy, and quickness of parts.

9. A defect which belongs to the memory of man, as

finite. These are defects we may observe in the memory of one man compared with another. There is another defect which we may conceive to be in the memory of man in general;—compared with some superior created intellectual beings, which in this faculty may so far excel man, that they may have constantly in view the whole scene of all their former actions, wherein no one of the thoughts they have ever had may slip out of their sight. The omniscience of God, who knows all things, past, present, and to come, and to whom the thoughts of men's hearts always lie open, may satisfy us of the possibility of this. For who can doubt but God may communicate to those glorious spirits, his immediate attendants, any of his perfections; in what proportions he pleases, as far as created finite beings can be capable? It is reported of that prodigy of parts, Monsieur Pascal, that till the decay of his health had impaired his memory, he forgot nothing of what he had done, read, or thought, in any part of his rational age. This is a privilege so little known to most men, that it seems almost incredible to those who, after the ordinary way, measure all others

by themselves; but yet, when considered, may help us to enlarge our thoughts towards greater perfections of it, in superior ranks of spirits. For this of Monsieur Pascal was still with the narrowness that human minds are confined to here,—of having great variety of ideas only by succession, not all at once. Whereas the several degrees of angels may probably have larger views; and some of them be endowed with capacities able to retain together, and constantly set before them, as in one picture, all their past knowledge at once. This, we may conceive, would be no small advantage to the knowledge of a thinking man,—if all his past thoughts and reasonings could be always present to him. And therefore we may suppose it one of those ways, wherein the knowledge of separate spirits may exceedingly surpass ours.

10. Brutes have memory. This faculty of laying up and retaining the ideas that are brought into the mind, several other animals seem to have to a great degree, as well as man. For, to pass by other instances, birds learning of tunes, and the endeavours one may observe in

them to hit the notes right, put it past doubt with me, that they have perception, and retain ideas in their memories, and use them for patterns. For it seems to me impossible that they should endeavour to conform their voices to notes (as it is plain they do) of which they had no ideas. For, though I should grant sound may mechanically cause a certain motion of the animal spirits in the brains of those birds, whilst the tune is actually playing; and that motion may be continued on to the muscles of the wings, and so the bird mechanically be driven away by certain noises, because this may tend to the bird's preservation; yet that can never be supposed a reason why it should cause mechanically—either whilst the tune is playing, much less after it has ceased—such a motion of the organs in the bird's voice as should conform it to the notes of a foreign sound, which imitation can be of no use to the bird's preservation. But, which is more, it cannot with any appearance of reason be supposed (much less proved) that birds, without sense and memory, can approach their notes nearer and nearer by degrees to a tune played yesterday; which if they

have no idea of in their memory, is now nowhere, nor can be a pattern for them to imitate, or which any repeated essays can bring them nearer to. Since there is no reason why the sound of a pipe should leave traces in their brains, which, not at first, but by their after-endeavours, should produce the like sounds; and why the sounds they make themselves, should not make traces which they should follow, as well as those of the pipe, is impossible to conceive.

Chapter XI

Of Discerning, and other operations of the Mind

1. No knowledge without discernment. Another faculty we may take notice of in our minds is that of discerning and distinguishing between the several ideas it has. It is not enough to have a confused perception of something in general. Unless the mind had a distinct perception of different objects and their qualities, it would be capable of very little knowledge, though the bodies that affect us were as busy about us as they are now,

and the mind were continually employed in thinking. On this faculty of distinguishing one thing from another depends the evidence and certainty of several, even very general, propositions, which have passed for innate truths;—because men, overlooking the true cause why those propositions find universal assent, impute it wholly to native uniform impressions; whereas it in truth depends upon this clear discerning faculty of the mind, whereby it perceives two ideas to be the same, or different. But of this more hereafter.

2. The difference of wit and judgment. How much the imperfection of accurately discriminating ideas one from another lies, either in the dulness or faults of the organs of sense; or want of acuteness, exercise, or attention in the understanding; or hastiness and precipitancy, natural to some tempers, I will not here examine: it suffices to take notice, that this is one of the operations that the mind may reflect on and observe in itself. It is of that consequence to its other knowledge, that so far as this faculty is in itself dull, or not rightly made use of, for the distinguishing one thing from another,—so

far our notions are confused, and our reason and judgment disturbed or misled. If in having our ideas in the memory ready at hand consists quickness of parts; in this, of having them unconfused, and being able nicely to distinguish one thing from another, where there is but the least difference, consists, in a great measure, the exactness of judgment, and clearness of reason, which is to be observed in one man above another. And hence perhaps may be given some reason of that common observation,—that men who have a great deal of wit, and prompt memories, have not always the clearest judgment or deepest reason. For wit lying most in the assemblage of ideas, and putting those together with quickness and variety, wherein can be found any resemblance or congruity, thereby to make up pleasant pictures and agreeable visions in the fancy; judgment, on the contrary, lies quite on the other side, in separating carefully, one from another, ideas wherein can be found the least difference, thereby to avoid being misled by similitude, and by affinity to take one thing for another. This is a way of proceeding quite contrary to metaphor and

allusion; wherein for the most part lies that entertainment and pleantry of wit, which strikes so lively on the fancy, and therefore is so acceptable to all people, because its beauty appears at first sight, and there is required no labor of thought to examine what truth or reason there is in it. The mind, without looking any further, rests satisfied with the agreeableness of the picture and the gaiety of the fancy. And it is a kind of affront to go about to examine it, by the severe rules of truth and good reason; whereby it appears that it consists in something that is not perfectly conformable to them.

3. Clearness done hinders confusion. To the well distinguishing our ideas, it chiefly contributes that they be clear and determinate. And when they are so, it will not breed any confusion or mistake about them, though the senses should (as sometimes they do) convey them from the same object differently on different occasions, and so seem to err. For, though a man in a fever should from sugar have a bitter taste, which at another time would produce a sweet one, yet the idea of bitter in that

man's mind would be as clear and distinct from the idea of sweet as if he had tasted only gall. Nor does it make any more confusion between the two ideas of sweet and bitter, that the same sort of body produces at one time one, and at another time another idea by the taste, than it makes a confusion in two ideas of white and sweet, or white and round, that the same piece of sugar produces them both in the mind at the same time. And the ideas of orange-colour and azure, that are produced in the mind by the same parcel of the infusion of lignum nephriticum, are no less distinct ideas than those of the same colours taken from two very different bodies.

4. Comparing. The *comparing* them one with another, in respect of extent, degrees, time, place, or any other circumstances, is another operation of the mind about its ideas, and is that upon which depends all that large tribe of ideas comprehended under relation; which, of how vast an extent it is, I shall have occasion to consider hereafter.

5. Brutes compare but imperfectly. How far brutes partake in this faculty, is not easy to determine. I imagine

they have it not in any great degree: for, though they probably have several ideas distinct enough, yet it seems to me to be the prerogative of human understanding, when it has sufficiently distinguished any ideas, so as to perceive them to be perfectly different, and so consequently two, to cast about and consider in what circumstances they are capable to be compared. And therefore, I think, beasts compare not their ideas further than some sensible circumstances annexed to the objects themselves. The other power of comparing, which may be observed in men, belonging to general ideas, and useful only to abstract reasonings, we may probably conjecture beasts have not.

6. Compounding. The next operation we may observe in the mind about its ideas is *composition*; whereby it puts together several of those simple ones it has received from sensation and reflection, and combines them into complex ones. Under this of composition may be reckoned also that of enlarging, wherein, though the composition does not so much appear as in more complex ones, yet it is nevertheless a putting several ideas

together, though of the same kind. Thus, by adding several units together, we make the idea of a dozen; and putting together the repeated ideas of several perches, we frame that of a furlong.

7. Brutes compound but little. In this also, I suppose, brutes come far short of man. For, though they take in, and retain together, several combinations of simple ideas, as possibly the shape, smell, and voice of his master make up the complex idea a dog has of him, or rather are so many distinct marks whereby he knows him; yet I do not think they do of themselves ever compound them and make complex ideas. And perhaps even where we think they have complex ideas, it is only one simple one that directs them in the knowledge of several things, which possibly they distinguish less by their sight than we imagine. For I have been credibly informed that a bitch will nurse, play with, and be fond of young foxes, as much as, and in place of her puppies, if you can but get them once to suck her so long that her milk may go through them. And those animals which have a numerous brood of young ones at once, appear not to have

any knowledge of their number; for though they are mightily concerned for any of their young that are taken from them whilst they are in sight or hearing, yet if one or two of them be stolen from them in their absence, or without noise, they appear not to miss them, or to have any sense that their number is lessened.

8. Naming. When children have, by repeated sensations, got ideas fixed in their memories, they begin by degrees to learn the use of signs. And when they have got the skill to apply the organs of speech to the framing of articulate sounds, they begin to make use of words, to signify their ideas to others. These verbal signs they sometimes borrow from others, and sometimes make themselves, as one may observe among the new and unusual names children often give to things in the first use of language.

9. Abstraction. The use of words then being to stand as outward marks of our internal ideas, and those ideas being taken from particular things, if every particular idea that we take in should have a distinct name, names must be endless. To prevent this, the mind makes the particu-

lar ideas received from particular objects to become general; which is done by considering them as they are in the mind such appearances,—separate from all other existences, and the circumstances of real existence, as time, place, or any other concomitant ideas. This is called *abstraction*, whereby ideas taken from particular beings become general representatives of all of the same kind; and their names general names, applicable to whatever exists conformable to such abstract ideas. Such precise, naked appearances in the mind, without considering how, whence, or with what others they came there, the understanding lays up (with names commonly annexed to them) as the standards to rank real existences into sorts, as they agree with these patterns, and to denominate them accordingly. Thus the same colour being observed to-day in chalk or snow, which the mind yesterday received from milk, it considers that appearance alone, makes it a representative of all of that kind; and having given it the name whiteness, it by that sound signifies the same quality wheresoever to be imagined or met with; and thus universals, whether ideas or terms, are made.

10. Brutes abstract not. If it may be doubted whether beasts compound and enlarge their ideas that way to any degree; this, I think, I may be positive in,—that the power of abstracting is not at all in them; and that the having of general ideas is that which puts a perfect distinction betwixt man and brutes, and is an excellency which the faculties of brutes do by no means attain to. For it is evident we observe no footsteps in them of making use of general signs for universal ideas; from which we have reason to imagine that they have not the faculty of abstracting, or making general ideas, since they have no use of words, or any other general signs.

11. Brutes abstract not, yet are not bare machines. Nor can it be imputed to their want of fit organs to frame articulate sounds, that they have no use or knowledge of general words; since many of them, we find, can fashion such sounds, and pronounce words distinctly enough, but never with any such application. And, on the other side, men who, through some defect in the organs, want words, yet fail not to express their universal ideas by

signs, which serve them instead of general words, a faculty which we see beasts come short in. And, therefore, I think, we may suppose, that it is in this that the species of brutes are discriminated from man: and it is that proper difference wherein they are wholly separated, and which at last widens to so vast a distance. For if they have any ideas at all, and are not bare machines, (as some would have them,) we cannot deny them to have some reason. It seems as evident to me, that they do some of them in certain instances reason, as that they have sense; but it is only in particular ideas, just as they received them from their senses. They are the best of them tied up within those narrow bounds, and have not (as I think) the faculty to enlarge them by any kind of abstraction.

12. Idiots and madmen. How far idiots are concerned in the want or weakness of any, or all of the foregoing faculties, an exact observation of their several ways of faulting would no doubt discover. For those who either perceive but dully, or retain the ideas that come into their minds but ill, who cannot readily excite or

compound them, will have little matter to think on. Those who cannot distinguish, compare, and abstract, would hardly be able to understand and make use of language, or judge or reason to any tolerable degree; but only a little and imperfectly about things present, and very familiar to their senses. And indeed any of the forementioned faculties, if wanting, or out of order, produce suitable defects in men's understandings and knowledge.

13. Difference between idiots and madmen. In fine, the defect in naturals seems to proceed from want of quickness, activity, and motion in the intellectual faculties, whereby they are deprived of reason; whereas madmen, on the other side, seem to suffer by the other extreme. For they do not appear to me to have lost the faculty of reasoning, but having joined together some ideas very wrongly, they mistake them for truths; and they err as men do that argue right from wrong principles. For, by the violence of their imaginations, having taken their fancies for realities, they make right deductions from them. Thus you shall find a distracted man fancying

himself a king, with a right inference require suitable attendance, respect, and obedience: others who have thought themselves made of glass, have used the caution necessary to preserve such brittle bodies. Hence it comes to pass that a man who is very sober, and of a right understanding in all other things, may in one particular be as frantic as any in Bedlam; if either by any sudden very strong impression, or long fixing his fancy upon one sort of thoughts, incoherent ideas have been cemented together so powerfully, as to remain united. But there are degrees of madness, as of folly; the disorderly jumbling ideas together is in some more, and some less. In short, herein seems to lie the difference between idiots and madmen: that madmen put wrong ideas together, and so make wrong propositions, but argue and reason right from them; but idiots make very few or no propositions, and reason scarce at all.

14. Method followed in this explication of faculties. These, I think, are the first faculties and operations of the mind, which it makes use of in understanding; and though they are exercised about all its ideas in general,

yet the instances I have hitherto given have been chiefly in simple ideas. And I have subjoined the explication of these faculties of the mind to that of simple ideas, before I come to what I have to say concerning complex ones, for these following reasons:—

First, Because several of these faculties being exercised at first principally about simple ideas, we might, by following nature in its ordinary method, trace and discover them, in their rise, progress, and gradual improvements.

Secondly, Because observing the faculties of the mind, how they operate about simple ideas,—which are usually, in most men's minds, much more clear, precise, and distinct than complex ones,—we may the better examine and learn how the mind extracts, denominates, compares, and exercises, in its other operations about those which are complex, wherein we are much more liable to mistake.

Thirdly, Because these very operations of the mind about ideas received from sensations, are themselves, when reflected on, another set of ideas, derived from

that other source of our knowledge, which I call reflection; and therefore fit to be considered in this place after the simple ideas of sensation. Of compounding, comparing, abstracting, &c., I have but just spoken, having occasion to treat of them more at large in other places.

15. The true beginning of human knowledge. And thus I have given a short, and, I think, true history of the first beginnings of human knowledge;—whence the mind has its first objects; and by what steps it makes its progress to the laying in and storing up those ideas, out of which is to be framed all the knowledge it is capable of: wherein I must appeal to experience and observation whether I am in the right: the best way to come to truth being to examine things as really they are, and not to conclude they are, as we fancy of ourselves, or have been taught by others to imagine.

16. Appeal to experience. To deal truly, this is the only way that I can discover, whereby the ideas of things are brought into the understanding. If other men have either innate ideas or infused principles, they have reason

to enjoy them; and if they are sure of it, it is impossible for others to deny them the privilege that they have above their neighbours. I can speak but of what I find in myself, and is agreeable to those notions, which, if we will examine the whole course of men in their several ages, countries, and educations, seem to depend on those foundations which I have laid, and to correspond with this method in all the parts and degrees thereof.

17. Dark room. I pretend not to teach, but to inquire; and therefore cannot but confess here again,—that external and internal sensation are the only passages I can find of knowledge to the understanding. These alone, as far as I can discover, are the windows by which light is let into this dark room. For, methinks, the understanding is not much unlike a closet wholly shut from light, with only some little openings left, to let in external visible resemblances, or ideas of things without: would the pictures coming into such a dark room but stay there, and lie so orderly as to be found upon occasion, it would very much resemble the understanding of a man, in reference to all objects of sight, and the ideas of them.

These are my guesses concerning the means whereby the understanding comes to have and retain simple ideas, and the modes of them, with some other operations about them.

I proceed now to examine some of these simple ideas and their modes a little more particularly.

Chapter XII

Of Complex Ideas

1. Made by the mind out of simple ones. We have hitherto considered those ideas, in the reception whereof the mind is only passive, which are those simple ones received from sensation and reflection before mentioned, whereof the mind cannot make one to itself, nor have any idea which does not wholly consist of them. But as the mind is wholly passive in the reception of all its simple ideas, so it exerts several acts of its own, whereby out of its simple ideas, as the materials and foundations of the rest, the others are framed. The acts of the mind, wherein it exerts its power over its simple ideas, are

chiefly these three: (1) Combining several simple ideas into one compound one; and thus all complex ideas are made. (2) The second is bringing two ideas, whether simple or complex, together, and setting them by one another, so as to take a view of them at once, without uniting them into one; by which way it gets all its ideas of relations. (3) The third is separating them from all other ideas that accompany them in their real existence: this is called abstraction: and thus all its general ideas are made. This shows man's power, and its ways of operation, to be much the same in the material and intellectual world. For the materials in both being such as he has no power over, either to make or destroy, all that man can do is either to unite them together, or to set them by one another, or wholly separate them. I shall here begin with the first of these in the consideration of complex ideas, and come to the other two in their due places. As simple ideas are observed to exist in several combinations united together, so the mind has a power to consider several of them united together as one idea; and that not only as they are united in exter-

nal objects, but as itself has joined them together. Ideas thus made up of several simple ones put together, I call complex;—such as are beauty, gratitude, a man, an army, the universe; which, though complicated of various simple ideas, or complex ideas made up of simple ones, yet are, when the mind pleases, considered each by itself, as one entire thing, and signified by one name.

2. Made voluntarily. In this faculty of repeating and joining together its ideas, the mind has great power in varying and multiplying the objects of its thoughts, infinitely beyond what sensation or reflection furnished it with: but all this still confined to those simple ideas which it received from those two sources, and which are the ultimate materials of all its compositions. For simple ideas are all from things themselves, and of these the mind can have no more, nor other than what are suggested to it. It can have no other ideas of sensible qualities than what come from without by the senses; nor any ideas of other kind of operations of a thinking substance, than what it finds in itself. But when it has once got these simple ideas, it is not confined barely to

observation, and what offers itself from without; it can, by its own power, put together those ideas it has, and make new complex ones, which it never received so united.

3. Complex ideas are either of modes, substances, or relations. *Complex ideas*, however compounded and decomposed, though their number be infinite, and the variety endless, wherewith they fill and entertain the thoughts of men; yet I think they may be all reduced under these three heads:—

1. MODES.

2. SUBSTANCES.

3. RELATIONS.

4. Ideas of modes. First, Modes I call such complex ideas which, however compounded, contain not in them the supposition of subsisting by themselves, but are considered as dependences on, or affections of substances;—such as are the ideas signified by the words triangle, gratitude, murder, &c. And if in this I use the word

mode in somewhat a different sense from its ordinary signification, I beg pardon; it being unavoidable in discourses, differing from the ordinary received notions, either to make new words, or to use old words in somewhat a new signification; the later whereof, in our present case, is perhaps the more tolerable of the two.

5. Simple and mixed modes of simple ideas. Of these modes, there are two sorts which deserve distinct consideration:

First, there are some which are only variations, or different combinations of the same simple idea, without the mixture of any other;—as a dozen, or score; which are nothing but the ideas of so many distinct units added together, and these I call simple modes as being contained within the bounds of one simple idea.

Secondly, there are others compounded of simple ideas of several kinds, put together to make one complex one;—v.g. beauty, consisting of a certain composition of colour and figure, causing delight to the beholder; theft, which being the concealed change of the possession of anything, without the consent of the propri-

etor, contains, as is visible, a combination of several ideas of several kinds: and these I call mixed modes.

6. Ideas of substances, single or collective. Secondly, the ideas of Substances are such combinations of simple ideas as are taken to represent distinct particular things subsisting by themselves; the supposed or confused idea of substance, such as it is, is always the first and chief. Thus if to substance be joined the simple idea of a certain dull whitish colour, with certain degrees of weight, hardness, ductility, and fusibility, we have the idea of lead; and a combination of the ideas of a certain sort of figure, with the powers of motion, thought and reasoning, joined to substance, the ordinary idea of a man. Now of substances also, there are two sorts of ideas:—one of single substances, as they exist separately, as of a man or a sheep; the other of several of those put together, as an army of men, or flock of sheep—which collective ideas of several substances thus put together are as much each of them one single idea as that of a man or an unit.

7. Ideas of relation. Thirdly, the last sort of complex

ideas is that we call Relation, which consists in the consideration and comparing one idea with another.

Of these several kinds we shall treat in their order.

8. The abstrusest ideas we can have are all from two sources. If we trace the progress of our minds, and with attention observe how it repeats, adds together, and unites its simple ideas received from sensation or reflection, it will lead us further than at first perhaps we should have imagined. And, I believe, we shall find, if we warily observe the originals of our notions, that even the most abstruse ideas, how remote soever they may seem from sense, or from any operations of our own minds, are yet only such as the understanding frames to itself, by repeating and joining together ideas that it had either from objects of sense, or from its own operations about them: so that those even large and abstract ideas are derived from sensation or reflection, being no other than what the mind, by the ordinary use of its own faculties, employed about ideas received from objects of sense, or from the operations it observes in itself about them, may, and does, attain unto.

This I shall endeavour to show in the ideas we have of space, time, and infinity, and some few others that seem the most remote, from those originals.

Chapter XIII

Complex Ideas of Simple Modes:— and First, of the Simple Modes of the Idea of Space

1. Simple modes of simple ideas. Though in the foregoing part I have often mentioned simple ideas, which are truly the materials of all our knowledge; yet having treated of them there, rather in the way that they come into the mind, than as distinguished from others more compounded, it will not be perhaps amiss to take a view of some of them again under this consideration, and examine those different modifications of the same idea; which the mind either finds in things existing, or is able to make within itself without the help of any extrinsical object, or any foreign suggestion.

Those modifications of any one simple idea (which, as

has been said, I call simple modes) are as perfectly different and distinct ideas in the mind as those of the greatest distance or contrariety. For the idea of two is as distinct from that of one, as blueness from heat, or either of them from any number: and yet it is made up only of that simple idea of an unit repeated; and repetitions of this kind joined together make those distinct simple modes, of a dozen, a gross, a million.

2. Idea of Space. I shall begin with the simple idea of space. I have showed above, chap. V, that we get the idea of space, both by our sight and touch; which, I think, is so evident, that it would be as needless to go to prove that men perceive, by their sight, a distance between bodies of different colours, or between the parts of the same body, as that they see colours themselves: nor is it less obvious, that they can do so in the dark by feeling and touch.

3. Space and extension. This space, considered barely in length between any two beings, without considering anything else between them, is called distance: if considered in length, breadth, and thickness, I think it may

be called capacity. (The term extension is usually applied to it in what manner soever considered.)

4. Immensity. Each different distance is a different modification of space; and each idea of any different distance, or space, is a simple mode of this idea. Men, for the use and by the custom of measuring, settle in their minds the ideas of certain stated lengths,—such as are an inch, foot, yard, fathom, mile, diameter of the earth, &c., which are so many distinct ideas made up only of space. When any such stated lengths or measures of space are made familiar to men's thoughts, they can, in their minds, repeat them as often as they will, without mixing or joining to them the idea of body, or anything else; and frame to themselves the ideas of long, square, or cubic feet, yards or fathoms, here amongst the bodies of the universe, or else beyond the utmost bounds of all bodies; and, by adding these still one to another, enlarge their ideas of space as much as they please. The power of repeating or doubling any idea we have of any distance and adding it to the former as often as we will, without being ever able to come to any stop or stint, let

us enlarge it as much as we will, is that which gives us the idea of immensity.

5. Figure. There is another modification of this idea, which is nothing but the relation which the parts of the termination of extension, or circumscribed space, have amongst themselves. This the touch discovers in sensible bodies, whose extremities come within our reach; and the eye takes both from bodies and colours, whose boundaries are within its view: where, observing how the extremities terminate,—either in straight lines which meet at discernible angles, or in crooked lines wherein no angles can be perceived; by considering these as they relate to one another, in all parts of the extremities of any body or space, it has that idea we call figure, which affords to the mind infinite variety. For, besides the vast number of different figures that do really exist, in the coherent masses of matter, the stock that the mind has in its power, by varying the idea of space, and thereby making still new compositions, by repeating its own ideas, and joining them as it pleases, is perfectly inexhaustible. And so it can multiply figures in infinitum.

6. Endless variety of figures. For the mind having a power to repeat the idea of any length directly stretched out, and join it to another in the same direction, which is to double the length of that straight line; or else join another with what inclination it thinks fit, and so make what sort of angle it pleases: and being able also to shorten any line it imagines, by taking from it one half, one fourth, or what part it pleases, without being able to come to an end of any such divisions, it can make an angle of any bigness. So also the lines that are its sides, of what length it pleases, which joining again to other lines, of different lengths, and at different angles, till it has wholly enclosed any space, it is evident that it can multiply figures, both in their shape and capacity, in infinitum; all which are but so many different simple modes of space. The same that it can do with straight lines, it can also do with crooked, or crooked and straight together; and the same it can do in lines, it can also in superficies; by which we may be led into farther thoughts of the endless variety of figures that the mind has a power to make, and thereby to multiply the simple modes of space.

7. Place. Another idea coming under this head, and belonging to this tribe, is that we call place. As in simple space, we consider the relation of distance between any two bodies or points; so in our idea of place, we consider the relation of distance betwixt anything, and any two or more points, which are considered as keeping the same distance one with another, and so considered as at rest. For when we find anything at the same distance now which it was yesterday, from any two or more points, which have not since changed their distance one with another, and with which we then compared it, we say it hath kept the same place: but if it hath sensibly altered its distance with either of those points, we say it hath changed its place: though, vulgarly speaking, in the common notion of place, we do not always exactly observe the distance from these precise points, but from larger portions of sensible objects, to which we consider the thing placed to bear relation, and its distance from which we have some reason to observe.

8. Place relative to particular bodies. Thus, a company of chess-men, standing on the same squares of the chess-

board where we left them, we say they are all in the same place, or unmoved, though perhaps the chess-board hath been in the mean time carried out of one room into another; because we compared them only to the parts of the chess-board, which keep the same distance one with another. The chess-board, we also say, is in the same place it was, if it remain in the same part of the cabin, though perhaps the ship which it is in sails all the while. And the ship is said to be in the same place, supposing it kept the same distance with the parts of the neighbouring land; though perhaps the earth hath turned round, and so both chess-men, and board, and ship, have every one changed place, in respect of remoter bodies, which have kept the same distance one with another. But yet the distance from certain parts of the board being that which determines the place of the chessmen; and the distance from the fixed parts of the cabin (with which we made the comparison) being that which determined the place of the chess-board; and the fixed parts of the earth that by which we determined the place of the ship,—these things may be said to be in the same place in those re-

spects: though their distance from some other things, which in this matter we did not consider, being varied, they have undoubtedly changed place in that respect; and we ourselves shall think so, when we have occasion to compare them with those other.

9. Place relative to a present purpose. But this modification of distance we call place, being made by men for their common use, that by it they might be able to design the particular position of things, where they had occasion for such designation; men consider and determine of this place by reference to those adjacent things which best served to their present purpose, without considering other things which, to another purpose, would better determine the place of the same thing. Thus in the chess-board, the use of the designation of the place of each chess-man being determined only within that chequered piece of wood, it would cross that purpose to measure it by anything else; but when these very chess-men are put up in a bag, if any one should ask where the black king is, it would be proper to determine the place by the part of the room it was in, and

not by the chess-board; there being another use of designing the place it is now in, than when in play it was on the chess-board, and so must be determined by other bodies. So if any one should ask, in what place are the verses which report the story of Nisus and Euryalus, it would be very improper to determine this place, by saying, they were in such a part of the earth, or in Bodley's library: but the right designation of the place would be by the parts of Virgil's works; and the proper answer would be, that these verses were about the middle of the ninth book of his *Aeneids*, and that they have been always constantly in the same place ever since Virgil was printed: which is true, though the book itself hath moved a thousand times, the use of the idea of place here being, to know in what part of the book that story is, that so, upon occasion, we may know where to find it, and have recourse to it for use.

10. Place of the universe. That our idea of place is nothing else but such a relative position of anything as I have before mentioned, I think is plain, and will be easily admitted, when we consider that we can have no

idea of the place of the universe, though we can of all the parts of it; because beyond that we have not the idea of any fixed, distinct, particular beings, in reference to which we can imagine it to have any relation of distance; but all beyond it is one uniform space or expansion, wherein the mind finds no variety, no marks. For to say that the world is somewhere, means no more than that it does exist; this, though a phrase borrowed from place, signifying only its existence, not location: and when one can find out, and frame in his mind, clearly and distinctly, the place of the universe, he will be able to tell us whether it moves or stands still in the undistinguishable inane of infinite space: though it be true that the word place has sometimes a more confused sense, and stands for that space which anybody takes up; and so the universe is in a place.

The idea, therefore, of place we have by the same means that we get the idea of space, (whereof this is but a particular limited consideration,) viz, by our sight and touch; by either of which we receive into our minds the ideas of extension or distance.

11. Extension and body not the same. There are some that would persuade us, that body and extension are the same thing, who either change the signification of words, which I would not suspect them of,—they having so severely condemned the philosophy of others, because it hath been too much placed in the uncertain meaning, or deceitful obscurity of doubtful or insignificant terms. If, therefore, they mean by body and extension the same that other people do, viz. by body something that is solid and extended, whose parts are separable and movable different ways; and by extension, only the space that lies between the extremities of those solid coherent parts, and which is possessed by them,—they confound very different ideas one with another; for I appeal to every man's own thoughts whether the idea of space be not as distinct from that of solidity, as it is from the idea of scarlet colour? It is true, solidity cannot exist without extension, neither can scarlet colour exist without extension, but this hinders not, but that they are distinct ideas. Many ideas require others, as necessary to their existence or conception, which yet

are very distinct ideas. Motion can neither be, nor be conceived, without space; and yet motion is not space, nor space motion; space can exist without it, and they are very distinct ideas; and so, I think, are those of space and solidity. Solidity is so inseparable an idea from body, that upon that depends its filling of space, its contact, impulse, and communication of motion upon impulse. And if it be a reason to prove that spirit is different from body, because thinking includes not the idea of extension in it; the same reason will be as valid, I suppose, to prove that space is not body, because it includes not the idea of solidity in it; space and solidity being as distinct ideas as thinking and extension, and as wholly separable in the mind one from another. Body then and extension, it is evident, are two distinct ideas. For,

12. Extension not solidity. First, Extension includes no solidity, nor resistance to the motion of body, as body does.

13. The parts of space inseparable, both really and mentally. Secondly, The parts of pure space are inseparable

one from the other; so that the continuity cannot be separated, neither really nor mentally. For I demand of any one to remove any part of it from another, with which it is continued, even so much as in thought. To divide and separate actually is, as I think, by removing the parts one from another, to make two superficies, where before there was a continuity: and to divide mentally is, to make in the mind two superficies, where before there was a continuity, and consider them as removed one from the other; which can only be done in things considered by the mind as capable of being separated; and by separation, of acquiring new distinct superficies, which they then have not, but are capable of. But neither of these ways of separation, whether real or mental, is, as I think, compatible to pure space.

It is true, a man may consider so much of such a space as is answerable or commensurate to a foot, without considering the rest, which is, indeed, a partial consideration, but not so much as mental separation or division; since a man can no more mentally divide, without considering two superficies separate one from the other,

than he can actually divide, without making two superficies disjoined one from the other: but a partial consideration is not separating. A man may consider light in the sun without its heat, or mobility in body without its extension, without thinking of their separation. One is only a partial consideration, terminating in one alone; and the other is a consideration of both, as existing separately.

14. The parts of space, immovable. Thirdly, The parts of pure space are immovable, which follows from their inseparability; motion being nothing but change of distance between any two things; but this cannot be between parts that are inseparable, which, therefore, must needs be at perpetual rest one amongst another.

Thus the determined idea of simple space distinguishes it plainly and sufficiently from body; since its parts are inseparable, immovable, and without resistance to the motion of body.

15. The definition of extension explains it not. If any one ask me what this space I speak of is, I will tell him when he tells me what his extension is. For to say, as is

usually done, that extension is to have partes extra partes, is to say only, that extension is extension. For what am I the better informed in the nature of extension, when I am told that extension is to have parts that are extended, exterior to parts that are extended, i.e. extension consists of extended parts? As if one, asking what a fibre was, I should answer him,—that it was a thing made up of several fibres. Would he thereby be enabled to understand what a fibre was better than he did before? Or rather, would he not have reason to think that my design was to make sport with him, rather than seriously to instruct him?

16. Division of beings into bodies and spirits proves not space and body the same. Those who contend that space and body are the same, bring this dilemma:—either this space is something or nothing; if nothing be between two bodies, they must necessarily touch; if it be allowed to be something, they ask, Whether it be body or spirit? To which I answer by another question, Who told them that there was, or could be, nothing but solid beings, which could not think, and thinking beings that were

not extended?—which is all they mean by the terms body and spirit.

17. Substance which we know not, no proof against space without body. If it be demanded (as usually it is) whether this space, void of body, be substance or accident, I shall readily answer I know not; nor shall be ashamed to own my ignorance, till they that ask show me a clear distinct idea of substance.

18. Different meanings of substance. I endeavour as much as I can to deliver myself from those fallacies which we are apt to put upon ourselves, by taking words for things. It helps not our ignorance to feign a knowledge where we have none, by making a noise with sounds, without clear and distinct significations. Names made at pleasure, neither alter the nature of things, nor make us understand them, but as they are signs of and stand for determined ideas. And I desire those who lay so much stress on the sound of these two syllables, substance, to consider whether applying it, as they do, to the infinite, incomprehensible God, to finite spirits, and to body, it be in the same sense; and whether it stands for the

same idea, when each of those three so different beings are called substances. If so, whether it will thence follow—that God, spirits, and body, agreeing in the same common nature of substance, differ not any otherwise than in a bare different modification of that substance; as a tree and a pebble, being in the same sense body, and agreeing in the common nature of body, differ only in a bare modification of that common matter, which will be a very harsh doctrine. If they say, that they apply it to God, finite spirit, and matter, in three different significations and that it stands for one idea when God is said to be a substance; for another when the soul is called substance; and for a third when body is called so;—if the name substance stands for three several distinct ideas, they would do well to make known those distinct ideas, or at least to give three distinct names to them, to prevent in so important a notion the confusion and errors that will naturally follow from the promiscuous use of so doubtful a term; which is so far from being suspected to have three distinct, that in ordinary use it has scarce one clear distinct signification. And if they can thus

make three distinct ideas of substance, what hinders why another may not make a fourth?

19. Substance and accidents of little use in philosophy. They who first ran into the notion of accidents, as a sort of real beings that needed something to inhere in, were forced to find out the word substance to support them. Had the poor Indian philosopher (who imagined that the earth also wanted something to bear it up) but thought of this word substance, he needed not to have been at the trouble to find an elephant to support it, and a tortoise to support his elephant: the word substance would have done it effectually. And he that inquired might have taken it for as good an answer from an Indian philosopher,—that substance, without knowing what it is, is that which supports the earth, as we take it for a sufficient answer and good doctrine from our European philosophers,—that substance, without knowing what it is, is that which supports accidents. So that of substance, we have no idea of what it is, but only a confused, obscure one of what it does.

20. Sticking on and under-propping. Whatever a learned

man may do here, an intelligent American, who inquired into the nature of things, would scarce take it for a satisfactory account, if, desiring to learn our architecture, he should be told that a pillar is a thing supported by a basis, and a basis something that supported a pillar. Would he not think himself mocked, instead of taught, with such an account as this? And a stranger to them would be very liberally instructed in the nature of books, and the things they contained, if he should be told that all learned books consisted of paper and letters, and that letters were things inhering in paper, and paper a thing that held forth letters: a notable way of having clear ideas of letters and paper. But were the Latin words, *inhaerentia* and *substantio*, put into the plain English ones that answer them, and were called sticking on and under-propping, they would better discover to us the very great clearness there is in the doctrine of substance and accidents, and show of what use they are in deciding of questions in philosophy.

21. A vacuum beyond the utmost bounds of body. But to return to our idea of space. If body be not supposed

infinite, (which I think no one will affirm), I would ask, whether, if God placed a man at the extremity of corporeal beings, he could not stretch his hand beyond his body? If he could, then he would put his arm where there was before space without body; and if there he spread his fingers, there would still be space between them without body. If he could not stretch out his hand, it must be because of some external hindrance; (for we suppose him alive, with such a power of moving the parts of his body that he hath now, which is not in itself impossible, if God so pleased to have it; or at least it is not impossible for God so to move him): and then I ask,—whether that which hinders his hand from moving outwards be substance or accident, something or nothing? And when they have resolved that, they will be able to resolve themselves,—what that is, which is or may be between two bodies at a distance, that is not body, and has no solidity. In the mean time, the argument is at least as good, that, where nothing hinders, (as beyond the utmost bounds of all bodies), a body put in motion may move on, as where there is nothing be-

tween, there two bodies must necessarily touch. For pure space between is sufficient to take away the necessity of mutual contact; but bare space in the way is not sufficient to stop motion. The truth is, these men must either own that they think body infinite, though they are loth to speak it out, or else affirm that space is not body. For I would fain meet with that thinking man that can in his thoughts set any bounds to space, more than he can to duration; or by thinking hope to arrive at the end of either. And therefore, if his idea of eternity be infinite, so is his idea of immensity; they are both finite or infinite alike.

22. The power of annihilation proves a vacuum. Farther, those who assert the impossibility of space existing without matter, must not only make body infinite, but must also deny a power in God to annihilate any part of matter. No one, I suppose, will deny that God can put an end to all motion that is in matter, and fix all the bodies of the universe in a perfect quiet and rest, and continue them so long as he pleases. Whoever then will allow that God can, during such a general rest, an-

nihilate either this book or the body of him that reads it, must necessarily admit the possibility of a vacuum. For, it is evident that the space that was filled by the parts of the annihilated body will still remain, and be a space without body. For the circumambient bodies being in perfect rest, are a wall of adamant, and in that state make it a perfect impossibility for any other body to get into that space. And indeed the necessary motion of one particle of matter into the place from whence another particle of matter is removed, is but a consequence from the supposition of plenitude; which will therefore need some better proof than a supposed matter of fact, which experiment can never make out;—our own clear and distinct ideas plainly satisfying us, that there is no necessary connexion between space and solidity, since we can conceive the one without the other. And those who dispute for or against a vacuum, do thereby confess they have distinct ideas of vacuum and plenum, i.e. that they have an idea of extension void of solidity, though they deny its existence; or else they dispute about nothing at all. For they who so much

alter the signification of words, as to call extension body, and consequently make the whole essence of body to be nothing but pure extension without solidity, must talk absurdly whenever they speak of vacuum; since it is impossible for extension to be without extension. For vacuum, whether we affirm or deny its existence, signifies space without body; whose very existence no one can deny to be possible, who will not make matter infinite, and take from God a power to annihilate any particle of it.

23. Motion proves a vacuum. But not to go so far as beyond the utmost bounds of body in the universe, nor appeal to God's omnipotency to find a vacuum, the motion of bodies that are in our view and neighbourhood seems to me plainly to evince it. For I desire any one so to divide a solid body, of any dimension he pleases, as to make it possible for the solid parts to move up and down freely every way within the bounds of that superficies, if there be not left in it a void space as big as the least part into which he has divided the said solid body. And if, where the least particle of the body divided is as

big as a mustard-seed, a void space equal to the bulk of a mustard-seed be requisite to make room for the free motion of the parts of the divided body within the bounds of its superficies, where the particles of matter are 100,000,000 less than a mustard-seed, there must also be a space void of solid matter as big as 100,000,000 part of a mustard-seed; for if it hold in the one it will hold in the other, and so on in infinitum. And let this void space be as little as it will, it destroys the hypothesis of plenitude. For if there can be a space void of body equal to the smallest separate particle of matter now existing in nature, it is still space without body; and makes as great a difference between space and body as if it were mega chasma, a distance as wide as any in nature. And therefore, if we suppose not the void space necessary to motion equal to the least parcel of the divided solid matter, but to $1/10$ or $1/1000$ of it, the same consequence will always follow of space without matter.

24. The ideas of space and body distinct. But the question being here,—Whether the idea of space or exten-

sion be the same with the idea of body? it is not necessary to prove the real existence of a vacuum, but the idea of it; which it is plain men have when they inquire and dispute whether there be a vacuum or no. For if they had not the idea of space without body, they could not make a question about its existence: and if their idea of body did not include in it something more than the bare idea of space, they could have no doubt about the plenitude of the world; and it would be as absurd to demand, whether there were space without body, as whether there were space without space, or body without body, since these were but different names of the same idea.

25. Extension being inseparable from body, proves it not the same. It is true, the idea of extension joins itself so inseparably with all visible, and most tangible qualities, that it suffers us to see no one, or feel very few external objects, without taking in impressions of extension too. This readiness of extension to make itself be taken notice of so constantly with other ideas, has been the occasion, I guess, that some have made the

whole essence of body to consist in extension; which is not much to be wondered at, since some have had their minds, by their eyes and touch, (the busiest of all our senses,) so filled with the idea of extension, and, as it were, wholly possessed with it, that they allowed no existence to anything that had not extension. I shall not now argue with those men, who take the measure and possibility of all being only from their narrow and gross imaginations: but having here to do only with those who conclude the essence of body to be extension, because they say they cannot imagine any sensible quality of any body without extension,—I shall desire them to consider, that, had they reflected on their ideas of tastes and smells as much as on those of sight and touch; nay, had they examined their ideas of hunger and thirst, and several other pains, they would have found that they included in them no idea of extension at all, which is but an affection of body, as well as the rest, discoverable by our senses, which are scarce acute enough to look into the pure essences of things.

26. Essences of things. If those ideas which are con-

stantly joined to all others, must therefore be concluded to be the essence of those things which have constantly those ideas joined to them, and are inseparable from them; then unity is without doubt the essence of everything. For there is not any object of sensation or reflection which does not carry with it the idea of one: but the weakness of this kind of argument we have already shown sufficiently.

27. Ideas of space and solidity distinct. To conclude: whatever men shall think concerning the existence of a vacuum, this is plain to me—that we have as clear an idea of space distinct from solidity, as we have of solidity distinct from motion, or motion from space. We have not any two more distinct ideas; and we can as easily conceive space without solidity, as we can conceive body or space without motion, though it be never so certain that neither body nor motion can exist without space. But whether any one will take space to be only a relation resulting from the existence of other beings at a distance; or whether they will think the words of the most knowing King Solomon, “The heaven, and the

heaven of heavens, cannot contain thee”; or those more emphatical ones of the inspired philosopher St. Paul, “In him we live, move, and have our being,” are to be understood in a literal sense, I leave every one to consider: only our idea of space is, I think, such as I have mentioned, and distinct from that of body. For, whether we consider, in matter itself, the distance of its coherent solid parts, and call it, in respect of those solid parts, extension; or whether, considering it as lying between the extremities of any body in its several dimensions, we call it length, breadth, and thickness; or else, considering it as lying between any two bodies or positive beings, without any consideration whether there be any matter or not between, we call it distance;—however named or considered, it is always the same uniform simple idea of space, taken from objects about which our senses have been conversant; whereof, having settled ideas in our minds, we can revive, repeat, and add them one to another as often as we will, and consider the space or distance so imagined, either as filled with solid parts, so that another body cannot come there without displac-

ing and thrusting out the body that was there before; or else as void of solidity, so that a body of equal dimensions to that empty or pure space may be placed in it, without the removing or expulsion of anything that was there. But, to avoid confusion in discourses concerning this matter, it were possibly to be wished that the name extension were applied only to matter, or the distance of the extremities of particular bodies; and the term expansion to space in general, with or without solid matter possessing it,—so as to say space is expanded and body extended. But in this every one has his liberty: I propose it only for the more clear and distinct way of speaking.

28. Men differ little in clear, simple ideas. The knowing precisely what our words stand for, would, I imagine, in this as well as a great many other cases, quickly end the dispute. For I am apt to think that men, when they come to examine them, find their simple ideas all generally to agree, though in discourse with one another they perhaps confound one another with different names. I imagine that men who abstract their thoughts, and do

well examine the ideas of their own minds, cannot much differ in thinking; however they may perplex themselves with words, according to the way of speaking to the several schools or sects they have been bred up in: though amongst unthinking men, who examine not scrupulously and carefully their own ideas, and strip them not from the marks men use for them, but confound them with words, there must be endless dispute, wrangling, and jargon; especially if they be learned, bookish men, devoted to some sect, and accustomed to the language of it, and have learned to talk after others. But if it should happen that any two thinking men should really have different ideas, I do not see how they could discourse or argue with another. Here I must not be mistaken, to think that every floating imagination in men's brains is presently of that sort of ideas I speak of. It is not easy for the mind to put off those confused notions and prejudices it has imbibed from custom, inadvertency, and common conversation. It requires pains and assiduity to examine its ideas, till it resolves them into those clear and distinct simple ones, out of which

they are compounded; and to see which, amongst its simple ones, have or have not a necessary connexion and dependence one upon another. Till a man doth this in the primary and original notions of things, he builds upon floating and uncertain principles, and will often find himself at a loss.

Chapter XIV

Idea of Duration and its Simple Modes

1. Duration is fleeting extension. There is another sort of distance, or length, the idea whereof we get not from the permanent parts of space, but from the fleeting and perpetually perishing parts of succession. This we call duration; the simple modes whereof are any different lengths of it whereof we have distinct ideas, as hours, days, years, &c., time and eternity.

2. Its idea from reflection on the train of our ideas. The answer of a great man, to one who asked what time was: *Si non rogas intelligo*, (which amounts to this; The more I set myself to think of it, the less I understand

it,) might perhaps persuade one that time, which reveals all other things, is itself not to be discovered. Duration, time, and eternity, are, not without reason, thought to have something very abstruse in their nature. But however remote these may seem from our comprehension, yet if we trace them right to their originals, I doubt not but one of those sources of all our knowledge, viz. sensation and reflection, will be able to furnish us with these ideas, as clear and distinct as many others which are thought much less obscure; and we shall find that the idea of eternity itself is derived from the same common original with the rest of our ideas.

3. Nature and origin of the idea of duration. To understand time and eternity aright, we ought with attention to consider what idea it is we have of duration, and how we came by it. It is evident to any one who will but observe what passes in his own mind, that there is a train of ideas which constantly succeed one another in his understanding, as long as he is awake. Reflection on these appearances of several ideas one after another in our minds, is that which furnishes us with the idea of

succession: and the distance between any parts of that succession, or between the appearance of any two ideas in our minds, is that we call duration. For whilst we are thinking, or whilst we receive successively several ideas in our minds, we know that we do exist; and so we call the existence, or the continuation of the existence of ourselves, or anything else, commensurate to the succession of any ideas in our minds, the duration of ourselves, or any such other thing co-existent with our thinking.

4. Proof that its idea is got from reflection on the train of our ideas. That we have our notion of succession and duration from this original, viz. from reflection on the train of ideas, which we find to appear one after another in our own minds, seems plain to me, in that we have no perception of duration but by considering the train of ideas that take their turns in our understandings. When that succession of ideas ceases, our perception of duration ceases with it; which every one clearly experiments in himself, whilst he sleeps soundly, whether an hour or a day, a month or a year; of which duration

of things, while he sleeps or thinks not, he has no perception at all, but it is quite lost to him; and the moment wherein he leaves off to think, till the moment he begins to think again, seems to him to have no distance. And so I doubt not it would be to a waking man, if it were possible for him to keep only one idea in his mind, without variation and the succession of others. And we see, that one who fixes his thoughts very intently on one thing, so as to take but little notice of the succession of ideas that pass in his mind, whilst he is taken up with that earnest contemplation, lets slip out of his account a good part of that duration, and thinks that time shorter than it is. But if sleep commonly unites the distant parts of duration, it is because during that time we have no succession of ideas in our minds. For if a man, during his sleep, dreams, and variety of ideas make themselves perceptible in his mind one after another, he hath then, during such dreaming, a sense of duration, and of the length of it. By which it is to me very clear, that men derive their ideas of duration from their reflections on the train of the ideas they observe

to succeed one another in their own understandings; without which observation they can have no notion of duration, whatever may happen in the world.

5. The idea of duration applicable to things whilst we sleep. Indeed a man having, from reflecting on the succession and number of his own thoughts, got the notion or idea of duration, he can apply that notion to things which exist while he does not think; as he that has got the idea of extension from bodies by his sight or touch, can apply it to distances, where no body is seen or felt. And therefore, though a man has no perception of the length of duration which passed whilst he slept or thought not; yet, having observed the revolution of days and nights, and found the length of their duration to be in appearance regular and constant, he can, upon the supposition that that revolution has proceeded after the same manner whilst he was asleep or thought not, as it used to do at other times, he can, I say, imagine and make allowance for the length of duration whilst he slept. But if Adam and Eve, (when they were alone in the world), instead of their ordinary night's sleep, had

passed the whole twenty-four hours in one continued sleep, the duration of that twenty-four hours had been irrecoverably lost to them, and been for ever left out of their account of time.

6. The idea of succession not from motion. Thus by reflecting on the appearing of various ideas one after another in our understandings, we get the notion of succession; which, if any one should think we did rather get from our observation of motion by our senses, he will perhaps be of my mind when he considers, that even motion produces in his mind an idea of succession no otherwise than as it produces there a continued train of distinguishable ideas. For a man looking upon a body really moving, perceives yet no motion at all unless that motion produces a constant train of successive ideas: v.g. a man becalmed at sea, out of sight of land, in a fair day, may look on the sun, or sea, or ship, a whole hour together, and perceive no motion at all in either; though it be certain that two, and perhaps all of them, have moved during that time a great way. But as soon as he perceives either of them to have changed distance with

some other body, as soon as this motion produces any new idea in him, then he perceives that there has been motion. But wherever a man is, with all things at rest about him, without perceiving any motion at all,—if during this hour of quiet he has been thinking, he will perceive the various ideas of his own thoughts in his own mind, appearing one after another, and thereby observe and find succession where he could observe no motion.

7. Very slow motions unperceived. And this, I think, is the reason why motions very slow, though they are constant, are not perceived by us; because in their remove from one sensible part towards another, their change of distance is so slow, that it causes no new ideas in us, but a good while one after another. And so not causing a constant train of new ideas to follow one another immediately in our minds, we have no perception of motion; which consisting in a constant succession, we cannot perceive that succession without a constant succession of varying ideas arising from it.

8. Very swift motions unperceived. On the contrary, things that move so swift as not to affect the senses

distinctly with several distinguishable distances of their motion, and so cause not any train of ideas in the mind, are not also perceived. For anything that moves round about in a circle, in less times than our ideas are wont to succeed one another in our minds, is not perceived to move; but seems to be a perfect entire circle of that matter or colour, and not a part of a circle in motion.

9. The train of ideas has a certain degree of quickness. Hence I leave it to others to judge, whether it be not probable that our ideas do, whilst we are awake, succeed one another in our minds at certain distances; not much unlike the images in the inside of a lantern, turned round by the heat of a candle. This appearance of theirs in train, though perhaps it may be sometimes faster and sometimes slower, yet, I guess, varies not very much in a waking man: there seem to be certain bounds to the quickness and slowness of the succession of those ideas one to another in our minds, beyond which they can neither delay nor hasten.

10. Real succession in swift motions without sense of succession. The reason I have for this odd conjecture is,

from observing that, in the impressions made upon any of our senses, we can but to a certain degree perceive any succession; which, if exceeding quick, the sense of succession is lost, even in cases where it is evident that there is a real succession. Let a cannon-bullet pass through a room, and in its way take with it any limb, or fleshy parts of a man, it is as clear as any demonstration can be, that it must strike successively the two sides of the room: it is also evident that it must touch one part of the flesh first, and another after, and so in succession: and yet, I believe, nobody who ever felt the pain of such a shot, or heard the blow against the two distant walls, could perceive any succession either in the pain or sound of so swift a stroke. Such a part of duration as this, wherein we perceive no succession, is that which we call an instant, and is that which takes up the time of only one idea in our minds, without the succession of another; wherein, therefore, we perceive no succession at all.

11. In slow motions. This also happens where the motion is so slow as not to supply a constant train of fresh

ideas to the senses, as fast as the mind is capable of receiving new ones into it; and so other ideas of our own thoughts, having room to come into our minds between those offered to our senses by the moving body, there the sense of motion is lost; and the body, though it really moves, yet, not changing perceivable distance with some other bodies as fast as the ideas of our own minds do naturally follow one another in train, the thing seems to stand still; as is evident in the hands of clocks, and shadows of sun-dials, and other constant but slow motions, where, though, after certain intervals, we perceive, by the change of distance, that it hath moved, yet the motion itself we perceive not.

12. This train, the measure of other successions. So that to me it seems, that the constant and regular succession of ideas in a waking man, is, as it were, the measure and standard of all other successions. Whereof, if any one either exceeds the pace of our ideas, as where two sounds or pains, &c., take up in their succession the duration of but one idea; or else where any motion or succession is so slow, as that it keeps not pace with

the ideas in our minds, or the quickness in which they take their turns, as when any one or more ideas in their ordinary course come into our mind, between those which are offered to the sight by the different perceptible distances of a body in motion, or between sounds or smells following one another,—there also the sense of a constant continued succession is lost, and we perceive it not, but with certain gaps of rest between.

13. The mind cannot fix long on one invariable idea. If it be so, that the ideas of our minds, whilst we have any there, do constantly change and shift in a continual succession, it would be impossible, may any one say, for a man to think long of any one thing. By which, if it be meant that a man may have one self-same single idea a long time alone in his mind, without any variation at all, I think, in matter of fact, it is not possible. For which (not knowing how the ideas of our minds are framed, of what materials they are made, whence they have their light, and how they come to make their appearances) I can give no other reason but experience: and I would have any one try, whether he can keep one

unvaried single idea in his mind, without any other, for any considerable time together.

14. Proof. For trial, let him take any figure, any degree of light or whiteness, or what other he pleases, and he will, I suppose, find it difficult to keep all other ideas out of his mind; but that some, either of another kind, or various considerations of that idea, (each of which considerations is a new idea), will constantly succeed one another in his thoughts, let him be as wary as he can.

15. The extent of our power over the succession of our ideas. All that is in a man's power in this case, I think, is only to mind and observe what the ideas are that take their turns in his understanding; or else to direct the sort, and call in such as he hath a desire or use of: but hinder the constant succession of fresh ones, I think he cannot, though he may commonly choose whether he will heedfully observe and consider them.

16. Ideas, however made, include no sense of motion. Whether these several ideas in a man's mind be made by certain motions, I will not here dispute; but this I am

sure, that they include no idea of motion in their appearance; and if a man had not the idea of motion otherwise, I think he would have none at all, which is enough to my present purpose; and sufficiently shows that the notice we take of the ideas of our own minds, appearing there one after another, is that which gives us the idea of succession and duration, without which we should have no such ideas at all. It is not then motion, but the constant train of ideas in our minds whilst we are waking, that furnishes us with the idea of duration; whereof motion no otherwise gives us any perception than as it causes in our minds a constant succession of ideas, as I have before showed: and we have as clear an idea of succession and duration, by the train of other ideas succeeding one another in our minds, without the idea of any motion, as by the train of ideas caused by the uninterrupted sensible change of distance between two bodies, which we have from motion; and therefore we should as well have the idea of duration were there no sense of motion at all.

17. Time is duration set out by measures. Having thus

got the idea of duration, the next thing natural for the mind to do, is to get some measure of this common duration, whereby it might judge of its different lengths, and consider the distinct order wherein several things exist; without which a great part of our knowledge would be confused, and a great part of history be rendered very useless. This consideration of duration, as set out by certain periods, and marked by certain measures or epochs, is that, I think, which most properly we call time.

18. A good measure of time must divide its whole duration into equal periods. In the measuring of extension, there is nothing more required but the application of the standard or measure we make use of to the thing of whose extension we would be informed. But in the measuring of duration this cannot be done, because no two different parts of succession can be put together to measure one another. And nothing being a measure of duration but duration, as nothing is of extension but extension, we cannot keep by us any standing, unvarying measure of duration, which consists in a constant

fleeting succession, as we can of certain lengths of extension, as inches, feet, yards, &c., marked out in permanent parcels of matter. Nothing then could serve well for a convenient measure of time, but what has divided the whole length of its duration into apparently equal portions, by constantly repeated periods. What portions of duration are not distinguished, or considered as distinguished and measured, by such periods, come not so properly under the notion of time; as appears by such phrases as these, viz. "Before all time," and "When time shall be no more."

19. The revolutions of the sun and moon, the properest measures of time for mankind. The diurnal and annual revolutions of the sun, as having been, from the beginning of nature, constant, regular, and universally observable by all mankind, and supposed equal to one another, have been with reason made use of for the measure of duration. But the distinction of days and years having depended on the motion of the sun, it has brought this mistake with it, that it has been thought that motion and duration were the measure one of another. For

men, in the measuring of the length of time, having been accustomed to the ideas of minutes, hours, days, months, years, &c., which they found themselves upon any mention of time or duration presently to think on, all which portions of time were measured out by the motion of those heavenly bodies, they were apt to confound time and motion; or at least to think that they had a necessary connexion one with another. Whereas any constant periodical appearance, or alteration of ideas, in seemingly equidistant spaces of duration, if constant and universally observable, would have as well distinguished the intervals of time, as those that have been made use of. For, supposing the sun, which some have taken to be a fire, had been lighted up at the same distance of time that it now every day comes about to the same meridian, and then gone out again about twelve hours after, and that in the space of an annual revolution it had sensibly increased in brightness and heat, and so decreased again,—would not such regular appearances serve to measure out the distances of duration to all that could observe it, as well without as with

motion? For if the appearances were constant, universally observable, in equidistant periods, they would serve mankind for measure of time as well were the motion away.

20. But not by their motion, but periodical appearances. For the freezing of water, or the blowing of a plant, returning at equidistant periods in all parts of the earth, would as well serve men to reckon their years by as the motions of the sun: and in effect we see, that some people in America counted their years by the coming of certain birds amongst them at their certain seasons, and leaving them at others. For a fit of an ague; the sense of hunger or thirst; a smell or a taste; or any other idea returning constantly at equidistant periods, and making itself universally be taken notice of, would not fail to measure out the course of succession, and distinguish the distances of time. Thus we see that men born blind count time well enough by years, whose revolutions yet they cannot distinguish by motions that they perceive not. And I ask whether a blind man, who distinguished his years either by the heat of summer, or

cold of winter; by the smell of any flower of the spring, or taste of any fruit of the autumn, would not have a better measure of time than the Romans had before the reformation of their calendar by Julius Caesar, or many other people whose years, notwithstanding the motion of the sun, which they pretended to make use of, are very irregular? And it adds no small difficulty to chronology, that the exact lengths of the years that several nations counted by, are hard to be known, they differing very much one from another, and I think I may say all of them from the precise motion of the sun. And if the sun moved from the creation to the flood constantly in the equator, and so equally dispersed its light and heat to all the habitable parts of the earth, in days all of the same length, without its annual variations to the tropics, as a late ingenious author supposes, I do not think it very easy to imagine, that (notwithstanding the motion of the sun) men should in the antediluvian world, from the beginning, count by years, or measure their time by periods that had no sensible marks very obvious to distinguish them by.

21. No two parts of duration can be certainly known to be equal. But perhaps it will be said,—without a regular motion, such as of the sun, or some other, how could it ever be known that such periods were equal? To which I answer,—the equality of any other returning appearances might be known by the same way that that of days was known, or presumed to be so at first; which was only by judging of them by the train of ideas which had passed in men's minds in the intervals; by which train of ideas discovering inequality in the natural days, but none in the artificial days, the artificial days, or nuchtheerha, were guessed to be equal, which was sufficient to make them serve for a measure; though exacter search has since discovered inequality in the diurnal revolutions of the sun, and we know not whether the annual also be not unequal. These yet, by their presumed and apparent equality, serve as well to reckon time by (though not to measure the parts of duration exactly) as if they could be proved to be exactly equal. We must, therefore, carefully distinguish betwixt duration itself, and the measures we make use of to judge of

its length. Duration, in itself, is to be considered as going on in one constant, equal, uniform course: but none of the measures of it which we make use of can be known to do so, nor can we be assured that their assigned parts or periods are equal in duration one to another; for two successive lengths of duration, however measured, can never be demonstrated to be equal. The motion of the sun, which the world used so long and so confidently for an exact measure of duration, has, as I said, been found in its several parts unequal. And though men have, of late, made use of a pendulum, as a more steady and regular motion than that of the sun, or, (to speak more truly), of the earth;—yet if any one should be asked how he certainly knows that the two successive swings of a pendulum are equal, it would be very hard to satisfy him that they are infallibly so; since we cannot be sure that the cause of that motion, which is unknown to us, shall always operate equally; and we are sure that the medium in which the pendulum moves is not constantly the same: either of which varying, may alter the equality of such periods, and

thereby destroy the certainty and exactness of the measure by motion, as well as any other periods of other appearances; the notion of duration still remaining clear, though our measures of it cannot (any of them) be demonstrated to be exact. Since then no two portions of succession can be brought together, it is impossible ever certainly to know their equality. All that we can do for a measure of time is, to take such as have continual successive appearances at seemingly equidistant periods; of which seeming equality we have no other measure, but such as the train of our own ideas have lodged in our memories, with the concurrence of other probable reasons, to persuade us of their equality.

22. Time not the measure of motion. One thing seems strange to me,—that whilst all men manifestly measured time by the motion of the great and visible bodies of the world, time yet should be defined to be the “measure of motion”: whereas it is obvious to every one who reflects ever so little on it, that to measure motion, space is as necessary to be considered as time; and those who look a little farther will find also the bulk of the

thing moved necessary to be taken into the computation, by any one who will estimate or measure motion so as to judge right of it. Nor indeed does motion any otherwise conduce to the measuring of duration, than as it constantly brings about the return of certain sensible ideas, in seeming equidistant periods. For if the motion of the sun were as unequal as of a ship driven by unsteady winds, sometimes very slow, and at others irregularly very swift; or if, being constantly equally swift, it yet was not circular, and produced not the same appearances,—it would not at all help us to measure time, any more than the seeming unequal motion of a comet does.

23. Minutes, hours, days, and years not necessary measures of duration. Minutes, hours, days, and years are, then, no more necessary to time or duration, than inches, feet, yards, and miles, marked out in any matter, are to extension. For, though we in this part of the universe, by the constant use of them, as of periods set out by the revolutions of the sun, or as known parts of such periods, have fixed the ideas of such lengths of duration

in our minds, which we apply to all parts of time whose lengths we would consider; yet there may be other parts of the universe, where they no more use there measures of ours, than in Japan they do our inches, feet, or miles; but yet something analogous to them there must be. For without some regular periodical returns, we could not measure ourselves, or signify to others, the length of any duration; though at the same time the world were as full of motion as it is now, but no part of it disposed into regular and apparently equidistant revolutions. But the different measures that may be made use of for the account of time, do not at all alter the notion of duration, which is the thing to be measured; no more than the different standards of a foot and a cubit alter the notion of extension to those who make use of those different measures.

24. Our measure of time applicable to duration before time. The mind having once got such a measure of time as the annual revolution of the sun, can apply that measure to duration wherein that measure itself did not exist, and with which, in the reality of its being, it had

nothing to do. For should one say, that Abraham was born in the two thousand seven hundred and twelfth year of the Julian period, it is altogether as intelligible as reckoning from the beginning of the world, though there were so far back no motion of the sun, nor any motion at all. For, though the Julian period be supposed to begin several hundred years before there were really either days, nights, or years, marked out by any revolutions of the sun,—yet we reckon as right, and thereby measure durations as well, as if really at that time the sun had existed, and kept the same ordinary motion it doth now. The idea of duration equal to an annual revolution of the sun, is as easily applicable in our thoughts to duration, where no sun or motion was, as the idea of a foot or yard, taken from bodies here, can be applied in our thoughts to duration, where no sun or motion was, as the idea of a foot or yard, taken from bodies here, can be applied in our thoughts to distances beyond the confines of the world, where are no bodies at all.

25. As we can measure space in our thoughts where there is no body. For supposing it were 5639 miles, or

millions of miles, from this place to the remotest body of the universe, (for being finite, it must be at a certain distance), as we suppose it to be 5639 years from this time to the first existence of any body in the beginning of the world;—we can, in our thoughts, apply this measure of a year to duration before the creation, or beyond the duration of bodies or motion, as we can this measure of a mile to space beyond the utmost bodies; and by the one measure duration, where there was no motion, as well as by the other measure space in our thoughts, where there is no body.

26. The assumption that the world is neither boundless nor eternal. If it be objected to me here, that, in this way of explaining of time, I have begged what I should not, viz. that the world is neither eternal nor infinite; I answer, That to my present purpose it is not needful, in this place, to make use of arguments to evince the world to be finite both in duration and extension. But it being at least as conceivable as the contrary, I have certainly the liberty to suppose it, as well as any one hath to suppose the contrary; and I doubt not, but that every

one that will go about it, may easily conceive in his mind the beginning of motion, though not of all duration, and so may come to a step and non ultra in his consideration of motion. So also, in his thoughts, he may set limits to body, and the extension belonging to it; but not to space, where no body is, the utmost bounds of space and duration being beyond the reach of thought, as well as the utmost bounds of number are beyond the largest comprehension of the mind; and all for the same reason, as we shall see in another place.

27. Eternity. By the same means, therefore, and from the same original that we come to have the idea of time, we have also that idea which we call Eternity; viz. having got the idea of succession and duration, by reflecting on the train of our own ideas, caused in us either by the natural appearances of those ideas coming constantly of themselves into our waking thoughts, or else caused by external objects successively affecting our senses; and having from the revolutions of the sun got the ideas of certain lengths of duration,—we can in our thoughts add such lengths of duration to one another,

as often as we please, and apply them, so added, to durations past or to come. And this we can continue to do on, without bounds or limits, and proceed in infinitum, and apply thus the length of the annual motion of the sun to duration, supposed before the sun's or any other motion had its being; which is no more difficult or absurd, than to apply the notion I have of the moving of a shadow one hour to-day upon the sun-dial to the duration of something last night, v.g. the burning of a candle, which is now absolutely separate from all actual motion; and it is as impossible for the duration of that flame for an hour last night to co-exist with any motion that now is, or for ever shall be, as for any part of duration, that was before the beginning of the world, to co-exist with the motion of the sun now. But yet this hinders not but that, having the idea of the length of the motion of the shadow on a dial between the marks of two hours, I can as distinctly measure in my thoughts the duration of that candle-light last night, as I can the duration of anything that does now exist: and it is no more than to think, that, had the sun shone then on

the dial, and moved after the same rate it doth now, the shadow on the dial would have passed from one hour-line to another whilst that flame of the candle lasted.

28. Our measures of duration dependent on our ideas. The notion of an hour, day, or year, being only the idea I have of the length of certain periodical regular motions, neither of which motions do ever all at once exist, but only in the ideas I have of them in my memory derived from my senses or reflection; I can with the same ease, and for the same reason, apply it in my thoughts to duration antecedent to all manner of motion, as well as to anything that is but a minute or a day antecedent to the motion that at this very moment the sun is in. All things past are equally and perfectly at rest; and to this way of consideration of them are all one, whether they were before the beginning of the world, or but yesterday: the measuring of any duration by some motion depending not at all on the real co-existence of that thing to that motion, or any other periods of revolution, but the having a clear idea of the length of some periodical known motion, or other in-

terval of duration, in my mind, and applying that to the duration of the thing I would measure.

29. The duration of anything need not be co-existent with the motion we measure it by. Hence we see that some men imagine the duration of the world, from its first existence to this present year 1689, to have been 5639 years, or equal to 5639 annual revolutions of the sun, and others a great deal more; as the Egyptians of old, who in the time of Alexander counted 23,000 years from the reign of the sun; and the Chinese now, who account the world 3,269,000 years old, or more; which longer duration of the world, according to their computation, though I should not believe to be true, yet I can equally imagine it with them, and as truly understand, and say one is longer than the other, as I understand, that Methusalem's life was longer than Enoch's. And if the common reckoning Of 5639 should be true, (as it may be as well as any other assigned,) it hinders not at all my imagining what others mean, when they make the world one thousand years older, since every one may with the same facility imagine (I do not say believe)

the world to be 50,000 years old, as 5639; and may as well conceive the duration of 50,000 years as 5639. Whereby it appears that, to the measuring the duration of anything by time, it is not requisite that that thing should be co-existent to the motion we measure by, or any other periodical revolution; but it suffices to this purpose, that we have the idea of the length of any regular periodical appearances, which we can in our minds apply to duration, with which the motion or appearance never co-existed.

30. Infinity in duration. For, as in the history of the creation delivered by Moses, I can imagine that light existed three days before the sun was, or had any motion, barely by thinking that the duration of light before the sun was created was so long as (if the sun had moved then as it doth now) would have been equal to three of his diurnal revolutions; so by the same way I can have an idea of the chaos, or angels, being created before there was either light or any continued motion, a minute, an hour, a day, a year, or one thousand years. For, if I can but consider duration equal to one minute,

before either the being or motion of any body, I can add one minute more till I come to sixty; and by the same way of adding minutes, hours, or years (i.e. such or such parts of the sun's revolutions, or any other period whereof I have the idea) proceed in infinitum, and suppose a duration exceeding as many such periods as I can reckon, let me add whilst I will, which I think is the notion we have of eternity; of whose infinity we have no other notion than we have of the infinity of number, to which we can add for ever without end.

31. Origin of our ideas of duration, and of the measures of it. And thus I think it is plain, that from those two fountains of all knowledge before mentioned, viz. reflection and sensation, we got the ideas of duration, and the measures of it. For, First, by observing what passes in our minds, how our ideas there in train constantly some vanish and others begin to appear, we come by the idea of succession.

Secondly, by observing a distance in the parts of this succession, we get the idea of duration.

Thirdly, by sensation observing certain appearances,

at certain regular and seeming equidistant periods, we get the ideas of certain lengths or measures of duration, as minutes, hours, days, years, &c.

Fourthly, by being able to repeat those measures of time, or ideas of stated length of duration, in our minds, as often as we will, we can come to imagine duration, where nothing does really endure or exist; and thus we imagine to-morrow, next year, or seven years hence.

Fifthly, by being able to repeat ideas of any length of time, as of a minute, a year, or an age, as often as we will in our own thoughts, and adding them one to another, without ever coming to the end of such addition, any nearer than we can to the end of number, to which we can always add; we come by the idea of eternity, as the future eternal duration of our souls, as well as the eternity of that infinite Being which must necessarily have always existed. Sixthly, by considering any part of infinite duration, as set out by periodical measures, we come by the idea of what we call time in general.

Chapter XV
Ideas of Duration and Expansion,
considered together

1. Both capable of greater and less. Though we have in the precedent chapters dwelt pretty long on the considerations of space and duration, yet, they being ideas of general concernment, that have something very abstruse and peculiar in their nature, the comparing them one with another may perhaps be of use for their illustration; and we may have the more clear and distinct conception of them by taking a view of them together. Distance or space, in its simple abstract conception, to avoid confusion, I call expansion, to distinguish it from extension, which by some is used to express this distance only as it is in the solid parts of matter, and so includes, or at least intimates, the idea of body: whereas the idea of pure distance includes no such thing. I prefer also the word expansion to space, because space is often applied to distance of fleeting successive parts, which never exist together, as well as to those which are permanent.

In both these (viz. expansion and duration) the mind has this common idea of continued lengths, capable of greater or less quantities. For a man has as clear an idea of the difference of the length of an hour and a day, as of an inch and a foot.

2. Expansion not bounded by matter. The mind, having got the idea of the length of any part of expansion, let it be a span, or a pace, or what length you will, can, as has been said, repeat that idea, and so, adding it to the former, enlarge its idea of length, and make it equal to two spans, or two paces; and so, as often as it will, till it equals the distance of any parts of the earth one from another, and increase thus till it amounts to the distance of the sun or remotest star. By such a progression as this, setting out from the place where it is, or any other place, it can proceed and pass beyond all those lengths, and find nothing to stop its going on, either in or without body. It is true, we can easily in our thoughts come to the end of solid extension; the extremity and bounds of all body we have no difficulty to arrive at: but when the mind is there, it finds nothing to hinder

its progress into this endless expansion; of that it can neither find nor conceive any end. Nor let any one say, that beyond the bounds of body, there is nothing at all; unless he will confine God within the limits of matter. Solomon, whose understanding was filled and enlarged with wisdom, seems to have other thoughts when he says, "Heaven, and the heaven of heavens, cannot contain thee." And he, I think, very much magnifies to himself the capacity of his own understanding, who persuades himself that he can extend his thoughts further than God exists, or imagine any expansion where He is not.

3. Nor duration by motion. Just so is it in duration. The mind having got the idea of any length of duration, can double, multiply, and enlarge it, not only beyond its own, but beyond the existence of all corporeal beings, and all the measures of time, taken from the great bodies of all the world and their motions. But yet every one easily admits, that, though we make duration boundless, as certainly it is, we cannot yet extend it beyond all being. God, every one easily allows, fills eternity; and

it is hard to find a reason why any one should doubt that He likewise fills immensity. His infinite being is certainly as boundless one way as another; and methinks it ascribes a little too much to matter to say, where there is no body, there is nothing.

4. Why men more easily admit infinite duration than infinite expansion. Hence I think we may learn the reason why every one familiarly and without the least hesitation speaks of and supposes Eternity, and sticks not to ascribe infinity to duration; but it is with more doubting and reserve that many admit or suppose the infinity of space. The reason whereof seems to me to be this,—That duration and extension being used as names of affections belonging to other beings, we easily conceive in God infinite duration, and we cannot avoid doing so: but, not attributing to Him extension, but only to matter, which is finite, we are apter to doubt of the existence of expansion without matter; of which alone we commonly suppose it an attribute. And, therefore, when men pursue their thoughts of space, they are apt to stop at the confines of body: as if space were there at an

end too, and reached no further. Or if their ideas, upon consideration, carry them further, yet they term what is beyond the limits of the universe, imaginary space: as if it were nothing, because there is no body existing in it. Whereas duration, antecedent to all body, and to the motions which it is measured by, they never term imaginary: because it is never supposed void of some other real existence. And if the names of things may at all direct our thoughts towards the original of men's ideas, (as I am apt to think they may very much,) one may have occasion to think by the name duration, that the continuation of existence, with a kind of resistance to any destructive force, and the continuation of solidity (which is apt to be confounded with, and if we will look into the minute anatomical parts of matter, is little different from, hardness) were thought to have some analogy, and gave occasion to words so near of kin as *durare* and *durum esse*. And that *durare* is applied to the idea of hardness, as well as that of existence, we see in Horace, *Epod. xvi. ferro duravit secula*. But, be that as it will, this is certain, that whoever pursues his own thoughts,

will find them sometimes launch out beyond the extent of body, into the infinity of space or expansion; the idea whereof is distinct and separate from body and all other things: which may, (to those who please), be a subject of further meditation.

5. Time to duration is as place to expansion. Time in general is to duration as place to expansion. They are so much of those boundless oceans of eternity and immensity as is set out and distinguished from the rest, as it were by landmarks; and so are made use of to denote the position of finite real beings, in respect one to another, in those uniform infinite oceans of duration and space. These, rightly considered, are only ideas of determinate distances from certain known points, fixed in distinguishable sensible things, and supposed to keep the same distance one from another. From such points fixed in sensible beings we reckon, and from them we measure our portions of those infinite quantities; which, so considered, are that which we call time and place. For duration and space being in themselves uniform and boundless, the order and position of things, without

such known settled points, would be lost in them; and all things would lie jumbled in an incurable confusion.

6. Time and place are taken for so much of either as are set out by the existence and motion of bodies. Time and place, taken thus for determinate distinguishable portions of those infinite abysses of space and duration, set out or supposed to be distinguished from the rest, by marks and known boundaries, have each of them a two-fold acceptance.

First, Time in general is commonly taken for so much of infinite duration as is measured by, and co-existent with, the existence and motions of the great bodies of the universe, as far as we know anything of them: and in this sense time begins and ends with the frame of this sensible world, as in these phrases before mentioned, "Before all time," or, "When time shall be no more." Place likewise is taken sometimes for that portion of infinite space which is possessed by and comprehended within the material world; and is thereby distinguished from the rest of expansion; though this may be more properly called extension than place. Within these two

are confined, and by the observable parts of them are measured and determined, the particular time or duration, and the particular extension and place, of all corporeal beings.

7. Sometimes for so much of either as we design by measures taken from the bulk or motion of bodies. Secondly, sometimes the word time is used in a larger sense, and is applied to parts of that infinite duration, not that were really distinguished and measured out by this real existence, and periodical motions of bodies, that were appointed from the beginning to be for signs and for seasons and for days and years, and are accordingly our measures of time; but such other portions too of that infinite uniform duration, which we upon any occasion do suppose equal to certain lengths of measured time; and so consider them as bounded and determined. For, if we should suppose the creation, or fall of the angels, was at the beginning of the Julian period, we should speak properly enough, and should be understood if we said, it is a longer time since the creation of angels than the creation of the world, by 7640 years:

whereby we would mark out so much of that undistinguished duration as we suppose equal to, and would have admitted, 7640 annual revolutions of the sun, moving at the rate it now does. And thus likewise we sometimes speak of place, distance, or bulk, in the great inane, beyond the confines of the world, when we consider so much of that space as is equal to, or capable to receive, a body of any assigned dimensions, as a cubic foot; or do suppose a point in it, at such a certain distance from any part of the universe.

8. They belong to all finite beings. Where and when are questions belonging to all finite existences, and are by us always reckoned from some known parts of this sensible world, and from some certain epochs marked out to us by the motions observable in it. Without some such fixed parts or periods, the order of things would be lost, to our finite understandings, in the boundless invariable oceans of duration and expansion, which comprehend in them all finite beings, and in their full extent belong only to the Deity. And therefore we are not to wonder that we comprehend them not, and do so

often find our thoughts at a loss, when we would consider them, either abstractly in themselves, or as any way attributed to the first incomprehensible Being. But when applied to any particular finite beings, the extension of any body is so much of that infinite space as the bulk of the body takes up. And place is the position of any body, when considered at a certain distance from some other. As the idea of the particular duration of anything is, an idea of that portion of infinite duration which passes during the existence of that thing; so the time when the thing existed is, the idea of that space of duration which passed between some known and fixed period of duration, and the being of that thing. One shows the distance of the extremities of the bulk or existence of the same thing, as that it is a foot square, or lasted two years; the other shows the distance of it in place, or existence from other fixed points of space or duration, as that it was in the middle of Lincoln's Inn Fields, or the first degree of Taurus, and in the year of our Lord 1671, or the 1000th year of the Julian period. All which distances we measure by preconceived ideas

of certain lengths of space and duration,—as inches, feet, miles, and degrees, and in the other, minutes, days, and years, &c.

9. All the parts of extension are extension, and all the parts of duration are duration. There is one thing more wherein space and duration have a great conformity, and that is, though they are justly reckoned amongst our simple ideas, yet none of the distinct ideas we have of either is without all manner of composition: it is the very nature of both of them to consist of parts: but their parts being all of the same kind, and without the mixture of any other idea, hinder them not from having a place amongst simple ideas. Could the mind, as in number, come to so small a part of extension or duration as excluded divisibility, that would be, as it were, the indivisible unit or idea; by repetition of which, it would make its more enlarged ideas of extension and duration. But, since the mind is not able to frame an idea of any space without parts, instead thereof it makes use of the common measures, which, by familiar use in each country, have imprinted themselves on the memory (as inches and feet; or

cubits and parasangs; and so seconds, minutes, hours, days, and years in duration);—the mind makes use, I say, of such ideas as these, as simple ones: and these are the component parts of larger ideas, which the mind upon occasion makes by the addition of such known lengths which it is acquainted with. On the other side, the ordinary smallest measure we have of either is looked on as an unit in number, when the mind by division would reduce them into less fractions. Though on both sides, both in addition and division, either of space or duration, when the idea under consideration becomes very big or very small its precise bulk becomes very obscure and confused; and it is the number of its repeated additions or divisions that alone remains clear and distinct; as will easily appear to any one who will let his thoughts loose in the vast expansion of space, or divisibility of matter. Every part of duration is duration too; and every part of extension is extension, both of them capable of addition or division in infinitum. But the least portions of either of them, whereof we have clear and distinct ideas, may perhaps be fittest to be considered by us, as the simple

ideas of that kind out of which our complex modes of space, extension, and duration are made up, and into which they can again be distinctly resolved. Such a small part in duration may be called a moment, and is the time of one idea in our minds, in the train of their ordinary succession there. The other, wanting a proper name, I know not whether I may be allowed to call a sensible point, meaning thereby the least particle of matter or space we can discern, which is ordinarily about a minute, and to the sharpest eyes seldom less than thirty seconds of a circle, whereof the eye is the centre.

10. Their parts inseparable. Expansion and duration have this further agreement, that, though they are both considered by us as having parts, yet their parts are not separable one from another, no not even in thought: though the parts of bodies from whence we take our measure of the one; and the parts of motion, or rather the succession of ideas in our minds, from whence we take the measure of the other, may be interrupted and separated; as the one is often by rest, and the other is by sleep, which we call rest too.

11. Duration is as a line, expansion as a solid. But there is this manifest difference between them,—That the ideas of length which we have of expansion are turned every way, and so make figure, and breadth, and thickness; but duration is but as it were the length of one straight line, extended in infinitum, not capable of multiplicity, variation, or figure; but is one common measure of all existence whatsoever, wherein all things, whilst they exist, equally partake. For this present moment is common to all things that are now in being, and equally comprehends that part of their existence, as much as if they were all but one single being; and we may truly say, they all exist in the same moment of time. Whether angels and spirits have any analogy to this, in respect to expansion, is beyond my comprehension: and perhaps for us, who have understandings and comprehensions suited to our own preservation, and the ends of our own being, but not to the reality and extent of all other beings, it is near as hard to conceive any existence, or to have an idea of any real being, with a perfect negation of all manner of expansion, as it is to have the idea

of any real existence with a perfect negation of all manner of duration. And therefore, what spirits have to do with space, or how they communicate in it, we know not. All that we know is, that bodies do each singly possess its proper portion of it, according to the extent of solid parts; and thereby exclude all other bodies from having any share in that particular portion of space, whilst it remains there.

12. Duration has never two parts together, expansion altogether. Duration, and time which is a part of it, is the idea we have of perishing distance, of which no two parts exist together, but follow each other in succession; an expansion is the idea of lasting distance, all whose parts exist together, and are not capable of succession. And therefore, though we cannot conceive any duration without succession, nor can put it together in our thoughts that any being does now exist tomorrow, or possess at once more than the present moment of duration; yet we can conceive the eternal duration of the Almighty far different from that of man, or any other finite being. Because man comprehends not in his

knowledge or power all past and future things: his thoughts are but of yesterday, and he knows not what tomorrow will bring forth. What is once past he can never recall; and what is yet to come he cannot make present. What I say of man, I say of all finite beings; who, though they may far exceed man in knowledge and power, yet are no more than the meanest creature, in comparison with God himself Finite or any magnitude holds not any proportion to infinite. God's infinite duration, being accompanied with infinite knowledge and infinite power, He sees all things, past and to come; and they are no more distant from His knowledge, no further removed from His sight, than the present: they all lie under the same view: and there is nothing which He cannot make exist each moment He pleases. For the existence of all things, depending upon His good pleasure, all things exist every moment that He thinks fit to have them exist. To conclude: expansion and duration do mutually embrace and comprehend each other; every part of space being in every part of duration, and every part of duration in every part of expansion. Such a com-

bination of two distinct ideas is, I suppose, scarce to be found in all that great variety we do or can conceive, and may afford matter to further speculation.

Chapter XVI Idea of Number

1. Number the simplest and most universal idea. Amongst all the ideas we have, as there is none suggested to the mind by more ways, so there is none more simple, than that of unity, or one: it has no shadow of variety or composition in it: every object our senses are employed about; every idea in our understandings; every thought of our minds, brings this idea along with it. And therefore it is the most intimate to our thoughts, as well as it is, in its agreement to all other things, the most universal idea we have. For number applies itself to men, angels, actions, thoughts; everything that either doth exist, or can be imagined.

2. Its modes made by addition. By repeating this idea in our minds, and adding the repetitions together, we come

by the complex ideas of the modes of it. Thus, by adding one to one, we have the complex idea of a couple; by putting twelve units together, we have the complex idea of a dozen; and so of a score, or a million, or any other number.

3. Each mode distinct. The simple modes of number are of all other the most distinct; every the least variation, which is an unit, making each combination as clearly different from that which approacheth nearest to it, as the most remote; two being as distinct from one, as two hundred; and the idea of two as distinct from the idea of three, as the magnitude of the whole earth is from that of a mite. This is not so in other simple modes, in which it is not so easy, nor perhaps possible for us to distinguish betwixt two approaching ideas, which yet are really different. For who will undertake to find a difference between the white of this paper and that of the next degree to it: or can form distinct ideas of every the least excess in extension?

4. Therefore demonstrations in numbers the most precise. The clearness and distinctness of each mode of

number from all others, even those that approach nearest, makes me apt to think that demonstrations in numbers, if they are not more evident and exact than in extension, yet they are more general in their use, and more determinate in their application. Because the ideas of numbers are more precise and distinguishable than in extension; where every equality and excess are not so easy to be observed or measured; because our thoughts cannot in space arrive at any determined smallness beyond which it cannot go, as an unit; and therefore the quantity or proportion of any the least excess cannot be discovered; which is clear otherwise in number, where, as has been said, 91 is as distinguishable from 90 as from 9000, though 91 be the next immediate excess to 90. But it is not so in extension, where, whatsoever is more than just a foot or an inch, is not distinguishable from the standard of a foot or an inch; and in lines which appear of an equal length, one may be longer than the other by innumerable parts: nor can any one assign an angle, which shall be the next biggest to a right one.

5. Names necessary to numbers. By the repeating, as has been said, the idea of an unit, and joining it to another unit, we make thereof one collective idea, marked by the name two. And whosoever can do this, and proceed on, still adding one more to the last collective idea which he had of any number, and gave a name to it, may count, or have ideas, for several collections of units, distinguished one from another, as far as he hath a series of names for following numbers, and a memory to retain that series, with their several names: all numeration being but still the adding of one unit more, and giving to the whole together, as comprehended in one idea, a new or distinct name or sign, whereby to know it from those before and after, and distinguish it from every smaller or greater multitude of units. So that he that can add one to one, and so to two, and so go on with his tale, taking still with him the distinct names belonging to every progression; and so again, by subtracting an unit from each collection, retreat and lessen them, is capable of all the ideas of numbers within the compass of his language, or for which he hath names,

though not perhaps of more. For, the several simple modes of numbers being in our minds but so many combinations of units, which have no variety, nor are capable of any other difference but more or less, names or marks for each distinct combination seem more necessary than in any other sort of ideas. For, without such names or marks, we can hardly well make use of numbers in reckoning, especially where the combination is made up of any great multitude of units; which put together, without a name or mark to distinguish that precise collection, will hardly be kept from being a heap in confusion.

6. Another reason for the necessity of names to numbers. This I think to be the reason why some Americans I have spoken with, (who were otherwise of quick and rational parts enough,) could not, as we do, by any means count to 1000; nor had any distinct idea of that number, though they could reckon very well to 20. Because their language being scanty, and accommodated only to the few necessities of a needy, simple life, unacquainted either with trade or mathematics, had no words

in it to stand for 1000; so that when they were discoursed with of those greater numbers, they would show the hairs of their head, to express a great multitude, which they could not number; which inability, I suppose, proceeded from their want of names. The Tououpinambos had no names for numbers above 5; any number beyond that they made out by showing their fingers, and the fingers of others who were present. And I doubt not but we ourselves might distinctly number in words a great deal further than we usually do, would we find out but some fit denominations to signify them by; whereas, in the way we take now to name them, by millions of millions of millions, &c., it is hard to go beyond eighteen, or at most, four and twenty, decimal progressions, without confusion. But to show how much distinct names conduce to our well reckoning, or having useful ideas of numbers, let us see all these following figures in one continued line, as the marks of one number: v. g.

Nonillions Octillions Septillions Sextillions Quintrillions
857324 162486 345896 437918 423147
Quartrillions Trillions Billions Millions Units
248106 235421 261734 368149 623137

The ordinary way of naming this number in English, will be the often repeating of millions, of millions, of millions, of millions, of millions, of millions, (which is the denomination of the second six figures). In which way, it will be very hard to have any distinguishing notions of this number. But whether, by giving every six figures a new and orderly denomination, these, and perhaps a great many more figures in progression, might not easily be counted distinctly, and ideas of them both got more easily to ourselves, and more plainly signified to others, I leave it to be considered. This I mention only to show how necessary distinct names are to numbering, without pretending to introduce new ones of my invention.

7. Why children number not earlier. Thus children, either for want of names to mark the several progressions

of numbers, or not having yet the faculty to collect scattered ideas into complex ones, and range them in a regular order, and so retain them in their memories, as is necessary to reckoning, do not begin to number very early, nor proceed in it very far or steadily, till a good while after they are well furnished with good store of other ideas: and one may often observe them discourse and reason pretty well, and have very clear conceptions of several other things, before they can tell twenty. And some, through the default of their memories, who cannot retain the several combinations of numbers, with their names, annexed in their distinct orders, and the dependence of so long a train of numeral progressions, and their relation one to another, are not able all their lifetime to reckon, or regularly go over any moderate series of numbers. For he that will count twenty, or have any idea of that number, must know that nineteen went before, with the distinct name or sign of every one of them, as they stand marked in their order; for wherever this fails, a gap is made, the chain breaks, and the progress in numbering can go no further. So that to

reckon right, it is required, (1) That the mind distinguish carefully two ideas, which are different one from another only by the addition or subtraction of one unit: (2) That it retain in memory the names or marks of the several combinations, from an unit to that number; and that not confusedly, and at random, but in that exact order that the numbers follow one another. In either of which, if it trips, the whole business of numbering will be disturbed, and there will remain only the confused idea of multitude, but the ideas necessary to distinct numeration will not be attained to.

8. Number measures all measureables. This further is observable in number, that it is that which the mind makes use of in measuring all things that by us are measurable, which principally are expansion and duration; and our idea of infinity, even when applied to those, seems to be nothing but the infinity of number. For what else are our ideas of Eternity and Immensity, but the repeated additions of certain ideas of imagined parts of duration and expansion, with the infinity of number; in which we can come to no end of addition? For such

an inexhaustible stock, number (of all other our ideas) most clearly furnishes us with, as is obvious to every one. For let a man collect into one sum as great a number as he pleases, this multitude, how great soever, lessens not one jot the power of adding to it, or brings him any nearer the end of the inexhaustible stock of number; where still there remains as much to be added, as if none were taken out. And this endless addition or addibility (if any one like the word better) of numbers, so apparent to the mind, is that, I think, which gives us the clearest and most distinct idea of infinity: of which more in the following chapter.

Chapter XVII

Of Infinity

1. Infinity, in its original intention, attributed to space, duration, and number. He that would know what kind of idea it is to which we give the name of infinity, cannot do it better than by considering to what infinity is by the mind more immediately attributed; and then how

the mind comes to frame it. Finite and infinite seem to me to be looked upon by the mind as the modes of quantity, and to be attributed primarily in their first designation only to those things which have parts, and are capable of increase or diminution by the addition or subtraction of any the least part: and such are the ideas of space, duration, and number, which we have considered in the foregoing chapters. It is true, that we cannot but be assured, that the great God, of whom and from whom are all things, is incomprehensibly infinite: but yet, when we apply to that first and supreme Being our idea of infinite, in our weak and narrow thoughts, we do it primarily in respect to his duration and ubiquity; and, I think, more figuratively to his power, wisdom, and goodness, and other attributes, which are properly inexhaustible and incomprehensible, &c. For, when we call them infinite, we have no other idea of this infinity but what carries with it some reflection on, and imitation of, that number or extent of the acts or objects of God's power, wisdom, and goodness, which can never be supposed so great, or so many, which these

attributes will not always surmount and exceed, let us multiply them in our thoughts as far as we can, with all the infinity of endless number. I do not pretend to say how these attributes are in God, who is infinitely beyond the reach of our narrow capacities: they do, without doubt, contain in them all possible perfection: but this, I say, is our way of conceiving them, and these our ideas of their infinity.

2. The idea of finite easily got. Finite then, and infinite, being by the mind looked on as modifications of expansion and duration, the next thing to be considered, is,—How the mind comes by them. As for the idea of finite, there is no great difficulty. The obvious portions of extension that affect our senses, carry with them into the mind the idea of finite: and the ordinary periods of succession, whereby we measure time and duration, as hours, days, and years, are bounded lengths. The difficulty is, how we come by those boundless ideas of eternity and immensity; since the objects we converse with come so much short of any approach or proportion to that largeness.

3. How we come by the idea of infinity. Every one that has any idea of any stated lengths of space, as a foot, finds that he can repeat that idea; and joining it to the former, make the idea of two feet; and by the addition of a third, three feet; and so on, without ever coming to an end of his additions, whether of the same idea of a foot, or, if he pleases, of doubling it, or any other idea he has of any length, as a mile, or diameter of the earth, or of the orbis magnus: for whichever of these he takes, and how often soever he doubles, or any otherwise multiplies it, he finds, that, after he has continued his doubling in his thoughts, and enlarged his idea as much as he pleases, he has no more reason to stop, nor is one jot nearer the end of such addition, than he was at first setting out: the power of enlarging his idea of space by further additions remaining still the same, he hence takes the idea of infinite space.

4. Our idea of space boundless. This, I think, is the way whereby the mind gets the idea of infinite space. It is a quite different consideration, to examine whether the mind has the idea of such a boundless space actually

existing; since our ideas are not always proofs of the existence of things: but yet, since this comes here in our way, I suppose I may say, that we are apt to think that space in itself is actually boundless, to which imagination the idea of space or expansion of itself naturally leads us. For, it being considered by us, either as the extension of body, or as existing by itself, without any solid matter taking it up, (for of such a void space we have not only the idea, but I have proved, as I think, from the motion of body, its necessary existence), it is impossible the mind should be ever able to find or suppose any end of it, or be stopped anywhere in its progress in this space, how far soever it extends its thoughts. Any bounds made with body, even adamant walls, are so far from putting a stop to the mind in its further progress in space and extension that it rather facilitates and enlarges it. For so far as that body reaches, so far no one can doubt of extension; and when we are come to the utmost extremity of body, what is there that can there put a stop, and satisfy the mind that it is at the end of space, when it perceives that it is not; nay, when

it is satisfied that body itself can move into it? For, if it be necessary for the motion of body, that there should be an empty space, though ever so little, here amongst bodies; and if it be possible for body to move in or through that empty space;—nay, it is impossible for any particle of matter to move but into an empty space; the same possibility of a body's moving into a void space, beyond the utmost bounds of body, as well as into a void space interspersed amongst bodies, will always remain clear and evident: the idea of empty pure space, whether within or beyond the confines of all bodies, being exactly the same, differing not in nature, though in bulk; and there being nothing to hinder body from moving into it. So that wherever the mind places itself by any thought, either amongst, or remote from all bodies, it can, in this uniform idea of space, nowhere find any bounds, any end; and so must necessarily conclude it, by the very nature and idea of each part of it, to be actually infinite.

5. And so of duration. As, by the power we find in ourselves of repeating, as often as we will, any idea of

space, we get the idea of immensity; so, by being able to repeat the idea of any length of duration we have in our minds, with all the endless addition of number, we come by the idea of eternity. For we find in ourselves, we can no more come to an end of such repeated ideas than we can come to the end of number; which every one perceives he cannot. But here again it is another question, quite different from our having an idea of eternity, to know whether there were any real being, whose duration has been eternal. And as to this, I say, he that considers something now existing, must necessarily come to Something eternal. But having spoke of this in another place, I shall say here no more of it, but proceed on to some other considerations of our idea of infinity.

6. Why other ideas are not capable of infinity. If it be so, that our idea of infinity be got from the power we observe in ourselves of repeating, without end, our own ideas, it may be demanded,—Why we do not attribute infinity to other ideas, as well as those of space and duration; since they may be as easily, and as often, repeated in our minds as the other: and yet nobody ever

thinks of infinite sweetness, or infinite whiteness, though he can repeat the idea of sweet or white, as frequently as those of a yard or a day? To which I answer,—All the ideas that are considered as having parts, and are capable of increase by the addition of any equal or less parts, afford us, by their repetition, the idea of infinity; because, with this endless repetition, there is continued an enlargement of which there can be no end. But in other ideas it is not so. For to the largest idea of extension or duration that I at present have, the addition of any the least part makes an increase; but to the perfectest idea I have of the whitest whiteness, if I add another of a less or equal whiteness, (and of a whiter than I have, I cannot add the idea), it makes no increase, and enlarges not my idea at all; and therefore the different ideas of whiteness, &c. are called degrees. For those ideas that consist of parts are capable of being augmented by every addition of the least part; but if you take the idea of white, which one parcel of snow yielded yesterday to our sight, and another idea of white from another parcel of snow you see to-day, and put them together in

your mind, they embody, as it were, and run into one, and the idea of whiteness is not at all increased; and if we add a less degree of whiteness to a greater, we are so far from increasing, that we diminish it. Those ideas that consist not of parts cannot be augmented to what proportion men please, or be stretched beyond what they have received by their senses; but space, duration, and number, being capable of increase by repetition, leave in the mind an idea of endless room for more; nor can we conceive anywhere a stop to a further addition or progression: and so those ideas alone lead our minds towards the thought of infinity.

7. Difference between infinity of space, and space infinite. Though our idea of infinity arise from the contemplation of quantity, and the endless increase the mind is able to make in quantity, by the repeated additions of what portions thereof it pleases; yet I guess we cause great confusion in our thoughts, when we join infinity to any supposed idea of quantity the mind can be thought to have, and so discourse or reason about an infinite quantity, as an infinite space, or an infinite duration.

For, as our idea of infinity being, as I think, an endless growing idea, but the idea of any quantity the mind has, being at that time terminated in that idea, (for be it as great as it will, it can be no greater than it is,)—to join infinity to it, is to adjust a standing measure to a growing bulk; and therefore I think it is not an insignificant subtilty, if I say, that we are carefully to distinguish between the idea of the infinity of space, and the idea of a space infinite. The first is nothing but a supposed endless progression of the mind, over what repeated ideas of space it pleases; but to have actually in the mind the idea of a space infinite, is to suppose the mind already passed over, and actually to have a view of all those repeated ideas of space which an endless repetition can never totally represent to it; which carries in it a plain contradiction.

8. We have no idea of infinite space. This, perhaps, will be a little plainer, if we consider it in numbers. The infinity of numbers, to the end of whose addition every one perceives there is no approach, easily appears to any one that reflects on it. But, how clear soever this

idea of the infinity of number be, there is nothing yet more evident than the absurdity of the actual idea of an infinite number. Whatsoever positive ideas we have in our minds of any space, duration, or number, let them be ever so great, they are still finite; but when we suppose an inexhaustible remainder, from which we remove all bounds, and wherein we allow the mind an endless progression of thought, without ever completing the idea, there we have our idea of infinity: which, though it seems to be pretty clear when we consider nothing else in it but the negation of an end, yet, when we would frame in our minds the idea of an infinite space or duration, that idea is very obscure and confused, because it is made up of two parts, very different, if not inconsistent. For, let a man frame in his mind an idea of any space or number, as great as he will; it is plain the mind rests and terminates in that idea, which is contrary to the idea of infinity, which consists in a supposed endless progression. And therefore I think it is that we are so easily confounded, when we come to argue and reason about infinite space or duration, &c.

Because the parts of such an idea not being perceived to be, as they are, inconsistent, the one side or other always perplexes, whatever consequences we draw from the other; as an idea of motion not passing on would perplex any one who should argue from such an idea, which is not better than an idea of motion at rest. And such another seems to me to be the idea of a space, or (which is the same thing) a number infinite, i.e. of a space or number which the mind actually has, and so views and terminates in; and of a space or number, which, in a constant and endless enlarging and progression, it can in thought never attain to. For, how large soever an idea of space I have in my mind, it is no larger than it is that instant that I have it, though I be capable the next instant to double it, and so on in infinitum; for that alone is infinite which has no bounds; and that the idea of infinity, in which our thoughts can find none.

9. Number affords us the clearest idea of infinity. But of all other ideas, it is number, as I have said, which I think furnishes us with the clearest and most distinct idea of infinity we are capable of. For, even in space and

duration, when the mind pursues the idea of infinity, it there makes use of the ideas and repetitions of numbers, as of millions and millions of miles, or years, which are so many distinct ideas,—kept best by number from running into a confused heap, wherein the mind loses itself; and when it has added together as many millions, &c., as it pleases, of known lengths of space or duration, the clearest idea it can get of infinity, is the confused incomprehensible remainder of endless addible numbers, which affords no prospect of stop or boundary.

10. Our different conceptions of the infinity of number contrasted with those of duration and expansion. It will, perhaps, give us a little further light into the idea we have of infinity, and discover to us, that it is nothing but the infinity of number applied to determinate parts, of which we have in our minds the distinct ideas, if we consider that number is not generally thought by us infinite, whereas duration and extension are apt to be so; which arises from hence,—that in number we are at one end, as it were: for there being in number nothing

less than an unit, we there stop, and are at an end; but in addition, or increase of number, we can set no bounds: and so it is like a line, whereof one end terminating with us, the other is extended still forwards, beyond all that we can conceive. But in space and duration it is otherwise. For in duration we consider it as if this line of number were extended both ways—to an unconceivable, undeterminate, and infinite length; which is evident to any one that will but reflect on what consideration he hath of Eternity; which, I suppose, will find to be nothing else but the turning this infinity of number both ways, a parte ante, and a parte post, as they speak. For, when we would consider eternity, a parte ante, what do we but, beginning from ourselves and the present time we are in, repeat in our minds the ideas of years, or ages, or any other assignable portion of duration past, with a prospect of proceeding in such addition with all the infinity of number: and when we would consider eternity, a parte post, we just after the same rate begin from ourselves, and reckon by multiplied periods yet to come, still extending that line of

number as before. And these two being put together, are that infinite duration we call Eternity: which, as we turn our view either way, forwards or backwards, appears infinite, because we still turn that way the infinite end of number, i.e. the power still of adding more.

11. How we conceive the infinity of space. The same happens also in space, wherein, conceiving ourselves to be, as it were, in the centre, we do on all sides pursue those indeterminable lines of number; and reckoning any way from ourselves, a yard, mile, diameter of the earth, or orbis magnus,—by the infinity of number, we add others to them, as often as we will. And having no more reason to set bounds to those repeated ideas than we have to set bounds to number, we have that indeterminable idea of immensity.

12. Infinite divisibility. And since in any bulk of matter our thoughts can never arrive at the utmost divisibility, therefore there is an apparent infinity to us also in that, which has the infinity also of number; but with this difference,—that, in the former considerations of the infinity of space and duration, we only use addition of

numbers; whereas this is like the division of an unit into its fractions, wherein the mind also can proceed in infinitum, as well as in the former additions; it being indeed but the addition still of new numbers: though in the addition of the one, we can have no more the positive idea of a space infinitely great, than, in the division of the other, we can have the [positive] idea of a body infinitely little;—our idea of infinity being, as I may say, a growing or fugitive idea, still in a boundless progression, that can stop nowhere.

13. No positive idea of infinity. Though it be hard, I think, to find anyone so absurd as to say he has the positive idea of an actual infinite number;—the infinity whereof lies only in a power still of adding any combination of units to any former number, and that as long and as much as one will; the like also being in the infinity of space and duration, which power leaves always to the mind room for endless additions;—yet there be those who imagine they have positive ideas of infinite duration and space. It would, I think, be enough to destroy any such positive idea of infinite, to ask him that has

it,—whether he could add to it or no; which would easily show the mistake of such a positive idea. We can, I think, have no positive idea of any space or duration which is not made up of, and commensurate to, repeated numbers of feet or yards, or days and years; which are the common measures, whereof we have the ideas in our minds, and whereby we judge of the greatness of this sort of quantities. And therefore, since an infinite idea of space or duration must needs be made up of infinite parts, it can have no other infinity than that of number capable still of further addition; but not an actual positive idea of a number infinite. For, I think it is evident, that the addition of finite things together (as are all lengths whereof we have the positive ideas) can never otherwise produce the idea of infinite than as number does; which, consisting of additions of finite units one to another, suggests the idea of infinite, only by a power we find we have of still increasing the sum, and adding more of the same kind; without coming one jot nearer the end of such progression.

14. How we cannot have a positive idea of infinity in

quantity. They who would prove their idea of infinite to be positive, seem to me to do it by a pleasant argument, taken from the negation of an end; which being negative, the negation of it is positive. He that considers that the end is, in body, but the extremity or superficies of that body, will not perhaps be forward to grant that the end is a bare negative: and he that perceives the end of his pen is black or white, will be apt to think that the end is something more than a pure negation. Nor is it, when applied to duration, the bare negation of existence, but more properly the last moment of it. But if they will have the end to be nothing but the bare negation of existence, I am sure they cannot deny but the beginning is the first instant of being, and is not by any body conceived to be a bare negation; and therefore, by their own argument, the idea of eternal, a parte ante, or of a duration without a beginning, is but a negative idea.

15. What is positive, what negative, in our idea of infinite. The idea of infinite has, I confess, something of positive in all those things we apply to it. When we

would think of infinite space or duration, we at first step usually make some very large idea, as perhaps of millions of ages, or miles, which possibly we double and multiply several times. All that we thus amass together in our thoughts is positive, and the assemblage of a great number of positive ideas of space or duration. But what still remains beyond this we have no more a positive distinct notion of than a mariner has of the depth of the sea; where, having let down a large portion of his sounding-line, he reaches no bottom. Whereby he knows the depth to be so many fathoms, and more; but how much the more is, he hath no distinct notion at all: and could he always supply new line, and find the plummet always sink, without ever stopping, he would be something in the posture of the mind reaching after a complete and positive idea of infinity. In which case, let this line be ten, or ten thousand fathoms long, it equally discovers what is beyond it, and gives only this confused and comparative idea, that this is not all, but one may yet go farther. So much as the mind comprehends of any space, it has a positive idea of: but in endeavour-

ing to make it infinite,—it being always enlarging, always advancing,—the idea is still imperfect and incomplete. So much space as the mind takes a view of in its contemplation of greatness, is a clear picture, and positive in the understanding: but infinite is still greater. 1. Then the idea of so much is positive and clear. 2. The idea of greater is also clear; but it is but a comparative idea, the idea of so much greater as cannot be comprehended. 3. And this is plainly negative: not positive. For he has no positive clear idea of the largeness of any extension, (which is that sought for in the idea of infinite), that has not a comprehensive idea of the dimensions of it: and such, nobody, I think, pretends to in what is infinite. For to say a man has a positive clear idea of any quantity, without knowing how great it is, is as reasonable as to say, he has the positive clear idea of the number of the sands on the sea-shore, who knows not how many there be, but only that they are more than twenty. For just such a perfect and positive idea has he of an infinite space or duration, who says it is larger than the extent or duration of ten, one hundred,

one thousand, or any other number of miles, or years, whereof he has or can have a positive idea; which is all the idea, I think, we have of infinite. So that what lies beyond our positive idea towards infinity, lies in obscurity, and has the indeterminate confusion of a negative idea, wherein I know I neither do nor can comprehend all I would, it being too large for a finite and narrow capacity. And that cannot but be very far from a positive complete idea, wherein the greatest part of what I would comprehend is left out, under the indeterminate intimation of being still greater. For to say, that, having in any quantity measured so much, or gone so far, you are not yet at the end, is only to say that that quantity is greater. So that the negation of an end in any quantity is, in other words, only to say that it is bigger; and a total negation of an end is but carrying this bigger still with you, in all the progressions of your thoughts shall make in quantity; and adding this idea of still greater to all the ideas you have, or can be supposed to have, of quantity. Now, whether such an idea as that be positive, I leave any one to consider.

16. We have no positive idea of an infinite duration. I ask those who say they have a positive idea of eternity, whether their idea of duration includes in it succession, or not? If it does not, they ought to show the difference of their notion of duration, when applied to an eternal Being, and to a finite; since, perhaps, there may be others as well as I, who will own to them their weakness of understanding in this point, and acknowledge that the notion they have of duration forces them to conceive, that whatever has duration, is of a longer continuance to-day than it was yesterday. If, to avoid succession in external existence, they return to the punctum stans of the schools, I suppose they will thereby very little mend the matter, or help us to a more clear and positive idea of infinite duration; there being nothing more inconceivable to me than duration without succession. Besides, that punctum stans, if it signify anything, being not quantum, finite or infinite cannot belong to it. But, if our weak apprehensions cannot separate succession from any duration whatsoever, our idea of eternity can be nothing but of infinite succession of moments of

duration wherein anything does exist; and whether any one has, or can have, a positive idea of an actual infinite number, I leave him to consider, till his infinite number be so great that he himself can add no more to it; and as long as he can increase it, I doubt he himself will think the idea he hath of it a little too scanty for positive infinity.

17. No complete idea of eternal being. I think it unavoidable for every considering, rational creature, that will but examine his own or any other existence, to have the notion of an eternal, wise Being, who had no beginning: and such an idea of infinite duration I am sure I have. But this negation of a beginning, being but the negation of a positive thing, scarce gives me a positive idea of infinity; which, whenever I endeavour to extend my thoughts to, I confess myself at a loss, and I find I cannot attain any clear comprehension of it.

18. No positive idea of infinite space. He that thinks he has a positive idea of infinite space, will, when he considers it, find that he can no more have a positive idea of the greatest, than he has of the least space. For in

this latter, which seems the easier of the two, and more within our comprehension, we are capable only of a comparative idea of smallness, which will always be less than any one whereof we have the positive idea. All our positive ideas of any quantity, whether great or little, have always bounds, though our comparative idea, whereby we can always add to the one, and take from the other, hath no bounds. For that which remains, either great or little, not being comprehended in that positive idea which we have, lies in obscurity; and we have no other idea of it, but of the power of enlarging the one and diminishing the other, without ceasing. A pestle and mortar will as soon bring any particle of matter to indivisibility, as the acutest thought of a mathematician; and a surveyor may as soon with his chain measure out infinite space, as a philosopher by the quickest flight of mind reach it, or by thinking comprehend it; which is to have a positive idea of it. He that thinks on a cube of an inch diameter, has a clear and positive idea of it in his mind, and so can frame one of $1/2$, $1/4$, $1/8$, and so on, till he has the idea in his thoughts of

something very little; but yet reaches not the idea of that incomprehensible littleness which division can produce. What remains of smallness is as far from his thoughts as when he first began; and therefore he never comes at all to have a clear and positive idea of that smallness which is consequent to infinite divisibility.

19. What is positive, what negative, in our idea of infinite. Every one that looks towards infinity does, as I have said, at first glance make some very large idea of that which he applies it to, let it be space or duration; and possibly he wearies his thoughts, by multiplying in his mind that first large idea: but yet by that he comes no nearer to the having a positive clear idea of what remains to make up a positive infinite, than the country fellow had of the water which was yet to come, and pass the channel of the river where he stood:

*Rusticus expectat dum defluat amnis, at ille
Labitur, et labetur in omne volubilis oevum.*

20. Some think they have a positive idea of eternity,

and not of infinite space. There are some I have met that put so much difference between infinite duration and infinite space, that they persuade themselves that they have a positive idea of eternity, but that they have not, nor can have any idea of infinite space. The reason of which mistake I suppose to be this—that finding, by a due contemplation of causes and effects, that it is necessary to admit some Eternal Being, and so to consider the real existence of that Being as taken up and commensurate to their idea of eternity; but, on the other side, not finding it necessary, but, on the contrary, apparently absurd, that body should be infinite, they forwardly conclude that they can have no idea of infinite space, because they can have no idea of infinite matter. Which consequence, I conceive, is very ill collected, because the existence of matter is no ways necessary to the existence of space, no more than the existence of motion, or the sun, is necessary to duration, though duration used to be measured by it. And I doubt not but that a man may have the idea of ten thousand miles square, without any body so big, as well as the

idea of ten thousand years, without any body so old. It seems as easy to me to have the idea of space empty of body, as to think of the capacity of a bushel without corn, or the hollow of a nut-shell without a kernel in it: it being no more necessary that there should be existing a solid body, infinitely extended, because we have an idea of the infinity of space, than it is necessary that the world should be eternal, because we have an idea of infinite duration. And why should we think our idea of infinite space requires the real existence of matter to support it, when we find that we have as clear an idea of an infinite duration to come, as we have of infinite duration past? Though I suppose nobody thinks it conceivable that anything does or has existed in that future duration. Nor is it possible to join our idea of future duration with present or past existence, any more than it is possible to make the ideas of yesterday, today, and to-morrow to be the same; or bring ages past and future together, and make them contemporary. But if these men are of the mind, that they have clearer ideas of infinite duration than of infinite space, because

it is past doubt that God has existed from all eternity, but there is no real matter co-extended with infinite space; yet those philosophers who are of opinion that infinite space is possessed by God's infinite omnipresence, as well as infinite duration by his eternal existence, must be allowed to have as clear an idea of infinite space as of infinite duration; though neither of them, I think, has any positive idea of infinity in either case. For whatsoever positive ideas a man has in his mind of any quantity, he can repeat it, and add it to the former, as easy as he can add together the ideas of two days, or two paces, which are positive ideas of lengths he has in his mind, and so on as long as he pleases: whereby, if a man had a positive idea of infinite, either duration or space, he could add two infinities together; nay, make one infinite infinitely bigger than another—absurdities too gross to be confuted.

21. Supposed positive ideas of infinity, cause of mistakes. But yet if after all this, there be men who persuade themselves that they have clear positive comprehensive ideas of infinity, it is fit they enjoy their privi-

lege: and I should be very glad (with some others that I know, who acknowledge they have none such) to be better informed by their communication. For I have been hitherto apt to think that the great and inextricable difficulties which perpetually involve all discourses concerning infinity,—whether of space, duration, or divisibility, have been the certain marks of a defect in our ideas of infinity, and the disproportion the nature thereof has to the comprehension of our narrow capacities. For, whilst men talk and dispute of infinite space or duration, as if they had as complete and positive ideas of them as they have of the names they use for them, or as they have of a yard, or an hour, or any other determinate quantity; it is no wonder if the incomprehensible nature of the thing they discourse of, or reason about, leads them into perplexities and contradictions, and their minds be overlaid by an object too large and mighty to be surveyed and managed by them.

22. All these are modes of ideas got from sensation and reflection. If I have dwelt pretty long on the consideration of duration, space, and number, and what arises

from the contemplation of them,—Infinity, it is possibly no more than the matter requires; there being few simple ideas whose modes give more exercise to the thoughts of men than those do. I pretend not to treat of them in their full latitude. It suffices to my design to show how the mind receives them, such as they are, from sensation and reflection; and how even the idea we have of infinity, how remote soever it may seem to be from any object of sense, or operation of our mind, has, nevertheless, as all our other ideas, its original there. Some mathematicians perhaps, of advanced speculations, may have other ways to introduce into their minds ideas of infinity. But this hinders not but that they themselves, as well as all other men, got the first ideas which they had of infinity from sensation and reflection, in the method we have here set down.

Chapter XVIII

Other Simple Modes

1. Other simple modes of simple ideas of sensation.

Though I have, in the foregoing chapters, shown how, from simple ideas taken in by sensation, the mind comes to extend itself even to infinity; which, however it may of all others seem most remote from any sensible perception, yet at last hath nothing in it but what is made out of simple ideas: received into the mind by the senses, and afterwards there put together, by the faculty the mind has to repeat its own ideas;—Though, I say, these might be instances enough of simple modes of the simple ideas of sensation, and suffice to show how the mind comes by them, yet I shall, for method's sake, though briefly, give an account of some few more, and then proceed to more complex ideas.

2. Simple modes of motion. To slide, roll, tumble, walk, creep, run, dance, leap, skip, and abundance of others that might be named, are words which are no sooner heard but every one who understands English has presently in his mind distinct ideas, which are all but the different modifications of motion. Modes of motion answer those of extension; swift and slow are two different ideas of motion, the measures whereof are made of

the distances of time and space put together; so they are complex ideas, comprehending time and space with motion.

3. Modes of sounds. The like variety have we in sounds. Every articulate word is a different modification of sound; by which we see that, from the sense of hearing, by such modifications, the mind may be furnished with distinct ideas, to almost an infinite number. Sounds also, besides the distinct cries of birds and beasts, are modified by diversity of notes of different length put together, which make that complex idea called a tune, which a musician may have in his mind when he hears or makes no sound at all, by reflecting on the ideas of those sounds, so put together silently in his own fancy.

4. Modes of colours. Those of colours are also very various: some we take notice of as the different degrees, or as they were termed shades, of the same colour. But since we very seldom make assemblages of colours, either for use or delight, but figure is taken in also, and has its part in it, as in painting, weaving, needleworks, &c.; those which are taken notice of do most commonly

belong to mixed modes, as being made up of ideas of divers kinds, viz. figure and colour, such as beauty, rainbow, &c.

5. Modes of tastes. All compounded tastes and smells are also modes, made up of the simple ideas of those senses. But they, being such as generally we have no names for, are less taken notice of, and cannot be set down in writing; and therefore must be left without enumeration to the thoughts and experience of my reader.

6. Some simple modes have no names. In general it may be observed, that those simple modes which are considered but as different degrees of the same simple idea, though they are in themselves many of them very distinct ideas, yet have ordinarily no distinct names, nor are much taken notice of, as distinct ideas, where the difference is but very small between them. Whether men have neglected these modes, and given no names to them, as wanting measures nicely to distinguish them; or because, when they were so distinguished, that knowledge would not be of general or necessary use, I leave it

to the thoughts of others. It is sufficient to my purpose to show, that all our simple ideas come to our minds only by sensation and reflection; and that when the mind has them, it can variously repeat and compound them, and so make new complex ideas. But, though white, red, or sweet, &c. have not been modified, or made into complex ideas, by several combinations, so as to be named, and thereby ranked into species; yet some others of the simple ideas, viz. those of unity, duration, and motion, &c., above instanced in, as also power and thinking, have been thus modified to a great variety of complex ideas, with names belonging to them.

7. Why some modes have, and others have not, names. The reason whereof, I suppose, has been this,—That the great concernment of men being with men one amongst another, the knowledge of men, and their actions, and the signifying of them to one another, was most necessary; and therefore they made ideas of actions very nicely modified, and gave those complex ideas names, that they might the more easily record and discourse of those things they were daily conversant in,

without long ambages and circumlocutions; and that the things they were continually to give and receive information about might be the easier and quicker understood. That this is so, and that men in framing different complex ideas, and giving them names, have been much governed by the end of speech in general, (which is a very short and expedite way of conveying their thoughts one to another), is evident in the names which in several arts have been found out, and applied to several complex ideas of modified actions, belonging to their several trades, for dispatch sake, in their direction or discourses about them. Which ideas are not generally framed in the minds of men not conversant about these operations. And thence the words that stand for them, by the greatest part of men of the same language, are not understood: v.g. coltshire, drilling, filtration, cohobation, are words standing for certain complex ideas, which being seldom in the minds of any but those few whose particular employments do at every turn suggest them to their thoughts, those names of them are not generally understood but by smiths and chymists; who,

having framed the complex ideas which these words stand for, and having given names to them, or received them from others, upon hearing of these names in communication, readily conceive those ideas in their minds;—as by cohobation all the simple ideas of distilling, and the pouring the liquor distilled from anything back upon the remaining matter, and distilling it again. Thus we see that there are great varieties of simple ideas, as of tastes and smells, which have no names; and of modes many more; which either not having been generally enough observed, or else not being of any great use to be taken notice of in the affairs and converse of men, they have not had names given to them, and so pass not for species. This we shall have occasion hereafter to consider more at large, when we come to speak of words.

Chapter XIX

Of the Modes of Thinking

1. Sensation, remembrance, contemplation, &c., modes of thinking. When the mind turns its view inwards upon

itself, and contemplates its own actions, thinking is the first that occurs. In it the mind observes a great variety of modifications, and from thence receives distinct ideas. Thus the perception or thought which actually accompanies, and is annexed to, any impression on the body, made by an external object, being distinct from all other modifications of thinking, furnishes the mind with a distinct idea, which we call sensation;—which is, as it were, the actual entrance of any idea into the understanding by the senses. The same idea, when it again recurs without the operation of the like object on the external sensory, is remembrance: if it be sought after by the mind, and with pain and endeavour found, and brought again in view, it is recollection: if it be held there long under attentive consideration, it is contemplation: when ideas float in our mind, without any reflection or regard of the understanding, it is that which the French call *reverie*; our language has scarce a name for it: when the ideas that offer themselves (for, as I have observed in another place, whilst we are awake, there will always be a train of ideas succeeding one an-

other in our minds) are taken notice of, and, as it were, registered in the memory, it is attention: when the mind with great earnestness, and of choice, fixes its view on any idea, considers it on all sides, and will not be called off by the ordinary solicitation of other ideas, it is that we call intention or study: sleep, without dreaming, is rest from all these: and dreaming itself is the having of ideas (whilst the outward senses are stopped, so that they receive not outward objects with their usual quickness) in the mind, not suggested by any external objects, or known occasion; nor under any choice or conduct of the understanding at all: and whether that which we call ecstasy be not dreaming with the eyes open, I leave to be examined.

2. Other modes of thinking. These are some few instances of those various modes of thinking, which the mind may observe in itself, and so have as distinct ideas of as it hath of white and red, a square or a circle. I do not pretend to enumerate them all, nor to treat at large of this set of ideas, which are got from reflection: that would be to make a volume. It suffices to my present

purpose to have shown here, by some few examples, of what sort these ideas are, and how the mind comes by them; especially since I shall have occasion hereafter to treat more at large of reasoning, judging, volition, and knowledge, which are some of the most considerable operations of the mind, and modes of thinking.

3. The various degrees of attention in thinking. But perhaps it may not be an unpardonable digression, nor wholly impertinent to our present design, if we reflect here upon the different state of the mind in thinking, which those instances of attention, reverie, and dreaming, &c., before mentioned, naturally enough suggest. That there are ideas, some or other, always present in the mind of a waking man, every one's experience convinces him; though the mind employs itself about them with several degrees of attention. Sometimes the mind fixes itself with so much earnestness on the contemplation of some objects, that it turns their ideas on all sides; marks their relations and circumstances; and views every part so nicely and with such intention, that it shuts out all other thoughts, and takes no notice of the

ordinary impressions made then on the senses, which at another season would produce very sensible perceptions: at other times it barely observes the train of ideas that succeed in the understanding, without directing and pursuing any of them: and at other times it lets them pass almost quite unregarded, as faint shadows that make no impression.

4. Hence it is probable that thinking is the action, not the essence of the soul. This difference of intention, and remission of the mind in thinking, with a great variety of degrees between earnest study and very near minding nothing at all, every one, I think, has experimented in himself. Trace it a little further, and you find the mind in sleep retired as it were from the senses, and out of the reach of those motions made on the organs of sense, which at other times produce very vivid and sensible ideas. I need not, for this, instance in those who sleep out whole stormy nights, without hearing the thunder, or seeing the lightning, or feeling the shaking of the house, which are sensible enough to those who are waking. But in this retirement of the mind from the

senses, it often retains a yet more loose and incoherent manner of thinking, which we call dreaming. And, last of all, sound sleep closes the scene quite, and puts an end to all appearances. This, I think almost every one has experience of in himself, and his own observation without difficulty leads him thus far. That which I would further conclude from hence is, that since the mind can sensibly put on, at several times, several degrees of thinking, and be sometimes, even in a waking man, so remiss, as to have thoughts dim and obscure to that degree that they are very little removed from none at all; and at last, in the dark retirements of sound sleep, loses the sight perfectly of all ideas whatsoever: since, I say, this is evidently so in matter of fact and constant experience, I ask whether it be not probable, that thinking is the action and not the essence of the soul? Since the operations of agents will easily admit of intention and remission: but the essences of things are not conceived capable of any such variation. But this by the by.

Chapter XX
Of Modes of Pleasure and Pain

1. Pleasure and pain, simple ideas. Amongst the simple ideas which we receive both from sensation and reflection, pain and pleasure are two very considerable ones. For as in the body there is sensation barely in itself, or accompanied with pain or pleasure, so the thought or perception of the mind is simply so, or else accompanied also with pleasure or pain, delight or trouble, call it how you please. These, like other simple ideas, cannot be described, nor their names defined; the way of knowing them is, as of the simple ideas of the senses, only by experience. For, to define them by the presence of good or evil, is no otherwise to make them known to us than by making us reflect on what we feel in ourselves, upon the several and various operations of good and evil upon our minds, as they are differently applied to or considered by us.

2. Good and evil, what. Things then are good or evil, only in reference to pleasure or pain. That we call good,

which is apt to cause or increase pleasure, or diminish pain in us; or else to procure or preserve us the possession of any other good or absence of any evil. And, on the contrary, we name that evil which is apt to produce or increase any pain, or diminish any pleasure in us: or else to procure us any evil, or deprive us of any good. By pleasure and pain, I must be understood to mean of body or mind, as they are commonly distinguished; though in truth they be only different constitutions of the mind, sometimes occasioned by disorder in the body, sometimes by thoughts of the mind.

3. Our passions moved by good and evil. Pleasure and pain and that which causes them,—good and evil, are the hinges on which our passions turn. And if we reflect on ourselves, and observe how these, under various considerations, operate in us; what modifications or tempers of mind, what internal sensations (if I may so call them) they produce in us we may thence form to ourselves the ideas of our passions.

4. Love. Thus any one reflecting upon the thought he has of the delight which any present or absent thing is

apt to produce in him, has the idea we call love. For when a man declares in autumn when he is eating them, or in spring when there are none, that he loves grapes, it is no more but that the taste of grapes delights him: let an alteration of health or constitution destroy the delight of their taste, and he then can be said to love grapes no longer.

5. Hatred. On the contrary, the thought of the pain which anything present or absent is apt to produce in us, is what we call hatred. Were it my business here to inquire any further than into the bare ideas of our passions, as they depend on different modifications of pleasure and pain, I should remark, that our love and hatred of inanimate insensible beings is commonly founded on that pleasure and pain which we receive from their use and application any way to our senses, though with their destruction. But hatred or love, to beings capable of happiness or misery, is often the uneasiness or delight which we find in ourselves, arising from a consideration of their very being or happiness. Thus the being and welfare of a man's children or friends, producing

constant delight in him, he is said constantly to love them. But it suffices to note, that our ideas of love and hatred are but the dispositions of the mind, in respect of pleasure and pain in general, however caused in us.

6. Desire. The uneasiness a man finds in himself upon the absence of anything whose present enjoyment carries the idea of delight with it, is that we call desire; which is greater or less, as that uneasiness is more or less vehement. Where, by the by, it may perhaps be of some use to remark, that the chief, if not only spur to human industry and action is uneasiness. For whatsoever good is proposed, if its absence carries no displeasure or pain with it, if a man be easy and content without it, there is no desire of it, nor endeavour after it; there is no more but a bare velleity, the term used to signify the lowest degree of desire, and that which is next to none at all, when there is so little uneasiness in the absence of anything, that it carries a man no further than some faint wishes for it, without any more effectual or vigorous use of the means to attain it. Desire also is stopped or abated by the opinion of the im-

possibility or unattainableness of the good proposed, as far as the uneasiness is cured or allayed by that consideration. This might carry our thoughts further, were it seasonable in this place.

7. Joy is a delight of the mind, from the consideration of the present or assured approaching possession of a good; and we are then possessed of any good, when we have it so in our power that we can use it when we please. Thus a man almost starved has joy at the arrival of relief, even before he has the pleasure of using it: and a father, in whom the very well-being of his children causes delight, is always, as long as his children are in such a state, in the possession of that good; for he needs but to reflect on it, to have that pleasure.

8. Sorrow is uneasiness in the mind, upon the thought of a good lost, which might have been enjoyed longer; or the sense of a present evil.

9. Hope is that pleasure in the mind, which every one finds in himself, upon the thought of a probable future enjoyment of a thing which is apt to delight him.

10. Fear is an uneasiness of the mind, upon the thought

of future evil likely to befall us.

11. Despair is the thought of the unattainableness of any good, which works differently in men's minds, sometimes producing uneasiness or pain, sometimes rest and indolency.

12. Anger is uneasiness or discomposure of the mind, upon the receipt of any injury, with a present purpose of revenge.

13. Envy is an uneasiness of the mind, caused by the consideration of a good we desire obtained by one we think should not have had it before us.

14. What passions all men have. These two last, envy and anger, not being caused by pain and pleasure simply in themselves, but having in them some mixed considerations of ourselves and others, are not therefore to be found in all men, because those other parts, of valuing their merits, or intending revenge, is wanting in them. But all the rest, terminating purely in pain and pleasure, are, I think, to be found in all men. For we love, desire, rejoice, and hope, only in respect of pleasure; we hate, fear, and grieve, only in respect of pain ultimately.

In fine, all these passions are moved by things, only as they appear to be the causes of pleasure and pain, or to have pleasure or pain some way or other annexed to them. Thus we extend our hatred usually to the subject (at least, if a sensible or voluntary agent) which has produced pain in us; because the fear it leaves is a constant pain: but we do not so constantly love what has done us good; because pleasure operates not so strongly on us as pain, and because we are not so ready to have hope it will do so again. But this by the by.

15. Pleasure and pain, what. By pleasure and pain, delight and uneasiness, I must all along be understood (as I have above intimated) to mean not only bodily pain and pleasure, but whatsoever delight or uneasiness is felt by us, whether arising from any grateful or unacceptable sensation or reflection.

16. Removal or lessening of either. It is further to be considered, that, in reference to the passions, the removal or lessening of a pain is considered, and operates, as a pleasure: and the loss or diminishing of a pleasure, as a pain.

17. Shame. The passions too have most of them, in most persons, operations on the body, and cause various changes in it; which not being always sensible, do not make a necessary part of the idea of each passion. For shame, which is an uneasiness of the mind upon the thought of having done something which is indecent, or will lessen the valued esteem which others have for us, has not always blushing accompanying it.

18. These instances to show how our ideas of the passions are got from sensation and reflection. I would not be mistaken here, as if I meant this as a Discourse of the Passions; they are many more than those I have here named: and those I have taken notice of would each of them require a much larger and more accurate discourse. I have only mentioned these here, as so many instances of modes of pleasure and pain resulting in our minds from various considerations of good and evil. I might perhaps have instanced in other modes of pleasure and pain, more simple than these; as the pain of hunger and thirst, and the pleasure of eating and drinking to remove them: the pain of teeth set on edge; the pleasure

of music; pain from captious uninstrusive wrangling, and the pleasure of rational conversation with a friend, or of well-directed study in the search and discovery of truth. But the passions being of much more concernment to us, I rather made choice to instance in them, and show how the ideas we have of them are derived from sensation or reflection.

Chapter XXI

Of Power

1. This idea how got. The mind being every day informed, by the senses, of the alteration of those simple ideas it observes in things without; and taking notice how one comes to an end, and ceases to be, and another begins to exist which was not before; reflecting also on what passes within itself, and observing a constant change of its ideas, sometimes by the impression of outward objects on the senses, and sometimes by the determination of its own choice; and concluding from what it has so constantly observed to have been, that the like

changes will for the future be made in the same things, by like agents, and by the like ways,—considers in one thing the possibility of having any of its simple ideas changed, and in another the possibility of making that change; and so comes by that idea which we call power. Thus we say, Fire has a power to melt gold, i.e. to destroy the consistency of its insensible parts, and consequently its hardness, and make it fluid; and gold has a power to be melted; that the sun has a power to blanch wax, and wax a power to be blanched by the sun, whereby the yellowness is destroyed, and whiteness made to exist in its room. In which, and the like cases, the power we consider is in reference to the change of perceivable ideas. For we cannot observe any alteration to be made in, or operation upon anything, but by the observable change of its sensible ideas; nor conceive any alteration to be made, but by conceiving a change of some of its ideas.

2. Power, active and passive. Power thus considered is two-fold, viz. as able to make, or able to receive any change. The one may be called active, and the other

passive power. Whether matter be not wholly destitute of active power, as its author, God, is truly above all passive power; and whether the intermediate state of created spirits be not that alone which is capable of both active and passive power, may be worth consideration. I shall not now enter into that inquiry, my present business being not to search into the original of power, but how we come by the idea of it. But since active powers make so great a part of our complex ideas of natural substances, (as we shall see hereafter,) and I mention them as such, according to common apprehension; yet they being not, perhaps, so truly active powers as our hasty thoughts are apt to represent them, I judge it not amiss, by this intimation, to direct our minds to the consideration of God and spirits, for the clearest idea of active power.

3. Power includes relation. I confess power includes in it some kind of relation, (a relation to action or change,) as indeed which of our ideas, of what kind soever, when attentively considered, does not? For, our ideas of extension, duration, and number, do they not all contain

in them a secret relation of the parts? Figure and motion have something relative in them much more visibly. And sensible qualities, as colours and smells, &c., what are they but the powers of different bodies, in relation to our perception, &c.? And, if considered in the things themselves, do they not depend on the bulk, figure, texture, and motion of the parts? All which include some kind of relation in them. Our idea therefore of power, I think, may well have a place amongst other simple ideas, and be considered as one of them; being one of those that make a principal ingredient in our complex ideas of substances, as we shall hereafter have occasion to observe.

4. The clearest idea of active power had from spirit. We are abundantly furnished with the idea of passive power by almost all sorts of sensible things. In most of them we cannot avoid observing their sensible qualities, nay, their very substances, to be in a continual flux. And therefore with reason we look on them as liable still to the same change. Nor have we of active power (which is the more proper signification of the word power) fewer

instances. Since whatever change is observed, the mind must collect a power somewhere able to make that change, as well as a possibility in the thing itself to receive it. But yet, if we will consider it attentively, bodies, by our senses, do not afford us so clear and distinct an idea of active power, as we have from reflection on the operations of our minds. For all power relating to action, and there being but two sorts of action whereof we have an idea, viz. thinking and motion, let us consider whence we have the clearest ideas of the powers which produce these actions. (1) Of thinking, body affords us no idea at all; it is only from reflection that we have that. (2) Neither have we from body any idea of the beginning of motion. A body at rest affords us no idea of any active power to move; and when it is set in motion itself, that motion is rather a passion than an action in it. For, when the ball obeys the motion of a billiard-stick, it is not any action of the ball, but bare passion. Also when by impulse it sets another ball in motion that lay in its way, it only communicates the motion it had received from another, and loses in itself

so much as the other received: which gives us but a very obscure idea of an active power of moving in body, whilst we observe it only to transfer, but not produce any motion. For it is but a very obscure idea of power which reaches not the production of the action, but the continuation of the passion. For so is motion in a body impelled by another; the continuation of the alteration made in it from rest to motion being little more an action, than the continuation of the alteration of its figure by the same blow is an action. The idea of the beginning of motion we have only from reflection on what passes in ourselves; where we find by experience, that, barely by willing it, barely by a thought of the mind, we can move the parts of our bodies, which were before at rest. So that it seems to me, we have, from the observation of the operation of bodies by our senses, but a very imperfect obscure idea of active power; since they afford us not any idea in themselves of the power to begin any action, either motion or thought. But if, from the impulse bodies are observed to make one upon another, any one thinks he has a clear idea of power, it serves as

well to my purpose; sensation being one of those ways whereby the mind comes by its ideas: only I thought it worth while to consider here, by the way, whether the mind doth not receive its idea of active power clearer from reflection on its own operations, than it doth from any external sensation.

5. Will and understanding two powers in mind or spirit. This, at least, I think evident,—That we find in ourselves a power to begin or forbear, continue or end several actions of our minds, and motions of our bodies, barely by a thought or preference of the mind ordering, or as it were commanding, the doing or not doing such or such a particular action. This power which the mind has thus to order the consideration of any idea, or the forbearing to consider it; or to prefer the motion of any part of the body to its rest, and vice versa, in any particular instance, is that which we call the Will. The actual exercise of that power, by directing any particular action, or its forbearance, is that which we call volition or willing. The forbearance of that action, consequent to such order or command of the mind, is called volun-

tary. And whatsoever action is performed without such a thought of the mind, is called involuntary. The power of perception is that which we call the Understanding. Perception, which we make the act of the understanding, is of three sorts:—1. The perception of ideas in our minds. 2. The perception of the signification of signs. 3. The perception of the connexion or repugnancy, agreement or disagreement, that there is between any of our ideas. All these are attributed to the understanding, or perceptive power, though it be the two latter only that use allows us to say we understand.

6. Faculties, not real beings. These powers of the mind, viz. of perceiving, and of preferring, are usually called by another name. And the ordinary way of speaking is, that the understanding and will are two faculties of the mind; a word proper enough, if it be used, as all words should be, so as not to breed any confusion in men's thoughts, by being supposed (as I suspect it has been) to stand for some real beings in the soul that performed those actions of understanding and volition. For when we say the will is the commanding and superior faculty

of the soul; that it is or is not free; that it determines the inferior faculties; that it follows the dictates of the understanding, &c.,—though these and the like expressions, by those that carefully attend to their own ideas, and conduct their thoughts more by the evidence of things than the sound of words, may be understood in a clear and distinct sense—yet I suspect, I say, that this way of speaking of faculties has misled many into a confused notion of so many distinct agents in us, which had their several provinces and authorities, and did command, obey, and perform several actions, as so many distinct beings; which has been no small occasion of wrangling, obscurity, and uncertainty, in questions relating to them.

7. Whence the ideas of liberty and necessity. Every one, I think, finds in himself a power to begin or forbear, continue or put an end to several actions in himself. From the consideration of the extent of this power of the mind over the actions of the man, which everyone finds in himself, arise the ideas of liberty and necessity.

8. Liberty, what. All the actions that we have any idea

of reducing themselves, as has been said, to these two, viz. thinking and motion; so far as a man has power to think or not to think, to move or not to move, according to the preference or direction of his own mind, so far is a man free. Wherever any performance or forbearance are not equally in a man's power; wherever doing or not doing will not equally follow upon the preference of his mind directing it, there he is not free, though perhaps the action may be voluntary. So that the idea of liberty is, the idea of a power in any agent to do or forbear any particular action, according to the determination or thought of the mind, whereby either of them is preferred to the other: where either of them is not in the power of the agent to be produced by him according to his volition, there he is not at liberty; that agent is under necessity. So that liberty cannot be where there is no thought, no volition, no will; but there may be thought, there may be will, there may be volition, where there is no liberty. A little consideration of an obvious instance or two may make this clear.

9. Supposes understanding and will. A tennis-ball,

whether in motion by the stroke of a racket, or lying still at rest, is not by any one taken to be a free agent. If we inquire into the reason, we shall find it is because we conceive not a tennis-ball to think, and consequently not to have any volition, or preference of motion to rest, or vice versa; and therefore has not liberty, is not a free agent; but all its both motion and rest come under our idea of necessary, and are so called. Likewise a man falling into the water, (a bridge breaking under him), has not herein liberty, is not a free agent. For though he has volition, though he prefers his not falling to falling; yet the forbearance of that motion not being in his power, the stop or cessation of that motion follows not upon his volition; and therefore therein he is not free. So a man striking himself, or his friend, by a convulsive motion of his arm, which it is not in his power, by volition or the direction of his mind, to stop or forbear, nobody thinks he has in this liberty; every one pities him, as acting by necessity and constraint.

10. Belongs not to volition. Again: suppose a man be carried, whilst fast asleep, into a room where is a person

he longs to see and speak with; and be there locked fast in, beyond his power to get out: he awakes, and is glad to find himself in so desirable company, which he stays willingly in, i.e. prefers his stay to going away. I ask, is not this stay voluntary? I think nobody will doubt it: and yet, being locked fast in, it is evident he is not at liberty not to stay, he has not freedom to be gone. So that liberty is not an idea belonging to volition, or preferring; but to the person having the power of doing, or forbearing to do, according as the mind shall choose or direct. Our idea of liberty reaches as far as that power, and no farther. For wherever restraint comes to check that power, or compulsion takes away that indifferency of ability to act, or to forbear acting, there liberty, and our notion of it, presently ceases.

11. Voluntary opposed to involuntary, not to necessary. We have instances enough, and often more than enough, in our own bodies. A man's heart beats, and the blood circulates, which it is not in his power by any thought or volition to stop; and therefore in respect of these motions, where rest depends not on his choice,

nor would follow the determination of his mind, if it should prefer it, he is not a free agent. Convulsive motions agitate his legs, so that though he wills it ever so much, he cannot by any power of his mind stop their motion, (as in that odd disease called chorea sancti viti), but he is perpetually dancing; he is not at liberty in this action, but under as much necessity of moving, as a stone that falls, or a tennis-ball struck with a racket. On the other side, a palsy or the stocks hinder his legs from obeying the determination of his mind, if it would thereby transfer his body to another place. In all these there is want of freedom; though the sitting still, even of a paralytic, whilst he prefers it to a removal, is truly voluntary. Voluntary, then, is not opposed to necessary, but to involuntary. For a man may prefer what he can do, to what he cannot do; the state he is in, to its absence or change; though necessity has made it in itself unalterable.

12. Liberty, what. As it is in the motions of the body, so it is in the thoughts of our minds: where any one is such, that we have power to take it up, or lay it by,

according to the preference of the mind, there we are at liberty. A waking man, being under the necessity of having some ideas constantly in his mind, is not at liberty to think or not to think; no more than he is at liberty, whether his body shall touch any other or no: but whether he will remove his contemplation from one idea to another is many times in his choice; and then he is, in respect of his ideas, as much at liberty as he is in respect of bodies he rests on; he can at pleasure remove himself from one to another. But yet some ideas to the mind, like some motions to the body, are such as in certain circumstances it cannot avoid, nor obtain their absence by the utmost effort it can use. A man on the rack is not at liberty to lay by the idea of pain, and divert himself with other contemplations: and sometimes a boisterous passion hurries our thoughts, as a hurricane does our bodies, without leaving us the liberty of thinking on other things, which we would rather choose. But as soon as the mind regains the power to stop or continue, begin or forbear, any of these motions of the body without, or thoughts within, according as

it thinks fit to prefer either to the other, we then consider the man as a free agent again.

13. Necessity, what. Wherever thought is wholly wanting, or the power to act or forbear according to the direction of thought, there necessity takes place. This, in an agent capable of volition, when the beginning or continuation of any action is contrary to that preference of his mind, is called compulsion; when the hindering or stopping any action is contrary to his volition, it is called restraint. Agents that have no thought, no volition at all, are in everything necessary agents.

14. Liberty belongs not to the will. If this be so, (as I imagine it is,) I leave it to be considered, whether it may not help to put an end to that long agitated, and, I think, unreasonable, because unintelligible question, viz. Whether man's will be free or no? For if I mistake not, it follows from what I have said, that the question itself is altogether improper; and it is as insignificant to ask whether man's will be free, as to ask whether his sleep be swift, or his virtue square: liberty being as little applicable to the will, as swiftness of motion is to

sleep, or squareness to virtue. Every one would laugh at the absurdity of such a question as either of these: because it is obvious that the modifications of motion belong not to sleep, nor the difference of figure to virtue; and when one well considers it, I think he will as plainly perceive that liberty, which is but a power, belongs only to agents, and cannot be an attribute or modification of the will, which is also but a power.

15. Volition. Such is the difficulty of explaining and giving clear notions of internal actions by sounds, that I must here warn my reader, that ordering, directing, choosing, preferring, &c., which I have made use of, will not distinctly enough express volition, unless he will reflect on what he himself does when he wills. For example, preferring, which seems perhaps best to express the act of volition, does it not precisely. For though a man would prefer flying to walking, yet who can say he ever wills it? Volition, it is plain, is an act of the mind knowingly exerting that dominion it takes itself to have over any part of the man, by employing it in, or withholding it from, any particular action. And what is the

will, but the faculty to do this? And is that faculty anything more in effect than a power; the power of the mind to determine its thought, to the producing, continuing, or stopping any action, as far as it depends on us? For can it be denied that whatever agent has a power to think on its own actions, and to prefer their doing or omission either to other, has that faculty called will? Will, then, is nothing but such a power. Liberty, on the other side, is the power a man has to do or forbear doing any particular action according as its doing or forbearance has the actual preference in the mind; which is the same thing as to say, according as he himself wills it.

16. Powers, belonging to agents. It is plain then that the will is nothing but one power or ability, and freedom another power or ability so that, to ask, whether the will has freedom, is to ask whether one power has another power, one ability another ability; a question at first sight too grossly absurd to make a dispute, or need an answer. For, who is it that sees not that powers belong only to agents, and are attributes only of sub-

stances, and not of powers themselves? So that this way of putting the question (viz. whether the will be free) is in effect to ask, whether the will be a substance, an agent, or at least to suppose it, since freedom can properly be attributed to nothing else. If freedom can with any propriety of speech be applied to power, it may be attributed to the power that is in a man to produce, or forbear producing, motion in parts of his body, by choice or preference; which is that which denominates him free, and is freedom itself. But if any one should ask, whether freedom were free, he would be suspected not to understand well what he said; and he would be thought to deserve Midas's ears, who, knowing that rich was a denomination for the possession of riches, should demand whether riches themselves were rich.

17. How the will, instead of the man, is called free. However, the name faculty, which men have given to this power called the will, and whereby they have been led into a way of talking of the will as acting, may, by an appropriation that disguises its true sense, serve a little to palliate the absurdity; yet the will, in truth, signifies

nothing but a power or ability to prefer or choose: and when the will, under the name of a faculty, is considered as it is, barely as an ability to do something, the absurdity in saying it is free, or not free, will easily discover itself. For, if it be reasonable to suppose and talk of faculties as distinct beings that can act, (as we do, when we say the will orders, and the will is free,) it is fit that we should make a speaking faculty, and a walking faculty, and a dancing faculty, by which these actions are produced, which are but several modes of motion; as well as we make the will and understanding to be faculties, by which the actions of choosing and perceiving are produced, which are but several modes of thinking. And we may as properly say that it is the singing faculty sings, and the dancing faculty dances, as that the will chooses, or that the understanding conceives; or, as is usual, that the will directs the understanding, or the understanding obeys or obeys not the will: it being altogether as proper and intelligible to say that the power of speaking directs the power of singing, or the power of singing obeys or disobeys the power of speaking.

18. This way of talking causes confusion of thought. This way of talking, nevertheless, has prevailed, and, as I guess, produced great confusion. For these being all different powers in the mind, or in the man, to do several actions, he exerts them as he thinks fit: but the power to do one action is not operated on by the power of doing another action. For the power of thinking operates not on the power of choosing, nor the power of choosing on the power of thinking; no more than the power of dancing operates on the power of singing, or the power of singing on the power of dancing, as any one who reflects on it will easily perceive. And yet this is it which we say when we thus speak, that the will operates on the understanding, or the understanding on the will.

19. Powers are relations, not agents. I grant, that this or that actual thought may be the occasion of volition, or exercising the power a man has to choose; or the actual choice of the mind, the cause of actual thinking on this or that thing: as the actual singing of such a tune may be the cause of dancing such a dance, and the

actual dancing of such a dance the occasion of singing such a tune. But in all these it is not one power that operates on another: but it is the mind that operates, and exerts these powers; it is the man that does the action; it is the agent that has power, or is able to do. For powers are relations, not agents: and that which has the power or not the power to operate, is that alone which is or is not free, and not the power itself. For freedom, or not freedom, can belong to nothing but what has or has not a power to act.

20. Liberty belongs not to the will. The attributing to faculties that which belonged not to them, has given occasion to this way of talking: but the introducing into discourses concerning the mind, with the name of faculties, a notion of their operating, has, I suppose, as little advanced our knowledge in that part of ourselves, as the great use and mention of the like invention of faculties, in the operations of the body, has helped us in the knowledge of physic. Not that I deny there are faculties, both in the body and mind: they both of them have their powers of operating, else neither the one nor

the other could operate. For nothing can operate that is not able to operate; and that is not able to operate that has no power to operate. Nor do I deny that those words, and the like, are to have their place in the common use of languages that have made them current. It looks like too much affectation wholly to lay them by: and philosophy itself, though it likes not a gaudy dress, yet, when it appears in public, must have so much complacency as to be clothed in the ordinary fashion and language of the country, so far as it can consist with truth and perspicuity. But the fault has been, that faculties have been spoken of and represented as so many distinct agents. For, it being asked, what it was that digested the meat in our stomachs? it was a ready and very satisfactory answer to say, that it was the digestive faculty. What was it that made anything come out of the body? the expulsive faculty. What moved? the motive faculty. And so in the mind, the intellectual faculty, or the understanding, understood; and the elective faculty, or the will, willed or commanded. This is, in short, to say, that the ability to digest, digested; and

the ability to move, moved; and the ability to understand, understood. For faculty, ability, and power, I think, are but different names of the same things: which ways of speaking, when put into more intelligible words, will, I think, amount to thus much;—That digestion is performed by something that is able to digest, motion by something able to move, and understanding by something able to understand. And, in truth, it would be very strange if it should be otherwise; as strange as it would be for a man to be free without being able to be free.

21. But to the agent, or man. To return, then, to the inquiry about liberty, I think the question is not proper, whether the will be free, but whether a man be free. Thus, I think, First, That so far as any one can, by the direction or choice of his mind, preferring the existence of any action to the non-existence of that action, and vice versa, make it to exist or not exist, so far he is free. For if I can, by a thought directing the motion of my finger, make it move when it was at rest, or vice versa, it is evident, that in respect of that I am free: and if I

can, by a like thought of my mind, preferring one to the other, produce either words or silence, I am at liberty to speak or hold my peace: and as far as this power reaches, of acting or not acting, by the determination of his own thought preferring either, so far is a man free. For how can we think any one freer, than to have the power to do what he will? And so far as any one can, by preferring any action to its not being, or rest to any action, produce that action or rest, so far can he do what he will. For such a preferring of action to its absence, is the willing of it: and we can scarce tell how to imagine any being freer, than to be able to do what he wills. So that in respect of actions within the reach of such a power in him, a man seems as free as it is possible for freedom to make him.

22. In respect of willing, a man is not free. But the inquisitive mind of man, willing to shift off from himself, as far as he can, all thoughts of guilt, though it be by putting himself into a worse state than that of fatal necessity, is not content with this: freedom, unless it reaches further than this, will not serve the turn: and it

passes for a good plea, that a man is not free at all, if he be not as free to will as he is to act what he wills. Concerning a man's liberty, there yet, therefore, is raised this further question, Whether a man be free to will? Which I think is what is meant, when it is disputed whether the will be free. And as to that I imagine.

23. How a man cannot be free to will. Secondly, That willing, or volition, being an action, and freedom consisting in a power of acting or not acting, a man in respect of willing or the act of volition, when any action in his power is once proposed to his thoughts, as presently to be done, cannot be free. The reason whereof is very manifest. For, it being unavoidable that the action depending on his will should exist or not exist, and its existence or not existence following perfectly the determination and preference of his will, he cannot avoid willing the existence or non-existence of that action; it is absolutely necessary that he will the one or the other; i.e. prefer the one to the other: since one of them must necessarily follow; and that which does follow follows by the choice and determination of his mind; that is, by

his willing it: for if he did not will it, it would not be. So that, in respect of the act of willing, a man in such a case is not free: liberty consisting in a power to act or not to act; which, in regard of volition, a man, upon such a proposal has not. For it is unavoidably necessary to prefer the doing or forbearance of an action in a man's power, which is once so proposed to his thoughts; a man must necessarily will the one or the other of them; upon which preference or volition, the action or its forbearance certainly follows, and is truly voluntary. But the act of volition, or preferring one of the two, being that which he cannot avoid, a man, in respect of that act of willing, is under a necessity, and so cannot be free; unless necessity and freedom can consist together, and a man can be free and bound at once. Besides to make a man free after this manner, by making the action of willing to depend on his will, there must be another antecedent will, to determine the acts of this will, and another to determine that, and so in infinitum: for wherever one stops, the actions of the last will cannot be free. Nor is any being, as far I can com-

prehend beings above me, capable of such a freedom of will, that it can forbear to will, i.e. to prefer the being or not being of anything in its power, which it has once considered as such.

24. Liberty is freedom to execute what is willed. This, then, is evident, That a man is not at liberty to will, or not to will, anything in his power that he once considers of: liberty consisting in a power to act or to forbear acting, and in that only. For a man that sits still is said yet to be at liberty; because he can walk if he wills it. A man that walks is at liberty also, not because he walks or moves; but because he can stand still if he wills it. But if a man sitting still has not a power to remove himself, he is not at liberty; so likewise a man falling down a precipice, though in motion, is not at liberty, because he cannot stop that motion if he would. This being so, it is plain that a man that is walking, to whom it is proposed to give off walking, is not at liberty, whether he will determine himself to walk, or give off walking or not: he must necessarily prefer one or the other of them; walking or not walking. And so it is in

regard of all other actions in our power so proposed, which are the far greater number. For, considering the vast number of voluntary actions that succeed one another every moment that we are awake in the course of our lives, there are but few of them that are thought on or proposed to the will, till the time they are to be done; and in all such actions, as I have shown, the mind, in respect of willing, has not a power to act or not to act, wherein consists liberty. The mind, in that case, has not a power to forbear willing; it cannot avoid some determination concerning them, let the consideration be as short, the thought as quick as it will, it either leaves the man in the state he was before thinking, or changes it; continues the action, or puts an end to it. Whereby it is manifest, that it orders and directs one, in preference to, or with neglect of the other, and thereby either the continuation or change becomes unavoidably voluntary.

25. The will determined by something without it. Since then it is plain that, in most cases, a man is not at liberty, whether he will or no, (for, when an action in

his power is proposed to his thoughts, he cannot forbear volition; he must determine one way or the other); the next thing demanded is,—Whether a man be at liberty to will which of the two he pleases, motion or rest? This question carries the absurdity of it so manifestly in itself, that one might thereby sufficiently be convinced that liberty concerns not the will. For, to ask whether a man be at liberty to will either motion or rest, speaking or silence, which he pleases, is to ask whether a man can will what he wills, or be pleased with what he is pleased with? A question which, I think, needs no answer: and they who can make a question of it must suppose one will to determine the acts of another, and another to determine that, and so on in infinitum.

26. The ideas of liberty and volition must be defined. To avoid these and the like absurdities, nothing can be of greater use than to establish in our minds determined ideas of the things under consideration. If the ideas of liberty and volition were well fixed in our understandings, and carried along with us in our minds, as they ought, through all the questions that are raised

about them, I suppose a great part of the difficulties that perplex men's thoughts, and entangle their understandings, would be much easier resolved; and we should perceive where the confused signification of terms, or where the nature of the thing caused the obscurity.

27. Freedom. First, then, it is carefully to be remembered, That freedom consists in the dependence of the existence, or not existence of any action, upon our volition of it; and not in the dependence of any action, or its contrary, on our preference. A man standing on a cliff, is at liberty to leap twenty yards downwards into the sea, not because he has a power to do the contrary action, which is to leap twenty yards upwards, for that he cannot do; but he is therefore free, because he has a power to leap or not to leap. But if a greater force than his, either holds him fast, or tumbles him down, he is no longer free in that case; because the doing or forbearance of that particular action is no longer in his power. He that is a close prisoner in a room twenty feet square, being at the north side of his chamber, is at liberty to walk twenty feet southward, because he can

walk or not walk it; but is not, at the same time, at liberty to do the contrary, i.e. to walk twenty feet northward.

In this, then, consists freedom, viz. in our being able to act or not to act, according as we shall choose or will.

28. What volition and action mean. Secondly, we must remember, that volition or willing is an act of the mind directing its thought to the production of any action, and thereby exerting its power to produce it. To avoid multiplying of words, I would crave leave here, under the word action, to comprehend the forbearance too of any action proposed: sitting still, or holding one's peace, when walking or speaking are proposed, though mere forbearances, requiring as much the determination of the will, and being as often weighty in their consequences, as the contrary actions, may, on that consideration, well enough pass for actions too: but this I say, that I may not be mistaken, if (for brevity's sake) I speak thus.

29. What determines the will. Thirdly, the will being nothing but a power in the mind to direct the operative

faculties of a man to motion or rest, as far as they depend on such direction; to the question, What is it determines the will? the true and proper answer is, The mind. For that which determines the general power of directing, to this or that particular direction, is nothing but the agent itself exercising the power it has that particular way. If this answer satisfies not, it is plain the meaning of the question, What determines the will? is this,—What moves the mind, in every particular instance, to determine its general power of directing, to this or that particular motion or rest? And to this I answer,—The motive for continuing in the same state or action, is only the present satisfaction in it; the motive to change is always some uneasiness: nothing setting us upon the change of state, or upon any new action, but some uneasiness. This is the great motive that works on the mind to put it upon action, which for shortness' sake we will call determining of the will, which I shall more at large explain.

30. Will and desire must not be confounded. But, in the way to it, it will be necessary to premise, that, though

I have above endeavoured to express the act of volition, by choosing, preferring, and the like terms, that signify desire as well as volition, for want of other words to mark that act of the mind whose proper name is willing or volition; yet, it being a very simple act, whosoever desires to understand what it is, will better find it by reflecting on his own mind, and observing what it does when it wills, than by any variety of articulate sounds whatsoever. This caution of being careful not to be misled by expressions that do not enough keep up the difference between the will and several acts of the mind that are quite distinct from it, I think the more necessary, because I find the will often confounded with several of the affections, especially desire, and one put for the other; and that by men who would not willingly be thought not to have had very distinct notions of things, and not to have writ very clearly about them. This, I imagine, has been no small occasion of obscurity and mistake in this matter; and therefore is, as much as may be, to be avoided. For he that shall turn his thoughts inwards upon what passes in his mind when he wills,

shall see that the will or power of volition is conversant about nothing but our own actions; terminates there; and reaches no further; and that volition is nothing but that particular determination of the mind, whereby, barely by a thought the mind endeavours to give rise, continuation, or stop, to any action which it takes to be in its power. This, well considered, plainly shows that the will is perfectly distinguished from desire; which, in the very same action, may have a quite contrary tendency from that which our will sets us upon. A man, whom I cannot deny, may oblige me to use persuasions to another, which, at the same time I am speaking, I may wish may not prevail on him. In this case, it is plain the will and desire run counter. I will the action; that tends one way, whilst my desire tends another, and that the direct contrary way. A man who, by a violent fit of the gout in his limbs, finds a doziness in his head, or a want of appetite in his stomach removed, desires to be eased too of the pain of his feet or hands, (for wherever there is pain, there is a desire to be rid of it), though yet, whilst he apprehends that the removal of

the pain may translate the noxious humour to a more vital part, his will is never determined to any one action that may serve to remove this pain. Whence it is evident that desiring and willing are two distinct acts of the mind; and consequently, that the will, which is but the power of volition, is much more distinct from desire.

31. Uneasiness determines the will. To return, then, to the inquiry, what is it that determines the will in regard to our actions? And that, upon second thoughts, I am apt to imagine is not, as is generally supposed, the greater good in view; but some (and for the most part the most pressing) uneasiness a man is at present under. This is that which successively determines the will, and sets us upon those actions we perform. This uneasiness we may call, as it is, desire; which is an uneasiness of the mind for want of some absent good. All pain of the body, of what sort soever, and disquiet of the mind, is uneasiness: and with this is always joined desire, equal to the pain or uneasiness felt; and is scarce distinguishable from it. For desire being nothing but an uneasiness in the want of an absent good, in reference to any pain felt,

ease is that absent good; and till that ease be attained, we may call it desire; nobody feeling pain that he wishes not to be eased of, with a desire equal to that pain, and inseparable from it. Besides this desire of ease from pain, there is another of absent positive good; and here also the desire and uneasiness are equal. As much as we desire any absent good, so much are we in pain for it. But here all absent good does not, according to the greatness it has, or is acknowledged to have, cause pain equal to that greatness; as all pain causes desire equal to itself: because the absence of good is not always a pain, as the presence of pain is. And therefore absent good may be looked on and considered without desire. But so much as there is anywhere of desire, so much there is of uneasiness.

32. Desire is uneasiness. That desire is a state of uneasiness, every one who reflects on himself will quickly find. Who is there that has not felt in desire what the wise man says of hope, (which is not much different from it), that it being “deferred makes the heart sick”; and that still proportionable to the greatness of the desire,

which sometimes raises the uneasiness to that pitch, that it makes people cry out, "Give me children." give me the thing desired, "or I die." Life itself, and all its enjoyments, is a burden cannot be borne under the lasting and unremoved pressure of such an uneasiness.

33. The uneasiness of desire determines the will. Good and evil, present and absent, it is true, work upon the mind. But that which immediately determines the will, from time to time, to every voluntary action, is the uneasiness of desire, fixed on some absent good: either negative, as indolence to one in pain; or positive, as enjoyment of pleasure. That it is this uneasiness that determines the will to the successive voluntary actions, whereof the greatest part of our lives is made up, and by which we are conducted through different courses to different ends, I shall endeavour to show, both from experience, and the reason of the thing.

34. This is the spring of action. When a man is perfectly content with the state he is in—which is when he is perfectly without any uneasiness—what industry, what action, what will is there left, but to continue in

it? Of this every man's observation will satisfy him. And thus we see our all-wise Maker, suitably to our constitution and frame, and knowing what it is that determines the will, has put into man the uneasiness of hunger and thirst, and other natural desires, that return at their seasons, to move and determine their wills, for the preservation of themselves, and the continuation of their species. For I think we may conclude, that, if the bare contemplation of these good ends to which we are carried by these several uneasinesses had been sufficient to determine the will, and set us on work, we should have had none of these natural pains, and perhaps in this world little or no pain at all. "It is better to marry than to burn," says St. Paul, where we may see what it is that chiefly drives men into the enjoyments of a conjugal life. A little burning felt pushes us more powerfully than greater pleasures in prospect draw or allure.

35. The greatest positive good determines not the will, but present uneasiness alone. It seems so established and settled a maxim, by the general consent of all mankind, that good, the greater good, determines the will,

that I do not at all wonder that, when I first published my thoughts on this subject I took it for granted; and I imagine that, by a great many, I shall be thought more excusable for having then done so, than that now I have ventured to recede from so received an opinion. But yet, upon a stricter inquiry, I am forced to conclude that good, the greater good, though apprehended and acknowledged to be so, does not determine the will, until our desire, raised proportionably to it, makes us uneasy in the want of it. Convince a man never so much, that plenty has its advantages over poverty; make him see and own, that the handsome conveniences of life are better than nasty penury: yet, as long as he is content with the latter, and finds no uneasiness in it, he moves not; his will never is determined to any action that shall bring him out of it. Let a man be ever so well persuaded of the advantages of virtue, that it is as necessary to a man who has any great aims in this world, or hopes in the next, as food to life: yet, till he hungers or thirsts after righteousness, till he feels an uneasiness in the want of it, his will will not be determined to any

action in pursuit of this confessed greater good; but any other uneasiness he feels in himself shall take place, and carry his will to other actions. On the other side, let a drunkard see that his health decays, his estate wastes; discredit and diseases, and the want of all things, even of his beloved drink, attends him in the course he follows: yet the returns of uneasiness to miss his companions, the habitual thirst after his cups at the usual time, drives him to the tavern, though he has in his view the loss of health and plenty, and perhaps of the joys of another life: the least of which is no inconsiderable good, but such as he confesses is far greater than the tickling of his palate with a glass of wine, or the idle chat of a soaking club. It is not want of viewing the greater good; for he sees and acknowledges it, and, in the intervals of his drinking hours, will take resolutions to pursue the greater good; but when the uneasiness to miss his accustomed delight returns, the great acknowledged good loses its hold, and the present uneasiness determines the will to the accustomed action; which thereby gets stronger footing to prevail against the next occasion,

though he at the same time makes secret promises to himself that he will do so no more; this is the last time he will act against the attainment of those greater goods. And thus he is, from time to time, in the state of that unhappy complainer, *Video meliora, proboque, deteriora sequor*: which sentence, allowed for true, and made good by constant experience, may in this, and possibly no other way, be easily made intelligible.

36. Because the removal of uneasiness is the first step to happiness. If we inquire into the reason of what experience makes so evident in fact, and examine, why it is uneasiness alone operates on the will, and determines it in its choice, we shall find that, we being capable but of one determination of the will to one action at once, the present uneasiness that we are under does naturally determine the will, in order to that happiness which we all aim at in all our actions. For, as much as whilst we are under any uneasiness, we cannot apprehend ourselves happy, or in the way to it; pain and uneasiness being, by every one, concluded and felt to be inconsistent with happiness, spoiling the relish even of those

good things which we have: a little pain serving to mar all the pleasure we rejoiced in. And, therefore, that which of course determines the choice of our will to the next action will always be—the removing of pain, as long as we have any left, as the first and necessary step towards happiness.

37. Because uneasiness alone is present. Another reason why it is uneasiness alone determines the will, is this: because that alone is present and, it is against the nature of things, that what is absent should operate where it is not. It may be said that absent good may, by contemplation, be brought home to the mind and made present. The idea of it indeed may be in the mind, and viewed as present there; but nothing will be in the mind as a present good, able to counterbalance the removal of any uneasiness which we are under, till it raises our desire; and the uneasiness of that has the prevalency in determining the will. Till then, the idea in the mind of whatever is good is there only, like other ideas, the object of bare unactive speculation; but operates not on the will, nor sets us on work; the reason whereof I shall

show by and by. How many are to be found that have had lively representations set before their minds of the unspeakable joys of heaven, which they acknowledge both possible and probable too, who yet would be content to take up with their happiness here? And so the prevailing uneasiness of their desires, let loose after the enjoyments of this life, take their turns in the determining their wills; and all that while they take not one step, are not one jot moved, towards the good things of another life, considered as ever so great.

38. Because all who allow the joys of heaven possible, pursue them not. Were the will determined by the views of good, as it appears in contemplation greater or less to the understanding, which is the state of all absent good, and that which, in the received opinion, the will is supposed to move to, and to be moved by,—I do not see how it could ever get loose from the infinite eternal joys of heaven, once proposed and considered as possible. For, all absent good, by which alone, barely proposed, and coming in view, the will is thought to be determined, and so to set us on action, being only possible,

but not infallibly certain, it is unavoidable that the infinitely greater possible good should regularly and constantly determine the will in all the successive actions it directs; and then we should keep constantly and steadily in our course towards heaven, without ever standing still, or directing our actions to any other end: the eternal condition of a future state infinitely outweighing the expectation of riches, or honour, or any other worldly pleasure which we can propose to ourselves, though we should grant these the more probable to be obtained: for nothing future is yet in possession, and so the expectation even of these may deceive us. If it were so that the greater good in view determines the will, so great a good, once proposed, could not but seize the will, and hold it fast to the pursuit of this infinitely greatest good, without ever letting it go again: for the will having a power over, and directing the thoughts, as well as other actions, would, if it were so, hold the contemplation of the mind fixed to that good.

39. But any great uneasiness is never neglected. This would be the state of the mind, and regular tendency of

the will in all its determinations, were it determined by that which is considered and in view the greater good. But that it is not so, is visible in experience; the infinitely greatest confessed good being often neglected, to satisfy the successive uneasiness of our desires pursuing trifles. But, though the greatest allowed, even everlasting unspeakable, good, which has sometimes moved and affected the mind, does not stedfastly hold the will, yet we see any very great and prevailing uneasiness having once laid hold on the will, let it not go; by which we may be convinced, what it is that determines the will. Thus any vehement pain of the body; the ungovernable passion of a man violently in love; or the impatient desire of revenge, keeps the will steady and intent; and the will, thus determined, never lets the understanding lay by the object, but all the thoughts of the mind and powers of the body are uninterruptedly employed that way, by the determination of the will, influenced by that topping uneasiness, as long as it lasts; whereby it seems to me evident, that the will, or power of setting us upon one action in preference to all others, is deter-

mined in us by uneasiness: and whether this be not so, I desire every one to observe in himself.

40. Desire accompanies all uneasiness. I have hitherto chiefly instanced in the uneasiness of desire, as that which determines the will: because that is the chief and most sensible; and the will seldom orders any action, nor is there any voluntary action performed, without some desire accompanying it; which I think is the reason why the will and desire are so often confounded. But yet we are not to look upon the uneasiness which makes up, or at least accompanies, most of the other passions, as wholly excluded in the case. Aversion, fear, anger, envy, shame, &c. have each their uneasinesses too, and thereby influence the will. These passions are scarce any of them, in life and practice, simple and alone, and wholly unmixed with others; though usually, in discourse and contemplation, that carries the name which operates strongest, and appears most in the present state of the mind. Nay, there is, I think, scarce any of the passions to be found without desire joined with it. I am sure wherever there is uneasiness, there is desire. For

we constantly desire happiness; and whatever we feel of uneasiness, so much it is certain we want of happiness; even in our own opinion, let our state and condition otherwise be what it will. Besides, the present moment not being our eternity, whatever our enjoyment be, we look beyond the present, and desire goes with our foresight, and that still carries the will with it. So that even in joy itself, that which keeps up the action whereon the enjoyment depends, is the desire to continue it, and fear to lose it: and whenever a greater uneasiness than that takes place in the mind, the will presently is by that determined to some new action, and the present delight neglected.

41. The most pressing uneasiness naturally determines the will. But we being in this world beset with sundry uneasinesses, distracted with different desires, the next inquiry naturally will be,—Which of them has the precedence in determining the will to the next action? and to that the answer is,—That ordinarily which is the most pressing of those that are judged capable of being then removed. For, the will being the power of directing our

operative faculties to some action, for some end, cannot at any time be moved towards what is judged at that time unattainable: that would be to suppose an intelligent being designedly to act for an end, only to lose its labour; for so it is to act for what is judged not attainable; and therefore very great uneasinesses move not the will, when they are judged not capable of a cure: they in that case put us not upon endeavours. But, these set apart, the most important and urgent uneasiness we at that time feel, is that which ordinarily determines the will, successively, in that train of voluntary actions which makes up our lives. The greatest present uneasiness is the spur to action, that is constantly most felt, and for the most part determines the will in its choice of the next action. For this we must carry along with us, that the proper and only object of the will is some action of ours, and nothing else. For we producing nothing by our willing it, but some action in our power, it is there the will terminates, and reaches no further.

42. All desire happiness. If it be further asked,—What it is moves desire? I answer,—happiness, and that alone.

Happiness and misery are the names of two extremes, the utmost bounds whereof we know not; it is what "eye hath not seen, ear hath not heard, nor hath it entered into the heart of man to conceive." But of some degrees of both we have very lively impressions; made by several instances of delight and joy on the one side, and torment and sorrow on the other; which, for shortness' sake, I shall comprehend under the names of pleasure and pain; there being pleasure and pain of the mind as well as the body,—"With him is fulness of joy, and pleasure for evermore." Or, to speak truly, they are all of the mind; though some have their rise in the mind from thought, others in the body from certain modifications of motion.

43. Happiness and misery, good and evil, what they are. Happiness, then, in its full extent, is the utmost pleasure we are capable of, and misery the utmost pain; and the lowest degree of what can be called happiness is so much ease from all pain, and so much present pleasure, as without which any one cannot be content. Now, because pleasure and pain are produced in us by the

operation of certain objects, either on our minds or our bodies, and in different degrees; therefore, what has an aptness to produce pleasure in us is that we call good, and what is apt to produce pain in us we call evil; for no other reason but for its aptness to produce pleasure and pain in us, wherein consists our happiness and misery. Further, though what is apt to produce any degree of pleasure be in itself good; and what is apt to produce any degree of pain be evil; yet it often happens that we do not call it so when it comes in competition with a greater of its sort; because, when they come in competition, the degrees also of pleasure and pain have justly a preference. So that if we will rightly estimate what we call good and evil, we shall find it lies much in comparison: for the cause of every less degree of pain, as well as every greater degree of pleasure, has the nature of good, and vice versa.

44. What good is desired, what not. Though this be that which is called good and evil, and all good be the proper object of desire in general; yet all good, even seen and confessed to be so, does not necessarily move

every particular man's desire; but only that part, or so much of it as is considered and taken to make a necessary part of his happiness. All other good, however great in reality or appearance, excites not a man's desires who looks not on it to make a part of that happiness where-with he, in his present thoughts, can satisfy himself. Happiness, under this view, every one constantly pursues, and desires what makes any part of it: other things, acknowledged to be good, he can look upon without desire, pass by, and be content without. There is nobody, I think, so senseless as to deny that there is pleasure in knowledge: and for the pleasures of sense, they have too many followers to let it be questioned whether men are taken with them or no. Now, let one man place his satisfaction in sensual pleasures, another in the delight of knowledge: though each of them cannot but confess, there is great pleasure in what the other pursues; yet, neither of them making the other's delight a part of his happiness, their desires are not moved, but each is satisfied without what the other enjoys; and so his will is not determined to the pursuit of it. But yet,

as soon as the studious man's hunger and thirst make him uneasy, he, whose will was never determined to any pursuit of good cheer, poignant sauces, delicious wine, by the pleasant taste he has found in them, is, by the uneasiness of hunger and thirst, presently determined to eating and drinking, though possibly with great indifferency, what wholesome food comes in his way. And, on the other side, the epicure buckles to study, when shame, or the desire to recommend himself to his mistress, shall make him uneasy in the want of any sort of knowledge. Thus, how much soever men are in earnest and constant in pursuit of happiness, yet they may have a clear view of good, great and confessed good, without being concerned for it, or moved by it, if they think they can make up their happiness without it. Though as to pain, that they are always concerned for; they can feel no uneasiness without being moved. And therefore, being uneasy in the want of whatever is judged necessary to their happiness, as soon as any good appears to make a part of their portion of happiness, they begin to desire it.

45. Why the greatest good is not always desired. This, I think, any one may observe in himself and others,—That the greater visible good does not always raise men's desires in proportion to the greatness it appears, and is acknowledged, to have: though every little trouble moves us, and sets us on work to get rid of it. The reason whereof is evident from the nature of our happiness and misery itself. All present pain, whatever it be, makes a part of our present misery. but all absent good does not at any time make a necessary part of our present happiness, nor the absence of it make a part of our misery. If it did, we should be constantly and infinitely miserable; there being infinite degrees of happiness which are not in our possession. All uneasiness therefore being removed, a moderate portion of good serves at present to content men; and a few degrees of pleasure, in a succession of ordinary enjoyments, make up a happiness wherein they can be satisfied. If this were not so, there could be no room for those indifferent and visibly trifling actions, to which our wills are so often determined, and wherein we voluntarily waste so much of our lives; which re-

missness could by no means consist with a constant determination of will or desire to the greatest apparent good. That this is so, I think few people need go far from home to be convinced. And indeed in this life there are not many whose happiness reaches so far as to afford them a constant train of moderate mean pleasures, without any mixture of uneasiness; and yet they could be content to stay here for ever: though they cannot deny, but that it is possible there may be a state of eternal durable joys after this life, far surpassing all the good that is to be found here. Nay, they cannot but see that it is more possible than the attainment and continuation of that pittance of honour, riches, or pleasure which they pursue, and for which they neglect that eternal state. But yet, in full view of this difference, satisfied of the possibility of a perfect, secure, and lasting happiness in a future state, and under a clear conviction that it is not to be had here,—whilst they bound their happiness within some little enjoyment or aim of this life, and exclude the joys of heaven from making any necessary part of it,—their desires are not moved

by this greater apparent good, nor their wills determined to any action, or endeavour for its attainment.

46. Why not being desired, it moves not the will. The ordinary necessities of our lives fill a great part of them with the uneasinesses of hunger, thirst, heat, cold, weariness, with labour, and sleepiness, in their constant returns, &c. To which, if, besides accidental harms, we add the fantastical uneasiness (as itch after honour, power, or riches, &c.) which acquired habits, by fashion, example, and education, have settled in us, and a thousand other irregular desires, which custom has made natural to us, we shall find that a very little part of our life is so vacant from these uneasinesses, as to leave us free to the attraction of remoter absent good. We are seldom at ease, and free enough from the solicitation of our natural or adopted desires, but a constant succession of uneasinesses out of that stock which natural wants or acquired habits have heaped up, take the will in their turns; and no sooner is one action dispatched, which by such a determination of the will we are set upon, but another uneasiness is ready to set us on work.

For, the removing of the pains we feel, and are at present pressed with, being the getting out of misery, and consequently the first thing to be done in order to happiness,—absent good, though thought on, confessed, and appearing to be good, not making any part of this unhappiness in its absence, is justled out, to make way for the removal of those uneasinesses we feel; till due and repeated contemplation has brought it nearer to our mind, given some relish of it, and raised in us some desire: which then beginning to make a part of our present uneasiness, stands upon fair terms with the rest to be satisfied, and so, according to its greatness and pressure, comes in its turn to determine the will.

47. Due consideration raises desire. And thus, by a due consideration, and examining any good proposed, it is in our power to raise our desires in a due proportion to the value of that good, whereby in its turn and place it may come to work upon the will, and be pursued. For good, though appearing and allowed ever so great, yet till it has raised desires in our minds, and thereby made us uneasy in its want, it reaches not our wills; we are

not within the sphere of its activity, our wills being under the determination only of those uneasinesses which are present to us, which (whilst we have any) are always soliciting, and ready at hand to give the will its next determination. The balancing, when there is any in the mind, being only, which desire shall be next satisfied, which uneasiness first removed. Whereby it comes to pass that, as long as any uneasiness, any desire, remains in our mind, there is no room for good, barely as such, to come at the will, or at all to determine it. Because, as has been said, the first step in our endeavours after happiness being to get wholly out of the confines of misery, and to feel no part of it, the will can be at leisure for nothing else, till every uneasiness we feel be perfectly removed. which, in the multitude of wants and desires we are beset with in this imperfect state, we are not like to be ever freed from in this world.

48. The power to suspend the prosecution of any desire makes way for consideration. There being in us a great many uneasinesses, always soliciting and ready to determine the will, it is natural, as I have said, that the

greatest and most pressing should determine the will to the next action; and so it does for the most part, but not always. For, the mind having in most cases, as is evident in experience, a power to suspend the execution and satisfaction of any of its desires; and so all, one after another; is at liberty to consider the objects of them, examine them on all sides, and weigh them with others. In this lies the liberty man has; and from the not using of it right comes all that variety of mistakes, errors, and faults which we run into in the conduct of our lives, and our endeavours after happiness; whilst we precipitate the determination of our wills, and engage too soon, before due examination. To prevent this, we have a power to suspend the prosecution of this or that desire; as every one daily may experiment in himself. This seems to me the source of all liberty; in this seems to consist that which is (as I think improperly) called free-will. For, during this suspension of any desire, before the will be determined to action, and the action (which follows that determination) done, we have opportunity to examine, view, and judge of the good or

evil of what we are going to do; and when, upon due examination, we have judged, we have done our duty, all that we can, or ought to do, in pursuit of our happiness; and it is not a fault, but a perfection of our nature, to desire, will, and act according to the last result of a fair examination.

49. To be determined by our own judgment, is no restraint to liberty. This is so far from being a restraint or diminution of freedom, that it is the very improvement and benefit of it; it is not an abridgment, it is the end and use of our liberty; and the further we are removed from such a determination, the nearer we are to misery and slavery. A perfect indifference in the mind, not determinable by its last judgment of the good or evil that is thought to attend its choice, would be so far from being an advantage and excellency of any intellectual nature, that it would be as great an imperfection, as the want of indifference. To act, or not to act, till determined by the will, would be an imperfection on the other side. A man is at liberty to lift up his hand to his head, or let it rest quiet: he is perfectly indifferent in

either; and it would be an imperfection in him, if he wanted that power, if he were deprived of that indifference. But it would be as great an imperfection, if he had the same indifference, whether he would prefer the lifting up his hand, or its remaining in rest, when it would save his head or eyes from a blow he sees coming: it is as much a perfection, that desire, or the power of preferring, should be determined by good, as that the power of acting should be determined by the will; and the certainer such determination is, the greater is the perfection. Nay, were we determined by anything but the last result of our own minds, judging of the good or evil of any action, we were not free; the very end of our freedom being, that we may attain the good we choose. And therefore, every man is put under a necessity, by his constitution as an intelligent being, to be determined in willing by his own thought and judgment what is best for him to do: else he would be under the determination of some other than himself, which is want of liberty. And to deny that a man's will, in every determination, follows his own judgment, is to say, that a man

wills and acts for an end that he would not have, at the time that he wills and acts for it. For if he prefers it in his present thoughts before any other, it is plain he then thinks better of it, and would have it before any other; unless he can have and not have it, will and not will it, at the same time; a contradiction too manifest to be admitted.

50. The freest agents are so determined. If we look upon those superior beings above us, who enjoy perfect happiness, we shall have reason to judge that they are more steadily determined in their choice of good than we; and yet we have no reason to think they are less happy, or less free, than we are. And if it were fit for such poor finite creatures as we are to pronounce what infinite wisdom and goodness could do, I think we might say, that God himself cannot choose what is not good; the freedom of the Almighty hinders not his being determined by what is best.

51. A constant determination to a pursuit of happiness no abridgment of liberty. But to give a right view of this mistaken part of liberty let me ask,—Would any one be

a changeling, because he is less determined by wise considerations than a wise man? Is it worth the name of freedom to be at liberty to play the fool, and draw shame and misery upon a man's self? If to break loose from the conduct of reason, and to want that restraint of examination and judgment which keeps us from choosing or doing the worse, be liberty, true liberty, madmen and fools are the only freemen: but yet, I think, nobody would choose to be mad for the sake of such liberty, but he that is mad already. The constant desire of happiness, and the constraint it puts upon us to act for it, nobody, I think, accounts an abridgment of liberty, or at least an abridgment of liberty to be complained of. God Almighty himself is under the necessity of being happy; and the more any intelligent being is so, the nearer is its approach to infinite perfection and happiness. That, in this state of ignorance, we short-sighted creatures might not mistake true felicity, we are endowed with a power to suspend any particular desire, and keep it from determining the will, and engaging us in action. This is standing still, where we are not suffi-

ciently assured of the way: examination is consulting a guide. The determination of the will upon inquiry, is following the direction of that guide: and he that has a power to act or not to act, according as such determination directs, is a free agent: such determination abridges not that power wherein liberty consists. He that has his chains knocked off, and the prison doors set open to him, is perfectly at liberty, because he may either go or stay, as he best likes; though his preference be determined to stay, by the darkness of the night, or illness of the weather, or want of other lodging. He ceases not to be free; though the desire of some convenience to be had there absolutely determines his preference, and makes him stay in his prison.

52. The necessity of pursuing true happiness the foundation of liberty. As therefore the highest perfection of intellectual nature lies in a careful and constant pursuit of true and solid happiness; so the care of ourselves, that we mistake not imaginary for real happiness, is the necessary foundation of our liberty. The stronger ties we have to an unalterable pursuit of happiness in gen-

eral, which is our greatest good, and which, as such, our desires always follow, the more are we free from any necessary determination of our will to any particular action, and from a necessary compliance with our desire, set upon any particular, and then appearing preferable good, till we have duly examined whether it has a tendency to, or be inconsistent with, our real happiness: and therefore, till we are as much informed upon this inquiry as the weight of the matter, and the nature of the case demands, we are, by the necessity of preferring and pursuing true happiness as our greatest good, obliged to suspend the satisfaction of our desires in particular cases.

53. Power to suspend. This is the hinge on which turns the liberty of intellectual beings, in their constant endeavours after, and a steady prosecution of true felicity,—That they can suspend this prosecution in particular cases, till they have looked before them, and informed themselves whether that particular thing which is then proposed or desired lie in the way to their main end, and make a real part of that which is their greatest

good. For, the inclination and tendency of their nature to happiness is an obligation and motive to them, to take care not to mistake or miss it; and so necessarily puts them upon caution, deliberation, and wariness, in the direction of their particular actions, which are the means to obtain it. Whatever necessity determines to the pursuit of real bliss, the same necessity, with the same force, establishes suspense, deliberation, and scrutiny of each successive desire, whether the satisfaction of it does not interfere with our true happiness, and mislead us from it. This, as seems to me, is the great privilege of finite intellectual beings; and I desire it may be well considered, whether the great inlet and exercise of all the liberty men have, are capable of, or can be useful to them, and that whereon depends the turn of their actions, does not lie in this,—That they can suspend their desires, and stop them from determining their wills to any action, till they have duly and fairly examined the good and evil of it, as far forth as the weight of the thing requires. This we are able to do; and when we have done it, we have done our duty, and all that is in

our power; and indeed all that needs. For, since the will supposes knowledge to guide its choice, all that we can do is to hold our wills undetermined, till we have examined the good and evil of what we desire. What follows after that, follows in a chain of consequences, linked one to another, all depending on the last determination of the judgment, which, whether it shall be upon a hasty and precipitate view, or upon a due and mature examination, is in our power; experience showing us, that in most cases, we are able to suspend the present satisfaction of any desire.

54. Government of our passions the right improvement of liberty. But if any extreme disturbance (as sometimes it happens) possesses our whole mind, as when the pain of the rack, an impetuous uneasiness, as of love, anger, or any other violent passion, running away with us, allows us not the liberty of thought, and we are not masters enough of our own minds to consider thoroughly and examine fairly;—God, who knows our frailty, pities our weakness, and requires of us no more than we are able to do, and sees what was and what was not in

our power, will judge as a kind and merciful Father. But the forbearance of a too hasty compliance with our desires, the moderation and restraint of our passions, so that our understandings may be free to examine, and reason unbiased give its judgment, being that whereon a right direction of our conduct to true happiness depends; it is in this we should employ our chief care and endeavours. In this we should take pains to suit the relish of our minds to the true intrinsic good or ill that is in things; and not permit an allowed or supposed possible great and weighty good to slip out of our thoughts, without leaving any relish, any desire of itself there, till, by a due consideration of its true worth, we have formed appetites in our minds suitable to it, and made ourselves uneasy in the want of it, or in the fear of losing it. And how much this is in every one's power, by making resolutions to himself, such as he may keep, is easy for every one to try. Nor let any one say, he cannot govern his passions, nor hinder them from breaking out, and carrying him into action; for what he can do before a prince or a great man, he can do alone, or in

the presence of God, if he will.

55. How men come to pursue different, and often evil, courses. From what has been said, it is easy to give an account how it comes to pass, that, though all men desire happiness, yet their wills carry them so contrarily; and consequently some of them to what is evil. And to this I say, that the various and contrary choices that men make in the world do not argue that they do not all pursue good; but that the same thing is not good to every man alike. This variety of pursuits shows, that every one does not place his happiness in the same thing, or choose the same way to it. Were all the concerns of man terminated in this life, why one followed study and knowledge, and another hawking and hunting: why one chose luxury and debauchery, and another sobriety and riches, would not be because every one of these did not aim at his own happiness; but because their happiness was placed in different things. And therefore it was a right answer of the physician to his patient that had sore eyes:—If you have more pleasure in the taste of wine than in the use of your sight, wine is good for

you; but if the pleasure of seeing be greater to you than that of drinking, wine is naught.

56. All men seek happiness, but not of the same sort. The mind has a different relish, as well as the palate; and you will as fruitlessly endeavour to delight all men with riches or glory (which yet some men place their happiness in) as you would to satisfy all men's hunger with cheese or lobsters; which, though very agreeable and delicious fare to some, are to others extremely nauseous and offensive: and many persons would with reason prefer the griping of an hungry belly to those dishes which are a feast to others. Hence it was, I think, that the philosophers of old did in vain inquire, whether summum bonum consisted in riches, or bodily delights, or virtue, or contemplation: and they might have as reasonably disputed, whether the best relish were to be found in apples, plums, or nuts, and have divided themselves into sects upon it. For, as pleasant tastes depend not on the things themselves, but on their agreeableness to this or that particular palate, wherein there is great variety; so the greatest happiness consists in the

having those things which produce the greatest pleasure, and in the absence of those which cause any disturbance, any pain. Now these, to different men, are very different things. If, therefore, men in this life only have hope; if in this life only they can enjoy, it is not strange nor unreasonable, that they should seek their happiness by avoiding all things that disease them here, and by pursuing all that delight them; wherein it will be no wonder to find variety and difference. For if there be no prospect beyond the grave, the inference is certainly right—"Let us eat and drink," let us enjoy what we "for to-morrow we shall die." This, I think, may serve to show us the reason, why, though all men's desires tend to happiness, yet they are not moved by the same object. Men may choose different things, and yet all choose right; supposing them only like a company of poor insects; whereof some are bees, delighted with flowers and their sweetness; others beetles, delighted with other kinds of viands, which having enjoyed for a season, they would cease to be, and exist no more for ever.

57. Power to suspend volition explains responsibility

for ill choice. These things, duly weighed, will give us, as I think, a clear view into the state of human liberty. Liberty, it is plain, consists in a power to do, or not to do; to do, or forbear doing, as we will. This cannot be denied. But this seeming to comprehend only the actions of a man consecutive to volition, it is further inquired,—Whether he be at liberty to will or no? And to this it has been answered, that, in most cases, a man is not at liberty to forbear the act of volition: he must exert an act of his will, whereby the action proposed is made to exist or not to exist. But yet there is a case wherein a man is at liberty in respect of willing; and that is the choosing of a remote good as an end to be pursued. Here a man may suspend the act of his choice from being determined for or against the thing proposed, till he has examined whether it be really of a nature, in itself and consequences, to make him happy or not. For, when he has once chosen it, and thereby it is become a part of his happiness, it raises desire, and that proportionably gives him uneasiness; which determines his will, and sets him at work in pursuit of his

choice on all occasions that offer. And here we may see how it comes to pass that a man may justly incur punishment, though it be certain that, in all the particular actions that he wills, he does, and necessarily does, will that which he then judges to be good. For, though his will be always determined by that which is judged good by his understanding, yet it excuses him not; because, by a too hasty choice of his own making, he has imposed on himself wrong measures of good and evil; which, however false and fallacious, have the same influence on all his future conduct, as if they were true and right. He has vitiated his own palate, and must be answerable to himself for the sickness and death that follows from it. The eternal law and nature of things must not be altered to comply with his ill-ordered choice. If the neglect or abuse of the liberty he had, to examine what would really and truly make for his happiness, misleads him, the miscarriages that follow on it must be imputed to his own election. He had a power to suspend his determination; it was given him, that he might examine, and take care of his own happiness, and look that

he were not deceived. And he could never judge, that it was better to be deceived than not, in a matter of so great and near concernment.

58. Why men choose what makes them miserable. What has been said may also discover to us the reason why men in this world prefer different things, and pursue happiness by contrary courses. But yet, since men are always constant and in earnest in matters of happiness and misery, the question still remains, How men come often to prefer the worse to the better; and to choose that, which, by their own confession, has made them miserable?

59. The causes of this. To account for the various and contrary ways men take, though all aim at being happy, we must consider whence the various uneasinesses that determine the will, in the preference of each voluntary action, have their rise:

(1) From bodily pain. Some of them come from causes not in our power; such as are often the pains of the body from want, disease, or outward injuries, as the rack, &c.; which, when present and violent, operate for

the most part forcibly on the will, and turn the courses of men's lives from virtue, piety, and religion, and what before they judged to lead to happiness; every one not endeavouring, or, through disuse, not being able, by the contemplation of remote and future good, to raise in himself desires of them strong enough to counterbalance the uneasiness he feels in those bodily torments, and to keep his will steady in the choice of those actions which lead to future happiness. A neighbouring country has been of late a tragical theatre from which we might fetch instances, if there needed any, and the world did not in all countries and ages furnish examples enough to confirm that received observation, *Necessitas cogit ad turpia*; and therefore there is great reason for us to pray, "Lead us not into temptation."

(2) From wrong desires arising from wrong judgments. Other uneasinesses arise from our desires of absent good; which desires always bear proportion to, and depend on, the judgment we make, and the relish we have of any absent good; in both which we are apt to be variously misled, and that by our own fault.

60. Our judgment of present good or evil always right. In the first place, I shall consider the wrong judgments men make of future good and evil, whereby their desires are misled. For, as to present happiness and misery, when that alone comes into consideration, and the consequences are quite removed, a man never chooses amiss: he knows what best pleases him, and that he actually prefers. Things in their present enjoyment are what they seem: the apparent and real good are, in this case, always the same. For, the pain or pleasure being just so great and no greater than it is felt, the present good or evil is really so much as it appears. And therefore were every action of ours concluded within itself, and drew no consequences after it, we should undoubtedly never err in our choice of good: we should always infallibly prefer the best. Were the pains of honest industry, and of starving with hunger and cold set together before us, nobody would be in doubt which to choose: were the satisfaction of a lust and the joys of heaven offered at once to any one's present possession, he would not balance, or err in the determination of his choice.

61. Our wrong judgments have regard to future good and evil only. But since our voluntary actions carry not all the happiness and misery that depend on them along with them in their present performance, but are the precedent causes of good and evil, which they draw after them, and bring upon us, when they themselves are past and cease to be; our desires look beyond our present enjoyments, and carry the mind out to absent good, according to the necessity which we think there is of it, to the making or increase of our happiness. It is our opinion of such a necessity that gives it its attraction: without that, we are not moved by absent good. For, in this narrow scantling of capacity which we are accustomed to and sensible of here, wherein we enjoy but one pleasure at once, which, when all uneasiness is away, is, whilst it lasts, sufficient to make us think ourselves happy, it is not all remote and even apparent good that affects us. Because the indolency and enjoyment we have, sufficing for our present happiness, we desire not to venture the change; since we judge that we are happy already, being content, and that is enough. For who is

content is happy. But as soon as any new uneasiness comes in, this happiness is disturbed, and we are set afresh on work in the pursuit of happiness.

62. From a wrong judgment of what makes a necessary part of their happiness. Their aptness therefore to conclude that they can be happy without it, is one great occasion that men often are not raised to the desire of the greatest absent good. For, whilst such thoughts possess them, the joys of a future state move them not; they have little concern or uneasiness about them; and the will, free from the determination of such desires, is left to the pursuit of nearer satisfactions, and to the removal of those uneasinesses which it then feels, in its want of and longings after them. Change but a man's view of these things; let him see that virtue and religion are necessary to his happiness; let him look into the future state of bliss or misery, and see there God, the righteous judge, ready to "render to every man according to his deeds; to them who by patient continuance in well-doing seek for glory, and honour, and immortality, eternal life; but unto every soul that doth

evil, indignation and wrath, tribulation and anguish." To him, I say, who hath a prospect of the different state of perfect happiness or misery that attends all men after this life, depending on their behaviour here, the measures of good and evil that govern his choice are mightily changed. For, since nothing of pleasure and pain in this life can bear any proportion to the endless happiness or exquisite misery of an immortal soul hereafter, actions in his power will have their preference, not according to the transient pleasure or pain that accompanies or follows them here, but as they serve to secure that perfect durable happiness hereafter.

63. A more particular account of wrong judgments. But, to account more particularly for the misery that men often bring on themselves, notwithstanding that they do all in earnest pursue happiness, we must consider how things come to be represented to our desires under deceitful appearances: and that is by the judgment pronouncing wrongly concerning them. To see how far this reaches, and what are the causes of wrong judgment, we must remember that things are judged good

or bad in a double sense:—

First, That which is properly good or bad, is nothing but barely pleasure or pain.

Secondly, But because not only present pleasure and pain, but that also which is apt by its efficacy or consequences to bring it upon us at a distance, is a proper object of our desires, and apt to move a creature that has foresight; therefore things also that draw after them pleasure and pain, are considered as good and evil.

64. No one chooses misery willingly, but only by wrong judgment. The wrong judgment that misleads us, and makes the will often fasten on the worse side, lies in misreporting upon the various comparisons of these. The wrong judgment I am here speaking of is not what one man may think of the determination of another, but what every man himself must confess to be wrong. For, since I lay it for a certain ground, that every intelligent being really seeks happiness, which consists in the enjoyment of pleasure, without any considerable mixture of uneasiness; it is impossible anyone should willingly put into his own draught any bitter ingredi-

ent, or leave out anything in his power that would tend to his satisfaction, and the completing of his happiness, but only by a wrong judgment. I shall not here speak of that mistake which is the consequence of invincible error, which scarce deserves the name of wrong judgment; but of that wrong judgment which every man himself must confess to be so.

65. Men may err in comparing present and future. (1) Therefore, as to present pleasure and pain, the mind, as has been said, never mistakes that which is really good or evil; that which is the greater pleasure, or the greater pain, is really just as it appears. But, though present pleasure and pain show their difference and degrees so plainly as not to leave room to mistake; yet, when we compare present pleasure or pain with future, (which is usually the case in most important determinations of the will,) we often make wrong judgments of them; taking our measures of them in different positions of distance. Objects near our view are apt to be thought greater than those of a larger size that are more remote. And so it is with pleasures and pains: the present is apt to carry

it; and those at a distance have the disadvantage in the comparison. Thus most men, like spendthrift heirs, are apt to judge a little in hand better than a great deal to come; and so, for small matters in possession, part with greater ones in reversion. But that this is a wrong judgment every one must allow, let his pleasure consist in whatever it will: since that which is future will certainly come to be present; and then, having the same advantage of nearness, will show itself in its full dimensions, and discover his wilful mistake who judged of it by unequal measures. Were the pleasure of drinking accompanied, the very moment a man takes off his glass, with that sick stomach and aching head which, in some men, are sure to follow not many hours after, I think nobody, whatever pleasure he had in his cups, would, on these conditions, ever let wine touch his lips; which yet he daily swallows, and the evil side comes to be chosen only by the fallacy of a little difference in time. But, if pleasure or pain can be so lessened only by a few hours' removal, how much more will it be so by a further distance, to a man that will not, by a right judg-

ment, do what time will, i.e. bring it home upon himself, and consider it as present, and there take its true dimensions? This is the way we usually impose on ourselves, in respect of bare pleasure and pain, or the true degrees of happiness or misery: the future loses its just proportion, and what is present obtains the preference as the greater. I mention not here the wrong judgment, whereby the absent are not only lessened, but reduced to perfect nothing; when men enjoy what they can in present, and make sure of that, concluding amiss that no evil will thence follow. For that lies not in comparing the greatness of future good and evil, which is that we are here speaking of; but in another sort of wrong judgment, which is concerning good or evil, as it is considered to be the cause and procurement of pleasure or pain that will follow from it.

66. Causes of our judging amiss when we compare present pleasure and pain with future. The cause of our judging amiss, when we compare our present pleasure or pain with future, seems to me to be the weak and narrow constitution of our minds. We cannot well enjoy

two pleasures at once; much less any pleasure almost, whilst pain possesses us. The present pleasure, if it be not very languid, and almost none at all, fills our narrow souls, and so takes up the whole mind that it scarce leaves any thought of things absent: or if among our pleasures there are some which are not strong enough to exclude the consideration of things at a distance, yet we have so great an abhorrence of pain, that a little of it extinguishes all our pleasures. A little bitter mingled in our cup, leaves no relish of the sweet. Hence it comes that, at any rate, we desire to be rid of the present evil, which we are apt to think nothing absent can equal; because, under the present pain, we find not ourselves capable of any the least degree of happiness. Men's daily complaints are a loud proof of this: the pain that any one actually feels is still of all other the worst; and it is with anguish they cry out,—“Any rather than this: nothing can be so intolerable as what I now suffer.” And therefore our whole endeavours and thoughts are intent to get rid of the present evil, before all things, as the first necessary condition to our happiness; let what

will follow. Nothing, as we passionately think, can exceed, or almost equal, the uneasiness that sits so heavy upon us. And because the abstinence from a present pleasure that offers itself is a pain, nay, oftentimes a very great one, the desire being inflamed by a near and tempting object, it is no wonder that that operates after the same manner pain does, and lessens in our thoughts what is future; and so forces us, as it were blindfold, into its embraces.

67. Absent good unable to counterbalance present uneasiness. Add to this, that absent good, or, which is the same thing, future pleasure,—especially if of a sort we are unacquainted with,—seldom is able to counterbalance any uneasiness, either of pain or desire, which is present. For, its greatness being no more than what shall be really tasted when enjoyed, men are apt enough to lessen that; to make it give place to any present desire; and conclude with themselves that, when it comes to trial, it may possibly not answer the report or opinion that generally passes of it: they having often found that, not only what others have magnified, but even

what they themselves have enjoyed with great pleasure and delight at one time, has proved insipid or nauseous at another; and therefore they see nothing in it for which they should forego a present enjoyment. But that this is a false way of judging, when applied to the happiness of another life, they must confess; unless they will say, God cannot make those happy he designs to be so. For that being intended for a state of happiness, it must certainly be agreeable to everyone's wish and desire: could we suppose their relishes as different there as they are here, yet the manna in heaven will suit every one's palate. Thus much of the wrong judgment we make of present and future pleasure and pain, when they are compared together, and so the absent considered as future.

68. Wrong judgment in considering consequences of actions. (II) As to things good or bad in their consequences, and by the aptness that is in them to procure us good or evil in the future, we judge amiss several ways.

1. When we judge that so much evil does not really depend on them as in truth there does.

2. When we judge that, though the consequence be of that moment, yet it is not of that certainty, but that it may otherwise fall out, or else by some means be avoided; as by industry, address, change, repentance, &c.

That these are wrong ways of judging, were easy to show in every particular, if I would examine them at large singly: but I shall only mention this in general, viz. that it is a very wrong and irrational way of proceeding, to venture a greater good for a less, upon uncertain guesses; and before a due examination be made, proportionable to the weightiness of the matter, and the concernment it is to us not to mistake. This I think every one must confess, especially if he considers the usual cause of this wrong judgment, whereof these following are some:—

69. Causes of this. (i) Ignorance: He that judges without informing himself to the utmost that he is capable, cannot acquit himself of judging amiss.

(ii) Inadvertency: When a man overlooks even that which he does know. This is an affected and present ignorance, which misleads our judgments as much as

the other. Judging is, as it were, balancing an account, and determining on which side the odds lie. If therefore either side be huddled up in haste, and several of the sums that should have gone into the reckoning be overlooked and left out, this precipitancy causes as wrong a judgment as if it were a perfect ignorance. That which most commonly causes this is, the prevalency of some present pleasure or pain, heightened by our feeble passionate nature, most strongly wrought on by what is present. To check this precipitancy, our understanding and reason were given us, if we will make a right use of them, to search and see, and then judge thereupon. Without liberty, the understanding would be to no purpose: and without understanding, liberty (if it could be) would signify nothing. If a man sees what would do him good or harm, what would make him happy or miserable, without being able to move himself one step towards or from it, what is he the better for seeing? And he that is at liberty to ramble in perfect darkness, what is his liberty better than if he were driven up and down as a bubble by the force of the wind? The being acted by

a blind impulse from without, or from within, is little odds. The first, therefore, and great use of liberty is to hinder blind precipitancy; the principal exercise of freedom is to stand still, open the eyes, look about, and take a view of the consequence of what we are going to do, as much as the weight of the matter requires. How much sloth and negligence, heat and passion, the prevalency of fashion or acquired indispositions do severally contribute, on occasion, to these wrong judgments, I shall not here further inquire. I shall only add one other false judgment, which I think necessary to mention, because perhaps it is little taken notice of, though of great influence.

70. Wrong judgment of what is necessary to our happiness. All men desire happiness, that is past doubt: but, as has been already observed, when they are rid of pain, they are apt to take up with any pleasure at hand, or that custom has endeared to them; to rest satisfied in that; and so being happy, till some new desire, by making them uneasy, disturbs that happiness, and shows them that they are not so, they look no further; nor is the will

determined to any action in pursuit of any other known or apparent good. For since we find that we cannot enjoy all sorts of good, but one excludes another; we do not fix our desires on every apparent greater good, unless it be judged to be necessary to our happiness: if we think we can be happy without it, it moves us not. This is another occasion to men of judging wrong; when they take not that to be necessary to their happiness which really is so. This mistake misleads us, both in the choice of the good we aim at, and very often in the means to it, when it is a remote good. But, which way ever it be, either by placing it where really it is not, or by neglecting the means as not necessary to it;—when a man misses his great end, happiness, he will acknowledge he judged not right. That which contributes to this mistake is the real or supposed unpleasantness of the actions which are the way to this end; it seeming so preposterous a thing to men, to make themselves unhappy in order to happiness, that they do not easily bring themselves to it.

71. We can change the agreeableness or disagreeableness in things. The last inquiry, therefore, concerning

this matter is,—Whether it be in a man's power to change the pleasantness and unpleasantness that accompanies any sort of action? And as to that, it is plain, in many cases he can. Men may and should correct their palates, and give relish to what either has, or they suppose has none. The relish of the mind is as various as that of the body, and like that too may be altered; and it is a mistake to think that men cannot change the displeasingness or indifferency that is in actions into pleasure and desire, if they will do but what is in their power. A due consideration will do it in some cases; and practice, application, and custom in most. Bread or tobacco may be neglected where they are shown to be useful to health, because of an indifferency or disrelish to them; reason and consideration at first recommends, and begins their trial, and use finds, or custom makes them pleasant. That this is so in virtue too, is very certain. Actions are pleasing or displeasing, either in themselves, or considered as a means to a greater and more desirable end. The eating of a well-seasoned dish, suited to a man's palate, may move the mind by the delight itself that accompa-

nies the eating, without reference to any other end; to which the consideration of the pleasure there is in health and strength (to which that meat is subservient) may add a new gusto, able to make us swallow an ill-relished potion. In the latter of these, any action is rendered more or less pleasing, only by the contemplation of the end, and the being more or less persuaded of its tendency to it, or necessary connexion with it: but the pleasure of the action itself is best acquired or increased by use and practice. Trials often reconcile us to that, which at a distance we looked on with aversion; and by repetitions wear us into a liking of what possibly, in the first essay, displeased us. Habits have powerful charms, and put so strong attractions of easiness and pleasure into what we accustom ourselves to, that we cannot forbear to do, or at least be easy in the omission of, actions, which habitual practice has suited, and thereby recommends to us. Though this be very visible, and every one's experience shows him he can do so; yet it is a part in the conduct of men towards their happiness, neglected to a degree, that it will be possibly enter-

tained as a paradox, if it be said, that men can make things or actions more or less pleasing to themselves; and thereby remedy that, to which one may justly impute a great deal of their wandering. Fashion and the common opinion having settled wrong notions, and education and custom ill habits, the just values of things are misplaced, and the palates of men corrupted. Pains should be taken to rectify these; and contrary habits change our pleasures, and give a relish to that which is necessary or conducive to our happiness. This every one must confess he can do; and when happiness is lost, and misery overtakes him, he will confess he did amiss in neglecting it, and condemn himself for it; and I ask every one, whether he has not often done so?

72. Preference of vice to virtue a manifest wrong judgment. I shall not now enlarge any further on the wrong judgments and neglect of what is in their power, whereby men mislead themselves. This would make a volume, and is not my business. But whatever false notions, or shameful neglect of what is in their power, may put men out of their way to happiness, and distract them, as we see,

into so different courses of life, this yet is certain, that morality, established upon its true foundations, cannot but determine the choice in any one that will but consider: and he that will not be so far a rational creature as to reflect seriously upon infinite happiness and misery, must needs condemn himself as not making that use of his understanding he should. The rewards and punishments of another life, which the Almighty has established, as the enforcements of his law, are of weight enough to determine the choice, against whatever pleasure or pain this life can show, when the eternal state is considered but in its bare possibility, which nobody can make any doubt of. He that will allow exquisite and endless happiness to be but the possible consequence of a good life here, and the contrary state the possible reward of a bad one, must own himself to judge very much amiss if he does not conclude,—That a virtuous life, with the certain expectation of everlasting bliss, which may come, is to be preferred to a vicious one, with the fear of that dreadful state of misery, which it is very possible may overtake the guilty; or, at best, the

terrible uncertain hope of annihilation. This is evidently so, though the virtuous life here had nothing but pain, and the vicious continual pleasure: which yet is, for the most part, quite otherwise, and wicked men have not much the odds to brag of, even in their present possession; nay, all things rightly considered, have, I think, even the worse part here. But when infinite happiness is put into one scale, against infinite misery in the other; if the worst that comes to the pious man, if he mistakes, be the best that the wicked can attain to, if he be in the right, who can without madness run the venture? Who in his wits would choose to come within a possibility of infinite misery; which if he miss, there is yet nothing to be got by that hazard? Whereas, on the other side, the sober man ventures nothing against infinite happiness to be got, if his expectation comes not to pass. If the good man be in the right, he is eternally happy; if he mistakes, he's not miserable, he feels nothing. On the other side, if the wicked man be in the right, he is not happy; if he mistakes, he is infinitely miserable. Must it not be a most manifest wrong judg-

ment that does not presently see to which side, in this case, the preference is to be given? I have forborne to mention anything of the certainty or probability of a future state, designing here to show the wrong judgment that any one must allow he makes, upon his own principles, laid how he pleases, who prefers the short pleasures of a vicious life upon any consideration, whilst he knows, and cannot but be certain, that a future life is at least possible.

73. Recapitulation—liberty of indifferency. To conclude this inquiry into human liberty, which, as it stood before, I myself from the beginning fearing, and a very judicious friend of mine, since the publication, suspecting to have some mistake in it, though he could not particularly show it me, I was put upon a stricter review of this chapter. Wherein lighting upon a very easy and scarce observable slip I had made, in putting one seemingly indifferent word for another that discovery opened to me this present view, which here, in this second edition, I submit to the learned world, and which, in short, is this: Liberty is a power to act or not to act, according

as the mind directs. A power to direct the operative faculties to motion or rest in particular instances is that which we call the will. That which in the train of our voluntary actions determines the will to any change of operation is some present uneasiness, which is, or at least is always accompanied with that of desire. Desire is always moved by evil, to fly it: because a total freedom from pain always makes a necessary part of our happiness: but every good, nay, every greater good, does not constantly move desire, because it may not make, or may not be taken to make, part of our happiness. For all that we desire, is only to be happy. But, though this general desire of happiness operates constantly and invariably, yet the satisfaction of any particular desire can be suspended from determining the will to any subervient action, till we have maturely examined whether the particular apparent good which we then desire makes a part of our real happiness, or be consistent or inconsistent with it. The result of our judgment upon that examination is what ultimately determines the man; who could not be free if his will were determined by any-

thing but his own desire, guided by his own judgment. I know that liberty, by some, is placed in an indifferency of the man; antecedent to the determination of his will. I wish they who lay so much stress on such an antecedent indifferency, as they call it, had told us plainly, whether this supposed indifferency be antecedent to the thought and judgment of the understanding, as well as to the decree of the will. For it is pretty hard to state it between them, i.e. immediately after the judgment of the understanding, and before the determination of the will: because the determination of the will immediately follows the judgment of the understanding: and to place liberty in an indifferency, antecedent to the thought and judgment of the understanding, seems to me to place liberty in a state of darkness, wherein we can neither see nor say anything of it; at least it places it in a subject incapable of it, no agent being allowed capable of liberty, but in consequence of thought and judgment. I am not nice about phrases, and therefore consent to say with those that love to speak so, that liberty is placed in indifferency, but it is an indifferency which

remains after the judgment of the understanding, yea, even after the determination of the will: and that is an indifferency not of the man, (for after he has once judged which is best, viz. to do or forbear, he is no longer indifferent,) but an indifferency of the operative powers of the man, which remaining equally able to operate or to forbear operating after as before the decree of the will, are in a state, which, if one pleases, may be called indifferency; and as far as this indifferency reaches, a man is free, and no further: v.g. I have the ability to move my hand, or to let it rest; that operative power is indifferent to move or not to move my hand. I am then, in that respect perfectly free; my will determines that operative power to rest: I am yet free, because the indifferency of that my operative power to act, or not to act, still remains; the power of moving my hand is not at all impaired by the determination of my will, which at present orders rest; the indifferency of that power to act, or not to act, is just as it was before, as will appear, if the will puts it to the trial, by ordering the contrary. But if, during the rest of my hand, it be

seized with a sudden palsy, the indifferency of that operative power is gone, and with it my liberty; I have no longer freedom in that respect, but am under a necessity of letting my hand rest. On the other side, if my hand be put into motion by a convulsion, the indifferency of that operative faculty is taken away by that motion; and my liberty in that case is lost, for I am under a necessity of having my hand move. I have added this, to show in what sort of indifferency liberty seems to me to consist, and not in any other, real or imaginary.

74. Active and passive power, in motions and in thinking. True notions concerning the nature and extent of liberty are of so great importance, that I hope I shall be pardoned this digression, which my attempt to explain it has led me into. The ideas of will, volition, liberty, and necessity, in this Chapter of Power, came naturally in my way. In a former edition of this Treatise I gave an account of my thoughts concerning them, according to the light I then had. And now, as a lover of truth, and not a worshipper of my own doctrines, I own some change of my opinion; which I think I have discovered ground

for. In what I first writ, I with an unbiased indifferency followed truth, whither I thought she led me. But neither being so vain as to fancy infallibility, nor so disingenuous as to dissemble my mistakes for fear of blemishing my reputation, I have, with the same sincere design for truth only, not been ashamed to publish what a severer inquiry has suggested. It is not impossible but that some may think my former notions right; and some (as I have already found) these latter; and some neither. I shall not at all wonder at this variety in men's opinions: impartial deductions of reason in controverted points being so rare, and exact ones in abstract notions not so very easy, especially if of any length. And, therefore, I should think myself not a little beholden to any one, who would, upon these or any other grounds, fairly clear this subject of liberty from any difficulties that may yet remain.

Before I close this chapter, it may perhaps be to our purpose, and help to give us clearer conceptions about power, if we make our thoughts take a little more exact survey of action. I have said above, that we have ideas

but of two sorts of action, viz. motion and thinking. These, in truth, though called and counted actions, yet, if nearly considered, will not be found to be always perfectly so. For, if I mistake not, there are instances of both kinds, which, upon due consideration, will be found rather passions than actions; and consequently so far the effects barely of passive powers in those subjects, which yet on their accounts are thought agents. For, in these instances, the substance that hath motion or thought receives the impression, whereby it is put into that action, purely from without, and so acts merely by the capacity it has to receive such an impression from some external agent; and such power is not properly an active power, but a mere passive capacity in the subject. Sometimes the substance or agent puts itself into action by its own power, and this is properly active power. Whatsoever modification a substance has, whereby it produces any effect, that is called action: v.g. a solid substance, by motion, operates on or alters the sensible ideas of another substance, and therefore this modification of motion we call action. But yet this motion in

that solid substance is, when rightly considered, but a passion, if it received it only from some external agent. So that the active power of motion is in no substance which cannot begin motion in itself or in another substance when at rest. So likewise in thinking, a power to receive ideas or thoughts from the operation of any external substance is called a power of thinking: but this is but a passive power, or capacity. But to be able to bring into view ideas out of sight at one's own choice, and to compare which of them one thinks fit, this is an active power. This reflection may be of some use to preserve us from mistakes about powers and actions, which grammar, and the common frame of languages, may be apt to lead us into. Since what is signified by verbs that grammarians call active, does not always signify action: v.g. this proposition: I see the moon, or a star, or I feel the heat of the sun, though expressed by a verb active, does not signify any action in me, whereby I operate on those substances, but only the reception of the ideas of light, roundness, and heat; wherein I am not active, but barely passive, and cannot, in that position of my eyes

or body, avoid receiving them. But when I turn my eyes another way, or remove my body out of the sunbeams, I am properly active; because of my own choice, by a power within myself, I put myself into that motion. Such an action is the product of active power.

75. Summary of our original ideas. And thus I have, in a short draught, given a view of our original ideas, from whence all the rest are derived, and of which they are made up; which, if I would consider as a philosopher, and examine on what causes they depend, and of what they are made, I believe they all might be reduced to these very few primary and original ones, viz.

Extension,

Solidity,

Mobility, or the power of being moved; which by our senses we receive from body:

Perceptivity, or the power of perception, or thinking;

Motivity, or the power of moving: which by reflection we receive from our minds.

I crave leave to make use of these two new words, to avoid the danger of being mistaken in the use of those

which are equivocal.

To which if we add

Existence,

Duration,

Number,

which belong both to the one and the other, we have, perhaps, all the original ideas on which the rest depend. For by these, I imagine, might be explained the nature of colours, sounds, tastes, smells, and all other ideas we have, if we had but faculties acute enough to perceive the severally modified extensions and motions of these minute bodies, which produce those several sensations in us. But my present purpose being only to inquire into the knowledge the mind has of things, by those ideas and appearances which God has fitted it to receive from them, and how the mind comes by that knowledge; rather than into their causes or manner of production, I shall not, contrary to the design of this Essay, set myself to inquire philosophically into the peculiar constitution of bodies, and the configuration of parts, whereby they have the power to produce in us the ideas

of their sensible qualities. I shall not enter any further into that disquisition; it sufficing to my purpose to observe, that gold or saffron has a power to produce in us the idea of yellow, and snow or milk, the idea of white, which we can only have by our sight; without examining the texture of the parts of those bodies, or the particular figures or motion of the particles which rebound from them, to cause in us that particular sensation: though, when we go beyond the bare ideas in our minds, and would inquire into their causes, we cannot conceive anything else to be in any sensible object, whereby it produces different ideas in us, but the different bulk, figure, number, texture, and motion of its insensible parts.

Chapter XXII

Of Mixed Modes

1. Mixed modes, what. Having treated of simple modes in the foregoing chapters, and given several instances of some of the most considerable of them, to show what they are, and how we come by them; we are now in the

next place to consider those we call mixed modes; such are the complex ideas we mark by the names obligation, drunkenness, a lie, &c.; which consisting of several combinations of simple ideas of different kinds, I have called mixed modes, to distinguish them from the more simple modes, which consist only of simple ideas of the same kind. These mixed modes, being also such combinations of simple ideas as are not looked upon to be characteristical marks of any real beings that have a steady existence, but scattered and independent ideas put together by the mind, are thereby distinguished from the complex ideas of substances.

2. Made by the mind. That the mind, in respect of its simple ideas, is wholly passive, and receives them all from the existence and operations of things, such as sensation or reflection offers them, without being able to make any one idea, experience shows us. But if we attentively consider these ideas I call mixed modes, we are now speaking of, we shall find their original quite different. The mind often exercises an active power in making these several combinations. For, it being once

furnished with simple ideas, it can put them together in several compositions, and so make variety of complex ideas, without examining whether they exist so together in nature. And hence I think it is that these ideas are called notions: as if they had their original, and constant existence, more in the thoughts of men, than in the reality of things; and to form such ideas, it sufficed that the mind put the parts of them together, and that they were consistent in the understanding, without considering whether they had any real being: though I do not deny but several of them might be taken from observation, and the existence of several simple ideas so combined, as they are put together in the understanding. For the man who first framed the idea of hypocrisy, might have either taken it at first from the observation of one who made show of good qualities which he had not; or else have framed that idea in his mind without having any such pattern to fashion it by. For it is evident that, in the beginning of languages and societies of men, several of those complex ideas, which were consequent to the constitutions established amongst them,

must needs have been in the minds of men, before they existed anywhere else; and that many names that stood for such complex ideas were in use, and so those ideas framed, before the combinations they stood for ever existed.

3. Sometimes got by the explication of their names. Indeed, now that languages are made, and abound with words standing for such combinations, an usual way of getting these complex ideas is, by the explication of those terms that stand for them. For, consisting of a company of simple ideas combined, they may, by words standing for those simple ideas, be represented to the mind of one who understands those words, though that complex combination of simple ideas were never offered to his mind by the real existence of things. Thus a man may come to have the idea of sacrilege or murder, by enumerating to him the simple ideas which these words stand for; without ever seeing either of them committed.

4. The name ties the parts of mixed modes into one idea. Every mixed mode consisting of many distinct simple ideas, it seems reasonable to inquire, Whence it has its

unity; and how such a precise multitude comes to make but one idea; since that combination does not always exist together in nature? To which I answer, it is plain it has its unity from an act of the mind, combining those several simple ideas together, and considering them as one complex one, consisting of those parts; and the mark of this union, or that which is looked on generally to complete it, is one name given to that combination. For it is by their names that men commonly regulate their account of their distinct species of mixed modes, seldom allowing or considering any number of simple ideas to make one complex one, but such collections as there be names for. Thus, though the killing of an old man be as fit in nature to be united into one complex idea, as the killing a man's father; yet, there being no name standing precisely for the one, as there is the name of parricide to mark the other, it is not taken for a particular complex idea, nor a distinct species of actions from that of killing a young man, or any other man.

5. The cause of making mixed modes. If we should inquire a little further, to see what it is that occasions

men to make several combinations of simple ideas into distinct, and, as it were, settled modes, and neglect others, which in the nature of things themselves, have as much an aptness to be combined and make distinct ideas, we shall find the reason of it to be the end of language; which being to mark, or communicate men's thoughts to one another with all the dispatch that may be, they usually make such collections of ideas into complex modes, and affix names to them, as they have frequent use of in their way of living and conversation, leaving others, which they have but seldom an occasion to mention, loose and without names that tie them together: they rather choosing to enumerate (when they have need) such ideas as make them up, by the particular names that stand for them, than to trouble their memories by multiplying of complex ideas with names to them, which they seldom or never have any occasion to make use of.

6. Why words in one language have none answering in another. This shows us how it comes to pass that there are in every language many particular words which can-

not be rendered by any one single word of another. For the several fashions, customs, and manners of one nation, making several combinations of ideas familiar and necessary in one, which another people have had never an occasion to make, or perhaps so much as take notice of, names come of course to be annexed to them, to avoid long periphrases in things of daily conversation; and so they become so many distinct complex ideas in their minds. Thus *ostrhakismos* amongst the Greeks, and *proscriptio* amongst the Romans, were words which other languages had no names that exactly answered; because they stood for complex ideas which were not in the minds of the men of other nations. Where there was no such custom, there was no notion of any such actions; no use of such combinations of ideas as were united, and, as it were, tied together, by those terms: and therefore in other countries there were no names for them.

7. And languages change. Hence also we may see the reason, why languages constantly change, take up new and lay by old terms. Because change of customs and opinions bringing with it new combinations of ideas,

which it is necessary frequently to think on and talk about, new names, to avoid long descriptions, are annexed to them; and so they become new species of complex modes. What a number of different ideas are by this means wrapped up in one short sound, and how much of our time and breath is thereby saved, any one will see, who will but take the pains to enumerate all the ideas that either reprieve or appeal stand for; and instead of either of those names, use a periphrasis, to make any one understand their meaning.

8. Mixed modes, where they exist. Though I shall have occasion to consider this more at large when I come to treat of Words and their use, yet I could not avoid to take this much notice here of the names of mixed modes; which being fleeting and transient combinations of simple ideas, which have but a short existence anywhere but in the minds of men, and there too have no longer any existence than whilst they are thought on, have not so much anywhere the appearance of a constant and lasting existence as in their names: which are therefore, in this sort of ideas, very apt to be taken for the ideas

themselves. For, if we should inquire where the idea of a triumph or apotheosis exists, it is evident they could neither of them exist altogether anywhere in the things themselves, being actions that required time to their performance, and so could never all exist together; and as to the minds of men, where the ideas of these actions are supposed to be lodged, they have there too a very uncertain existence: and therefore we are apt to annex them to the names that excite them in us.

9. How we get the ideas of mixed modes. There are therefore three ways whereby we get these complex ideas of mixed modes:—(1) By experience and observation of things themselves: thus, by seeing two men wrestle or fence, we get the idea of wrestling or fencing. (2) By invention, or voluntary putting together of several simple ideas in our own minds: so he that first invented printing or etching, had an idea of it in his mind before it ever existed. (3) Which is the most usual way, by explaining the names of actions we never saw, or motions we cannot see; and by enumerating, and thereby, as it were, setting before our imaginations all those ideas

which go to the making them up, and are the constituent parts of them. For, having by sensation and reflection stored our minds with simple ideas, and by use got the names that stand for them, we can by those means represent to another any complex idea we would have him conceive; so that it has in it no simple ideas but what he knows, and has with us the same name for. For all our complex ideas are ultimately resolvable into simple ideas, of which they are compounded and originally made up, though perhaps their immediate ingredients, as I may so say, are also complex ideas. Thus, the mixed mode which the word *lie* stands for is made of these simple ideas:—(1) Articulate sounds. (2) Certain ideas in the mind of the speaker. (3) Those words the signs of those ideas. (4) Those signs put together, by affirmation or negation, otherwise than the ideas they stand for are in the mind of the speaker. I think I need not go any further in the analysis of that complex idea we call a *lie*: what I have said is enough to show that it is made up of simple ideas. And it could not be but an offensive tediousness to my reader, to trouble him with a more

minute enumeration of every particular simple idea that goes to this complex one; which, from what has been said, he cannot but be able to make out to himself. The same may be done in all our complex ideas whatsoever; which, however compounded and decompounded, may at last be resolved into simple ideas, which are all the materials of knowledge or thought we have, or can have. Nor shall we have reason to fear that the mind is hereby stinted to too scanty a number of ideas, if we consider what an inexhaustible stock of simple modes number and figure alone afford us. How far then mixed modes, which admit of the various combinations of different simple ideas, and their infinite modes, are from being few and scanty, we may easily imagine. So that, before we have done, we shall see that nobody need be afraid he shall not have scope and compass enough for his thoughts to range in, though they be, as I pretend, confined only to simple ideas, received from sensation or reflection, and their several combinations.

10. Motion, thinking, and power have been most modified. It is worth our observing, which of all our simple

ideas have been most modified, and had most mixed ideas made out of them, with names given to them. And those have been these three:—thinking and motion (which are the two ideas which comprehend in them all action,) and power, from whence these actions are conceived to flow. These simple ideas, I say, of thinking, motion, and power, have been those which have been most modified; and out of whose modifications have been made most complex modes, with names to them. For action being the great business of mankind, and the whole matter about which all laws are conversant, it is no wonder that the several modes of thinking and motion should be taken notice of, the ideas of them observed, and laid up in the memory, and have names assigned to them; without which laws could be but ill made, or vice and disorders repressed. Nor could any communication be well had amongst men without such complex ideas, with names to them: and therefore men have settled names, and supposed settled ideas in their minds, of modes of actions, distinguished by their causes, means, objects, ends, instruments, time, place, and other

circumstances; and also of their powers fitted for those actions: v.g. boldness is the power to speak or do what we intend, before others, without fear or disorder; and the Greeks call the confidence of speaking by a peculiar name, parrhesia: which power or ability in man of doing anything, when it has been acquired by frequent doing the same thing, is that idea we name habit; when it is forward, and ready upon every occasion to break into action, we call it disposition. Thus, testiness is a disposition or aptness to be angry. To conclude: Let us examine any modes of action, v.g. consideration and assent, which are actions of the mind; running and speaking, which are actions of the body; revenge and murder, which are actions of both together, and we shall find them but so many collections of simple ideas, which, together, make up the complex ones signified by those names.

11. Several words seeming to signify action, signify but the effect. Power being the source from whence all action proceeds, the substances wherein these powers are, when they exert this power into act, are called causes,

and the substances which thereupon are produced, or the simple ideas which are introduced into any subject by the exerting of that power, are called effects. The efficacy whereby the new substance or idea is produced is called, in the subject exerting that power, action; but in the subject wherein any simple idea is changed or produced, it is called passion: which efficacy, however various, and the effects almost infinite, yet we can, I think, conceive it, in intellectual agents, to be nothing else but modes of thinking and willing; in corporeal agents, nothing else but modifications of motion. I say, I think we cannot conceive it to be any other but these two. For whatever sort of action besides these produce any effects, I confess myself to have no notion nor idea of; and so it is quite remote from my thoughts, apprehensions, and knowledge; and as much in the dark to me as five other senses, or as the ideas of colours to a blind man. And therefore many words which seem to express some action, signify nothing of the action or *modus operandi* at all, but barely the effect, with some circumstances of the subject wrought on, or cause op-

erating: v.g. creation, annihilation, contain in them no idea of the action or manner whereby they are produced, but barely of the cause, and the thing done. And when a countryman says the cold freezes water, though the word freezing seems to import some action, yet truly it signifies nothing but the effect, viz. that water that was before fluid is become hard and consistent, without containing any idea of the action whereby it is done.

12. Mixed modes made also of other ideas than those of power and action. I think I shall not need to remark here that, though power and action make the greatest part of mixed modes, marked by names, and familiar in the minds and mouths of men, yet other simple ideas, and their several combinations, are not excluded: much less, I think, will it be necessary for me to enumerate all the mixed modes which have been settled, with names to them. That would be to make a dictionary of the greatest part of the words made use of in divinity, ethics, law, and politics, and several other sciences. All that is requisite to my present design, is to show what sort of ideas those are which I call mixed modes; how the

mind comes by them; and that they are compositions made up of simple ideas got from sensation and reflection; which I suppose I have done.

Chapter XXIII Of our Complex Ideas of Substances

1. Ideas of particular substances, how made. The mind being, as I have declared, furnished with a great number of the simple ideas, conveyed in by the senses as they are found in exterior things, or by reflection on its own operations, takes notice also that a certain number of these simple ideas go constantly together; which being presumed to belong to one thing, and words being suited to common apprehensions, and made use of for quick dispatch, are called, so united in one subject, by one name; which, by inadvertency, we are apt afterward to talk of and consider as one simple idea, which indeed is a complication of many ideas together: because, as I have said, not imagining how these simple ideas can subsist by themselves, we accustom ourselves to sup-

pose some substratum wherein they do subsist, and from which they do result, which therefore we call substance.

2. Our obscure idea of substance in general. So that if any one will examine himself concerning his notion of pure substance in general, he will find he has no other idea of it at all, but only a supposition of he knows not what support of such qualities which are capable of producing simple ideas in us; which qualities are commonly called accidents. If any one should be asked, what is the subject wherein colour or weight inheres, he would have nothing to say, but the solid extended parts; and if he were demanded, what is it that solidity and extension adhere in, he would not be in a much better case than the Indian before mentioned who, saying that the world was supported by a great elephant, was asked what the elephant rested on; to which his answer was—a great tortoise: but being again pressed to know what gave support to the broad-backed tortoise, replied—something, he knew not what. And thus here, as in all other cases where we use words without having clear and distinct ideas, we talk like children: who, being questioned what

such a thing is, which they know not, readily give this satisfactory answer, that it is something: which in truth signifies no more, when so used, either by children or men, but that they know not what; and that the thing they pretend to know, and talk of, is what they have no distinct idea of at all, and so are perfectly ignorant of it, and in the dark. The idea then we have, to which we give the general name substance, being nothing but the supposed, but unknown, support of those qualities we find existing, which we imagine cannot subsist sine re substante, without something to support them, we call that support substantia; which, according to the true import of the word, is, in plain English, standing under or upholding.

3. Of the sorts of substances. An obscure and relative idea of substance in general being thus made we come to have the ideas of particular sorts of substances, by collecting such combinations of simple ideas as are, by experience and observation of men's senses, taken notice of to exist together; and are therefore supposed to flow from the particular internal constitution, or un-

known essence of that substance. Thus we come to have the ideas of a man, horse, gold, water, &c.; of which substances, whether any one has any other clear idea, further than of certain simple ideas co-existent together, I appeal to every one's own experience. It is the ordinary qualities observable in iron, or a diamond, put together, that make the true complex idea of those substances, which a smith or a jeweller commonly knows better than a philosopher; who, whatever substantial forms he may talk of, has no other idea of those substances, than what is framed by a collection of those simple ideas which are to be found in them: only we must take notice, that our complex ideas of substances, besides all those simple ideas they are made up of, have always the confused idea of something to which they belong, and in which they subsist: and therefore when we speak of any sort of substance, we say it is a thing having such or such qualities; as body is a thing that is extended, figured, and capable of motion; spirit, a thing capable of thinking; and so hardness, friability, and power to draw iron, we say, are qualities to be found in a

loadstone. These, and the like fashions of speaking, intimate that the substance is supposed always something besides the extension, figure, solidity, motion, thinking, or other observable ideas, though we know not what it is.

4. No clear or distinct idea of substance in general. Hence, when we talk or think of any particular sort of corporeal substances, as horse, stone, &c., though the idea we have of either of them be but the complication or collection of those several simple ideas of sensible qualities, which we used to find united in the thing called horse or stone; yet, because we cannot conceive how they should subsist alone, nor one in another, we suppose them existing in and supported by some common subject; which support we denote by the name substance, though it be certain we have no clear or distinct idea of that thing we suppose a support.

5. As clear an idea of spiritual substance as of corporeal substance. The same thing happens concerning the operations of the mind, viz. thinking, reasoning, fearing, &c., which we concluding not to subsist of themselves,

nor apprehending how they can belong to body, or be produced by it, we are apt to think these the actions of some other substance, which we call spirit; whereby yet it is evident that, having no other idea or notion of matter, but something wherein those many sensible qualities which affect our senses do subsist; by supposing a substance wherein thinking, knowing, doubting, and a power of moving, &c., do subsist, we have as clear a notion of the substance of spirit, as we have of body; the one being supposed to be (without knowing what it is) the substratum to those simple ideas we have from without; and the other supposed (with a like ignorance of what it is) to be the substratum to those operations we experiment in ourselves within. It is plain then, that the idea of corporeal substance in matter is as remote from our conceptions and apprehensions, as that of spiritual substance, or spirit: and therefore, from our not having any notion of the substance of spirit, we can no more conclude its non-existence, than we can, for the same reason, deny the existence of body; it being as rational to affirm there is no body, because we have no

clear and distinct idea of the substance of matter, as to say there is no spirit, because we have no clear and distinct idea of the substance of a spirit.

6. Our ideas of particular sorts of substances. Whatever therefore be the secret abstract nature of substance in general, all the ideas we have of particular distinct sorts of substances are nothing but several combinations of simple ideas, coexisting in such, though unknown, cause of their union, as makes the whole subsist of itself. It is by such combinations of simple ideas, and nothing else, that we represent particular sorts of substances to ourselves; such are the ideas we have of their several species in our minds; and such only do we, by their specific names, signify to others, v.g. man, horse, sun, water, iron: upon hearing which words, every one who understands the language, frames in his mind a combination of those several simple ideas which he has usually observed, or fancied to exist together under that denomination; all which he supposes to rest in and be, as it were, adherent to that unknown common subject, which inheres not in anything else. Though, in the meantime,

it be manifest, and every one, upon inquiry into his own thoughts, will find, that he has no other idea of any substance, v.g. let it be gold, horse, iron, man, vitriol, bread, but what he has barely of those sensible qualities, which he supposes to inhere; with a supposition of such a substratum as gives, as it were, a support to those qualities or simple ideas, which he has observed to exist united together. Thus, the idea of the sun,—what is it but an aggregate of those several simple ideas, bright, hot, roundish, having a constant regular motion, at a certain distance from us, and perhaps some other: as he who thinks and discourses of the sun has been more or less accurate in observing those sensible qualities, ideas, or properties, which are in that thing which he calls the sun.

7. Their active and passive powers a great part of our complex ideas of substances. For he has the perfectest idea of any of the particular sorts of substances, who has gathered, and put together, most of those simple ideas which do exist in it; among which are to be reckoned its active powers, and passive capacities, which,

though not simple ideas, yet in this respect, for brevity's sake, may conveniently enough be reckoned amongst them. Thus, the power of drawing iron is one of the ideas of the complex one of that substance we call a loadstone; and a power to be so drawn is a part of the complex one we call iron: which powers pass for inherent qualities in those subjects. Because every substance, being as apt, by the powers we observe in it, to change some sensible qualities in other subjects, as it is to produce in us those simple ideas which we receive immediately from it, does, by those new sensible qualities introduced into other subjects, discover to us those powers which do thereby mediately affect our senses, as regularly as its sensible qualities do it immediately: v.g. we immediately by our senses perceive in fire its heat and colour; which are, if rightly considered, nothing but powers in it to produce those ideas in us: we also by our senses perceive the colour and brittleness of charcoal, whereby we come by the knowledge of another power in fire, which it has to change the colour and consistency of wood. By the former, fire immediately,

by the latter, it mediately discovers to us these several powers; which therefore we look upon to be a part of the qualities of fire, and so make them a part of the complex idea of it. For all those powers that we take cognizance of, terminating only in the alteration of some sensible qualities in those subjects on which they operate, and so making them exhibit to us new sensible ideas, therefore it is that I have reckoned these powers amongst the simple ideas which make the complex ones of the sort? of substances; though these powers considered in themselves, are truly complex ideas. And in this looser sense I crave leave to be understood, when I name any of these potentialities among the simple ideas which we recollect in our minds when we think of particular substances. For the powers that are severally in them are necessary to be considered, if we will have true distinct notions of the several sorts of substances.

8. And why. Nor are we to wonder that powers make a great part of our complex ideas of substances; since their secondary qualities are those which in most of them serve principally to distinguish substances one from another,

and commonly make a considerable part of the complex idea of the several sorts of them. For, our senses failing us in the discovery of the bulk, texture, and figure of the minute parts of bodies, on which their real constitutions and differences depend, we are fain to make use of their secondary qualities as the characteristic notes and marks whereby to frame ideas of them in our minds, and distinguish them one from another: all which secondary qualities, as has been shown, are nothing but bare powers. For the colour and taste of opium are, as well as its soporific or anodyne virtues, mere powers, depending on its primary qualities, whereby it is fitted to produce different operations on different parts of our bodies.

9. Three sorts of ideas make our complex ones of corporeal substances. The ideas that make our complex ones of corporeal substances, are of these three sorts. First, the ideas of the primary qualities of things, which are discovered by our senses, and are in them even when we perceive them not; such are the bulk, figure, number, situation, and motion of the parts of bodies; which are really in them, whether we take notice of them or not.

Secondly, the sensible secondary qualities, which, depending on these, are nothing but the powers those substances have to produce several ideas in us by our senses; which ideas are not in the things themselves, otherwise than as anything is in its cause. Thirdly, the aptness we consider in any substance, to give or receive such alterations of primary qualities, as that the substance so altered should produce in us different ideas from what it did before; these are called active and passive powers: all which powers, as far as we have any notice or notion of them, terminate only in sensible simple ideas. For whatever alteration a loadstone has the power to make in the minute particles of iron, we should have no notion of any power it had at all to operate on iron, did not its sensible motion discover it: and I doubt not, but there are a thousand changes, that bodies we daily handle have a power to use in one another, which we never suspect, because they never appear in sensible effects.

10. Powers thus make a great part of our complex ideas of particular substances. Powers therefore justly make a

great part of our complex ideas of substances. He that will examine his complex idea of gold, will find several of its ideas that make it up to be only powers; as the power of being melted, but of not spending itself in the fire; of being dissolved in aqua regia, are ideas as necessary to make up our complex idea of gold, as its colour and weight: which, if duly considered, are also nothing but different powers. For, to speak truly, yellowness is not actually in gold, but is a power in gold to produce that idea in us by our eyes, when placed in a due light: and the heat, which we cannot leave out of our ideas of the sun, is no more really in the sun, than the white colour it introduces into wax. These are both equally powers in the sun, operating, by the motion and figure of its sensible parts, so on a man, as to make him have the idea of heat; and so on wax, as to make it capable to produce in a man the idea of white.

11. The now secondary qualities of bodies would disappear, if we could discover the primary ones of their minute parts. Had we senses acute enough to discern the minute particles of bodies, and the real constitution on which

their sensible qualities depend, I doubt not but they would produce quite different ideas in us: and that which is now the yellow colour of gold, would then disappear, and instead of it we should see an admirable texture of parts, of a certain size and figure. This microscopes plainly discover to us; for what to our naked eyes produces a certain colour, is, by thus augmenting the acuteness of our senses, discovered to be quite a different thing; and the thus altering, as it were, the proportion of the bulk of the minute parts of a coloured object to our usual sight, produces different ideas from what it did before. Thus, sand or pounded glass, which is opaque, and white to the naked eye, is pellucid in a microscope; and a hair seen in this way, loses its former colour, and is, in a great measure, pellucid, with a mixture of some bright sparkling colours, such as appear from the refraction of diamonds, and other pellucid bodies. Blood, to the naked eye, appears all red; but by a good microscope, wherein its lesser parts appear, shows only some few globules of red, swimming in a pellucid liquor, and how these red globules would appear, if glasses could be found

that could yet magnify them a thousand or ten thousand times more, is uncertain.

12. Our faculties for discovery of the qualities and powers of substances suited to our state. The infinite wise Contriver of us, and all things about us, hath fitted our senses, faculties, and organs, to the conveniences of life, and the business we have to do here. We are able, by our senses, to know and distinguish things: and to examine them so far as to apply them to our uses, and several ways to accommodate the exigences of this life. We have insight enough into their admirable contrivances and wonderful effects, to admire and magnify the wisdom, power, and goodness of their Author. Such a knowledge as this, which is suited to our present condition, we want not faculties to attain. But it appears not that God intended we should have a perfect, clear, and adequate knowledge of them: that perhaps is not in the comprehension of any finite being. We are furnished with faculties (dull and weak as they are) to discover enough in the creatures to lead us to the knowledge of the Creator, and the knowledge of our duty; and we are

fitted well enough with abilities to provide for the conveniences of living: these are our business in this world. But were our senses altered, and made much quicker and acuter, the appearance and outward scheme of things would have quite another face to us; and, I am apt to think, would be inconsistent with our being, or at least well-being, in this part of the universe which we inhabit. He that considers how little our constitution is able to bear a remove into parts of this air, not much higher than that we commonly breath in, will have reason to be satisfied, that in this globe of earth allotted for our mansion, the all-wise Architect has suited our organs, and the bodies that are to affect them, one to another. If our sense of hearing were but a thousand times quicker than it is, how would a perpetual noise distract us. And we should in the quietest retirement be less able to sleep or meditate than in the middle of a sea-fight. Nay, if that most instructive of our senses, seeing, were in any man a thousand or a hundred thousand times more acute than it is by the best microscope, things several millions of times less than the smallest object of

his sight now would then be visible to his naked eyes, and so he would come nearer to the discovery of the texture and motion of the minute parts of corporeal things; and in many of them, probably get ideas of their internal constitutions: but then he would be in a quite different world from other people: nothing would appear the same to him and others: the visible ideas of everything would be different. So that I doubt, whether he and the rest of men could discourse concerning the objects of sight, or have any communication about colours, their appearances being so wholly different. And perhaps such a quickness and tenderness of sight could not endure bright sunshine, or so much as open daylight; nor take in but a very small part of any object at once, and that too only at a very near distance. And if by the help of such microscopical eyes (if I may so call them) a man could penetrate further than ordinary into the secret composition and radical texture of bodies, he would not make any great advantage by the change, if such an acute sight would not serve to conduct him to the market and exchange; if he could not see things he

was to avoid, at a convenient distance; nor distinguish things he had to do with by those sensible qualities others do. He that was sharp-sighted enough to see the configuration of the minute particles of the spring of a clock, and observe upon what peculiar structure and impulse its elastic motion depends, would no doubt discover something very admirable: but if eyes so framed could not view at once the hand, and the characters of the hour-plate, and thereby at a distance see what o'clock it was, their owner could not be much benefited by that acuteness; which, whilst it discovered the secret contrivance of the parts of the machine, made him lose its use.

13. Conjecture about the corporeal organs of some spirits. And here give me leave to propose an extravagant conjecture of mine, viz. That since we have some reason (if there be any credit to be given to the report of things that our philosophy cannot account for) to imagine, that Spirits can assume to themselves bodies of different bulk, figure, and conformation of parts—whether one great advantage some of them have over us may not lie in this, that they can so frame and shape to them-

selves organs of sensation or perception, as to suit them to their present design, and the circumstances of the object they would consider. For how much would that man exceed all others in knowledge, who had but the faculty so to alter the structure of his eyes, that one sense, as to make it capable of all the several degrees of vision which the assistance of glasses (casually at first lighted on) has taught us to conceive? What wonders would he discover, who could so fit his eyes to all sorts of objects, as to see when he pleased the figure and motion of the minute particles in the blood, and other juices of animals, as distinctly as he does, at other times, the shape and motion of the animals themselves? But to us, in our present state, unalterable organs, so contrived as to discover the figure and motion of the minute parts of bodies, whereon depend those sensible qualities we now observe in them, would perhaps be of no advantage. God has no doubt made them so as is best for us in our present condition. He hath fitted us for the neighbourhood of the bodies that surround us, and we have to do with; and though we cannot, by the facul-

ties we have, attain to a perfect knowledge of things, yet they will serve us well enough for those ends above-mentioned, which are our great concernment. I beg my reader's pardon for laying before him so wild a fancy concerning the ways of perception of beings above us; but how extravagant soever it be, I doubt whether we can imagine anything about the knowledge of angels but after this manner, some way or other in proportion to what we find and observe in ourselves. And though we cannot but allow that the infinite power and wisdom of God may frame creatures with a thousand other faculties and ways of perceiving things without them than what we have, yet our thoughts can go no further than our own: so impossible it is for us to enlarge our very guesses beyond the ideas received from our own sensation and reflection. The supposition, at least, that angels do sometimes assume bodies, needs not startle us; since some of the most ancient and most learned Fathers of the church seemed to believe that they had bodies: and this is certain, that their state and way of existence is unknown to us.

14. Our specific ideas of substances. But to return to the matter in hand,—the ideas we have of substances, and the ways we come by them. I say, our specific ideas of substances are nothing else but a collection of a certain number of simple ideas, considered as united in one thing. These ideas of substances, though they are commonly simple apprehensions, and the names of them simple terms, yet in effect are complex and compounded. Thus the idea which an Englishman signifies by the name swan, is white colour, long neck, red beak, black legs, and whole feet, and all these of a certain size, with a power of swimming in the water, and making a certain kind of noise, and perhaps, to a man who has long observed this kind of birds, some other properties: which all terminate in sensible simple ideas, all united in one common subject.

15. Our ideas of spiritual substances, as clear as of bodily substances. Besides the complex ideas we have of material sensible substances, of which I have last spoken,—by the simple ideas we have taken from those operations of our own minds, which we experiment daily in our-

selves, as thinking, understanding, willing, knowing, and power of beginning motion, &c., co-existing in some substance, we are able to frame the complex idea of an immaterial spirit. And thus, by putting together the ideas of thinking, perceiving, liberty, and power of moving themselves and other things, we have as clear a perception and notion of immaterial substances as we have of material. For putting together the ideas of thinking and willing, or the power of moving or quieting corporeal motion, joined to substance, of which we have no distinct idea, we have the idea of an immaterial spirit; and by putting together the ideas of coherent solid parts, and a power of being moved, joined with substance, of which likewise we have no positive idea, we have the idea of matter. The one is as clear and distinct an idea as the other: the idea of thinking, and moving a body, being as clear and distinct ideas as the ideas of extension, solidity, and being moved. For our idea of substance is equally obscure, or none at all, in both; it is but a supposed I know not what, to support those ideas we call accidents. It is for want reflection that we are

apt to think that our senses show us nothing but material things. Every act of sensation, when duly considered, gives us an equal view of both parts of nature, the corporeal and spiritual. For whilst I know, by seeing or hearing, &c., that there is some corporeal being without me, the object of that sensation, I do more certainly know, that there is some spiritual being within me that sees and hears. This, I must be convinced, cannot be the action of bare insensible matter; nor ever could be, without an immaterial thinking being.

16. No idea of abstract substance either in body or spirit. By the complex idea of extended, figured, coloured, and all other sensible qualities, which is all that we know of it, we are as far from the idea of the substance of body, as if we knew nothing at all: nor after all the acquaintance and familiarity which we imagine we have with matter, and the many qualities men assure themselves they perceive and know in bodies, will it perhaps upon examination be found, that they have any more or clearer primary ideas belonging to body, than they have belonging to immaterial spirit.

17. Cohesion of solid parts and impulse, the primary ideas peculiar to body. The primary ideas we have peculiar to body, as contradistinguished to spirit, are the cohesion of solid, and consequently separable, parts, and a power of communicating motion by impulse. These, I think, are the original ideas proper and peculiar to body; for figure is but the consequence of finite extension.

18. Thinking and motivity the primary ideas peculiar to spirit. The ideas we have belonging and peculiar to spirit, are thinking, and will, or a power of putting body into motion by thought, and, which is consequent to it, liberty. For, as body cannot but communicate its motion by impulse to another body, which it meets with at rest, so the mind can put bodies into motion, or forbear to do so, as it pleases. The ideas of existence, duration, and mobility, are common to them both.

19. Spirits capable of motion. There is no reason why it should be thought strange, that I make mobility belong to spirit; for having no other idea of motion, but change of distance with other beings that are considered as at rest; and finding that spirits, as well as bodies, cannot

operate but where they are; and that spirits do operate at several times in several places, I cannot but attribute change of place to all finite spirits: (for of the Infinite Spirit I speak not here). For my soul, being a real being as well as my body, is certainly as capable of changing distance with any other body, or being, as body itself; and so is capable of motion. And if a mathematician can consider a certain distance, or a change of that distance between two points, one may certainly conceive a distance, and a change of distance, between two spirits; and so conceive their motion, their approach or removal, one from another.

20. Proof of this. Every one finds in himself that his soul can think, will, and operate on his body in the place where that is, but cannot operate on a body, or in a place, an hundred miles distant from it. Nobody can imagine that his soul can think or move a body at Oxford, whilst he is at London; and cannot but know, that, being united to his body, it constantly changes place all the whole journey between Oxford and London, as the coach or horse does that carries him, and I think may

be said to be truly all that while in motion: or if that will not be allowed to afford us a clear idea enough of its motion, its being separated from the body in death, I think, will; for to consider it as going out of the body, or leaving it, and yet to have no idea of its motion, seems to me impossible.

21. God immoveable, because infinite. If it be said by any one that it cannot change place, because it hath none, for the spirits are not in loco, but ubi; I suppose that way of talking will not now be of much weight to many, in an age that is not much disposed to admire, or suffer themselves to be deceived by such unintelligible ways of speaking. But if any one thinks there is any sense in that distinction, and that it is applicable to our present purpose, I desire him to put it into intelligible English; and then from thence draw a reason to show that immaterial spirits are not capable of motion. Indeed motion cannot be attributed to God; not because he is an immaterial, but because he is an infinite spirit.

22. Our complex idea of an immaterial spirit and our complex idea of body compared. Let us compare, then,

our complex idea of an immaterial spirit with our complex idea of body, and see whether there be any more obscurity in one than in the other, and in which most. Our idea of body, as I think, is an extended solid substance, capable of communicating motion by impulse: and our idea of soul, as an immaterial spirit, is of a substance that thinks, and has a power of exciting motion in body, by willing, or thought. These, I think, are our complex ideas of soul and body, as contradistinguished; and now let us examine which has most obscurity in it, and difficulty to be apprehended. I know that people whose thoughts are immersed in matter, and have so subjected their minds to their senses that they seldom reflect on anything beyond them, are apt to say, they cannot comprehend a thinking thing, which perhaps is true: but I affirm, when they consider it well, they can no more comprehend an extended thing.

23. Cohesion of solid parts in body as hard to be conceived as thinking in a soul. If any one says he knows not what it is thinks in him, he means he knows not what the substance is of that thinking thing: No more,

say I, knows he what the substance is of that solid thing. Further, if he says he knows not how he thinks, I answer, Neither knows he how he is extended, how the solid parts of body are united, or cohere together to make extension. For though the pressure of the particles of air may account for the cohesion of several parts of matter that are grosser than the particles of air, and have pores less than the corpuscles of air, yet the weight or pressure of the air will not explain, nor can be a cause of the coherence of the particles of air themselves. And if the pressure of the aether, or any subtiler matter than the air, may unite, and hold fast together, the parts of a particle of air, as well as other bodies, yet it cannot make bonds for itself, and hold together the parts that make up every the least corpuscle of that *materia subtilis*. So that that hypothesis, how ingeniously soever explained, by showing that the parts of sensible bodies are held together by the pressure of other external insensible bodies, reaches not the parts of the aether itself; and by how much the more evident it proves, that the parts of other bodies are held together by the

external pressure of the aether, and can have no other conceivable cause of their cohesion and union, by so much the more it leaves us in the dark concerning the cohesion of the parts of the corpuscles of the aether itself: which we can neither conceive without parts, they being bodies, and divisible, nor yet how their parts cohere, they wanting that cause of cohesion which is given of the cohesion of the parts of all other bodies.

24. Not explained by an ambient fluid. But, in truth, the pressure of any ambient fluid, how great soever, can be no intelligible cause of the cohesion of the solid parts of matter. For, though such a pressure may hinder the avulsion of two polished superficies, one from another, in a line perpendicular to them, as in the experiment of two polished marbles; yet it can never in the least hinder the separation by a motion, in a line parallel to those surfaces. Because the ambient fluid, having a full liberty to succeed in each point of space, deserted by a lateral motion, resists such a motion of bodies, so joined, no more than it would resist the motion of that body were it on all sides environed by that fluid, and touched

no other body; and therefore, if there were no other cause of cohesion, all parts of bodies must be easily separable by such a lateral sliding motion. For if the pressure of the aether be the adequate cause of cohesion, wherever that cause operates not, there can be no cohesion. And since it cannot operate against a lateral separation, (as has been shown), therefore in every imaginary plane, intersecting any mass of matter, there could be no more cohesion than of two polished surfaces, which will always, notwithstanding any imaginable pressure of a fluid, easily slide one from another.

So that perhaps, how clear an idea soever we think we have of the extension of body, which is nothing but the cohesion of solid parts, he that shall well consider it in his mind, may have reason to conclude, That it is as easy for him to have a clear idea how the soul thinks as how body is extended. For, since body is no further, nor otherwise, extended, than by the union and cohesion of its solid parts, we shall very ill comprehend the extension of body, without understanding wherein consists the union and cohesion of its parts; which seems to me

as incomprehensible as the manner of thinking, and how it is performed.

25. We can as little understand how the parts cohere in extension, as how our spirits perceive or move. I allow it is usual for most people to wonder how any one should find a difficulty in what they think they every day observe. Do we not see (will they be ready to say) the parts of bodies stick firmly together? Is there anything more common? And what doubt can there be made of it? And the like, I say, concerning thinking and voluntary motion. Do we not every moment experiment it in ourselves, and therefore can it be doubted? The matter of fact is clear, I confess; but when we would a little nearer look into it, and consider how it is done, there I think we are at a loss, both in the one and the other; and can as little understand how the parts of body cohere, as how we ourselves perceive or move. I would have any one intelligibly explain to me, how the parts of gold, or brass, (that but now in fusion were as loose from one another as the particles of water, or the sands of an hour-glass), come in a few moments to be so united,

and adhere so strongly one to another, that the utmost force of men's arms cannot separate them? A considering man will, I suppose, be here at a loss to satisfy his own, or another man's understanding.

26. The cause of coherence of atoms in extended substances incomprehensible. The little bodies that compose that fluid we call water, are so extremely small, that I have never heard of any one, who, by a microscope, (and yet I have heard of some that have magnified to ten thousand; nay, to much above a hundred thousand times), pretended to perceive their distinct bulk, figure, or motion; and the particles of water are also so perfectly loose one from another, that the least force sensibly separates them. Nay, if we consider their perpetual motion, we must allow them to have no cohesion one with another; and yet let but a sharp cold come, and they unite, they consolidate; these little atoms cohere, and are not, without great force, separable. He that could find the bonds that tie these heaps of loose little bodies together so firmly; he that could make known the cement that makes them stick so fast one to

another, would discover a great and yet unknown secret: and yet when that was done, would he be far enough from making the extension of body (which is the cohesion of its solid parts) intelligible, till he could show wherein consisted the union, or consolidation of the parts of those bonds, or of that cement, or of the least particle of matter that exists. Whereby it appears that this primary and supposed obvious quality of body will be found, when examined, to be as incomprehensible as anything belonging to our minds, and a solid extended substance as hard to be conceived as a thinking immaterial one, whatever difficulties some would raise against it.

27. The supposed pressure brought to explain cohesion is unintelligible. For, to extend our thoughts a little further, that pressure which is brought to explain the cohesion of bodies is as unintelligible as the cohesion itself. For if matter be considered, as no doubt it is, finite, let any one send his contemplation to the extremities of the universe, and there see what conceivable hoops, what bond he can imagine to hold this mass

of matter in so close a pressure together; from whence steel has its firmness, and the parts of a diamond their hardness and indissolubility. If matter be finite, it must have its extremes; and there must be something to hinder it from scattering asunder. If, to avoid this difficulty, any one will throw himself into the supposition and abyss of infinite matter, let him consider what light he thereby brings to the cohesion of body, and whether he be ever the nearer making it intelligible, by resolving it into a supposition the most absurd and most incomprehensible of all other: so far is our extension of body (which is nothing but the cohesion of solid parts) from being clearer, or more distinct, when we would inquire into the nature, cause, or manner of it, than the idea of thinking.

28. Communication of motion by impulse, or by thought, equally unintelligible. Another idea we have of body is, the power of communication of motion by impulse; and of our souls, the power of exciting motion by thought. These ideas, the one of body, the other of our minds, every day's experience clearly furnishes us with: but if

here again we inquire how this is done, we are equally in the dark. For, in the communication of motion by impulse, wherein as much motion is lost to one body as is got to the other, which is the ordinarist case, we can have no other conception, but of the passing of motion out of one body into another; which, I think, is as obscure and inconceivable as how our minds move or stop our bodies by thought, which we every moment find they do. The increase of motion by impulse, which is observed or believed sometimes to happen, is yet harder to be understood. We have by daily experience clear evidence of motion produced both by impulse and by thought; but the manner how, hardly comes within our comprehension: we are equally at a loss in both. So that, however we consider motion, and its communication, either from body or spirit, the idea which belongs to spirit is at least as clear as that which belongs to body. And if we consider the active power of moving, or, as I may call it, motivity, it is much clearer in spirit than body; since two bodies, placed by one another at rest, will never afford us the idea of a power in the one

to move the other, but by a borrowed motion: whereas the mind every day affords us ideas of an active power of moving of bodies; and therefore it is worth our consideration, whether active power be not the proper attribute of spirits, and passive power of matter. Hence may be conjectured that created spirits are not totally separate from matter, because they are both active and passive. Pure spirit, viz. God, is only active; pure matter is only passive; those beings that are both active and passive, we may judge to partake of both. But be that as it will, I think, we have as many and as clear ideas belonging to spirit as we have belonging to body, the substance of each being equally unknown to us; and the idea of thinking in spirit, as clear as of extension in body; and the communication of motion by thought, which we attribute to spirit, is as evident as that by impulse, which we ascribe to body. Constant experience makes us sensible of both these, though our narrow understandings can comprehend neither. For, when the mind would look beyond those original ideas we have from sensation or reflection, and penetrate into their

causes, and manner of production, we find still it discovers nothing but its own short-sightedness.

29. Summary. To conclude. Sensation convinces us that there are solid extended substances; and reflection, that there are thinking ones: experience assures us of the existence of such beings, and that the one hath a power to move body by impulse, the other by thought; this we cannot doubt of. Experience, I say, every moment furnishes us with the clear ideas both of the one and the other. But beyond these ideas, as received from their proper sources, our faculties will not reach. If we would inquire further into their nature, causes, and manner, we perceive not the nature of extension clearer than we do of thinking. If we would explain them any further, one is as easy as the other; and there is no more difficulty to conceive how a substance we know not should, by thought, set body into motion, than how a substance we know not should, by impulse, set body into motion. So that we are no more able to discover wherein the ideas belonging to body consist, than those belonging to spirit. From whence it seems probable to me, that

the simple ideas we receive from sensation and reflection are the boundaries of our thoughts; beyond which the mind, whatever efforts it would make, is not able to advance one jot; nor can it make any discoveries, when it would pry into the nature and hidden causes of those ideas.

30. Our idea of spirit and our idea of body compared. So that, in short, the idea we have of spirit, compared with the idea we have of body, stands thus: the substance of spirits is unknown to us; and so is the substance of body equally unknown to us. Two primary qualities or properties of body, viz. solid coherent parts and impulse, we have distinct clear ideas of: so likewise we know, and have distinct clear ideas, of two primary qualities or properties of spirit, viz. thinking, and a power of action; i.e. a power of beginning or stopping several thoughts or motions. We have also the ideas of several qualities inherent in bodies, and have the clear distinct ideas of them; which qualities are but the various modifications of the extension of cohering solid parts, and their motion. We have likewise the ideas of the several

modes of thinking viz. believing, doubting, intending, fearing, hoping; all which are but the several modes of thinking. We have also the ideas of willing, and moving the body consequent to it, and with the body itself too; for, as has been shown, spirit is capable of motion.

31. The notion of spirit involves no more difficulty in it than that of body. Lastly, if this notion of immaterial spirit may have, perhaps, some difficulties in it not easily to be explained, we have therefore no more reason to deny or doubt the existence of such spirits, than we have to deny or doubt the existence of body; because the notion of body is cumbered with some difficulties very hard, and perhaps impossible to be explained or understood by us. For I would fain have instanced anything in our notion of spirit more perplexed, or nearer a contradiction, than the very notion of body includes in it; the divisibility in infinitum of any finite extension involving us, whether we grant or deny it, in consequences impossible to be explicated or made in our apprehensions consistent; consequences that carry greater difficulty, and more apparent absurdity, than anything

can follow from the notion of an immaterial knowing substance.

32. We know nothing of things beyond our simple ideas of them. Which we are not at all to wonder at, since we having but some few superficial ideas of things, discovered to us only by the senses from without, or by the mind, reflecting on what it experiments in itself within, have no knowledge beyond that, much less of the internal constitution, and true nature of things, being destitute of faculties to attain it. And therefore experimenting and discovering in ourselves knowledge, and the power of voluntary motion, as certainly as we experiment, or discover in things without us, the cohesion and separation of solid parts, which is the extension and motion of bodies; we have as much reason to be satisfied with our notion of immaterial spirit, as with our notion of body, and the existence of the one as well as the other. For it being no more a contradiction that thinking should exist separate and independent from solidity, than it is a contradiction that solidity should exist separate and independent from thinking, they be-

ing both but simple ideas, independent one from another: and having as clear and distinct ideas in us of thinking, as of solidity, I know not why we may not as well allow a thinking thing without solidity, i.e. immaterial, to exist, as a solid thing without thinking, i.e. matter, to exist; especially since it is not harder to conceive how thinking should exist without matter, than how matter should think. For whensoever we would proceed beyond these simple ideas we have from sensation and reflection, and dive further into the nature of things, we fall presently into darkness and obscurity, perplexedness and difficulties, and can discover nothing further but our own blindness and ignorance. But whichever of these complex ideas be clearest, that of body, or immaterial spirit, this is evident, that the simple ideas that make them up are no other than what we have received from sensation or reflection: and so is it of all our other ideas of substances, even of God himself.

33. Our complex idea of God. For if we examine the idea we have of the incomprehensible Supreme Being, we shall find that we come by it the same way; and that the

complex ideas we have both of God, and separate spirits, are made of the simple ideas we receive from reflection: v.g. having, from what we experiment in ourselves, got the ideas of existence and duration; of knowledge and power; of pleasure and happiness; and of several other qualities and powers, which it is better to have than to be without; when we would frame an idea the most suitable we can to the Supreme Being, we enlarge every one of these with our idea of infinity; and so putting them together, make our complex idea of God. For that the mind has such a power of enlarging some of its ideas, received from sensation and reflection, has been already shown.

34. Our complex idea of God as infinite. If I find that I know some few things, and some of them, or all, perhaps imperfectly, I can frame an idea of knowing twice as many; which I can double again, as often as I can add to number; and thus enlarge my idea of knowledge, by extending its comprehension to all things existing, or possible. The same also I can do of knowing them more perfectly; i.e. all their qualities, powers, causes, conse-

quences, and relations, &c., till all be perfectly known that is in them, or can any way relate to them: and thus frame the idea of infinite or boundless knowledge. The same may also be done of power, till we come to that we call infinite; and also of the duration of existence, without beginning or end, and so frame the idea of an eternal being. The degrees or extent wherein we ascribe existence, power, wisdom, and all other perfections (which we can have any ideas of) to that sovereign Being, which we call God, being all boundless and infinite, we frame the best idea of him our minds are capable of: all which is done, I say, by enlarging those simple ideas we have taken from the operations of our own minds, by reflection; or by our senses, from exterior things, to that vastness to which infinity can extend them.

35. God in his own essence incognisable. For it is infinity, which, joined to our ideas of existence, power, knowledge, &c., makes that complex idea, whereby we represent to ourselves, the best we can, the Supreme Being. For, though in his own essence (which certainly we do not know, not knowing the real essence of a pebble, or

a fly, or of our own selves) God be simple and uncompounded; yet I think I may say we have no other idea of him, but a complex one of existence, knowledge, power, happiness, &c., infinite and eternal: which are all distinct ideas, and some of them, being relative, are again compounded of others: all which being, as has been shown, originally got from sensation and reflection, go to make up the idea or notion we have of God.

36. No ideas in our complex ideas of spirits, but those got from sensation or reflection. This further is to be observed, that there is no idea we attribute to God, bating infinity, which is not also a part of our complex idea of other spirits. Because, being capable of no other simple ideas, belonging to anything but body, but those which by reflection we receive from the operation of our own minds, we can attribute to spirits no other but what we receive from thence: and all the difference we can put between them, in our contemplation of spirits, is only in the several extents and degrees of their knowledge, power, duration, happiness, &c. For that in our ideas, as well of spirits as of other things, we are re-

strained to those we receive from sensation and reflection, is evident from hence,—That, in our ideas of spirits, how much soever advanced in perfection beyond those of bodies, even to that of infinite, we cannot yet have any idea of the manner wherein they discover their thoughts one to another: though we must necessarily conclude that separate spirits, which are beings that have perfecter knowledge and greater happiness than we, must needs have also a perfecter way of communicating their thoughts than we have, who are fain to make use of corporeal signs, and particular sounds; which are therefore of most general use, as being the best and quickest we are capable of. But of immediate communication having no experiment in ourselves, and consequently no notion of it at all, we have no idea how spirits, which use not words, can with quickness, or much less how spirits that have no bodies can be masters of their own thoughts, and communicate or conceal them at pleasure, though we cannot but necessarily suppose they have such a power.

37. Recapitulation. And thus we have seen what kind

of ideas we have of substances of all kinds, wherein they consist, and how we came by them. From whence, I think, it is very evident,

First, That all our ideas of the several sorts of substances are nothing but collections of simple ideas: with a supposition of something to which they belong, and in which they subsist: though of this supposed something we have no clear distinct idea at all.

Secondly, That all the simple ideas, that thus united in one common substratum, make up our complex ideas of several sorts of substances, are no other but such as we have received from sensation or reflection. So that even in those which we think we are most intimately acquainted with, and that come nearest the comprehension of our most enlarged conceptions, we cannot go beyond those simple ideas. And even in those which seem most remote from all we have to do with, and do infinitely surpass anything we can perceive in ourselves by reflection; or discover by sensation in other things, we can attain to nothing but those simple ideas, which we originally received from sensation or reflection; as is

evident in the complex ideas we have of angels, and particularly of God himself.

Thirdly, That most of the simple ideas that make up our complex ideas of substances, when truly considered, are only powers, however we are apt to take them for positive qualities; v.g. the greatest part of the ideas that make our complex idea of gold are yellowness, great weight, ductility, fusibility, and solubility in aqua regia, &c., all united together in an unknown substratum: all which ideas are nothing else but so many relations to other substances; and are not really in the gold, considered barely in itself, though they depend on those real and primary qualities of its internal constitution, whereby it has a fitness differently to operate, and be operated on by several other substances.

Chapter XXIV

Of Collective Ideas of Substances

1. A collective idea is one idea. Besides these complex ideas of several single substances, as of man, horse, gold,

violet, apple, &c., the mind hath also complex collective ideas of substances; which I so call, because such ideas are made up of many particular substances considered together, as united into one idea, and which so joined are looked on as one; v.g. the idea of such a collection of men as make an army, though consisting of a great number of distinct substances, is as much one idea as the idea of a man: and the great collective idea of all bodies whatsoever, signified by the name world, is as much one idea as the idea of any the least particle of matter in it; it sufficing to the unity of any idea, that it be considered as one representation or picture, though made up of ever so many particulars.

2. Made by the power of composing in the mind. These collective ideas of substances the mind makes, by its power of composition, and uniting severally either simple or complex ideas into one, as it does, by the same faculty, make the complex ideas of particular substances, consisting of an aggregate of divers simple ideas, united in one substance. And as the mind, by putting together the repeated ideas of unity, makes the collective mode,

or complex idea, of any number, as a score, or a gross, &c.,—so, by putting together several particular substances, it makes collective ideas of substances, as a troop, an army, a swarm, a city, a fleet; each of which every one finds that he represents to his own mind by one idea, in one view; and so under that notion considers those several things as perfectly one, as one ship, or one atom. Nor is it harder to conceive how an army of ten thousand men should make one idea, than how a man should make one idea; it being as easy to the mind to unite into one the idea of a great number of men, and consider it as one, as it is to unite into one particular all the distinct ideas that make up the composition of a man, and consider them all together as one.

3. Artificial things that are made up of distinct substances are our collective ideas. Amongst such kind of collective ideas are to be counted most part of artificial things, at least such of them as are made up of distinct substances: and, in truth, if we consider all these collective ideas aright, as army, constellation, universe, as they are united into so many single ideas, they are but

the artificial draughts of the mind; bringing things very remote, and independent on one another, into one view, the better to contemplate and discourse of them, united into one conception, and signified by one name. For there are no things so remote, nor so contrary, which the mind cannot, by this art of composition, bring into one idea; as is visible in that signified by the name universe.

Chapter XXV

Of Relation

1. Relation, what. Besides the ideas, whether simple or complex, that the mind has of things as they are in themselves, there are others it gets from their comparison one with another. The understanding, in the consideration of anything, is not confined to that precise object: it can carry an idea as it were beyond itself, or at least look beyond it, to see how it stands in conformity to any other. When the mind so considers one thing, that it does as it were bring it to, and set it by another,

and carries its view from one to the other—this is, as the words import, relation and respect; and the denominations given to positive things, intimating that respect, and serving as marks to lead the thoughts beyond the subject itself denominated to something distinct from it, are what we call relatives; and the things so brought together, related. Thus, when the mind considers Caius as such a positive being, it takes nothing into that idea but what really exists in Caius; v.g. when I consider him as a man, I have nothing in my mind but the complex idea of the species, man. So likewise, when I say Caius is a white man, I have nothing but the bare consideration of a man who hath that white colour. But when I give Caius the name husband, I intimate some other person; and when I give him the name whiter, I intimate some other thing: in both cases my thought is led to something beyond Caius, and there are two things brought into consideration. And since any idea, whether simple or complex, may be the occasion why the mind thus brings two things together, and as it were takes a view of them at once, though still considered as distinct:

therefore any of our ideas may be the foundation of relation. As in the above-mentioned instance, the contract and ceremony of marriage with Sempronia is the occasion of the denomination and relation of husband; and the colour white the occasion why he is said to be whiter than free-stone.

2. Ideas of relations without correlative terms, not easily apprehended. These and the like relations, expressed by relative terms that have others answering them, with a reciprocal intimation, as father and son, bigger and less, cause and effect, are very obvious to every one, and everybody at first sight perceives the relation. For father and son, husband and wife, and such other correlative terms, seem so nearly to belong one to another, and, through custom, do so readily chime and answer one another in people's memories, that, upon the naming of either of them, the thoughts are presently carried beyond the thing so named; and nobody overlooks or doubts of a relation, where it is so plainly intimated. But where languages have failed to give correlative names, there the relation is not always so easily taken notice

of. Concubine is, no doubt, a relative name, as well as a wife: but in languages where this and the like words have not a correlative term, there people are not so apt to take them to be so, as wanting that evident mark of relation which is between correlatives, which seem to explain one another, and not to be able to exist, but together. Hence it is, that many of those names, which, duly considered, do include evident relations, have been called external denominations. But all names that are more than empty sounds must signify some idea, which is either in the thing to which the name is applied, and then it is positive, and is looked on as united to and existing in the thing to which the denomination is given; or else it arises from the respect the mind finds in it to something distinct from it, with which it considers it, and then it includes a relation.

3. Some seemingly absolute terms contain relations. Another sort of relative terms there is, which are not looked on to be either relative, or so much as external denominations: which yet, under the form and appearance of signifying something absolute in the subject, do

conceal a tacit, though less observable, relation. Such are the seemingly positive terms of old, great, imperfect, &c., whereof I shall have occasion to speak more at large in the following chapters.

4. Relation different from the things related. This further may be observed, That the ideas of relation may be the same in men who have far different ideas of the things that are related, or that are thus compared: v.g. those who have far different ideas of a man, may yet agree in the notion of a father; which is a notion superinduced to the substance, or man, and refers only to an act of that thing called man whereby he contributed to the generation of one of his own kind, let man be what it will.

5. Change of relation may be without any change in the things related. The nature therefore of relation consists in the referring or comparing two things one to another; from which comparison one or both comes to be denominated. And if either of those things be removed, or cease to be, the relation ceases, and the denomination consequent to it, though the other receive

in itself no alteration at all: v.g. Caius, whom I consider to-day as a father, ceases to be so to-morrow, only by the death of his son, without any alteration made in himself. Nay, barely by the mind's changing the object to which it compares anything, the same thing is capable of having contrary denominations at the same time: v.g. Caius, compared to several persons, may be truly be said to be older and younger, stronger and weaker, &c.

6. Relation only betwixt two things. Whatsoever doth or can exist, or be considered as one thing is positive: and so not only simple ideas and substances, but modes also, are positive beings: though the parts of which they consist are very often relative one to another: but the whole together considered as one thing, and producing in us the complex idea of one thing, which idea is in our minds, as one picture, though an aggregate of divers parts, and under one name, it is a positive or absolute thing, or idea. Thus a triangle, though the parts thereof compared one to another be relative, yet the idea of the whole is a positive absolute idea. The same may be said of a family, a tune, &c.; for there can be no relation but

betwixt two things considered as two things. There must always be in relation two ideas or things, either in themselves really separate, or considered as distinct, and then a ground or occasion for their comparison.

7. All things capable of relation. Concerning relation in general, these things may be considered:

First, That there is no one thing, whether simple idea, substance, mode, or relation, or name of either of them, which is not capable of almost an infinite number of considerations in reference to other things: and therefore this makes no small part of men's thoughts and words: v.g. one single man may at once be concerned in, and sustain all these following relations, and many more, viz. father, brother, son, grandfather, grandson, father-in-law, son-in-law, husband, friend, enemy, subject, general, judge, patron, client, professor, European, Englishman, islander, servant, master, possessor, captain, superior, inferior, bigger, less, older, younger, contemporary, like, unlike, &c., to an almost infinite number: he being capable of as many relations as there can be occasions of comparing him to other things, in any

manner of agreement, disagreement, or respect whatsoever. For, as I said, relation is a way of comparing or considering two things together, and giving one or both of them some appellation from that comparison; and sometimes giving even the relation itself a name.

8. Our ideas of relations often clearer than of the subjects related. Secondly, This further may be considered concerning relation, that though it be not contained in the real existence of things, but something extraneous and superinduced, yet the ideas which relative words stand for are often clearer and more distinct than of those substances to which they do belong. The notion we have of a father or brother is a great deal clearer and more distinct than that we have of a man; or, if you will, paternity is a thing whereof it is easier to have a clear idea, than of humanity; and I can much easier conceive what a friend is, than what God; because the knowledge of one action, or one simple idea, is oftentimes sufficient to give me the notion of a relation; but to the knowing of any substantial being, an accurate collection of sundry ideas is necessary. A man, if he compares

two things together, can hardly be supposed not to know what it is wherein he compares them: so that when he compares any things together, he cannot but have a very clear idea of that relation. The ideas, then, of relations, are capable at least of being more perfect and distinct in our minds than those of substances. Because it is commonly hard to know all the simple ideas which are really in any substance, but for the most part easy enough to know the simple ideas that make up any relation I think on, or have a name for: v.g. comparing two men in reference to one common parent, it is very easy to frame the ideas of brothers, without having yet the perfect idea of a man. For significant relative words, as well as others, standing only for ideas; and those being all either simple, or made up of simple ones, it suffices for the knowing the precise idea the relative term stands for, to have a clear conception of that which is the foundation of the relation; which may be done without having a perfect and clear idea of the thing it is attributed to. Thus, having the notion that one laid the egg out of which the other was hatched, I have a clear idea of the relation of dam

and chick between the two cassiowaries in St. James's Park; though perhaps I have but a very obscure and imperfect idea of those birds themselves.

9. Relations all terminate in simple ideas. Thirdly, Though there be a great number of considerations wherein things may be compared one with another, and so a multitude of relations, yet they all terminate in, and are concerned about those simple ideas, either of sensation or reflection, which I think to be the whole materials of all our knowledge. To clear this, I shall show it in the most considerable relations that we have any notion of; and in some that seem to be the most remote from sense or reflection: which yet will appear to have their ideas from thence, and leave it past doubt that the notions we have of them are but certain simple ideas, and so originally derived from sense or reflection.

10. Terms leading the mind beyond the subject denominated, are relative. Fourthly, That relation being the considering of one thing with another which is extrinsic to it, it is evident that all words that necessarily lead the mind to any other ideas than are supposed really to

exist in that thing to which the words are applied are relative words: v.g. a man, black, merry, thoughtful, thirsty, angry, extended; these and the like are all absolute, because they neither signify nor intimate anything but what does or is supposed really to exist in the man thus denominated; but father, brother, king, husband, blacker, merrier, &c., are words which, together with the thing they denominate, imply also something else separate and exterior to the existence of that thing.

11. All relatives made up of simple ideas. Having laid down these premises concerning relation in general, I shall now proceed to show, in some instances, how all the ideas we have of relation are made up, as the others are, only of simple ideas; and that they all, how refined or remote from sense soever they seem, terminate at last in simple ideas. I shall begin with the most comprehensive relation, wherein all things that do, or can exist, are concerned, and that is the relation of cause and effect: the idea whereof, how derived from the two fountains of all our knowledge, sensation and reflection, I shall in the next place consider.

Chapter XXVI

Of Cause and Effect, and other Relations

1. Whence the ideas of cause and effect got. In the notice that our senses take of the constant vicissitude of things, we cannot but observe that several particular, both qualities and substances, begin to exist; and that they receive this their existence from the due application and operation of some other being. From this observation we get our ideas of cause and effect. That which produces any simple or complex idea we denote by the general name, cause, and that which is produced, effect. Thus, finding that in that substance which we call wax, fluidity, which is a simple idea that was not in it before, is constantly produced by the application of a certain degree of heat we call the simple idea of heat, in relation to fluidity in wax, the cause of it, and fluidity the effect. So also, finding that the substance, wood, which is a certain collection of simple ideas so called, by the application of fire, is turned into another substance, called ashes; i.e., another complex idea, consisting of a

collection of simple ideas, quite different from that complex idea which we call wood; we consider fire, in relation to ashes, as cause, and the ashes, as effect. So that whatever is considered by us to conduce or operate to the producing any particular simple idea, or collection of simple ideas, whether substance or mode, which did not before exist, hath thereby in our minds the relation of a cause, and so is denominated by us.

2. Creation, generation, making, alteration. Having thus, from what our senses are able to discover in the operations of bodies on one another, got the notion of cause and effect, viz. that a cause is that which makes any other thing, either simple idea, substance, or mode, begin to be; and an effect is that which had its beginning from some other thing; the mind finds no great difficulty to distinguish the several originals of things into two sorts:—

First, When the thing is wholly made new, so that no part thereof did ever exist before; as when a new particle of matter doth begin to exist, in *rerum natura*, which had before no being, and this we call creation.

Secondly, When a thing is made up of particles, which did all of them before exist; but that very thing, so constituted of pre-existing particles, which, considered all together, make up such a collection of simple ideas, had not any existence before, as this man, this egg, rose, or cherry, &c. And this, when referred to a substance, produced in the ordinary course of nature by internal principle, but set on work by, and received from, some external agent, or cause, and working by insensible ways which we perceive not, we call generation. When the cause is extrinsical, and the effect produced by a sensible separation, or juxta-position of discernible parts, we call it making; and such are all artificial things. When any simple idea is produced, which was not in that subject before, we call it alteration. Thus a man is generated, a picture made; and either of them altered, when any new sensible quality or simple idea is produced in either of them, which was not there before: and the things thus made to exist, which were not there before, are effects; and those things which operated to the existence, causes. In which, and all other cases, we

may observe, that the notion of cause and effect has its rise from ideas received by sensation or reflection; and that this relation, how comprehensive soever, terminates at last in them. For to have the idea of cause and effect, it suffices to consider any simple idea or substance, as beginning to exist, by the operation of some other, without knowing the manner of that operation.

3. Relations of time. Time and place are also the foundations of very large relations; and all finite beings at least are concerned in them. But having already shown in another place how we get those ideas, it may suffice here to intimate, that most of the denominations of things received from time are only relations. Thus, when any one says that Queen Elizabeth lived sixty-nine, and reigned forty-five years, these words import only the relation of that duration to some other, and mean no more but this, That the duration of her existence was equal to sixty-nine, and the duration of her government to forty-five annual revolutions of the sun; and so are all words, answering, How Long? Again, William the Conqueror invaded England about the year 1066; which means this,

That, taking the duration from our Saviour's time till now for one entire great length of time, it shows at what distance this invasion was from the two extremes; and so do all words of time answering to the question, When, which show only the distance of any point of time from the period of a longer duration, from which we measure, and to which we thereby consider it as related.

4. Some ideas of time supposed positive and found to be relative. There are yet, besides those, other words of time, that ordinarily are thought to stand for positive ideas, which yet will, when considered, be found to be relative; such as are, young, old, &c., which include and intimate the relation anything has to a certain length of duration, whereof we have the idea in our minds. Thus, having settled in our thoughts the idea of the ordinary duration of a man to be seventy years, when we say a man is young, we mean that his age is yet but a small part of that which usually men attain to; and when we denominate him old, we mean that his duration is run out almost to the end of that which men do not usually exceed. And so it is but comparing the particular age or

duration of this or that man, to the idea of that duration which we have in our minds, as ordinarily belonging to that sort of animals: which is plain in the application of these names to other things; for a man is called young at twenty years, and very young at seven years old: but yet a horse we call old at twenty, and a dog at seven years, because in each of these we compare their age to different ideas of duration, which are settled in our minds as belonging to these several sorts of animals, in the ordinary course of nature. But the sun and stars, though they have outlasted several generations of men, we call not old, because we do not know what period God hath set to that sort of beings. This term belonging properly to those things which we can observe in the ordinary course of things, by a natural decay, to come to an end in a certain period of time; and so have in our minds, as it were, a standard to which we can compare the several parts of their duration; and, by the relation they bear thereunto, call them young or old; which we cannot, therefore, do to a ruby or a diamond, things whose usual periods we know not.

5. Relations of place and extension. The relation also that things have to one another in their places and distances is very obvious to observe; as above, below, a mile distant from Charing-cross, in England, and in London. But as in duration, so in extension and bulk, there are some ideas that are relative which we signify by names that are thought positive; as great and little are truly relations. For here also, having, by observation, settled in our minds the ideas of the bigness of several species of things from those we have been most accustomed to, we make them as it were the standards, whereby to denominate the bulk of others. Thus we call a great apple, such a one as is bigger than the ordinary sort of those we have been used to; and a little horse, such a one as comes not up to the size of that idea which we have in our minds to belong ordinarily to horses; and that will be a great horse to a Welchman, which is but a little one to a Fleming; they two having, from the different breed of their countries, taken several-sized ideas to which they compare, and in relation to which they denominate their great and their little.

6. Absolute terms often stand for relations. So likewise weak and strong are but relative denominations of power, compared to some ideas we have at that time of greater or less power. Thus, when we say a weak man, we mean one that has not so much strength or power to move as usually men have, or usually those of his size have; which is a comparing his strength to the idea we have of the usual strength of men, or men of such a size. The like when we say the creatures are all weak things; weak there is but a relative term, signifying the disproportion there is in the power of God and the creatures. And so abundance of words, in ordinary speech, stand only for relations (and perhaps the greatest part) which at first sight seem to have no such signification: v.g. the ship has necessary stores. Necessary and stores are both relative words; one having a relation to the accomplishing the voyage intended, and the other to future use. All which relations, how they are confined to, and terminate in ideas derived from sensation or reflection, is too obvious to need any explication.

Chapter XXVII Of Identity and Diversity

1. Wherein identity consists. Another occasion the mind often takes of comparing, is the very being of things, when, considering anything as existing at any determined time and place, we compare it with itself existing at another time, and thereon form the ideas of identity and diversity. When we see anything to be in any place in any instant of time, we are sure (be it what it will) that it is that very thing, and not another which at that same time exists in another place, how like and undistinguishable soever it may be in all other respects: and in this consists identity, when the ideas it is attributed to vary not at all from what they were that moment wherein we consider their former existence, and to which we compare the present. For we never finding, nor conceiving it possible, that two things of the same kind should exist in the same place at the same time, we rightly conclude, that, whatever exists anywhere at any time, excludes all of the same kind, and is there itself

alone. When therefore we demand whether anything be the same or no, it refers always to something that existed such a time in such a place, which it was certain, at that instant, was the same with itself, and no other. From whence it follows, that one thing cannot have two beginnings of existence, nor two things one beginning; it being impossible for two things of the same kind to be or exist in the same instant, in the very same place; or one and the same thing in different places. That, therefore, that had one beginning, is the same thing; and that which had a different beginning in time and place from that, is not the same, but diverse. That which has made the difficulty about this relation has been the little care and attention used in having precise notions of the things to which it is attributed.

2. Identity of substances. We have the ideas but of three sorts of substances: 1. God. 2. Finite intelligences. 3. Bodies.

First, God is without beginning, eternal, unalterable, and everywhere, and therefore concerning his identity there can be no doubt.

Secondly, Finite spirits having had each its determinate time and place of beginning to exist, the relation to that time and place will always determine to each of them its identity, as long as it exists.

Thirdly, The same will hold of every particle of matter, to which no addition or subtraction of matter being made, it is the same. For, though these three sorts of substances, as we term them, do not exclude one another out of the same place, yet we cannot conceive but that they must necessarily each of them exclude any of the same kind out of the same place: or else the notions and names of identity and diversity would be in vain, and there could be no such distinctions of substances, or anything else one from another. For example: could two bodies be in the same place at the same time; then those two parcels of matter must be one and the same, take them great or little; nay, all bodies must be one and the same. For, by the same reason that two particles of matter may be in one place, all bodies may be in one place: which, when it can be supposed, takes away the distinction of identity and diversity of one and more,

and renders it ridiculous. But it being a contradiction that two or more should be one, identity and diversity are relations and ways of comparing well founded, and of use to the understanding.

Identity of modes and relations. All other things being but modes or relations ultimately terminated in substances, the identity and diversity of each particular existence of them too will be by the same way determined: only as to things whose existence is in succession, such as are the actions of finite beings, v.g. motion and thought, both which consist in a continued train of succession, concerning their diversity there can be no question: because each perishing the moment it begins, they cannot exist in different times, or in different places, as permanent beings can at different times exist in distant places; and therefore no motion or thought, considered as at different times, can be the same, each part thereof having a different beginning of existence.

3. Principium Individuationis. From what has been said, it is easy to discover what is so much inquired after, the

principium individuationis; and that, it is plain, is existence itself; which determines a being of any sort to a particular time and place, incommunicable to two beings of the same kind. This, though it seems easier to conceive in simple substances or modes; yet, when reflected on, is not more difficult in compound ones, if care be taken to what it is applied: v.g. let us suppose an atom, i.e. a continued body under one immutable superficies, existing in a determined time and place; it is evident, that, considered in any instant of its existence, it is in that instant the same with itself. For, being at that instant what it is, and nothing else, it is the same, and so must continue as long as its existence is continued; for so long it will be the same, and no other. In like manner, if two or more atoms be joined together into the same mass, every one of those atoms will be the same, by the foregoing rule: and whilst they exist united together, the mass, consisting of the same atoms, must be the same mass, or the same body, let the parts be ever so differently jumbled. But if one of these atoms be taken away, or one new one added, it is no longer the

same mass or the same body. In the state of living creatures, their identity depends not on a mass of the same particles, but on something else. For in them the variation of great parcels of matter alters not the identity: an oak growing from a plant to a great tree, and then lopped, is still the same oak; and a colt grown up to a horse, sometimes fat, sometimes lean, is all the while the same horse: though, in both these cases, there may be a manifest change of the parts; so that truly they are not either of them the same masses of matter, though they be truly one of them the same oak, and the other the same horse. The reason whereof is, that, in these two cases—a mass of matter and a living body—identity is not applied to the same thing.

4. Identity of vegetables. We must therefore consider wherein an oak differs from a mass of matter, and that seems to me to be in this, that the one is only the cohesion of particles of matter any how united, the other such a disposition of them as constitutes the parts of an oak; and such an organization of those parts as is fit to receive and distribute nourishment, so as to con-

tinue and frame the wood, bark, and leaves, &c., of an oak, in which consists the vegetable life. That being then one plant which has such an organization of parts in one coherent body, partaking of one common life, it continues to be the same plant as long as it partakes of the same life, though that life be communicated to new particles of matter vitally united to the living plant, in a like continued organization conformable to that sort of plants. For this organization, being at any one instant in any one collection of matter, is in that particular concrete distinguished from all other, and is that individual life, which existing constantly from that moment both forwards and backwards, in the same continuity of insensibly succeeding parts united to the living body of the plant, it has that identity which makes the same plant, and all the parts of it, parts of the same plant, during all the time that they exist united in that continued organization, which is fit to convey that common life to all the parts so united.

5. Identity of animals. The case is not so much different in brutes but that any one may hence see what

makes an animal and continues it the same. Something we have like this in machines, and may serve to illustrate it. For example, what is a watch? It is plain it is nothing but a fit organization or construction of parts to a certain end, which, when a sufficient force is added to it, it is capable to attain. If we would suppose this machine one continued body, all whose organized parts were repaired, increased, or diminished by a constant addition or separation of insensible parts, with one common life, we should have something very much like the body of an animal; with this difference, That, in an animal the fitness of the organization, and the motion wherein life consists, begin together, the motion coming from within; but in machines the force coming sensibly from without, is often away when the organ is in order, and well fitted to receive it.

6. The identity of man. This also shows wherein the identity of the same man consists; viz. in nothing but a participation of the same continued life, by constantly fleeting particles of matter, in succession vitally united to the same organized body. He that shall place the iden-

tity of man in anything else, but, like that of other animals, in one fitly organized body, taken in any one instant, and from thence continued, under one organization of life, in several successively fleeting particles of matter united to it, will find it hard to make an embryo, one of years, mad and sober, the same man, by any supposition, that will not make it possible for Seth, Ismael, Socrates, Pilate, St. Austin, and Caesar Borgia, to be the same man. For if the identity of soul alone makes the same man; and there be nothing in the nature of matter why the same individual spirit may not be united to different bodies, it will be possible that those men, living in distant ages, and of different tempers, may have been the same man: which way of speaking must be from a very strange use of the word man, applied to an idea out of which body and shape are excluded. And that way of speaking would agree yet worse with the notions of those philosophers who allow of transmigration, and are of opinion that the souls of men may, for their miscarriages, be detruded into the bodies of beasts, as fit habitations, with organs suited

to the satisfaction of their brutal inclinations. But yet I think nobody, could he be sure that the soul of Heliogabalus were in one of his hogs, would yet say that hog were a man or Heliogabalus.

7. Idea of identity suited to the idea it is applied to. It is not therefore unity of substance that comprehends all sorts of identity, or will determine it in every case; but to conceive and judge of it aright, we must consider what idea the word it is applied to stands for: it being one thing to be the same substance, another the same man, and a third the same person, if person, man, and substance, are three names standing for three different ideas;—for such as is the idea belonging to that name, such must be the identity; which, if it had been a little more carefully attended to, would possibly have prevented a great deal of that confusion which often occurs about this matter, with no small seeming difficulties, especially concerning personal identity, which therefore we shall in the next place a little consider.

8. Same man. An animal is a living organized body; and consequently the same animal, as we have observed, is

the same continued life communicated to different particles of matter, as they happen successively to be united to that organized living body. And whatever is talked of other definitions, ingenious observation puts it past doubt, that the idea in our minds, of which the sound man in our mouths is the sign, is nothing else but of an animal of such a certain form. Since I think I may be confident, that, whoever should see a creature of his own shape or make, though it had no more reason all its life than a cat or a parrot, would call him still a man; or whoever should hear a cat or a parrot discourse, reason, and philosophize, would call or think it nothing but a cat or a parrot; and say, the one was a dull irrational man, and the other a very intelligent rational parrot. A relation we have in an author of great note, is sufficient to countenance the supposition of a rational parrot.

His words are: “I had a mind to know, from Prince Maurice’s own mouth, the account of a common, but much credited story, that I had heard so often from many others, of an old parrot he had in Brazil, during his government there, that spoke, and asked, and an-

swered common questions, like a reasonable creature: so that those of his train there generally concluded it to be witchery or possession; and one of his chaplains, who lived long afterwards in Holland, would never from that time endure a parrot, but said they all had a devil in them. I had heard many particulars of this story, and as severed by people hard to be discredited, which made me ask Prince Maurice what there was of it. He said, with his usual plainness and dryness in talk, there was something true, but a great deal false of what had been reported. I desired to know of him what there was of the first. He told me short and coldly, that he had heard of such an old parrot when he had been at Brazil; and though he believed nothing of it, and it was a good way off, yet he had so much curiosity as to send for it: that it was a very great and a very old one; and when it came first into the room where the prince was, with a great many Dutchmen about him, it said presently, What a company of white men are here! They asked it, what it thought that man was, pointing to the prince. It answered, Some General or other. When they brought it

close to him, he asked it, *D’ou venez-vous?* It answered, De Marinnan. The Prince, *A qui estes-vous?* The Parrot, *A un Portugais.* The Prince, *Que fais-tu la?* Parrot, *Je garde les poulles.* The Prince laughed, and said, *Vous gardez les poulles?* The Parrot answered, *Oui, moi; et je scai bien faire;* and made the chuck four or five times that people use to make to chickens when they call them. I set down the words of this worthy dialogue in French, just as Prince Maurice said them to me. I asked him in what language the parrot spoke, and he said in Brazilian. I asked whether he understood Brazilian; he said No, but he had taken care to have two interpreters by him, the one a Dutchman that spoke Brazilian, and the other a Brazilian that spoke Dutch; that he asked them separately and privately, and both of them agreed in telling him just the same thing that the parrot had said. I could not but tell this odd story, because it is so much out of the way, and from the first hand, and what may pass for a good one; for I dare say this Prince at least believed himself in all he told me, having ever passed for a very honest and pious man: I leave it to naturalists

to reason, and to other men to believe, as they please upon it; however, it is not, perhaps, amiss to relieve or enliven a busy scene sometimes with such digressions, whether to the purpose or no." I have taken care that the reader should have the story at large in the author's own words, because he seems to me not to have thought it incredible; for it cannot be imagined that so able a man as he, who had sufficiency enough to warrant all the testimonies he gives of himself, should take so much pains, in a place where it had nothing to do, to pin so close, not only on a man whom he mentions as his friend, but on a Prince in whom he acknowledges very great honesty and piety, a story which, if he himself thought incredible, he could not but also think ridiculous. The Prince, it is plain, who vouches this story, and our author, who relates it from him, both of them call this talker a parrot: and I ask any one else who thinks such a story fit to be told, whether, if this parrot, and all of its kind, had always talked, as we have a prince's word for it this one did,—whether, I say, they would not have passed for a race of rational animals; but yet,

whether, for all that, they would have been allowed to be men, and not parrots? For I presume it is not the idea of a thinking or rational being alone that makes the idea of a man in most people's sense: but of a body, so and so shaped, joined to it: and if that be the idea of a man, the same successive body not shifted all at once, must, as well as the same immaterial spirit, go to the making of the same man.

9. Personal identity. This being premised, to find wherein personal identity consists, we must consider what person stands for;—which, I think, is a thinking intelligent being, that has reason and reflection, and can consider itself as itself, the same thinking thing, in different times and places; which it does only by that consciousness which is inseparable from thinking, and, as it seems to me, essential to it: it being impossible for any one to perceive without perceiving that he does perceive. When we see, hear, smell, taste, feel, meditate, or will anything, we know that we do so. Thus it is always as to our present sensations and perceptions: and by this every one is to himself that which he calls self:—it

not being considered, in this case, whether the same self be continued in the same or divers substances. For, since consciousness always accompanies thinking, and it is that which makes every one to be what he calls self, and thereby distinguishes himself from all other thinking things, in this alone consists personal identity, i.e. the sameness of a rational being: and as far as this consciousness can be extended backwards to any past action or thought, so far reaches the identity of that person; it is the same self now it was then; and it is by the same self with this present one that now reflects on it, that that action was done.

10. Consciousness makes personal identity. But it is further inquired, whether it be the same identical substance. This few would think they had reason to doubt of, if these perceptions, with their consciousness, always remained present in the mind, whereby the same thinking thing would be always consciously present, and, as would be thought, evidently the same to itself. But that which seems to make the difficulty is this, that this consciousness being interrupted always by forget-

fulness, there being no moment of our lives wherein we have the whole train of all our past actions before our eyes in one view, but even the best memories losing the sight of one part whilst they are viewing another; and we sometimes, and that the greatest part of our lives, not reflecting on our past selves, being intent on our present thoughts, and in sound sleep having no thoughts at all, or at least none with that consciousness which remarks our waking thoughts,—I say, in all these cases, our consciousness being interrupted, and we losing the sight of our past selves, doubts are raised whether we are the same thinking thing, i.e. the same substance or no. Which, however reasonable or unreasonable, concerns not personal identity at all. The question being what makes the same person; and not whether it be the same identical substance, which always thinks in the same person, which, in this case, matters not at all: different substances, by the same consciousness (where they do partake in it) being united into one person, as well as different bodies by the same life are united into one animal, whose identity is preserved in that change

of substances by the unity of one continued life. For, it being the same consciousness that makes a man be himself to himself, personal identity depends on that only, whether it be annexed solely to one individual substance, or can be continued in a succession of several substances. For as far as any intelligent being can repeat the idea of any past action with the same consciousness it had of it at first, and with the same consciousness it has of any present action; so far it is the same personal self. For it is by the consciousness it has of its present thoughts and actions, that it is self to itself now, and so will be the same self, as far as the same consciousness can extend to actions past or to come. and would be by distance of time, or change of substance, no more two persons, than a man be two men by wearing other clothes to-day than he did yesterday, with a long or a short sleep between: the same consciousness uniting those distant actions into the same person, whatever substances contributed to their production.

11. Personal identity in change of substance. That this is so, we have some kind of evidence in our very bodies,

all whose particles, whilst vitally united to this same thinking conscious self, so that we feel when they are touched, and are affected by, and conscious of good or harm that happens to them, as a part of ourselves; i.e. of our thinking conscious self. Thus, the limbs of his body are to every one a part of Himself; he sympathizes and is concerned for them. Cut off a hand, and thereby separate it from that consciousness he had of its heat, cold, and other affections, and it is then no longer a part of that which is himself, any more than the remotest part of matter. Thus, we see the substance whereof personal self consisted at one time may be varied at another, without the change of personal identity; there being no question about the same person, though the limbs which but now were a part of it, be cut off.

12. Personality in change of substance. But the question is, Whether if the same substance which thinks be changed, it can be the same person; or, remaining the same, it can be different persons? And to this I answer: First, This can be no question at all to those who place thought in a purely material animal constitution, void

of an immaterial substance. For, whether their supposition be true or no, it is plain they conceive personal identity preserved in something else than identity of substance; as animal identity is preserved in identity of life, and not of substance. And therefore those who place thinking in an immaterial substance only, before they can come to deal with these men, must show why personal identity cannot be preserved in the change of immaterial substances, or variety of particular immaterial substances, as well as animal identity is preserved in the change of material substances, or variety of particular bodies: unless they will say, it is one immaterial spirit that makes the same life in brutes, as it is one immaterial spirit that makes the same person in men; which the Cartesians at least will not admit, for fear of making brutes thinking things too.

13. Whether in change of thinking substances there can be one person. But next, as to the first part of the question, Whether, if the same thinking substance (supposing immaterial substances only to think) be changed, it can be the same person? I answer, that cannot be

resolved but by those who know what kind of substances they are that do think; and whether the consciousness of past actions can be transferred from one thinking substance to another. I grant were the same consciousness the same individual action it could not: but it being a present representation of a past action, why it may not be possible, that that may be represented to the mind to have been which really never was, will remain to be shown. And therefore how far the consciousness of past actions is annexed to any individual agent, so that another cannot possibly have it, will be hard for us to determine, till we know what kind of action it is that cannot be done without a reflex act of perception accompanying it, and how performed by thinking substances, who cannot think without being conscious of it. But that which we call the same consciousness, not being the same individual act, why one intellectual substance may not have represented to it, as done by itself, what it never did, and was perhaps done by some other agent—why, I say, such a representation may not possibly be without reality of matter of fact, as well as sev-

eral representations in dreams are, which yet whilst dreaming we take for true—will be difficult to conclude from the nature of things. And that it never is so, will by us, till we have clearer views of the nature of thinking substances, be best resolved into the goodness of God; who, as far as the happiness or misery of any of his sensible creatures is concerned in it, will not, by a fatal error of theirs, transfer from one to another that consciousness which draws reward or punishment with it. How far this may be an argument against those who would place thinking in a system of fleeting animal spirits, I leave to be considered. But yet, to return to the question before us, it must be allowed, that, if the same consciousness (which, as has been shown, is quite a different thing from the same numerical figure or motion in body) can be transferred from one thinking substance to another, it will be possible that two thinking substances may make but one person. For the same consciousness being preserved, whether in the same or different substances, the personal identity is preserved.

14. Whether, the same immaterial substance remain-

ing, there can be two persons. As to the second part of the question, Whether the same immaterial substance remaining, there may be two distinct persons; which question seems to me to be built on this,—Whether the same immaterial being, being conscious of the action of its past duration, may be wholly stripped of all the consciousness of its past existence, and lose it beyond the power of ever retrieving it again: and so as it were beginning a new account from a new period, have a consciousness that cannot reach beyond this new state. All those who hold pre-existence are evidently of this mind; since they allow the soul to have no remaining consciousness of what it did in that pre-existent state, either wholly separate from body, or informing any other body; and if they should not, it is plain experience would be against them. So that personal identity, reaching no further than consciousness reaches, a pre-existent spirit not having continued so many ages in a state of silence, must needs make different persons. Suppose a Christian Platonist or a Pythagorean should, upon God's having ended all his works of creation the seventh day, think

his soul hath existed ever since; and should imagine it has revolved in several human bodies; as I once met with one, who was persuaded his had been the soul of Socrates (how reasonably I will not dispute; this I know, that in the post he filled, which was no inconsiderable one, he passed for a very rational man, and the press has shown that he wanted not parts or learning;)—would any one say, that he, being not conscious of any of Socrates's actions or thoughts, could be the same person with Socrates? Let any one reflect upon himself, and conclude that he has in himself an immaterial spirit, which is that which thinks in him, and, in the constant change of his body keeps him the same: and is that which he calls himself: let him also suppose it to be the same soul that was in Nestor or Thersites, at the siege of Troy, (for souls being, as far as we know anything of them, in their nature indifferent to any parcel of matter, the supposition has no apparent absurdity in it), which it may have been, as well as it is now the soul of any other man: but he now having no consciousness of any of the actions either of Nestor or Thersites, does or

can he conceive himself the same person with either of them? Can he be concerned in either of their actions? attribute them to himself, or think them his own, more than the actions of any other men that ever existed? So that this consciousness, not reaching to any of the actions of either of those men, he is no more one self with either of them than if the soul or immaterial spirit that now informs him had been created, and began to exist, when it began to inform his present body; though it were never so true, that the same spirit that informed Nestor's or Thersites' body were numerically the same that now informs his. For this would no more make him the same person with Nestor, than if some of the particles of matter that were once a part of Nestor were now a part of this man; the same immaterial substance, without the same consciousness, no more making the same person, by being united to any body, than the same particle of matter, without consciousness, united to any body, makes the same person. But let him once find himself conscious of any of the actions of Nestor, he then finds himself the same person with Nestor.

15. The body, as well as the soul, goes to the making of a man. And thus may we be able, without any difficulty, to conceive the same person at the resurrection, though in a body not exactly in make or parts the same which he had here,—the same consciousness going along with the soul that inhabits it. But yet the soul alone, in the change of bodies, would scarce to any one but to him that makes the soul the man, be enough to make the same man. For should the soul of a prince, carrying with it the consciousness of the prince's past life, enter and inform the body of a cobbler, as soon as deserted by his own soul, every one sees he would be the same person with the prince, accountable only for the prince's actions: but who would say it was the same man? The body too goes to the making the man, and would, I guess, to everybody determine the man in this case, wherein the soul, with all its princely thoughts about it, would not make another man: but he would be the same cobbler to every one besides himself. I know that, in the ordinary way of speaking, the same person, and the same man, stand for one and the same thing. And

indeed every one will always have a liberty to speak as he pleases, and to apply what articulate sounds to what ideas he thinks fit, and change them as often as he pleases. But yet, when we will inquire what makes the same spirit, man, or person, we must fix the ideas of spirit, man, or person in our minds; and having resolved with ourselves what we mean by them, it will not be hard to determine, in either of them, or the like, when it is the same, and when not.

16. Consciousness alone unites actions into the same person. But though the same immaterial substance or soul does not alone, wherever it be, and in whatsoever state, make the same man; yet it is plain, consciousness, as far as ever it can be extended—should it be to ages past—unites existences and actions very remote in time into the same person, as well as it does the existences and actions of the immediately preceding moment: so that whatever has the consciousness of present and past actions, is the same person to whom they both belong. Had I the same consciousness that I saw the ark and Noah's flood, as that I saw an overflowing of the Thames

last winter, or as that I write now, I could no more doubt that I who write this now, that saw' the Thames overflowed last winter, and that viewed the flood at the general deluge, was the same self,—place that self in what substance you please—than that I who write this am the same myself now whilst I write (whether I consist of all the same substance, material or immaterial, or no) that I was yesterday. For as to this point of being the same self, it matters not whether this present self be made up of the same or other substances—I being as much concerned, and as justly accountable for any action that was done a thousand years since, appropriated to me now by this self-consciousness, as I am for what I did the last moment.

17. Self depends on consciousness, not on substance. Self is that conscious thinking thing,—whatever substance made up of, (whether spiritual or material, simple or compounded, it matters not)—which is sensible or conscious of pleasure and pain, capable of happiness or misery, and so is concerned for itself, as far as that consciousness extends. Thus every one finds that, whilst

comprehended under that consciousness, the little finger is as much a part of himself as what is most so. Upon separation of this little finger, should this consciousness go along with the little finger, and leave the rest of the body, it is evident the little finger would be the person, the same person; and self then would have nothing to do with the rest of the body. As in this case it is the consciousness that goes along with the substance, when one part is separate from another, which makes the same person, and constitutes this inseparable self: so it is in reference to substances remote in time. That with which the consciousness of this present thinking thing can join itself, makes the same person, and is one self with it, and with nothing else; and so attributes to itself, and owns all the actions of that thing, as its own, as far as that consciousness reaches, and no further; as every one who reflects will perceive.

18. Persons, not substances, the objects of reward and punishment. In this personal identity is founded all the right and justice of reward and punishment; happiness and misery being that for which every one is concerned

for himself, and not mattering what becomes of any substance, not joined to, or affected with that consciousness. For, as it is evident in the instance I gave but now, if the consciousness went along with the little finger when it was cut off, that would be the same self which was concerned for the whole body yesterday, as making part of itself, whose actions then it cannot but admit as its own now. Though, if the same body should still live, and immediately from the separation of the little finger have its own peculiar consciousness, whereof the little finger knew nothing, it would not at all be concerned for it, as a part of itself, or could own any of its actions, or have any of them imputed to him.

19. Which shows wherein personal identity consists. This may show us wherein personal identity consists: not in the identity of substance, but, as I have said, in the identity of consciousness, wherein if Socrates and the present mayor of Queinborough agree, they are the same person: if the same Socrates waking and sleeping do not partake of the same consciousness, Socrates waking and sleeping is not the same person. And to punish

Socrates waking for what sleeping Socrates thought, and waking Socrates was never conscious of, would be no more of right, than to punish one twin for what his brother-twin did, whereof he knew nothing, because their outsides were so like, that they could not be distinguished; for such twins have been seen.

20. Absolute oblivion separates what is thus forgotten from the person, but not from the man. But yet possibly it will still be objected,—Suppose I wholly lose the memory of some parts of my life, beyond a possibility of retrieving them, so that perhaps I shall never be conscious of them again; yet am I not the same person that did those actions, had those thoughts that I once was conscious of, though I have now forgot them? To which I answer, that we must here take notice what the word I is applied to; which, in this case, is the man only. And the same man being presumed to be the same person, I is easily here supposed to stand also for the same person. But if it be possible for the same man to have distinct incommunicable consciousness at different times, it is past doubt the same man would at different times

make different persons; which, we see, is the sense of mankind in the solemnest declaration of their opinions, human laws not punishing the mad man for the sober man's actions, nor the sober man for what the mad man did,—thereby making them two persons: which is somewhat explained by our way of speaking in English when we say such an one is “not himself,” or is “beside himself”; in which phrases it is insinuated, as if those who now, or at least first used them, thought that self was changed; the selfsame person was no longer in that man.

21. Difference between identity of man and of person. But yet it is hard to conceive that Socrates, the same individual man, should be two persons. To help us a little in this, we must consider what is meant by Socrates, or the same individual man.

First, it must be either the same individual, immaterial, thinking substance; in short, the same numerical soul, and nothing else.

Secondly, or the same animal, without any regard to an immaterial soul.

Thirdly, or the same immaterial spirit united to the

same animal. Now, take which of these suppositions you please, it is impossible to make personal identity to consist in anything but consciousness; or reach any further than that does.

For, by the first of them, it must be allowed possible that a man born of different women, and in distant times, may be the same man. A way of speaking which, whoever admits, must allow it possible for the same man to be two distinct persons, as any two that have lived in different ages without the knowledge of one another's thoughts. By the second and third, Socrates, in this life and after it, cannot be the same man any way, but by the same consciousness; and so making human identity to consist in the same thing wherein we place personal identity, there will be no difficulty to allow the same man to be the same person. But then they who place human identity in consciousness only, and not in something else, must consider how they will make the infant Socrates the same man with Socrates after the resurrection. But whatsoever to some men makes a man, and consequently the same individual man, wherein per-

haps few are agreed, personal identity can by us be placed in nothing but consciousness, (which is that alone which makes what we call self,) without involving us in great absurdities.

22. But is not a man drunk and sober the same person? why else is he punished for the fact he commits when drunk, though he be never afterwards conscious of it? Just as much the same person as a man that walks, and does other things in his sleep, is the same person, and is answerable for any mischief he shall do in it. Human laws punish both, with a justice suitable to their way of knowledge;—because, in these cases, they cannot distinguish certainly what is real, what counterfeit: and so the ignorance in drunkenness or sleep is not admitted as a plea. For, though punishment be annexed to personality, and personality to consciousness, and the drunkard perhaps be not conscious of what he did, yet human judicatures justly punish him; because the fact is proved against him, but want of consciousness cannot be proved for him. But in the Great Day, wherein the secrets of all hearts shall be laid open, it may be

reasonable to think, no one shall be made to answer for what he knows nothing of, but shall receive his doom, his conscience accusing or excusing him.

23. Consciousness alone unites remote existences into one person. Nothing but consciousness can unite remote existences into the same person: the identity of substance will not do it; for whatever substance there is, however framed, without consciousness there is no person: and a carcass may be a person, as well as any sort of substance be so, without consciousness.

Could we suppose two distinct incommunicable consciousnesses acting the same body, the one constantly by day, the other by night; and, on the other side, the same consciousness, acting by intervals, two distinct bodies: I ask, in the first case, whether the day and the night—man would not be two as distinct persons as Socrates and Plato? And whether, in the second case, there would not be one person in two distinct bodies, as much as one man is the same in two distinct clothings? Nor is it at all material to say, that this same, and this distinct consciousness, in the cases above mentioned, is

owing to the same and distinct immaterial substances, bringing it with them to those bodies; which, whether true or no, alters not the case: since it is evident the personal identity would equally be determined by the consciousness, whether that consciousness were annexed to some individual immaterial substance or no. For, granting that the thinking substance in man must be necessarily supposed immaterial, it is evident that immaterial thinking thing may sometimes part with its past consciousness, and be restored to it again: as appears in the forgetfulness men often have of their past actions; and the mind many times recovers the memory of a past consciousness, which it had lost for twenty years together. Make these intervals of memory and forgetfulness to take their turns regularly by day and night, and you have two persons with the same immaterial spirit, as much as in the former instance two persons with the same body. So that self is not determined by identity or diversity of substance, which it cannot be sure of, but only by identity of consciousness.

24. Not the substance with which the consciousness

may be united. Indeed it may conceive the substance whereof it is now made up to have existed formerly, united in the same conscious being: but, consciousness removed, that substance is no more itself, or makes no more a part of it, than any other substance; as is evident in the instance we have already given of a limb cut off, of whose heat, or cold, or other affections, having no longer any consciousness, it is no more of a man's self than any other matter of the universe. In like manner it will be in reference to any immaterial substance, which is void of that consciousness whereby I am myself to myself: if there be any part of its existence which I cannot upon recollection join with that present consciousness whereby I am now myself, it is, in that part of its existence, no more myself than any other immaterial being. For, whatsoever any substance has thought or done, which I cannot recollect, and by my consciousness make my own thought and action, it will no more belong to me, whether a part of me thought or did it, than if it had been thought or done by any other immaterial being anywhere existing.

25. Consciousness unites substances, material or spiritual, with the same personality. I agree, the more probable opinion is, that this consciousness is annexed to, and the affection of, one individual immaterial substance.

But let men, according to their diverse hypotheses, resolve of that as they please. This every intelligent being, sensible of happiness or misery, must grant—that there is something that is himself, that he is concerned for, and would have happy; that this self has existed in a continued duration more than one instant, and therefore it is possible may exist, as it has done, months and years to come, without any certain bounds to be set to its duration; and may be the same self, by the same consciousness continued on for the future. And thus, by this consciousness he finds himself to be the same self which did such and such an action some years since, by which he comes to be happy or miserable now. In all which account of self, the same numerical substance is not considered as making the same self, but the same continued consciousness, in which several substances may have been united, and again separated from it,

which, whilst they continued in a vital union with that wherein this consciousness then resided, made a part of that same self. Thus any part of our bodies, vitally united to that which is conscious in us, makes a part of ourselves: but upon separation from the vital union by which that consciousness is communicated, that which a moment since was part of ourselves, is now no more so than a part of another man's self is a part of me: and it is not impossible but in a little time may become a real part of another person. And so we have the same numerical substance become a part of two different persons; and the same person preserved under the change of various substances. Could we suppose any spirit wholly stripped of all its memory or consciousness of past actions, as we find our minds always are of a great part of ours, and sometimes of them all; the union or separation of such a spiritual substance would make no variation of personal identity, any more than that of any particle of matter does. Any substance vitally united to the present thinking being is a part of that very same self which now is; anything united to it by a conscious-

ness of former actions, makes also a part of the same self, which is the same both then and now.

26. "Person" a forensic term. Person, as I take it, is the name for this self. Wherever a man finds what he calls himself, there, I think, another may say is the same person. It is a forensic term, appropriating actions and their merit; and so belongs only to intelligent agents, capable of a law, and happiness, and misery. This personality extends itself beyond present existence to what is past, only by consciousness,—whereby it becomes concerned and accountable; owns and imputes to itself past actions, just upon the same ground and for the same reason as it does the present. All which is founded in a concern for happiness, the unavoidable concomitant of consciousness; that which is conscious of pleasure and pain, desiring that that self that is conscious should be happy. And therefore whatever past actions it cannot reconcile or appropriate to that present self by consciousness, it can be no more concerned in than if they had never been done: and to receive pleasure or pain, i.e. reward or punishment, on the account of any such

action, is all one as to be made happy or miserable in its first being, without any demerit at all. For, supposing a man punished now for what he had done in another life, whereof he could be made to have no consciousness at all, what difference is there between that punishment and being created miserable? And therefore, conformable to this, the apostle tells us, that, at the great day, when every one shall "receive according to his doings, the secrets of all hearts shall be laid open." The sentence shall be justified by the consciousness all persons shall have, that they themselves, in what bodies soever they appear, or what substances soever that consciousness adheres to, are the same that committed those actions, and deserve that punishment for them.

27. Suppositions that look strange are pardonable in our ignorance. I am apt enough to think I have, in treating of this subject, made some suppositions that will look strange to some readers, and possibly they are so in themselves. But yet, I think they are such as are pardonable, in this ignorance we are in of the nature of that thinking thing that is in us, and which we look on

as ourselves. Did we know what it was, or how it was tied to a certain system of fleeting animal spirits; or whether it could or could not perform its operations of thinking and memory out of a body organized as ours is; and whether it has pleased God that no one such spirit shall ever be united to any but one such body, upon the right constitution of whose organs its memory should depend; we might see the absurdity of some of those suppositions I have made. But taking, as we ordinarily now do (in the dark concerning these matters), the soul of a man for an immaterial substance, independent from matter, and indifferent alike to it all; there can, from the nature of things, be no absurdity at all to suppose that the same soul may at different times be united to different bodies, and with them make up for that time one man: as well as we suppose a part of a sheep's body yesterday should be a part of a man's body to-morrow, and in that union make a vital part of Meliboeus himself, as well as it did of his ram.

28. The difficulty from ill use of names. To conclude: Whatever substance begins to exist, it must, during its

existence, necessarily be the same: whatever compositions of substances begin to exist, during the union of those substances, the concrete must be the same: whatsoever mode begins to exist, during its existence it is the same: and so if the composition be of distinct substances and different modes, the same rule holds. Whereby it will appear, that the difficulty or obscurity that has been about this matter rather rises from the names ill-used, than from any obscurity in things themselves. For whatever makes the specific idea to which the name is applied, if that idea be steadily kept to, the distinction of anything into the same and divers will easily be conceived, and there can arise no doubt about it.

29. Continuance of that which we have made to be our complex idea of man makes the same man. For, supposing a rational spirit be the idea of a man, it is easy to know what is the same man, viz. the same spirit—whether separate or in a body—will be the same man. Supposing a rational spirit vitally united to a body of a certain conformation of parts to make a man; whilst that rational spirit, with that vital conformation of parts,

though continued in a fleeting successive body, remains, it will be the same man. But if to any one the idea of a man be but the vital union of parts in a certain shape; as long as that vital union and shape remain in a concrete, no otherwise the same but by a continued succession of fleeting particles, it will be the same man. For, whatever be the composition whereof the complex idea is made, whenever existence makes it one particular thing under any denomination the same existence continued preserves it the same individual under the same denomination.

Chapter XXVIII Of Other Relations

1. Ideas of proportional relations. Besides the before-mentioned occasions of time, place, and causality of comparing or referring things one to another, there are, as I have said, infinite others, some whereof I shall mention.

First, The first I shall name is some one simple idea,

which, being capable of parts or degrees, affords an occasion of comparing the subjects wherein it is to one another, in respect of that simple idea, v.g. whiter, sweeter, equal, more, &c. These relations depending on the equality and excess of the same simple idea, in several subjects, may be called, if one will, proportional; and that these are only conversant about those simple ideas received from sensation or reflection is so evident that nothing need be said to evince it.

2. Natural relation. Secondly, Another occasion of comparing things together, or considering one thing, so as to include in that consideration some other thing, is the circumstances of their origin or beginning; which being not afterwards to be altered, make the relations depending thereon as lasting as the subjects to which they belong, v.g. father and son, brothers, cousin-germans, &c., which have their relations by one community of blood, wherein they partake in several degrees: countrymen, i.e. those who were born in the same country or tract of ground; and these I call natural relations: wherein we may observe, that mankind

have fitted their notions and words to the use of common life, and not to the truth and extent of things. For it is certain, that, in reality, the relation is the same betwixt the begetter and the begotten, in the several races of other animals as well as men; but yet it is seldom said, this bull is the grandfather of such a calf, or that two pigeons are cousin-germans. It is very convenient that, by distinct names, these relations should be observed and marked out in mankind, there being occasion, both in laws and other communications one with another, to mention and take notice of men under these relations: from whence also arise the obligations of several duties amongst men: whereas, in brutes, men having very little or no cause to mind these relations, they have not thought fit to give them distinct and peculiar names. This, by the way, may give us some light into the different state and growth of languages; which being suited only to the convenience of communication, are proportioned to the notions men have, and the commerce of thoughts familiar amongst them; and not to the reality or extent of things, nor to the various re-

spects might be found among them; nor the different abstract considerations might be framed about them. Where they had no philosophical notions, there they had no terms to express them: and it is no wonder men should have framed no names for those things they found no occasion to discourse of. From whence it is easy to imagine why, as in some countries, they may have not so much as the name for a horse; and in others, where they are more careful of the pedigrees of their horses, than of their own, that there they may have not only names for particular horses, but also of their several relations of kindred one to another.

3. Ideas of instituted or voluntary relations. Thirdly, Sometimes the foundation of considering things, with reference to one another, is some act whereby any one comes by a moral right, power, or obligation to do something. Thus, a general is one that hath power to command an army; and an army under a general is a collection of armed men, obliged to obey one man. A citizen, or a burgher, is one who has a right to certain privileges in this or that place. All this sort depending upon men's

wills, or agreement in society, I call instituted, or voluntary; and may be distinguished from the natural, in that they are most, if not all of them, some way or other alterable, and separable from the persons to whom they have sometimes belonged, though neither of the substances, so related, be destroyed. Now, though these are all reciprocal, as well as the rest, and contain in them a reference of two things one to the other; yet, because one of the two things often wants a relative name, importing that reference, men usually take no notice of it, and the relation is commonly overlooked: v.g. a patron and client are easily allowed to be relations, but a constable or dictator are not so readily at first hearing considered as such. Because there is no peculiar name for those who are under the command of a dictator or constable, expressing a relation to either of them; though it be certain that either of them hath a certain power over some others, and so is so far related to them, as well as a patron is to his client, or general to his army.

4. Ideas of moral relations. Fourthly, There is another sort of relation, which is the conformity or disagreement

men's voluntary actions have to a rule to which they are referred, and by which they are judged of; which, I think, may be called moral relation, as being that which denominates our moral actions, and deserves well to be examined; there being no part of knowledge wherein we should be more careful to get determined ideas, and avoid, as much as may be, obscurity and confusion. Human actions, when with their various ends, objects, manners, and circumstances, they are framed into distinct complex ideas, are, as has been shown so many mixed modes, a great part whereof have names annexed to them. Thus, supposing gratitude to be a readiness to acknowledge and return kindness received; polygamy to be the having more wives than one at once: when we frame these notions thus in our minds, we have there so many determined ideas of mixed modes. But this is not all that concerns our actions: it is not enough to have determined ideas of them, and to know what names belong to such and such combinations of ideas. We have a further and greater concernment, and that is, to know whether such actions, so made up, are morally good or bad.

5. Moral good and evil. Good and evil, as hath been shown, (Bk. II. chap. xx. SS 2, and chap. xxi. SS 43,) are nothing but pleasure or pain, or that which occasions or procures pleasure or pain to us. Moral good and evil, then, is only the conformity or disagreement of our voluntary actions to some law, whereby good or evil is drawn on us, from the will and power of the law-maker; which good and evil, pleasure or pain, attending our observance or breach of the law by the decree of the lawmaker, is that we call reward and punishment.

6. Moral rules. Of these moral rules or laws, to which men generally refer, and by which they judge of the rectitude or pravity of their actions, there seem to me to be three sorts, with their three different enforcements, or rewards and punishments. For, since it would be utterly in vain to suppose a rule set to the free actions of men, without annexing to it some enforcement of good and evil to determine his will, we must, wherever we suppose a law, suppose also some reward or punishment annexed to that law. It would be in vain for one intelligent being to set a rule to the actions of an-

other, if he had it not in his power to reward the compliance with, and punish deviation from his rule, by some good and evil, that is not the natural product and consequence of the action itself. For that, being a natural convenience or inconvenience, would operate of itself, without a law. This, if I mistake not, is the true nature of all law, properly so called.

7. Laws. The laws that men generally refer their actions to, to judge of their rectitude or obliquity, seem to me to be these three:—1. The divine law. 2. The civil law. 3. The law of opinion or reputation, if I may so call it. By the relation they bear to the first of these, men judge whether their actions are sins or duties; by the second, whether they be criminal or innocent; and by the third, whether they be virtues or vices.

8. Divine law the measure of sin and duty. First, the divine law, whereby that law which God has set to the actions of men,—whether promulgated to them by the light of nature, or the voice of revelation. That God has given a rule whereby men should govern themselves, I think there is nobody so brutish as to deny. He has a

right to do it; we are his creatures: he has goodness and wisdom to direct our actions to that which is best: and he has power to enforce it by rewards and punishments of infinite weight and duration in another life; for nobody can take us out of his hands. This is the only true touchstone of moral rectitude; and, by comparing them to this law, it is that men judge of the most considerable moral good or evil of their actions; that is, whether, as duties or sins, they are like to procure them happiness or misery from the hands of the *Almighty*.

9. Civil law the measure of crimes and innocence. Secondly, the civil law—the rule set by the commonwealth to the actions of those who belong to it—is another rule to which men refer their actions; to judge whether they be criminal or no. This law nobody overlooks: the rewards and punishments that enforce it being ready at hand, and suitable to the power that makes it: which is the force of the Commonwealth, engaged to protect the lives, liberties, and possessions of those who live according to its laws, and has power to take away life, liberty, or goods, from him who disobeys; which is the

punishment of offences committed against his law.

10. Philosophical law the measure of virtue and vice. Thirdly, the law of opinion or reputation. Virtue and vice are names pretended and supposed everywhere to stand for actions in their own nature right and wrong: and as far as they really are so applied, they so far are coincident with the divine law above mentioned. But yet, whatever is pretended, this is visible, that these names, virtue and vice, in the particular instances of their application, through the several nations and societies of men in the world, are constantly attributed only to such actions as in each country and society are in reputation or discredit. Nor is it to be thought strange, that men everywhere should give the name of virtue to those actions, which amongst them are judged praiseworthy; and call that vice, which they account blamable: since otherwise they would condemn themselves, if they should think anything right, to which they allowed not commendation, anything wrong, which they let pass without blame. Thus the measure of what is everywhere called and esteemed virtue and vice is this

approbation or dislike, praise or blame, which, by a secret and tacit consent, establishes itself in the several societies, tribes, and clubs of men in the world: whereby several actions come to find credit or disgrace amongst them, according to the judgment, maxims, or fashion of that place. For, though men uniting into politic societies, have resigned up to the public the disposing of all their force, so that they cannot employ it against any fellow-citizens any further than the law of the country directs: yet they retain still the power of thinking well or ill, approving or disapproving of the actions of those whom they live amongst, and converse with: and by this approbation and dislike they establish amongst themselves what they will call virtue and vice.

11. The measure that men commonly apply to determine what they call virtue and vice. That this is the common measure of virtue and vice, will appear to any one who considers, that, though that passes for vice in one country which is counted a virtue, or at least not vice, in another, yet everywhere virtue and praise, vice and blame, go together. Virtue is everywhere, that which

is thought praiseworthy; and nothing else but that which has the allowance of public esteem is called virtue. Virtue and praise are so united, that they are called often by the same name. *Sunt sua praemia laudi*, says Virgil; and so Cicero, *Nihil habet natura praestantius, quam honestatem, quam laudem, quam dignitatem, quam decus*, which he tells you are all names for the same thing. This is the language of the heathen philosophers, who well understood wherein their notions of virtue and vice consisted. And though perhaps, by the different temper, education, fashion, maxims, or interest of different sorts of men, it fell out, that what was thought praiseworthy in one place, escaped not censure in another; and so in different societies, virtues and vices were changed: yet, as to the main, they for the most part kept the same everywhere. For, since nothing can be more natural than to encourage with esteem and reputation that wherein every one finds his advantage, and to blame and discountenance the contrary; it is no wonder that esteem and discredit, virtue and vice, should, in a great measure, everywhere correspond with the un-

changeable rule of right and wrong, which the law of God hath established; there being nothing that so directly and visibly secures and advances the general good of mankind in this world, as obedience to the laws he has set them, and nothing that breeds such mischiefs and confusion, as the neglect of them. And therefore men, without renouncing all sense and reason, and their own interest, which they are so constantly true to, could not generally mistake, in placing their commendation and blame on that side that really deserved it not. Nay, even those men whose practice was otherwise, failed not to give their approbation right, few being depraved to that degree as not to condemn, at least in others, the faults they themselves were guilty of; whereby, even in the corruption of manners, the true boundaries of the law of nature, which ought to be the rule of virtue and vice, were pretty well preferred. So that even the exhortations of inspired teachers, have not feared to appeal to common repute: "Whatsoever is lovely, whatsoever is of good report, if there be any virtue, if there be any praise," &c. (Phil. 4. 8.)

12. Its enforcement is commendation and discredit. If any one shall imagine that I have forgot my own notion of a law, when I make the law, whereby men judge of virtue and vice, to be nothing else but the consent of private men, who have not authority enough to make a law: especially wanting that which is so necessary and essential to a law, a power to enforce it: I think I may say, that he who imagines commendation and disgrace not to be strong motives to men to accommodate themselves to the opinions and rules of those with whom they converse, seems little skilled in the nature or history of mankind: the greatest part whereof we shall find to govern themselves chiefly, if not solely, by this law of fashion; and so they do that which keeps them in reputation with their company, little regard the laws of God, or the magistrate. The penalties that attend the breach of God's laws some, nay perhaps most men, seldom seriously reflect on: and amongst those that do, many, whilst they break the law, entertain thoughts of future reconciliation, and making their peace for such breaches. And as to the punishments due from the laws of the com-

monwealth, they frequently flatter themselves with the hopes of impunity. But no man escapes the punishment of their censure and dislike, who offends against the fashion and opinion of the company he keeps, and would recommend himself to. Nor is there one of ten thousand, who is stiff and insensible enough, to bear up under the constant dislike and condemnation of his own club. He must be of a strange and unusual constitution, who can content himself to live in constant disgrace and disrepute with his own particular society. Solitude many men have sought, and been reconciled to: but nobody that has the least thought or sense of a man about him, can live in society under the constant dislike and ill opinion of his familiars, and those he converses with. This is a burden too heavy for human sufferance: and he must be made up of irreconcilable contradictions, who can take pleasure in company, and yet be insensible of contempt and disgrace from his companions.

13. These three laws the rules of moral good and evil. These three then, first, the law of God; secondly, the

law of politic societies; thirdly, the law of fashion, or private censure, are those to which men variously compare their actions: and it is by their conformity to one of these laws that they take their measures, when they would judge of their moral rectitude, and denominate their actions good or bad.

14. Morality is the relation of voluntary actions to these rules. Whether the rule to which, as to a touchstone, we bring our voluntary actions, to examine them by, and try their goodness, and accordingly to name them, which is, as it were, the mark of the value we set upon them: whether, I say, we take that rule from the fashion of the country, or the will of a law-maker, the mind is easily able to observe the relation any action hath to it, and to judge whether the action agrees or disagrees with the rule; and so hath a notion of moral goodness or evil, which is either conformity or not conformity of any action to that rule: and therefore is often called moral rectitude. This rule being nothing but a collection of several simple ideas, the conformity thereto is but so ordering the action, that the simple ideas belonging to

it may correspond to those which the law requires. And thus we see how moral beings and notions are founded on, and terminated in, these simple ideas we have received from sensation or reflection. For example: let us consider the complex idea we signify by the word murder: and when we have taken it asunder, and examined all the particulars, we shall find them to amount to a collection of simple ideas derived from reflection or sensation, viz. First, from reflection on the operations of our own minds, we have the ideas of willing, considering, purposing beforehand, malice, or wishing ill to another; and also of life, or perception, and self-motion. Secondly, from sensation we have the collection of those simple sensible ideas which are to be found in a man, and of some action, whereby we put an end to perception and motion in the man; all which simple ideas are comprehended in the word murder. This collection of simple ideas, being found by me to agree or disagree with the esteem of the country I have been bred in, and to be held by most men there worthy praise or blame, I call the action virtuous or vicious: if I have the will of a

supreme invisible Lawgiver for my rule, then, as I supposed the action commanded or forbidden by God, I call it good or evil, sin or duty: and if I compare it to the civil law, the rule made by the legislative power of the country, I call it lawful or unlawful, a crime or no crime. So that whencesoever we take the rule of moral actions; or by what standard soever we frame in our minds the ideas of virtues or vices, they consist only, and are made up of collections of simple ideas, which we originally received from sense or reflection: and their rectitude or obliquity consists in the agreement or disagreement with those patterns prescribed by some law.

15. Moral actions may be regarded either absolutely, or as ideas of relation. To conceive rightly of moral actions, we must take notice of them under this two-fold consideration. First, as they are in themselves, each made up of such a collection of simple ideas. Thus drunkenness, or lying, signify such or such a collection of simple ideas, which I call mixed modes: and in this sense they are as much positive absolute ideas, as the drinking of a horse, or speaking of a parrot. Secondly, our actions are

considered as good, bad, or indifferent; and in this respect they are relative, it being their conformity to, or disagreement with some rule that makes them to be regular or irregular, good or bad; and so, as far as they are compared with a rule, and thereupon denominated, they come under relation. Thus the challenging and fighting with a man, as it is a certain positive mode, or particular sort of action, by particular ideas, distinguished from all others, is called duelling: which, when considered in relation to the law of God, will deserve the name of sin; to the law of fashion, in some countries, valour and virtue; and to the municipal laws of some governments, a capital crime. In this case, when the positive mode has one name, and another name as it stands in relation to the law, the distinction may as easily be observed as it is in substances, where one name, v.g. man, is used to signify the thing; another, v.g. father, to signify the relation.

16. The denominations of actions often mislead us. But because very frequently the positive idea of the action, and its moral relation, are comprehended together un-

der one name, and the game word made use of to express both the mode or action, and its moral rectitude or obliquity: therefore the relation itself is less taken notice of; and there is often no distinction made between the positive idea of the action, and the reference it has to a rule. By which confusion of these two distinct considerations under one term, those who yield too easily to the impressions of sounds, and are forward to take names for things, are often misled in their judgment of actions. Thus, the taking from another what is his, without his knowledge or allowance, is properly called stealing: but that name, being commonly understood to signify also the moral pravity of the action, and to denote its contrariety to the law, men are apt to condemn whatever they hear called stealing, as an ill action, disagreeing with the rule of right. And yet the private taking away his sword from a madman, to prevent his doing mischief, though it be properly denominated stealing, as the name of such a mixed mode; yet when compared to the law of God, and considered in its relation to that supreme rule, it is no sin or transgression, though

the name stealing ordinarily carries such an intimation with it.

17. Relations innumerable, and only the most considerable here mentioned. And thus much for the relation of human actions to a law, which, therefore, I call moral relations.

It would make a volume to go over all sorts of relations: it is not, therefore, to be expected that I should here mention them all. It suffices to our present purpose to show by these, what the ideas are we have of this comprehensive consideration called relation. Which is so various, and the occasions of it so many, (as many as there can be of comparing things one to another,) that it is not very easy to reduce it to rules, or under just heads. Those I have mentioned, I think, are some of the most considerable; and such as may serve to let us see from whence we get our ideas of relations, and wherein they are founded. But before I quit this argument, from what has been said give me leave to observe:

18. All relations terminate in simple ideas. First, That it is evident, that all relation terminates in, and is ulti-

mately founded on, those simple ideas we have got from sensation or reflection: so that all we have in our thoughts ourselves, (if we think of anything, or have any meaning), or would signify to others, when we use words standing for relations, is nothing but some simple ideas, or collections of simple ideas, compared one with another. This is so manifest in that sort called proportional, that nothing can be more. For when a man says "honey is sweeter than wax," it is plain that his thoughts in this relation terminate in this simple idea, sweetness; which is equally true of all the rest: though, where they are compounded, or decompounded, the simple ideas they are made up of, are, perhaps, seldom taken notice of: v.g. when the word father is mentioned: first, there is meant that particular species, or collective idea, signified by the word man; secondly, those sensible simple ideas, signified by the word generation; and, thirdly, the effects of it, and all the simple ideas signified by the word child. So the word friend, being taken for a man who loves and is ready to do good to another, has all these following ideas to the making of it up: first, all the

simple ideas, comprehended in the word man, or intelligent being; secondly, the idea of love; thirdly, the idea of readiness or disposition; fourthly, the idea of action, which is any kind of thought or motion; fifthly, the idea of good, which signifies anything that may advance his happiness, and terminates at last, if examined, in particular simple ideas, of which the word good in general signifies any one: but, if removed from all simple ideas quite, it signifies nothing at all. And thus also all moral words terminate at last, though perhaps more remotely, in a collection of simple ideas: the immediate signification of relative words, being very often other supposed known relations; which, if traced one to another, still end in simple ideas.

19. We have ordinarily as clear a notion of the relation, as of the simple ideas in things on which it is founded. Secondly, That in relations, we have for the most part, if not always, as clear a notion of the relation as we have of those simple ideas wherein it is founded: agreement or disagreement, whereon relation depends, being things whereof we have commonly as clear ideas as of

any other whatsoever; it being but the distinguishing simple ideas, or their degrees one from another, without which we could have no distinct knowledge at all. For, if I have a clear idea of sweetness, light, or extension, I have, too, of equal, or more, or less, of each of these: if I know what it is for one man to be born of a woman, viz. Sempronia, I know what it is for another man to be born of the same woman Sempronia; and so have as clear a notion of brothers as of births, and perhaps clearer. For if I believed that Sempronia digged Titus out of the parsley-bed, (as they used to tell children), and thereby became his mother; and that afterwards, in the same manner, she digged Caius out of the parsley-bed, I had as clear a notion of the relation of brothers between them, as if I had all the skill of a midwife: the notion that the same woman contributed, as mother, equally to their births, (though I were ignorant or mistaken in the manner of it), being that on which I grounded the relation; and that they agreed in that circumstance of birth, let it be what it will. The comparing them then in their descent from the same person, without knowing

the particular circumstances of that descent, is enough to found my notion of their having, or not having the relation of brothers. But though the ideas of particular relations are capable of being as clear and distinct in the minds of those who will duly consider them as those of mixed modes, and more determinate than those of substances: yet the names belonging to relation are often of as doubtful and uncertain signification as those of substances or mixed modes; and much more than those of simple ideas. Because relative words, being the marks of this comparison, which is made only by men's thoughts, and is an idea only in men's minds, men frequently apply them to different comparisons of things, according to their own imaginations; which do not always correspond with those of others using the same name.

20. The notion of relation is the same, whether the rule any action is compared to be true or false. Thirdly, That in these I call moral relations, I have a true notion of relation, by comparing the action with the rule, whether the rule be true or false. For if I measure any-

thing by a yard, I know whether the thing I measure be longer or shorter than that supposed yard, though perhaps the yard I measure by be not exactly the standard: which indeed is another inquiry. For though the rule be erroneous, and I mistaken in it; yet the agreement or disagreement observable in that which I compare with, makes me perceive the relation. Though, measuring by a wrong rule, I shall thereby be brought to judge amiss of its moral rectitude; because I have tried it by that which is not the true rule: yet I am not mistaken in the relation which that action bears to that rule I compare it to, which is agreement or disagreement.

Chapter XXIX

Of Clear and Obscure, Distinct and Confused Ideas

1. Ideas, some clear and distinct, others obscure and confused. Having shown the original of our ideas, and taken a view of their several sorts; considered the difference between the simple and the complex; and observed how the complex ones are divided into those of

modes, substances, and relations—all which, I think, is necessary to be done by any one who would acquaint himself thoroughly with the progress of the mind, in its apprehension and knowledge of things—it will, perhaps, be thought I have dwelt long enough upon the examination of ideas. I must nevertheless, crave leave to offer some few other considerations concerning them.

The first is, that some are clear and others obscure; some distinct and others confused.

2. Clear and obscure explained by sight. The perception of the mind being most aptly explained by words relating to the sight, we shall best understand what is meant by clear and obscure in our ideas, by reflecting on what we call clear and obscure in the objects of sight. Light being that which discovers to us visible objects, we give the name of obscure to that which is not placed in a light sufficient to discover minutely to us the figure and colours which are observable in it, and which, in a better light, would be discernible. In like manner, our simple ideas are clear, when they are such as the objects themselves from whence they were taken did or might,

in a well-ordered sensation or perception, present them. Whilst the memory retains them thus, and can produce them to the mind whenever it has occasion to consider them, they are clear ideas. So far as they either want anything of the original exactness, or have lost any of their first freshness, and are, as it were, faded or tarnished by time, so far are they obscure. Complex ideas, as they are made up of simple ones, so they are clear, when the ideas that go to their composition are clear, and the number and order of those simple ideas that are the ingredients of any complex one is determinate and certain.

3. Causes of obscurity. The causes of obscurity, in simple ideas, seem to be either dull organs; or very slight and transient impressions made by the objects; or else a weakness in the memory, not able to retain them as received. For to return again to visible objects, to help us to apprehend this matter. If the organs, or faculties of perception, like wax over-hardened with cold, will not receive the impression of the seal, from the usual impulse wont to imprint it; or, like wax of a temper too soft, will

not hold it well, when well imprinted; or else supposing the wax of a temper fit, but the seal not applied with a sufficient force to make a clear impression: in any of these cases, the print left by the seal will be obscure. This, I suppose, needs no application to make it plainer.

4. Distinct and confused, what. As a clear idea is that whereof the mind has such a full and evident perception, as it does receive from an outward object operating duly on a well-disposed organ, so a distinct idea is that wherein the mind perceives a difference from all other; and a confused idea is such an one as is not sufficiently distinguishable from another, from which it ought to be different.

5. Objection. If no idea be confused, but such as is not sufficiently distinguishable from another from which it should be different, it will be hard, may any one say, to find anywhere a confused idea. For, let any idea be as it will, it can be no other but such as the mind perceives it to be; and that very perception sufficiently distinguishes it from all other ideas, which cannot be other, i.e. different, without being perceived to be so. No idea,

therefore, can be undistinguishable from another from which it ought to be different, unless you would have it different from itself: for from all other it is evidently different.

6. Confusion of ideas is in reference to their names. To remove this difficulty, and to help us to conceive aright what it is that makes the confusion ideas are at any time chargeable with, we must consider, that things ranked under distinct names are supposed different enough to be distinguished, that so each sort by its peculiar name may be marked, and discoursed of apart upon any occasion: and there is nothing more evident, than that the greatest part of different names are supposed to stand for different things. Now every idea a man has, being visibly what it is, and distinct from all other ideas but itself; that which makes it confused, is, when it is such that it may as well be called by another name as that which it is expressed by; the difference which keeps the things (to be ranked under those two different names) distinct, and makes some of them belong rather to the one and some of them to the other of

those names, being left out; and so the distinction, which was intended to be kept up by those different names, is quite lost.

7. Defaults which make this confusion. The defaults which usually occasion this confusion, I think, are chiefly these following:

Complex Ideas made up of too few simple ones. First, when any complex idea (for it is complex ideas that are most liable to confusion) is made up of too small a number of simple ideas, and such only as are common to other things, whereby the differences that make it deserve a different name, are left out. Thus, he that has an idea made up of barely the simple ones of a beast with spots, has but a confused idea of a leopard; it not being thereby sufficiently distinguished from a lynx, and several other sorts of beasts that are spotted. So that such an idea, though it hath the peculiar name leopard, is not distinguishable from those designed by the names lynx or panther, and may as well come under the name lynx as leopard. How much the custom of defining of words by general terms contributes to make

the ideas we would express by them confused and undetermined, I leave others to consider. This is evident, that confused ideas are such as render the use of words uncertain, and take away the benefit of distinct names. When the ideas, for which we use different terms, have not a difference answerable to their distinct names, and so cannot be distinguished by them, there it is that they are truly confused.

8. Their simple ones jumbled disorderly together. Secondly, Another fault which makes our ideas confused is, when, though the particulars that make up any idea are in number enough, yet they are so jumbled together, that it is not easily discernible whether it more belongs to the name that is given it than to any other. There is nothing properer to make us conceive this confusion than a sort of pictures, usually shown as surprising pieces of art, wherein the colours, as they are laid by the pencil on the table itself, mark out very odd and unusual figures, and have no discernible order in their position. This draught, thus made up of parts wherein no symmetry nor order appears, is in itself no more a confused

thing, than the picture of a cloudy sky; wherein, though there be as little order of colours or figures to be found, yet nobody thinks it a confused picture. What is it, then, that makes it be thought confused, since the want of symmetry does not? As it is plain it does not: for another draught made barely in imitation of this could not be called confused. I answer, That which makes it be thought confused is, the applying it to some name to which it does no more discernibly belong than to some other: v.g. when it is said to be the picture of a man, or Caesar, then any one with reason counts it confused; because it is not discernible in that state to belong more to the name man, or Caesar, than to the name baboon, or Pompey: which are supposed to stand for different ideas from those signified by man, or Caesar. But when a cylindrical mirror, placed right, had reduced those irregular lines on the table into their due order and proportion, then the confusion ceases, and the eye presently sees that it is a man, or Caesar; i.e. that it belongs to those names; and that it is sufficiently distinguishable from a baboon, or Pompey; i.e. from the ideas sig-

nified by those names. Just thus it is with our ideas, which are as it were the pictures of things. No one of these mental draughts, however the parts are put together, can be called confused (for they are plainly discernible as they are) till it be ranked under some ordinary name to which it cannot be discerned to belong, any more than it does to some other name of an allowed different signification.

9. Their simple ones mutable and undetermined. Thirdly, A third defect that frequently gives the name of confused to our ideas, is, when any one of them is uncertain and undetermined. Thus we may observe men who, not forbearing to use the ordinary words of their language till they have learned their precise signification, change the idea they make this or that term stand for, almost as often as they use it. He that does this out of uncertainty of what he should leave out, or put into his idea of church, or idolatry, every time he thinks of either, and holds not steady to any one precise combination of ideas that makes it up, is said to have a confused idea of idolatry or the church: though this be still for

the same reason as the former, viz. because a mutable idea (if we will allow it to be one idea) cannot belong to one name rather than another, and so loses the distinction that distinct names are designed for.

10. Confusion without reference to names, hardly conceivable. By what has been said, we may observe how much names, as supposed steady signs of things, and by their difference to stand for, and keep things distinct that in themselves are different, are the occasion of denominating ideas distinct or confused, by a secret and unobserved reference the mind makes of its ideas to such names. This perhaps will be fuller understood, after what I say of Words in the third Book has been read and considered. But without taking notice of such a reference of ideas to distinct names, as the signs of distinct things, it will be hard to say what a confused idea is. And therefore when a man designs, by any name, a sort of things, or any one particular thing, distinct from all others, the complex idea he annexes to that name is the more distinct, the more particular the ideas are, and the greater and more determinate the number and order

of them is, whereof it is made up. For, the more it has of these, the more it has still of the perceivable differences, whereby it is kept separate and distinct from all ideas belonging to other names, even those that approach nearest to it, and thereby all confusion with them is avoided.

11. Confusion concerns always two ideas. Confusion making it a difficulty to separate two things that should be separated, concerns always two ideas; and those most which most approach one another. Whenever, therefore, we suspect any idea to be confused, we must examine what other it is in danger to be confounded with, or which it cannot easily be separated from; and that will always be found an idea belonging to another name, and so should be a different thing, from which yet it is not sufficiently distinct: being either the same with it, or making a part of it, or at least as properly called by that name as the other it is ranked under; and so keeps not that difference from that other idea which the different names import.

12. Causes of confused ideas. This, I think, is the confusion proper to ideas; which still carries with it a secret

reference to names. At least, if there be any other confusion of ideas, this is that which most of all disorders men's thoughts and discourses: ideas, as ranked under names, being those that for the most part men reason of within themselves, and always those which they commune about with others. And therefore where there are supposed two different ideas, marked by two different names, which are not as distinguishable as the sounds that stand for them, there never fails to be confusion; and where any ideas are distinct as the ideas of those two sounds they are marked by, there can be between them no confusion. The way to prevent it is to collect and unite into one complex idea, as precisely as is possible, all those ingredients whereby it is differenced from others; and to them, so united in a determinate number and order, apply steadily the same name. But this neither accommodating men's ease or vanity, nor serving any design but that of naked truth, which is not always the thing aimed at, such exactness is rather to be wished than hoped for. And since the loose application of names, to undetermined, variable, and almost no ideas, serves

both to cover our own ignorance, as well as to perplex and confound others, which goes for learning and superiority in knowledge, it is no wonder that most men should use it themselves, whilst they complain of it in others. Though I think no small part of the confusion to be found in the notions of men might, by care and ingenuity, be avoided, yet I am far from concluding it everywhere wilful. Some ideas are so complex, and made up of so many parts, that the memory does not easily retain the very same precise combination of simple ideas under one name: much less are we able constantly to divine for what precise complex idea such a name stands in another man's use of it. From the first of these, follows confusion in a man's own reasonings and opinions within himself; from the latter, frequent confusion in discoursing and arguing with others. But having more at large treated of Words, their defects, and abuses, in the following Book, I shall here say no more of it.

13. Complex ideas may be distinct in one part, and confused in another. Our complex ideas, being made up of collections, and so variety of simple ones, may ac-

cordingly be very clear and distinct in one part, and very obscure and confused in another. In a man who speaks of a chiliaedron, or a body of a thousand sides, the ideas of the figure may be very confused, though that of the number be very distinct; so that he being able to discourse and demonstrate concerning that part of his complex idea which depends upon the number of thousand, he is apt to think he has a distinct idea of a chiliaedron; though it be plain he has no precise idea of its figure, so as to distinguish it, by that, from one that has but 999 sides: the not observing whereof causes no small error in men's thoughts, and confusion in their discourses.

14. This, if not heeded, causes confusion in our arguings. He that thinks he has a distinct idea of the figure of a chiliaedron, let him for trial sake take another parcel of the same uniform matter, viz. gold or wax of an equal bulk, and make it into a figure of 999 sides. He will, I doubt not, be able to distinguish these two ideas one from another, by the number of sides; and reason and argue distinctly about them, whilst he keeps his thoughts

and reasoning to that part only of these ideas which is contained in their numbers; as that the sides of the one could be divided into two equal numbers, and of the others not, &c. But when he goes about to distinguish them by their figure, he will there be presently at a loss, and not be able, I think, to frame in his mind two ideas, one of them distinct from the other, by the bare figure of these two pieces of gold; as he could, if the same parcels of gold were made one into a cube, the other a figure of five sides. In which incomplete ideas, we are very apt to impose on ourselves, and wrangle with others, especially where they have particular and familiar names. For, being satisfied in that part of the idea which we have clear; and the name which is familiar to us, being applied to the whole, containing that part also which is imperfect and obscure, we are apt to use it for that confused part, and draw deductions from it in the obscure part of its signification, as confidently as we do from the other.

15. Instance in eternity. Having frequently in our mouths the name Eternity, we are apt to think we have

a positive comprehensive idea of it, which is as much as to say, that there is no part of that duration which is not clearly contained in our idea. It is true that he that thinks so may have a clear idea of duration; he may also have a clear idea of a very great length of duration; he may also have a clear idea of the comparison of that great one with still a greater: but it not being possible for him to include in his idea of any duration, let it be as great as it will, the whole extent together of a duration, where he supposes no end, that part of his idea, which is still beyond the bounds of that large duration he represents to his own thoughts, is very obscure and undetermined. And hence it is that in disputes and reasonings concerning eternity, or any other infinite, we are very apt to blunder, and involve ourselves in manifest absurdities.

16. Infinite divisibility of matter. In matter, we have no clear ideas of the smallness of parts much beyond the smallest that occur to any of our senses: and therefore, when we talk of the divisibility of matter in infinitum, though we have clear ideas of division and divisibility,

and have also clear ideas of parts made out of a whole by division; yet we have but very obscure and confused ideas of corpuscles, or minute bodies, so to be divided, when, by former divisions, they are reduced to a smallness much exceeding the perception of any of our senses; and so all that we have clear and distinct ideas of is of what division in general or abstractedly is, and the relation of totum and pars: but of the bulk of the body, to be thus infinitely divided after certain progressions, I think, we have no clear nor distinct idea at all. For I ask any one, whether, taking the smallest atom of dust he ever saw, he has any distinct idea (bating still the number, which concerns not extension) betwixt the 1,000,000th and the 1,000,000,000th part of it. Or if he think he can refine his ideas to that degree, without losing sight of them, let him add ten cyphers to each of those numbers. Such a degree of smallness is not unreasonable to be supposed; since a division carried on so far brings it no nearer the end of infinite division, than the first division into two halves does. I must confess, for my part, I have no clear distinct ideas of the differ-

ent bulk or extension of those bodies, having but a very obscure one of either of them. So that, I think, when we talk of division of bodies in infinitum, our idea of their distinct bulks, which is the subject and foundation of division, comes, after a little progression, to be confounded, and almost lost in obscurity. For that idea which is to represent only bigness must be very obscure and confused, which we cannot distinguish from one ten times as big, but only by number: so that we have clear distinct ideas, we may say, of ten and one, but no distinct ideas of two such extensions. It is plain from hence, that, when we talk of infinite divisibility of body or extension, our distinct and clear ideas are only of numbers: but the clear distinct ideas of extension after some progress of division, are quite lost; and of such minute parts we have no distinct ideas at all; but it returns, as all our ideas of infinite do, at last to that of number always to be added; but thereby never amounts to any distinct idea of actual infinite parts. We have, it is true, a clear idea of division, as often as we think of it; but thereby we have no more a clear idea of infinite

parts in matter, than we have a clear idea of an infinite number, by being able still to add new numbers to any assigned numbers we have: endless divisibility giving us no more a clear and distinct idea of actually infinite parts, than endless addibility (if I may so speak) gives us a clear and distinct idea of an actually infinite number: they both being only in a power still of increasing the number, be it already as great as it will. So that of what remains to be added (wherein consists the infinity) we have but an obscure, imperfect, and confused idea; from or about which we can argue or reason with no certainty or clearness, no more than we can in arithmetic, about a number of which we have no such distinct idea as we have of 4 or 100; but only this relative obscure one, that, compared to any other, it is still bigger: and we have no more a clear positive idea of it, when we say or conceive it is bigger, or more than 400,000,000, than if we should say it is bigger than 40 or 4: 400,000,000 having no nearer a proportion to the end of addition or number than 4. For he that adds only 4 to 4, and so proceeds, shall as soon come to the end of all addition,

as he that adds 400,000,000 to 400,000,000. And so likewise in eternity; he that has an idea of but four years, has as much a positive complete idea of eternity, as he that has one of 400,000,000 of years: for what remains of eternity beyond either of these two numbers of years, is as clear to the one as the other; i.e. neither of them has any clear positive idea of it at all. For he that adds only 4 years to 4, and so on, shall as soon reach eternity as he that adds 400,000,000 of years, and so on; or, if he please, doubles the increase as often as he will: the remaining abyss being still as far beyond the end of all these progressions as it is from the length of a day or an hour. For nothing finite bears any proportion to infinite; and therefore our ideas, which are all finite, cannot bear any. Thus it is also in our idea of extension, when we increase it by addition, as well as when we diminish it by division, and would enlarge our thoughts to infinite space. After a few doublings of those ideas of extension, which are the largest we are accustomed to have, we lose the clear distinct idea of that space: it becomes a confusedly great one, with a surplus of still

greater; about which, when we would argue or reason, we shall always find ourselves at a loss; confused ideas, in our arguings and deductions from that part of them which is confused, always leading us into confusion.

Chapter XXX Of Real and Fantastical Ideas

1. Ideas considered in reference to their archetypes. Besides what we have already mentioned concerning ideas, other considerations belong to them, in reference to things from whence they are taken, or which they may be supposed to represent; and thus, I think, they may come under a three-fold distinction, and are:—

First, either real or fantastical;

Secondly, adequate or inadequate;

Thirdly, true or false.

First, by real ideas, I mean such as have a foundation in nature; such as have a conformity with the real being and existence of things, or with their archetypes. Fantastical or chimerical, I call such as have no founda-

tion in nature, nor have any conformity with that reality of being to which they are tacitly referred, as to their archetypes. If we examine the several sorts of ideas before mentioned, we shall find that,

2. Simple ideas are all real appearances of things. First, Our simple ideas are all real, all agree to the reality of things: not that they are all of them the images or representations of what does exist; the contrary whereof, in all but the primary qualities of bodies, hath been already shown. But, though whiteness and coldness are no more in snow than pain is; yet those ideas of whiteness and coldness, pain, &c., being in us the effects of powers in things without us, ordained by our Maker to produce in us such sensations; they are real ideas in us, whereby we distinguish the qualities that are really in things themselves. For, these several appearances being designed to be the mark whereby we are to know and distinguish things which we have to do with, our ideas do as well serve us to that purpose, and are as real distinguishing characters, whether they be only constant effects, or else exact resemblances of something in

the things themselves: the reality lying in that steady correspondence they have with the distinct constitutions of real beings. But whether they answer to those constitutions, as to causes or patterns, it matters not; it suffices that they are constantly produced by them. And thus our simple ideas are all real and true, because they answer and agree to those powers of things which produce them in our minds; that being all that is requisite to make them real, and not fictions at pleasure. For in simple ideas (as has been shown) the mind is wholly confined to the operation of things upon it, and can make to itself no simple idea, more than what it has received.

3. Complex ideas are voluntary combinations. Though the mind be wholly passive in respect of its simple ideas; yet, I think, we may say it is not so in respect of its complex ideas. For those being combinations of simple ideas put together, and united under one general name, it is plain that the mind of man uses some kind of liberty in forming those complex ideas: how else comes it to pass that one man's idea of gold, or justice, is differ-

ent from another's, but because he has put in, or left out of his, some simple idea which the other has not? The question then is, Which of these are real, and which barely imaginary combinations? What collections agree to the reality of things, and what not? And to this I say that,

4. Mixed modes and relations, made of consistent ideas, are real. Secondly, Mixed modes and relations, having no other reality but what they have in the minds of men, there is nothing more required to this kind of ideas to make them real, but that they be so framed, that there be a possibility of existing conformable to them. These ideas themselves, being archetypes, cannot differ from their archetypes, and so cannot be chimerical, unless any one will jumble together in them inconsistent ideas. Indeed, as any of them have the names of a known language assigned to them, by which he that has them in his mind would signify them to others, so bare possibility of existing is not enough; they must have a conformity to the ordinary signification of the name that is given them, that they may not be thought

fantastical: as if a man would give the name of justice to that idea which common use calls liberality. But this fantasticalness relates more to propriety of speech, than reality of ideas. For a man to be undisturbed in danger, sedately to consider what is fittest to be done, and to execute it steadily, is a mixed mode, or a complex idea of an action which may exist. But to be undisturbed in danger, without using one's reason or industry, is what is also possible to be; and so is as real an idea as the other. Though the first of these, having the name courage given to it, may, in respect of that name, be a right or wrong idea; but the other, whilst it has not a common received name of any known language assigned to it, is not capable of any deformity, being made with no reference to anything but itself.

5. Complex ideas of substances are real, when they agree with the existence of things. Thirdly, Our complex ideas of substances, being made all of them in reference to things existing without us, and intended to be representations of substances as they really are, are no further real than as they are such combinations of simple

ideas as are really united, and co-exist in things without us. On the contrary, those are fantastical which are made up of such collections of simple ideas as were really never united, never were found together in any substance: v.g. a rational creature, consisting of a horse's head, joined to a body of human shape, or such as the centaurs are described: or, a body yellow, very malleable, fusible, and fixed, but lighter than common water: or an uniform, unorganized body, consisting, as to sense, all of similar parts, with perception and voluntary motion joined to it. Whether such substances as these can possibly exist or no, it is probable we do not know: but be that as it will, these ideas of substances, being made conformable to no pattern existing that we know; and consisting of such collections of ideas as no substance ever showed us united together, they ought to pass with us for barely imaginary: but much more are those complex ideas so, which contain in them any inconsistency or contradiction of their parts.

Chapter XXXI

Of Adequate and Inadequate Ideas

1. Adequate ideas are such as perfectly represent their archetypes. Of our real ideas, some are adequate, and some are inadequate. Those I call adequate, which perfectly represent those archetypes which the mind supposes them taken from: which it intends them to stand for, and to which it refers them. Inadequate ideas are such, which are but a partial or incomplete representation of those archetypes to which they are referred. Upon which account it is plain,

2. Simple ideas all adequate. First, that all our simple ideas are adequate. Because, being nothing but the effects of certain powers in things, fitted and ordained by God to produce such sensations in us, they cannot but be correspondent and adequate to those powers: and we are sure they agree to the reality of things. For, if sugar produce in us the ideas which we call whiteness and sweetness, we are sure there is a power in sugar to produce those ideas in our minds, or else they could not

have been produced by it. And so each sensation answering the power that operates on any of our senses, the idea so produced is a real idea, (and not a fiction of the mind, which has no power to produce any simple idea); and cannot but be adequate, since it ought only to answer that power: and so all simple ideas are adequate. It is true, the things producing in us these simple ideas are but few of them denominated by us, as if they were only the causes of them; but as if those ideas were real beings in them. For, though fire be called painful to the touch, whereby is signified the power of producing in us the idea of pain, yet it is denominated also light and hot; as if light and heat were really something in the fire, more than a power to excite these ideas in us; and therefore are called qualities in or of the fire. But these being nothing, in truth, but powers to excite such ideas in us, I must in that sense be understood, when I speak of secondary qualities as being in things; or of their ideas as being the objects that excite them in us. Such ways of speaking, though accommodated to the vulgar notions, without which one cannot be well understood, yet truly signify nothing but those powers which

are in things to excite certain sensations or ideas in us. Since were there no fit organs to receive the impressions fire makes on the sight and touch, nor a mind joined to those organs to receive the ideas of light and heat by those impressions from the fire or sun, there would yet be no more light or heat in the world than there would be pain if there were no sensible creature to feel it, though the sun should continue just as it is now, and Mount AEtina flame higher than ever it did. Solidity and extension, and the termination of it, figure, with motion and rest, whereof we have the ideas, would be really in the world as they are, whether there were any sensible being to perceive them or no: and therefore we have reason to look on those as the real modifications of matter, and such as are the exciting causes of all our various sensations from bodies. But this being an inquiry not belonging to this place, I shall enter no further into it, but proceed to show what complex ideas are adequate, and what not.

3. Modes are all adequate. Secondly, our complex ideas of modes, being voluntary collections of simple ideas, which the mind puts together, without reference to any

real archetypes, or standing patterns, existing anywhere, are and cannot but be adequate ideas. Because they, not being intended for copies of things really existing, but for archetypes made by the mind, to rank and denominate things by, cannot want anything; they having each of them that combination of ideas, and thereby that perfection, which the mind intended they should: so that the mind acquiesces in them, and can find nothing wanting. Thus, by having the idea of a figure with three sides meeting at three angles, I have a complete idea, wherein I require nothing else to make it perfect. That the mind is satisfied with the perfection of this its idea is plain, in that it does not conceive that any understanding hath, or can have, a more complete or perfect idea of that thing it signifies by the word triangle, supposing it to exist, than itself has, in that complex idea of three sides and three angles, in which is contained all that is or can be essential to it, or necessary to complete it, wherever or however it exists. But in our ideas of substances it is otherwise. For there, desiring to copy things as they really do exist, and to represent to our-

selves that constitution on which all their properties depend, we perceive our ideas attain not that perfection we intend: we find they still want something we should be glad were in them; and so are all inadequate. But mixed modes and relations, being archetypes without patterns, and so having nothing to represent but themselves, cannot but be adequate, everything being so to itself. He that at first put together the idea of danger perceived, absence of disorder from fear, sedate consideration of what was justly to be done, and executing that without disturbance, or being deterred by the danger of it, had certainly in his mind that complex idea made up of that combination: and intending it to be nothing else but what is, nor to have in it any other simple ideas but what it hath, it could not also but be an adequate idea: and laying this up in his memory, with the name courage annexed to it, to signify to others, and denominate from thence any action he should observe to agree with it, had thereby a standard to measure and denominate actions by, as they agreed to it. This idea, thus made and laid up for a pattern, must

necessarily be adequate, being referred to nothing else but itself, nor made by any other original but the good liking and will of him that first made this combination.

4. Modes, in reference to settled names, may be inadequate. Indeed another coming after, and in conversation learning from him the word courage, may make an idea, to which he gives the name courage, different from what the first author applied it to, and has in his mind when he uses it. And in this case, if he designs that his idea in thinking should be conformable to the other's idea, as the name he uses in speaking is conformable in sound to his from whom he learned it, his idea may be very wrong and inadequate: because in this case, making the other man's idea the pattern of his idea in thinking, as the other man's word or sound is the pattern of his in speaking, his idea is so far defective and inadequate, as it is distant from the archetype and pattern he refers it to, and intends to express and signify by the name he uses for it; which name he would have to be a sign of the other man's idea, (to which, in its proper use, it is primarily annexed), and of his own, as agree-

ing to it: to which if his own does not exactly correspond, it is faulty and inadequate.

5. Because then meant, in propriety of speech, to correspond to the ideas in some other mind. Therefore these complex ideas of modes, which they are referred by the mind, and intended to correspond to the ideas in the mind of some other intelligent being, expressed by the names we apply to them, they may be very deficient, wrong, and inadequate; because they agree not to that which the mind designs to be their archetype and pattern: in which respect only any idea of modes can be wrong, imperfect, or inadequate. And on this account our ideas of mixed modes are the most liable to be faulty of any other; but this refers more to proper speaking than knowing right.

6. Ideas of substances, as referred to real essences, not adequate. Thirdly, what ideas we have of substances, I have above shown. Now, those ideas have in the mind a double reference: 1. Sometimes they are referred to a supposed real essence of each species of things. 2. Sometimes they are only designed to be pictures and repre-

sentations in the mind of things that do exist, by ideas of those qualities that are discoverable in them. In both which ways these copies of those originals and archetypes are imperfect and inadequate. First, it is usual for men to make the names of substances stand for things as supposed to have certain real essences, whereby they are of this or that species: and names standing for nothing but the ideas that are in men's minds, they must constantly refer their ideas to such real essences, as to their archetypes. That men (especially such as have been bred up in the learning taught in this part of the world) do suppose certain specific essences of substances, which each individual in its several kinds is made conformable to and partakes of, is so far from needing proof that it will be thought strange if any one should do otherwise. And thus they ordinarily apply the specific names they rank particular substances under, to things as distinguished by such specific real essences. Who is there almost, who would not take it amiss if it should be doubted whether he called himself a man, with any other meaning than as having the real essence of a man? And yet if

you demand what those real essences are, it is plain men are ignorant, and know them not. From whence it follows, that the ideas they have in their minds, being referred to real essences, as to archetypes which are unknown, must be so far from being adequate that they cannot be supposed to be any representation of them at all. The complex ideas we have of substances are, as it has been shown, certain collections of simple ideas that have been observed or supposed constantly to exist together. But such a complex idea cannot be the real essence of any substance; for then the properties we discover in that body would depend on that complex idea, and be deducible from it, and their necessary connexion with it be known; as all properties of a triangle depend on, and, as far as they are discoverable, are deducible from the complex idea of three lines including a space. But it is plain that in our complex ideas of substances are not contained such ideas, on which all the other qualities that are to be found in them do depend. The common idea men have of iron is, a body of a certain colour, weight, and hardness; and a property that they

look on as belonging to it, is malleableness. But yet this property has no necessary connexion with that complex idea, or any part of it: and there is no more reason to think that malleableness depends on that colour, weight, and hardness, than that colour or that weight depends on its malleableness. And yet, though we know nothing of these real essences, there is nothing more ordinary than that men should attribute the sorts of things to such essences. The particular parcel of matter which makes the ring I have on my finger is forwardly by most men supposed to have a real essence, whereby it is gold; and from whence those qualities flow which I find in it, viz. its peculiar colour, weight, hardness, fusibility, fixedness, and change of colour upon a slight touch of mercury, &c. This essence, from which all these properties flow, when I inquire into it and search after it, I plainly perceive I cannot discover: the furthest I can go is, only to presume that, it being nothing but body, its real essence or internal constitution, on which these qualities depend, can be nothing but the figure, size, and connexion of its solid parts; of neither of which

having any distinct perception at all can I have any idea of its essence: which is the cause that it has that particular shining yellowness; a greater weight than anything I know of the same bulk; and a fitness to have its colour changed by the touch of quicksilver. If any one will say, that the real essence and internal constitution, on which these properties depend, is not the figure, size, and arrangement or connexion of its solid parts, but something else, called its particular form, I am further from having any idea of its real essence than I was before. For I have an idea of figure, size, and situation of solid parts in general, though I have none of the particular figure, size, or putting together of parts, whereby the qualities above mentioned are produced; which qualities I find in that particular parcel of matter that is on my finger, and not in another parcel of matter, with which I cut the pen I write with. But, when I am told that something besides the figure, size, and posture of the solid parts of that body in its essence, something called substantial form, of that I confess I have no idea at all, but only of the sound form; which is far enough

from an idea of its real essence or constitution. The like ignorance as I have of the real essence of this particular substance, I have also of the real essence of all other natural ones: of which essences I confess I have no distinct ideas at all; and, I am apt to suppose, others, when they examine their own knowledge, will find in themselves, in this one point, the same sort of ignorance.

7. Because men know not the real essences of substances. Now, then, when men apply to this particular parcel of matter on my finger a general name already in use, and denominate it gold, do they not ordinarily, or are they not understood to give it that name, as belonging to a particular species of bodies, having a real internal essence; by having of which essence this particular substance comes to be of that species, and to be called by that name? If it be so, as it is plain it is, the name by which things are marked as having that essence must be referred primarily to that essence; and consequently the idea to which that name is given must be referred also to that essence, and be intended to represent it. Which essence, since they who so use the names know

not, their ideas of substances must be all inadequate in that respect, as not containing in them that real essence which the mind intends they should.

8. Ideas of substances, when regarded as collections of their qualities, are all inadequate. Secondly, those who, neglecting that useless supposition of unknown real essences, whereby they are distinguished, endeavour to copy the substances that exist in the world, by putting together the ideas of those sensible qualities which are found coexisting in them, though they come much nearer a likeness of them than those who imagine they know not what real specific essences: yet they arrive not at perfectly adequate ideas of those substances they would thus copy into the their minds: nor do those copies exactly and fully contain all that is to be found in their archetypes. Because those qualities and powers of substances, whereof we make their complex ideas, are so many and various, that no man's complex idea contains them all. That our complex ideas of substances do not contain in them all the simple ideas that are united in the things themselves is evident, in that men do rarely

put into their complex idea of any substance all the simple ideas they do know to exist in it. Because, endeavouring to make the signification of their names as clear and as little cumbersome as they can, they make their specific ideas of the sorts of substance, for the most part, of a few of those simple ideas which are to be found in them: but these having no original precedency, or right to be put in, and make the specific idea, more than others that are left out, it is plain that both these ways our ideas of substances are deficient and inadequate. The simple ideas whereof we make our complex ones of substances are all of them (bating only the figure and bulk of some sorts) powers; which being relations to other substances, we can never be sure that we know all the powers that are in any one body, till we have tried what changes it is fitted to give to or receive from other substances in their several ways of application: which being impossible to be tried upon any one body, much less upon all, it is impossible we should have adequate ideas of any substance made up of a collection of all its properties.

9. Their powers usually make up our complex ideas of substances. Whosoever first lighted on a parcel of that sort of substance we denote by the word gold, could not rationally take the bulk and figure he observed in that lump to depend on its real essence, or internal constitution. Therefore those never went into his idea of that species of body; but its peculiar colour, perhaps, and weight, were the first he abstracted from it, to make the complex idea of that species. Which both are but powers; the one to affect our eyes after such a manner, and to produce in us that idea we call yellow; and the other to force upwards any other body of equal bulk, they being put into a pair of equal scales, one against another. Another perhaps added to these the ideas of fusibility and fixedness, two other passive powers, in relation to the operation of fire upon it; another, its ductility and solubility in aqua regia, two other powers, relating to the operation of other bodies, in changing its outward figure, or separation of it into insensible parts. These, or parts of these, put together, usually make the complex idea in men's minds of that sort of body we call gold.

10. Substances have innumerable powers not contained in our complex ideas of them. But no one who hath considered the properties of bodies in general, or this sort in particular, can doubt that this, called gold, has infinite other properties not contained in that complex idea. Some who have examined this species more accurately could, I believe, enumerate ten times as many properties in gold, all of them as inseparable from its internal constitution, as its colour or weight: and it is probable, if any one knew all the properties that are by divers men known of this metal, there would be an hundred times as many ideas go to the complex idea of gold as any one man yet has in his; and yet perhaps that not be the thousandth part of what is to be discovered in it. The changes that that one body is apt to receive, and make in other bodies, upon a due application, exceeding far not only what we know, but what we are apt to imagine. Which will not appear so much a paradox to any one who will but consider how far men are yet from knowing all the properties of that one, no very compound figure, a triangle; though it be no small number

that are already by mathematicians discovered of it.

11. Ideas of substances, being got only by collecting their qualities, are all inadequate. So that all our complex ideas of substances are imperfect and inadequate. Which would be so also in mathematical figures, if we were to have our complex ideas of them, only by collecting their properties in reference to other figures. How uncertain and imperfect would our ideas be of an ellipsis, if we had no other idea of it, but some few of its properties? Whereas, having in our plain idea the whole essence of that figure, we from thence discover those properties, and demonstratively see how they flow, and are inseparable from it.

12. Simple ideas, *ektupa*, and adequate. Thus the mind has three sorts of abstract ideas or nominal essences:

First, simple ideas, which are *ektupa* or copies; but yet certainly adequate. Because, being intended to express nothing but the power in things to produce in the mind such a sensation, that sensation when it is produced, cannot but be the effect of that power. So the paper I write on, having the power in the light (I speak

according to the common notion of light) to produce in men the sensation which I call white, it cannot but be the effect of such a power in something without the mind; since the mind has not the power to produce any such idea in itself: and being meant for nothing else but the effect of such a power, that simple idea is real and adequate; the sensation of white, in my mind, being the effect of that power which is in the paper to produce it, is perfectly adequate to that power; or else that power would produce a different idea.

13. Ideas of substances are *ektupa*, and inadequate.

Secondly, the complex ideas of substances are ectypes, copies too; but not perfect ones, not adequate: which is very evident to the mind, in that it plainly perceives, that whatever collection of simple ideas it makes of any substance that exists, it cannot be sure that it exactly answers all that are in that substance. Since, not having tried all the operations of all other substances upon it, and found all the alterations it would receive from, or cause in, other substances, it cannot have an exact adequate collection of all its active and passive capaci-

ties; and so not have an adequate complex idea of the powers of any substance existing, and its relations; which is that sort of complex idea of substances we have. And, after all, if we would have, and actually had, in our complex idea, an exact collection of all the secondary qualities or powers of any substance, we should not yet thereby have an idea of the essence of that thing. For, since the powers or qualities that are observable by us are not the real essence of that substance, but depend on it, and flow from it, any collection whatsoever of these qualities cannot be the real essence of that thing. Whereby it is plain, that our ideas of substances are not adequate; are not what the mind intends them to be. Besides, a man has no idea of substance in general, nor knows what substance is in itself.

14. Ideas of modes and relations are archetypes and cannot be adequate. Thirdly, complex ideas of modes and relations are originals, and archetypes; are not copies, nor made after the pattern of any real existence, to which the mind intends them to be conformable, and exactly to answer. These being such collections of simple

ideas that the mind itself puts together, and such collections that each of them contains in it precisely all that the mind intends that it should, they are archetypes and essences of modes that may exist; and so are designed only for, and belong only to such modes as, when they do exist, have an exact conformity with those complex ideas. The ideas, therefore, of modes and relations cannot but be adequate.

Chapter XXXII

Of True and False Ideas

1. Truth and falsehood properly belong to propositions, not to ideas. Though truth and falsehood belong, in propriety of speech, only to propositions: yet ideas are oftentimes termed true or false (as what words are there that are not used with great latitude, and with some deviation from their strict and proper significations?) Though I think that when ideas themselves are termed true or false, there is still some secret or tacit proposition, which is the foundation of that denomination: as

we shall see, if we examine the particular occasions wherein they come to be called true or false. In all which we shall find some kind of affirmation or negation, which is the reason of that denomination. For our ideas, being nothing but bare appearances, or perceptions in our minds, cannot properly and simply in themselves be said to be true or false, no more than a single name of anything can be said to be true or false.

2. Ideas and words may be said to be true, inasmuch as they really are ideas and words. Indeed both ideas and words may be said to be true, in a metaphysical sense of the word truth; as all other things that any way exist are said to be true, i.e. really to be such as they exist. Though in things called true, even in that sense, there is perhaps a secret reference to our ideas, looked upon as the standards of that truth; which amounts to a mental proposition, though it be usually not taken notice of.

3. No idea, as an appearance in the mind, either true or false. But it is not in that metaphysical sense of truth which we inquire here, when we examine, whether our ideas are capable of being true or false, but in the more

ordinary acceptation of those words: and so I say that the ideas in our minds, being only so many perceptions or appearances there, none of them are false; the idea of a centaur having no more falsehood in it when it appears in our minds, than the name centaur has falsehood in it, when it is pronounced by our mouths, or written on paper. For truth or falsehood lying always in some affirmation or negation, mental or verbal, our ideas are not capable, any of them, of being false, till the mind passes some judgment on them; that is, affirms or denies something of them.

4. Ideas referred to anything extraneous to them may be true or false. Whenever the mind refers any of its ideas to anything extraneous to them, they are then capable to be called true or false. Because the mind, in such a reference, makes a tacit supposition of their conformity to that thing; which supposition, as it happens to be true or false, so the ideas themselves come to be denominated. The most usual cases wherein this happens, are these following:

5. Other men's ideas; real existence; and supposed real

essences, are what men usually refer their ideas to. First, when the mind supposes any idea it has conformable to that in other men's minds, called by the same common name; v.g. when the mind intends or judges its ideas of justice, temperance, religion, to be the same with what other men give those names to.

Secondly, when the mind supposes any idea it has in itself to be conformable to some real existence. Thus the two ideas of a man and a centaur, supposed to be the ideas of real substances, are the one true and the other false; the one having a conformity to what has really existed, the other not.

Thirdly, when the mind refers any of its ideas to that real constitution and essence of anything, whereon all its properties depend: and thus the greatest part, if not all our ideas of substances, are false.

6. The cause of such reference. These suppositions the mind is very apt tacitly to make concerning its own ideas. But yet, if we will examine it, we shall find it is chiefly, if not only, concerning its abstract complex ideas. For the natural tendency of the mind being towards

knowledge; and finding that, if it should proceed by and dwell upon only particular things, its progress would be very slow, and its work endless; therefore, to shorten its way to knowledge, and make each perception more comprehensive, the first thing it does, as the foundation of the easier enlarging its knowledge, either by contemplation of the things themselves that it would know, or conference with others about them, is to bind them into bundles, and rank them so into sorts, that what knowledge it gets of any of them it may thereby with assurance extend to all of that sort; and so advance by larger steps in that which is its great business, knowledge. This, as I have elsewhere shown, is the reason why we collect things under comprehensive ideas, with names annexed to them, into genera and species; i.e. into kinds and sorts.

7. Names of things supposed to carry in them knowledge of their essences. If therefore we will warily attend to the motions of the mind, and observe what course it usually takes in its way to knowledge, we shall I think find, that the mind having got an idea which it thinks

it may have use of either in contemplation or discourse, the first thing it does is to abstract it, and then get a name to it; and so lay it up in its storehouse, the memory, as containing the essence of a sort of things, of which that name is always to be the mark. Hence it is, that we may often observe that, when any one sees a new thing of a kind that he knows not, he presently asks, what it is; meaning by that inquiry nothing but the name. As if the name carried with it the knowledge of the species, or the essence of it; whereof it is indeed used as the mark, and is generally supposed annexed to it.

8. How men suppose that their ideas must correspond to things, and to the customary meanings of names. But this abstract idea, being something in the mind, between the thing that exists, and the name that is given to it; it is in our ideas that both the rightness of our knowledge, and the propriety and intelligibleness of our speaking, consists. And hence it is that men are so forward to suppose, that the abstract ideas they have in their minds are such as agree to the things existing without them, to which they are referred; and are the

same also to which the names they give them do by the use and propriety of that language belong. For without this double conformity of their ideas, they find they should both think amiss of things in themselves, and talk of them unintelligibly to others.

9. Simple ideas may be false, in reference to others of the same name, but are least liable to be so. First, then, I say, that when the truth of our ideas is judged of by the conformity they have to the ideas which other men have, and commonly signify by the same name, they may be any of them false. But yet simple ideas are least of all liable to be so mistaken. Because a man, by his senses and every day's observation, may easily satisfy himself what the simple ideas are which their several names that are in common use stand for; they being but few in number, and such as, if he doubts or mistakes in, he may easily rectify by the objects they are to be found in. Therefore it is seldom that any one mistakes in his names of simple ideas, or applies the name red to the idea green, or the name sweet to the idea bitter: much less are men apt to confound the names of ideas belong-

ing to different senses, and call a colour by the name of a taste, &c. Whereby it is evident that the simple ideas they call by any name are commonly the same that others have and mean when they use the same names.

10. Ideas of mixed modes most liable to be false in this sense. Complex ideas are much more liable to be false in this respect; and the complex ideas of mixed modes, much more than those of substances; because in substances (especially those which the common and unborrowed names of any language are applied to) some remarkable sensible qualities, serving ordinarily to distinguish one sort from another, easily preserve those who take any care in the use of their words, from applying them to sorts of substances to which they do not at all belong. But in mixed modes we are much more uncertain; it being not so easy to determine of several actions, whether they are to be called justice or cruelty, liberality or prodigality. And so in referring our ideas to those of other men, called by the same names, ours may be false; and the idea in our minds, which we express by the word justice, may perhaps be that which ought to have another name.

11. Or at least to be thought false. But whether or no our ideas of mixed modes are more liable than any sort to be different from those of other men, which are marked by the same names, this at least is certain, That this sort of falsehood is much more familiarly attributed to our ideas of mixed modes than to any other. When a man is thought to have a false idea of justice, or gratitude, or glory, it is for no other reason, but that his agrees not with the ideas which each of those names are the signs of in other men.

12. And why. The reason whereof seems to me to be this: That the abstract ideas of mixed modes, being men's voluntary combinations of such a precise collection of simple ideas, and so the essence of each species being made by men alone, whereof we have no other sensible standard existing anywhere but the name itself, or the definition of that name; we having nothing else to refer these our ideas of mixed modes to, as a standard to which we would conform them, but the ideas of those who are thought to use those names in their most proper significations; and, so as our ideas conform or differ

from them, they pass for true or false. And thus much concerning the truth and falsehood of our ideas, in reference to their names.

13. As referred to real existence, none of our ideas can be false but those of substances. Secondly, as to the truth and falsehood of our ideas, in reference to the real existence of things. When that is made the standard of their truth, none of them can be termed false but only our complex ideas of substances.

14. Simple ideas in this sense not false, and why. First, our simple ideas, being barely such perceptions as God has fitted us to receive, and given power to external objects to produce in us by established laws and ways, suitable to his wisdom and goodness, though incomprehensible to us, their truth consists in nothing else but in such appearances as are produced in us, and must be suitable to those powers he has placed in external objects or else they could not be produced in us: and thus answering those powers, they are what they should be, true ideas. Nor do they become liable to any imputation of falsehood, if the mind (as in most men I believe it

does) judges these ideas to be in the things themselves. For God in his wisdom having set them as marks of distinction in things, whereby we may be able to discern one thing from another, and so choose any of them for our uses as we have occasion; it alters not the nature of our simple idea, whether we think that the idea of blue be in the violet itself, or in our mind only; and only the power of producing it by the texture of its parts, reflecting the particles of light after a certain manner, to be in the violet itself. For that texture in the object, by a regular and constant operation producing the same idea of blue in us, it serves us to distinguish, by our eyes, that from any other thing; whether that distinguishing mark, as it is really in the violet, be only a peculiar texture of parts, or else that very colour, the idea whereof (which is in us) is the exact resemblance. And it is equally from that appearance to be denominated blue, whether it be that real colour, or only a peculiar texture in it, that causes in us that idea: since the name, blue, notes properly nothing but that mark of distinction that is in a violet, discernible only by our

eyes, whatever it consists in; that being beyond our capacities distinctly to know, and perhaps would be of less use to us, if we had faculties to discern.

15. Though one man's idea of blue should be different from another's. Neither would it carry any imputation of falsehood to our simple ideas, if by the different structure of our organs it were so ordered, that the same object should produce in several men's minds different ideas at the same time; v.g. if the idea that a violet produced in one man's mind by his eyes were the same that a marigold produced in another man's, and vice versa. For, since this could never be known, because one man's mind could not pass into another man's body, to perceive what appearances were produced by those organs; neither the ideas hereby, nor the names, would be at all confounded, or any falsehood be in either. For all things that had the texture of a violet, producing constantly the idea that he called blue, and those which had the texture of a marigold, producing constantly the idea which he as constantly called yellow, whatever those appearances were in his mind; he would be able as regu-

larly to distinguish things for his use by those appearances, and understand and signify those distinctions marked by the name blue and yellow, as if the appearances or ideas in his mind received from those two flowers were exactly the same with the ideas in other men's minds. I am nevertheless very apt to think that the sensible ideas produced by any object in different men's minds, are most commonly very near and undiscernibly alike. For which opinion, I think, there might be many reasons offered: but that being besides my present business, I shall not trouble my reader with them; but only mind him, that the contrary supposition, if it could be proved, is of little use, either for the improvement of our knowledge, or conveniency of life, and so we need not trouble ourselves to examine it.

16. Simple ideas can none of them be false in respect of real existence. From what has been said concerning our simple ideas, I think it evident that our simple ideas can none of them be false in respect of things existing without us. For the truth of these appearances or perceptions in our minds consisting, as has been said, only in

their being answerable to the powers in external objects to produce by our senses such appearances in us, and each of them being in the mind such as it is, suitable to the power that produced it, and which alone it represents, it cannot upon that account, or as referred to such a pattern, be false. Blue and yellow, bitter or sweet, can never be false ideas: these perceptions in the mind are just such as they are there, answering the powers appointed by God to produce them; and so are truly what they are, and are intended to be. Indeed the names may be misapplied, but that in this respect makes no falsehood in the ideas; as if a man ignorant in the English tongue should call purple scarlet.

17. Modes not false cannot be false in reference to essences of things. Secondly, neither can our complex ideas of modes, in reference to the essence of anything really existing, be false; because whatever complex ideas I have of any mode, it hath no reference to any pattern existing, and made by nature; it is not supposed to contain in it any other ideas than what it hath; nor to represent anything but such a complication of ideas as it does.

Thus, when I have the idea of such an action of a man who forbears to afford himself such meat, drink, and clothing, and other conveniences of life, as his riches and estate will be sufficient to supply and his station requires, I have no false idea; but such an one as represents an action, either as I find or imagine it, and so is capable of neither truth nor falsehood. But when I give the name frugality or virtue to this action, then it may be called a false idea, if thereby it be supposed to agree with that idea to which, in propriety of speech, the name of frugality doth belong, or to be conformable to that law which is the standard of virtue and vice.

18. Ideas of substances may be false in reference to existing things. Thirdly, our complex ideas of substances, being all referred to patterns in things themselves, may be false. That they are all false, when looked upon as the representations of the unknown essences of things, is so evident that there needs nothing to be said of it. I shall therefore pass over that chimerical supposition, and consider them as collections of simple ideas in the mind, taken from combinations of simple ideas existing

together constantly in things, of which patterns they are the supposed copies; and in this reference of them to the existence of things, they are false ideas:—

(1) When they put together simple ideas, which in the real existence of things have no union; as when to the shape and size that exist together in a horse, is joined in the same complex idea the power of barking like a dog: which three ideas, however put together into one in the mind, were never united in nature; and this, therefore, may be called a false idea of a horse. (2) Ideas of substances are, in this respect, also false, when, from any collection of simple ideas that do always exist together, there is separated, by a direct negation, any other simple idea which is constantly joined with them. Thus, if to extension, solidity, fusibility, the peculiar weightiness, and yellow colour of gold, any one join in his thoughts the negation of a greater degree of fixedness than is in lead or copper, he may be said to have a false complex idea, as well as when he joins to those other simple ones the idea of perfect absolute fixedness. For either way, the complex idea of gold being made up

of such simple ones as have no union in nature, may be termed false. But, if he leave out of this his complex idea that of fixedness quite, without either actually joining to or separating it from the rest in his mind, it is, I think, to be looked on as an inadequate and imperfect idea, rather than a false one; since, though it contains not all the simple ideas that are united in nature, yet it puts none together but what do really exist together.

19. Truth or falsehood always supposes affirmation or negation. Though, in compliance with the ordinary way of speaking, I have shown in what sense and upon what ground our ideas may be sometimes called true or false; yet if we will look a little nearer into the matter, in all cases where any idea is called true or false, it is from some judgment that the mind makes, or is supposed to make, that is true or false. For truth or falsehood, being never without some affirmation or negation, express or tacit, it is not to be found but where signs are joined or separated, according to the agreement or disagreement of the things they stand for. The signs we chiefly use are either ideas or words; wherewith we make either

mental or verbal propositions. Truth lies in so joining or separating these representatives, as the things they stand for do in themselves agree or disagree; and falsehood in the contrary, as shall be more fully shown hereafter.

20. Ideas in themselves neither true nor false. Any idea, then, which we have in our minds, whether conformable or not to the existence of things, or to any idea in the minds of other men, cannot properly for this alone be called false. For these representations, if they have nothing in them but what is really existing in things without, cannot be thought false, being exact representations of something: nor yet if they have anything in them differing from the reality of things, can they properly be said to be false representations, or ideas of things they do not represent. But the mistake and falsehood is:

21. But are false—when judged agreeable to another man's idea, without being so. First, when the mind having any idea, it judges and concludes it the same that is in other men's minds, signified by the same name; or that it is conformable to the ordinary received signification or definition of that word, when indeed it is not:

which is the most usual mistake in mixed modes, though other ideas also are liable to it.

22. When judged to agree to real existence, when they do not. (2) When it having a complex idea made up of such a collection of simple ones as nature never puts together, it judges it to agree to a species of creatures really existing; as when it joins the weight of tin to the colour, fusibility, and fixedness of gold.

23. When judged adequate, without being so. (3) When in its complex idea it has united a certain number of simple ideas that do really exist together in some sort of creatures, but has also left out others as much inseparable, it judges this to be a perfect complete idea of a sort of things which really it is not; v.g. having joined the ideas of substance, yellow, malleable, most heavy, and fusible, it takes that complex idea to be the complete idea of gold, when yet its peculiar fixedness, and solubility in aqua regia, are as inseparable from those other ideas, or qualities, of that body as they are one from another.

24. When judged to represent the real essence. (4) The

mistake is yet greater, when I judge that this complex idea contains in it the real essence of any body existing; when at least it contains but some few of those properties which flow from its real essence and constitution. I say only some few of those properties; for those properties consisting mostly in the active and passive powers it has in reference to other things, all that are vulgarly known of any one body, of which the complex idea of that kind of things is usually made, are but a very few, in comparison of what a man that has several ways tried and examined it knows of that one sort of things; and all that the most expert man knows are but a few, in comparison of what are really in that body, and depend on its internal or essential constitution. The essence of a triangle lies in a very little compass, consists in a very few ideas: three lines including a space make up that essence: but the properties that flow from this essence are more than can be easily known or enumerated. So I imagine it is in substances; their real essences lie in a little compass, though the properties flowing from that internal constitution are endless.

25. Ideas, when called false. To conclude, a man having no notion of anything without him, but by the idea he has of it in his mind, (which idea he has a power to call by what name he pleases), he may indeed make an idea neither answering the reason of things, nor agreeing to the idea commonly signified by other people's words; but cannot make a wrong or false idea of a thing which is no otherwise known to him but by the idea he has of it: v.g. when I frame an idea of the legs, arms, and body of a man, and join to this a horse's head and neck, I do not make a false idea of anything; because it represents nothing without me. But when I call it a man or Tartar, and imagine it to represent some real being without me, or to be the same idea that others call by the same name; in either of these cases I may err. And upon this account it is that it comes to be termed a false idea; though indeed the falsehood lies not in the idea, but in that tacit mental proposition, wherein a conformity and resemblance is attributed to it which it has not. But yet, if, having framed such an idea in my mind without thinking either that existence, or the name man or Tar-

tar, belongs to it, I will call it man or Tartar, I may be justly thought fantastical in the naming; but not erroneous in my judgment; nor the idea any way false.

26. More properly to be called right or wrong. Upon the whole, matter, I think that our ideas, as they are considered by the mind,—either in reference to the proper signification of their names; or in reference to the reality of things,—may very fitly be called right or wrong ideas, according as they agree or disagree to those patterns to which they are referred. But if any one had rather call them true or false, it is fit he use a liberty, which every one has, to call things by those names he thinks best; though, in propriety of speech, truth or falsehood will, I think, scarce agree to them, but as they, some way or other, virtually contain in them some mental proposition. The ideas that are in a man's mind, simply considered, cannot be wrong; unless complex ones, wherein inconsistent parts are jumbled together. All other ideas are in themselves right, and the knowledge about them right and true knowledge; but when we come to refer them to anything, as to their patterns and arche-

types, then they are capable of being wrong, as far as they disagree with such archetypes.

Chapter XXXIII

Of the Association of Ideas

1. Something unreasonable in most men. There is scarce any one that does not observe something that seems odd to him, and is in itself really extravagant, in the opinions, reasonings, and actions of other men. The least flaw of this kind, if at all different from his own, every one is quick-sighted enough to espy in another, and will by the authority of reason forwardly condemn; though he be guilty of much greater unreasonableness in his own tenets and conduct, which he never perceives, and will very hardly, if at all, be convinced of.

2. Not wholly from self-love. This proceeds not wholly from self-love, though that has often a great hand in it. Men of fair minds, and not given up to the overweening of self-flattery, are frequently guilty of it; and in many cases one with amazement hears the arguings, and is

astonished at the obstinacy of a worthy man, who yields not to the evidence of reason, though laid before him as clear as daylight.

3. Not from education. This sort of unreasonableness is usually imputed to education and prejudice, and for the most part truly enough, though that reaches not the bottom of the disease, nor shows distinctly enough whence it rises, or wherein it lies. Education is often rightly assigned for the cause, and prejudice is a good general name for the thing itself: but yet, I think, he ought to look a little further, who would trace this sort of madness to the root it springs from, and so explain it, as to show whence this flaw has its original in very sober and rational minds, and wherein it consists.

4. A degree of madness found in most men. I shall be pardoned for calling it by so harsh a name as madness, when it is considered that opposition to reason deserves that name, and is really madness; and there is scarce a man so free from it, but that if he should always, on all occasions, argue or do as in some cases he constantly does, would not be thought fitter for Bedlam than civil

conversation. I do not here mean when he is under the power of an unruly passion, but in the steady calm course of his life. That which will yet more apologize for this harsh name, and ungrateful imputation on the greatest part of mankind, is, that, inquiring a little by the bye into the nature of madness (Bk. ii. ch. xi. SS 13), I found it to spring from the very same root, and to depend on the very same cause we are here speaking of. This consideration of the thing itself, at a time when I thought not the least on the subject which I am now treating of, suggested it to me. And if this be a weakness to which all men are so liable, if this be a taint which so universally infects mankind, the greater care should be taken to lay it open under its due name, thereby to excite the greater care in its prevention and cure.

5. From a wrong connexion of ideas. Some of our ideas have a natural correspondence and connexion one with another: it is the office and excellency of our reason to trace these, and hold them together in that union and correspondence which is founded in their peculiar beings. Besides this, there is another connexion of ideas

wholly owing to chance or custom. Ideas that in themselves are not all of kin, come to be so united in some men's minds, that it is very hard to separate them; they always keep in company, and the one no sooner at any time comes into the understanding, but its associate appears with it; and if they are more than two which are thus united, the whole gang, always inseparable, show themselves together.

6. This connexion made by custom. This strong combination of ideas, not allied by nature, the mind makes in itself either voluntarily or by chance; and hence it comes in different men to be very different, according to their different inclinations, education, interests, &c. Custom settles habits of thinking in the understanding, as well as of determining in the will, and of motions in the body: all which seems to be but trains of motions in the animal spirits, which, once set a going, continue in the same steps they have used to; which, by often treading, are worn into a smooth path, and the motion in it becomes easy, and as it were natural. As far as we can comprehend thinking, thus ideas seem to be produced

in our minds; or, if they are not, this may serve to explain their following one another in an habitual train, when once they are put into their track, as well as it does to explain such motions of the body. A musician used to any tune will find that, let it but once begin in his head, the ideas of the several notes of it will follow one another orderly in his understanding, without any care or attention, as regularly as his fingers move orderly over the keys of the organ to play out the tune he has begun, though his unattentive thoughts be elsewhere a wandering. Whether the natural cause of these ideas, as well as of that regular dancing of his fingers be the motion of his animal spirits, I will not determine, how probable soever, by this instance, it appears to be so: but this may help us a little to conceive of intellectual habits, and of the tying together of ideas.

7. Some antipathies an effect of it. That there are such associations of them made by custom, in the minds of most men, I think nobody will question, who has well considered himself or others; and to this, perhaps, might be justly attributed most of the sympathies and antipa-

thies observable in men, which work as strongly, and produce as regular effects as if they were natural; and are therefore called so, though they at first had no other original but the accidental connexion of two ideas, which either the strength of the first impression, or future indulgence so united, that they always afterwards kept company together in that man's mind, as if they were but one idea. I say most of the antipathies, I do not say all; for some of them are truly natural, depend upon our original constitution, and are born with us; but a great part of those which are counted natural, would have been known to be from unheeded, though perhaps early, impressions, or wanton fancies at first, which would have been acknowledged the original of them, if they had been warily observed. A grown person surfeiting with honey no sooner hears the name of it, but his fancy immediately carries sickness and qualms to his stomach, and he cannot bear the very idea of it; other ideas of dislike, and sickness, and vomiting, presently accompany it, and he is disturbed; but he knows from whence to date this weakness, and can tell how he got

this indisposition. Had this happened to him by an overdose of honey when a child, all the same effects would have followed; but the cause would have been mistaken, and the antipathy counted natural.

8. Influence of association to be watched educating young children. I mention this, not out of any great necessity there is in this present argument to distinguish nicely between natural and acquired antipathies; but I take notice of it for another purpose, viz. that those who have children, or the charge of their education, would think it worth their while diligently to watch, and carefully to prevent the undue connexion of ideas in the minds of young people. This is the time most susceptible of lasting impressions; and though those relating to the health of the body are by discreet people minded and fenced against, yet I am apt to doubt, that those which relate more peculiarly to the mind, and terminate in the understanding or passions, have been much less heeded than the thing deserves: nay, those relating purely to the understanding, have, as I suspect, been by most men wholly overlooked.

9. Wrong connexion of ideas a great cause of errors. This wrong connexion in our minds of ideas in themselves loose and independent of one another, has such an influence, and is of so great force to set us awry in our actions, as well moral as natural, passions, reasonings, and notions themselves, that perhaps there is not any one thing that deserves more to be looked after.

10. An instance. The ideas of goblins and sprites have really no more to do with darkness than light: yet let but a foolish maid inculcate these often on the mind of a child, and raise them there together, possibly he shall never be able to separate them again so long as he lives, but darkness shall ever afterwards bring with it those frightful ideas, and they shall be so joined, that he can no more bear the one than the other.

11. Another instance. A man receives a sensible injury from another, thinks on the man and that action over and over, and by ruminating on them strongly, or much, in his mind, so cements those two ideas together, that he makes them almost one; never thinks on the man, but the pain and displeasure he suffered comes into his

mind with it, so that he scarce distinguishes them, but has as much an aversion for the one as the other. Thus hatreds are often begotten from slight and innocent occasions, and quarrels propagated and continued in the world.

12. A third instance. A man has suffered pain or sickness in any place; he saw his friend die in such a room: though these have in nature nothing to do one with another, yet when the idea of the place occurs to his mind, it brings (the impression being once made) that of the pain and displeasure with it: he confounds them in his mind, and can as little bear the one as the other.

13. Why time cures some disorders in the mind, which reason cannot cure. When this combination is settled, and while it lasts, it is not in the power of reason to help us, and relieve us from the effects of it. Ideas in our minds, when they are there, will operate according to their natures and circumstances. And here we see the cause why time cures certain affections, which reason, though in the right, and allowed to be so, has not power over, nor is able against them to prevail with

those who are apt to hearken to it in other cases. The death of a child that was the daily delight of its mother's eyes, and joy of her soul, rends from her heart the whole comfort of her life, and gives her all the torment imaginable: use the consolations of reason in this case, and you were as good preach ease to one on the rack, and hope to allay, by rational discourses, the pain of his joints tearing asunder. Till time has by disuse separated the sense of that enjoyment and its loss, from the idea of the child returning to her memory, all representations, though ever so reasonable, are in vain; and therefore some in whom the union between these ideas is never dissolved, spend their lives in mourning, and carry an incurable sorrow to their graves.

14. Another instance of the effect of the association of ideas. A friend of mine knew one perfectly cured of madness by a very harsh and offensive operation. The gentleman who was thus recovered, with great sense of gratitude and acknowledgment owned the cure all his life after, as the greatest obligation he could have received; but, whatever gratitude and reason suggested

to him, he could never bear the sight of the operator: that image brought back with it the idea of that agony which he suffered from his hands, which was too mighty and intolerable for him to endure.

15. More instances. Many children, imputing the pain they endured at school to their books they were corrected for, so join those ideas together, that a book becomes their aversion, and they are never reconciled to the study and use of them all their lives after; and thus reading becomes a torment to them, which otherwise possibly they might have made the great pleasure of their lives. There are rooms convenient enough, that some men cannot study in, and fashions of vessels, which, though ever so clean and commodious, they cannot drink out of, and that by reason of some accidental ideas which are annexed to them, and make them offensive; and who is there that hath not observed some man to flag at the appearance, or in the company of some certain person not otherwise superior to him, but because, having once on some occasion got the ascendant, the idea of authority and distance goes along with that of the per-

son, and he that has been thus subjected, is not able to separate them.

16. A curious instance. Instances of this kind are so plentiful everywhere, that if I add one more, it is only for the pleasant oddness of it. It is of a young gentleman, who, having learnt to dance, and that to great perfection, there happened to stand an old trunk in the room where he learnt. The idea of this remarkable piece of household stuff had so mixed itself with the turns and steps of all his dances, that though in that chamber he could dance excellently well, yet it was only whilst that trunk was there; nor could he perform well in any other place, unless that or some such other trunk had its due position in the room. If this story shall be suspected to be dressed up with some comical circumstances, a little beyond precise nature, I answer for myself that I had it some years since from a very sober and worthy man, upon his own knowledge, as I report it; and I dare say there are very few inquisitive persons who read this, who have not met with accounts, if not examples, of this nature, that may parallel, or at least justify this.

17. Influence of association on intellectual habits. Intellectual habits and defects this way contracted, are not less frequent and powerful, though less observed. Let the ideas of being and matter be strongly joined, either by education or much thought; whilst these are still combined in the mind, what notions, what reasonings, will there be about separate spirits? Let custom from the very childhood have joined figure and shape to the idea of God, and what absurdities will that mind be liable to about the Deity? Let the idea of infallibility be inseparably joined to any person, and these two constantly together possess the mind; and then one body in two places at once, shall unexamined be swallowed for a certain truth, by an implicit faith, whenever that imagined infallible person dictates and demands assent without inquiry.

18. Observable in the opposition between different sects of philosophy and of religion. Some such wrong and unnatural combinations of ideas will be found to establish the irreconcilable opposition between different sects of philosophy and religion; for we cannot imagine every

one of their followers to impose wilfully on himself, and knowingly refuse truth offered by plain reason. Interest, though it does a great deal in the case, yet cannot be thought to work whole societies of men to so universal a perverseness, as that every one of them to a man should knowingly maintain falsehood: some at least must be allowed to do what all pretend to, i.e. to pursue truth sincerely; and therefore there must be something that blinds their understandings, and makes them not see the falsehood of what they embrace for real truth. That which thus captivates their reasons, and leads men of sincerity blindfold from common sense, will, when examined, be found to be what we are speaking of: some independent ideas, of no alliance to one another, are, by education, custom, and the constant din of their party, so coupled in their minds, that they always appear there together; and they can no more separate them in their thoughts than if they were but one idea, and they operate as if they were so. This gives sense to jargon, demonstration to absurdities, and consistency to nonsense, and is the foundation of the greatest, I had almost said

of all the errors in the world; or, if it does not reach so far, it is at least the most dangerous one, since, so far as it obtains, it hinders men from seeing and examining. When two things, in themselves disjoined, appear to the sight constantly united; if the eye sees these things riveted which are loose, where will you begin to rectify the mistakes that follow in two ideas that they have been accustomed so to join in their minds as to substitute one for the other, and, as I am apt to think, often without perceiving it themselves? This, whilst they are under the deceit of it, makes them incapable of conviction, and they applaud themselves as zealous champions for truth, when indeed they are contending for error; and the confusion of two different ideas, which a customary connexion of them in their minds hath to them made in effect but one, fills their heads with false views, and their reasonings with false consequences.

19. Conclusion. Having thus given an account of the original, sorts, and extent of our IDEAS, with several other considerations about these (I know not whether I may say) instruments, or materials of our knowledge,

the method I at first proposed to myself would now require that I should immediately proceed to show, what use the understanding makes of them, and what *knowledge* we have by them. This was that which, in the first general view I had of this subject, was all that I thought I should have to do: but, upon a nearer approach, I find that there is so close a connexion between ideas and WORDS, and our abstract ideas and general words have so constant a relation one to another, that it is impossible to speak clearly and distinctly of our knowledge, which all consists in propositions, without considering, first, the nature, use, and signification of Language; which, therefore, must be the business of the next Book.

BOOK III

Of Words

Chapter I

Of Words or Language in General

1. Man fitted to form articulate sounds. God, having designed man for a sociable creature, made him not only with an inclination, and under a necessity to have fellowship with those of his own kind, but furnished him also with language, which was to be the great instrument and common tie of society. Man, therefore, had by nature his organs so fashioned, as to be fit to frame articulate sounds, which we call words. But this was not enough to produce language; for parrots, and several other birds, will be taught to make articulate sounds distinct enough, which yet by no means are capable of language.

2. To use these sounds as signs of ideas. Besides articulate sounds, therefore, it was further necessary that he should be able to use these sounds as signs of internal conceptions; and to make them stand as marks for the

ideas within his own mind, whereby they might be made known to others, and the thoughts of men's minds be conveyed from one to another.

3. To make them general signs. But neither was this sufficient to make words so useful as they ought to be. It is not enough for the perfection of language, that sounds can be made signs of ideas, unless those signs can be so made use of as to comprehend several particular things: for the multiplication of words would have perplexed their use, had every particular thing need of a distinct name to be signified by. To remedy this inconvenience, language had yet a further improvement in the use of general terms, whereby one word was made to mark a multitude of particular existences: which advantageous use of sounds was obtained only by the difference of the ideas they were made signs of: those names becoming general, which are made to stand for general ideas, and those remaining particular, where the ideas they are used for are particular.

4. To make them signify the absence of positive ideas. Besides these names which stand for ideas, there be other

words which men make use of, not to signify any idea, but the want or absence of some ideas, simple or complex, or all ideas together; such as are *nihil* in Latin, and in English, ignorance and barrenness. All which negative or privative words cannot be said properly to belong to, or signify no ideas: for then they would be perfectly insignificant sounds; but they relate to positive ideas, and signify their absence.

5. Words ultimately derived from such as signify sensible ideas. It may also lead us a little towards the original of all our notions and knowledge, if we remark how great a dependence our words have on common sensible ideas; and how those which are made use of to stand for actions and notions quite removed from sense, have their rise from thence, and from obvious sensible ideas are transferred to more abstruse significations, and made to stand for ideas that come not under the cognizance of our senses; v.g. to imagine, apprehend, comprehend, adhere, conceive, instil, disgust, disturbance, tranquillity, &c., are all words taken from the operations of sensible things, and applied to certain modes of thinking.

Spirit, in its primary signification, is breath; angel, a messenger: and I doubt not but, if we could trace them to their sources, we should find, in all languages, the names which stand for things that fall not under our senses to have had their first rise from sensible ideas. By which we may give some kind of guess what kind of notions they were, and whence derived, which filled their minds who were the first beginners of languages, and how nature, even in the naming of things, unawares suggested to men the originals and principles of all their knowledge: whilst, to give names that might make known to others any operations they felt in themselves, or any other ideas that came not under their senses, they were fain to borrow words from ordinary known ideas of sensation, by that means to make others the more easily to conceive those operations they experimented in themselves, which made no outward sensible appearances; and then, when they had got known and agreed names to signify those internal operations of their own minds, they were sufficiently furnished to make known by words all their other ideas; since they could consist of nothing

but either of outward sensible perceptions, or of the inward operations of their minds about them; we having, as has been proved, no ideas at all, but what originally come either from sensible objects without, or what we feel within ourselves, from the inward workings of our own spirits, of which we are conscious to ourselves within.

6. Distribution of subjects to be treated of. But to understand better the use and force of Language, as subservient to instruction and knowledge, it will be convenient to consider:

First, To what it is that names, in the use of language, are immediately applied.

Secondly, Since all (except proper) names are general, and so stand not particularly for this or that single thing, but for sorts and ranks of things, it will be necessary to consider, in the next place, what the sorts and kinds, or, if you rather like the Latin names, what the Species and Genera of things are, wherein they consist, and how they come to be made. These being (as they ought) well looked into, we shall the better come to find the right

use of words; the natural advantages and defects of language; and the remedies that ought to be used, to avoid the inconveniences of obscurity or uncertainty in the signification of words: without which it is impossible to discourse with any clearness or order concerning knowledge: which, being conversant about propositions, and those most commonly universal ones, has greater connexion with words than perhaps is suspected.

These considerations, therefore, shall be the matter of the following chapters.

Chapter II Of the Signification of Words

1. Words are sensible signs, necessary for communication of ideas. Man, though he have great variety of thoughts, and such from which others as well as himself might receive profit and delight; yet they are all within his own breast, invisible and hidden from others, nor can of themselves be made to appear. The comfort and advantage of society not being to be had without com-

munication of thoughts, it was necessary that man should find out some external sensible signs, whereof those invisible ideas, which his thoughts are made up of, might be made known to others. For this purpose nothing was so fit, either for plenty or quickness, as those articulate sounds, which with so much ease and variety he found himself able to make. Thus we may conceive how words, which were by nature so well adapted to that purpose, came to be made use of by men as the signs of their ideas; not by any natural connexion that there is between particular articulate sounds and certain ideas, for then there would be but one language amongst all men; but by a voluntary imposition, whereby such a word is made arbitrarily the mark of such an idea. The use, then, of words, is to be sensible marks of ideas; and the ideas they stand for are their proper and immediate signification.

2. Words, in their immediate signification, are the sensible signs of his ideas who uses them. The use men have of these marks being either to record their own thoughts, for the assistance of their own memory or, as

it were, to bring out their ideas, and lay them before the view of others: words, in their primary or immediate signification, stand for nothing but the ideas in the mind of him that uses them, how imperfectly soever or carelessly those ideas are collected from the things which they are supposed to represent. When a man speaks to another, it is that he may be understood: and the end of speech is, that those sounds, as marks, may make known his ideas to the hearer. That then which words are the marks of are the ideas of the speaker: nor can any one apply them as marks, immediately, to anything else but the ideas that he himself hath: for this would be to make them signs of his own conceptions, and yet apply them to other ideas; which would be to make them signs and not signs of his ideas at the same time, and so in effect to have no signification at all. Words being voluntary signs, they cannot be voluntary signs imposed by him on things he knows not. That would be to make them signs of nothing, sounds without signification. A man cannot make his words the signs either of qualities in things, or of conceptions in the mind of another,

whereof he has none in his own. Till he has some ideas of his own, he cannot suppose them to correspond with the conceptions of another man; nor can he use any signs for them of another man; nor can he use any signs for them: for thus they would be the signs of he knows not what, which is in truth to be the signs of nothing. But when he represents to himself other men's ideas by some of his own, if he consent to give them the same names that other men do, it is still to his own ideas; to ideas that he has, and not to ideas that he has not.

3. Examples of this. This is so necessary in the use of language, that in this respect the knowing and the ignorant, the learned and the unlearned, use the words they speak (with any meaning) all alike. They, in every man's mouth, stand for the ideas he has, and which he would express by them. A child having taken notice of nothing in the metal he hears called gold, but the bright shining yellow colour, he applies the word gold only to his own idea of that colour, and nothing else; and therefore calls the same colour in a peacock's tail gold. Another that hath better observed, adds to shining yellow

great weight: and then the sound gold, when he uses it, stands for a complex idea of a shining yellow and a very weighty substance. Another adds to those qualities fusibility: and then the word gold signifies to him a body, bright, yellow, fusible, and very heavy. Another adds malleability. Each of these uses equally the word gold, when they have occasion to express the idea which they have applied it to: but it is evident that each can apply it only to his own idea; nor can he make it stand as a sign of such a complex idea as he has not.

4. Words are often secretly referred first to the ideas supposed to be in other men's minds. But though words, as they are used by men, can properly and immediately signify nothing but the ideas that are in the mind of the speaker; yet they in their thoughts give them a secret reference to two other things.

First, They suppose their words to be marks of the ideas in the minds also of other men, with whom they communicate: for else they should talk in vain, and could not be understood, if the sounds they applied to one idea were such as by the hearer were applied to another,

which is to speak two languages. But in this men stand not usually to examine, whether the idea they, and those they discourse with have in their minds be the same: but think it enough that they use the word, as they imagine, in the common acceptation of that language; in which they suppose that the idea they make it a sign of is precisely the same to which the understanding men of that country apply that name.

5. To the reality of things. Secondly, Because men would not be thought to talk barely of their own imagination, but of things as really they are; therefore they often suppose the words to stand also for the reality of things. But this relating more particularly to substances and their names, as perhaps the former does to simple ideas and modes, we shall speak of these two different ways of applying words more at large, when we come to treat of the names of mixed modes and substances in particular: though give me leave here to say, that it is a perverting the use of words, and brings unavoidable obscurity and confusion into whenever we make them stand for anything but those ideas we have in our own minds.

6. Words by use readily excite ideas of their objects. Concerning words, also, it is further to be considered:

First, that they being immediately the signs of men's ideas, and by that means the instruments whereby men communicate their conceptions, and express to one another those thoughts and imaginations they have within their own their own breasts; there comes, by constant use, to be such a connexion between certain sounds and the ideas they stand for, that the names heard, almost as readily excite certain ideas as if the objects themselves, which are apt to produce them, did actually affect the senses. Which is manifestly so in all obvious sensible qualities, and in all substances that frequently and familiarly occur to us.

7. Words are often used without signification, and why.

Secondly, That though the proper and immediate signification of words are ideas in the mind of the speaker, yet, because by familiar use from our cradles, we come to learn certain articulate sounds very perfectly, and have them readily on our tongues, and always at hand in our memories, but yet are not always careful to ex-

amine or settle their significations perfectly; it often happens that men, even when they would apply themselves to an attentive consideration, do set their thoughts more on words than things. Nay, because words are many of them learned before the ideas are known for which they stand: therefore some, not only children but men, speak several words no otherwise than parrots do, only because they have learned them, and have been accustomed to those sounds. But so far as words are of use and signification, so far is there a constant connexion between the sound and the idea, and a designation that the one stands for the other; without which application of them, they are nothing but so much insignificant noise.

8. Their signification perfectly arbitrary, not the consequence of a natural connexion. Words, by long and familiar use, as has been said, come to excite in men certain ideas so constantly and readily, that they are apt to suppose a natural connexion between them. But that they signify only men's peculiar ideas, and that by a perfect arbitrary imposition, is evident, in that they

often fail to excite in others (even that use the same language) the same ideas we take them to be signs of: and every man has so inviolable a liberty to make words stand for what ideas he pleases, that no one hath the power to make others have the same ideas in their minds that he has, when they use the same words that he does. And therefore the great Augustus himself, in the possession of that power which ruled the world, acknowledged he could not make a new Latin word: which was as much as to say, that he could not arbitrarily appoint what idea any sound should be a sign of, in the mouths and common language of his subjects. It is true, common use, by a tacit consent, appropriates certain sounds to certain ideas in all languages, which so far limits the signification of that sound, that unless a man applies it to the same idea, he does not speak properly: and let me add, that unless a man's words excite the same ideas in the hearer which he makes them stand for in speaking, he does not speak intelligibly. But whatever be the consequence of any man's using of words differently, either from their general meaning, or the

particular sense of the person to whom he addresses them; this is certain, their signification, in his use of them, is limited to his ideas, and they can be signs of nothing else.

Chapter III Of General Terms

1. The greatest part of words are general terms. All things that exist being particulars, it may perhaps be thought reasonable that words, which ought to be conformed to things, should be so too,—I mean in their signification: but yet we find quite the contrary. The far greatest part of words that make all languages are general terms: which has not been the effect of neglect or chance, but of reason and necessity.

2. That every particular thing should have a name for itself is impossible. First, It is impossible that every particular thing should have a distinct peculiar name. For, the signification and use of words depending on that connexion which the mind makes between its ideas and

the sounds it uses as signs of them, it is necessary, in the application of names to things, that the mind should have distinct ideas of the things, and retain also the particular name that belongs to every one, with its peculiar appropriation to that idea. But it is beyond the power of human capacity to frame and retain distinct ideas of all the particular things we meet with: every bird and beast men saw; every tree and plant that affected the senses, could not find a place in the most capacious understanding. If it be looked on as an instance of a prodigious memory, that some generals have been able to call every soldier in their army by his proper name, we may easily find a reason why men have never attempted to give names to each sheep in their flock, or crow that flies over their heads; much less to call every leaf of plants, or grain of sand that came in their way, by a peculiar name.

3. And would be useless, if it were possible. Secondly, If it were possible, it would yet be useless; because it would not serve to the chief end of language. Men would in vain heap up names of particular things, that would not

serve them to communicate their thoughts. Men learn names, and use them in talk with others, only that they may be understood: which is then only done when, by use or consent, the sound I make by the organs of speech, excites in another man's mind who hears it, the idea I apply it to in mine, when I speak it. This cannot be done by names applied to particular things; whereof I alone having the ideas in my mind, the names of them could not be significant or intelligible to another, who was not acquainted with all those very particular things which had fallen under my notice.

4. A distinct name for every particular thing, not fitted for enlargement of knowledge. Thirdly, But yet, granting this also feasible, (which I think is not), yet a distinct name for every particular thing would not be of any great use for the improvement of knowledge: which, though founded in particular things, enlarges itself by general views; to which things reduced into sorts, under general names, are properly subservient. These, with the names belonging to them, come within some compass, and do not multiply every moment, beyond what

either the mind can contain, or use requires. And therefore, in these, men have for the most part stopped: but yet not so as to hinder themselves from distinguishing particular things by appropriated names, where convenience demands it. And therefore in their own species, which they have most to do with, and wherein they have often occasion to mention particular persons, they make use of proper names; and there distinct individuals have distinct denominations.

5. What things have proper names, and why. Besides persons, countries also, cities, rivers, mountains, and other the like distinctions of place have usually found peculiar names, and that for the same reason; they being such as men have often an occasion to mark particularly, and, as it were, set before others in their discourses with them. And I doubt not but, if we had reason to mention particular horses as often as we have to mention particular men, we should have proper names for the one, as familiar as for the other, and Bucephalus would be a word as much in use as Alexander. And therefore we see that, amongst jockeys, horses have their

proper names to be known and distinguished by, as commonly as their servants: because, amongst them, there is often occasion to mention this or that particular horse when he is out of sight.

6. How general words are made. The next thing to be considered is,—How general words come to be made. For, since all things that exist are only particulars, how come we by general terms; or where find we those general natures they are supposed to stand for? Words become general by being made the signs of general ideas: and ideas become general, by separating from them the circumstances of time and place, and any other ideas that may determine them to this or that particular existence. By this way of abstraction they are made capable of representing more individuals than one; each of which having in it a conformity to that abstract idea, is (as we call it) of that sort.

7. Shown by the way we enlarge our complex ideas from infancy. But, to deduce this a little more distinctly, it will not perhaps be amiss to trace our notions and names from their beginning, and observe by what de-

grees we proceed, and by what steps we enlarge our ideas from our first infancy. There is nothing more evident, than that the ideas of the persons children converse with (to instance in them alone) are, like the persons themselves, only particular. The ideas of the nurse and the mother are well framed in their minds; and, like pictures of them there, represent only those individuals. The names they first gave to them are confined to these individuals; and the names of nurse and mamma, the child uses, determine themselves to those persons. Afterwards, when time and a larger acquaintance have made them observe that there are a great many other things in the world, that in some common agreements of shape, and several other qualities, resemble their father and mother, and those persons they have been used to, they frame an idea, which they find those many particulars do partake in; and to that they give, with others, the name man, for example. And thus they come to have a general name, and a general idea. Wherein they make nothing new; but only leave out of the complex idea they had of Peter and James, Mary and Jane,

that which is peculiar to each, and retain only what is common to them all.

8. And further enlarge our complex ideas, by still leaving out properties contained in them. By the same way that they come by the general name and idea of man, they easily advance to more general names and notions. For, observing that several things that differ from their idea of man, and cannot therefore be comprehended under that name, have yet certain qualities wherein they agree with man, by retaining only those qualities, and uniting them into one idea, they have again another and more general idea; to which having given a name they make a term of a more comprehensive extension: which new idea is made, not by any new addition, but only as before, by leaving out the shape, and some other properties signified by the name man, and retaining only a body, with life, sense, and spontaneous motion, comprehended under the name animal.

9. General natures are nothing but abstract and partial ideas of more complex ones. That this is the way whereby men first formed general ideas, and general names to

them, I think is so evident, that there needs no other proof of it but the considering of a man's self, or others, and the ordinary proceedings of their minds in knowledge. And he that thinks general natures or notions are anything else but such abstract and partial ideas of more complex ones, taken at first from particular existences, will, I fear, be at a loss where to find them. For let any one effect, and then tell me, wherein does his idea of man differ from that of Peter and Paul, or his idea of horse from that of Bucephalus, but in the leaving out something that is peculiar to each individual, and retaining so much of those particular complex ideas of several particular existences as they are found to agree in? Of the complex ideas signified by the names man and horse, leaving out but those particulars wherein they differ, and retaining only those wherein they agree, and of those making a new distinct complex idea, and giving the name animal to it, one has a more general term, that comprehends with man several other creatures. Leave out of the idea of animal, sense and spontaneous motion, and the remaining complex idea, made up of

the remaining simple ones of body, life, and nourishment, becomes a more general one, under the more comprehensive term, vivens. And, not to dwell longer upon this particular, so evident in itself; by the same way the mind proceeds to body, substance, and at last to being, thing, and such universal terms, which stand for any of our ideas whatsoever. To conclude: this whole mystery of genera and species, which make such a noise in the schools, and are with justice so little regarded out of them, is nothing else but abstract ideas, more or less comprehensive, with names annexed to them. In all which this is constant and unvariable, That every more general term stands for such an idea, and is but a part of any of those contained under it.

10. Why the genus is ordinarily made use of in definitions. This may show us the reason why, in the defining of words, which is nothing but declaring their signification, we make use of the genus, or next general word that comprehends it. Which is not out of necessity, but only to save the labour of enumerating the several simple ideas which the next general word or genus stands for;

or, perhaps, sometimes the shame of not being able to do it. But though defining by genus and differentia (I crave leave to use these terms of art, though originally Latin, since they most properly suit those notions they are applied to), I say, though defining by the genus be the shortest way, yet I think it may be doubted whether it be the best. This I am sure, it is not the only, and so not absolutely necessary. For, definition being nothing but making another understand by words what idea the term defined stands for, a definition is best made by enumerating those simple ideas that are combined in the signification of the term defined: and if, instead of such an enumeration, men have accustomed themselves to use the next general term, it has not been out of necessity, or for greater clearness, but for quickness and dispatch sake. For I think that, to one who desired to know what idea the word man stood for; if it should be said, that man was a solid extended substance, having life, sense, spontaneous motion, and the faculty of reasoning, I doubt not but the meaning of the term man would be as well understood, and the idea it stands

for be at least as clearly made known, as when it is defined to be a rational animal: which, by the several definitions of animal, vivens, and corpus, resolves itself into those enumerated ideas. I have, in explaining the term man, followed here the ordinary definition of the schools; which, though perhaps not the most exact, yet serves well enough to my present purpose. And one may, in this instance, see what gave occasion to the rule, that a definition must consist of genus and differentia; and it suffices to show us the little necessity there is of such a rule, or advantage in the strict observing of it. For, definitions, as has been said, being only the explaining of one word by several others, so that the meaning or idea it stands for may be certainly known; languages are not always so made according to the rules of logic, that every term can have its signification exactly and clearly expressed by two others. Experience sufficiently satisfies us to the contrary; or else those who have made this rule have done ill, that they have given us so few definitions conformable to it. But of definitions more in the next chapter.

11. General and universal are creatures of the understanding, and belong not to the real existence of things. To return to general words: it is plain, by what has been said, that general and universal belong not to the real existence of things; but are the inventions and creatures of the understanding, made by it for its own use, and concern only signs, whether words or ideas. Words are general, as has been said, when used for signs of general ideas, and so are applicable indifferently to many particular things; and ideas are general when they are set up as the representatives of many particular things: but universality belongs not to things themselves, which are all of them particular in their existence, even those words and ideas which in their signification are general. When therefore we quit particulars, the generals that rest are only creatures of our own making; their general nature being nothing but the capacity they are put into, by the understanding, of signifying or representing many particulars. For the signification they have is nothing but a relation that, by the mind of man, is added to them.

12. Abstract ideas are the essences of genera and species. The next thing therefore to be considered is, What kind of signification it is that general words have. For, as it is evident that they do not signify barely one particular thing; for then they would not be general terms, but proper names, so, on the other side, it is as evident they do not signify a plurality; for man and men would then signify the same; and the distinction of numbers (as the grammarians call them) would be superfluous and useless. That then which general words signify is a sort of things; and each of them does that, by being a sign of an abstract idea in the mind; to which idea, as things existing are found to agree, so they come to be ranked under that name, or, which is all one, be of that sort. Whereby it is evident that the essences of the sorts, or, if the Latin word pleases better, species of things, are nothing else but these abstract ideas. For the having the essence of any species, being that which makes anything to be of that species; and the conformity to the idea to which the name is annexed being that which gives a right to that name; the having the essence, and

the having that conformity, must needs be the same thing: since to be of any species, and to have a right to the name of that species, is all one. As, for example, to be a man, or of the species man, and to have right to the name man, is the same thing. Again, to be a man, or of the species man, and have the essence of a man, is the same thing. Now, since nothing can be a man, or have a right to the name man, but what has a conformity to the abstract idea the name man stands for, nor anything be a man, or have a right to the species man, but what has the essence of that species; it follows, that the abstract idea for which the name stands, and the essence of the species, is one and the same. From whence it is easy to observe, that the essences of the sorts of things, and, consequently, the sorting of things, is the workmanship of the understanding that abstracts and makes those general ideas.

13. They are the workmanship of the understanding, but have their foundation in the similitude of things. I would not here be thought to forget, much less to deny, that Nature, in the production of things, makes several

of them alike: there is nothing more obvious, especially in the race of animals, and all things propagated by seed. But yet I think we may say, the sorting of them under names is the workmanship of the understanding, taking occasion, from the similitude it observes amongst them, to make abstract general ideas, and set them up in the mind, with names annexed to them, as patterns or forms, (for, in that sense, the word form has a very proper signification,) to which as particular things existing are found to agree, so they come to be of that species, have that denomination, or are put into that classis. For when we say this is a man, that a horse; this justice, that cruelty; this a watch, that a jack; what do we else but rank things under different specific names, as agreeing to those abstract ideas, of which we have made those names the signs? And what are the essences of those species set out and marked by names, but those abstract ideas in the mind; which are, as it were, the bonds between particular things that exist, and the names they are to be ranked under? And when general names have any connexion with particular beings, these abstract

ideas are the medium that unites them: so that the essences of species, as distinguished and denominated by us, neither are nor can be anything but those precise abstract ideas we have in our minds. And therefore the supposed real essences of substances, if different from our abstract ideas, cannot be the essences of the species we rank things into. For two species may be one, as rationally as two different essences be the essence of one species: and I demand what are the alterations [which] may, or may not be made in a horse or lead, without making either of them to be of another species? In determining the species of things by our abstract ideas, this is easy to resolve: but if any one will regulate himself herein by supposed real essences, he will, I suppose, be at a loss: and he will never be able to know when anything precisely ceases to be of the species of a horse or lead.

14. Each distinct abstract idea is a distinct essence. Nor will any one wonder that I say these essences, or abstract ideas (which are the measures of name, and the boundaries of species) are the workmanship of the un-

derstanding, who considers that at least the complex ones are often, in several men, different collections of simple ideas; and therefore that is covetousness to one man, which is not so to another. Nay, even in substances, where their abstract ideas seem to be taken from the things themselves, they are not constantly the same; no, not in that species which is most familiar to us, and with which we have the most intimate acquaintance: it having been more than once doubted, whether the foetus born of a woman were a man, even so far as that it hath been debated, whether it were or were not to be nourished and baptized: which could not be, if the abstract idea or essence to which the name man belonged were of nature's making; and were not the uncertain and various collection of simple ideas, which the understanding put together, and then, abstracting it, affixed a name to it. So that, in truth, every distinct abstract idea is a distinct essence; and the names that stand for such distinct ideas are the names of things essentially different. Thus a circle is as essentially different from an oval as a sheep from a goat; and rain is as

essentially different from snow as water from earth: that abstract idea which is the essence of one being impossible to be communicated to the other. And thus any two abstract ideas, that in any part vary one from another, with two distinct names annexed to them, constitute two distinct sorts, or, if you please, species, as essentially different as any two of the most remote or opposite in the world.

15. Several significations of the word "essence." But since the essences of things are thought by some (and not without reason) to be wholly unknown, it may not be amiss to consider the several significations of the word essence.

Real essences. First, Essence may be taken for the very being of anything, whereby it is what it is. And thus the real internal, but generally (in substances) unknown constitution of things, whereon their discoverable qualities depend, may be called their essence. This is the proper original signification of the word, as is evident from the formation of it; *essentia*, in its primary notation, signifying properly, being. And in this sense

it is still used, when we speak of the essence of particular things, without giving them any name.

Nominal essences. Secondly, The learning and disputes of the schools having been much busied about genus and species, the word essence has almost lost its primary signification: and, instead of the real constitution of things, has been almost wholly applied to the artificial constitution of genus and species. It is true, there is ordinarily supposed a real constitution of the sorts of things; and it is past doubt there must be some real constitution, on which any collection of simple ideas co-existing must depend. But, it being evident that things are ranked under names into sorts or species, only as they agree to certain abstract ideas, to which we have annexed those names, the essence of each genus, or sort, comes to be nothing but that abstract idea which the general, or sortal (if I may have leave so to call it from sort, as I do general from genus), name stands for. And this we shall find to be that which the word essence imports in its most familiar use.

These two sorts of essences, I suppose, may not un-

fitly be termed, the one the real, the other nominal essence.

16. Constant connexion between the name and nominal essence. Between the nominal essence and the name there is so near a connexion, that the name of any sort of things cannot be attributed to any particular being but what has this essence, whereby it answers that abstract idea whereof that name is the sign.

17. Supposition, that species are distinguished by their real essences, useless. Concerning the real essences of corporeal substances (to mention these only) there are, if I mistake not, two opinions. The one is of those who, using the word essence for they know not what, suppose a certain number of those essences, according to which all natural things are made, and wherein they do exactly every one of them partake, and so become of this or that species. The other and more rational opinion is of those who look on all natural things to have a real, but unknown, constitution of their insensible parts; from which flow those sensible qualities which serve us to distinguish them one from another, according as we

have occasion to rank them into sorts, under common denominations. The former of these opinions, which supposes these essences as a certain number of forms or moulds, wherein all natural things that exist are cast, and do equally partake, has, I imagine, very much perplexed the knowledge of natural things. The frequent productions of monsters, in all the species of animals, and of changelings, and other strange issues of human birth, carry with them difficulties, not possible to consist with this hypothesis; since it is as impossible that two things partaking exactly of the same real essence should have different properties, as that two figures partaking of the same real essence of a circle should have different properties. But were there no other reason against it, yet the supposition of essences that cannot be known; and the making of them, nevertheless, to be that which distinguishes the species of things, is so wholly useless and unserviceable to any part of our knowledge, that that alone were sufficient to make us lay it by, and content ourselves with such essences of the sorts or species of things as come within the reach

of our knowledge: which, when seriously considered, will be found, as I have said, to be nothing else but, those abstract complex ideas to which we have annexed distinct general names.

18. Real and nominal essence the same in simple ideas and modes, different in substances. Essences being thus distinguished into nominal and real, we may further observe, that, in the species of simple ideas and modes, they are always the same; but in substances always quite different. Thus, a figure including a space between three lines, is the real as well as nominal essence of a triangle; it being not only the abstract idea to which the general name is annexed, but the very essentia or being of the thing itself; that foundation from which all its properties flow, and to which they are all inseparably annexed. But it is far otherwise concerning that parcel of matter which makes the ring on my finger; wherein these two essences are apparently different. For, it is the real constitution of its insensible parts, on which depend all those properties of colour, weight, fusibility, fixedness, &c., which are to be found in it; which constitution we

know not, and so, having no particular idea of, having no name that is the sign of it. But yet it is its colour, weight, fusibility, fixedness, &c., which makes it to be gold, or gives it a right to that name, which is therefore its nominal essence. Since nothing can be called gold but what has a conformity of qualities to that abstract complex idea to which that name is annexed. But this distinction of essences, belonging particularly to substances, we shall, when we come to consider their names, have an occasion to treat of more fully.

19. Essences ingenerable and incorruptible. That such abstract ideas, with names to them, as we have been speaking of are essences, may further appear by what we are told concerning essences, viz. that they are all ingenerable and incorruptible. Which cannot be true of the real constitutions of things, which begin and perish with them. All things that exist, besides their Author, are all liable to change; especially those things we are acquainted with, and have ranked into bands under distinct names or ensigns. Thus, that which was grass to-day is to-morrow the flesh of a sheep; and, within a few

days after, becomes part of a man: in all which and the like changes, it is evident their real essence—i.e. that constitution whereon the properties of these several things depended—is destroyed, and perishes with them. But essences being taken for ideas established in the mind, with names annexed to them, they are supposed to remain steadily the same, whatever mutations the particular substances are liable to. For, whatever becomes of Alexander and Bucephalus, the ideas to which man and horse are annexed, are supposed nevertheless to remain the same; and so the essences of those species are preserved whole and undestroyed, whatever changes happen to any or all of the individuals of those species. By this means the essence of a species rests safe and entire, without the existence of so much as one individual of that kind. For, were there now no circle existing anywhere in the world, (as perhaps that figure exists not anywhere exactly marked out), yet the idea annexed to that name would not cease to be what it is; nor cease to be as a pattern to determine which of the particular figures we meet with have or have not a right

to the name circle, and so to show which of them, by having that essence, was of that species. And though there neither were nor had been in nature such a beast as an unicorn, or such a fish as a mermaid; yet, supposing those names to stand for complex abstract ideas that contained no inconsistency in them, the essence of a mermaid is as intelligible as that of a man; and the idea of an unicorn as certain, steady, and permanent as that of a horse. From what has been said, it is evident, that the doctrine of the immutability of essences proves them to be only abstract ideas; and is founded on the relation established between them and certain sounds as signs of them; and will always be true, as long as the same name can have the same signification.

20. Recapitulation. To conclude. This is that which in short I would say, viz. that all the great business of genera and species, and their essences, amounts to no more but this:—That men making abstract ideas, and settling them in their minds with names annexed to them, do thereby enable themselves to consider things, and discourse of them, as it were in bundles, for the

easier and readier improvement and communication of their knowledge, which would advance but slowly were their words and thoughts confined only to particulars.

Chapter IV

Of the Names of Simple Ideas

1. Names of simple ideas, modes, and substances, have each something peculiar. Though all words, as I have shown, signify nothing immediately but the ideas in the mind of the speaker; yet, upon a nearer survey, we shall find the names of simple ideas, mixed modes (under which I comprise relations too), and natural substances, have each of them something peculiar and different from the other. For example:

2. Names of simple ideas, and of substances intimate real existence. First, the names of simple ideas and substances, with the abstract ideas in the mind which they immediately signify, intimate also some real existence, from which was derived their original pattern. But the names of mixed modes terminate in the idea that is in

the mind, and lead not the thoughts any further; as we shall see more at large in the following chapter.

3. Names of simple ideas and modes signify always both real and nominal essences. Secondly, The names of simple ideas and modes signify always the real as well as nominal essence of their species. But the names of natural substances signify rarely, if ever, anything but barely the nominal essences of those species; as we shall show in the chapter that treats of the names of substances in particular.

4. Names of simple ideas are undefinable. Thirdly, The names of simple ideas are not capable of any definition; the names of all complex ideas are. It has not, that I know, been yet observed by anybody what words are, and what are not, capable of being defined; the want whereof is (as I am apt to think) not seldom the occasion of great wrangling and obscurity in men's discourses, whilst some demand definitions of terms that cannot be defined; and others think they ought not to rest satisfied in an explication made by a more general word, and its restriction, (or to speak in terms of art, by a genus and differ-

ence), when, even after such definition, made according to rule, those who hear it have often no more a clear conception of the meaning of the word than they had before. This at least I think, that the showing what words are, and what are not, capable of definitions, and wherein consists a good definition, is not wholly besides our present purpose; and perhaps will afford so much light to the nature of these signs and our ideas, as to deserve a more particular consideration.

5. If all names were definable, it would be a process in infinitum. I will not here trouble myself to prove that all terms are not definable, from that progress in infinitum, which it will visibly lead us into, if we should allow that all names could be defined. For, if the terms of one definition were still to be defined by another, where at last should we stop? But I shall, from the nature of our ideas, and the signification of our words, show why some names can, and others cannot be defined; and which they are.

6. What a definition is. I think it is agreed, that a definition is nothing else but the showing the meaning of one word by several other not synonymous terms. The

meaning of words being only the ideas they are made to stand for by him that uses them, the meaning of any term is then showed, or the word is defined, when, by other words, the idea it is made the sign of, and annexed to, in the mind of the speaker, is as it were represented, or set before the view of another; and thus its signification is ascertained. This is the only use and end of definitions; and therefore the only measure of what is, or is not a good definition.

7. Simple ideas, why undefinable. This being premised, I say that the names of simple ideas, and those only, are incapable of being defined. The reason whereof is this, That the several terms of a definition, signifying several ideas, they can all together by no means represent an idea which has no composition at all: and therefore a definition, which is properly nothing but the showing the meaning of one word by several others not signifying each the same thing, can in the names of simple ideas have no place.

8. Instances: scholastic definitions of motion. The not observing this difference in our ideas, and their names,

has produced that eminent trifling in the schools, which is so easy to be observed in the definitions they give us of some few of these simple ideas. For, as to the greatest part of them, even those masters of definitions were fain to leave them untouched, merely by the impossibility they found in it. What more exquisite jargon could the wit of man invent, than this definition:—"The act of a being in power, as far forth as in power"; which would puzzle any rational man, to whom it was not already known by its famous absurdity, to guess what word it could ever be supposed to be the explication of. If Tully, asking a Dutchman what beweegeinge was, should have received this explication in his own language, that it was "actus entis in potentia quatenus in potentia"; I ask whether any one can imagine he could thereby have understood what the word beweegeinge signified, or have guessed what idea a Dutchman ordinarily had in his mind, and would signify to another, when he used that sound?

9. Modern definitions of motion. Nor have the modern philosophers, who have endeavoured to throw off the jargon of the schools, and speak intelligibly, much bet-

ter succeeded in defining simple ideas, whether by explaining their causes, or any otherwise. The atomists, who define motion to be “a passage from one place to another,” what do they more than put one synonymous word for another? For what is passage other than motion? And if they were asked what passage was, how would they better define it than by motion? For is it not at least as proper and significant to say, Passage is a motion from one place to another, as to say, Motion is a passage, &c.? This is to translate, and not to define, when we change two words of the same signification one for another; which, when one is better understood than the other, may serve to discover what idea the unknown stands for; but is very far from a definition, unless we will say every English word in the dictionary is the definition of the Latin word it answers, and that motion is a definition of motus. Nor will the “successive application of the parts of the superficies of one body to those of another,” which the Cartesians give us, prove a much better definition of motion, when well examined.

10. Definitions of light. “The act of perspicuous, as far

forth as perspicuous,” is another Peripatetic definition of a simple idea; which, though not more absurd than the former of motion, yet betrays its uselessness and insignificancy more plainly; because experience will easily convince any one that it cannot make the meaning of the word light (which it pretends to define) at all understood by a blind man, but the definition of motion appears not at first sight so useless, because it escapes this way of trial. For this simple idea, entering by the touch as well as sight, it is impossible to show an example of any one who has no other way to get the idea of motion, but barely by the definition of that name. Those who tell us that light is a great number of little globules, striking briskly on the bottom of the eye, speak more intelligibly than the Schools: but yet these words never so well understood would make the idea the word light stands for no more known to a man that understands it not before, than if one should tell him that light was nothing but a company of little tennis-balls, which fairies all day long struck with rackets against some men’s foreheads, whilst they passed by others. For

granting this explication of the thing to be true, yet the idea of the cause of light, if we had it never so exact, would no more give us the idea of light itself, as it is such a particular perception in us, than the idea of the figure and motion of a sharp piece of steel would give us the idea of that pain which it is able to cause in us. For the cause of any sensation, and the sensation itself, in all the simple ideas of one sense, are two ideas; and two ideas so different and distant one from another, that no two can be more so. And therefore, should Descartes's globules strike never so long on the retina of a man who was blind by a gutta serena, he would thereby never have any idea of light, or anything approaching it, though he understood never so well what little globules were, and what striking on another body was. And therefore the Cartesians very well distinguish between that light which is the cause of that sensation in us, and the idea which is produced in us by it, and is that which is properly light.

11. Simple ideas, why undefinable, further explained. Simple ideas, as has been shown, are only to be got by

those impressions objects themselves make on our minds, by the proper inlets appointed to each sort. If they are not received this way, all the words in the world, made use of to explain or define any of their names, will never be able to produce in us the idea it stands for. For, words being sounds, can produce in us no other simple ideas than of those very sounds; nor excite any in us, but by that voluntary connexion which is known to be between them and those simple ideas which common use has made them the signs of. He that thinks otherwise, let him try if any words can give him the taste of a pineapple, and make him have the true idea of the relish of that celebrated delicious fruit. So far as he is told it has a resemblance with any tastes whereof he has the ideas already in his memory, imprinted there by sensible objects, not strangers to his palate, so far may he approach that resemblance in his mind. But this is not giving us that idea by a definition, but exciting in us other simple ideas by their known names; which will be still very different from the true taste of that fruit itself. In light and colours, and all other simple ideas, it is the same

thing: for the signification of sounds is not natural, but only imposed and arbitrary. And no definition of light or redness is more fitted or able to produce either of those ideas in us, than the sound light or red, by itself. For, to hope to produce an idea of light or colour by a sound, however formed, is to expect that sounds should be visible, or colours audible; and to make the ears do the office of all the other senses. Which is all one as to say, that we might taste, smell, and see by the ears: a sort of philosophy worthy only of Sancho Panza, who had the faculty to see Dulcinea by hearsay. And therefore he that has not before received into his mind, by the proper inlet, the simple idea which any word stands for, can never come to know the signification of that word by any other words or sounds whatsoever, put together according to any rules of definition. The only way is, by applying to his senses the proper object; and so producing that idea in him, for which he has learned the name already. A studious blind man, who had mightily beat his head about visible objects, and made use of the explication of his books and friends, to understand those

names of light and colours which often came in his way, bragged one day, That he now understood what scarlet signified. Upon which, his friend demanding what scarlet was? The blind man answered, It was like the sound of a trumpet. Just such an understanding of the name of any other simple idea will he have, who hopes to get it only from a definition, or other words made use of to explain it.

12. The contrary shown in complex ideas, by instances of a statue and rainbow. The case is quite otherwise in complex ideas; which, consisting of several simple ones, it is in the power of words, standing for the several ideas that make that composition, to imprint complex ideas in the mind which were never there before, and so make their names be understood. In such collections of ideas, passing under one name, definition, or the teaching the signification of one word by several others, has place, and may make us understand the names of things which never came within the reach of our senses; and frame ideas suitable to those in other men's minds, when they use those names: provided that none of the terms

of the definition stand for any such simple ideas, which he to whom the explication is made has never yet had in his thought. Thus the word statue may be explained to a blind man by other words, when picture cannot; his senses having given him the idea of figure, but not of colours, which therefore words cannot excite in him. This gained the prize to the painter against the statuary: each of which contending for the excellency of his art, and the statuary bragging that his was to be preferred, because it reached further, and even those who had lost their eyes could yet perceive the excellency of it. The painter agreed to refer himself to the judgment of a blind man; who being brought where there was a statue made by the one, and a picture drawn by the other; he was first led to the statue, in which he traced with his hands all the lineaments of the face and body, and with great admiration applauded the skill of the workman. But being led to the picture, and having his hands laid upon it, was told, that now he touched the head, and then the forehead, eyes, nose, &c., as his hand moved over the parts of the picture on the cloth,

without finding any the least distinction: whereupon he cried out, that certainly that must needs be a very admirable and divine piece of workmanship, which could represent to them all those parts, where he could neither feel nor perceive anything.

13. Colours indefinable to the born-blind. He that should use the word rainbow to one who knew all those colours, but yet had never seen that phenomenon, would, by enumerating the figure, largeness, position, and order of the colours, so well define that word that it might be perfectly understood. But yet that definition, how exact and perfect soever, would never make a blind man understand it; because several of the simple ideas that make that complex one, being such as he never received by sensation and experience, no words are able to excite them in his mind.

14. Complex ideas definable only when the simple ideas of which they consist have been got from experience. Simple ideas, as has been shown, can only be got by experience from those objects which are proper to produce in us those perceptions. When, by this means, we

have our minds stored with them, and know the names for them, then we are in a condition to define, and by definition to understand, the names of complex ideas that are made up of them. But when any term stands for a simple idea that a man has never yet had in his mind, it is impossible by any words to make known its meaning to him. When any term stands for an idea a man is acquainted with, but is ignorant that that term is the sign of it, then another name of the same idea, which he has been accustomed to, may make him understand its meaning. But in no case whatsoever is any name of any simple idea capable of a definition.

15. Names of simple ideas of less doubtful meaning than those of mixed modes and substances. Fourthly, But though the names of simple ideas have not the help of definition to determine their signification, yet that hinders not but that they are generally less doubtful and uncertain than those of mixed modes and substances; because they, standing only for one simple perception, men for the most part easily and perfectly agree in their signification; and there is little room for mistake and

wrangling about their meaning. He that knows once that whiteness is the name of that colour he has observed in snow or milk, will not be apt to misapply that word, as long as he retains that idea; which when he has quite lost, he is not apt to mistake the meaning of it, but perceives he understands it not. There is neither a multiplicity of simple ideas to be put together, which makes the doubtfulness in the names of mixed modes; nor a supposed, but an unknown, real essence, with properties depending thereon, the precise number whereof is also unknown, which makes the difficulty in the names of substances. But, on the contrary, in simple ideas the whole signification of the name is known at once, and consists not of parts, whereof more or less being put in, the idea may be varied, and so the signification of name be obscure, or uncertain.

16. Simple ideas have few ascents in *linea praedicamentali*. Fifthly, This further may be observed concerning simple ideas and their names, that they have but few ascents in *linea praedicamentali*, (as they call it,) from the lowest species to the summum genus. The

reason whereof is, that the lowest species being but one simple idea, nothing can be left out of it, that so the difference being taken away, it may agree with some other thing in one idea common to them both; which, having one name, is the genus of the other two: v.g. there is nothing that can be left out of the idea of white and red to make them agree in one common appearance, and so have one general name; as rationality being left out of the complex idea of man, makes it agree with brute in the more general idea and name of animal. And therefore when, to avoid unpleasant enumerations, men would comprehend both white and red, and several other such simple ideas, under one general name, they have been fain to do it by a word which denotes only the way they get into the mind. For when white, red, and yellow are all comprehended under the genus or name colour, it signifies no more but such ideas as are produced in the mind only by the sight, and have entrance only through the eyes. And when they would frame yet a more general term to comprehend both colours and sounds, and the like simple ideas, they do it by a word

that signifies all such as come into the mind only by one sense. And so the general term quality, in its ordinary acceptation, comprehends colours, sounds, tastes, smells, and tangible qualities, with distinction from extension, number, motion, pleasure, and pain, which make impressions on the mind and introduce their ideas by more senses than one.

17. Names of simple ideas not arbitrary, but perfectly taken from the existence of things. Sixthly, The names of simple ideas, substances, and mixed modes have also this difference: that those of mixed modes stand for ideas perfectly arbitrary; those of substances are not perfectly so, but refer to a pattern, though with some latitude; and those of simple ideas are perfectly taken from the existence of things, and are not arbitrary at all. Which, what difference it makes in the significations of their names, we shall see in the following chapters.

Simple modes. The names of simple modes differ little from those of simple ideas.

Chapter V

Of the Names of Mixed Modes and Relations

1. Mixed modes stand for abstract ideas, as other general names. The names of mixed modes, being general, they stand, as has been shewed, for sorts or species of things, each of which has its peculiar essence. The essences of these species also, as has been shewed, are nothing but the abstract ideas in the mind, to which the name is annexed. Thus far the names and essences of mixed modes have nothing but what is common to them with other ideas: but if we take a little nearer survey of them, we shall find that they have something peculiar, which perhaps may deserve our attention.

2. First, The abstract ideas they stand for are made by the understanding. The first particularity I shall observe in them, is, that the abstract ideas, or, if you please, the essences, of the several species of mixed modes, are made by the understanding, wherein they differ from those of simple ideas: in which sort the mind has no power to make any one, but only receives such as are presented

to it by the real existence of things operating upon it.

3. Secondly, made arbitrarily, and without patterns. In the next place, these essences of the species of mixed modes are not only made by the mind, but made very arbitrarily, made without patterns, or reference to any real existence. Wherein they differ from those of substances, which carry with them the supposition of some real being, from which they are taken, and to which they are conformable. But, in its complex ideas of mixed modes, the mind takes a liberty not to follow the existence of things exactly. It unites and retains certain collections, as so many distinct specific ideas; whilst others, that as often occur in nature, and are as plainly suggested by outward things, pass neglected, without particular names or specifications. Nor does the mind, in these of mixed modes, as in the complex idea of substances, examine them by the real existence of things; or verify them by patterns containing such peculiar compositions in nature. To know whether his idea of adultery or incest be right, will a man seek it anywhere amongst things existing? Or is it true because any one

has been witness to such an action? No: but it suffices here, that men have put together such a collection into one complex idea, that makes the archetype and specific idea, whether ever any such action were committed in rerum natura or no.

4. How this is done. To understand this right, we must consider wherein this making of these complex ideas consists; and that is not in the making any new idea, but putting together those which the mind had before. Wherein the mind does these three things: First, It chooses a certain number; Secondly, It gives them connexion, and makes them into one idea; Thirdly, It ties them together by a name. If we examine how the mind proceeds in these, and what liberty it takes in them, we shall easily observe how these essences of the species of mixed modes are the workmanship of the mind; and, consequently, that the species themselves are of men's making. Evidently arbitrary, in that the idea is often before the existence. Nobody can doubt but that these ideas of mixed modes are made by a voluntary collection of ideas, put together in the mind, indepen-

dent from any original patterns in nature, who will but reflect that this sort of complex ideas may be made, abstracted, and have names given them, and so a species be constituted, before any one individual of that species ever existed. Who can doubt but the ideas of sacrilege or adultery might be framed in the minds of men, and have names given them, and so these species of mixed modes be constituted, before either of them was ever committed; and might be as well discoursed of and reasoned about, and as certain truths discovered of them, whilst yet they had no being but in the understanding, as well as now, that they have but too frequently a real existence? Whereby it is plain how much the sorts of mixed modes are the creatures of the understanding, where they have a being as subservient to all the ends of real truth and knowledge, as when they really exist. And we cannot doubt but law-makers have often made laws about species of actions which were only the creatures of their own understandings; beings that had no other existence but in their own minds. And I think nobody can deny but that the resurrection

was a species of mixed modes in the mind, before it really existed.

6. Instances: murder, incest, stabbing. To see how arbitrarily these essences of mixed modes are made by the mind, we need but take a view of almost any of them. A little looking into them will satisfy us, that it is the mind that combines several scattered independent ideas into one complex one; and, by the common name it gives them, makes them the essence of a certain species, without regulating itself by any connexion they have in nature. For what greater connexion in nature has the idea of a man than the idea of a sheep with killing, that this is made a particular species of action, signified by the word murder, and the other not? Or what union is there in nature between the idea of the relation of a father with killing than that of a son or neighbour, that those are combined into one complex idea, and thereby made the essence of the distinct species parricide, whilst the other makes no distinct species at all? But, though they have made killing a man's father or mother a distinct species from killing his son or

daughter, yet, in some other cases, son and daughter are taken in too, as well as father and mother: and they are all equally comprehended in the same species, as in that of incest. Thus the mind in mixed modes arbitrarily unites into complex ideas such as it finds convenient; whilst others that have altogether as much union in nature are left loose, and never combined into one idea, because they have no need of one name. It is evident then that the mind, by its free choice, gives a connexion to a certain number of ideas, which in nature have no more union with one another than others that it leaves out: why else is the part of the weapon the beginning of the wound is made with taken notice of, to make the distinct species called stabbing, and the figure and matter of the weapon left out? I do not say this is done without reason, as we shall see more by and by; but this I say, that it is done by the free choice of the mind, pursuing its own ends; and that, therefore, these species of mixed modes are the workmanship of the understanding. And there is nothing more evident than that, for the most part, in the framing of these ideas, the

mind searches not its patterns in nature, nor refers the ideas it makes to the real existence of things, but puts such together as may best serve its own purposes, without tying itself to a precise imitation of anything that really exists.

7. But still subservient to the end of language, and not made at random. But, though these complex ideas or essences of mixed modes depend on the mind, and are made by it with great liberty, yet they are not made at random, and jumbled together without any reason at all. Though these complex ideas be not always copied from nature, yet they are always suited to the end for which abstract ideas are made: and though they be combinations made of ideas that are loose enough, and have as little union in themselves as several others to which the mind never gives a connexion that combines them into one idea; yet they are always made for the convenience of communication, which is the chief end of language. The use of language is, by short sounds, to signify with ease and dispatch general conceptions; wherein not only abundance of particulars may be contained,

but also a great variety of independent ideas collected into one complex one. In the making therefore of the species of mixed modes, men have had regard only to such combinations as they had occasion to mention one to another. Those they have combined into distinct complex ideas, and given names to; whilst others, that in nature have as near a union, are left loose and unregarded. For, to go no further than human actions themselves, if they would make distinct abstract ideas of all the varieties which might be observed in them, the number must be infinite, and the memory confounded with the plenty, as well as overcharged to little purpose. It suffices that men make and name so many complex ideas of these mixed modes as they find they have occasion to have names for, in the ordinary occurrence of their affairs. If they join to the idea of killing the idea of father or mother, and so make a distinct species from killing a man's son or neighbour, it is because of the different heinousness of the crime, and the distinct punishment is, due to the murdering a man's father and mother, different to what ought to be inflicted on the

murderer of a son or neighbour; and therefore they find it necessary to mention it by a distinct name, which is the end of making that distinct combination. But though the ideas of mother and daughter are so differently treated, in reference to the idea of killing, that the one is joined with it to make a distinct abstract idea with a name, and so a distinct species, and the other not; yet, in respect of carnal knowledge, they are both taken in under incest: and that still for the same convenience of expressing under one name, and reckoning of one species, such unclean mixtures as have a peculiar turpitude beyond others; and this to avoid circumlocutions and tedious descriptions.

8. Whereof the intranslatable words of divers languages are a proof. A moderate skill in different languages will easily satisfy one of the truth of this, it being so obvious to observe great store of words in one language which have not any that answer them in another. Which plainly shows that those of one country, by their customs and manner of life, have found occasion to make several complex ideas, and given names to them, which

others never collected into specific ideas. This could not have happened if these species were the steady workmanship of nature, and not collections made and abstracted by the mind, in order to naming, and for the convenience of communication. The terms of our law, which are not empty sounds, will hardly find words that answer them in the Spanish or Italian, no scanty languages; much less, I think, could any one translate them into the Caribbee or Westoe tongues: and the versura of the Romans, or corban of the Jews, have no words in other languages to answer them; the reason whereof is plain, from what has been said. Nay, if we look a little more nearly into this matter, and exactly compare different languages, we shall find that, though they have words which in translations and dictionaries are supposed to answer one another, yet there is scarce one of ten amongst the names of complex ideas, especially of mixed modes, that stands for the same precise idea which the word does that in dictionaries it is rendered by. There are no ideas more common and less compounded than the measures of time, extension and weight; and

the Latin names, hora, pes, libra, are without difficulty rendered by the English names, hour, foot, and pound: but yet there is nothing more evident than that the ideas a Roman annexed to these Latin names, were very far different from those which an Englishman expresses by those English ones. And if either of these should make use of the measures that those of the other language designed by their names, he would be quite out in his account. These are too sensible proofs to be doubted; and we shall find this much more so in the names of more abstract and compounded ideas, such as are the greatest part of those which make up moral discourses: whose names, when men come curiously to compare with those they are translated into, in other languages, they will find very few of them exactly to correspond in the whole extent of their significations.

9. This shows species to be made for communication. The reason why I take so particular notice of this is, that we may not be mistaken about genera and species, and their essences, as if they were things regularly and constantly made by nature, and had a real existence in

things; when they appear, upon a more wary survey, to be nothing else but an artifice of the understanding, for the easier signifying such collections of ideas as it should often have occasion to communicate by one general term; under which divers particulars, as far forth as they agreed to that abstract idea, might be comprehended. And if the doubtful signification of the word species may make it sound harsh to some, that I say the species of mixed modes are “made by the understanding”; yet, I think, it can by nobody be denied that it is the mind makes those abstract complex ideas to which specific names are given. And if it be true, as it is, that the mind makes the patterns for sorting and naming of things, I leave it to be considered who makes the boundaries of the sort or species; since with me species and sort have no other difference than that of a Latin and English idiom.

10. In mixed modes it is the name that ties the combination of simple ideas together, and makes it a species. The near relation that there is between species, essences, and their general name, at least in mixed modes, will further appear when we consider, that it is the name

that seems to preserve those essences, and give them their lasting duration. For, the connexion between the loose parts of those complex ideas being made by the mind, this union, which has no particular foundation in nature, would cease again, were there not something that did, as it were, hold it together, and keep the parts from scattering. Though therefore it be the mind that makes the collection, it is the name which is as it were the knot that ties them fast together. What a vast variety of different ideas does the word triumphus hold together, and deliver to us as one species! Had this name been never made, or quite lost, we might, no doubt, have had descriptions of what passed in that solemnity: but yet, I think, that which holds those different parts together, in the unity of one complex idea, is that very word annexed to it; without which the several parts of that would no more be thought to make one thing, than any other show, which having never been made but once, had never been united into one complex idea, under one denomination. How much, therefore, in mixed modes, the unity necessary to any essence depends on

the mind; and how much the continuation and fixing of that unity depends on the name in common use annexed to it, I leave to be considered by those who look upon essences and species as real established things in nature.

11. Suitable to this, we find that men speaking of mixed modes, seldom imagine or take any other for species of them, but such as are set out by name: because they, being of man's making only, in order to naming, no such species are taken notice of, or supposed to be, unless a name be joined to it, as the sign of man's having combined into one idea several loose ones; and by that name giving a lasting union to the parts which would otherwise cease to have any, as soon as the mind laid by that abstract idea, and ceased actually to think on it. But when a name is once annexed to it, wherein the parts of that complex idea have a settled and permanent union, then is the essence, as it were, established, and the species looked on as complete. For to what purpose should the memory charge itself with such compositions, unless it were by abstraction to make them gen-

eral? And to what purpose make them general, unless it were that they might have general names for the convenience of discourse and communication? Thus we see, that killing a man with a sword or a hatchet are looked on as no distinct species of action; but if the point of the sword first enter the body, it passes for a distinct species, where it has a distinct name, as in England, in whose language it is called stabbing: but in another country, where it has not happened to be specified under a peculiar name, it passes not for a distinct species. But in the species of corporeal substances, though it be the mind that makes the nominal essence, yet, since those ideas which are combined in it are supposed to have an union in nature whether the mind joins them or not, therefore those are looked on as distinct species, without any operation of the mind, either abstracting, or giving a name to that complex idea.

12. For the originals of our mixed modes, we look no further than the mind; which also shows them to be the workmanship of the understanding. Conformable also to what has been said concerning the essences of the

species of mixed modes, that they are the creatures of the understanding rather than the works of nature; conformable, I say, to this, we find that their names lead our thoughts to the mind, and no further. When we speak of justice, or gratitude, we frame to ourselves no imagination of anything existing, which we would conceive; but our thoughts terminate in the abstract ideas of those virtues, and look not further; as they do when we speak of a horse, or iron, whose specific ideas we consider not as barely in the mind, but as in things themselves, which afford the original patterns of those ideas. But in mixed modes, at least the most considerable parts of them, which are moral beings, we consider the original patterns as being in the mind, and to those we refer for the distinguishing of particular beings under names. And hence I think it is that these essences of the species of mixed modes are by a more particular name called notions; as, by a peculiar right, appertaining to the understanding.

13. Their being made by the understanding without patterns, shows the reason why they are so compounded.

Hence, likewise, we may learn why the complex ideas of mixed modes are commonly more compounded and decompounded than those of natural substances. Because they being the workmanship of the understanding, pursuing only its own ends, and the conveniency of expressing in short those ideas it would make known to another, it does with great liberty unite often into one abstract idea things that, in their nature, have no coherence; and so under one term bundle together a great variety of compounded and decompounded ideas. Thus the name of procession: what a great mixture of independent ideas of persons, habits, tapers, orders, motions, sounds, does it contain in that complex one, which the mind of man has arbitrarily put together, to express by that one name? Whereas the complex ideas of the sorts of substances are usually made up of only a small number of simple ones; and in the species of animals, these two, viz. shape and voice, commonly make the whole nominal essence.

14. Names of mixed modes stand always for their real essences, which are the workmanship of our minds.

Another thing we may observe from what has been said is, That the names of mixed modes always signify (when they have any determined signification) the real essences of their species. For, these abstract ideas being the workmanship of the mind, and not referred to the real existence of things, there is no supposition of anything more signified by that name, but barely that complex idea the mind itself has formed; which is all it would have expressed by it; and is that on which all the properties of the species depend, and from which alone they all flow: and so in these the real and nominal essence is the same; which, of what concernment it is to the certain knowledge of general truth, we shall see hereafter.

15. Why their names are usually got before their ideas. This also may show us the reason why for the most part the names of fixed modes are got before the ideas they stand for are perfectly known. Because there being no species of these ordinarily taken notice of but what have names, and those species, or rather their essences, being abstract complex ideas, made arbitrarily by the mind, it is convenient, if not necessary, to know the names,

before one endeavour to frame these complex ideas: unless a man will fill his head with a company of abstract complex ideas, which, others having no names for, he has nothing to do with, but to lay by and forget again. I confess that, in the beginning of languages, it was necessary to have the idea before one gave it the name: and so it is still, where, making a new complex idea, one also, by giving it a new name, makes a new word. But this concerns not languages made, which have generally pretty well provided for ideas which men have frequent occasion to have and communicate; and in such, I ask whether it be not the ordinary method, that children learn the names of mixed modes before they have their ideas? What one of a thousand ever frames the abstract ideas of glory and ambition, before he has heard the names of them? In simple ideas and substances I grant it is otherwise, which, being such ideas as have a real existence and union in nature, the ideas and names are got one before the other, as it happens.

16. Reason of my being so large on this subject. What has been said here of mixed modes is, with very little

difference, applicable also to relations; which, since every man himself may observe, I may spare myself the pains to enlarge on: especially, since what I have here said concerning Words in this third Book, will possibly be thought by some to be much more than what so slight a subject required. I allow it might be brought into a narrower compass; but I was willing to stay my reader on an argument that appears to me new and a little out of the way, (I am sure it is one I thought not of when I began to write,) that, by searching it to the bottom, and turning it on every side, some part or other might meet with every one's thoughts, and give occasion to the most averse or negligent to reflect on a general miscarriage, which, though of great consequence, is little taken notice of. When it is considered what a pudder is made about essences, and how much all sorts of knowledge, discourse, and conversation are pestered and disordered by the careless and confused use and application of words, it will perhaps be thought worth while thoroughly to lay it open. And I shall be pardoned if I have dwelt long on an argument which I think, therefore, needs to be incul-

Chapter VI

Of the Names of Substances

cated, because the faults men are usually guilty of in this kind, are not only the greatest hindrances of true knowledge, but are so well thought of as to pass for it. Men would often see what a small pittance of reason and truth, or possibly none at all, is mixed with those huffing opinions they are swelled with; if they would but look beyond fashionable sounds, and observe what ideas are or are not comprehended under those words with which they are so armed at all points, and with which they so confidently lay about them. I shall imagine I have done some service to truth, peace, and learning, if, by any enlargement on this subject, I can make men reflect on their own use of language; and give them reason to suspect, that, since it is frequent for others, it may also be possible for them, to have sometimes very good and approved words in their mouths and writings, with very uncertain, little, or no signification. And therefore it is not unreasonable for them to be wary herein themselves, and not to be unwilling to have them examined by others. With this design, therefore, I shall go on with what I have further to say concerning this matter.

1. The common names of substances stand for sorts. The common names of substances, as well as other general terms, stand for sorts: which is nothing else but the being made signs of such complex ideas wherein several particular substances do or might agree, by virtue of which they are capable of being comprehended in one common conception, and signified by one name. I say do or might agree: for though there be but one sun existing in the world, yet the idea of it being abstracted, so that more substances (if there were several) might each agree in it, it is as much a sort as if there were as many suns as there are stars. They want not their reasons who think there are, and that each fixed star would answer the idea the name sun stands for, to one who was placed in a due distance: which, by the way, may show us how much the sorts, or, if you please, genera and species of things (for those Latin terms signify to me no more than the English word sort) depend on

such collections of ideas as men have made, and not on the real nature of things; since it is not impossible but that, in propriety of speech, that might be a sun to one which is a star to another.

2. The essence of each sort of substance is our abstract idea to which the name is annexed. The measure and boundary of each sort or species, whereby it is constituted that particular sort, and distinguished from others, is that we call its essence, which is nothing but that abstract idea to which the name is annexed; so that everything contained in that idea is essential to that sort. This, though it be all the essence of natural substances that we know, or by which we distinguish them into sorts, yet I call it by a peculiar name, the nominal essence, to distinguish it from the real constitution of substances, upon which depends this nominal essence, and all the properties of that sort; which, therefore, as has been said, may be called the real essence: v.g. the nominal essence of gold is that complex idea the word gold stands for, let it be, for instance, a body yellow, of a certain weight, malleable, fusible, and fixed.

But the real essence is the constitution of the insensible parts of that body, on which those qualities and all the other properties of gold depend. How far these two are different, though they are both called essence, is obvious at first sight to discover.

3. The nominal and real essence different. For, though perhaps voluntary motion, with sense and reason, joined to a body of a certain shape, be the complex idea to which I and others annex the name man, and so be the nominal essence of the species so called: yet nobody will say that complex idea is the real essence and source of all those operations which are to be found in any individual of that sort. The foundation of all those qualities which are the ingredients of our complex idea, is something quite different: and had we such a knowledge of that constitution of man, from which his faculties of moving, sensation, and reasoning, and other powers flow, and on which his so regular shape depends, as it is possible angels have, and it is certain his Maker has, we should have a quite other idea of his essence than what now is contained in our definition of that species, be it

what it will: and our idea of any individual man would be as far different from what it is now, as is his who knows all the springs and wheels and other contrivances within of the famous clock at Strasburg, from that which a gazing countryman has of it, who barely sees the motion of the hand, and hears the clock strike, and observes only some of the outward appearances.

4. Nothing essential to individuals. That essence, in the ordinary use of the word, relates to sorts, and that it is considered in particular beings no further than as they are ranked into sorts, appears from hence: that, take but away the abstract ideas by which we sort individuals, and rank them under common names, and then the thought of anything essential to any of them instantly vanishes: we have no notion of the one without the other, which plainly shows their relation. It is necessary for me to be as I am; God and nature has made me so: but there is nothing I have is essential to me. An accident or disease may very much alter my colour or shape; a fever or fall may take away my reason or memory, or both; and an apoplexy leave neither sense, nor under-

standing, no, nor life. Other creatures of my shape may be made with more and better, or fewer and worse faculties than I have; and others may have reason and sense in a shape and body very different from mine. None of these are essential to the one or the other, or to any individual whatever, till the mind refers it to some sort or species of things; and then presently, according to the abstract idea of that sort, something is found essential. Let any one examine his own thoughts, and he will find that as soon as he supposes or speaks of essential, the consideration of some species, or the complex idea signified by some general name, comes into his mind; and it is in reference to that that this or that quality is said to be essential. So that if it be asked, whether it be essential to me or any other particular corporeal being, to have reason? I say, no; no more than it is essential to this white thing I write on to have words in it. But if that particular being be to be counted of the sort man, and to have the name man given it, then reason is essential to it; supposing reason to be a part of the complex idea the name man stands for: as it is essential to

this thing I write on to contain words, if I will give it the name treatise, and rank it under that species. So that essential and not essential relate only to our abstract ideas, and the names annexed to them; which amounts to no more than this, That whatever particular thing has not in it those qualities which are contained in the abstract idea which any general term stands for, cannot be ranked under that species, nor be called by that name; since that abstract idea is the very essence of that species.

5. The only essences perceived by us in individual substances are those qualities which entitle them to receive their names. Thus, if the idea of body with some people be bare extension or space, then solidity is not essential to body: if others make the idea to which they give the name body to be solidity and extension, then solidity is essential to body. That therefore, and that alone, is considered as essential, which makes a part of the complex idea the name of a sort stands for: without which no particular thing can be reckoned of that sort, nor be entitled to that name. Should there be found a parcel of

matter that had all the other qualities that are in iron, but wanted obedience to the loadstone, and would neither be drawn by it nor receive direction from it, would any one question whether it wanted anything essential? It would be absurd to ask, Whether a thing really existing wanted anything essential to it. Or could it be demanded, Whether this made an essential or specific difference or no, since we have no other measure of essential or specific but our abstract ideas? And to talk of specific differences in nature, without reference to general ideas in names, is to talk unintelligibly. For I would ask any one, What is sufficient to make an essential difference in nature between any two particular beings, without any regard had to some abstract idea, which is looked upon as the essence and standard of a species? All such patterns and standards being quite laid aside, particular beings, considered barely in themselves, will be found to have all their qualities equally essential; and everything in each individual will be essential to it; or, which is more, nothing at all. For, though it may be reasonable to ask, Whether obeying the mag-

net be essential to iron? yet I think it is very improper and insignificant to ask, whether it be essential to the particular parcel of matter I cut my pen with; without considering it under the name, iron, or as being of a certain species. And if, as has been said, our abstract ideas, which have names annexed to them, are the boundaries of species, nothing can be essential but what is contained in those ideas.

6. Even the real essences of individual substances imply potential sorts. It is true, I have often mentioned a real essence, distinct in substances from those abstract ideas of them, which I call their nominal essence. By this real essence I mean, that real constitution of anything, which is the foundation of all those properties that are combined in, and are constantly found to co-exist with the nominal essence; that particular constitution which everything has within itself, without any relation to anything without it. But essence, even in this sense, relates to a sort, and supposes a species. For, being that real constitution on which the properties depend, it necessarily supposes a sort of things, properties belong-

ing only to species, and not to individuals: v.g. supposing the nominal essence of gold to be a body of such a peculiar colour and weight, with malleability and fusibility, the real essence is that constitution of the parts of matter on which these qualities and their union depend; and is also the foundation of its solubility in aqua regia and other properties, accompanying that complex idea. Here are essences and properties, but all upon supposition of a sort or general abstract idea, which is considered as immutable; but there is no individual parcel of matter to which any of these qualities are so annexed as to be essential to it or inseparable from it. That which is essential belongs to it as a condition whereby it is of this or that sort: but take away the consideration of its being ranked under the name of some abstract idea, and then there is nothing necessary to it, nothing inseparable from it. Indeed, as to the real essences of substances, we only suppose their being, without precisely knowing what they are; but that which annexes them still to the species is the nominal essence, of which they are the supposed foundation and cause.

7. The nominal essence bounds the species for us. The next thing to be considered is, by which of those essences it is that substances are determined into sorts or species; and that, it is evident, is by the nominal essence. For it is that alone that the name, which is the mark of the sort, signifies. It is impossible, therefore, that anything should determine the sorts of things, which we rank under general names, but that idea which that name is designed as a mark for; which is that, as has been shown, which we call nominal essence. Why do we say this is a horse, and that a mule; this is an animal, that an herb? How comes any particular thing to be of this or that sort, but because it has that nominal essence; or, which is all one, agrees to that abstract idea, that name is annexed to? And I desire any one but to reflect on his own thoughts, when he hears or speaks any of those or other names of substances, to know what sort of essences they stand for.

8. The nature of species, as formed by us. And that the species of things to us are nothing but the ranking them under distinct names, according to the complex ideas in

us, and not according to precise, distinct, real essences in them, is plain from hence:—That we find many of the individuals that are ranked into one sort, called by one common name, and so received as being of one species, have yet qualities, depending on their real constitutions, as far different one from another as from others from which they are accounted to differ specifically. This, as it is easy to be observed by all who have to do with natural bodies, so chemists especially are often, by sad experience, convinced of it, when they, sometimes in vain, seek for the same qualities in one parcel of sulphur, antimony, or vitriol, which they have found in others. For, though they are bodies of the same species, having the same nominal essence, under the same name, yet do they often, upon severe ways of examination, betray qualities so different one from another, as to frustrate the expectation and labour of very wary chemists. But if things were distinguished into species, according to their real essences, it would be as impossible to find different properties in any two individual substances of the same species, as it is to find different properties in

two circles, or two equilateral triangles. That is properly the essence to us, which determines every particular to this or that classis; or, which is the same thing, to this or that general name: and what can that be else, but that abstract idea to which that name is annexed; and so has, in truth, a reference, not so much to the being of particular things, as to their general denominations?

9. Not the real essence, or texture of parts, which we know not. Nor indeed can we rank and sort things, and consequently (which is the end of sorting) denominate them, by their real essences; because we know them not. Our faculties carry us no further towards the knowledge and distinction of substances, than a collection of those sensible ideas which we observe in them; which, however made with the greatest diligence and exactness we are capable of, yet is more remote from the true internal constitution from which those qualities flow, than, as I said, a countryman's idea is from the inward contrivance of that famous clock at Strasburg, whereof he only sees the outward figure and motions. There is not so contemptible a plant or animal, that does not

confound the most enlarged understanding. Though the familiar use of things about us take off our wonder, yet it cures not our ignorance. When we come to examine the stones we tread on, or the iron we daily handle, we presently find we know not their make; and can give no reason of the different qualities we find in them. It is evident the internal constitution, whereon their properties depend, is unknown to us: for to go no further than the grossest and most obvious we can imagine amongst them, What is that texture of parts, that real essence, that makes lead and antimony fusible, wood and stones not? What makes lead and iron malleable, antimony and stones not? And yet how infinitely these come short of the fine contrivances and inconceivable real essences of plants or animals, every one knows. The workmanship of the all-wise and powerful God in the great fabric of the universe, and every part thereof, further exceeds the capacity and comprehension of the most inquisitive and intelligent man, than the best contrivance of the most ingenious man doth the conceptions of the most ignorant of rational creatures. Therefore we

in vain pretend to range things into sorts, and dispose them into certain classes under names, by their real essences, that are so far from our discovery or comprehension. A blind man may as soon sort things by their colours, and he that has lost his smell as well distinguish a lily and a rose by their odours, as by those internal constitutions which he knows not. He that thinks he can distinguish sheep and goats by their real essences, that are unknown to him, may be pleased to try his skill in those species called cassiowary and querechinchio; and by their internal real essences determine the boundaries of those species, without knowing the complex idea of sensible qualities that each of those names stand for, in the countries where those animals are to be found.

10. Not the substantial form, which we know less. Those, therefore, who have been taught that the several species of substances had their distinct internal substantial forms, and that it was those forms which made the distinction of substances into their true species and genera, were led yet further out of the way by having their minds set

upon fruitless inquiries after “substantial forms”; wholly unintelligible, and whereof we have scarce so much as any obscure or confused conception in general.

11. That the nominal essence is that only whereby we distinguish species of substances, further evident, from our ideas of finite spirits and of God. That our ranking and distinguishing natural substances into species consists in the nominal essences the mind makes, and not in the real essences to be found in the things themselves, is further evident from our ideas of spirits. For the mind getting, only by reflecting on its own operations, those simple ideas which it attributes to spirits, it hath or can have no other notion of spirit but by attributing all those operations it finds in itself to a sort of beings; without consideration of matter. And even the most advanced notion we have of GOD is but attributing the same simple ideas which we have got from reflection on what we find in ourselves, and which we conceive to have more perfection in them than would be in their absence; attributing, I say, those simple ideas to Him in an unlimited degree. Thus, having got from

reflecting on ourselves the idea of existence, knowledge, power and pleasure—each of which we find it better to have than to want; and the more we have of each the better—joining all these together, with infinity to each of them, we have the complex idea of an eternal, omniscient, omnipotent, infinitely wise and happy being. And though we are told that there are different species of angels; yet we know not how to frame distinct specific ideas of them: not out of any conceit that the existence of more species than one of spirits is impossible; but because having no more simple ideas (nor being able to frame more) applicable to such beings, but only those few taken from ourselves, and from the actions of our own minds in thinking, and being delighted, and moving several parts of our bodies; we can no otherwise distinguish in our conceptions the several species of spirits, one from another, but by attributing those operations and powers we find in ourselves to them in a higher or lower degree; and so have no very distinct specific ideas of spirits, except only of GOD, to whom we attribute both duration and all those other ideas with

infinity; to the other spirits, with limitation: nor, as I humbly conceive, do we, between GOD and them in our ideas, put any difference, by any number of simple ideas which we have of one and not of the other, but only that of infinity. All the particular ideas of existence, knowledge, will, power, and motion, &c., being ideas derived from the operations of our minds, we attribute all of them to all sorts of spirits, with the difference only of degrees; to the utmost we can imagine, even infinity, when we would frame as well as we can an idea of the First Being; who yet, it is certain, is infinitely more remote, in the real excellency of his nature, from the highest and perfectest of all created beings, than the greatest man, nay, purest seraph, is from the most contemptible part of matter; and consequently must infinitely exceed what our narrow understandings can conceive of Him.

12. Of finite spirits there are probably numberless species, in a continuous series or gradation. It is not impossible to conceive, nor repugnant to reason, that there may be many species of spirits, as much separated and

diversified one from another by distinct properties whereof we have no ideas, as the species of sensible things are distinguished one from another by qualities which we know and observe in them. That there should be more species of intelligent creatures above us, than there are of sensible and material below us, is probable to me from hence: that in all the visible corporeal world, we see no chasms or gaps. All quite down from us the descent is by easy steps, and a continued series of things, that in each remove differ very little one from the other. There are fishes that have wings, and are not strangers to the airy region: and there are some birds that are inhabitants of the water, whose blood is cold as fishes, and their flesh so like in taste that the scrupulous are allowed them on fish-days. There are animals so near of kin both to birds and beasts that they are in the middle between both: amphibious animals link the terrestrial and aquatic together; seals live at land and sea, and porpoises have the warm blood and entrails of a hog; not to mention what is confidently reported of mermaids, or sea-men. There are some brutes that seem to

have as much knowledge and reason as some that are called men: and the animal and vegetable kingdoms are so nearly joined, that, if you will take the lowest of one and the highest of the other, there will scarce be perceived any great difference between them: and so on, till we come to the lowest and the most inorganic parts of matter, we shall find everywhere that the several species are linked together, and differ but in almost insensible degrees. And when we consider the infinite power and wisdom of the Maker, we have reason to think that it is suitable to the magnificent harmony of the universe, and the great design and infinite goodness of the Architect, that the species of creatures should also, by gentle degrees, ascend upward from us toward his infinite perfection, as we see they gradually descend from us downwards: which if it be probable, we have reason then to be persuaded that there are far more species of creatures above us than there are beneath; we being, in degrees of perfection, much more remote from the infinite being of GOD than we are from the lowest state of being, and that which approaches nearest to nothing.

And yet of all those distinct species, for the reasons above said, we have no clear distinct ideas.

13. The nominal essence that of the species, as conceived by us, proved from water and ice. But to return to the species of corporeal substances. If I should ask any one whether ice and water were two distinct species of things, I doubt not but I should be answered in the affirmative: and it cannot be denied but he that says they are two distinct species is in the right. But if an Englishman bred in Jamaica, who perhaps had never seen nor heard of ice, coming into England in the winter, find the water he put in his basin at night in a great part frozen in the morning, and, not knowing any peculiar name it had, should call it hardened water; I ask whether this would be a new species to him, different from water? And I think it would be answered here, It would not be to him a new species, no more than congealed jelly, when it is cold, is a distinct species from the same jelly fluid and warm; or than liquid gold in the furnace is a distinct species from hard gold in the hands of a workman. And if this be so, it is plain that our

distinct species are nothing but distinct complex ideas, with distinct names annexed to them. It is true every substance that exists has its peculiar constitution, whereon depend those sensible qualities and powers we observe in it; but the ranking of things into species (which is nothing but sorting them under several titles) is done by us according to the ideas that we have of them: which, though sufficient to distinguish them by names, so that we may be able to discourse of them when we have them not present before us; yet if we suppose it to be done by their real internal constitutions, and that things existing are distinguished by nature into species, by real essences, according as we distinguish them into species by names, we shall be liable to great mistakes.

14. Difficulties in the supposition of a certain number of real essences. To distinguish substantial beings into species, according to the usual supposition, that there are certain precise essences or forms of things, whereby all the individuals existing are, by nature distinguished into species, these things are necessary:—

15. A crude supposition. First, To be assured that nature, in the production of things, always designs them to partake of certain regulated established essences, which are to be the models of all things to be produced. This, in that crude sense it is usually proposed, would need some better explication, before it can fully be assented to.

16. Monstrous births. Secondly, It would be necessary to know whether nature always attains that essence it designs in the production of things. The irregular and monstrous births, that in divers sorts of animals have been observed, will always give us reason to doubt of one or both of these.

17. Are monsters really a distinct species? Thirdly, It ought to be determined whether those we call monsters be really a distinct species, according to the scholastic notion of the word species; since it is certain that everything that exists has its particular constitution. And yet we find that some of these monstrous productions have few or none of those qualities which are supposed to result from, and accompany, the essence of that species from whence, they derive their originals, and to

which, by their descent, they seem to belong.

18. Men can have no ideas of real essences. Fourthly, The real essences of those things which we distinguish into species, and as so distinguished we name, ought to be known; i.e. we ought to have ideas of them. But since we are ignorant in these four points, the supposed real essences of things stand us not in stead for the distinguishing substances into species.

19. Our nominal essences of substances not perfect collections of the properties that flow from their real essences. Fifthly, The only imaginable help in this case would be, that, having framed perfect complex ideas of the properties of things flowing from their different real essences, we should thereby distinguish them into species. But neither can this be done. For, being ignorant of the real essence itself, it is impossible to know all those properties that flow from it, and are so annexed to it, that any one of them being away, we may certainly conclude that that essence is not there, and so the thing is not of that species. We can never know what is the precise number of properties depending on

the real essence of gold, any one of which failing, the real essence of gold, and consequently gold, would not be there, unless we knew the real essence of gold itself, and by that determined that species. By the word gold here, I must be understood to design a particular piece of matter; v.g. the last guinea that was coined. For, if it should stand here, in its ordinary signification, for that complex idea which I or any one else calls gold, i.e. for the nominal essence of gold, it would be jargon. So hard is it to show the various meaning and imperfection of words, when we have nothing else but words to do it by.

20. Hence names independent of real essences. By all which it is clear, that our distinguishing substances into species by names, is not at all founded on their real essences; nor can we pretend to range and determine them exactly into species, according to internal essential differences.

21. But stand for such a collection of simple substances, as we have made the name stand for. But since, as has been remarked, we have need of general words, though we know not the real essences of things; all we can do

is, to collect such a number of simple ideas as, by examination, we find to be united together in things existing, and thereof to make one complex idea. Which, though it be not the real essence of any substance that exists, is yet the specific essence to which our name belongs, and is convertible with it; by which we may at least try the truth of these nominal essences. For example: there be that say that the essence of body is extension; if it be so, we can never mistake in putting the essence of anything for the thing itself. Let us then in discourse put extension for body, and when we would say that body moves, let us say that extension moves, and see how ill it will look. He that should say that one extension by impulse moves another extension, would, by the bare expression, sufficiently show the absurdity of such a notion. The essence of anything in respect of us, is the whole complex idea comprehended and marked by that name; and in substances, besides the several distinct simple ideas that make them up, the confused one of substance, or of an unknown support and cause of their union, is always a part: and therefore the essence of

body is not bare extension, but an extended solid thing; and so to say, an extended solid thing moves, or impels another, is all one, and as intelligible, as to say, body moves or impels. Likewise, to say that a rational animal is capable of conversation, is all one as to say a man; but no one will say that rationality is capable of conversation, because it makes not the whole essence to which we give the name man.

22. Our abstract ideas are to us the measures of the species we make: instance in that of man. There are creatures in the world that have shapes like ours, but are hairy, and want language and reason. There are naturals amongst us that have perfectly our shape, but want reason, and some of them language too. There are creatures, as it is said, (*sit fides penes authorem*, but there appears no contradiction that there should be such), that, with language and reason and a shape in other things agreeing with ours, have hairy tails; others where the males have no beards, and others where the females have. If it be asked whether these be all men or no, all of human species? it is plain, the question refers

only to the nominal essence: for those of them to whom the definition of the word man, or the complex idea signified by the name, agrees, are men, and the other not. But if the inquiry be made concerning the supposed real essence; and whether the internal constitution and frame of these several creatures be specifically different, it is wholly impossible for us to answer, no part of that going into our specific idea: only we have reason to think, that where the faculties or outward frame so much differs, the internal constitution is not exactly the same. But what difference in the real internal constitution makes a specific difference it is in vain to inquire; whilst our measures of species be, as they are, only our abstract ideas, which we know; and not that internal constitution, which makes no part of them. Shall the difference of hair only on the skin be a mark of a different internal specific constitution between a changeling and a drill, when they agree in shape, and want of reason and speech? And shall not the want of reason and speech be a sign to us of different real constitutions and species between a changeling and a rea-

sonable man? And so of the rest, if we pretend that distinction of species or sorts is fixedly established by the real frame and secret constitutions of things.

23. Species in animals not distinguished by generation. Nor let any one say, that the power of propagation in animals by the mixture of male and female, and in plants by seeds, keeps the supposed real species distinct and entire. For, granting this to be true, it would help us in the distinction of the species of things no further than the tribes of animals and vegetables. What must we do for the rest? But in those too it is not sufficient: for if history lie not, women have conceived by drills; and what real species, by that measure, such a production will be in nature will be a new question: and we have reason to think this is not impossible, since mules and jumarts, the one from the mixture of an ass and a mare, the other from the mixture of a bull and a mare, are so frequent in the world. I once saw a creature that was the issue of a cat and a rat, and had the plain marks of both about it; wherein nature appeared to have followed the pattern of neither sort alone, but to have

jumbled them both together. To which he that shall add the monstrous productions that are so frequently to be met with in nature, will find it hard, even in the race of animals, to determine by the pedigree of what species every animal's issue is; and be at a loss about the real essence, which he thinks certainly conveyed by generation, and has alone a right to the specific name. But further, if the species of animals and plants are to be distinguished only by propagation, must I go to the Indies to see the sire and dam of the one, and the plant from which the seed was gathered that produced the other, to know whether this be a tiger or that tea?

24. Not by substantial forms. Upon the whole matter, it is evident that it is their own collections of sensible qualities that men make the essences of their several sorts of substances; and that their real internal structures are not considered by the greatest part of men in the sorting them. Much less were any substantial forms ever thought on by any but those who have in this one part of the world learned the language of the schools: and yet those ignorant men, who pretend not any in-

sight into the real essences, nor trouble themselves about substantial forms, but are content with knowing things one from another by their sensible qualities, are often better acquainted with their differences; can more nicely distinguish them from their uses; and better know what they expect from each, than those learned quick-sighted men, who look so deep into them, and talk so confidently of something more hidden and essential.

25. The specific essences that are commonly made by men. But supposing that the real essences of substances were discoverable by those that would severely apply themselves to that inquiry, yet we could not reasonably think that the ranking of things under general names was regulated by those internal real constitutions, or anything else but their obvious appearances; since languages, in all countries, have been established long before sciences. So that they have not been philosophers or logicians, or such who have troubled themselves about forms and essences, that have made the general names that are in use amongst the several nations of men: but those more or less comprehensive terms have, for the

most part, in all languages, received their birth and signification from ignorant and illiterate people, who sorted and denominated things by those sensible qualities they found in them; thereby to signify them, when absent, to others, whether they had an occasion to mention a sort or a particular thing.

26. Therefore very various and uncertain in the ideas of different men. Since then it is evident that we sort and name substances by their nominal and not by their real essences, the next thing to be considered is how, and by whom these essences come to be made. As to the latter, it is evident they are made by the mind, and not by nature: for were they Nature's workmanship, they could not be so various and different in several men as experience tells us they are. For if we will examine it, we shall not find the nominal essence of any one species of substances in all men the same: no, not of that which of all others we are the most intimately acquainted with. It could not possibly be that the abstract idea to which the name man is given should be different in several men, if it were of Nature's making; and that to one it

should be animal rationale, and to another, animal implume bipes latis unguibus. He that annexes the name to a complex idea, made up of sense and spontaneous motion, joined to a body of such a shape, has thereby one essence of the species man; and he that, upon further examination, adds rationality, has another essence of the species he calls man: by which means the same individual will be a true man to the one which is not so to the other. I think there is scarce any one will allow this upright figure, so well known, to be the essential difference of the species man; and yet how far men determine of the sorts of animals rather by their shape than descent, is very visible; since it has been more than once debated, whether several human foetuses should be preserved or received to baptism or no, only because of the difference of their outward configuration from the ordinary make of children, without knowing whether they were not as capable of reason as infants cast in another mould: some whereof, though of an approved shape, are never capable of as much appearance of reason all their lives as is to be found in an

ape, or an elephant, and never give any signs of being acted by a rational soul. Whereby it is evident, that the outward figure, which only was found wanting, and not the faculty of reason, which nobody could know would be wanting in its due season, was made essential to the human species. The learned divine and lawyer must, on such occasions, renounce his sacred definition of animal rationale, and substitute some other essence of the human species. Monsieur Menage furnishes us with an example worth the taking notice of on this occasion: "When the abbot of Saint Martin," says he, "was born, he had so little of the figure of a man, that it bespoke him rather a monster. It was for some time under deliberation, whether he should be baptized or no. However, he was baptized, and declared a man provisionally till time should show what he would prove. Nature had moulded him so untowardly, that he was called all his life the Abbot Malotru; i.e. ill-shaped. He was of Caen." (Menagiana, 278, 430.) This child, we see, was very near being excluded out of the species of man, barely by his shape. He escaped very narrowly as he was; and it is

certain, a figure a little more oddly turned had cast him, and he had been executed, as a thing not to be allowed to pass for a man. And yet there can be no reason given why, if the lineaments of his face had been a little altered, a rational soul could not have been lodged in him; why a visage somewhat longer, or a nose flatter, or a wider mouth, could not have consisted, as well as the rest of his ill figure, with such a soul, such parts, as made him, disfigured as he was, capable to be a dignitary in the church.

27. Nominal essences of particular substances are undetermined by nature, and therefore various as men vary. Wherein, then, would I gladly know, consist the precise and unmovable boundaries of that species? It is plain, if we examine, there is no such thing made by Nature, and established by her amongst men. The real essence of that or any other sort of substances, it is evident, we know not; and therefore are so undetermined in our nominal essences, which we make ourselves, that, if several men were to be asked concerning some oddly shaped foetus, as soon as born, whether it were a man or no, it

is past doubt one should meet with different answers. Which could not happen, if the nominal essences, whereby we limit and distinguish the species of substances, were not made by man with some liberty; but were exactly copied from precise boundaries set by nature, whereby it distinguished all substances into certain species. Who would undertake to resolve what species that monster was of which is mentioned by Licetus (Bk. i. c. 3), with a man's head and hog's body? Or those other which to the bodies of men had the heads of beasts, as dogs, horses, &c. If any of these creatures had lived, and could have spoke, it would have increased the difficulty. Had the upper part to the middle been of human shape, and all below swine, had it been murder to destroy it? Or must the bishop have been consulted, whether it were man enough to be admitted to the font or no? As I have been told it happened in France some years since, in somewhat a like case. So uncertain are the boundaries of species of animals to us, who have no other measures than the complex ideas of our own collecting: and so far are we from certainly knowing what

a man is; though perhaps it will be judged great ignorance to make any doubt about it. And yet I think I may say, that the certain boundaries of that species are so far from being determined, and the precise number of simple ideas which make the nominal essence so far from being settled and perfectly known, that very material doubts may still arise about it. And I imagine none of the definitions of the word man which we yet have, nor descriptions of that sort of animal, are so perfect and exact as to satisfy a considerate inquisitive person; much less to obtain a general consent, and to be that which men would everywhere stick by, in the decision of cases, and determining of life and death, baptism or no baptism, in productions that might happen.

28. But not so arbitrary as mixed modes. But though these nominal essences of substances are made by the mind, they are not yet made so arbitrarily as those of mixed modes. To the making of any nominal essence, it is necessary, First, that the ideas whereof it consists have such a union as to make but one idea, how compounded soever. Secondly, that the particular ideas so

united be exactly the same, neither more nor less. For if two abstract complex ideas differ either in number or sorts of their component parts, they make two different, and not one and the same essence. In the first of these, the mind, in making its complex ideas of substances, only follows nature; and puts none together which are not supposed to have a union in nature. Nobody joins the voice of a sheep with the shape of a horse; nor the colour of lead with the weight and fixedness of gold, to be the complex ideas of any real substances; unless he has a mind to fill his head with chimeras, and his discourse with unintelligible words. Men observing certain qualities always joined and existing together, therein copied nature; and of ideas so united made their complex ones of substances. For, though men may make what complex ideas they please, and give what names to them they will; yet, if they will be understood when they speak of things really existing, they must in some degree conform their ideas to the things they would speak of; or else men's language will be like that of Babel; and every man's words, being intelligible only to

himself, would no longer serve to conversation and the ordinary affairs of life, if the ideas they stand for be not some way answering the common appearances and agreement of substances as they really exist.

29. Our nominal essences of substances usually consist of a few obvious qualities observed in things. Secondly, Though the mind of man, in making its complex ideas of substances, never puts any together that do not really, or are not supposed to, co-exist; and so it truly borrows that union from nature: yet the number it combines depends upon the various care, industry, or fancy of him that makes it. Men generally content themselves with some few sensible obvious qualities; and often, if not always, leave out others as material and as firmly united as those that they take. Of sensible substances there are two sorts: one of organized bodies, which are propagated by seed; and in these the shape is that which to us is the leading quality, and most characteristical part, that determines the species. And therefore in vegetables and animals, an extended solid substance of such a certain figure usually serves the turn. For however

some men seem to prize their definition of animal rationale, yet should there a creature be found that had language and reason, but partaked not of the usual shape of a man, I believe it would hardly pass for a man, how much soever it were animal rationale. And if Balaam's ass had all his life discoursed as rationally as he did once with his master, I doubt yet whether any one would have thought him worthy the name man, or allowed him to be of the same species with himself. As in vegetables and animals it is the shape, so in most other bodies, not propagated by seed, it is the colour we must fix on, and are most led by. Thus where we find the colour of gold, we are apt to imagine all the other qualities comprehended in our complex idea to be there also: and we commonly take these two obvious qualities, viz. shape and colour, for so presumptive ideas of several species, that in a good picture, we readily say, this is a lion, and that a rose; this is a gold, and that a silver goblet, only by the different figures and colours represented to the eye by the pencil.

30. Yet, imperfect as they thus are, they serve for com-

mon converse. But though this serves well enough for gross and confused conceptions, and inaccurate ways of talking and thinking; yet men are far enough from having agreed on the precise number of simple ideas or qualities belonging to any sort of things, signified by its name. Nor is it a wonder; since it requires much time, pains, and skill, strict inquiry, and long examination to find out what, and how many, those simple ideas are, which are constantly and inseparably united in nature, and are always to be found together in the same subject. Most men, wanting either time, inclination, or industry enough for this, even to some tolerable degree, content themselves with some few obvious and outward appearances of things, thereby readily to distinguish and sort them for the common affairs of life: and so, without further examination, give them names, or take up the names already in use. Which, though in common conversation they pass well enough for the signs of some few obvious qualities co-existing, are yet far enough from comprehending, in a settled signification, a precise number of simple ideas, much less all those which

are united in nature. He that shall consider, after so much stir about genus and species, and such a deal of talk of specific differences, how few words we have yet settled definitions of, may with reason imagine, that those forms which there hath been so much noise made about are only chimeras, which give us no light into the specific natures of things. And he that shall consider how far the names of substances are from having significations wherein all who use them do agree, will have reason to conclude that, though the nominal essences of substances are all supposed to be copied from nature, yet they are all, or most of them, very imperfect. Since the composition of those complex ideas are, in several men, very different: and therefore that these boundaries of species are as men, and not as Nature, makes them, if at least there are in nature any such prefixed bounds. It is true that many particular substances are so made by Nature, that they have agreement and likeness one with another, and so afford a foundation of being ranked into sorts. But the sorting of things by us, or the making of determinate species, being in order to

naming and comprehending them under general terms, I cannot see how it can be properly said, that Nature sets the boundaries of the species of things: or, if it be so, our boundaries of species are not exactly conformable to those in nature. For we, having need of general names for present use, stay not for a perfect discovery of all those qualities which would best show us their most material differences and agreements; but we ourselves divide them, by certain obvious appearances, into species, that we may the easier under general names communicate our thoughts about them. For, having no other knowledge of any substance but of the simple ideas that are united in it; and observing several particular things to agree with others in several of those simple ideas; we make that collection our specific idea, and give it a general name; that in recording our thoughts, and in our discourse with others, we may in one short word designate all the individuals that agree in that complex idea, without enumerating the simple ideas that make it up; and so not waste our time and breath in tedious descriptions: which we see they are fain to do

who would discourse of any new sort of things they have not yet a name for.

31. Essences of species under the same name very different in different minds. But however these species of substances pass well enough in ordinary conversation, it is plain that this complex idea, wherein they observe several individuals to agree, is by different men made very differently; by some more, and others less accurately. In some, this complex idea contains a greater, and in others a smaller number of qualities; and so is apparently such as the mind makes it. The yellow shining colour makes gold to children; others add weight, malleableness, and fusibility; and others yet other qualities, which they find joined with that yellow colour, as constantly as its weight and fusibility. For in all these and the like qualities, one has as good a right to be put into the complex idea of that substance wherein they are all joined as another. And therefore different men, leaving out or putting in several simple ideas which others do not, according to their various examination, skill, or observation of that subject, have different essences of

gold, which must therefore be of their own and not of nature's making.

32. The more general our ideas of substances are, the more incomplete and partial they are. If the number of simple ideas that make the nominal essence of the lowest species, or first sorting, of individuals, depends on the mind of man, variously collecting them, it is much more evident that they do so in the more comprehensive classes, which, by the masters of logic, are called genera. These are complex ideas designedly imperfect: and it is visible at first sight, that several of those qualities that are to be found in the things themselves are purposely left out of generical ideas. For, as the mind, to make general ideas comprehending several particulars, leaves out those of time and place, and such other, that make them incommunicable to more than one individual; so to make other yet more general ideas, that may comprehend different sorts, it leaves out those qualities that distinguish them, and puts into its new collection only such ideas as are common to several sorts. The same convenience that made men express several

parcels of yellow matter coming from Guinea and Peru under one name, sets them also upon making of one name that may comprehend both gold and silver, and some other bodies of different sorts. This is done by leaving out those qualities, which are peculiar to each sort, and retaining a complex idea made up of those that are common to them all. To which the name metal being annexed, there is a genus constituted; the essence whereof being that abstract idea, containing only malleableness and fusibility, with certain degrees of weight and fixedness, wherein some bodies of several kinds agree, leaves out the colour and other qualities peculiar to gold and silver, and the other sorts comprehended under the name metal. Whereby it is plain that men follow not exactly the patterns set them by nature, when they make their general ideas of substances; since there is no body to be found which has barely malleableness and fusibility in it, without other qualities as inseparable as those. But men, in making their general ideas, seeking more the convenience of language, and quick dispatch by short and comprehensive signs, than

the true and precise nature of things as they exist, have, in the framing their abstract ideas, chiefly pursued that end; which was to be furnished with store of general and variously comprehensive names. So that in this whole business of genera and species, the genus, or more comprehensive, is but a partial conception of what is in the species; and the species but a partial idea of what is to be found in each individual. If therefore any one will think that a man, and a horse, and an animal, and a plant, &c., are distinguished by real essences made by nature, he must think nature to be very liberal of these real essences, making one for body, another for an animal, and another for a horse; and all these essences liberally bestowed upon Bucephalus. But if we would rightly consider what is done in all these genera and species, or sorts, we should find that there is no new thing made; but only more or less comprehensive signs, whereby we may be enabled to express in a few syllables great numbers of particular things, as they agree in more or less general conceptions, which we have framed to that purpose. In all which we may observe, that the

more general term is always the name of a less complex idea; and that each genus is but a partial conception of the species comprehended under it. So that if these abstract general ideas be thought to be complete, it can only be in respect of a certain established relation between them and certain names which are made use of to signify them; and not in respect of anything existing, as made by nature.

33. This all accommodated to the end of speech. This is adjusted to the true end of speech, which is to be the easiest and shortest way of communicating our notions. For thus he that would discourse of things, as they agreed in the complex idea of extension and solidity, needed but use the word body to denote all such. He that to these would join others, signified by the words life, sense, and spontaneous motion, needed but use the word animal to signify all which partaked of those ideas, and he that had made a complex idea of a body, with life, sense, and motion, with the faculty of reasoning, and a certain shape joined to it, needed but use the short monosyllable man, to express all particulars that correspond

to that complex idea. This is the proper business of genus and species: and this men do without any consideration of real essences, or substantial forms; which come not within the reach of our knowledge when we think of those things, nor within the signification of our words when we discourse with others.

34. Instance in Cassowaries. Were I to talk with any one of a sort of birds I lately saw in St. James's Park, about three or four feet high, with a covering of something between feathers and hair, of a dark brown colour, without wings, but in the place thereof two or three little branches coming down like sprigs of Spanish broom, long great legs, with feet only of three claws, and without a tail; I must make this description of it, and so may make others understand me. But when I am told that the name of it is cassuaris, I may then use that word to stand in discourse for all my complex idea mentioned in that description; though by that word, which is now become a specific name, I know no more of the real essence or constitution of that sort of animals than I did before; and knew probably as much of the nature of

that species of birds before I learned the name, as many Englishmen do of swans or herons, which are specific names, very well known, of sorts of birds common in England.

35. Men determine the sorts of substances, which may be sorted variously. From what has been said, it is evident that men make sorts of things. For, it being different essences alone that make different species, it is plain that they who make those abstract ideas which are the nominal essences do thereby make the species, or sort. Should there be a body found, having all the other qualities of gold except malleableness, it would no doubt be made a question whether it were gold or not, i.e. whether it were of that species. This could be determined only by that abstract idea to which every one annexed the name gold: so that it would be true gold to him, and belong to that species, who included not malleableness in his nominal essence, signified by the sound gold; and on the other side it would not be true gold, or of that species, to him who included malleableness in his specific idea. And who, I pray, is it that makes these diverse

species, even under one and the same name, but men that make two different abstract ideas, consisting not exactly of the same collection of qualities? Nor is it a mere supposition to imagine that a body may exist wherein the other obvious qualities of gold may be without malleableness; since it is certain that gold itself will be sometimes so eager, (as artists call it), that it will as little endure the hammer as glass itself. What we have said of the putting in, or leaving out of malleableness, in the complex idea the name gold is by any one annexed to, may be said of its peculiar weight, fixedness, and several other the like qualities: for whatever is left out, or put in, it is still the complex idea to which that name is annexed that makes the species: and as any particular parcel of matter answers that idea, so the name of the sort belongs truly to it; and it is of that species. And thus anything is true gold, perfect metal. All which determination of the species, it is plain, depends on the understanding of man, making this or that complex idea.

36. Nature makes the similitudes of substances. This, then, in short, is the case: Nature makes many particu-

lar things, which do agree one with another in many sensible qualities, and probably too in their internal frame and constitution: but it is not this real essence that distinguishes them into species; it is men who, taking occasion from the qualities they find united in them, and wherein they observe often several individuals to agree, range them into sorts, in order to their naming, for the convenience of comprehensive signs; under which individuals, according to their conformity to this or that abstract idea, come to be ranked as under ensigns: so that this is of the blue, that the red regiment; this is a man, that a drill: and in this, I think, consists the whole business of genus and species.

37. The manner of sorting particular beings the work of fallible men, though nature makes things alike. I do not deny but nature, in the constant production of particular beings, makes them not always new and various, but very much alike and of kin one to another: but I think it nevertheless true, that the boundaries of the species, whereby men sort them, are made by men; since the essences of the species, distinguished by different

names, are, as has been proved, of man's making, and seldom adequate to the internal nature of the things they are taken from. So that we may truly say, such a manner of sorting of things is the workmanship of men.

38. Each abstract idea, with a name to it, makes a nominal essence. One thing I doubt not but will seem very strange in this doctrine, which is, that from what has been said it will follow, that each abstract idea, with a name to it, makes a distinct species. But who can help it, if truth will have it so? For so it must remain till somebody can show us the species of things limited and distinguished by something else; and let us see that general terms signify not our abstract ideas, but something different from them. I would fain know why a shock and a hound are not as distinct species as a spaniel and an elephant. We have no other idea of the different essence of an elephant and a spaniel, than we have of the different essence of a shock and a hound; all the essential difference, whereby we know and distinguish them one from another, consisting only in the different collection of simple ideas, to which we have given those different names.

39. How genera and species are related to naming. How much the making of species and genera is in order to general names; and how much general names are necessary, if not to the being, yet at least to the completing of a species, and making it pass for such, will appear, besides what has been said above concerning ice and water, in a very familiar example. A silent and a striking watch are but one species to those who have but one name for them: but he that has the name watch for one, and clock for the other, and distinct complex ideas to which those names belong, to him they are different species. It will be said perhaps, that the inward contrivance and constitution is different between these two, which the watchmaker has a clear idea of. And yet it is plain they are but one species to him, when he has but one name for them. For what is sufficient in the inward contrivance to make a new species? There are some watches that are made with four wheels, others with five; is this a specific difference to the workman? Some have strings and physies, and others none; some have the balance loose, and others regulated by a spiral spring,

and others by hogs' bristles. Are any or all of these enough to make a specific difference to the workman, that knows each of these and several other different contrivances in the internal constitutions of watches? It is certain each of these hath a real difference from the rest; but whether it be an essential, a specific difference or no, relates only to the complex idea to which the name watch is given: as long as they all agree in the idea which that name stands for, and that name does not as a generical name comprehend different species under it, they are not essentially nor specifically different. But if any one will make minuter divisions, from differences that he knows in the internal frame of watches, and to such precise complex ideas give names that shall prevail; they will then be new species, to them who have those ideas with names to them, and can by those differences distinguish watches into these several sorts; and then watch will be a generical name. But yet they would be no distinct species to men ignorant of clock-work, and the inward contrivances of watches, who had no other idea but the outward shape and bulk, with the

marking of the hours by the hand. For to them all those other names would be but synonymous terms for the same idea, and signify no more, nor no other thing but a watch. Just thus I think it is in natural things. Nobody will doubt that the wheels or springs (if I may so say) within, are different in a rational man and a changeling; no more than that there is a difference in the frame between a drill and a changeling. But whether one or both these differences be essential or specific, is only to be known to us by their agreement or disagreement with the complex idea that the name man stands for: for by that alone can it be determined whether one, or both, or neither of those be a man.

40. Species of artificial things less confused than natural. From what has been before said, we may see the reason why, in the species of artificial things, there is generally less confusion and uncertainty than in natural. Because an artificial thing being a production of man, which the artificer designed, and therefore well knows the idea of, the name of it is supposed to stand for no other idea, nor to import any other essence, than

what is certainly to be known, and easy enough to be apprehended. For the idea or essence of the several sorts of artificial things, consisting for the most part in nothing but the determinate figure of sensible parts, and sometimes motion depending thereon, which the artificer fashions in matter, such as he finds for his turn; it is not beyond the reach of our faculties to attain a certain idea thereof; and so settle the signification of the names whereby the species of artificial things are distinguished, with less doubt, obscurity, and equivocation than we can in things natural, whose differences and operations depend upon contrivances beyond the reach of our discoveries.

41. Artificial things of distinct species. I must be excused here if I think artificial things are of distinct species as well as natural: since I find they are as plainly and orderly ranked into sorts, by different abstract ideas, with general names annexed to them, as distinct one from another as those of natural substances. For why should we not think a watch and pistol as distinct species one from another, as a horse and a dog; they being

expressed in our minds by distinct ideas, and to others by distinct appellations?

42. Substances alone, of all our several sorts of ideas, have proper names. This is further to be observed concerning substances, that they alone of all our several sorts of ideas have particular or proper names, whereby one only particular thing is signified. Because in simple ideas, modes, and relations, it seldom happens that men have occasion to mention often this or that particular when it is absent. Besides, the greatest part of mixed modes, being actions which perish in their birth, are not capable of a lasting duration, as substances which are the actors; and wherein the simple ideas that make up the complex ideas designed by the name have a lasting union.

43. Difficult to lead another by words into the thoughts of things stripped of those abstract ideas we give them. I must beg pardon of my reader for having dwelt so long upon this subject, and perhaps with some obscurity. But I desire it may be considered, how difficult it is to lead another by words into the thoughts of things, stripped of those specific differences we give them:

which things, if I name not, I say nothing; and if I do name them, I thereby rank them into some sort or other, and suggest to the mind the usual abstract idea of that species; and so cross my purpose. For, to talk of a man, and to lay by, at the same time, the ordinary signification of the name man, which is our complex idea usually annexed to it; and bid the reader consider man, as he is in himself, and as he is really distinguished from others in his internal constitution, or real essence, that is, by something he knows not what, looks like trifling: and yet thus one must do who would speak of the supposed real essences and species of things, as thought to be made by nature, if it be but only to make it understood, that there is no such thing signified by the general names which substances are called by. But because it is difficult by known familiar names to do this, give me leave to endeavour by an example to make the different consideration the mind has of specific names and ideas a little more clear; and to show how the complex ideas of modes are referred sometimes to archetypes in the minds of other intelligent beings, or, which is the

same, to the signification annexed by others to their received names; and sometimes to no archetypes at all. Give me leave also to show how the mind always refers its ideas of substances, either to the substances themselves, or to the signification of their names, as to the archetypes; and also to make plain the nature of species or sorting of things, as apprehended and made use of by us; and of the essences belonging to those species: which is perhaps of more moment to discover the extent and certainty of our knowledge than we at first imagine.

44. Instances of mixed modes named kinneah and niouph. Let us suppose Adam, in the state of a grown man, with a good understanding, but in a strange country, with all things new and unknown about him; and no other faculties to attain the knowledge of them but what one of this age has now. He observes Lamech more melancholy than usual, and imagines it to be from a suspicion he has of his wife Adah, (whom he most ardently loved) that she had too much kindness for another man. Adam discourses these his thoughts to Eve, and desires her to take care that Adah commit not folly:

and in these discourses with Eve he makes use of these two new words kinneah and niouph. In time, Adam's mistake appears, for he finds Lamech's trouble proceeded from having killed a man: but yet the two names kinneah and niouph, (the one standing for suspicion in a husband of his wife's disloyalty to him; and the other for the act of committing disloyalty), lost not their distinct significations. It is plain then, that here were two distinct complex ideas of mixed modes, with names to them, two distinct species of actions essentially different; I ask wherein consisted the essences of these two distinct species of actions? And it is plain it consisted in a precise combination of simple ideas, different in one from the other. I ask, whether the complex idea in Adam's mind, which he called kinneah, were adequate or not? And it is plain it was; for it being a combination of simple ideas, which he, without any regard to any archetype, without respect to anything as a pattern, voluntarily put together, abstracted, and gave the name kinneah to, to express in short to others, by that one sound, all the simple ideas contained and united in that

complex one; it must necessarily follow that it was an adequate idea. His own choice having made that combination, it had all in it he intended it should, and so could not but be perfect, could not but be adequate; it being referred to no other archetype which it was supposed to represent.

45. These words, kinneah and niouph, by degrees grew into common use, and then the case was somewhat altered. Adam's children had the same faculties, and thereby the same power that he had, to make what complex ideas of mixed modes they pleased in their own minds; to abstract them, and make what sounds they pleased the signs of them: but the use of names being to make our ideas within us known to others, that cannot be done, but when the same sign stands for the same idea in two who would communicate their thoughts and discourse together. Those, therefore, of Adam's children, that found these two words, kinneah and niouph, in familiar use, could not take them for insignificant sounds, but must needs conclude they stood for something; for certain ideas, abstract ideas. they being general names;

which abstract ideas were the essences of the species distinguished by those names. If, therefore, they would use these words as names of species already established and agreed on, they were obliged to conform the ideas in their minds, signified by these names, to the ideas that they stood for in other men's minds, as to their patterns and archetypes; and then indeed their ideas of these complex modes were liable to be inadequate, as being very apt (especially those that consisted of combinations of many simple ideas) not to be exactly conformable to the ideas in other men's minds, using the same names; though for this there be usually a remedy at hand, which is to ask the meaning of any word we understand not of him that uses it: it being as impossible to know certainly what the words jealousy and adultery stand for in another man's mind, with whom I would discourse about them; as it was impossible, in the beginning of language, to know what kinneah and niouph stood for in another man's mind, without explication; they being voluntary signs in every one.

46. Instances of a species of substance named Zahab.

Let us now also consider, after the same manner, the names of substances in their first application. One of Adam's children, roving in the mountains, lights on a glittering substance which pleases his eye. Home he carries it to Adam, who, upon consideration of it, finds it to be hard, to have a bright yellow colour, and an exceeding great weight. These perhaps, at first, are all the qualities he takes notice of in it; and abstracting this complex idea, consisting of a substance having that peculiar bright yellowness, and a weight very great in proportion to its bulk, he gives the name zahab, to denominate and mark all substances that have these sensible qualities in them. It is evident now, that, in this case, Adam acts quite differently from what he did before, in forming those ideas of mixed modes to which he gave the names kinneah and niouph. For there he put ideas together only by his own imagination, not taken from the existence of anything; and to them he gave names to denominate all things that should happen to agree to those his abstract ideas, without considering whether any such thing did exist or not; the standard

there was of his own making. But in the forming his idea of this new substance, he takes the quite contrary course; here he has a standard made by nature; and therefore, being to represent that to himself, by the idea he has of it, even when it is absent, he puts in no simple idea into his complex one, but what he has the perception of from the thing itself. He takes care that his idea be conformable to this archetype, and intends the name should stand for an idea so conformable.

47. This piece of matter, thus denominated zahab by Adam, being quite different from any he had seen before, nobody, I think, will deny to be a distinct species, and to have its peculiar essence: and that the name zahab is the mark of the species, and a name belonging to all things partaking in that essence. But here it is plain the essence Adam made the name zahab stand for was nothing but a body hard, shining, yellow, and very heavy. But the inquisitive mind of man, not content with the knowledge of these, as I may say, superficial qualities, puts Adam upon further examination of this matter. He therefore knocks, and beats it with flints, to

see what was discoverable in the inside: he finds it yield to blows, but not easily separate into pieces: he finds it will bend without breaking. Is not now ductility to be added to his former idea, and made part of the essence of the species that name zahab stands for? Further trials discover fusibility and fixedness. Are not they also, by the same reason that any of the others were, to be put into the complex idea signified by the name zahab? If not, what reason will there be shown more for the one than the other? If these must, then all the other properties, which any further trials shall discover in this matter, ought by the same reason to make a part of the ingredients of the complex idea which the name zahab stands for, and so be the essence of the species marked by that name. Which properties, because they are endless, it is plain that the idea made after this fashion, by this archetype, will be always inadequate.

48. The abstract ideas of substances always imperfect, and therefore various. But this is not all. It would also follow that the names of substances would not only have, as in truth they have, but would also be supposed

to have different significations, as used by different men, which would very much cumber the use of language. For if every distinct quality that were discovered in any matter by any one were supposed to make a necessary part of the complex idea signified by the common name given to it, it must follow, that men must suppose the same word to signify different things in different men: since they cannot doubt but different men may have discovered several qualities, in substances of the same denomination, which others know nothing of.

49. Therefore to fix their nominal species, a real essence is supposed. To avoid this therefore, they have supposed a real essence belonging to every species, from which these properties all flow, and would have their name of the species stand for that. But they, not having any idea of that real essence in substances, and their words signifying nothing but the ideas they have, that which is done by this attempt is only to put the name or sound in the place and stead of the thing having that real essence, without knowing what the real essence is, and this is that which men do when they speak of species of

things, as supposing them made by nature, and distinguished by real essences.

50. Which supposition is of no use. For, let us consider, when we affirm that “all gold is fixed,” either it means that fixedness is a part of the definition, i.e., part of the nominal essence the word gold stands for; and so this affirmation, “all gold is fixed,” contains nothing but the signification of the term gold. Or else it means, that fixedness, not being a part of the definition of the gold, is a property of that substance itself: in which case it is plain that the word gold stands in the place of a substance, having the real essence of a species of things made by nature. In which way of substitution it has so confused and uncertain a signification, that, though this proposition—“gold is fixed”—be in that sense an affirmation of something real; yet it is a truth will always fail us in its particular application, and so is of no real use or certainty. For let it be ever so true, that all gold, i.e. all that has the real essence of gold, is fixed, what serves this for, whilst we know not, in this sense, what is or is not gold? For if we know not the real

essence of gold, it is impossible we should know what parcel of matter has that essence, and so whether it be true gold or no.

51. Conclusion. To conclude: what liberty Adam had at first to make any complex ideas of mixed modes by no other pattern but by his own thoughts, the same have all men ever since had. And the same necessity of conforming his ideas of substances to things without him, as to archetypes made by nature, that Adam was under, if he would not wilfully impose upon himself, the same are all men ever since under too. The same liberty also that Adam had of affixing any new name to any idea, the same has any one still, (especially the beginners of languages, if we can imagine any such); but only with this difference, that, in places where men in society have already established a language amongst them, the significations of words are very warily and sparingly to be altered. Because men being furnished already with names for their ideas, and common use having appropriated known names to certain ideas, an affected misapplication of them cannot but be very ridiculous. He

that hath new notions will perhaps venture sometimes on the coining of new terms to express them: but men think it a boldness, and it is uncertain whether common use will ever make them pass for current. But in communication with others, it is necessary that we conform the ideas we make the vulgar words of any language stand for to their known proper significations, (which I have explained at large already), or else to make known that new signification we apply them to.

Chapter VII Of Particles

1. Particles connect parts, or whole sentences together. Besides words which are names of ideas in the mind, there are a great many others that are made use of to signify the connexion that the mind gives to ideas, or to propositions, one with another. The mind, in communicating its thoughts to others, does not only need signs of the ideas it has then before it, but others also, to show or intimate some particular action of its own,

at that time, relating to those ideas. This it does several ways; as Is, and Is not, are the general marks, of the mind, affirming or denying. But besides affirmation or negation, without which there is in words no truth or falsehood, the mind does, in declaring its sentiments to others, connect not only the parts of propositions, but whole sentences one to another, with their several relations and dependencies, to make a coherent discourse.

2. In right use of particles consists the art of well-speaking. The words whereby it signifies what connexion it gives to the several affirmations and negations, that it unites in one continued reasoning or narration, are generally called particles: and it is in the right use of these that more particularly consists the clearness and beauty of a good style. To think well, it is not enough that a man has ideas clear and distinct in his thoughts, nor that he observes the agreement or disagreement of some of them; but he must think in train, and observe the dependence of his thoughts and reasonings upon one another. And to express well such methodical and rational thoughts, he must have words to show what

connexion, restriction, distinction, opposition, emphasis &c., he gives to each respective part of his discourse. To mistake in any of these, is to puzzle instead of informing his hearer: and therefore it is, that those words which are not truly by themselves the names of any ideas are of such constant and indispensable use in language, and do much contribute to men's well expressing themselves.

3. They show what relation the mind gives to its own thoughts. This part of grammar has been perhaps as much neglected as some others over-diligently cultivated. It is easy for men to write, one after another, of cases and genders, moods and tenses, gerunds and supines: in these and the like there has been great diligence used; and particles themselves, in some languages, have been, with great show of exactness, ranked into their several orders. But though prepositions and conjunctions, &c., are names well known in grammar, and the particles contained under them carefully ranked into their distinct subdivisions; yet he who would show the right use of particles, and what significancy and force they have,

must take a little more pains, enter into his own thoughts, and observe nicely the several postures of his mind in discoursing.

4. They are all marks of some action or intimation of the mind. Neither is it enough, for the explaining of these words, to render them, as is usual in dictionaries, by words of another tongue which come nearest to their signification: for what is meant by them is commonly as hard to be understood in one as another language. They are all marks of some action or intimation of the mind; and therefore to understand them rightly, the several views, postures, stands, turns, limitations, and exceptions, and several other thoughts of the mind, for which we have either none or very deficient names, are diligently to be studied. Of these there is a great variety, much exceeding the number of particles that most languages have to express them by: and therefore it is not to be wondered that most of these particles have divers and sometimes almost opposite significations. In the Hebrew tongue there is a particle consisting of but one single letter, of which there are reckoned up, as I re-

member, seventy, I am sure above fifty, several significations.

5. Instance in “but.” “But” is a particle, none more familiar in our language: and he that says it is a discrete conjunction, and that it answers to *sed* Latin, or *mais* in French, thinks he has sufficiently explained it. But yet it seems to me to intimate several relations the mind gives to the several propositions or parts of them which it joins by this monosyllable.

First, “But to say no more”: here it intimates a stop of the mind in the course it was going, before it came quite to the end of it.

Secondly, “I saw but two plants”; here it shows that the mind limits the sense to what is expressed, with a negation of all other.

Thirdly, “You pray; but it is not that God would bring you to the true religion.”

Fourthly, “But that he would confirm you in your own.” The first of these buts intimates a supposition in the mind of something otherwise than it should be: the latter shows that the mind makes a direct opposition

between that and what goes before it.

Fifthly, "All animals have sense, but a dog is an animal": here it signifies little more but that the latter proposition is joined to the former, as the minor of a syllogism.

6. This matter of the use of particles but lightly touched here. To these, I doubt not, might be added a great many other significations of this particle, if it were my business to examine it in its full latitude, and consider it in all the places it is to be found: which if one should do, I doubt whether in all those manners it is made use of, it would deserve the title of discrete, which grammarians give to it. But I intend not here a full explication of this sort of signs. The instances I have given in this one may give occasion to reflect on their use and force in language, and lead us into the contemplation of several actions of our minds in discoursing, which it has found a way to intimate to others by these particles, some whereof constantly, and others in certain constructions, have the sense of a whole sentence contained in them.

Chapter VIII

Of Abstract and Concrete Terms

1. Abstract terms not predictable one of another, and why. The ordinary words of language, and our common use of them, would have given us light into the nature of our ideas, if they had been but considered with attention. The mind, as has been shown, has a power to abstract its ideas, and so they become essences, general essences, whereby the sorts of things are distinguished. Now each abstract idea being distinct, so that of any two the one can never be the other, the mind will, by its intuitive knowledge, perceive their difference, and therefore in propositions no two whole ideas can ever be affirmed one of another. This we see in the common use of language, which permits not any two abstract words, or names of abstract ideas, to be affirmed one of another. For how near of kin soever they may seem to be, and how certain soever it is that man is an animal, or rational, or white, yet every one at first hearing perceives the falsehood of these propositions: humanity is

animality, or rationality, or whiteness: and this is as evident as any of the most allowed maxims. All our affirmations then are only in concrete, which is the affirming, not one abstract idea to be another, but one abstract idea to be joined to another; which abstract ideas, in substances, may be of any sort; in all the rest are little else but of relations; and in substances the most frequent are of powers: v.g. "a man is white," signifies that the thing that has the essence of a man has also in it the essence of whiteness, which is nothing but a power to produce the idea of whiteness in one whose eyes can discover ordinary objects: or, "a man is rational," signifies that the same thing that hath the essence of a man hath also in it the essence of rationality, i.e. a power of reasoning.

2. They show the difference of our ideas. This distinction of names shows us also the difference of our ideas: for if we observe them, we shall find that our simple ideas have all abstract as well as concrete names: the one whereof is (to speak the language of grammarians) a substantive, the other an adjective; as whiteness, white;

sweetness, sweet. The like also holds in our ideas of modes and relations; as justice, just; equality, equal: only with this difference, that some of the concrete names of relations amongst men chiefly are substantives; as, paternitas, pater; whereof it were easy to render a reason. But as to our ideas of substances, we have very few or no abstract names at all. For though the Schools have introduced animalitas, humanitas, corporietas, and some others; yet they hold no proportion with that infinite number of names of substances, to which they never were ridiculous enough to attempt the coining of abstract ones: and those few that the schools forged, and put into the mouths of their scholars, could never yet get admittance into common use, or obtain the license of public approbation. Which seems to me at least to intimate the confession of all mankind, that they have no ideas of the real essences of substances, since they have not names for such ideas: which no doubt they would have had, had not their consciousness to themselves of their ignorance of them kept them from so idle an attempt. And therefore, though they

had ideas enough to distinguish gold from a stone, and metal from wood; yet they but timorously ventured on such terms, as *aurietas* and *saxietas*, *metallietas* and *lignietas*, or the like names, which should pretend to signify the real essences of those substances whereof they knew they had no ideas. And indeed it was only the doctrine of substantial forms, and the confidence of mistaken pretenders to a knowledge that they had not, which first coined and then introduced *animalitas* and *humanitas*, and the like; which yet went very little further than their own Schools, and could never get to be current amongst understanding men. Indeed, *humanitas* was a word in familiar use amongst the Romans; but in a far different sense, and stood not for the abstract essence of any substance; but was the abstracted name of a mode, and its concrete *humanus*, not *homo*.

Chapter IX

Of the Imperfection of Words

1. Words are used for recording and communicating our thoughts. From what has been said in the foregoing chapters, it is easy to perceive what imperfection there is in language, and how the very nature of words makes it almost unavoidable for many of them to be doubtful and uncertain in their significations. To examine the perfection or imperfection of words, it is necessary first to consider their use and end: for as they are more or less fitted to attain that, so they are more or less perfect. We have, in the former part of this discourse often, upon occasion, mentioned a double use of words.

First, One for the recording of our own thoughts.

Secondly, The other for the communicating of our thoughts to others.

2. Any words will serve for recording. As to the first of these, for the recording our own thoughts for the help of our own memories, whereby, as it were, we talk to ourselves, any words will serve the turn. For since sounds

are voluntary and indifferent signs of any ideas, a man may use what words he pleases to signify his own ideas to himself: and there will be no imperfection in them, if he constantly use the same sign for the same idea: for then he cannot fail of having his meaning understood, wherein consists the right use and perfection of language.

3. Communication by words either for civil or philosophical purposes. Secondly, As to communication by words, that too has a double use.

I. Civil.

II. Philosophical.

First, by their civil use, I mean such a communication of thoughts and ideas by words, as may serve for the upholding common conversation and commerce, about the ordinary affairs and conveniences of civil life, in the societies of men, one amongst another.

Secondly, By the philosophical use of words, I mean such a use of them as may serve to convey the precise notions of things, and to express in general propositions certain and undoubted truths, which the mind may rest upon and be satisfied with in its search after true knowl-

edge. These two uses are very distinct; and a great deal less exactness will serve in the one than in the other, as we shall see in what follows.

4. The imperfection of words is the doubtfulness or ambiguity of their signification, which is caused by the sort of ideas they stand for. The chief end of language in communication being to be understood, words serve not well for that end, neither in civil nor philosophical discourse, when any word does not excite in the hearer the same idea which it stands for in the mind of the speaker. Now, since sounds have no natural connexion with our ideas, but have all their signification from the arbitrary imposition of men, the doubtfulness and uncertainty of their signification, which is the imperfection we here are speaking of, has its cause more in the ideas they stand for than in any incapacity there is in one sound more than in another to signify any idea: for in that regard they are all equally perfect.

That then which makes doubtfulness and uncertainty in the signification of some more than other words, is the difference of ideas they stand for.

5. Natural causes of their imperfection, especially in those that stand for mixed modes, and for our ideas of substances. Words having naturally no signification, the idea which each stands for must be learned and retained, by those who would exchange thoughts, and hold intelligible discourse with others, in any language. But this is the hardest to be done where,

First, The ideas they stand for are very complex, and made up of a great number of ideas put together.

Secondly, Where the ideas they stand for have no certain connexion in nature; and so no settled standard anywhere in nature existing, to rectify and adjust them by.

Thirdly, When the signification of the word is referred to a standard, which standard is not easy to be known.

Fourthly, Where the signification of the word and the real essence of the thing are not exactly the same.

These are difficulties that attend the signification of several words that are intelligible. Those which are not intelligible at all, such as names standing for any simple ideas which another has not organs or faculties to at-

tain; as the names of colours to a blind man, or sounds to a deaf man, need not here be mentioned.

In all these cases we shall find an imperfection in words; which I shall more at large explain, in their particular application to our several sorts of ideas: for if we examine them, we shall find that the names of Mixed Modes are most liable to doubtfulness and imperfection, for the two first of these reasons; and the names of Substances chiefly for the two latter.

6. The names of mixed modes doubtful. First, because the ideas they stand for are so complex. First, The names of mixed modes are, many of them, liable to great uncertainty and obscurity in their signification

I. Because of that great composition these complex ideas are often made up of. To make words serviceable to the end of communication, it is necessary, as has been said, that they excite in the hearer exactly the same idea they stand for in the mind of the speaker. Without this, men fill one another's heads with noise and sounds; but convey not thereby their thoughts, and lay not before one another their ideas, which is the end of discourse

and language. But when a word stands for a very complex idea that is compounded and decompounded, it is not easy for men to form and retain that idea so exactly, as to make the name in common use stand for the same precise idea, without any the least variation. Hence it comes to pass that men's names of very compound ideas, such as for the most part are moral words, have seldom in two different men the same precise signification; since one man's complex idea seldom agrees with another's, and often differs from his own—from that which he had yesterday, or will have to-morrow.

7. Secondly, because they have no standards in nature. Because the names of mixed modes for the most part want standards in nature, whereby men may rectify and adjust their significations, therefore they are very various and doubtful. They are assemblages of ideas put together at the pleasure of the mind, pursuing its own ends of discourse, and suited to its own notions, whereby it designs not to copy anything really existing, but to denominate and rank things as they come to agree with those archetypes or forms it has made. He that first

brought the word sham, or wheedle, or banter, in use, put together as he thought fit those ideas he made it stand for; and as it is with any new names of modes that are now brought into any language, so it was with the old ones when they were first made use of. Names, therefore, that stand for collections of ideas which the mind makes at pleasure must needs be of doubtful signification, when such collections are nowhere to be found constantly united in nature, nor any patterns to be shown whereby men may adjust them. What the word murder, or sacrilege, &c., signifies can never be known from things themselves: there be many of the parts of those complex ideas which are not visible in the action itself; the intention of the mind, or the relation of holy things, which make a part of murder or sacrilege, have no necessary connexion with the outward and visible action of him that commits either: and the pulling the trigger of the gun with which the murder is committed, and is all the action that perhaps is visible, has no natural connexion with those other ideas that make up the complex one named murder. They have their union and

combination only from the understanding which unites them under one name: but, uniting them without any rule or pattern, it cannot be but that the signification of the name that stands for such voluntary collections should be often various in the minds of different men, who have scarce any standing rule to regulate themselves and their notions by, in such arbitrary ideas.

8. Common use, or propriety not a sufficient remedy. It is true, common use, that is, the rule of propriety may be supposed here to afford some aid, to settle the signification of language; and it cannot be denied but that in some measure it does. Common use regulates the meaning of words pretty well for common conversation; but nobody having an authority to establish the precise signification of words, nor determine to what ideas any one shall annex them, common use is not sufficient to adjust them to Philosophical Discourses; there being scarce any name of any very complex idea (to say nothing of others) which, in common use, has not a great latitude, and which, keeping within the bounds of propriety, may not be made the sign of far different ideas.

Besides, the rule and measure of propriety itself being nowhere established, it is often matter of dispute, whether this or that way of using a word be propriety of speech or no. From all which it is evident, that the names of such kind of very complex ideas are naturally liable to this imperfection, to be of doubtful and uncertain signification; and even in men that have a mind to understand one another, do not always stand for the same idea in speaker and hearer. Though the names glory and gratitude be the same in every man's mouth through a whole country, yet the complex collective idea which every one thinks on or intends by that name, is apparently very different in men using the same language.

9. The way of learning these names contributes also to their doubtfulness. The way also wherein the names of mixed modes are ordinarily learned, does not a little contribute to the doubtfulness of their signification. For if we will observe how children learn languages, we shall find that, to make them understand what the names of simple ideas or substances stand for, people ordinarily show them the thing whereof they would have them

have the idea; and then repeat to them the name that stands for it; as white, sweet, milk, sugar, cat, dog. But as for mixed modes, especially the most material of them, moral words, the sounds are usually learned first; and then, to know what complex ideas they stand for, they are either beholden to the explication of others, or (which happens for the most part) are left to their own observation and industry; which being little laid out in the search of the true and precise meaning of names, these moral words are in most men's mouths little more than bare sounds; or when they have any, it is for the most part but a very loose and undetermined, and, consequently, obscure and confused signification. And even those themselves who have with more attention settled their notions, do yet hardly avoid the inconvenience to have them stand for complex ideas different from those which other, even intelligent and studious men, make them the signs of. Where shall one find any, either controversial debate, or familiar discourse, concerning honour, faith, grace, religion, church, &c., wherein it is not easy to observe the different notions men have of

them? Which is nothing but this, that they are not agreed in the signification of those words, nor have in their minds the same complex ideas which they make them stand for, and so all the contests that follow thereupon are only about the meaning of a sound. And hence we see that, in the interpretation of laws, whether divine or human, there is no end; comments beget comments, and explications make new matter for explications; and of limiting, distinguishing, varying the signification of these moral words there is no end. These ideas of men's making are, by men still having the same power, multiplied in infinitum. Many a man who was pretty well satisfied of the meaning of a text of Scripture, or clause in the code, at first reading, has, by consulting commentators, quite lost the sense of it, and by these elucidations given rise or increase to his doubts, and drawn obscurity upon the place. I say not this that I think commentaries needless; but to show how uncertain the names of mixed modes naturally are, even in the mouths of those who had both the intention and the faculty of speaking as clearly as language was ca-

pable to express their thoughts.

10. Hence unavoidable obscurity in ancient authors. What obscurity this has unavoidably brought upon the writings of men who have lived in remote ages, and different countries, it will be needless to take notice. Since the numerous volumes of learned men, employing their thoughts that way, are proofs more than enough, to show what attention, study, sagacity, and reasoning are required to find out the true meaning of ancient authors. But, there being no writings we have any great concernment to be very solicitous about the meaning of, but those that contain either truths we are required to believe, or laws we are to obey, and draw inconveniences on us when we mistake or transgress, we may be less anxious about the sense of other authors; who, writing but their own opinions, we are under no greater necessity to know them, than they to know ours. Our good or evil depending not on their decrees, we may safely be ignorant of their notions: and therefore in the reading of them, if they do not use their words with a due clearness and perspicuity, we may lay them aside,

and without any injury done them, resolve thus with ourselves,

Si non vis intelligi, debes negligi.

11. Names of substances of doubtful signification, because the ideas they stand for relate to the reality of things. If the signification of the names of mixed modes be uncertain, because there be no real standards existing in nature to which those ideas are referred, and by which they may be adjusted, the names of substances are of a doubtful signification, for a contrary reason, viz. because the ideas they stand for are supposed conformable to the reality of things, and are referred to as standards made by Nature. In our ideas of substances we have not the liberty, as in mixed modes, to frame what combinations we think fit, to be the characteristical notes to rank and denominate things by. In these we must follow Nature, suit our complex ideas to real existences, and regulate the signification of their names by the things themselves, if we will have our names to be

signs of them, and stand for them. Here, it is true, we have patterns to follow; but patterns that will make the signification of their names very uncertain: for names must be of a very unsteady and various meaning, if the ideas they stand for be referred to standards without us, that either cannot be known at all, or can be known but imperfectly and uncertainly.

12. Names of substances referred, to real essences that cannot be known. The names of substances have, as has been shown, a double reference in their ordinary use.

First, Sometimes they are made to stand for, and so their signification is supposed to agree to, the real constitution of things, from which all their properties flow, and in which they all centre. But this real constitution, or (as it is apt to be called) essence, being utterly unknown to us, any sound that is put to stand for it must be very uncertain in its application; and it will be impossible to know what things are or ought to be called a horse, or antimony, when those words are put for real essences that we have no ideas of at all. And therefore in this supposition, the names of substances being re-

ferred to standards that cannot be known, their significations can never be adjusted and established by those standards.

13. To co-existing qualities, which are known but imperfectly.

Secondly, The simple ideas that are found to co-exist in substances being that which their names immediately signify, these, as united in the several sorts of things, are the proper standards to which their names are referred, and by which their significations may be best rectified. But neither will these archetypes so well serve to this purpose as to leave these names without very various and uncertain significations. Because these simple ideas that co-exist, and are united in the same subject, being very numerous, and having all an equal right to go into the complex specific idea which the specific name is to stand for, men, though they propose to themselves the very same subject to consider, yet frame very different ideas about it; and so the name they use for it unavoidably comes to have, in several men, very different significations. The simple qualities

which make up the complex ideas, being most of them powers, in relation to changes which they are apt to make in, or receive from other bodies, are almost infinite. He that shall but observe what a great variety of alterations any one of the baser metals is apt to receive, from the different application only of fire; and how much a greater number of changes any of them will receive in the hands of a chymist, by the application of other bodies, will not think it strange that I count the properties of any sort of bodies not easy to be collected, and completely known, by the ways of inquiry which our faculties are capable of. They being therefore at least so many, that no man can know the precise and definite number, they are differently discovered by different men, according to their various skill, attention, and ways of handling; who therefore cannot choose but have different ideas of the same substance, and therefore make the signification of its common name very various and uncertain. For the complex ideas of substances, being made up of such simple ones as are supposed to co-exist in nature, every one has a right to put into his complex

idea those qualities he has found to be united together. For, though in the substance of gold one satisfies himself with colour and weight, yet another thinks solubility in aqua regia as necessary to be joined with that colour in his idea of gold, as any one does its fusibility; solubility in aqua regia being a quality as constantly joined with its colour and weight as fusibility or any other; others put into it ductility or fixedness, &c., as they have been taught by tradition or experience. Who of all these has established the right signification of the word, gold? Or who shall be the judge to determine? Each has his standard in nature, which he appeals to, and with reason thinks he has the same right to put into his complex idea signified by the word gold, those qualities, which, upon trial, he has found united; as another who has not so well examined has to leave them out; or a third, who has made other trials, has to put in others. For the union in nature of these qualities being the true ground of their union in one complex idea, who can say one of them has more reason to be put in or left out than another? From hence it will unavoid-

ably follow, that the complex ideas of substances in men using the same names for them, will be very various, and so the significations of those names very uncertain.

14. Thirdly, to co-existing qualities which are known but imperfectly. Besides, there is scarce any particular thing existing, which, in some of its simple ideas, does not communicate with a greater, and in others a less number of particular beings: who shall determine in this case which are those that are to make up the precise collection that is to be signified by the specific name? or can with any just authority prescribe, which obvious or common qualities are to be left out; or which more secret, or more particular, are to be put into the signification of the name of any substance? All which together, seldom or never fall to produce that various and doubtful signification in the names of substances, which causes such uncertainty, disputes, or mistakes, when we come to a philosophical use of them.

15. With this imperfection, they may serve for civil, but not well for philosophical use. It is true, as to civil and common conversation, the general names of sub-

stances, regulated in their ordinary signification by some obvious qualities, (as by the shape and figure in things of known seminal propagation, and in other substances, for the most part by colour, joined with some other sensible qualities), do well enough to design the things men would be understood to speak of: and so they usually conceive well enough the substances meant by the word gold or apple, to distinguish the one from the other. But in philosophical inquiries and debates, where general truths are to be established, and consequences drawn from positions laid down, there the precise signification of the names of substances will be found not only not to be well established, but also very hard to be so. For example: he that shall make malleability, or a certain degree of fixedness, a part of his complex idea of gold, may make propositions concerning gold, and draw consequences from them, that will truly and clearly follow from gold, taken in such a signification: but yet such as another man can never be forced to admit, nor be convinced of their truth, who makes not malleableness, or the same degree of fixedness, part of that com-

plex idea that the name gold, in his use of it, stands for.

16. Instance, liquor. This is a natural and almost unavoidable imperfection in almost all the names of substances, in all languages whatsoever, which men will easily find when, once passing from confused or loose notions, they come to more strict and close inquiries. For then they will be convinced how doubtful and obscure those words are in their signification, which in ordinary use appeared very clear and determined. I was once in a meeting of very learned and ingenious physicians, where by chance there arose a question, whether any liquor passed through the filaments of the nerves. The debate having been managed a good while, by variety of arguments on both sides, I (who had been used to suspect, that the greatest part of disputes were more about the signification of words than a real difference in the conception of things) desired, that, before they went any further on in this dispute, they would first examine and establish amongst them, what the word liquor signified. They at first were a little surprised at the proposal; and had they been persons less ingenious,

they might perhaps have taken it for a very frivolous or extravagant one: since there was no one there that thought not himself to understand very perfectly what the word liquor stood for; which I think, too, none of the most perplexed names of substances. However, they were pleased to comply with my motion; and upon examination found that the signification of that word was not so settled or certain as they had all imagined; but that each of them made it a sign of a different complex idea. This made them perceive that the main of their dispute was about the signification of that term; and that they differed very little in their opinions concerning some fluid and subtle matter, passing through the conduits of the nerves; though it was not so easy to agree whether it was to be called liquor or no, a thing, which, when considered, they thought it not worth the contending about.

17. Instance, gold. How much this is the case in the greatest part of disputes that men are engaged so hotly in, I shall perhaps have an occasion in another place to take notice. Let us only here consider a little more ex-

actly the forementioned instance of the word gold, and we shall see how hard it is precisely to determine its signification. I think all agree to make it stand for a body of a certain yellow shining colour; which being the idea to which children have annexed that name, the shining yellow part of a peacock's tail is properly to them gold. Others finding fusibility joined with that yellow colour in certain parcels of matter, make of that combination a complex idea to which they give the name gold, to denote a sort of substances; and so exclude from being gold all such yellow shining bodies as by fire will be reduced to ashes; and admit to be of that species, or to be comprehended under that name gold, only such substances as, having that shining yellow colour, will by fire be reduced to fusion, and not to ashes. Another, by the same reason, adds the weight, which, being a quality as straightly joined with that colour as its fusibility, he thinks has the same reason to be joined in its idea, and to be signified by its name: and therefore the other made up of body, of such a colour and fusibility, to be imperfect; and so on of all the rest: wherein no

one can show a reason why some of the inseparable qualities, that are always united in nature, should be put into the nominal essence, and others left out: or why the word gold, signifying that sort of body the ring on his finger is made of, should determine that sort rather by its colour, weight, and fusibility, than by its colour, weight, and solubility in aqua regia: since the dissolving it by that liquor is as inseparable from it as the fusion by fire; and they are both of them nothing but the relation which that substance has to two other bodies, which have a power to operate differently upon it. For by what right is it that fusibility comes to be a part of the essence signified by the word gold, and solubility but a property of it? Or why is its colour part of the essence, and its malleableness but a property? That which I mean is this, That these being all but properties, depending on its real constitution, and nothing but powers, either active or passive, in reference to other bodies, no one has authority to determine the signification of the word gold (as referred to such a body existing in nature) more to one collection of ideas to be

found in that body than to another: whereby the signification of that name must unavoidably be very uncertain. Since, as has been said, several people observe several properties in the same substance; and I think I may say nobody all. And therefore we have but very imperfect descriptions of things, and words have very uncertain significations.

18. The names of simple ideas the least doubtful. From what has been said, it is easy to observe what has been before remarked, viz. that the names of simple ideas are, of all others, the least liable to mistakes, and that for these reasons. First, Because the ideas they stand for, being each but one single perception, are much easier got, and more clearly retained, than the more complex ones, and therefore are not liable to the uncertainty which usually attends those compounded ones of substances and mixed modes, in which the precise number of simple ideas that make them up are not easily agreed, so readily kept in mind. And, Secondly, Because they are never referred to any other essence, but barely that perception they immediately signify: which reference is

that which renders the signification of the names of substances naturally so perplexed, and gives occasion to so many disputes. Men that do not perversely use their words, or on purpose set themselves to cavil, seldom mistake, in any language which they are acquainted with, the use and signification of the name of simple ideas. White and sweet, yellow and bitter, carry a very obvious meaning with them, which every one precisely comprehends, or easily perceives he is ignorant of, and seeks to be informed. But what precise collection of simple ideas modesty or frugality stand for, in another's use, is not so certainly known. And however we are apt to think we well enough know what is meant by gold or iron; yet the precise complex idea others make them the signs of is not so certain: and I believe it is very seldom that, in speaker and hearer, they stand for exactly the same collection. Which must needs produce mistakes and disputes, when they are made use of in discourses, wherein men have to do with universal propositions, and would settle in their minds universal truths, and consider the consequences that follow from them.

19. And next to them, simple modes. By the same rule, the names of simple modes are, next to those of simple ideas, least liable to doubt and uncertainty; especially those of figure and number, of which men have so clear and distinct ideas. Who ever that had a mind to understand them mistook the ordinary meaning of seven, or a triangle? And in general the least compounded ideas in every kind have the least dubious names.

20. The most doubtful are the names of very compounded mixed modes and substances. Mixed modes, therefore, that are made up but of a few and obvious simple ideas, have usually names of no very uncertain signification. But the names of mixed modes which comprehend a great number of simple ideas, are commonly of a very doubtful and undetermined meaning, as has been shown. The names of substances, being annexed to ideas that are neither the real essences, nor exact representations of the patterns they are referred to, are liable to yet greater imperfection and uncertainty, especially when we come to a philosophical use of them.

21. Why this imperfection charged upon words. The

great disorder that happens in our names of substances, proceeding, for the most part, from our want of knowledge, and inability to penetrate into their real constitutions, it may probably be wondered why I charge this as an imperfection rather upon our words than understandings. This exception has so much appearance of justice, that I think myself obliged to give a reason why I have followed this method. I must confess, then, that, when I first began this Discourse of the Understanding, and a good while after, I had not the least thought that any consideration of words was at all necessary to it. But when, having passed over the original and composition of our ideas, I began to examine the extent and certainty of our knowledge, I found it had so near a connexion with words, that, unless their force and manner of signification were first well observed, there could be very little said clearly and pertinently concerning knowledge: which being conversant about truth, had constantly to do with propositions. And though it terminated in things, yet it was for the most part so much by the intervention of words, that they seemed scarce

separable from our general knowledge. At least they interpose themselves so much between our understandings, and the truth which it would contemplate and apprehend, that, like the medium through which visible objects pass, the obscurity and disorder do not seldom cast a mist before our eyes, and impose upon our understandings. If we consider, in the fallacies men put upon themselves, as well as others, and the mistakes in men's disputes and notions, how great a part is owing to words, and their uncertain or mistaken significations, we shall have reason to think this no small obstacle in the way to knowledge; which I conclude we are the more carefully to be warned of, because it has been so far from being taken notice of as an inconvenience, that the arts of improving it have been made the business of men's study, and obtained the reputation of learning and subtilty, as we shall see in the following chapter. But I am apt to imagine, that, were the imperfections of language, as the instrument of knowledge, more thoroughly weighed, a great many of the controversies that make such a noise in the world, would of themselves

cease; and the way to knowledge, and perhaps peace too, lie a great deal opener than it does.

22. This should teach us moderation in imposing our own sense of old authors. Sure I am that the signification of words in all languages, depending very much on the thoughts, notions, and ideas of him that uses them, must unavoidably be of great uncertainty to men of the same language and country. This is so evident in the Greek authors, that he that shall peruse their writings will find in almost every one of them, a distinct language, though the same words. But when to this natural difficulty in every country, there shall be added different countries and remote ages, wherein the speakers and writers had very different notions, tempers, customs, ornaments, and figures of speech, &c., every one of which influenced the signification of their words then, though to us now they are lost and unknown; it would become us to be charitable one to another in our interpretations or misunderstandings of those ancient writings; which, though of great concernment to be understood, are liable to the unavoidable difficulties of speech,

which (if we except the names of simple ideas, and some very obvious things) is not capable, without a constant defining the terms, of conveying the sense and intention of the speaker, without any manner of doubt and uncertainty to the hearer. And in discourses of religion, law, and morality, as they are matters of the highest concernment, so there will be the greatest difficulty.

23. Especially of the Old and New Testament Scriptures. The volumes of interpreters and commentators on the Old and New Testament are but too manifest proofs of this. Though everything said in the text be infallibly true, yet the reader may be, nay, cannot choose but be, very fallible in the understanding of it. Nor is it to be wondered, that the will of God, when clothed in words, should be liable to that doubt and uncertainty which unavoidably attends that sort of conveyance, when even his Son, whilst clothed in flesh, was subject to all the frailties and inconveniences of human nature, sin excepted. And we ought to magnify his goodness, that he hath spread before all the world such legible characters of his works and providence, and given all mankind so

sufficient a light of reason, that they to whom this written word never came, could not (whenever they set themselves to search) either doubt of the being of a God, or of the obedience due to him. Since then the precepts of Natural Religion are plain, and very intelligible to all mankind, and seldom come to be controverted; and other revealed truths, which are conveyed to us by books and languages, are liable to the common and natural obscurities and difficulties incident to words; methinks it would become us to be more careful and diligent in observing the former, and less magisterial, positive, and imperious, in imposing our own sense and interpretations of the latter.

Chapter X Of the Abuse of Words

1. Woeful abuse of words. Besides the imperfection that is naturally in language, and the obscurity and confusion that is so hard to be avoided in the use of words, there are several wilful faults and neglects which men

are guilty of in this way of communication, whereby they render these signs less clear and distinct in their signification than naturally they need to be.

2. Words are often employed without any, or without clear ideas. First, In this kind the first and most palpable abuse is, the using of words without clear and distinct ideas; or, which is worse, signs without anything signified. Of these there are two sorts:—

I. Some words introduced without clear ideas annexed to them, even in their first original. One may observe, in all languages, certain words that, if they be examined, will be found in their first original, and their appropriated use, not to stand for any clear and distinct ideas. These, for the most part, the several sects of philosophy and religion have introduced. For their authors or promoters, either affecting something singular, and out of the way of common apprehensions, or to support some strange opinions, or cover some weakness of their hypothesis, seldom fail to coin new words, and such as, when they come to be examined, may justly be called insignificant terms. For, having either had no determi-

nate collection of ideas annexed to them when they were first invented; or at least such as, if well examined, will be found inconsistent, it is no wonder, if, afterwards, in the vulgar use of the same party, they remain empty sounds, with little or no signification, amongst those who think it enough to have them often in their mouths, as the distinguishing characters of their Church or School, without much troubling their heads to examine what are the precise ideas they stand for. I shall not need here to heap up instances; every man's reading and conversation will sufficiently furnish him. Or if he wants to be better stored, the great mintmasters of this kind of terms, I mean the Schoolmen and Metaphysicians (under which I think the disputing natural and moral philosophers of these latter ages may be comprehended) have wherewithal abundantly to content him.

3. II. Other words, to which ideas were annexed at first, used afterwards without distinct meanings. Others there be who extend this abuse yet further, who take so little care to lay by words, which, in their primary notation have scarce any clear and distinct ideas which they are

annexed to, that, by an unpardonable negligence, they familiarly use words which the propriety of language has affixed to very important ideas, without any distinct meaning at all. Wisdom, glory, grace, &c., are words frequent enough in every man's mouth; but if a great many of those who use them should be asked what they mean by them, they would be at a stand, and not know what to answer: a plain proof, that, though they have learned those sounds, and have them ready at their tongues ends, yet there are no determined ideas laid up in their minds, which are to be expressed to others by them.

4. This occasioned by men learning names before they have the ideas the names belong to. Men having been accustomed from their cradles to learn words which are easily got and retained, before they knew or had framed the complex ideas to which they were annexed, or which were to be found in the things they were thought to stand for, they usually continue to do so all their lives; and without taking the pains necessary to settle in their minds determined ideas, they use their words for such

unsteady and confused notions as they have, contenting themselves with the same words other people use; as if their very sound necessarily carried with it constantly the same meaning. This, though men make a shift with in the ordinary occurrences of life, where they find it necessary to be understood, and therefore they make signs till they are so; yet this insignificancy in their words, when they come to reason concerning either their tenets or interest, manifestly fills their discourse with abundance of empty unintelligible noise and jargon, especially in moral matters, where the words for the most part standing for arbitrary and numerous collections of ideas, not regularly and permanently united in nature, their bare sounds are often only thought on, or at least very obscure and uncertain notions annexed to them. Men take the words they find in use amongst their neighbors; and that they may not seem ignorant what they stand for, use them confidently, without much troubling their heads about a certain fixed meaning; whereby, besides the ease of it, they obtain this advantage, That, as in such discourses they seldom are in the

right, so they are as seldom to be convinced that they are in the wrong; it being all one to go about to draw those men out of their mistakes who have no settled notions, as to dispossess a vagrant of his habitation who has no settled abode. This I guess to be so; and every one may observe in himself and others whether it be so or not.

5. Unsteady application of them. Secondly, Another great abuse of words is inconstancy in the use of them. It is hard to find a discourse written on any subject, especially of controversy, wherein one shall not observe, if he read with attention, the same words (and those commonly the most material in the discourse, and upon which the argument turns) used sometimes for one collection of simple ideas, and sometimes for another; which is a perfect abuse of language. Words being intended for signs of my ideas, to make them known to others, not by any natural signification, but by a voluntary imposition, it is plain cheat and abuse, when I make them stand sometimes for one thing and sometimes for another; the wilful doing whereof can be imputed to nothing but great folly, or greater dishonesty. And a man, in

his accounts with another may, with as much fairness make the characters of numbers stand sometimes for one and sometimes for another collection of units: v.g. this character 3, stand sometimes for three, sometimes for four, and sometimes for eight, as in his discourse or reasoning make the same words stand for different collections of simple ideas. If men should do so in their reckonings, I wonder who would have to do with them? One who would speak thus in the affairs and business of the world, and call 8 sometimes seven, and sometimes nine, as best served his advantage, would presently have clapped upon him, one of the two names men are commonly disgusted with. And yet in arguings and learned contests, the same sort of proceedings passes commonly for wit and learning; but to me it appears a greater dishonesty than the misplacing of counters in the casting up a debt; and the cheat the greater, by how much truth is of greater concernment and value than money.

6. III. Affected obscurity, as in the Peripatetick and other sects of philosophy. Thirdly, Another abuse of language is an affected obscurity; by either applying old

words to new and unusual significations; or introducing new and ambiguous terms, without defining either; or else putting them so together, as may confound their ordinary meaning. Though the Peripatetick philosophy has been most eminent in this way, yet other sects have not been wholly clear of it. There are scarce any of them that are not cumbered with some difficulties (such is the imperfection of human knowledge,) which they have been fain to cover with obscurity of terms, and to confound the signification of words, which, like a mist before people's eyes, might hinder their weak parts from being discovered. That body and extension in common use, stand for two distinct ideas, is plain to any one that will but reflect a little. For were their signification precisely the same, it would be as proper, and as intelligible to say, "the body of an extension," as the "extension of a body"; and yet there are those who find it necessary to confound their signification. To this abuse, and the mischiefs of confounding the signification of words, logic, and the liberal sciences as they have been handled in the schools, have given reputation; and the admired Art

of Disputing hath added much to the natural imperfection of languages, whilst it has been made use of and fitted to perplex the signification of words, more than to discover the knowledge and truth of things: and he that will look into that sort of learned writings, will find the words there much more obscure, uncertain, and undetermined in their meaning, than they are in ordinary conversation.

7. Logic and dispute have much contributed to this. This is unavoidably to be so, where men's parts and learning are estimated by their skill in disputing. And if reputation and reward shall attend these conquests, which depend mostly on the fineness and niceties of words, it is no wonder if the wit of man so employed, should perplex, involve, and subtilize the signification of sounds, so as never to want something to say in opposing or defending any question; the victory being adjudged not to him who had truth on his side, but the last word in the dispute.

8. Calling it "subtlety." This, though a very useless skin, and that which I think the direct opposite to the

ways of knowledge, hath yet passed hitherto under the laudable and esteemed names of subtlety and acuteness, and has had the applause of the schools, and encouragement of one part of the learned men of the world. And no wonder, since the philosophers of old, (the disputing and wrangling philosophers I mean, such as Lucian wittily and with reason taxes), and the Schoolmen since, aiming at glory and esteem, for their great and universal knowledge, easier a great deal to be pretended to than really acquired, found this a good expedient to cover their ignorance, with a curious and inexplicable web of perplexed words, and procure to themselves the admiration of others, by unintelligible terms, the apter to produce wonder because they could not be understood: whilst it appears in all history, that these profound doctors were no wiser nor more useful than their neighbours, and brought but small advantage to human life or the societies wherein they lived: unless the coining of new words, where they produced no new things to apply them to, or the perplexing or obscuring the signification of old ones, and so bringing all things into

question and dispute, were a thing profitable to the life of man, or worthy commendation and reward.

9. This learning very little benefits society. For, notwithstanding these learned disputants, these all-knowing doctors, it was to the unscholastic statesman that the governments of the world owed their peace, defence, and liberties; and from the illiterate and contemned mechanic (a name of disgrace) that they received the improvements of useful arts. Nevertheless, this artificial ignorance, and learned gibberish, prevailed mightily in these last ages, by the interest and artifice of those who found no easier way to that pitch of authority and dominion they have attained, than by amusing the men of business, and ignorant, with hard words, or employing the ingenious and idle in intricate disputes about unintelligible terms, and holding them perpetually entangled in that endless labyrinth. Besides, there is no such way to gain admittance, or give defence to strange and absurd doctrines, as to guard them round about with legions of obscure, doubtful, and undefined words. Which yet make these retreats more like the dens of robbers, or

holes of foxes, than the fortresses of fair warriors: which, if it be hard to get them out of, it is not for the strength that is in them, but the briars and thorns, and the obscurity of the thickets they are beset with. For untruth being unacceptable to the mind of man, there is no other defence left for absurdity but obscurity.

10. But destroys the instruments of knowledge and communication. Thus learned ignorance, and this art of keeping even inquisitive men from true knowledge, hath been propagated in the world, and hath much perplexed, whilst it pretended to inform the understanding. For we see that other well-meaning and wise men, whose education and parts had not acquired that acuteness, could intelligibly express themselves to one another; and in its plain use make a benefit of language. But though unlearned men well enough understood the words white and black, &c., and had constant notions of the ideas signified by those words; yet there were philosophers found who had learning and subtlety enough to prove that snow was black; i.e. to prove that white was black. Whereby they had the advantage to destroy the instru-

ments and means of discourse, conversation, instruction, and society; whilst, with great art and subtlety, they did no more but perplex and confound the signification of words, and thereby render language less useful than the real defects of it had made it; a gift which the illiterate had not attained to.

11. As useful as to confound the sounds that the letters of the alphabet stand for. These learned men did equally instruct men's understandings, and profit their lives, as he who should alter the signification of known characters, and, by a subtle device of learning, far surpassing the capacity of the illiterate, dull, and vulgar, should in his writing show that he could put A for B, and D for E, &c., to the no small admiration and benefit of his reader. It being as senseless to put black, which is a word agreed on to stand for one sensible idea, to put it, I say, for another, or the contrary idea; i.e. to call snow black, as to put this mark A, which is a character agreed on to stand for one modification of sound, made by a certain motion of the organs of speech, for B, which is agreed on to stand for another modification of sound,

made by another certain mode of the organs of speech.

12. This art has perplexed religion and justice. Nor hath this mischief stopped in logical niceties, or curious empty speculations; it hath invaded the great concernments of human life and society; obscured and perplexed the material truths of law and divinity; brought confusion, disorder, and uncertainty into the affairs of mankind; and if not destroyed, yet in a great measure rendered useless, these two great rules, religion and justice. What have the greatest part of the comments and disputes upon the laws of God and man served for, but to make the meaning more doubtful, and perplex the sense? What have been the effect of those multiplied curious distinctions, and acute niceties, but obscurity and uncertainty, leaving the words more unintelligible, and the reader more at a loss? How else comes it to pass that princes, speaking or writing to their servants, in their ordinary commands are easily understood; speaking to their people, in their laws, are not so? And, as I remarked before, doth it not often happen that a man of an ordinary capacity very well understands a text, or a law, that he

reads, till he consults an expositor, or goes to counsel; who, by that time he hath done explaining them, makes the words signify either nothing at all, or what he pleases.

13. And ought not to pass for learning. Whether any by-interests of these professions have occasioned this, I will not here examine; but I leave it to be considered, whether it would not be well for mankind, whose concernment it is to know things as they are, and to do what they ought, and not to spend their lives in talking about them, or tossing words to and fro;—whether it would not be well, I say, that the use of words were made plain and direct; and that language, which was given us for the improvement of knowledge and bond of society, should not be employed to darken truth and unsettle people's rights; to raise mists, and render unintelligible both morality and religion? Or that at least, if this will happen, it should not be thought learning or knowledge to do so?

14. IV. By taking words for things. Fourthly, Another great abuse of words, is the taking them for things. This, though it in some degree concerns all names in

general, yet more particularly affects those of substances. To this abuse those men are most subject who most confine their thoughts to anyone system, and give themselves up into a firm belief of the perfection of any received hypothesis: whereby they come to be persuaded that the terms of that sect are so suited to the nature of things, that they perfectly correspond with their real existence. Who is there that has been bred up in the Peripatetick philosophy, who does not think the Ten Names, under which are ranked the Ten Predicaments, to be exactly conformable to the nature of things? Who is there of that school that is not persuaded that substantial forms, vegetative souls, abhorrence of a vacuum, intentional species, &c., are something real? These words men have learned from their very entrance upon knowledge, and have found their masters and systems lay great stress upon them: and therefore they cannot quit the opinion, that they are conformable to nature, and are the representations of something that really exists. The Platonists have their soul of the world, and the Epicureans their endeavour towards motion in their atoms when

at rest. There is scarce any sect in philosophy has not a distinct set of terms that others understand not. But yet this gibberish, which, in the weakness of human understanding, serves so well to palliate men's ignorance, and cover their errors, comes, by familiar use amongst those of the same tribe, to seem the most important part of language, and of all other the terms the most significant: and should aerial and aetherial vehicles come once, by the prevalency of that doctrine, to be generally received anywhere, no doubt those terms would make impressions on men's minds, so as to establish them in the persuasion of the reality of such things, as much as Peripatetick forms and intentional species have heretofore done.

15. Instance, in matter. How much names taken for things are apt to mislead the understanding, the attentive reading of philosophical writers would abundantly discover; and that perhaps in words little suspected of any such misuse. I shall instance in one only, and that a very familiar one. How many intricate disputes have there been about matter, as if there were some such thing

really in nature, distinct from body; as it is evident the word matter stands for an idea distinct from the idea of body? For if the ideas these two terms stood for were precisely the same, they might indifferently in all places be put for one another. But we see that though it be proper to say, There is one matter of all bodies, one cannot say, There is one body of all matters: we familiarly say one body is bigger than another; but it sounds harsh (and I think is never used) to say one matter is bigger than another. Whence comes this, then? Viz. from hence: that, though matter and body be not really distinct, but wherever there is the one there is the other; yet matter and body stand for two different conceptions, whereof the one is incomplete, and but a part of the other. For body stands for a solid extended figured substance, whereof matter is but a partial and more confused conception; it seeming to me to be used for the substance and solidity of body, without taking in its extension and figure: and therefore it is that, speaking of matter, we speak of it always as one, because in truth it expressly contains nothing but the idea of a solid

substance, which is everywhere the same, everywhere uniform. This being our idea of matter, we no more conceive or speak of different matters in the world than we do of different solidities; though we both conceive and speak of different bodies, because extension and figure are capable of variation. But, since solidity cannot exist without extension and figure, the taking matter to be the name of something really existing under that precision, has no doubt produced those obscure and unintelligible discourses and disputes, which have filled the heads and books of philosophers concerning *materia prima*; which imperfection or abuse, how far it may concern a great many other general terms I leave to be considered. This, I think, I may at least say, that we should have a great many fewer disputes in the world, if words were taken for what they are, the signs of our ideas only; and not for things themselves. For, when we argue about matter, or any the like term, we truly argue only about the idea we express by that sound, whether that precise idea agree to anything really existing in nature or no. And if men would tell what ideas

they make their words stand for, there could not be half that obscurity or wrangling in the search or support of truth that there is.

16. This makes errors lasting. But whatever inconvenience follows from this mistake of words, this I am sure, that, by constant and familiar use, they charm men into notions far remote from the truth of things. It would be a hard matter to persuade any one that the words which his father, or schoolmaster, the parson of the parish, or such a reverend doctor used, signified nothing that really existed in nature: which perhaps is none of the least causes that men are so hardly drawn to quit their mistakes, even in opinions purely philosophical, and where they have no other interest but truth. For the words they have a long time been used to, remaining firm in their minds, it is no wonder that the wrong notions annexed to them should not be removed.

17. V. By setting them in the place of what they cannot signify. Fifthly Another abuse of words is the setting them in the place of things which they do or can

by no means signify. We may observe that in the general names of substances whereof the nominal essences are only known to us when we put them into propositions, and affirm or deny anything about them, we do most commonly tacitly suppose or intend, they should stand for the real essence of a certain sort of substances. For, when a man says gold is malleable, he means and would insinuate something more than this. That what I call gold is malleable, (though truly it amounts to no more,) but would have this understood, viz. That gold, i.e. what has the real essence of gold, is malleable; which amounts to thus much, that malleableness depends on, and is inseparable from the real essence of gold. But a man, not knowing wherein that real essence consists, the connexion in his mind of malleableness is not truly with an essence he knows not, but only with the sound gold he puts for it. Thus, when we say that animal rationale is, and animal implume bipes latis unguibus is not a good definition of a man; it is plain we suppose the name man in this case to stand for the real essence of a species, and would signify that “a rational animal”

better described that real essence than “a two-legged animal with broad nails, and without feathers.” For else, why might not Plato as properly make the word *anthropos*, or man, stand for his complex idea, made up of the idea of a body, distinguished from others by a certain shape and other outward appearances, as Aristotle make the complex idea to which he gave the name *anthropos*, or man, of body and the faculty of reasoning joined together; unless the name *anthropos*, or man, were supposed to stand for something else than what it signifies; and to be put in the place of some other thing than the idea a man professes he would express by it?

18. V.g. Putting them for the real essences of substances. It is true the names of substances would be much more useful, and propositions made in them much more certain, were the real essences of substances the ideas in our minds which those words signified. And it is for want of those real essences that our words convey so little knowledge or certainty in our discourses about them; and therefore the mind, to remove that imperfection as much as it can, makes them, by a secret suppo-

sition, to stand for a thing having that real essence, as if thereby it made some nearer approaches to it. For, though the word man or gold signify nothing truly but a complex idea of properties united together in one sort of substances; yet there is scarce anybody, in the use of these words, but often supposes each of those names to stand for a thing having the real essence on which these properties depend. Which is so far from diminishing the imperfection of our words, that by a plain abuse it adds to it, when we would make them stand for something, which, not being in our complex idea, the name we use can no ways be the sign of.

19. Hence we think change of our complex ideas of substances not to change their species. This shows us the reason why in mixed modes any of the ideas that make the composition of the complex one being left out or changed, it is allowed to be another thing, i.e. to be of another species, as is plain in chance-medley, manslaughter, murder, parricide, &c. The reason whereof is, because the complex idea signified by that name is the real as well as nominal essence; and there is no secret

reference of that name to any other essence but that. But in substances, it is not so. For though in that called gold, one puts into his complex idea what another leaves out, and vice versa: yet men do not usually think that therefore the species is changed: because they secretly in their minds refer that name, and suppose it annexed to a real immutable essence of a thing existing, on which those properties depend. He that adds to his complex idea of gold that of fixedness and solubility in aqua regia, which he put not in it before, is not thought to have changed the species; but only to have a more perfect idea, by adding another simple idea, which is always in fact joined with those other, of which his former complex idea consisted. But this reference of the name to a thing, whereof we have not the idea, is so far from helping at all, that it only serves the more to involve us in difficulties. For by this tacit reference to the real essence of that species of bodies, the word gold (which, by standing for a more or less perfect collection of simple ideas, serves to design that sort of body well enough in civil discourse) comes to have no significa-

tion at all, being put for somewhat whereof we have no idea at all, and so can signify nothing at all, when the body itself is away. For however it may be thought all one, yet, if well considered, it will be found a quite different thing, to argue about gold in name, and about a parcel in the body itself, v.g. a piece of leaf-gold laid before us; though in discourse we are fain to substitute the name for the thing.

20. The cause of this abuse, a supposition of nature's working always regularly, in setting boundaries to species. That which I think very much disposes men to substitute their names for the real essences of species, is the supposition before mentioned, that nature works regularly in the production of things, and sets the boundaries to each of those species, by giving exactly the same real internal constitution to each individual which we rank under one general name. Whereas anyone who observes their different qualities can hardly doubt, that many of the individuals, called by the same name, are, in their internal constitution, as different one from another as several of those which are ranked under differ-

ent specific names. This supposition, however, that the same precise and internal constitution goes always with the same specific name, makes men forward to take those names for the representatives of those real essences; though indeed they signify nothing but the complex ideas they have in their minds when they use them. So that, if I may so say, signifying one thing, and being supposed for, or put in the place of another, they cannot but, in such a kind of use, cause a great deal of uncertainty in men's discourses; especially in those who have thoroughly imbibed the doctrine of substantial forms, whereby they firmly imagine the several species of things to be determined and distinguished.

21. This abuse contains two false suppositions. But however postposterous and absurd it be to make our names stand for ideas we have not, or (which is all one) essences that we know not, it being in effect to make our words the signs of nothing; yet it is evident to any one who ever so little reflects on the use men make of their words, that there is nothing more familiar. When a man asks whether this or that thing he sees, let it be a drill,

or a monstrous foetus, be a man or no; it is evident the question is not, Whether that particular thing agree to his complex idea expressed by the name man: but whether it has in it the real essence of a species of things which he supposes his name man to stand for. In which way of using the names of substances, there are these false suppositions contained:—

First, that there are certain precise essences according to which nature makes all particular things, and by which they are distinguished into species. That everything has a real constitution, whereby it is what it is, and on which its sensible qualities depend, is past doubt: but I think it has been proved that this makes not the distinction of species as we rank them, nor the boundaries of their names.

Secondly, this tacitly also insinuates, as if we had ideas of these proposed essences. For to what purpose else is it, to inquire whether this or that thing have the real essence of the species man, if we did not suppose that there were such a specific essence known? Which yet is utterly false. And therefore such application of names

as would make them stand for ideas which we have not, must needs cause great disorder in discourses and reasonings about them, and be a great inconvenience in our communication by words.

22. VI. By proceeding upon the supposition that the words we use have a certain and evident signification which other men cannot but understand. Sixthly, there remains yet another more general, though perhaps less observed, abuse of words; and that is, that men having by a long and familiar use annexed to them certain ideas, they are apt to imagine so near and necessary a connexion between the names and the signification they use them in, that they forwardly suppose one cannot but understand what their meaning is; and therefore one ought to acquiesce in the words delivered, as if it were past doubt that, in the use of those common received sounds, the speaker and hearer had necessarily the same precise ideas. Whence presuming, that when they have in discourse used any term, they have thereby, as it were, set before others the very thing they talked of. And so likewise taking the words of others as naturally standing

for just what they themselves have been accustomed to apply them to, they never trouble themselves to explain their own, or understand clearly others' meaning. From whence commonly proceeds noise, and wrangling, without improvement or information; whilst men take words to be the constant regular marks of agreed notions, which in truth are no more but the voluntary and unsteady signs of their own ideas. And yet men think it strange, if in discourse, or (where it is often absolutely necessary) in dispute, one sometimes asks the meaning of their terms: though the arguings one may every day observe in conversation make it evident, that there are few names of complex ideas which any two men use for the same just precise collection. It is hard to name a word which is hard to name a word which will not be a clear instance of this. Life is a term, none more familiar. Any one almost would take it for an affront to be asked what he meant by it. And yet if it comes in question, whether a plant that lies ready formed in the seed have life; whether the embryo in an egg before incubation, or a man in a swoon without sense or motion, be alive

or no; it is easy to perceive that a clear, distinct, settled idea does not always accompany the use of so known a word as that of life is. Some gross and confused conceptions men indeed ordinarily have, to which they apply the common words of their language; and such a loose use of their words serves them well enough in their ordinary discourses or affairs. But this is not sufficient for philosophical inquiries. Knowledge and reasoning require precise determinate ideas. And though men will not be so importunately dull as not to understand what others say, without demanding an explication of their terms; nor so troublesomely critical as to correct others in the use of the words they receive from them: yet, where truth and knowledge are concerned in the case, I know not what fault it can be, to desire the explication of words whose sense seems dubious; or why a man should be ashamed to own his ignorance in what sense another man uses his words; since he has no other way of certainly knowing it but by being informed. This abuse of taking words upon trust has nowhere spread so far, nor with so ill effects, as amongst men of letters. The

multiplication and obstinacy of disputes, which have so laid waste the intellectual world, is owing to nothing more than to this ill use of words. For though it be generally believed that there is great diversity of opinions in the volumes and variety of controversies the world is distracted with; yet the most I can find that the contending learned men of different parties do, in their arguings one with another, is, that they speak different languages. For I am apt to imagine, that when any of them, quitting terms, think upon things, and know what they think, they think all the same: though perhaps what they would have be different.

23. The ends of language: First, to convey our ideas. To conclude this consideration of the imperfection and abuse of language. The ends of language in our discourse with others being chiefly these three: First, to make known one man's thoughts or ideas to another; Secondly, to do it with as much ease and quickness as possible; and, Thirdly, thereby to convey the knowledge of things: language is either abused or deficient, when it fails of any of these three.

First, Words fail in the first of these ends, and lay not

open one man's ideas to another's view: 1. When men have names in their mouths without any determinate ideas in their minds, whereof they are the signs: or, 2. When they apply the common received names of any language to ideas, to which the common use of that language does not apply them: or, 3. When they apply them very unsteadily, making them stand, now for one, and by and by for another idea.

24. To do it with quickness. Secondly, Men fail of conveying their thoughts with all the quickness and ease that may be, when they have complex ideas without having any distinct names for them. This is sometimes the fault of the language itself, which has not in it a sound yet applied to such a signification; and sometimes the fault of the man, who has not yet learned the name for that idea he would show another.

25. Therewith to convey the knowledge of things. Thirdly, There is no knowledge of things conveyed by men's words, when their ideas agree not to the reality of things. Though it be a defect that has its original in our ideas, which are not so conformable to the nature of

things as attention, study, and application might make them, yet it fails not to extend itself to our words too, when we use them as signs of real beings, which yet never had any reality or existence.

26. How men's words fail in all these: First, when used without any ideas. First, He that hath words of any language, without distinct ideas in his mind to which he applies them, does, so far as he uses them in discourse, only make a noise without any sense or signification; and how learned soever he may seem, by the use of hard words or learned terms, is not much more advanced thereby in knowledge, than he would be in learning, who had nothing in his study but the bare titles of books, without possessing the contents of them. For all such words, however put into discourse, according to the right construction of grammatical rules, or the harmony of well-turned periods, do yet amount to nothing but bare sounds, and nothing else.

27. When complex ideas are without names annexed to them. Secondly, He that has complex ideas, without particular names for them, would be in no better case than

a bookseller, who had in his warehouse volumes that lay there unbound, and without titles, which he could therefore make known to others only by showing the loose sheets, and communicate them only by tale. This man is hindered in his discourse, for want of words to communicate his complex ideas, which he is therefore forced to make known by an enumeration of the simple ones that compose them; and so is fain often to use twenty words, to express what another man signifies in one.

28. When the same sign is not put for the same idea. Thirdly, He that puts not constantly the same sign for the same idea, but uses the same words sometimes in one and sometimes in another signification, ought to pass in the schools and conversation for as fair a man, as he does in the market and exchange, who sells several things under the same name.

29. When words are diverted from their common use. Fourthly, He that applies the words of any language to ideas different from those to which the common use of that country applies them, however his own understanding may be filled with truth and light, will not by such

words be able to convey much of it to others, without defining his terms. For however the sounds are such as are familiarly known, and easily enter the ears of those who are accustomed to them; yet standing for other ideas than those they usually are annexed to, and are wont to excite in the mind of the hearers, they cannot make known the thoughts of him who thus uses them.

30. When they are names of fantastical imaginations. Fifthly, He that imagined to himself substances such as never have been, and filled his head with ideas which have not any correspondence with the real nature of things, to which yet he gives settled and defined names, may fill his discourse, and perhaps another man's head with the fantastical imaginations of his own brain, but will be very far from advancing thereby one jot in real and true knowledge.

31. Summary. He that hath names without ideas, wants meaning in his words, and speaks only empty sounds. He that hath complex ideas without names for them, wants liberty and dispatch in his expressions, and is necessitated to use periphrases. He that uses his words

loosely and unsteadily will either be not minded or not understood. He that applies his names to ideas different from their common use, wants propriety in his language, and speaks gibberish. And he that hath the ideas of substances disagreeing with the real existence of things, so far wants the materials of true knowledge in his understanding, and hath instead thereof chimeras.

32. How men's words fail when they stand for substances. In our notions concerning Substances, we are liable to all the former inconveniences: v.g. he that uses the word tarantula, without having any imagination or idea of what it stands for, pronounces a good word; but so long means nothing at all by it. 2. He that, in a newly-discovered country, shall see several sorts of animals and vegetables, unknown to him before, may have as true ideas of them, as of a horse or a stag; but can speak of them only by a description, till he shall either take the names the natives call them by, or give them names himself. 3. He that uses the word body sometimes for pure extension, and sometimes for extension and solidity together, will talk very fallaciously. 4. He that gives the name horse to that idea

which common usage calls mule, talks improperly, and will not be understood. 5. He that thinks the name centaur stands for some real being, imposes on himself, and mistakes words for things.

33. How when they stand for modes and relations. In Modes and Relations generally, we are liable only to the four first of these inconveniences; viz. 1. I may have in my memory the names of modes, as gratitude or charity, and yet not have any precise ideas annexed in my thoughts to those names. 2. I may have ideas, and not know the names that belong to them: v.g. I may have the idea of a man's drinking till his colour and humour be altered, till his tongue trips, and his eyes look red, and his feet fail him; and yet not know that it is to be called drunkenness. 3. I may have the ideas of virtues or vices, and names also, but apply them amiss: v.g. when I apply the name frugality to that idea which others call and signify by this sound, covetousness. 4. I may use any of those names with inconstancy. 5. But, in modes and relations, I cannot have ideas disagreeing to the existence of things: for modes being complex ideas, made by the mind at plea-

sure, and relation being but by way of considering or comparing two things together, and so also an idea of my own making, these ideas can scarce be found to disagree with anything existing; since they are not in the mind as the copies of things regularly made by nature, nor as properties inseparably flowing from the internal constitution or essence of any substance; but, as it were, patterns lodged in my memory, with names annexed to them, to denominate actions and relations by, as they come to exist. But the mistake is commonly in my giving a wrong name to my conceptions; and so using words in a different sense from other people: I am not understood, but am thought to have wrong ideas of them, when I give wrong names to them. Only if I put in my ideas of mixed modes or relations any inconsistent ideas together, I fill my head also with chimeras; since such ideas, if well examined, cannot so much as exist in the mind, much less any real being ever be denominated from them.

34. Seventhly, language is often abused by figurative speech. Since wit and fancy find easier entertainment in the world than dry truth and real knowledge, figurative

speeches and allusion in language will hardly be admitted as an imperfection or abuse of it. I confess, in discourses where we seek rather pleasure and delight than information and improvement, such ornaments as are borrowed from them can scarce pass for faults. But yet if we would speak of things as they are, we must allow that all the art of rhetoric, besides order and clearness; all the artificial and figurative application of words eloquence hath invented, are for nothing else but to insinuate wrong ideas, move the passions, and thereby mislead the judgment; and so indeed are perfect cheats: and therefore, however laudable or allowable oratory may render them in harangues and popular addresses, they are certainly, in all discourses that pretend to inform or instruct, wholly to be avoided; and where truth and knowledge are concerned, cannot but be thought a great fault, either of the language or person that makes use of them. What and how various they are, will be superfluous here to take notice; the books of rhetoric which abound in the world, will instruct those who want to be informed: only I cannot but observe how little the pres-

ervation and improvement of truth and knowledge is the care and concern of mankind; since the arts of fallacy are endowed and preferred. It is evident how much men love to deceive and be deceived, since rhetoric, that powerful instrument of error and deceit, has its established professors, is publicly taught, and has always been had in great reputation: and I doubt not but it will be thought great boldness, if not brutality, in me to have said thus much against it. Eloquence, like the fair sex, has too prevailing beauties in it to suffer itself ever to be spoken against. And it is in vain to find fault with those arts of deceiving, wherein men find pleasure to be deceived.

Chapter XI

Of the Remedies of the Foregoing Imperfections and Abuses of Words

1. Remedies are worth seeking The natural and improved imperfections of languages we have seen above at large: and speech being the great bond that holds society to-

gether, and the common conduit, whereby the improvements of knowledge are conveyed from one man and one generation to another, it would well deserve our most serious thoughts to consider, what remedies are to be found for the inconveniences above mentioned.

2. Are not easy to find. I am not so vain as to think that any one can pretend to attempt the perfect reforming the languages of the world, no not so much as of his own country, without rendering himself ridiculous. To require that men should use their words constantly in the same sense, and for none but determined and uniform ideas, would be to think that all men should have the same notions, and should talk of nothing but what they have clear and distinct ideas of: which is not to be expected by any one who hath not vanity enough to imagine he can prevail with men to be very knowing or very silent And he must be very little skilled in the world, who thinks that a voluble tongue shall accompany only a good understanding; or that men's talking much or little should hold proportion only to their knowledge.

3. But yet necessary to those who search after truth. But though the market and exchange must be left to their own ways of talking, and gossipings not be robbed of their ancient privilege: though the schools, and men of argument would perhaps take it amiss to have anything offered, to abate the length or lessen the number of their disputes; yet methinks those who pretend seriously to search after or maintain truth, should think themselves obliged to study how they might deliver themselves without obscurity, doubtfulness, or equivocation, to which men's words are naturally liable, if care be not taken.

4. Misuse of words the great cause of errors. For he that shall well consider the errors and obscurity, the mistakes and confusion, that are spread in the world by an ill use of words, will find some reason to doubt whether language, as it has been employed, has contributed more to the improvement or hindrance of knowledge amongst mankind. How many are there, that, when they would think on things, fix their thoughts only on words, especially when they would apply their minds to moral matters? And who then can wonder if the result of such

contemplations and reasonings, about little more than sounds, whilst the ideas they annex to them are very confused and very unsteady, or perhaps none at all; who can wonder, I say, that such thoughts and reasonings end in nothing but obscurity and mistake, without any clear judgment or knowledge?

5. Has made men more conceited and obstinate. This inconvenience, in an ill use of words, men suffer in their own private meditations: but much more manifest are the disorders which follow from it, in conversation, discourse, and arguing with others. For language being the great conduit, whereby men convey their discoveries, reasonings, and knowledge, from one to another, he that makes an ill use of it, though he does not corrupt the fountains of knowledge, which are in things themselves, yet he does, as much as in him lies, break or stop the pipes whereby it is distributed to the public use and advantage of mankind. He that uses words without any clear and steady meaning, what does he but lead himself and others into errors? And he that designedly does it, ought to be looked on as an enemy to truth and knowl-

edge. And yet who can wonder that all the sciences and parts of knowledge have been so overcharged with obscure and equivocal terms, and insignificant and doubtful expressions, capable to make the most attentive or quick-sighted very little, or not at all, the more knowing or orthodox: since subtlety, in those who make profession to teach or defend truth, hath passed so much for a virtue: a virtue, indeed, which, consisting for the most part in nothing but the fallacious and illusory use of obscure or deceitful terms, is only fit to make men more conceited in their ignorance, and more obstinate in their errors.

6. Addicted to wrangling about sounds. Let us look into the books of controversy of any kind, there we shall see that the effect of obscure, unsteady, or equivocal terms is nothing but noise and wrangling about sounds, without convincing or bettering a man's understanding. For if the idea be not agreed on, betwixt the speaker and hearer, for which the words stand, the argument is not about things, but names. As often as such a word whose signification is not ascertained betwixt them,

comes in use, their understandings have no other object wherein they agree, but barely the sound; the things that they think on at that time, as expressed by that word, being quite different.

7. Instance, bat and bird. Whether a bat be a bird or no, is not a question, Whether a bat be another thing than indeed it is, or have other qualities than indeed it has; for that would be extremely absurd to doubt of. But the question is, (1) Either between those that acknowledged themselves to have but imperfect ideas of one or both of this sort of things, for which these names are supposed to stand. And then it is a real inquiry concerning the nature of a bird or a bat, to make their yet imperfect ideas of it more complete; by examining whether all the simple ideas to which, combined together, they both give the name bird, be all to be found in a bat: but this is a question only of inquirers (not disputers) who neither affirm nor deny, but examine: Or, (2) It is a question between disputants; whereof the one affirms, and the other denies that a bat is a bird. And then the question is barely about the signification

of one or both these words; in that they not having both the same complex ideas to which they give these two names, one holds and the other denies, that these two names may be affirmed one of another. Were they agreed in the signification of these two names, it were impossible they should dispute about them. For they would presently and clearly see (were that adjusted between them), whether all the simple ideas of the more general name bird were found in the complex idea of a bat or no; and so there could be no doubt whether a bat were a bird or no. And here I desire it may be considered, and carefully examined, whether the greatest part of the disputes in the world are not merely verbal, and about the signification of words; and whether, if the terms they are made in were defined, and reduced in their signification (as they must be where they signify anything) to determined collections of the simple ideas they do or should stand for, those disputes would not end of themselves, and immediately vanish. I leave it then to be considered, what the learning of disputation is, and how well they are employed for the advantage of

themselves or others, whose business is only the vain ostentation of sounds; i.e. those who spend their lives in disputes and controversies. When I shall see any of those combatants strip all his terms of ambiguity and obscurity, (which every one may do in the words he uses himself), I shall think him a champion for knowledge, truth, and peace, and not the slave of vain-glory, ambition, or a party.

8. Remedies. To remedy the defects of speech before mentioned to some degree, and to prevent the inconveniences that follow from them, I imagine the observation of these following rules may be of use, till somebody better able shall judge it worth his while to think more maturely on this matter, and oblige the world with his thoughts on it.

First remedy: To use no word without an idea annexed to it. First, A man shall take care to use no word without a signification, no name without an idea for which he makes it stand. This rule will not seem altogether needless to any one who shall take the pains to recollect how often he has met with such words as instinct, sym-

pathy, and antipathy, &c., in the discourse of others, so made use of as he might easily conclude that those that used them had no ideas in their minds to which they applied them, but spoke them only as sounds, which usually served instead of reasons on the like occasions. Not but that these words, and the like, have very proper significations in which they may be used; but there being no natural connexion between any words and any ideas, these, and any other, may be learned by rote, and pronounced or writ by men who have no ideas in their minds to which they have annexed them, and for which they make them stand; which is necessary they should, if men would speak intelligibly even to themselves alone.

9. Second remedy: To have distinct, determinate ideas annexed to words, especially in mixed modes. Secondly, It is not enough a man uses his words as signs of some ideas: those he annexes them to, if they be simple, must be clear and distinct; if complex, must be determinate, i.e. the precise collection of simple ideas settled in the mind, with that sound annexed to it, as the sign of that precise determined collection, and no other. This is very

necessary in names of modes, and especially moral words; which, having no settled objects in nature, from whence their ideas are taken, as from their original, are apt to be very confused. Justice is a word in every man's mouth, but most commonly with a very undertermined, loose signification; which will always be so, unless a man has in his mind a distinct comprehension of the component parts that complex idea consists of: and if it be decomposed, must be able to resolve it still on, till he at last comes to the simple ideas that make it up: and unless this be done, a man makes an ill use of the word, let it be justice, for example, or any other. I do not say, a man needs stand to recollect, and make this analysis at large, every time the word justice comes in his way: but this at least is necessary, that he have so examined the signification of that name, and settled the idea of all its parts in his mind, that he can do it when he pleases. If any one who makes his complex idea of justice to be, such a treatment of the person or goods of another as is according to law, hath not a clear and distinct idea what law is, which makes a part of his complex idea of justice,

it is plain his idea of justice itself will be confused and imperfect. This exactness will, perhaps, be judged very troublesome; and therefore most men will think they may be excused from settling the complex ideas of mixed modes so precisely in their minds. But yet I must say, till this be done, it must not be wondered, that they have a great deal of obscurity and confusion in their own minds, and a great deal of wrangling in their discourse with others.

10. And distinct and conformable ideas in words that stand for substances. In the names of substances, for a right use of them, something more is required than barely determined ideas. In these the names must also be conformable to things as they exist; but of this I shall have occasion to speak more at large by and by. This exactness is absolutely necessary in inquiries after philosophical knowledge, and in controversies about truth. And though it would be well, too, if it extended itself to common conversation and the ordinary affairs of life; yet I think that is scarce to be expected. Vulgar notions suit vulgar discourses: and both, though confused

enough, yet serve pretty well the market and the wake. Merchants and lovers, cooks and tailors, have words wherewithal to dispatch their ordinary affairs: and so, I think, might philosophers and disputants too, if they had a mind to understand, and to be clearly understood.

11. Third remedy: To apply words to such ideas as common use has annexed them to. Thirdly, it is not enough that men have ideas, determined ideas, for which they make these signs stand; but they must also take care to apply their words as near as may be to such ideas as common use has annexed them to. For words, especially of languages already framed, being no man's private possession, but the common measure of commerce and communication, it is not for any one at pleasure to change the stamp they are current in, nor alter the ideas they are affixed to; or at least, when there is a necessity to do so, he is bound to give notice of it. Men's intentions in speaking are, or at least should be, to be understood; which cannot be without frequent explanations, demands, and other the like incommodious interruptions, where men do not follow common use. Propriety of

speech is that which gives our thoughts entrance into other men's minds with the greatest ease and advantage: and therefore deserves some part of our care and study, especially in the names of moral words. The proper signification and use of terms is best to be learned from those who in their writings and discourses appear to have had the clearest notions, and applied to them their terms with the exactest choice and fitness. This way of using a man's words, according to the propriety of the language, though it have not always the good fortune to be understood; yet most commonly leaves the blame of it on him who is so unskilful in the language he speaks, as not to understand it when made use of as it ought to be.

12. Fourth remedy: To declare the meaning in which we use them. Fourthly, But, because common use has not so visibly annexed any signification to words, as to make men know always certainly what they precisely stand for: and because men in the improvement of their knowledge, come to have ideas different from the vulgar and ordinary received ones, for which they must either

make new words, (which men seldom venture to do, for fear of being though guilty of affectation or novelty), or else must use old ones in a new signification: therefore, after the observation of the foregoing rules, it is sometimes necessary, for the ascertaining the signification of words, to declare their meaning; where either common use has left it uncertain and loose, (as it has in most names of very complex ideas); or where the term, being very material in the discourse, and that upon which it chiefly turns, is liable to any doubtfulness or mistake.

13. And that in three ways. As the ideas men's words stand for are of different sorts, so the way of making known the ideas they stand for, when there is occasion, is also different. For though defining be thought the proper way to make known the proper signification of words; yet there are some words that will not be defined, as there are others whose precise meaning cannot be made known but by definition: and perhaps a third, which partake somewhat of both the other, as we shall see in the names of simple ideas, modes, and substances.

14. I. In simple ideas, either by synonymous terms, or

by showing examples. First, when a man makes use of the name of any simple idea, which he perceives is not understood, or is in danger to be mistaken, he is obliged, by the laws of ingenuity and the end of speech, to declare his meaning, and make known what idea he makes it stand for. This, as has been shown, cannot be done by definition: and therefore, when a synonymous word fails to do it, there is but one of these ways left. First, Sometimes the naming the subject wherein that simple idea is to be found, will make its name to be understood by those who are acquainted with that subject, and know it by that name. So to make a countryman understand what feuilletmorte colour signifies, it may suffice to tell him, it is the colour of withered leaves falling in autumn. Secondly, but the only sure way of making known the signification of the name of any simple idea, is by presenting to his senses that subject which may produce it in his mind, and make him actually have the idea that word stands for.

15. II. In mixed modes, by definition. Secondly, Mixed modes, especially those belonging to morality, being most

of them such combinations of ideas as the mind puts together of its own choice, and whereof there are not always standing patterns to be found existing, the signification of their names cannot be made known, as those of simple ideas, by any showing: but, in recompense thereof, may be perfectly and exactly defined. For they being combinations of several ideas that the mind of man has arbitrarily put together, without reference to any archetypes, men may, if they please, exactly know the ideas that go to each composition, and so both use these words in a certain and undoubted signification, and perfectly declare, when there is occasion, what they stand for. This, if well considered, would lay great blame on those who make not their discourses about moral things very clear and distinct. For since the precise signification of the names of mixed modes, or, which is all one, the real essence of each species is to be known, they being not of nature's, but man's making, it is a great negligence and perverseness to discourse of moral things with uncertainty and obscurity; which is more pardonable in treating of natural substances, where

doubtful terms are hardly to be avoided, for a quite contrary reason, as we shall see by and by.

16. Morality capable of demonstration. Upon this ground it is that I am bold to think that morality is capable of demonstration, as well as mathematics: since the precise real essence of the things moral words stand for may be perfectly known, and so the congruity and incongruity of the things themselves be certainly discovered; in which consists perfect knowledge. Nor let any one object, that the names of substances are often to be made use of in morality, as well as those of modes, from which will arise obscurity. For, as to substances, when concerned in moral discourses, their divers natures are not so much inquired into as supposed: v.g. when we say that man is subject to law, we mean nothing by man but a corporeal rational creature: what the real essence or other qualities of that creature are in this case is no way considered. And, therefore, whether a child or changeling be a man, in a physical sense, may amongst the naturalists be as disputable as it will, it concerns not at all the moral man, as I may call him, which is this

immovable, unchangeable idea, a corporeal rational being. For, were there a monkey, or any other creature, to be found that had the use of reason to such a degree, as to be able to understand general signs, and to deduce consequences about general ideas, he would no doubt be subject to law, and in that sense be a man, how much soever he differed in shape from others of that name. The names of substances, if they be used in them as they should, can no more disturb moral than they do mathematical discourses; where, if the mathematician speaks of a cube or globe of gold, or of any other body, he has his clear, settled idea, which varies not, though it may by mistake be applied to a particular body to which it belongs not.

17. Definitions can make moral discourses clear. This I have here mentioned, by the by, to show of what consequence it is for men, in their names of mixed modes, and consequently in all their moral discourses, to define their words when there is occasion: since thereby moral knowledge may be brought to so great clearness and certainty. And it must be great want of ingenuousness

(to say no worse of it) to refuse to do it: since a definition is the only way whereby the precise meaning of moral words can be known; and yet a way whereby their meaning may be known certainly, and without leaving any room for any contest about it. And therefore the negligence or perverseness of mankind cannot be excused, if their discourses in morality be not much more clear than those in natural philosophy: since they are about ideas in the mind, which are none of them false or disproportionate; they having no external beings for the archetypes which they are referred to and must correspond with. It is far easier for men to frame in their minds an idea, which shall be the standard to which they will give the name justice; with which pattern so made, all actions that agree shall pass under that denomination, than, having seen Aristides, to frame an idea that shall in all things be exactly like him; who is as he is, let men make what idea they please of him. For the one, they need but know the combination of ideas that are put together in their own minds; for the other, they must inquire into the whole nature, and abstruse

hidden constitution, and various qualities of a thing existing without them.

18. And is the only way in which the meaning of mixed modes can be made known. Another reason that makes the defining of mixed modes so necessary, especially of moral words, is what I mentioned a little before, viz. that it is the only way whereby the signification of the most of them can be known with certainty. For the ideas they stand for, being for the most part such whose component parts nowhere exist together, but scattered and mingled with others, it is the mind alone that collects them, and gives them the union of one idea: and it is only by words enumerating the several simple ideas which the mind has united, that we can make known to others what their names stand for; the assistance of the senses in this case not helping us, by the proposal of sensible objects, to show the ideas which our names of this kind stand for, as it does often in the names of sensible simple ideas, and also to some degree in those of substances.

19. III. In substances, both by showing and by defining. Thirdly, for the explaining the signification of the

names of substances, as they stand for the ideas we have of their distinct species, both the forementioned ways, viz. of showing and defining, are requisite, in many cases, to be made use of. For, there being ordinarily in each sort some leading qualities, to which we suppose the other ideas which make up our complex idea of that species annexed, we forwardly give the specific name to that thing wherein that characteristical mark is found, which we take to be the most distinguishing idea of that species. These leading or characteristical (as I may call them) ideas, in the sorts of animals and vegetables, are (as has been before remarked, ch. vi. SS 29, and ch. ix. SS 15) mostly figure; and in inanimate bodies, colour; and in some, both together. Now, 20. Ideas of the leading qualities of substances are best got by showing. These leading sensible qualities are those which make the chief ingredients of our specific ideas, and consequently the most observable and invariable part in the definitions of our specific names, as attributed to sorts of substances coming under our knowledge. For though the sound man, in its own nature, be as apt to signify a complex idea made

up of animality and rationality, united in the same subject, as to signify any other combination; yet, used as a mark to stand for a sort of creatures we count of our own kind, perhaps the outward shape is as necessary to be taken into our complex idea, signified by the word man, as any other we find in it: and therefore, why Plato's animal *implume bipes latis unguibus* should not be a good definition of the name man, standing for that sort of creatures, will not be easy to show: for it is the shape, as the leading quality, that seems more to determine that species, than a faculty of reasoning, which appears not at first, and in some never. And if this be not allowed to be so, I do not know how they can be excused from murder who kill monstrous births, (as we call them), because of an unordinary shape, without knowing whether they have a rational soul or no; which can be no more discerned in a well-formed than ill-shaped infant, as soon as born. And who is it has informed us that a rational soul can inhabit no tenement, unless it has just such a sort of frontispiece; or can join itself to, and inform no sort of body, but one that is just of such an outward structure?

21. And can hardly be made known otherwise. Now these leading qualities are best made known by showing, and can hardly be made known otherwise. For the shape of a horse or cassowary will be but rudely and imperfectly imprinted on the mind by words; the sight of the animals doth it a thousand times better. And the idea of the particular colour of gold is not to be got by any description of it, but only by the frequent exercise of the eyes about it; as is evident in those who are used to this metal, who will frequently distinguish true from counterfeit, pure from adulterate, by the sight, where others (who have as good eyes, but yet by use have not got the precise nice idea of that peculiar yellow) shall not perceive any difference. The like may be said of those other simple ideas, peculiar in their kind to any substance; for which precise ideas there are no peculiar names. The particular ringing sound there is in gold, distinct from the sound of other bodies, has no particular name annexed to it, no more than the particular yellow that belongs to that metal.

22. The Ideas of the powers of substances are best known

by definition. But because many of the simple ideas that make up our specific ideas of substances are powers which lie not obvious to our senses in the things as they ordinarily appear; therefore, in the signification of our names of substances, some part of the signification will be better made known by enumerating those simple ideas, than by showing the substance itself. For, he that to the yellow shining colour of gold, got by sight, shall, from my enumerating them, have the ideas of great ductility, fusibility, fixedness, and solubility in aqua regia, will have a perfecter idea of gold than he can have by seeing a piece of gold, and thereby imprinting in his mind only its obvious qualities. But if the formal constitution of this shining, heavy, ductile thing, (from whence all these its properties flow), lay open to our senses, as the formal constitution or essence of a triangle does, the signification of the word gold might as easily be ascertained as that of triangle.

23. A reflection on the knowledge of corporeal things possessed by spirits separate from bodies. Hence we may take notice, how much the foundation of all our knowl-

edge of corporeal things lies in our senses. For how spirits, separate from bodies, (whose knowledge and ideas of these things are certainly much more perfect than ours), know them, we have no notion, no idea at all. The whole extent of our knowledge or imagination reaches not beyond our own ideas limited to our ways of perception. Though yet it be not to be doubted that spirits of a higher rank than those immersed in flesh may have as clear ideas of the radical constitution of substances as we have of a triangle, and so perceive how all their properties and operations flow from thence: but the manner how they come by that knowledge exceeds our conceptions.

24. IV Ideas of substances must be conformable to things. Fourthly, But, though definitions will serve to explain the names of substances as they stand for our ideas, yet they leave them not without great imperfection as they stand for things. For our names of substances being not put barely for our ideas, but being made use of ultimately to represent things, and so are put in their place, their signification must agree with the truth of things

as well as with men's ideas. And therefore, in substances, we are not always to rest in the ordinary complex idea commonly received as the signification of that word, but must go a little further, and inquire into the nature and properties of the things themselves, and thereby perfect, as much as we can, our ideas of their distinct species; or else learn them from such as are used to that sort of things, and are experienced in them. For, since it is intended their names should stand for such collections of simple ideas as do really exist in things themselves, as well as for the complex idea in other men's minds, which in their ordinary acceptation they stand for, therefore, to define their names right, natural history is to be inquired into, and their properties are, with care and examination, to be found out. For it is not enough, for the avoiding inconveniences in discourse and arguings about natural bodies and substantial things, to have learned, from the propriety of the language, the common, but confused, or very imperfect, idea to which each word is applied, and to keep them to that idea in our use of them; but we must, by acquainting ourselves

with the history of that sort of things, rectify and settle our complex idea belonging to each specific name; and in discourse with others, (if we find them mistake us), we ought to tell what the complex idea is that we make such a name stand for. This is the more necessary to be done by all those who search after knowledge and philosophical verity, in that children, being taught words, whilst they have but imperfect notions of things, apply them at random, and without much thinking, and seldom frame determined ideas to be signified by them. Which custom (it being easy, and serving well enough for the ordinary affairs of life and conversation) they are apt to continue when they are men: and so begin at the wrong end, learning words first and perfectly, but make the notions to which they apply those words afterwards very overtly. By this means it comes to pass, that men speaking the language of their country, i.e. according to grammar rules of that language, do yet speak very improperly of things themselves; and, by their arguing one with another, make but small progress in the discoveries of useful truths, and the knowledge of

things, as they are to be found in themselves, and not in our imaginations; and it matters not much for the improvement of our knowledge how they are called.

25. Not easy to be made so. It were therefore to be wished, That men versed in physical inquiries, and acquainted with the several sorts of natural bodies, would set down those simple ideas wherein they observe the individuals of each sort constantly to agree. This would remedy a great deal of that confusion which comes from several persons applying the same name to a collection of a smaller or greater number of sensible qualities, proportionably as they have been more or less acquainted with, or accurate in examining, the qualities of any sort of things which come under one denomination. But a dictionary of this sort, containing, as it were, a natural history, requires too many hands as well as too much time, cost, pains, and sagacity ever to be hoped for; and till that be done, we must content ourselves with such definitions of the names of substances as explain the sense men use them in. And it would be well, where there is occasion, if they would afford us so much. This

yet is not usually done; but men talk to one another, and dispute in words, whose meaning is not agreed between them, out of a mistake that the significations of common words are certainly established, and the precise ideas they stand for perfectly known; and that it is a shame to be ignorant of them. Both which suppositions are false; no names of complex ideas having so settled determined significations, that they are constantly used for the same precise ideas. Nor is it a shame for a man not to have a certain knowledge of anything, but by the necessary ways of attaining it; and so it is no discredit not to know what precise idea any sound stands for in another man's mind, without he declare it to me by some other way than barely using that sound, there being no other way, without such a declaration, certainly to know it. Indeed the necessity of communication by language brings men to an agreement in the signification of common words, within some tolerable latitude, that may serve for ordinary conversation: and so a man cannot be supposed wholly ignorant of the ideas which are annexed to words by common use, in a

language familiar to him. But common use being but a very uncertain rule, which reduces itself at last to the ideas of particular men, proves often but a very variable standard. But though such a Dictionary as I have above mentioned will require too much time, cost, and pains to be hoped for in this age; yet methinks it is not unreasonable to propose, that words standing for things which are known and distinguished by their outward shapes should be expressed by little draughts and prints made of them. A vocabulary made after this fashion would perhaps with more ease, and in less time, teach the true signification of many terms, especially in languages of remote countries or ages, and settle truer ideas in men's minds of several things, whereof we read the names in ancient authors, than all the large and laborious comments of learned critics. Naturalists, that treat of plants and animals, have found the benefit of this way: and he that has had occasion to consult them will have reason to confess that he has a clearer idea of apium or ibex, from a little print of that herb or beast, than he could have from a long definition of the names

of either of them. And so no doubt he would have of strigil and sistrum, if, instead of currycomb and cymbal, (which are the English names dictionaries render them by,) he could see stamped in the margin small pictures of these instruments, as they were in use amongst the ancients. Toga, tunica, pallium, are words easily translated by gown, coat, and cloak; but we have thereby no more true ideas of the fashion of those habits amongst the Romans, than we have of the faces of the tailors who made them. Such things as these, which the eye distinguishes by their shapes, would be best let into the mind by draughts made of them, and more determine the signification of such words, than any other words set for them, or made use of to define them. But this is only by the bye.

26. V. Fifth remedy: To use the same word constantly in the same sense. Fifthly, If men will not be at the pains to declare the meaning of their words, and definitions of their terms are not to be had, yet this is the least that can be expected, that, in all discourses wherein one man pretends to instruct or convince another, he

should use the same word constantly in the same sense. If this were done, (which nobody can refuse without great disingenuity,) many of the books extant might be spared; many of the controversies in dispute would be at an end; several of those great volumes, swollen with ambiguous words, now used in one sense, and by and by in another, would shrink into a very narrow compass; and many of the philosophers, (to mention no other) as well as poets works, might be contained in a nutshell.

27. When not so used, the variation is to be explained. But after all, the provision of words is so scanty in respect to that infinite variety of thoughts, that men, wanting terms to suit their precise notions, will, notwithstanding their utmost caution, be forced often to use the same word in somewhat different senses. And though in the continuation of a discourse, or the pursuit of an argument, there can be hardly room to digress into a particular definition, as often as a man varies the signification of any term; yet the import of the discourse will, for the most part, if there be no designed fallacy, sufficiently lead candid and intelligent readers

into the true meaning of it; but where there is not sufficient to guide the reader, there it concerns the writer to explain his meaning, and show in what sense he there uses that term.

BOOK IV

Of Knowledge and Probability

Chapter I

Of Knowledge in General

1. Our knowledge conversant about our ideas only. Since the mind, in all its thoughts and reasonings, hath no other immediate object but its own ideas, which it alone does or can contemplate, it is evident that our knowledge is only conversant about them.

2. Knowledge is the perception of the agreement or disagreement of two ideas. Knowledge then seems to me to be nothing but the perception of the connexion of and agreement, or disagreement and repugnancy of any of our ideas. In this alone it consists. Where this perception is, there is knowledge, and where it is not, there,

though we may fancy, guess, or believe, yet we always come short of knowledge. For when we know that white is not black, what do we else but perceive, that these two ideas do not agree? When we possess ourselves with the utmost security of the demonstration, that the three angles of a triangle are equal to two right ones, what do we more but perceive, that equality to two right ones does necessarily agree to, and is inseparable from, the three angles of a triangle?

3. This agreement or disagreement may be any of four sorts. But to understand a little more distinctly wherein this agreement or disagreement consists, I think we may reduce it all to these four sorts:

I. Identity, or diversity.

II. Relation.

III. Co-existence, or necessary connexion.

IV. Real existence.

4. Of identity, or diversity in ideas. First, As to the first sort of agreement or disagreement, viz. identity or diversity. It is the first act of the mind, when it has any sentiments or ideas at all, to perceive its ideas; and so

far as it perceives them, to know each what it is, and thereby also to perceive their difference, and that one is not another. This is so absolutely necessary, that without it there could be no knowledge, no reasoning, no imagination, no distinct thoughts at all. By this the mind clearly and infallibly perceives each idea to agree with itself, and to be what it is; and all distinct ideas to disagree, i.e. the one not to be the other: and this it does without pains, labour, or deduction; but at first view, by its natural power of perception and distinction. And though men of art have reduced this into those general rules, What is, is, and It is impossible for the same thing to be and not to be, for ready application in all cases, wherein there may be occasion to reflect on it: yet it is certain that the first exercise of this faculty is about particular ideas. A man infallibly knows, as soon as ever he has them in his mind, that the ideas he calls white and round are the very ideas they are; and that they are not other ideas which he calls red or square. Nor can any maxim or proposition in the world make him know it clearer or surer than he did before, and

without any such general rule. This then is the first agreement or disagreement which the mind perceives in its ideas; which it always perceives at first sight: and if there ever happen any doubt about it, it will always be found to be about the names, and not the ideas themselves, whose identity and diversity will always be perceived, as soon and clearly as the ideas themselves are; nor can it possibly be otherwise.

5. Of abstract relations between ideas. Secondly, the next sort of agreement or disagreement the mind perceives in any of its ideas may, I think, be called relative, and is nothing but the perception of the relation between any two ideas, of what kind soever, whether substances, modes, or any other. For, since all distinct ideas must eternally be known not to be the same, and so be universally and constantly denied one of another, there could be no room for any positive knowledge at all, if we could not perceive any relation between our ideas, and find out the agreement or disagreement they have one with another, in several ways the mind takes of comparing them.

6. Of their necessary co-existence in substances. Thirdly, The third sort of agreement or disagreement to be found in our ideas, which the perception of the mind is employed about, is co-existence or non-co-existence in the same subject; and this belongs particularly to substances. Thus when we pronounce concerning gold, that it is fixed, our knowledge of this truth amounts to no more but this, that fixedness, or a power to remain in the fire unconsumed, is an idea that always accompanies and is joined with that particular sort of yellowness, weight, fusibility, malleableness, and solubility in aqua regia, which make our complex idea signified by the word gold,

7. Of real existence agreeing to any idea. Fourthly, The fourth and last sort is that of actual real existence agreeing to any idea. Within these four sorts of agreement or disagreement is, I suppose, contained all the knowledge we have, or are capable of For all the inquiries we can make concerning any of our ideas, all that we know or can affirm concerning any of them, is, That it is, or is not, the same with some other; that it does or does not always coexist with some other idea in the same subject;

that it has this or that relation with some other idea; or that it has a real existence without the mind. Thus, "blue is not yellow," is of identity. "Two triangles upon equal bases between two parallels are equal," is of relation. "Iron is susceptible of magnetical impressions," is of co-existence. "God is," is of real existence. Though identity and co-existence are truly nothing but relations, yet they are such peculiar ways of agreement or disagreement of our ideas, that they deserve well to be considered as distinct heads, and not under relation in general; since they are so different grounds of affirmation and negation, as will easily appear to any one, who will but reflect on what is said in several places of this Essay.

I should now proceed to examine the several degrees of our knowledge, but that it is necessary first, to consider the different acceptations of the word knowledge.

8. Knowledge is either actual or habitual. There are several ways wherein the mind is possessed of truth; each of which is called knowledge.

I. There is actual knowledge, which is the present view the mind has of the agreement or disagreement of any

of its ideas, or of the relation they have one to another.

II. A man is said to know any proposition, which having been once laid before his thoughts, he evidently perceived the agreement or disagreement of the ideas whereof it consists; and so lodged it in his memory, that whenever that proposition comes again to be reflected on, he, without doubt or hesitation, embraces the right side, assents to, and is certain of the truth of it. This, I think, one may call habitual knowledge. And thus a man may be said to know all those truths which are lodged in his memory, by a foregoing clear and full perception, whereof the mind is assured past doubt as often as it has occasion to reflect on them. For our finite understandings being able to think clearly and distinctly but on one thing at once, if men had no knowledge of any more than what they actually thought on, they would all be very ignorant: and he that knew most, would know but one truth, that being all he was able to think on at one time.

9. Habitual knowledge is of two degrees. Of habitual knowledge there are, also, vulgarly speaking. two degrees:

First, The one is of such truths laid up in the memory as, whenever they occur to the mind, it actually perceives the relation is between those ideas. And this is in all those truths whereof we have an intuitive knowledge; where the ideas themselves, by an immediate view, discover their agreement or disagreement one with another.

Secondly, The other is of such truths whereof the mind having been convinced, it retains the memory of the conviction, without the proofs. Thus, a man that remembers certainly that he once perceived the demonstration, that the three angles of a triangle are equal to two right ones, is certain that he knows it, because he cannot doubt the truth of it. In his adherence to a truth, where the demonstration by which it was at first known is forgot, though a man may be thought rather to believe his memory than really to know, and this way of entertaining a truth seemed formerly to me like something between opinion and knowledge; a sort of assurance which exceeds bare belief, for that relies on the testimony of another;—yet upon a due examination I

find it comes not short of perfect certainty, and is in effect true knowledge. That which is apt to mislead our first thoughts into a mistake in this matter is, that the agreement or disagreement of the ideas in this case is not perceived, as it was at first, by an actual view of all the intermediate ideas whereby the agreement or disagreement of those in the proposition was at first perceived; but by other intermediate ideas, that show the agreement or disagreement of the ideas contained in the proposition whose certainty we remember. For example: in this proposition, that “the three angles of a triangle are equal to two right ones,” one who has seen and clearly perceived the demonstration of this truth knows it to be true, when that demonstration is gone out of his mind; so that at present it is not actually in view, and possibly cannot be recollected: but he knows it in a different way from what he did before. The agreement of the two ideas joined in that proposition is perceived; but it is by the intervention of other ideas than those which at first produced that perception. He remembers, i.e. he knows (for remembrance is but the reviving of

some past knowledge) that he was once certain of the truth of this proposition, that the three angles of a triangle are equal to two right ones. The immutability of the same relations between the same immutable things is now the idea that shows him, that if the three angles of a triangle were once equal to two right ones, they will always be equal to two right ones. And hence he comes to be certain, that what was once true in the case, is always true; what ideas once agreed will always agree; and consequently what he once knew to be true, he will always know to be true; as long as he can remember that he once knew it. Upon this ground it is, that particular demonstrations in mathematics afford general knowledge. If then the perception, that the same ideas will eternally have the same habitudes and relations, be not a sufficient ground of knowledge, there could be no knowledge of general propositions in mathematics; for no mathematical demonstration would be any other than particular: and when a man had demonstrated any proposition concerning one triangle or circle, his knowledge would not reach beyond that particular diagram. If he would extend

it further, he must renew his demonstration in another instance, before he could know it to be true in another like triangle, and so on: by which means one could never come to the knowledge of any general propositions. Nobody, I think, can deny, that Mr. Newton certainly knows any proposition that he now at any time reads in his book to be true; though he has not in actual view that admirable chain of intermediate ideas whereby he at first discovered it to be true. Such a memory as that, able to retain such a train of particulars, may be well thought beyond the reach of human faculties, when the very discovery, perception, and laying together that wonderful connexion of ideas, is found to surpass most readers' comprehension. But yet it is evident the author himself knows the proposition to be true, remembering he once saw the connexion of those ideas; as certainly as he knows such a man wounded another, remembering that he saw him run him through. But because the memory is not always so clear as actual perception, and does in all men more or less decay in length of time, this, amongst other differences, is one which shows that demonstrative knowledge

is much more imperfect than intuitive, as we shall see in the following chapter.

Chapter II

Of the Degrees of our Knowledge

1. Of the degrees, or differences in clearness, of our knowledge: 1. Intuitive. All our knowledge consisting, as I have said, in the view the mind has of its own ideas, which is the utmost light and greatest certainty we, with our faculties, and in our way of knowledge, are capable of, it may not be amiss to consider a little the degrees of its evidence. The different clearness of our knowledge seems to me to lie in the different way of perception the mind has of the agreement or disagreement of any of its ideas. For if we will reflect on our own ways of thinking, we will find, that sometimes the mind perceives the agreement or disagreement of two ideas immediately by themselves, without the intervention of any other: and this I think we may call intuitive knowledge. For in this the mind is at no pains of proving or

examining, but perceives the truth as the eye doth light, only by being directed towards it. Thus the mind perceives that white is not black, that a circle is not a triangle, that three are more than two and equal to one and two. Such kinds of truths the mind perceives at the first sight of the ideas together, by bare intuition; without the intervention of any other idea: and this kind of knowledge is the clearest and most certain that human frailty is capable of. This part of knowledge is irresistible, and, like bright sunshine, forces itself immediately to be perceived, as soon as ever the mind turns its view that way; and leaves no room for hesitation, doubt, or examination, but the mind is presently filled with the clear light of it. It is on this intuition that depends all the certainty and evidence of all our knowledge; Which certainty every one finds to be so great, that he cannot imagine, and therefore not require a greater: for a man cannot conceive himself capable of a greater certainty than to know that any idea in his mind is such as he perceives it to be; and that two ideas, wherein he perceives a difference, are different and not precisely the

same. He that demands a greater certainty than this, demands he knows not what, and shows only that he has a mind to be a sceptic, without being able to be so. Certainty depends so wholly on this intuition, that, in the next degree of knowledge which I call demonstrative, this intuition is necessary in all the connexions of the intermediate ideas, without which we cannot attain knowledge and certainty.

2. II. Demonstrative. The next degree of knowledge is, where the mind perceives the agreement or disagreement of any ideas, but not immediately. Though wherever the mind perceives the agreement or disagreement of any of its ideas, there be certain knowledge; yet it does not always happen, that the mind sees that agreement or disagreement, which there is between them, even where it is discoverable; and in that case remains in ignorance, and at most gets no further than a probable conjecture. The reason why the mind cannot always perceive presently the agreement or disagreement of two ideas, is, because those ideas, concerning whose agreement or disagreement the inquiry is made, cannot

by the mind be so put together as to show it. In this case then, when the mind cannot so bring its ideas together as by their immediate comparison, and as it were juxta-position or application one to another, to perceive their agreement or disagreement, it is fain, by the intervention of other ideas (one or more, as it happens) to discover the agreement or disagreement which it searches; and this is that which we call reasoning. Thus, the mind being willing to know the agreement or disagreement in bigness between the three angles of a triangle and two right ones, cannot by an immediate view and comparing them do it: because the three angles of a triangle cannot be brought at once, and be compared with any other one, or two, angles; and so of this the mind has no immediate, no intuitive knowledge. In this case the mind is fain to find out some other angles, to which the three angles of a triangle have an equality; and, finding those equal to two right ones. comes to know their equality to two right ones.

3. Demonstration depends on clearly perceived proofs. Those intervening ideas, which serve to show the agree-

ment of any two others, are called proofs; and where the agreement and disagreement is by this means plainly and clearly perceived, it is called demonstration; it being shown to the understanding, and the mind made to see that it is so. A quickness in the mind to find out these intermediate ideas, (that shall discover the agreement or disagreement of any other,) and to apply them right, is, I suppose, that which is called sagacity.

4. As certain, but not so easy and ready as intuitive knowledge. This knowledge, by intervening proofs, though it be certain, yet the evidence of it is not altogether so clear and bright, nor the assent so ready, as in intuitive knowledge. For, though in demonstration the mind does at last perceive the agreement or disagreement of the ideas it considers; yet it is not without pains and attention: there must be more than one transient view to find it. A steady application and pursuit are required to this discovery: and there must be a progression by steps and degrees, before the mind can in this way arrive at certainty, and come to perceive the agreement or repugnancy between two ideas that need

proofs and the use of reason to show it.

5. The demonstrated conclusion not without doubt, precedent to the demonstration. Another difference between intuitive and demonstrative knowledge is, that, though in the latter all doubt be removed when, by the intervention of the intermediate ideas, the agreement or disagreement is perceived, yet before the demonstration there was a doubt; which in intuitive knowledge cannot happen to the mind that has its faculty of perception left to a degree capable of distinct ideas; no more than it can be a doubt to the eye (that can distinctly see white and black), Whether this ink and this paper be all of a colour. If there be sight in the eyes, it will, at first glimpse, without hesitation, perceive the words printed on this paper different from the colour of the paper: and so if the mind have the faculty of distinct perception, it will perceive the agreement or disagreement of those ideas that produce intuitive knowledge. If the eyes have lost the faculty of seeing, or the mind of perceiving, we in vain inquire after the quickness of sight in one, or clearness of perception in the other.

6. Not so clear as intuitive knowledge. It is true, the perception produced by demonstration is also very clear; yet it is often with a great abatement of that evident lustre and full assurance that always accompany that which I call intuitive: like a face reflected by several mirrors one to another, where, as long as it retains the similitude and agreement with the object, it produces a knowledge; but it is still, in every successive reflection, with a lessening of that perfect clearness and distinctness which is in the first; till at last, after many removes, it has a great mixture of dimness, and is not at first sight so knowable, especially to weak eyes. Thus it is with knowledge made out by a long train of proof.

7. Each step in demonstrated knowledge must have intuitive evidence. Now, in every step reason makes in demonstrative knowledge, there is an intuitive knowledge of that agreement or disagreement it seeks with the next intermediate idea which it uses as a proof: for if it were not so, that yet would need a proof; since without the perception of such agreement or disagreement, there is no knowledge produced: if it be perceived

by itself, it is intuitive knowledge: if it cannot be perceived by itself, there is need of some intervening idea, as a common measure, to show their agreement or disagreement. By which it is plain that every step in reasoning that produces knowledge, has intuitive certainty; which when the mind perceives, there is no more required but to remember it, to make the agreement or disagreement of the ideas concerning which we inquire visible and certain. So that to make anything a demonstration, it is necessary to perceive the immediate agreement of the intervening ideas, whereby the agreement or disagreement of the two ideas under examination (whereof the one is always the first, and the other the last in the account) is found. This intuitive perception of the agreement or disagreement of the intermediate ideas, in each step and progression of the demonstration, must also be carried exactly in the mind, and a man must be sure that no part is left out: which, because in long deductions, and the use of many proofs, the memory does not always so readily and exactly retain; therefore it comes to pass, that this is more imper-

fect than intuitive knowledge, and men embrace often falsehood for demonstrations.

8. Hence the mistake, *ex praecognitis, et praeconcessis*. The necessity of this intuitive knowledge, in each step of scientific or demonstrative reasoning, gave occasion, I imagine, to that mistaken axiom, That all reasoning was *ex praecognitis et praeconcessis*: which, how far it is a mistake, I shall have occasion to show more at large, when I come to consider propositions, and particularly those propositions which are called maxims, and to show that it is by a mistake that they are supposed to be the foundations of all our knowledge and reasonings.

9. Demonstration not limited to ideas of mathematical quantity. It has been generally taken for granted, that mathematics alone are capable of demonstrative certainty: but to have such an agreement or disagreement as may intuitively be perceived, being, as I imagine, not the privilege of the ideas of number, extension, and figure alone, it may possibly be the want of due method and application in us, and not of sufficient evidence in things,

that demonstration has been thought to have so little to do in other parts of knowledge, and been scarce so much as aimed at by any but mathematicians. For whatever ideas we have wherein the mind can perceive the immediate agreement or disagreement that is between them, there the mind is capable of intuitive knowledge; and where it can perceive the agreement or disagreement of any two ideas, by an intuitive perception of the agreement or disagreement they have with any intermediate ideas, there the mind is capable of demonstration: which is not limited to ideas of extension, figure, number, and their modes.

10. Why it has been thought to be so limited. The reason why it has been generally sought for, and supposed to be only in those, I imagine has been, not only the general usefulness of those sciences: but because, in comparing their equality or excess, the modes of numbers have every the least difference very clear and perceivable: and though in extension every the least excess is not so perceptible, yet the mind has found out ways to examine, and discover demonstratively, the just equal-

ity of two angles, or extensions, or figures: and both these, i.e. numbers and figures, can be set down by visible and lasting marks, wherein the ideas under consideration are perfectly determined; which for the most part they are not, where they are marked only by names and words.

11. Modes of qualities not demonstrable like modes of quantity. But in other simple ideas, whose modes and differences are made and counted by degrees, and not quantity, we have not so nice and accurate a distinction of their differences as to perceive, or find ways to measure, their just equality, or the least differences. For those other simple ideas, being appearances of sensations produced in us, by the size, figure, number, and motion of minute corpuscles singly insensible; their different degrees also depend upon the variation of some or of all those causes: which, since it cannot be observed by us, in particles of matter whereof each is too subtile to be perceived, it is impossible for us to have any exact measures of the different degrees of these simple ideas. For, supposing the sensation or idea we

name whiteness be produced in us by a certain number of globules, which, having a verticity about their own centres, strike upon the retina of the eye, with a certain degree of rotation, as well as progressive swiftness; it will hence easily follow, that the more the superficial parts of any body are so ordered as to reflect the greater number of globules of light, and to give them the proper rotation, which is fit to produce this sensation of white in us, the more white will that body appear, that from an equal space sends to the retina the greater number of such corpuscles, with that peculiar sort of motion. I do not say that the nature of light consists in very small round globules; nor of whiteness in such a texture of parts as gives a certain rotation to these globules when it reflects them: for I am not now treating physically of light or colours. But this I think I may say, that I cannot (and I would be glad any one would make intelligible that he did), conceive how bodies without us can any ways affect our senses, but by the immediate contact of the sensible bodies themselves, as in tasting and feeling, or the impulse of some sensible particles coming

from them, as in seeing, hearing, and smelling; by the different impulse of which parts, caused by their different size, figure, and motion, the variety of sensations is produced in us.

12. Particles of light and simple ideas of colour. Whether then they be globules or no; or whether they have a verticity about their own centres that produces the idea of whiteness in us; this is certain, that the more particles of light are reflected from a body, fitted to give them that peculiar motion which produces the sensation of whiteness in us; and possibly too, the quicker that peculiar motion is,—the whiter does the body appear from which the greatest number are reflected, as is evident in the same piece of paper put in the sunbeams, in the shade, and in a dark hole; in each of which it will produce in us the idea of whiteness in far different degrees.

13. The secondary qualities of things not discovered by demonstration. Not knowing, therefore, what number of particles, nor what motion of them, is fit to produce any precise degree of whiteness, we cannot demonstrate the certain equality of any two degrees of whiteness;

because we have no certain standard to measure them by, nor means to distinguish every the least real difference, the only help we have being from our senses, which in this point fail us. But where the difference is so great as to produce in the mind clearly distinct ideas, whose differences can be perfectly retained, there these ideas or colours, as we see in different kinds, as blue and red, are as capable of demonstration as ideas of number and extension. What I have here said of whiteness and colours, I think holds true in all secondary qualities and their modes.

14. Sensitive knowledge of the particular existence of finite beings without us. These two, viz. intuition and demonstration, are the degrees of our knowledge; whatever comes short of one of these, with what assurance soever embraced, is but faith or opinion, but not knowledge, at least in all general truths. There is, indeed, another perception of the mind, employed about the particular existence of finite beings without us, which, going beyond bare probability, and yet not reaching perfectly to either of the foregoing degrees of certainty,

passes under the name of knowledge. There can be nothing more certain than that the idea we receive from an external object is in our minds: this is intuitive knowledge. But whether there be anything more than barely that idea in our minds; whether we can thence certainly infer the existence of anything without us, which corresponds to that idea, is that whereof some men think there may be a question made; because men may have such ideas in their minds, when no such thing exists, no such object affects their senses. But yet here I think we are provided with an evidence that puts us past doubting. For I ask any one, Whether he be not invincibly conscious to himself of a different perception, when he looks on the sun by day, and thinks on it by night; when he actually tastes wormwood, or smells a rose, or only thinks on that savour or odour? We as plainly find the difference there is between any idea revived in our minds by our own memory, and actually coming into our minds by our senses, as we do between any two distinct ideas. If any one say, a dream may do the same thing, and all these ideas may be produced in us with-

out any external objects; he may please to dream that I make him this answer:—

1. That it is no great matter, whether I remove his scruple or no: where all is but dream, reasoning and arguments are of no use, truth and knowledge nothing. 2. That I believe he will allow a very manifest difference between dreaming of being in the fire, and being actually in it. But yet if he be resolved to appear so sceptical as to maintain, that what I call being actually in the fire is nothing but a dream; and that we cannot thereby certainly know, that any such thing as fire actually exists without us: I answer, That we certainly finding that pleasure or pain follows upon the application of certain objects to us, whose existence we perceive, or dream that we perceive, by our senses; this certainty is as great as our happiness or misery, beyond which we have no concernment to know or to be. So that, I think, we may add to the two former sorts of knowledge this also, of the existence of particular external objects, by that perception and consciousness we have of the actual entrance of ideas from them, and allow these three degrees of knowledge, viz. intuitive, demonstrative, and sensitive:

in each of which there are different degrees and ways of evidence and certainty.

15. Knowledge not always clear, where the ideas that enter into it are clear. But since our knowledge is founded on and employed about our ideas only, will it not follow from thence that it is conformable to our ideas; and that where our ideas are clear and distinct, or obscure and confused, our knowledge will be so too? To which I answer, No: for our knowledge consisting in the perception of the agreement or disagreement of any two ideas, its clearness or obscurity consists in the clearness or obscurity of that perception, and not in the clearness or obscurity of the ideas themselves: v.g. a man that has as clear ideas of the angles of a triangle, and of equality to two right ones, as any mathematician in the world, may yet have but a very obscure perception of their agreement, and so have but a very obscure knowledge of it. But ideas which, by reason of their obscurity or otherwise, are confused, cannot produce any clear or distinct knowledge; because, as far as any ideas are confused, so far the mind cannot perceive clearly whether

they agree or disagree. Or to express the same thing in a way less apt to be misunderstood: he that hath not determined ideas to the words he uses, cannot make propositions of them of whose truth he can be certain.

Chapter III Of the Extent of Human Knowledge

1. Extent of our knowledge. Knowledge, as has been said, lying in the perception of the agreement or disagreement of any of our ideas, it follows from hence That, It extends no further than we have ideas. First, we can have knowledge no further than we have ideas.

2. It extends no further than we can perceive their agreement or disagreement. Secondly, That we can have no knowledge further than we can have perception of that agreement or disagreement. Which perception being: 1. Either by intuition, or the immediate comparing any two ideas; or, 2. By reason, examining the agreement or disagreement of two ideas, by the intervention of some others; or, 3. By sensation, perceiving the ex-

istence of particular things: hence it also follows:

3. Intuitive knowledge extends itself not to all the relations of all our ideas. Thirdly, That we cannot have an intuitive knowledge that shall extend itself to all our ideas, and all that we would know about them; because we cannot examine and perceive all the relations they have one to another, by juxta-position, or an immediate comparison one with another. Thus, having the ideas of an obtuse and an acute angled triangle, both drawn from equal bases, and between parallels, I can, by intuitive knowledge, perceive the one not to be the other, but cannot that way know whether they be equal or no; because their agreement or disagreement in equality can never be perceived by an immediate comparing them: the difference of figure makes their parts incapable of an exact immediate application; and therefore there is need of some intervening qualities to measure them by, which is demonstration, or rational knowledge.

4. Nor does demonstrative knowledge. Fourthly, It follows, also, from what is above observed, that our ratio-

nal knowledge cannot reach to the whole extent of our ideas: because between two different ideas we would examine, we cannot always find such mediums as we can connect one to another with an intuitive knowledge in all the parts of the deduction; and wherever that fails, we come short of knowledge and demonstration.

5. Sensitive knowledge narrower than either. Fifthly, Sensitive knowledge reaching no further than the existence of things actually present to our senses, is yet much narrower than either of the former.

6. Our knowledge, therefore, narrower than our ideas. Sixthly, From all which it is evident, that the extent of our knowledge comes not only short of the reality of things, but even of the extent of our own ideas. Though our knowledge be limited to our ideas, and cannot exceed them either in extent or perfection; and though these be very narrow bounds, in respect of the extent of All-being, and far short of what we may justly imagine to be in some even created understandings, not tied down to the dull and narrow information that is to be

received from some few, and not very acute, ways of perception, such as are our senses; yet it would be well with us if our knowledge were but as large as our ideas, and there were not many doubts and inquiries concerning the ideas we have, whereof we are not, nor I believe ever shall be in this world resolved. Nevertheless I do not question but that human knowledge, under the present circumstances of our beings and constitutions, may be carried much further than it has hitherto been, if men would sincerely, and with freedom of mind, employ all that industry and labour of thought, in improving the means of discovering truth, which they do for the colouring or support of falsehood, to maintain a system, interest, or party they are once engaged in. But yet after all, I think I may, without injury to human perfection, be confident, that our knowledge would never reach to all we might desire to know concerning those ideas we have; nor be able to surmount all the difficulties, and resolve all the questions that might arise concerning any of them. We have the ideas of a square, a circle, and equality; and yet, perhaps, shall never be

able to find a circle equal to a square, and certainly know that it is so. We have the ideas of matter and thinking, but possibly shall never be able to know whether any mere material being thinks or no; it being impossible for us, by the contemplation of our own ideas, without revelation, to discover whether Omnipotency has not given to some systems of matter, fitly disposed, a power to perceive and think, or else joined and fixed to matter, so disposed, a thinking immaterial substance: it being, in respect of our notions, not much more remote from our comprehension to conceive that GOD can, if he pleases, superadd to matter a faculty of thinking, than that he should superadd to it another substance with a faculty of thinking; since we know not wherein thinking consists, nor to what sort of substances the Almighty has been pleased to give that power, which cannot be in any created being, but merely by the good pleasure and bounty of the Creator.

Whether Matter may not be made by God to think is more than man can know. For I see no contradiction in it, that the first Eternal thinking Being, or Omnipotent

Spirit, should, if he pleased, give to certain systems of created senseless matter, put together as he thinks fit, some degrees of sense, perception, and thought: though, as I think I have proved, Bk. iv. ch. 10, SS 14, &c., it is no less than a contradiction to suppose matter (which is evidently in its own nature void of sense and thought) should be that Eternal first-thinking Being. What certainty of knowledge can any one have, that some perceptions, such as, v.g., pleasure and pain, should not be in some bodies themselves, after a certain manner modified and moved, as well as that they should be in an immaterial substance, upon the motion of the parts of body: Body, as far as we can conceive, being able only to strike and affect body, and motion, according to the utmost reach of our ideas, being able to produce nothing but motion; so that when we allow it to produce pleasure or pain, or the idea of a colour or sound, we are fain to quit our reason, go beyond our ideas, and attribute it wholly to the good pleasure of our Maker. For, since we must allow He has annexed effects to motion which we can no way conceive motion able to pro-

duce, what reason have we to conclude that He could not order them as well to be produced in a subject we cannot conceive capable of them, as well as in a subject we cannot conceive the motion of matter can any way operate upon? I say not this, that I would any way lessen the belief of the soul's immateriality: I am not here speaking of probability, but knowledge; and I think not only that it becomes the modesty of philosophy not to pronounce magisterially, where we want that evidence that can produce knowledge; but also, that it is of use to us to discern how far our knowledge does reach; for the state we are at present in, not being that of vision, we must in many things content ourselves with faith and probability: and in the present question, about the Immateriality of the Soul, if our faculties cannot arrive at demonstrative certainty, we need not think it strange. All the great ends of morality and religion are well enough secured, without philosophical proofs of the soul's immateriality; since it is evident, that he who made us at the beginning to subsist here, sensible intelligent beings, and for several years continued us in such

a state, can and will restore us to the like state of sensibility in another world, and make us capable there to receive the retribution he has designed to men, according to their doings in this life. And therefore it is not of such mighty necessity to determine one way or the other, as some, over-zealous for or against the immateriality of the soul, have been forward to make the world believe. Who, either on the one side, indulging too much their thoughts immersed altogether in matter, can allow no existence to what is not material: or who, on the other side, finding not cogitation within the natural powers of matter, examined over and over again by the utmost intention of mind, have the confidence to conclude—That Omnipotency itself cannot give perception and thought to a substance which has the modification of solidity. He that considers how hardly sensation is, in our thoughts, reconcilable to extended matter; or existence to anything that has no extension at all, will confess that he is very far from certainly knowing what his soul is. It is a point which seems to me to be put out of the reach of our knowledge: and he who will give him-

self leave to consider freely, and look into the dark and intricate part of each hypothesis, will scarce find his reason able to determine him fixedly for or against the soul's materiality. Since, on which side soever he views it, either as an unextended substance, or as a thinking extended matter, the difficulty to conceive either will, whilst either alone is in his thoughts, still drive him to the contrary side. An unfair way which some men take with themselves: who, because of the inconceivableness of something they find in one, throw themselves violently into the contrary hypothesis, though altogether as unintelligible to an unbiassed understanding. This serves not only to show the weakness and the scantiness of our knowledge, but the insignificant triumph of such sort of arguments; which, drawn from our own views, may satisfy us that we can find no certainty on one side of the question: but do not at all thereby help us to truth by running into the opposite opinion; which, on examination, will be found clogged with equal difficulties. For what safety, what advantage to any one is it, for the avoiding the seeming absurdities, and to him

unsurmountable rubs, he meets with in one opinion, to take refuge in the contrary, which is built on something altogether as inexplicable, and as far remote from his comprehension? It is past controversy, that we have in us something that thinks; our very doubts about what it is, confirm the certainty of its being, though we must content ourselves in the ignorance of what kind of being it is: and it is in vain to go about to be sceptical in this, as it is unreasonable in most other cases to be positive against the being of anything, because we cannot comprehend its nature. For I would fain know what substance exists, that has not something in it which manifestly baffles our understandings. Other spirits, who see and know the nature and inward constitution of things, how much must they exceed us in knowledge? To which, if we add larger comprehension, which enables them at one glance to see the connexion and agreement of very many ideas, and readily supplies to them the intermediate proofs, which we by single and slow steps, and long poring in the dark, hardly at last find out, and are often ready to forget one before we have

hunted out another; we may guess at some part of the happiness of superior ranks of spirits, who have a quicker and more penetrating sight, as well as a larger field of knowledge.

But to return to the argument in hand: our knowledge, I say, is not only limited to the paucity and imperfections of the ideas we have, and which we employ it about, but even comes short of that too: but how far it reaches, let us now inquire.

7. How far our knowledge reaches. The affirmations or negations we make concerning the ideas we have, may, as I have before intimated in general, be reduced to these four sorts, viz. identity, co-existence, relation, and real existence. I shall examine how far our knowledge extends in each of these:

8. Our knowledge of identity and diversity in ideas extends as far as our ideas themselves. First, as to identity and diversity. In this way of agreement or disagreement of our ideas, our intuitive knowledge is as far extended as our ideas themselves: and there can be no idea in the mind, which it does not, presently, by an intuitive knowl-

edge, perceive to be what it is, and to be different from any other.

9. Of their co-existence, extends only a very little way. Secondly, as to the second sort, which is the agreement or disagreement of our ideas in co-existence, in this our knowledge is very short; though in this consists the greatest and most material part of our knowledge concerning substances. For our ideas of the species of substances being, as I have showed, nothing but certain collections of simple ideas united in one subject, and so co-existing together; v.g. our idea of flame is a body hot, luminous, and moving upward; of gold, a body heavy to a certain degree, yellow, malleable, and fusible: for these, or some such complex ideas as these, in men's minds, do these two names of the different substances, flame and gold, stand for. When we would know anything further concerning these, or any other sort of substances, what do we inquire, but what other qualities or powers these substances have or have not? Which is nothing else but to know what other simple ideas do, or do not co-exist with those that make up that complex idea?

10. Because the connexion between simple ideas in substances is for the most part unknown. This, how weighty and considerable a part soever of human science, is yet very narrow, and scarce any at all. The reason whereof is, that the simple ideas whereof our complex ideas of substances are made up are, for the most part, such as carry with them, in their own nature, no visible necessary connexion or inconsistency with any other simple ideas, whose co-existence with them we would inform ourselves about.

11. Especially of the secondary qualities of bodies. The ideas that our complex ones of substances are made up of, and about which our knowledge concerning substances is most employed, are those of their secondary qualities; which depending all (as has been shown) upon the primary qualities of their minute and insensible parts; or, if not upon them, upon something yet more remote from our comprehension; it is impossible we should know which have a necessary union or inconsistency one with another. For, not knowing the root they spring from, not knowing what size, figure, and texture of parts they

are, on which depend, and from which result those qualities which make our complex idea of gold, it is impossible we should know what other qualities result from, or are incompatible with, the same constitution of the insensible parts of gold; and so consequently must always co-exist with that complex idea we have of it, or else are inconsistent with it.

12. Because necessary connexion between any secondary and the primary qualities is undiscoverable by us. Besides this ignorance of the primary qualities of the insensible parts of bodies, on which depend all their secondary qualities, there is yet another and more incurable part of ignorance, which sets us more remote from a certain knowledge of the co-existence or inco-existence (if I may so say) of different ideas in the same subject; and that is, that there is no discoverable connexion between any secondary quality and those primary qualities which it depends on.

13. We have no perfect knowledge of their primary qualities. That the size, figure, and motion of one body should cause a change in the size, figure, and motion of an-

other body, is not beyond our conception; the separation of the parts of one body upon the intrusion of another; and the change from rest to motion upon impulse; these and the like seem to have some connexion one with another. And if we knew these primary qualities of bodies, we might have reason to hope we might be able to know a great deal more of these operations of them one upon another: but our minds not being able to discover any connexion betwixt these primary qualities of bodies and the sensations that are produced in us by them, we can never be able to establish certain and undoubted rules of the consequence or co-existence of any secondary qualities, though we could discover the size, figure, or motion of those invisible parts which immediately produce them. We are so far from knowing what figure, size, or motion of parts produce a yellow colour, a sweet taste, or a sharp sound, that we can by no means conceive how any size, figure, or motion of any particles, can possibly produce in us the idea of any colour, taste, or sound whatsoever: there is no conceivable connexion between the one and the other.

14. And seek in vain for certain and universal knowledge of unperceived qualities in substances. In vain, therefore, shall we endeavour to discover by our ideas (the only true way of certain and universal knowledge) what other ideas are to be found constantly joined with that of our complex idea of any substance: since we neither know the real constitution of the minute parts on which their qualities do depend; nor, did we know them, could we discover any necessary connexion between them and any of the secondary qualities: which is necessary to be done before we can certainly know their necessary co-existence. So, that, let our complex idea of any species of substances be what it will, we can hardly, from the simple ideas contained in it, certainly determine the necessary co-existence of any other quality whatsoever. Our knowledge in all these inquiries reaches very little further than our experience. Indeed some few of the primary qualities have a necessary dependence and visible connexion one with another, as figure necessarily supposes extension; receiving or communicating motion by impulse, supposes solidity. But

though these, and perhaps some others of our ideas have: yet there are so few of them that have a visible connexion one with another, that we can by intuition or demonstration discover the co-existence of very few of the qualities that are to be found united in substances: and we are left only to the assistance of our senses to make known to us what qualities they contain. For of all the qualities that are co-existent in any subject, without this dependence and evident connexion of their ideas one with another, we cannot know certainly any two to co-exist, any further than experience, by our senses, informs us. Thus, though we see the yellow colour, and, upon trial, find the weight, malleableness, fusibility, and fixedness that are united in a piece of gold, yet; because no one of these ideas has any evident dependence or necessary connexion with the other, we cannot certainly know that where any four of these are, the fifth will be there also, how highly probable soever it may be; because the highest probability amounts not to certainty, without which there can be no true knowledge. For this co-existence can be no fur-

ther known than it is perceived; and it cannot be perceived but either in particular subjects, by the observation of our senses, or, in general, by the necessary connexion of the ideas themselves.

15. Of repugnancy to co-exist, our knowledge is larger. As to the incompatibility or repugnancy to coexistence, we may know that any subject may have of each sort of primary qualities but one particular at once: v.g. each particular extension, figure, number of parts, motion, excludes all other of each kind. The like also is certain of all sensible ideas peculiar to each sense; for whatever of each kind is present in any subject, excludes all other of that sort: v.g. no one subject can have two smells or two colours at the same time. To this, perhaps will be said, Has not an opal, or the infusion of lignum nephriticum, two colours at the same time? To which I answer, that these bodies, to eyes differently placed, may at the same time afford different colours: but I take liberty also to say, to eyes differently placed, it is different parts of the object that reflect the particles of light: and therefore it is not the same part of the object, and

so not the very same subject, which at the same time appears both yellow and azure. For, it is as impossible that the very same particle of any body should at the same time differently modify or reflect the rays of light, as that it should have two different figures and textures at the same time.

16. Our knowledge of the co-existence of powers in bodies extends but a very little way. But as to the powers of substances to change the sensible qualities of other bodies, which make a great part of our inquiries about them, and is no inconsiderable branch of our knowledge; I doubt as to these, whether our knowledge reaches much further than our experience; or whether we can come to the discovery of most of these powers, and be certain that they are in any subject, by the connexion with any of those ideas which to us make its essence. Because the active and passive powers of bodies, and their ways of operating, consisting in a texture and motion of parts which we cannot by any means come to discover; it is but in very few cases we can be able to perceive their dependence on, or repugnance to, any of

those ideas which make our complex one of that sort of things. I have here instanced in the corpuscularian hypothesis, as that which is thought to go furthest in an intelligible explication of those qualities of bodies; and I fear the weakness of human understanding is scarce able to substitute another, which will afford us a fuller and clearer discovery of the necessary connexion and coexistence of the powers which are to be observed united in several sorts of them. This at least is certain, that, whichever hypothesis be clearest and truest, (for of that it is not my business to determine,) our knowledge concerning corporeal substances will be very little advanced by any of them, till we are made to see what qualities and powers of bodies have a necessary connexion or repugnancy one with another; which in the present state of philosophy I think we know but to a very small degree: and I doubt whether, with those faculties we have, we shall ever be able to carry our general knowledge (I say not particular experience) in this part much further. Experience is that which in this part we must depend on. And it were to be wished that it were more

improved. We find the advantages some men's generous pains have this way brought to the stock of natural knowledge. And if others, especially the philosophers by fire, who pretend to it, had been so wary in their observations, and sincere in their reports as those who call themselves philosophers ought to have been, our acquaintance with the bodies here about us, and our insight into their powers and operations had been yet much greater.

17. Of the powers that co-exist in spirits yet narrower. If we are at a loss in respect of the powers and operations of bodies, I think it is easy to conclude we are much more in the dark in reference to spirits; whereof we naturally have no ideas but what we draw from that of our own, by reflecting on the operations of our own souls within us, as far as they can come within our observation. But how inconsiderable a rank the spirits that inhabit our bodies hold amongst those various and possibly innumerable kinds of nobler beings; and how far short they come of the endowments and perfections of cherubim and seraphim, and infinite sorts of spirits

above us, is what by a transient hint in another place I have offered to my reader's consideration.

18. Of relations between abstracted ideas it is not easy to say how far our knowledge extends. Thirdly, As to the third sort of our knowledge, viz. the agreement or disagreement of any of our ideas in any other relation: this, as it is the largest field of our knowledge, so it is hard to determine how far it may extend: because the advances that are made in this part of knowledge, depending on our sagacity in finding intermediate ideas, that may show the relations and habitudes of ideas whose co-existence is not considered, it is a hard matter to tell when we are at an end of such discoveries; and when reason has all the helps it is capable of, for the finding of proofs or examining the agreement or disagreement of remote ideas. They that are ignorant of Algebra cannot imagine the wonders in this kind are to be done by it: and what further improvements and helps advantageous to other parts of knowledge the sagacious mind of man may yet find out, it is not easy to determine. This at least I believe, that the ideas of quantity are not

those alone that are capable of demonstration and knowledge; and that other, and perhaps more useful, parts of contemplation, would afford us certainty, if vices, passions, and domineering interest did not oppose or menace such endeavours.

Morality capable of demonstration. The idea of a supreme Being, infinite in power, goodness, and wisdom, whose workmanship we are, and on whom we depend; and the idea of ourselves, as understanding, rational creatures, being such as are clear in us, would, I suppose, if duly considered and pursued, afford such foundations of our duty and rules of action as might place morality amongst the sciences capable of demonstration: wherein I doubt not but from self-evident propositions, by necessary consequences, as incontestible as those in mathematics, the measures of right and wrong might be made out, to any one that will apply himself with the same indifferency and attention to the one as he does to the other of these sciences. The relation of other modes may certainly be perceived, as well as those of number and extension: and I cannot see why they should not

also be capable of demonstration, if due methods were thought on to examine or pursue their agreement or disagreement. "Where there is no property there is no injustice," is a proposition as certain as any demonstration in Euclid: for the idea of property being a right to anything, and the idea to which the name "injustice" is given being the invasion or violation of that right, it is evident that these ideas, being thus established, and these names annexed to them, I can as certainly know this proposition to be true, as that a triangle has three angles equal to two right ones. Again: "No government allows absolute liberty." The idea of government being the establishment of society upon certain rules or laws which require conformity to them; and the idea of absolute liberty being for any one to do whatever he pleases; I am as capable of being certain of the truth of this proposition as of any in the mathematics.

19. Two things have made moral ideas to be thought incapable of demonstration: their unfitness for sensible representation, and their complexedness. That which in this respect has given the advantage to the ideas of

quantity, and made them thought more capable of certainty and demonstration, is,

First, That they can be set down and represented by sensible marks, which have a greater and nearer correspondence with them than any words or sounds whatsoever. Diagrams drawn on paper are copies of the ideas in the mind, and not liable to the uncertainty that words carry in their signification. An angle, circle, or square, drawn in lines, lies open to the view, and cannot be mistaken: it remains unchangeable, and may at leisure be considered and examined, and the demonstration be revised, and all the parts of it may be gone over more than once, without any danger of the least change in the ideas. This cannot be thus done in moral ideas: we have no sensible marks that resemble them, whereby we can set them down; we have nothing but words to express them by; which, though when written they remain the same, yet the ideas they stand for may change in the same man; and it is very seldom that they are not different in different persons.

Secondly, Another thing that makes the greater diffi-

culty in ethics is, That moral ideas are commonly more complex than those of the figures ordinarily considered in mathematics. From whence these two inconveniences follow:—First, that their names are of more uncertain signification, the precise collection of simple ideas they stand for not being so easily agreed on; and so the sign that is used for them in communication always, and in thinking often, does not steadily carry with it the same idea. Upon which the same disorder, confusion, and error follow, as would if a man, going to demonstrate something of an heptagon, should, in the diagram he took to do it, leave out one of the angles, or by oversight make the figure with one angle more than the name ordinarily imported, or he intended it should when at first he thought of his demonstration. This often happens, and is hardly avoidable in very complex moral ideas, where the same name being retained, one angle, i.e. one simple idea, is left out, or put in the complex one (still called by the same name) more at one time than another. Secondly, From the complexedness of these moral ideas there follows another inconvenience, viz. that the

mind cannot easily retain those precise combinations so exactly and perfectly as is necessary in the examination of the habitudes and correspondences, agreements or disagreements, of several of them one with another; especially where it is to be judged of by long deductions, and the intervention of several other complex ideas to show the agreement or disagreement of two remote ones.

The great help against this which mathematicians find in diagrams and figures, which remain unalterable in their draughts, is very apparent, and the memory would often have great difficulty otherwise to retain them so exactly, whilst the mind went over the parts of them step by step to examine their several correspondences. And though in casting up a long sum either in addition, multiplication, or division, every part be only a progression of the mind taking a view of its own ideas, and considering their agreement or disagreement, and the resolution of the question be nothing but the result of the whole, made up of such particulars, whereof the mind has a clear perception: yet, without setting down the several parts by marks, whose precise significations

are known, and by marks that last, and remain in view when the memory had let them go, it would be almost impossible to carry so many different ideas in the mind, without confounding or letting slip some parts of the reckoning, and thereby making all our reasonings about it useless. In which case the cyphers or marks help not the mind at all to perceive the agreement of any two or more numbers, their equalities or proportions; that the mind has only by intuition of its own ideas of the numbers themselves. But the numerical characters are helps to the memory, to record and retain the several ideas about which the demonstration is made, whereby a man may know how far his intuitive knowledge in surveying several of the particulars has proceeded; that so he may without confusion go on to what is yet unknown; and at last have in one view before him the result of all his perceptions and reasonings.

20. Remedies of our difficulties in dealing demonstratively with moral ideas. One part of these disadvantages in moral ideas which has made them be thought not capable of demonstration, may in a good measure be

remedied by definitions, setting down that collection of simple ideas, which every term shall stand for: and then using the terms steadily and constantly for that precise collection. And what methods algebra, or something of that kind, may hereafter suggest, to remove the other difficulties, it is not easy to foretell. Confident I am, that, if men would in the same method, and with the same indifferency, search after moral as they do mathematical truths, they would find them have a stronger connexion one with another, and a more necessary consequence from our clear and distinct ideas, and to come nearer perfect demonstration than is commonly imagined. But much of this is not to be expected, whilst the desire of esteem, riches, or power makes men espouse the well-endowed opinions in fashion, and then seek arguments either to make good their beauty, or varnish over and cover their deformity. Nothing being so beautiful to the eye as truth is to the mind; nothing so deformed and irreconcilable to the understanding as a lie. For though many a man can with satisfaction enough own a no very handsome wife to in his bosom; yet who

is bold enough openly to avow that he has espoused a falsehood, and received into his breast so ugly a thing as a lie? Whilst the parties of men cram their tenets down all men's throats whom they can get into their power, without permitting them to examine their truth or falsehood; and will not let truth have fair play in the world, nor men the liberty to search after it: what improvements can be expected of this kind? What greater light can be hoped for in the moral sciences? The subject part of mankind in most places might, instead thereof, with Egyptian bondage, expect Egyptian darkness, were not the candle of the Lord set up by himself in men's minds, which it is impossible for the breath or power of man wholly to extinguish.

21. Of the three real existences of which we have certain knowledge.

Fourthly, As to the fourth sort of our knowledge, viz. of the real actual existence of things, we have an intuitive knowledge of our own existence, and a demonstrative knowledge of the existence of a God: of the existence of anything else, we have no other but a sensitive

knowledge; which extends not beyond the objects present to our senses.

22. Our ignorance great. Our knowledge being so narrow, as I have shown, it will perhaps give us some light into the present state of our minds if we look a little into the dark side, and take a view of our ignorance; which, being infinitely larger than our knowledge, may serve much to the quieting of disputes, and improvement of useful knowledge; if, discovering how far we have clear and distinct ideas, we confine our thoughts within the contemplation of those things that are within the reach of our understandings, and launch not out into that abyss of darkness, (where we have not eyes to see, nor faculties to perceive anything), out of a presumption that nothing is beyond our comprehension. But to be satisfied of the folly of such a conceit, we need not go far. He that knows anything, knows this, in the first place, that he need not seek long for instances of his ignorance. The meanest and most obvious things that come in our way have dark sides, that the quickest sight cannot penetrate into. The clearest and most en-

larged understandings of thinking men find themselves puzzled and at a loss in every particle of matter. We shall the less wonder to find it so, when we consider the causes of our ignorance; which, from what has been said, I suppose will be found to be these three:—

Its causes. First, Want of ideas.

Secondly, Want of a discoverable connexion between the ideas we have.

Thirdly, Want of tracing and examining our ideas.

23. One cause of our ignorance want of ideas. First, There are some things, and those not a few, that we are ignorant of, for want of ideas.

I. Want of simple ideas that other creatures in other parts of the universe may have. First, all the simple ideas we have are confined (as I have shown) to those we receive from corporeal objects by sensation, and from the operations of our own minds as the objects of reflection. But how much these few and narrow inlets are disproportionate to the vast whole extent of all beings, will not be hard to persuade those who are not so foolish as to think their span the measure of all things.

What other simple ideas it is possible the creatures in other parts of the universe may have, by the assistance of senses and faculties more or perfecter than we have, or different from ours, it is not for us to determine. But to say or think there are no such, because we conceive nothing of them, is no better an argument than if a blind man should be positive in it, that there was no such thing as sight and colours, because he had no manner of idea of any such thing, nor could by any means frame to himself any notions about seeing. The ignorance and darkness that is in us no more hinders nor confines the knowledge that is in others, than the blindness of a mole is an argument against the quicksightedness of an eagle. He that will consider the infinite power, wisdom, and goodness of the Creator of all things will find reason to think it was not all laid out upon so inconsiderable, mean, and impotent a creature as he will find man to be; who in all probability is one of the lowest of all intellectual beings. What faculties, therefore, other species of creatures have to penetrate into the nature and inmost constitutions of things; what

ideas they may receive of them far different from ours, we know not. This we know and certainly find, that we want several other views of them besides those we have, to make discoveries of them more perfect. And we may be convinced that the ideas we can attain to by our faculties are very disproportionate to things themselves, when a positive, clear, distinct one of substance itself, which is the foundation of all the rest, is concealed from us. But want of ideas of this kind, being a part as well as cause of our ignorance, cannot be described. Only this I think I may confidently say of it, That the intellectual and sensible world are in this perfectly alike: that that part which we see of either of them holds no proportion with what we see not; and whatsoever we can reach with our eyes or our thoughts of either of them is but a point, almost nothing in comparison of the rest.

24. Want of simple ideas that men are capable of having, but have not, because of their remoteness. Secondly, Another great cause of ignorance is the want of ideas we are capable of. As the want of ideas which our

faculties are not able to give us shuts us wholly from those views of things which it is reasonable to think other beings, perfecter than we, have, of which we know nothing; so the want of ideas I now speak of keeps us in ignorance of things we conceive capable of being known to us. Bulk, figure, and motion we have ideas of. But though we are not without ideas of these primary qualities of bodies in general, yet not knowing what is the particular bulk, figure, and motion, of the greatest part of the bodies of the universe, we are ignorant of the several powers, efficacies, and ways of operation, whereby the effects which we daily see are produced. These are hid from us, in some things by being too remote, and in others by being too minute. When we consider the vast distance of the known and visible parts of the world, and the reasons we have to think that what lies within our ken is but a small part of the universe, we shall then discover a huge abyss of ignorance. What are the particular fabrics of the great masses of matter which make up the whole stupendous frame of corporeal beings; how far they are extended; what is their motion,

and how continued or communicated; and what influence they have one upon another, are contemplations that at first glimpse our thoughts lose themselves in. If we narrow our contemplations, and confine our thoughts to this little canton—I mean this system of our sun, and the grosser masses of matter that visibly move about it, What several sorts of vegetables, animals, and intellectual corporeal beings, infinitely different from those of our little spot of earth, may there probably be in the other planets, to the knowledge of which, even of their outward figures and parts, we can no way attain whilst we are confined to this earth; there being no natural means, either by sensation or reflection, to convey their certain ideas into our minds? They are out of the reach of those inlets of all our knowledge: and what sorts of furniture and inhabitants those mansions contain in them we cannot so much as guess, much less have clear and distinct ideas of them.

25. Because of their minuteness. If a great, nay, far the greatest part of the several ranks of bodies in the universe escape our notice by their remoteness, there are

others that are no less concealed from us by their minuteness. These insensible corpuscles, being the active parts of matter, and the great instruments of nature, on which depend not only all their secondary qualities, but also most of their natural operations, our want of precise distinct ideas of their primary qualities keeps us in an incurable ignorance of what we desire to know about them. I doubt not but if we could discover the figure, size, texture, and motion of the minute constituent parts of any two bodies, we should know without trial several of their operations one upon another; as we do now the properties of a square or a triangle. Did we know the mechanical affections of the particles of rhubarb, hemlock, opium, and a man, as a watchmaker does those of a watch, whereby it performs its operations; and of a file, which by rubbing on them will alter the figure of any of the wheels; we should be able to tell beforehand that rhubarb will purge, hemlock kill, and opium make a man sleep: as well as a watchmaker can, that a little piece of paper laid on the balance will keep the watch from going till it be removed; or that, some small part of

it being rubbed by a file, the machine would quite lose its motion, and the watch go no more. The dissolving of silver in aqua fortis, and gold in aqua regia, and not vice versa, would be then perhaps no more difficult to know than it is to a smith to understand why the turning of one key will open a lock, and not the turning of another. But whilst we are destitute of senses acute enough to discover the minute particles of bodies, and to give us ideas of their mechanical affections, we must be content to be ignorant of their properties and ways of operation; nor can we be assured about them any further than some few trials we make are able to reach. But whether they will succeed again another time, we cannot be certain. This hinders our certain knowledge of universal truths concerning natural bodies: and our reason carries us herein very little beyond particular matter of fact.

26. Hence no science of bodies within our reach. And therefore I am apt to doubt that, how far soever human industry may advance useful and experimental philosophy in physical things, scientific will still be out of our

reach: because we want perfect and adequate ideas of those very bodies which are nearest to us, and most under our command. Those which we have ranked into classes under names, and we think ourselves best acquainted with, we have but very imperfect and incomplete ideas of. Distinct ideas of the several sorts of bodies that fall under the examination of our senses perhaps we may have: but adequate ideas, I suspect, we have not of any one amongst them. And though the former of these will serve us for common use and discourse, yet whilst we want the latter, we are not capable of scientific knowledge; nor shall ever be able to discover general, instructive, unquestionable truths concerning them. Certainty and demonstration are things we must not, in these matters, pretend to. By the colour, figure, taste, and smell, and other sensible qualities, we have as clear and distinct ideas of sage and hemlock, as we have of a circle and a triangle: but having no ideas of the particular primary qualities of the minute parts of either of these plants, nor of other bodies which we would apply them to, we cannot tell what effects they

will produce; nor when we see those effects can we so much as guess, much less know, their manner of production. Thus, having no ideas of the particular mechanical affections of the minute parts of bodies that are within our view and reach, we are ignorant of their constitutions, powers, and operations: and of bodies more remote we are yet more ignorant, not knowing so much as their very outward shapes, or the sensible and grosser parts of their constitutions.

27. Much less a science of unembodied spirits. This at first will show us how disproportionate our knowledge is to the whole extent even of material beings; to which if we add the consideration of that infinite number of spirits that may be, and probably are, which are yet more remote from our knowledge, whereof we have no cognizance, nor can frame to ourselves any distinct ideas of their several ranks and sorts, we shall find this cause of ignorance conceal from us, in an impenetrable obscurity, almost the whole intellectual world; a greater certainty, and more beautiful world than the material. For, bating some very few, and those, if I may so call them, superficial ideas of spirit, which

by reflection we get of our own, and from thence the best we can collect of the Father of all spirits, the eternal independent Author of them, and us, and all things, we have no certain information, so much as of the existence of other spirits, but by revelation. Angels of all sorts are naturally beyond our discovery; and all those intelligences, whereof it is likely there are more orders than of corporeal substances, are things whereof our natural faculties give us no certain account at all. That there are minds and thinking beings in other men as well as himself, every man has a reason, from their words and actions, to be satisfied: and the knowledge of his own mind cannot suffer a man that considers, to be ignorant that there is a God. But that there are degrees of spiritual beings between us and the great God, who is there, that, by his own search and ability, can come to know? Much less have we distinct ideas of their different natures, conditions, states, powers, and several constitutions wherein they agree or differ from one another and from us. And, therefore, in what concerns their different species and properties we are in absolute ignorance.

28. Another cause, want of a discoverable connexion between ideas we have. Secondly, What a small part of the substantial beings that are in the universe the want of ideas leaves open to our knowledge, we have seen. In the next place, another cause of ignorance, of no less moment, is a want of a discoverable connexion between those ideas we have. For wherever we want that, we are utterly incapable of universal and certain knowledge; and are, in the former case, left only to observation and experiment: which, how narrow and confined it is, how far from general knowledge we need not be told. I shall give some few instances of this cause of our ignorance, and so leave it. It is evident that the bulk, figure, and motion of several bodies about us produce in us several sensations, as of colours, sounds, tastes, smells, pleasure, and pain, &c. These mechanical affections of bodies having no affinity at all with those ideas they produce in us, (there being no conceivable connexion between any impulse of any sort of body and any perception of a colour or smell which we find in our minds,) we can have no distinct knowledge of such operations beyond our experience; and

can reason no otherwise about them, than as effects produced by the appointment of an infinitely Wise Agent, which perfectly surpass our comprehensions. As the ideas of sensible secondary qualities which we have in our minds, can by us be no way deduced from bodily causes, nor any correspondence or connexion be found between them and those primary qualities which (experience shows us) produce them in us; so, on the other side, the operation of our minds upon our bodies is as inconceivable. How any thought should produce a motion in body is as remote from the nature of our ideas, as how any body should produce any thought in the mind. That it is so, if experience did not convince us, the consideration of the things themselves would never be able in the least to discover to us. These, and the like, though they have a constant and regular connexion in the ordinary course of things; yet that connexion being not discoverable in the ideas themselves, which appearing to have no necessary dependence one on another, we can attribute their connexion to nothing else but the arbitrary determination of that All-wise Agent who has made them to be, and to operate as they

do, in a way wholly above our weak understandings to conceive.

29. Instances. In some of our ideas there are certain relations, habitudes, and connexions, so visibly included in the nature of the ideas themselves, that we cannot conceive them separable from them by any power whatsoever. And in these only we are capable of certain and universal knowledge. Thus the idea of a right-lined triangle necessarily carries with it an equality of its angles to two right ones. Nor can we conceive this relation, this connexion of these two ideas, to be possibly mutable, or to depend on any arbitrary power, which of choice made it thus, or could make it otherwise. But the coherence and continuity of the parts of matter; the production of sensation in us of colours and sounds, &c., by impulse and motion; nay, the original rules and communication of motion being such, wherein we can discover no natural connexion with any ideas we have, we cannot but ascribe them to the arbitrary will and good pleasure of the Wise Architect. I need not, I think, here mention the resurrection of the dead, the future

state of this globe of earth, and such other things, which are by every one acknowledged to depend wholly on the determination of a free agent. The things that, as far as our observation reaches, we constantly find to proceed regularly, we may conclude do act by a law set them; but yet by a law that we know not: whereby, though causes work steadily, and effects constantly flow from them, yet their connexions and dependencies being not discoverable in our ideas, we can have but an experimental knowledge of them. From all which it is easy to perceive what a darkness we are involved in, how little it is of Being, and the things that are, that we are capable to know. And therefore we shall do no injury to our knowledge, when we modestly think with ourselves, that we are so far from being able to comprehend the whole nature of the universe and all the things contained in it, that we are not capable of a philosophical knowledge of the bodies that are about us, and make a part of us: concerning their secondary qualities, powers, and operations, we can have no universal certainty. Several effects come every day within the notice of our

senses, of which we have so far sensitive knowledge: but the causes, manner, and certainty of their production, for the two foregoing reasons, we must be content to be very ignorant of. In these we can go no further than particular experience informs us matter of fact, and by analogy to guess what effects the like bodies are, upon other trials, like to produce. But as to a perfect science of natural bodies, (not to mention spiritual beings,) we are, I think, so far from being capable of any such thing, that I conclude it lost labour to seek after it.

30. A third cause, want of tracing our ideas. Thirdly, Where we have adequate ideas, and where there is a certain and discoverable connexion between them, yet we are often ignorant, for want of tracing those ideas which we have or may have; and for want of finding out those intermediate ideas, which may show us what habitude of agreement or disagreement they have one with another. And thus many are ignorant of mathematical truths, not out of any imperfection of their faculties, or uncertainty in the things themselves, but for want of

application in acquiring, examining, and by due ways comparing those ideas. That which has most contributed to hinder the due tracing of our ideas, and finding out their relations, and agreements or disagreements, one with another, has been, I suppose, the ill use of words. It is impossible that men should ever truly seek or certainly discover the agreement or disagreement of ideas themselves, whilst their thoughts flutter about, or stick only in sounds of doubtful and uncertain significations. Mathematicians abstracting their thoughts from names, and accustoming themselves to set before their minds the ideas themselves that they would consider, and not sounds instead of them, have avoided thereby a great part of that perplexity, puddering, and confusion, which has so much hindered men's progress in other parts of knowledge. For whilst they stick in words of undetermined and uncertain signification, they are unable to distinguish true from false, certain from probable, consistent from inconsistent, in their own opinions. This having been the fate or misfortune of a great part of men of letters, the increase brought into the

stock of real knowledge has been very little, in proportion to the schools, disputes, and writings, the world has been filled with; whilst students, being lost in the great wood of words, knew not whereabouts they were, how far their discoveries were advanced, or what was wanting in their own, or the general stock of knowledge. Had men, in the discoveries of the material, done as they have in those of the intellectual world, involved all in the obscurity of uncertain and doubtful ways of talking, volumes writ of navigation and voyages, theories and stories of zones and tides, multiplied and disputed; nay, ships built, and fleets sent out, would never have taught us the way beyond the line; and the Antipodes would be still as much unknown, as when it was declared heresy to hold there were any. But having spoken sufficiently of words, and the ill or careless use that is commonly made of them, I shall not say anything more of it here.

31. Extent of human knowledge in respect to its universality. Hitherto we have examined the extent of our knowledge, in respect of the several sorts of beings that

are. There is another extent of it, in respect of universality, which will also deserve to be considered; and in this regard, our knowledge follows the nature of our ideas. If the ideas are abstract, whose agreement or disagreement we perceive, our knowledge is universal. For what is known of such general ideas, will be true of every particular thing in whom that essence, i.e. that abstract idea, is to be found: and what is once known of such ideas, will be perpetually and for ever true. So that as to all general knowledge we must search and find it only in our minds; and it is only the examining of our own ideas that furnisheth us with that. Truths belonging to essences of things (that is, to abstract ideas) are eternal; and are to be found out by the contemplation only of those essences: as the existence of things is to be known only from experience. But having more to say of this in the chapters where I shall speak of general and real knowledge, this may here suffice as to the universality of our knowledge in general.

Chapter IV Of the Reality of Knowledge

1. Objection. "Knowledge placed in our ideas may be all unreal or chimerical." I doubt not but my reader, by this time, may be apt to think that I have been all this while only building a castle in the air; and be ready to say to me:

"To what purpose all this stir? Knowledge, say you, is only the perception of the agreement or disagreement of our own ideas: but who knows what those ideas may be? Is there anything so extravagant as the imaginations of men's brains? Where is the head that has no chimeras in it? Or if there be a sober and a wise man, what difference will there be, by your rules, between his knowledge and that of the most extravagant fancy in the world? They both have their ideas, and perceive their agreement and disagreement one with another. If there be any difference between them, the advantage will be on the warm-headed man's side, as having the more ideas, and the more lively. And so, by your rules,

he will be the more knowing. If it be true, that all knowledge lies only in the perception of the agreement or disagreement of our own ideas, the visions of an enthusiast and the reasonings of a sober man will be equally certain. It is no matter how things are: so a man observe but the agreement of his own imaginations, and talk conformably, it is all truth, all certainty. Such castles in the air will be as strongholds of truth, as the demonstrations of Euclid. That an harpy is not a centaur is by this way as certain knowledge, and as much a truth, as that a square is not a circle."

"But of what use is all this fine knowledge of men's own imaginations, to a man that inquires after the reality of things? It matters not what men's fancies are, it is the knowledge of things that is only to be prized: it is this alone gives a value to our reasonings, and preference to one man's knowledge over another's, that it is of things as they really are, and not of dreams and fancies."

2. Answer: "Not so, where ideas agree with things." To which I answer, That if our knowledge of our ideas terminate in them, and reach no further, where there is

something further intended, our most serious thoughts will be of little more use than the reveries of a crazy brain; and the truths built thereon of no more weight than the discourses of a man who sees things clearly in a dream, and with great assurance utters them. But I hope, before I have done, to make it evident, that this way of certainty, by the knowledge of our own ideas, goes a little further than bare imagination: and I believe it will appear that all the certainty of general truths a man has lies in nothing else.

3. But what shall be the criterion of this agreement? It is evident the mind knows not things immediately, but only by the intervention of the ideas it has of them. Our knowledge, therefore is real only so far as there is a conformity between our ideas and the reality of things. But what shall be here the criterion? How shall the mind, when it perceives nothing but its own ideas, know that they agree with things themselves? This, though it seems not to want difficulty, yet, I think, there be two sorts of ideas that we may be assured agree with things.

4. As all simple ideas are really conformed to things.

First, The first are simple ideas, which since the mind, as has been shown, can by no means make to itself, must necessarily be the product of things operating on the mind, in a natural way, and producing therein those perceptions which by the Wisdom and Will of our Maker they are ordained and adapted to. From whence it follows, that simple ideas are not fictions of our fancies, but the natural and regular productions of things without us, really operating upon us; and so carry with them all the conformity which is intended; or which our state requires: for they represent to us things under those appearances which they are fitted to produce in us: whereby we are enabled to distinguish the sorts of particular substances, to discern the states they are in, and so to take them for our necessities, and apply them to our uses. Thus the idea of whiteness, or bitterness, as it is in the mind, exactly answering that power which is in any body to produce it there, has all the real conformity it can or ought to have, with things without us. And this conformity between our simple ideas and the existence of things, is sufficient for real knowledge.

5. All complex ideas, except ideas of substances, are their own archetypes. Secondly, All our complex ideas, except those of substances, being archetypes of the mind's own making, not intended to be the copies of anything, nor referred to the existence of anything, as to their originals, cannot want any conformity necessary to real knowledge. For that which is not designed to represent anything but itself, can never be capable of a wrong representation, nor mislead us from the true apprehension of anything, by its dislikeness to it: and such, excepting those of substances, are all our complex ideas. Which, as I have shown in another place, are combinations of ideas, which the mind, by its free choice, puts together, without considering any connexion they have in nature. And hence it is, that in all these sorts the ideas themselves are considered as the archetypes, and things no otherwise regarded, but as they are conformable to them. So that we cannot but be infallibly certain, that all the knowledge we attain concerning these ideas is real, and reaches things themselves. Because in all our thoughts, reasonings, and discourses of

this kind, we intend things no further than as they are conformable to our ideas. So that in these we cannot miss of a certain and undoubted reality.

6. Hence the reality of mathematical knowledge. I doubt not but it will be easily granted, that the knowledge we have of mathematical truths is not only certain, but real knowledge; and not the bare empty vision of vain, insignificant chimeras of the brain: and yet, if we will consider, we shall find that it is only of our own ideas. The mathematician considers the truth and properties belonging to a rectangle or circle only as they are in idea in his own mind. For it is possible he never found either of them existing mathematically, i.e. precisely true, in his life. But yet the knowledge he has of any truths or properties belonging to a circle, or any other mathematical figure, are nevertheless true and certain, even of real things existing: because real things are no further concerned, nor intended to be meant by any such propositions, than as things really agree to those archetypes in his mind. Is it true of the idea of a triangle, that its three angles are equal to two right ones? It is

true also of a triangle, wherever it really exists. Whatever other figure exists, that it is not exactly answerable to that idea of a triangle in his mind, is not at all concerned in that proposition. And therefore he is certain all his knowledge concerning such ideas is real knowledge: because, intending things no further than they agree with those his ideas, he is sure what he knows concerning those figures, when they have barely an ideal existence in his mind, will hold true of them also when they have a real existence in matter: his consideration being barely of those figures, which are the same wherever or however they exist.

7. And of moral. And hence it follows that moral knowledge is as capable of real certainty as mathematics. For certainty being but the perception of the agreement or disagreement of our ideas, and demonstration nothing but the perception of such agreement, by the intervention of other ideas or mediums; our moral ideas, as well as mathematical, being archetypes themselves, and so adequate and complete ideas; all the agreement or disagreement which we shall find in them will produce real

knowledge, as well as in mathematical figures.

8. Existence not required to make abstract knowledge real. For the attaining of knowledge and certainty, it is requisite that we have determined ideas: and, to make our knowledge real, it is requisite that the ideas answer their archetypes. Nor let it be wondered, that I place the certainty of our knowledge in the consideration of our ideas, with so little care and regard (as it may seem) to the real existence of things: since most of those discourses which take up the thoughts and engage the disputes of those who pretend to make it their business to inquire after truth and certainty, will, I presume, upon examination, be found to be general propositions, and notions in which existence is not at all concerned. All the discourses of the mathematicians about the squaring of a circle, conic sections, or any other part of mathematics, concern not the existence of any of those figures: but their demonstrations, which depend on their ideas, are the same, whether there be any square or circle existing in the world or no. In the same manner, the truth and certainty of moral discourses abstracts

from the lives of men, and the existence of those virtues in the world whereof they treat: nor are Tully's Offices less true, because there is nobody in the world that exactly practises his rules, and lives up to that pattern of a virtuous man which he has given us, and which existed nowhere when he writ but in idea. If it be true in speculation, i.e. in idea, that murder deserves death, it will also be true in reality of any action that exists conformable to that idea of murder. As for other actions, the truth of that proposition concerns them not. And thus it is of all other species of things, which have no other essences but those ideas which are in the minds of men.

9. Nor will it be less true or certain, because moral ideas are of our own making and naming. But it will here be said, that if moral knowledge be placed in the contemplation of our own moral ideas, and those, as other modes, be of our own making, What strange notions will there be of justice and temperance? What confusion of virtues and vice, if every one may make what ideas of them he pleases? No confusion or disorder in

the things themselves, nor the reasonings about them; no more than (in mathematics) there would be a disturbance in the demonstration, or a change in the properties of figures, and their relations one to another, if a man should make a triangle with four corners, or a trapezium with four right angles: that is, in plain English, change the names of the figures, and call that by one name, which mathematicians call ordinarily by another. For, let a man make to himself the idea of a figure with three angles, whereof one is a right one, and call it, if he please, equilaterum or trapezium, or anything else; the properties of, and demonstrations about that idea will be the same as if he called it a rectangular triangle. I confess the change of the name, by the impropriety of speech, will at first disturb him who knows not what idea it stands for: but as soon as the figure is drawn, the consequences and demonstrations are plain and clear. Just the same is it in moral knowledge: let a man have the idea of taking from others, without their consent, what their honest industry has possessed them of, and call this justice if he please. He that takes the name here

without the idea put to it will be mistaken, by joining another idea of his own to that name: but strip the idea of that name, or take it such as it is in the speaker's mind, and the same things will agree to it, as if you called it injustice. Indeed, wrong names in moral discourses breed usually more disorder, because they are not so easily rectified as in mathematics, where the figure, once drawn and seen, makes the name useless and of no force. For what need of a sign, when the thing signified is present and in view? But in moral names, that cannot be so easily and shortly done, because of the many decompositions that go to the making up the complex ideas of those modes. But yet for all this, the miscalling of any of those ideas, contrary to the usual signification of the words of that language, hinders not but that we may have certain and demonstrative knowledge of their several agreements and disagreements, if we will carefully, as in mathematics, keep to the same precise ideas, and trace them in their several relations one to another, without being led away by their names. If we but separate the idea under consideration from

the sign that stands for it, our knowledge goes equally on in the discovery of real truth and certainty, whatever sounds we make use of.

10. Misnaming disturbs not the certainty of the knowledge. One thing more we are to take notice of, That where God or any other law-maker, hath defined any moral names, there they have made the essence of that species to which that name belongs; and there it is not safe to apply or use them otherwise: but in other cases it is bare impropriety of speech to apply them contrary to the common usage of the country. But yet even this too disturbs not the certainty of that knowledge, which is still to be had by a due contemplation and comparing of those even nicknamed ideas.

11. Our complex ideas of substances have their archetypes without us; and here knowledge comes short. Thirdly, There is another sort of complex ideas, which, being referred to archetypes without us, may differ from them, and so our knowledge about them may come short of being real. Such are our ideas of substances, which, consisting of a collection of simple ideas, supposed taken

from the works of nature, may yet vary from them; by having more or different ideas united in them than are to be found united in the things themselves. From whence it comes to pass, that they may, and often do, fail of being exactly conformable to things themselves.

12. So far as our complex ideas agree with those archetypes without us, so far our knowledge concerning substances is real. I say, then, that to have ideas of substances which, by being conformable to things, may afford us real knowledge, it is not enough, as in modes, to put together such ideas as have no inconsistency, though they did never before so exist: v.g. the ideas of sacrilege or perjury, &c., were as real and true ideas before, as after the existence of any such fact. But our ideas of substances, being supposed copies, and referred to archetypes without us, must still be taken from something that does or has existed: they must not consist of ideas put together at the pleasure of our thoughts, without any real pattern they were taken from, though we can perceive no inconsistency in such a combination. The reason whereof is, because we, knowing not what

real constitution it is of substances whereon our simple ideas depend, and which really is the cause of the strict union of some of them one with another, and the exclusion of others there are very few of them that we can be sure are or are not inconsistent in nature, any further than experience and sensible observation reach. Herein, therefore, is founded the reality of our knowledge concerning substances—That all our complex ideas of them must be such, and such only, as are made up of such simple ones as have been discovered to co-exist in nature. And our ideas being thus true, though not perhaps very exact copies, are yet the subjects of real (as far as we have any) knowledge of them. Which (as has been already shown) will not be found to reach very far: but so far as it does, it will still be real knowledge. Whatever ideas we have, the agreement we find they have with others will still be knowledge. If those ideas be abstract, it will be general knowledge. But to make it real concerning substances, the ideas must be taken from the real existence of things. Whatever simple ideas have been found to co-exist in any substance, these we may

with confidence join together again, and so make abstract ideas of substances. For whatever have once had an union in nature, may be united again.

13. In our inquiries about substances, we must consider ideas, and not confine our thoughts to names or species supposed set out by names. This, if we rightly consider, and confine not our thoughts and abstract ideas to names, as if there were, or could be no other sorts of things than what known names had already determined, and, as it were, set out, we should think of things with greater freedom and less confusion than perhaps we do. It would possibly be thought a bold paradox, if not a very dangerous falsehood, if I should say that some changelings, who have lived forty years together, without any appearance of reason, are something between a man and a beast: which prejudice is founded upon nothing else but a false supposition, that these two names, man and beast, stand for distinct species so set out by real essences, that there can come no other species between them: whereas if we will abstract from those names, and the supposition of such specific

essences made by nature, wherein all things of the same denominations did exactly and equally partake; if we would not fancy that there were a certain number of these essences, wherein all things, as in moulds, were cast and formed; we should find that the idea of the shape, motion, and life of a man without reason, is as much a distinct idea, and makes as much a distinct sort of things from man and beast, as the idea of the shape of an ass with reason would be different from either that of man or beast, and be a species of an animal between, or distinct from both.

14. Objection against a changeling being something between a man and beast, answered. Here everybody will be ready to ask, If changelings may be supposed something between man and beast, pray what are they? I answer, changelings; which is as good a word to signify something different from the signification of man or beast, as the names man and beast are to have significations different one from the other. This, well considered, would resolve this matter, and show my meaning without any more ado. But I am not so unacquainted

with the zeal of some men, which enables them to spin consequences, and to see religion threatened, whenever any one ventures to quit their forms of speaking, as not to foresee what names such a proposition as this is like to be charged with: and without doubt it will be asked, If changelings are something between man and beast, what will become of them in the other world? To which I answer, I. It concerns me not to know or inquire. To their own master they stand or fall. It will make their state neither better nor worse, whether we determine anything of it or no. They are in the hands of a faithful Creator and a bountiful Father, who disposes not of his creatures according to our narrow thoughts or opinions, nor distinguishes them according to names and species of our contrivance. And we that know so little of this present world we are in, may, I think, content ourselves without being peremptory in defining the different states which creatures shall come into when they go off this stage. It may suffice us, that He hath made known to all those who are capable of instruction, discoursing, and reasoning, that they shall come to an

account, and receive according to what they have done in this body.

15. What will become of changelings in a future state? But, Secondly, I answer, The force of these men's question (viz. Will you deprive changelings of a future state?) is founded on one of these two suppositions, which are both false. The first is, That all things that have the outward shape and appearance of a man must necessarily be designed to an immortal future being after this life: or, secondly, That whatever is of human birth must be so. Take away these imaginations, and such questions will be groundless and ridiculous. I desire then those who think there is no more but an accidental difference between themselves and changelings, the essence in both being exactly the same, to consider, whether they can imagine immortality annexed to any outward shape of the body; the very proposing it is, I suppose, enough to make them disown it. No one yet, that ever I heard of, how much soever immersed in matter, allowed that excellency to any figure of the gross sensible outward parts, as to affirm eternal life due to

it, or a necessary consequence of it; or that any mass of matter should, after its dissolution here, be again restored hereafter to an everlasting state of sense, perception, and knowledge, only because it was moulded into this or that figure, and had such a particular frame of its visible parts. Such an opinion as this, placing immortality in a certain superficial figure, turns out of doors all consideration of soul or spirit; upon whose account alone some corporeal beings have hitherto been concluded immortal, and others not. This is to attribute more to the outside than inside of things; and to place the excellency of a man more in the external shape of his body, than internal perfections of his soul: which is but little better than to annex the great and inestimable advantage of immortality and life everlasting, which he has above other material beings, to annex it, I say, to the cut of his beard, or the fashion of his coat. For this or that outward mark of our bodies no more carries with it the hope of an eternal duration, than the fashion of a man's suit gives him reasonable grounds to imagine it will never wear out, or that it will make him

immortal. It will perhaps be said, that nobody thinks that the shape makes anything immortal, but it is the shape is the sign of a rational soul within, which is immortal. I wonder who made it the sign of any such thing: for barely saying it, will not make it so. It would require some proofs to persuade one of it. No figure that I know speaks any such language. For it may as rationally be concluded, that the dead body of a man, wherein there is to be found no more appearance or action of life than there is in a statue, has yet nevertheless a living soul in it, because of its shape; as that there is a rational soul in a changeling, because he has the outside of a rational creature, when his actions carry far less marks of reason with them, in the whole course of his life, than what are to be found in many a beast.

16. Monsters. But it is the issue of rational parents, and must therefore be concluded to have a rational soul. I know not by what logic you must so conclude. I am sure this is a conclusion that men nowhere allow of. For if they did, they would not make bold, as everywhere they do, to destroy ill-formed and mis-shaped produc-

tions. Ay, but these are monsters. Let them be so: what will your drivelling, unintelligent, intractable changeling be? Shall a defect in the body make a monster; a defect in the mind (the far more noble, and, in the common phrase, the far more essential part) not? Shall the want of a nose, or a neck, make a monster, and put such issue out of the rank of men; the want of reason and understanding, not? This is to bring all back again to what was exploded just now: this is to place all in the shape, and to take the measure of a man only by his outside. To show that according to the ordinary way of reasoning in this matter, people do lay the whole stress on the figure, and resolve the whole essence of the species of man (as they make it) into the outward shape, how unreasonable soever it be, and how much soever they disown it, we need but trace their thoughts and practice a little further, and then it will plainly appear. The well-shaped changeling is a man, has a rational soul, though it appear not: this is past doubt, say you: make the ears a little longer, and more pointed, and the nose a little flatter than ordinary, and then you begin to

boggle: make the face yet narrower, flatter, and longer, and then you are at a stand: add still more and more of the likeness of a brute to it, and let the head be perfectly that of some other animal, then presently it is a monster; and it is demonstration with you that it hath no rational soul, and must be destroyed. Where now (I ask) shall be the just measure; which the utmost bounds of that shape, that carries with it a rational soul? For, since there have been human foetuses produced, half beast and half man; and others three parts one, and one part the other; and so it is possible they may be in all the variety of approaches to the one or the other shape, and may have several degrees of mixture of the likeness of a man, or a brute;—I would gladly know what are those precise lineaments, which, according to this hypothesis, are or are not capable of a rational soul to be joined to them. What sort of outside is the certain sign that there is or is not such an inhabitant within? For till that be done, we talk at random of man: and shall always, I fear, do so, as long as we give ourselves up to certain sounds, and the imaginations of settled and fixed

species in nature, we know not what. But, after all, I desire it may be considered, that those who think they have answered the difficulty, by telling us, that a misshaped foetus is a monster, run into the same fault they are arguing against; by constituting a species between man and beast. For what else, I pray, is their monster in the case, (if the word monster signifies anything at all,) but something neither man nor beast, but partaking somewhat of either? And just so is the changeling before mentioned. So necessary is it to quit the common notion of species and essences, if we will truly look into the nature of things, and examine them by what our faculties can discover in them as they exist, and not by groundless fancies that have been taken up about them.

17. Words and species. I have mentioned this here, because I think we cannot be too cautious that words and species, in the ordinary notions which we have been used to of them, impose not on us. For I am apt to think therein lies one great obstacle to our clear and distinct knowledge, especially in reference to substances: and from thence has risen a great part of the difficulties

Chapter V

Of Truth in General

about truth and certainty. Would we accustom ourselves to separate our contemplations and reasonings from words, we might in a great measure remedy this inconvenience within our own thoughts: but yet it would still disturb us in our discourse with others, as long as we retained the opinion, that species and their essences were anything else but our abstract ideas (such as they are) with names annexed to them, to be the signs of them.

18. Recapitulation. Wherever we perceive the agreement or disagreement of any of our ideas, there is certain knowledge: and wherever we are sure those ideas agree with the reality of things, there is certain real knowledge. Of which agreement of our ideas with the reality of things, having here given the marks, I think, I have shown wherein it is that certainty, real certainty, consists. Which, whatever it was to others, was, I confess, to me heretofore, one of those desiderata which I found great want of.

1. What truth is. What is truth? was an inquiry many ages since; and it being that which all mankind either do, or pretend to search after, it cannot but be worth our while carefully to examine wherein it consists, and so acquaint ourselves with the nature of it, as to observe how the mind distinguishes it from falsehood.

2. A right joining or separating of signs, i.e. either ideas or words. Truth, then, seems to me, in the proper import of the word, to signify nothing but the joining or separating of Signs, as the Things signified by them do agree or disagree one with another. The joining or separating of signs here meant, is what by another name we call proposition. So that truth properly belongs only to propositions: whereof there are two sorts, viz. mental and verbal; as there are two sorts of signs commonly made use of, viz. ideas and words.

3. Which make mental or verbal propositions. To form a clear notion of truth, it is very necessary to consider

truth of thought, and truth of words, distinctly one from another: but yet it is very difficult to treat of them asunder. Because it is unavoidable, in treating of mental propositions, to make use of words: and then the instances given of mental propositions cease immediately to be barely mental, and become verbal. For a mental proposition being nothing but a bare consideration of the ideas, as they are in our minds, stripped of names, they lose the nature of purely mental propositions as soon as they are put into words.

4. Mental propositions are very hard to be treated of. And that which makes it yet harder to treat of mental and verbal propositions separately is, that most men, if not all, in their thinking and reasonings within themselves, make use of words instead of ideas; at least when the subject of their meditation contains in it complex ideas. Which is a great evidence of the imperfection and uncertainty of our ideas of that kind, and may, if attentively made use of, serve for a mark to show us what are those things we have clear and perfect established ideas of, and what not. For if we will curiously observe the

way our mind takes in thinking and reasoning, we shall find, I suppose, that when we make any propositions within our own thoughts about white or black, sweet or bitter, a triangle or a circle, we can and often do frame in our minds the ideas themselves, without reflecting on the names. But when we would consider, or make propositions about the more complex ideas, as of a man, vitriol, fortitude, glory, we usually put the name for the idea: because the ideas these names stand for, being for the most part imperfect, confused, and undetermined, we reflect on the names themselves, because they are more clear, certain, and distinct, and readier occur to our thoughts than the pure ideas: and so we make use of these words instead of the ideas themselves, even when we would meditate and reason within ourselves, and make tacit mental propositions. In substances, as has been already noticed, this is occasioned by the imperfections of our ideas: we making the name stand for the real essence, of which we have no idea at all. In modes, it is occasioned by the great number of simple ideas that go to the making them up. For many of them

being compounded, the name occurs much easier than the complex idea itself, which requires time and attention to be recollected, and exactly represented to the mind, even in those men who have formerly been at the pains to do it; and is utterly impossible to be done by those who, though they have ready in their memory the greatest part of the common words of that language, yet perhaps never troubled themselves in all their lives to consider what precise ideas the most of them stood for. Some confused or obscure notions have served their turns; and many who talk very much of religion and conscience, of church and faith, of power and right, of obstructions and humours, melancholy and choler, would perhaps have little left in their thoughts and meditations if one should desire them to think only of the things themselves and lay by those words with which they so often confound others, and not seldom themselves also.

5. Mental and verbal propositions contrasted. But to return to the consideration of truth: we must, I say, observe two sorts of propositions that we are capable of making:—

First, mental, wherein the ideas in our understandings are without the use of words put together, or separated, by the mind perceiving or judging of their agreement or disagreement.

Secondly, Verbal propositions, which are words, the signs of our ideas, put together or separated in affirmative or negative sentences. By which way of affirming or denying, these signs, made by sounds, are, as it were, put together or separated one from another. So that proposition consists in joining or separating signs; and truth consists in the putting together or separating those signs, according as the things which they stand for agree or disagree.

6. When mental propositions contain real truth, and when verbal. Every one's experience will satisfy him, that the mind, either by perceiving, or supposing, the agreement or disagreement of any of its ideas, does tacitly within itself put them into a kind of proposition affirmative or negative; which I have endeavoured to express by the terms putting together and separating. But this action of the mind, which is so familiar to ev-

ery thinking and reasoning man, is easier to be conceived by reflecting on what passes in us when we affirm or deny, than to be explained by words. When a man has in his head the idea of two lines, viz. the side and diagonal of a square, whereof the diagonal is an inch long, he may have the idea also of the division of that line into a certain number of equal parts: v.g. into five, ten, a hundred, a thousand, or any other number, and may have the idea of that inch line being divisible, or not divisible, into such equal parts, as a certain number of them will be equal to the sideline. Now, whenever he perceives, believes, or supposes such a kind of divisibility to agree or disagree to his idea of that line, he, as it were, joins or separates those two ideas, viz. the idea of that line, and the idea of that kind of divisibility; and so makes a mental proposition, which is true or false, according as such a kind of divisibility; a divisibility into such aliquot parts, does really agree to that line or no. When ideas are so put together, or separated in the mind, as they or the things they stand for do agree or not, that is, as I may call it, mental truth. But truth of

words is something more; and that is the affirming or denying of words one of another, as the ideas they stand for agree or disagree: and this again is two-fold; either purely verbal and trifling, which I shall speak of, (chap. viii.,) or real and instructive; which is the object of that real knowledge which we have spoken of already.

7. Objection against verbal truth, that "thus it may all be chimerical." But here again will be apt to occur the same doubt about truth, that did about knowledge: and it will be objected, that if truth be nothing but the joining and separating of words in propositions, as the ideas they stand for agree or disagree in men's minds, the knowledge of truth is not so valuable a thing as it is taken to be, nor worth the pains and time men employ in the search of it: since by this account it amounts to no more than the conformity of words to the chimeras of men's brains. Who knows not what odd notions many men's heads are filled with, and what strange ideas all men's brains are capable of? But if we rest here, we know the truth of nothing by this rule, but of the visionary words in our own imaginations; nor have other

truth, but what as much concerns harpies and centaurs, as men and horses. For those, and the like, may be ideas in our heads, and have their agreement or disagreement there, as well as the ideas of real beings, and so have as true propositions made about them. And it will be altogether as true a proposition to say all centaurs are animals, as that all men are animals; and the certainty of one as great as the other. For in both the propositions, the words are put together according to the agreement of the ideas in our minds: and the agreement of the idea of animal with that of centaur is as clear and visible to the mind, as the agreement of the idea of animal with that of man; and so these two propositions are equally true, equally certain. But of what use is all such truth to us?

8. Answered, "Real truth is about ideas agreeing to things." Though what has been said in the foregoing chapter to distinguish real from imaginary knowledge might suffice here, in answer to this doubt, to distinguish real truth from chimerical, or (if you please) barely nominal, they depending both on the same foundation;

yet it may not be amiss here again to consider, that though our words signify nothing but our ideas, yet being designed by them to signify things, the truth they contain when put into propositions will be only verbal, when they stand for ideas in the mind that have not an agreement with the reality of things. And therefore truth as well as knowledge may well come under the distinction of verbal and real; that being only verbal truth, wherein terms are joined according to the agreement or disagreement of the ideas they stand for; without regarding whether our ideas are such as really have, or are capable of having, an existence in nature. But then it is they contain real truth, when these signs are joined, as our ideas agree; and when our ideas are such as we know are capable of having an existence in nature: which in substances we cannot know, but by knowing that such have existed.

9. Truth and falsehood in general. Truth is the marking down in words the agreement or disagreement of ideas as it is. Falsehood is the marking down in words the agreement or disagreement of ideas otherwise than it is.

And so far as these ideas, thus marked by sounds, agree to their archetypes, so far only is the truth real. The knowledge of this truth consists in knowing what ideas the words stand for, and the perception of the agreement or disagreement of those ideas, according as it is marked by those words.

10. General propositions to be treated of more at large. But because words are looked on as the great conduits of truth and knowledge, and that in conveying and receiving of truth, and commonly in reasoning about it, we make use of words and propositions, I shall more at large inquire wherein the certainty of real truths contained in propositions consists, and where it is to be had; and endeavour to show in what sort of universal propositions we are capable of being certain of their real truth or falsehood.

I shall begin with general propositions, as those which most employ our thoughts, and exercise our contemplation. General truths are most looked after by the mind as those that most enlarge our knowledge; and by their comprehensiveness satisfying us at once of many par-

ticulars, enlarge our view, and shorten our way to knowledge.

11. Moral and metaphysical truth. Besides truth taken in the strict sense before mentioned, there are other sorts of truths: As, 1. Moral truth, which is speaking of things according to the persuasion of our own minds, though the proposition we speak agree not to the reality of things; 2. Metaphysical truth, which is nothing but the real existence of things, conformable to the ideas to which we have annexed their names. This, though it seems to consist in the very beings of things, yet, when considered a little nearly, will appear to include a tacit proposition, whereby the mind joins that particular thing to the idea it had before settled with the name to it. But these considerations of truth, either having been before taken notice of, or not being much to our present purpose, it may suffice here only to have mentioned them.

Chapter VI

Of Universal Propositions: Their Truth and Certainty

1. Treating of words necessary to knowledge. Though the examining and judging of ideas by themselves, their names being quite laid aside, be the best and surest way to clear and distinct knowledge: yet, through the prevailing custom of using sounds for ideas, I think it is very seldom practised. Every one may observe how common it is for names to be made use of, instead of the ideas themselves, even when men think and reason within their own breasts; especially if the ideas be very complex, and made up of a great collection of simple ones. This makes the consideration of words and propositions so necessary a part of the Treatise of Knowledge, that it is very hard to speak intelligibly of the one, without explaining the other.

2. General truths hardly to be understood, but in verbal propositions. All the knowledge we have, being only of particular or general truths, it is evident that what-

ever may be done in the former of these, the latter, which is that which with reason is most sought after, can never be well made known, and is very seldom apprehended, but as conceived and expressed in words. It is not, therefore, out of our way, in the examination of our knowledge, to inquire into the truth and certainty of universal propositions.

3. Certainty twofold—of truth and of knowledge. But that we may not be misled in this case by that which is the danger everywhere, I mean by the doubtfulness of terms, it is fit to observe that certainty is twofold: certainty of truth and certainty of knowledge. Certainty of truth is, when words are so put together in propositions as exactly to express the agreement or disagreement of the ideas they stand for, as really it is. Certainty of knowledge is to perceive the agreement or disagreement of ideas, as expressed in any proposition. This we usually call knowing, or being certain of the truth of any proposition.

4. No proposition can be certainly known to be true, where the real essence of each species mentioned is not

known. Now, because we cannot be certain of the truth of any general proposition, unless we know the precise bounds and extent of the species its terms stand for, it is necessary we should know the essence of each species, which is that which constitutes and bounds it.

This, in all simple ideas and modes, is not hard to do. For in these the real and nominal essence being the same, or, which is all one, the abstract idea which the general term stands for being the sole essence and boundary that is or can be supposed of the species, there can be no doubt how far the species extends, or what things are comprehended under each term; which, it is evident, are all that have an exact conformity with the idea it stands for, and no other.

But in substances, wherein a real essence, distinct from the nominal, is supposed to constitute, determine, and bound the species, the extent of the general word is very uncertain; because, not knowing this real essence, we cannot know what is, or what is not of that species; and, consequently, what may or may not with certainty be affirmed of it. And thus, speaking of a man, or gold,

or any other species of natural substances, as supposed constituted by a precise and real essence which nature regularly imparts to every individual of that kind, whereby it is made to be of that species, we cannot be certain of the truth of any affirmation or negation made of it. For man or gold, taken in this sense, and used for species of things constituted by real essences, different from the complex idea in the mind of the speaker, stand for we know not what; and the extent of these species, with such boundaries, are so unknown and undetermined, that it is impossible with any certainty to affirm, that all men are rational, or that all gold is yellow. But where the nominal essence is kept to, as the boundary of each species, and men extend the application of any general term no further than to the particular things in which the complex idea it stands for is to be found, there they are in no danger to mistake the bounds of each species, nor can be in doubt, on this account, whether any proposition be true or not. I have chosen to explain this uncertainty of propositions in this scholastic way, and have made use of the terms of essences,

and species, on purpose to show the absurdity and inconvenience there is to think of them as of any other sort of realities, than barely abstract ideas with names to them. To suppose that the species of things are anything but the sorting of them under general names, according as they agree to several abstract ideas of which we make those names the signs, is to confound truth, and introduce uncertainty into all general propositions that can be made about them. Though therefore these things might, to people not possessed with scholastic learning, be treated of in a better and clearer way; yet those wrong notions of essences or species having got root in most people's minds who have received any tincture from the learning which has prevailed in this part of the world, are to be discovered and removed, to make way for that use of words which should convey certainty with it.

5. This more particularly concerns substances. The names of substances, then, whenever made to stand for species which are supposed to be constituted by real essences which we know not, are not capable to convey

certainty to the understanding. Of the truth of general propositions made up of such terms we cannot be sure. The reason whereof is plain: for how can we be sure that this or that quality is in gold, when we know not what is or is not gold? Since in this way of speaking, nothing is gold but what partakes of an essence, which we, not knowing, cannot know where it is or is not, and so cannot be sure that any parcel of matter in the world is or is not in this sense gold; being incurably ignorant whether it has or has not that which makes anything to be called gold; i.e. that real essence of gold whereof we have no idea at all. This being as impossible for us to know as it is for a blind man to tell in what flower the colour of a pansy is or is not to be found, whilst he has no idea of the colour of a pansy at an. Or if we could (which is impossible) certainly know where a real essence, which we know not, is, v.g. in what parcels of matter the real essence of gold is, yet could we not be sure that this or that quality could with truth be affirmed of gold; since it is impossible for us to know that this or that quality or idea has a necessary connexion

with a real essence of which we have no idea at all, whatever species that supposed real essence may be imagined to constitute.

6. The truth of few universal propositions concerning substances is to be known. On the other side, the names of substances, when made use of as they should be, for the ideas men have in their minds, though they carry a clear and determinate signification with them, will not yet serve us to make many universal propositions of whose truth we can be certain. Not because in this use of them we are uncertain what things are signified by them, but because the complex ideas they stand for are such combinations of simple ones as carry not with them any discoverable connexion or repugnancy, but with a very few other ideas.

7. Because necessary co-existence of simple ideas in substances can in few cases be known. The complex ideas that our names of the species of substances properly stand for, are collections of such qualities as have been observed to co-exist in an unknown substratum, which we call substance; but what other qualities nec-

essarily co-exist with such combinations, we cannot certainly know, unless we can discover their natural dependence; which, in their primary qualities, we can go but a very little way in; and in all their secondary qualities we can discover no connexion at all: for the reasons mentioned, chap. iii. Viz. 1. Because we know not the real constitutions of substances, on which each secondary quality particularly depends. 2. Did we know that, it would serve us only for experimental (not universal) knowledge; and reach with certainty no further than that bare instance: because our understandings can discover no conceivable connexion between any secondary quality and any modification whatsoever of any of the primary ones. And therefore there are very few general propositions to be made concerning substances, which can carry with them undoubted certainty.

8. Instance in gold. "All gold is fixed," is a proposition whose truth we cannot be certain of, how universally soever it be believed. For if, according to the useless imagination of the Schools, any one supposes the term gold to stand for a species of things set out by nature,

by a real essence belonging to it, it is evident he knows not what particular substances are of that species; and so cannot with certainty affirm anything universally of gold. But if he makes gold stand for a species determined by its nominal essence, let the nominal essence, for example, be the complex idea of a body of a certain yellow colour, malleable, fusible, and heavier than any other known;—in this proper use of the word gold, there is no difficulty to know what is or is not gold. But yet no other quality can with certainty be universally affirmed or denied of gold, but what hath a discoverable connexion or inconsistency with that nominal essence. Fixedness, for example, having no necessary connexion that we can discover, with the colour, weight, or any other simple idea of our complex one, or with the whole combination together; it is impossible that we should certainly know the truth of this proposition, that all gold is fixed.

9. No discoverable necessary connexion between nominal essence of gold and other simple ideas. As there is no discoverable connexion between fixedness and the colour,

weight, and other simple ideas of that nominal essence of gold; so, if we make our complex idea of gold, a body yellow, fusible, ductile, weighty, and fixed, we shall be at the same uncertainty concerning solubility in aqua regia, and for the same reason. Since we can never, from consideration of the ideas themselves, with certainty affirm or deny of a body whose complex idea is made up of yellow, very weighty, ductile, fusible, and fixed, that it is soluble in aqua regia: and so on of the rest of its qualities. I would gladly meet with one general affirmation concerning any quality of gold, that any one can certainly know is true. It will, no doubt, be presently objected, Is not this an universal proposition, All gold is malleable? To which I answer, It is a very certain proposition, if malleableness be a part of the complex idea the word gold stands for. But then here is nothing affirmed of gold, but that that sound stands for an idea in which malleableness is contained: and such a sort of truth and certainty as this it is, to say a centaur is four-footed. But if malleableness make not a part of the specific essence the name of gold stands for, it is plain, all gold is

malleable, is not a certain proposition. Because, let the complex idea of gold be made up of whatsoever of its other qualities you please, malleableness will not appear to depend on that complex idea, nor follow from any simple one contained in it: the connexion that malleableness has (if it has any) with those other qualities being only by the intervention of the real constitution of its insensible parts; which, since we know not, it is impossible we should perceive that connexion, unless we could discover that which ties them together.

10. As far as any such co-existence can be known, so far universal propositions may be certain. But this will go but a little way. The more, indeed, of these coexisting qualities we unite into one complex idea, under one name, the more precise and determinate we make the signification of that word; but never yet make it thereby more capable of universal certainty, in respect of other qualities not contained in our complex idea: since we perceive not their connexion or dependence on one another; being ignorant both of that real constitution in which they are all founded, and also how they flow from

it. For the chief part of our knowledge concerning substances is not, as in other things, barely of the relation of two ideas that may exist separately; but is of the necessary connexion and co-existence of several distinct ideas in the same subject, or of their repugnancy so to co-exist. Could we begin at the other end, and discover what it was wherein that colour consisted, what made a body lighter or heavier, what texture of parts made it malleable, fusible, and fixed, and fit to be dissolved in this sort of liquor, and not in another;—if, I say, we had such an idea as this of bodies, and could perceive wherein all sensible qualities originally consist, and how they are produced; we might frame such abstract ideas of them as would furnish us with matter of more general knowledge, and enable us to make universal propositions, that should carry general truth and certainty with them. But whilst our complex ideas of the sorts of substances are so remote from that internal real constitution on which their sensible qualities depend, and are made up of nothing but an imperfect collection of those apparent qualities our senses can discover, there can be

few general propositions concerning substances of whose real truth we can be certainly assured; since there are but few simple ideas of whose connexion and necessary coexistence we can have certain and undoubted knowledge. I imagine, amongst all the secondary qualities of substances, and the powers relating to them, there cannot any two be named, whose necessary co-existence, or repugnance to coexist, can certainly be known; unless in those of the same sense, which necessarily exclude one another, as I have elsewhere shown. No one, I think, by the colour that is in any body, can certainly know what smell, taste, sound, or tangible qualities it has, nor what alterations it is capable to make or receive on or from other bodies. The same may be said of the sound or taste, &c. Our specific names of substances standing for any collections of such ideas, it is not to be wondered that we can with them make very few general propositions of undoubted real certainty. But yet so far as any complex idea of any sort of substances contains in it any simple idea, whose necessary existence with any other may be discovered, so far universal propo-

sitions may with certainty be made concerning it: v.g. could any one discover a necessary connexion between malleableness and the colour or weight of gold, or any other part of the complex idea signified by that name, he might make a certain universal proposition concerning gold in this respect; and the real truth of this proposition, that all gold is malleable, would be as certain as of this, the three angles of all right-lined triangles are all equal to two right ones.

11. The qualities which make our complex ideas of substances depend mostly on external, remote, and unperceived causes. Had we such ideas of substances as to know what real constitutions produce those sensible qualities we find in them, and how those qualities flowed from thence, we could, by the specific ideas of their real essences in our own minds, more certainly find out their properties, and discover what qualities they had or had not, than we can now by our senses: and to know the properties of gold, it would be no more necessary that gold should exist, and that we should make experiments upon it, than it is necessary for the knowing the prop-

erties of a triangle, that a triangle should exist in any matter, the idea in our minds would serve for the one as well as the other. But we are so far from being admitted into the secrets of nature, that we scarce so much as ever approach the first entrance towards them. For we are wont to consider the substances we meet with, each of them, as an entire thing by itself, having all its qualities in itself, and independent of other things; overlooking, for the most part, the operations of those invisible fluids they are encompassed with, and upon whose motions and operations depend the greatest part of those qualities which are taken notice of in them, and are made by us the inherent marks of distinction whereby we know and denominate them. Put a piece of gold anywhere by itself, separate from the reach and influence of all other bodies, it will immediately lose all its colour and weight, and perhaps malleableness too; which, for aught I know, would be changed into a perfect friability. Water, in which to us fluidity is an essential quality, left to itself, would cease to be fluid. But if inanimate bodies owe so much of their present state to other bod-

ies without them, that they would not be what they appear to us were those bodies that environ them removed; it is yet more so in vegetables, which are nourished, grow, and produce leaves, flowers, and seeds, in a constant succession. And if we look a little nearer into the state of animals, we shall find that their dependence, as to life, motion, and the most considerable qualities to be observed in them, is so wholly on extrinsical causes and qualities of other bodies that make no part of them, that they cannot subsist a moment without them: though yet those bodies on which they depend are little taken notice of, and make no part of the complex ideas we frame of those animals. Take the air but for a minute from the greatest part of living creatures, and they presently lose sense, life, and motion. This the necessity of breathing has forced into our knowledge. But how many other extrinsical and possibly very remote bodies do the springs of these admirable machines depend on, which are not vulgarly observed, or so much as thought on; and how many are there which the severest inquiry can never discover? The inhabitants of

this spot of the universe, though removed so many millions of miles from the sun, yet depend so much on the duly tempered motion of particles coming from or agitated by it, that were this earth removed but a small part of the distance out of its present situation, and placed a little further or nearer that source of heat, it is more than probable that the greatest part of the animals in it would immediately perish: since we find them so often destroyed by an excess or defect of the sun's warmth, which an accidental position in some parts of this our little globe exposes them to. The qualities observed in a loadstone must needs have their source far beyond the confines of that body; and the ravage made often on several sorts of animals by invisible causes, the certain death (as we are told) of some of them, by barely passing the line, or, as it is certain of other, by being removed into a neighbouring country; evidently show that the concurrence and operations of several bodies, with which they are seldom thought to have anything to do, is absolutely necessary to make them be what they appear to us, and to preserve those qualities by

which we know and distinguish them. We are then quite out of the way, when we think that things contain within themselves the qualities that appear to us in them; and we in vain search for that constitution within the body of a fly or an elephant, upon which depend those qualities and powers we observe in them. For which, perhaps, to understand them aright, we ought to look not only beyond this our earth and atmosphere, but even beyond the sun or remotest star our eyes have yet discovered. For how much the being and operation of particular substances in this our globe depends on causes utterly beyond our view, is impossible for us to determine. We see and perceive some of the motions and grosser operations of things here about us; but whence the streams come that keep all these curious machines in motion and repair, how conveyed and modified, is beyond our notice and apprehension: and the great parts and wheels, as I may say so, of this stupendous structure of the universe, may, for aught we know, have such a connexion and dependence in their influences and operations one upon another, that perhaps things

in this our mansion would put on quite another face, and cease to be what they are, if some one of the stars or great bodies incomprehensibly remote from us, should cease to be or move as it does. This is certain: things, however absolute and entire they seem in themselves, are but retainers to other parts of nature, for that which they are most taken notice of by us. Their observable qualities, actions, and powers are owing to something without them; and there is not so complete and perfect a part that we know of nature, which does not owe the being it has, and the excellences of it, to its neighbours; and we must not confine our thoughts within the surface of any body, but look a great deal further, to comprehend perfectly those qualities that are in it.

12. Our nominal essences of substances furnish few universal propositions about them that are certain. If this be so, it is not to be wondered that we have very imperfect ideas of substances, and that the real essences, on which depend their properties and operations, are unknown to us. We cannot discover so much as that size, figure, and texture of their minute and active parts,

which is really in them; much less the different motions and impulses made in and upon them by bodies from without, upon which depends, and by which is formed the greatest and most remarkable part of those qualities we observe in them, and of which our complex ideas of them are made up. This consideration alone is enough to put an end to all our hopes of ever having the ideas of their real essences; which whilst we want, the nominal essences we make use of instead of them will be able to furnish us but very sparingly with any general knowledge, or universal propositions capable of real certainty.

13. Judgment of probability concerning substances may reach further: but that is not knowledge. We are not therefore to wonder, if certainty be to be found in very few general propositions made concerning substances: our knowledge of their qualities and properties goes very seldom further than our senses reach and inform us. Possibly inquisitive and observing men may, by strength of judgment, penetrate further, and, on probabilities taken from wary observation, and hints well laid together, often guess right at what experience has not yet discov-

ered to them. But this is but guessing still; it amounts only to opinion, and has not that certainty which is requisite to knowledge. For all general knowledge lies only in our own thoughts, and consists barely in the contemplation of our own abstract ideas. Wherever we perceive any agreement or disagreement amongst them, there we have general knowledge; and by putting the names of those ideas together accordingly in propositions, can with certainty pronounce general truths. But because the abstract ideas of substances, for which their specific names stand, whenever they have any distinct and determinate signification, have a discoverable connexion or inconsistency with but a very few other ideas, the certainty of universal propositions concerning substances is very narrow and scanty, in that part which is our principal inquiry concerning them; and there are scarce any of the names of substances, let the idea it is applied to be what it will, of which we can generally, and with certainty, pronounce, that it has or has not this or that other quality belonging to it, and constantly co-existing or inconsistent with that idea, wherever it is to be found.

14. What is requisite for our knowledge of substances. Before we can have any tolerable knowledge of this kind, we must First know what changes the primary qualities of one body do regularly produce in the primary qualities of another, and how. Secondly, We must know what primary qualities of any body produce certain sensations or ideas in us. This is in truth no less than to know all the effects of matter, under its divers modifications of bulk, figure, cohesion of parts, motion and rest. Which, I think every body will allow, is utterly impossible to be known by us without revelation. Nor if it were revealed to us what sort of figure, bulk, and motion of corpuscles would produce in us the sensation of a yellow colour, and what sort of figure, bulk, and texture of parts in the superficies of any body were fit to give such corpuscles their due motion to produce that colour; would that be enough to make universal propositions with certainty, concerning the several sorts of them; unless we had faculties acute enough to perceive the precise bulk, figure, texture, and motion of bodies, in those minute parts, by which they operate on our

senses, so that we might by those frame our abstract ideas of them. I have mentioned here only corporeal substances, whose operations seem to lie more level to our understandings. For as to the operations of spirits, both their thinking and moving of bodies, we at first sight find ourselves at a loss; though perhaps, when we have applied our thoughts a little nearer to the consideration of bodies and their operations, and examined how far our notions, even in these, reach with any clearness beyond sensible matter of fact, we shall be bound to confess that, even in these too, our discoveries amount to very little beyond perfect ignorance and incapacity.

15. Whilst our complex ideas of substances contain not ideas of their real constitutions, we can make but few general certain propositions concerning them. This is evident, the abstract complex ideas of substances. for which their general names stand, not comprehending their real constitutions, can afford us very little universal certainty. Because our ideas of them are not made up of that on which those qualities we observe in them, and would inform ourselves about, do depend, or with

which they have any certain connexion: v.g. let the ideas to which we give the name man be, as it commonly is, a body of the ordinary shape, with sense, voluntary motion, and reason joined to it. This being the abstract idea, and consequently the essence of our species, man, we can make but very few general certain propositions concerning man, standing for such an idea. Because, not knowing the real constitution on which sensation, power of motion, and reasoning, with that peculiar shape, depend, and whereby they are united together in the same subject, there are very few other qualities with which we can perceive them to have a necessary connexion: and therefore we cannot with certainty affirm: That all men sleep by intervals; That no man can be nourished by wood or stones; That all men will be poisoned by hemlock: because these ideas have no connexion nor repugnancy with this our nominal essence of man, with this abstract idea that name stands for. We must, in these and the like, appeal to trial in particular subjects, which can reach but a little way. We must content ourselves with probability in the rest: but can have no general certainty, whilst our

specific idea of man contains not that real constitution which is the root wherein all his inseparable qualities are united, and from whence they flow. Whilst our idea the word man stands for is only an imperfect collection of some sensible qualities and powers in him, there is no discernible connexion or repugnance between our specific idea, and the operation of either the parts of hemlock or stones upon his constitution. There are animals that safely eat hemlock, and others that are nourished by wood and stones: but as long as we want ideas of those real constitutions of different sorts of animals whereon these and the like qualities and powers depend, we must not hope to reach certainty in universal propositions concerning them. Those few ideas only which have a discernible connexion with our nominal essence, or any part of it, can afford us such propositions. But these are so few, and of so little moment, that we may justly look on our certain general knowledge of substances as almost none at all.

16. Wherein lies the general certainty of propositions. To conclude: general propositions, of what kind soever,

are then only capable of certainty, when the terms used in them stand for such ideas, whose agreement or disagreement, as there expressed, is capable to be discovered by us. And we are then certain of their truth or falsehood, when we perceive the ideas the terms stand for to agree or not agree, according as they are affirmed or denied one of another. Whence we may take notice, that general certainty is never to be found but in our ideas. Whenever we go to seek it elsewhere, in experiment or observations without us, our knowledge goes not beyond particulars. It is the contemplation of our own abstract ideas that alone is able to afford us general knowledge.

Chapter VII

Of Maxims

1. Maxims or axioms are self-evident propositions. There are a sort of propositions, which, under the name of maxims and axioms, have passed for principles of science: and because they are self-evident, have been supposed in-

nate, without that anybody (that I know) ever went about to show the reason and foundation of their clearness or cogency. It may, however, be worth while to inquire into the reason of their evidence, and see whether it be peculiar to them alone; and also to examine how far they influence and govern our other knowledge.

2. Wherein that self-evidence consists. Knowledge, as has been shown, consists in the perception of the agreement or disagreement of ideas. Now, where that agreement or disagreement is perceived immediately by itself, without the intervention or help of any other, there our knowledge is self-evident. This will appear to be so to any who will but consider any of those propositions which, without any proof, he assents to at first sight: for in all of them he will find that the reason of his assent is from that agreement or disagreement which the mind, by an immediate comparing them, finds in those ideas answering the affirmation or negation in the proposition.

3. Self-evidence not peculiar to received axioms. This being so, in the next place, let us consider whether this

self-evidence be peculiar only to those propositions which commonly pass under the name of maxims, and have the dignity of axioms allowed them. And here it is plain, that several other truths, not allowed to be axioms, partake equally with them in this self-evidence. This we shall see, if we go over these several sorts of agreement or disagreement of ideas which I have above mentioned, viz. identity, relation, coexistence, and real existence; which will discover to us, that not only those few propositions which have had the credit of maxims are self-evident, but a great many, even almost an infinite number of other propositions are such.

4. I. As to identity and diversity, all propositions are equally self-evident. For, First, The immediate perception of the agreement or disagreement of identity being founded in the mind's having distinct ideas, this affords us as many self-evident propositions as we have distinct ideas. Every one that has any knowledge at all, has, as the foundation of it, various and distinct ideas: and it is the first act of the mind (without which it can never be capable of any knowledge) to know every one of its ideas

by itself, and distinguish it from others. Every one finds in himself, that he knows the ideas he has; that he knows also, when any one is in his understanding, and what it is; and that when more than one are there, he knows them distinctly and unconfusedly one from another; which always being so, (it being impossible but that he should perceive what he perceives,) he can never be in doubt when any idea is in his mind, that it is there, and is that idea it is; and that two distinct ideas, when they are in his mind, are there, and are not one and the same idea. So that all such affirmations and negations are made without any possibility of doubt, uncertainty, or hesitation, and must necessarily be assented to as soon as understood; that is, as soon as we have in our minds determined ideas, which the terms in the proposition stand for. And, therefore, whenever the mind with attention considers any proposition, so as to perceive the two ideas signified by the terms, and affirmed or denied one of the other to be the same or different; it is presently and infallibly certain of the truth of such a proposition; and this equally whether these

propositions be in terms standing for more general ideas, or such as are less so: v.g. whether the general idea of Being be affirmed of itself, as in this proposition, "whatsoever is, is"; or a more particular idea be affirmed of itself, as "a man is a man"; or, "whatsoever is white is white"; or whether the idea of being in general be denied of not-Being, which is the only (if I may so call it) idea different from it, as in this other proposition, "it is impossible for the same thing to be and not to be": or any idea of any particular being be denied of another different from it, as "a man is not a horse"; "red is not blue." The difference of the ideas, as soon as the terms are understood, makes the truth of the proposition presently visible, and that with an equal certainty and easiness in the less as well as the more general propositions; and all for the same reason, viz. because the mind perceives, in any ideas that it has, the same idea to be the same with itself; and two different ideas to be different, and not the same; and this it is equally certain of, whether these ideas be more or less general, abstract, and comprehensive. It is not, therefore, alone to these

two general propositions—"whatsoever is, is"; and "it is impossible for the same thing to be and not to be"—that this sort of self-evidence belongs by any peculiar right. The perception of being, or not being, belongs no more to these vague ideas, signified by the terms whatsoever, and thing, than it does to any other ideas. These two general maxims, amounting to no more, in short, but this, that the same is the same, and the same is not different, are truths known in more particular instances, as well as in those general maxims; and known also in particular instances, before these general maxims are ever thought on; and draw all their force from the discernment of the mind employed about particular ideas. There is nothing more visible than that the mind, without the help of any proof, or reflection on either of these general propositions, perceives so clearly, and knows so certainly, that the idea of white is the idea of white, and not the idea of blue; and that the idea of white, when it is in the mind, is there, and is not absent; that the consideration of these axioms can add nothing to the evidence or certainty of its knowledge. Just so it is

(as every one may experiment in himself) in all the ideas a man has in his mind: he knows each to be itself, and not to be another; and to be in his mind, and not away when it is there, with a certainty that cannot be greater; and, therefore, the truth of no general proposition can be known with a greater certainty, nor add anything to this. So that, in respect of identity, our intuitive knowledge reaches as far as our ideas. And we are capable of making as many self-evident propositions, as we have names for distinct ideas. And I appeal to every one's own mind, whether this proposition, "a circle is a circle," be not as self-evident a proposition as that consisting of more general terms, "whatsoever is, is"; and again, whether this proposition, "blue is not red," be not a proposition that the mind can no more doubt of, as soon as it understands the words, than it does of that axiom, "it is impossible for the same thing to be and not to be?" And so of all the like.

5. II. In co-existence we have few self-evident propositions. Secondly, as to co-existence, or such a necessary connexion between two ideas that, in the subject where

one of them is supposed, there the other must necessarily be also: of such agreement or disagreement as this, the mind has an immediate perception but in very few of them. And therefore in this sort we have but very little intuitive knowledge: nor are there to be found very many propositions that are self-evident, though some there are: v.g. the idea of filling a place equal to the contents of its superficies, being annexed to our idea of body, I think it is a self-evident proposition, that two bodies cannot be in the same place.

6. III. In other relations we may have many. Thirdly, As to the relations of modes, mathematicians have framed many axioms concerning that one relation of equality. As, "equals taken from equals, the remainder will be equal"; which, with the rest of that kind, however they are received for maxims by the mathematicians, and are unquestionable truths, yet, I think, that any one who considers them will not find that they have a clearer self-evidence than these,—that "one and one are equal to two"; that "if you take from the five fingers of one hand two, and from the five fingers of the other hand

two, the remaining numbers will be equal." These and a thousand other such propositions may be found in numbers, which, at the very first hearing, force the assent, and carry with them an equal, if not greater clearness, than those mathematical axioms.

7. IV. Concerning real existence, we have none. Fourthly, as to real existence, since that has no connexion with any other of our ideas, but that of ourselves, and of a First Being, we have in that, concerning the real existence of all other beings, not so much as demonstrative, much less a self-evident knowledge: and, therefore, concerning those there are no maxims.

8. These axioms do not much influence our other knowledge. In the next place let us consider, what influence these received maxims have upon the other parts of our knowledge. The rules established in the schools, that all reasonings are *Ex praeognitis et praeconcessis*, seem to lay the foundation of all other knowledge in these maxims, and to suppose them to be *praeognita*. Whereby, I think, are meant these two things: first, that these axioms are those truths that are first known to the mind;

and, secondly, that upon them the other parts of our knowledge depend.

9. Because maxims or axioms are not the truths we first knew. First, That they are not the truths first known to the mind is evident to experience, as we have shown in another place. (Bk. I. chap. i.) Who perceives not that a child certainly knows that a stranger is not its mother; that its sucking-bottle is not the rod, long before he knows that "it is impossible for the same thing to be and not to be?" And how many truths are there about numbers, which it is obvious to observe that the mind is perfectly acquainted with, and fully convinced of, before it ever thought on these general maxims, to which mathematicians, in their arguings, do sometimes refer them? Whereof the reason is very plain: for that which makes the mind assent to such propositions, being nothing else but the perception it has of the agreement or disagreement of its ideas, according as it finds them affirmed or denied one of another in words it understands; and every idea being known to be what it is, and every two distinct ideas being known not to be the

same; it must necessarily follow, that such self-evident truths must be first known which consist of ideas that are first in the mind. And the ideas first in the mind, it is evident, are those of particular things, from whence, by slow degrees, the understanding proceeds to some few general ones; which being taken from the ordinary and familiar objects of sense, are settled in the mind, with general names to them. Thus particular ideas are first received and distinguished, and so knowledge got about them; and next to them, the less general or specific, which are next to particular. For abstract ideas are not so obvious or easy to children, or the yet unexercised mind, as particular ones. If they seem so to grown men, it is only because by constant and familiar use they are made so. For, when we nicely reflect upon them, we shall find that general ideas are fictions and contrivances of the mind, that carry difficulty with them, and do not so easily offer themselves as we are apt to imagine. For example, does it not require some pains and skill to form the general idea of a triangle, (which is yet none of the most abstract, comprehensive, and difficult,) for

it must be neither oblique nor rectangle, neither equilateral, equicrural, nor scalenon; but all and none of these at once. In effect, it is something imperfect, that cannot exist; an idea wherein some parts of several different and inconsistent ideas are put together. It is true, the mind, in this imperfect state, has need of such ideas, and makes all the haste to them it can, for the conveniency of communication and enlargement of knowledge; to both which it is naturally very much inclined. But yet one has reason to suspect such ideas are marks of our imperfection; at least, this is enough to show that the most abstract and general ideas are not those that the mind is first and most easily acquainted with, nor such as its earliest knowledge is conversant about.

10. Because on perception of them the other parts of our knowledge do not depend. Secondly, from what has been said it plainly follows, that these magnified maxims are not the principles and foundations of all our other knowledge. For if there be a great many other truths, which have as much self-evidence as they, and a

great many that we know before them, it is impossible they should be the principles from which we deduce all other truths. Is it impossible to know that one and two are equal to three, but by virtue of this, or some such axiom, viz. "the whole is equal to all its parts taken together?" Many a one knows that one and two are equal to three, without having heard, or thought on, that or any other axiom by which it might be proved; and knows it as certainly as any other man knows, that "the whole is equal to all its parts," or any other maxim; and all from the same reason of self-evidence: the equality of those ideas being as visible and certain to him without that or any other axiom as with it, it needing no proof to make it perceived. Nor after the knowledge, that the whole is equal to all its parts, does he know that one and two are equal to three, better or more certainly than he did before. For if there be any odds in those ideas, the whole and parts are more obscure, or at least more difficult to be settled in the mind than those of one, two, and three. And indeed, I think, I may ask these men, who will needs have all knowledge, besides

those general principles themselves, to depend on general, innate, and self-evident principles. What principle is requisite to prove that one and one are two, that two and two are four, that three times two are six? Which being known without any proof, do evince, That either all knowledge does not depend on certain praecognita or general maxims, called principles; or else that these are principles: and if these are to be counted principles, a great part of numeration will be so. To which, if we add all the self-evident propositions which may be made about all our distinct ideas, principles will be almost infinite, at least innumerable, which men arrive to the knowledge of, at different ages; and a great many of these innate principles they never come to know all their lives. But whether they come in view of the mind earlier or later, this is true of them, that they are all known by their native evidence; are wholly independent; receive no light, nor are capable of any proof one from another; much less the more particular from the more general, or the more simple from the more compounded; the more simple and less abstract being the most familiar, and the

easier and earlier apprehended. But whichever be the clearest ideas, the evidence and certainty of all such propositions is in this, That a man sees the same idea to be the same idea, and infallibly perceives two different ideas to be different ideas. For when a man has in his understanding the ideas of one and of two, the idea of yellow, and the idea of blue, he cannot but certainly know that the idea of one is the idea of one, and not the idea of two; and that the idea of yellow is the idea of yellow, and not the idea of blue. For a man cannot confound the ideas in his mind, which he has distinct: that would be to have them confused and distinct at the same time, which is a contradiction: and to have none distinct, is to have no use of our faculties, to have no knowledge at all. And, therefore, what idea soever is affirmed of itself, or whatsoever two entire distinct ideas are denied one of another, the mind cannot but assent to such a proposition as infallibly true, as soon as it understands the terms, without hesitation or need of proof, or regarding those made in more general terms and called maxims.

11. What use these general maxims or axioms have. What shall we then say? Are these general maxims of no use? By no means; though perhaps their use is not that which it is commonly taken to be. But, since doubting in the least of what hath been by some men ascribed to these maxims may be apt to be cried out against, as overturning the foundations of all the sciences; it may be worth while to consider them with respect to other parts of our knowledge, and examine more particularly to what purposes they serve, and to what not.

(1) It is evident from what has been already said, that they are of no use to prove or confirm less general self-evident propositions.

(2) It is as plain that they are not, nor have been the foundations whereon any science hath been built. There is, I know, a great deal of talk, propagated from scholastic men, of sciences and the maxims on which they are built: but it has been my ill-luck never to meet with any such sciences; much less any one built upon these two maxims, what is, is; and it is impossible for the same thing to be and not to be. And I would be glad to be

shown where any such science, erected upon these or any other general axioms is to be found: and should be obliged to any one who would lay before me the frame and system of any science so built on these or any such like maxims, that could not be shown to stand as firm without any consideration of them. I ask, Whether these general maxims have not the same use in the study of divinity, and in theological questions, that they have in other sciences? They serve here, too, to silence wranglers, and put an end to dispute. But I think that nobody will therefore say, that the Christian religion is built upon these maxims, or that the knowledge we have of it is derived from these principals. It is from revelation we have received it, and without revelation these maxims had never been able to help us to it. When we find out an idea by whose intervention we discover the connexion of two others, this is a revelation from God to us by the voice of reason: for we then come to know a truth that we did not know before. When God declares any truth to us, this is a revelation to us by the voice of his Spirit, and we are advanced in our knowledge. But

in neither of these do we receive our light or knowledge from maxims. But in the one, the things themselves afford it: and we see the truth in them by perceiving their agreement or disagreement. In the other, God himself affords it immediately to us: and we see the truth of what he says in his unerring veracity.

(3) They are not of use to help men forward in the advancement of sciences, or new discoveries of yet unknown truths. Mr. Newton, in his never enough to be admired book, has demonstrated several propositions, which are so many new truths, before unknown to the world, and are further advances in mathematical knowledge: but, for the discovery of these, it was not the general maxims, "what is, is;" or, "the whole is bigger than a part," or the like, that helped him. These were not the clues that led him into the discovery of the truth and certainty of those propositions. Nor was it by them that he got the knowledge of those demonstrations, but by finding out intermediate ideas that showed the agreement or disagreement of the ideas, as expressed in the propositions he demonstrated. This is the great-

est exercise and improvement of human understanding in the enlarging of knowledge, and advancing the sciences; wherein they are far enough from receiving any help from the contemplation of these or the like magnified maxims. Would those who have this traditional admiration of these propositions, that they think no step can be made in knowledge without the support of an axiom, no stone laid in the building of the sciences without a general maxim, but distinguish between the method of acquiring knowledge, and of communicating it; between the method of raising any science, and that of teaching it to others, as far as it is advanced—they would see that those general maxims were not the foundations on which the first discoverers raised their admirable structures, not the keys that unlocked and opened those secrets of knowledge. Though afterwards, when schools were erected, and sciences had their professors to teach what others had found out, they often made use of maxims, i.e. laid down certain propositions which were self-evident, or to be received for true; which being settled in the minds of their scholars as unquestionable

verities they on occasion made use of, to convince them of truths in particular instances, that were not so familiar to their minds as those general axioms which had before been inculcated to them, and carefully settled in their minds. Though these particular instances, when well reflected on, are no less self-evident to the understanding than the general maxims brought to confirm them: and it was in those particular instances that the first discoverer found the truth, without the help of the general maxims: and so may any one else do, who with attention considers them. Maxims of use in the exposition of what has been discovered, and in silencing obstinate wranglers. To come, therefore, to the use that is made of maxims.

(1) They are of use, as has been observed, in the ordinary methods of teaching sciences as far as they are advanced: but of little or none in advancing them further.

(2) They are of use in disputes, for the silencing of obstinate wranglers, and bringing those contests to some conclusion. Whether a need of them to that end came

not in the manner following, I crave leave to inquire. The Schools having made disputation the touchstone of men's abilities, and the criterion of knowledge, adjudged victory to him that kept the field: and he that had the last word was concluded to have the better of the argument, if not of the cause. But because by this means there was like to be no decision between skilful combatants, whilst one never failed of a *medius terminus* to prove any proposition; and the other could as constantly, without or with a distinction, deny the major or minor; to prevent, as much as could be, running out of disputes into an endless train of syllogisms, certain general propositions—most of them, indeed, self-evident—were introduced into the Schools: which being such as all men allowed and agreed in, were looked on as general measures of truth, and served instead of principles (where the disputants had not lain down any other between them) beyond which there was no going, and which must not be receded from by either side. And thus these maxims, getting the name of principles, beyond which men in dispute could not retreat, were by mistake taken

to be the originals and sources from whence all knowledge began, and the foundations whereon the sciences were built. Because when in their disputes they came to any of these, they stopped there, and went no further; the matter was determined. But how much this is a mistake, hath been already shown.

How maxims came to be so much in vogue. This method of the Schools, which have been thought the fountains of knowledge, introduced, as I suppose, the like use of these maxims into a great part of conversation out of the Schools, to stop the mouths of cavillers, whom any one is excused from arguing any longer with, when they deny these general self-evident principles received by all reasonable men who have once thought of them: but yet their use herein is but to put an end to wrangling. They in truth, when urged in such cases, teach nothing: that is already done by the intermediate ideas made use of in the debate, whose connexion may be seen without the help of those maxims, and so the truth known before the maxim is produced, and the argument brought to a first principle. Men would give off a wrong

argument before it came to that, if in their disputes they proposed to themselves the finding and embracing of truth, and not a contest for victory. And thus maxims have their use to put a stop to their perverseness, whose ingenuity should have yielded sooner. But the method of the Schools having allowed and encouraged men to oppose and resist evident truth till they are baffled, i.e. till they are reduced to contradict themselves, or some established principles: it is no wonder that they should not in civil conversation be ashamed of that which in the Schools is counted a virtue and a glory, viz. obstinately to maintain that side of the question they have chosen, whether true or false, to the last extremity; even after conviction. A strange way to attain truth and knowledge: and that which I think the rational part of mankind, not corrupted by education, could scarce believe should ever be admitted amongst the lovers of truth, and students of religion or nature, or introduced into the seminaries of those who are to propagate the truths of religion or philosophy amongst the ignorant and unconvinced. How much such a way

of learning is like to turn young men's minds from the sincere search and love of truth; nay, and to make them doubt whether there is any such thing, or, at least, worth the adhering to, I shall not now inquire. This I think, that, bating those places, which brought the Peripatetick Philosophy into their schools, where it continued many ages, without teaching the world anything but the art of wrangling, these maxims were nowhere thought the foundations on which the sciences were built, nor the great helps to the advancement of knowledge.

Of great use to stop wranglers in disputes, but of little use to the discovery of truths. As to these general maxims, therefore, they are, as I have said, of great use in disputes, to stop the mouths of wranglers; but not of much use to the discovery of unknown truths, or to help the mind forwards in its search after knowledge. For who ever began to build his knowledge on the general proposition, what is, is; or, it is impossible for the same thing to be and not to be: and from either of these, as from a principle of science, deduced a system

of useful knowledge? Wrong opinions often involving contradictions, one of these maxims, as a touchstone, may serve well to show whither they lead. But yet, however fit to lay open the absurdity or mistake of a man's reasoning or opinion, they are of very little use for enlightening the understanding: and it will not be found that the mind receives much help from them in its progress in knowledge; which would be neither less, nor less certain, were these two general propositions never thought on. It is true, as I have said, they sometimes serve in argumentation to stop a wrangler's mouth, by showing the absurdity of what he saith, and by exposing him to the shame of contradicting what all the world knows, and he himself cannot but own to be true. But it is one thing to show a man that he is in an error, and another to put him in possession of truth; and I would fain know what truths these two propositions are able to teach, and by their influence make us know, which we did not know before, or could not know without them. Let us reason from them as well as we can, they are only about identical predications, and influence, if

any at all, none but such. Each particular proposition concerning identity or diversity is as clearly and certainly known in itself, if attended to, as either of these general ones: only these general ones, as serving in all cases, are therefore more inculcated and insisted on. As to other less general maxims, many of them are no more than bare verbal propositions, and teach us nothing but the respect and import of names one to another. "The whole is equal to all its parts": what real truth, I beseech you, does it teach us? What more is contained in that maxim, than what the signification of the word totum, or the whole, does of itself import? And he that knows that the word whole stands for what is made up of all its parts, knows very little less than that the whole is equal to all its parts. And, upon the same ground, I think that this proposition, "A hill is higher than a valley," and several the like, may also pass for maxims. But yet masters of mathematics, when they would, as teachers of what they know, initiate others in that science, do not without reason place this and some other such maxims at the entrance of their systems; that their

scholars, having in the beginning perfectly acquainted their thoughts with these propositions, made in such general terms, may be used to make such reflections, and have these more general propositions, as formed rules and sayings, ready to apply to all particular cases. Not that if they be equally weighed, they are more clear and evident than the particular instances they are brought to confirm; but that, being more familiar to the mind, the very naming them is enough to satisfy the understanding. But this, I say, is more from our custom of using them, and the establishment they have got in our minds by our often thinking of them, than from the different evidence of the things. But before custom has settled methods of thinking and reasoning in our minds, I am apt to imagine it is quite otherwise; and that the child, when a part of his apple is taken away, knows it better in that particular instance, than by this general proposition, "The whole is equal to all its parts"; and that, if one of these have need to be confirmed to him by the other, the general has more need to be let into his mind by the particular, than the par-

ticular by the general. For in particulars our knowledge begins, and so spreads itself, by degrees, to generals. Though afterwards the mind takes the quite contrary course, and having drawn its knowledge into as general propositions as it can, makes those familiar to its thoughts, and accustoms itself to have recourse to them, as to the standards of truth and falsehood. By which familiar use of them, as rules to measure the truth of other propositions, it comes in time to be thought, that more particular propositions have their truth and evidence from their conformity to these more general ones, which, in discourse and argumentation, are so frequently urged, and constantly admitted. And this I think to be the reason why, amongst so many self-evident propositions, the most general only have had the title of maxims.

12. Maxims, if care be not taken in the use of words, may prove contradictions. One thing further, I think, it may not be amiss to observe concerning these general maxims, That they are so far from improving or establishing our minds in true knowledge, that if our notions be wrong, loose, or unsteady, and we resign up our

thoughts to the sound of words, rather than fix them on settled, determined ideas of things; I say these general maxims will serve to confirm us in mistakes; and in such a way of use of words, which is most common, will serve to prove contradictions: v.g. he that with Descartes shall frame in his mind an idea of what he calls body to be nothing but extension, may easily demonstrate that there is no vacuum, i.e. no space void of body, by this maxim, What is, is. For the idea to which he annexes the name body, being bare extension, his knowledge that space cannot be without body, is certain. For he knows his own idea of extension clearly and distinctly, and knows that it is what it is, and not another idea, though it be called by these three names,—extension, body, space. Which three words, standing for one and the same idea, may, no doubt, with the same evidence and certainty be affirmed one of another, as each of itself: and it is as certain, that, whilst I use them all to stand for one and the same idea, this predication is as true and identical in its signification, that “space is body,” as this predication is true and identical, that “body is body,”

both in signification and sound.

13. Instance in vacuum. But if another should come and make to himself another idea, different from Descartes's, of the thing, which yet with Descartes he calls by the same name body, and make his idea, which he expresses by the word body, to be of a thing that hath both extension and solidity together; he will as easily demonstrate, that there may be a vacuum or space without a body, as Descartes demonstrated the contrary. Because the idea to which he gives the name space being barely the simple one of extension, and the idea to which he gives the name body being the complex idea of extension and resistibility or solidity, together in the same subject, these two ideas are not exactly one and the same, but in the understanding as distinct as the ideas of one and two, white and black, or as of corporeity and humanity, if I may use those barbarous terms: and therefore the predication of them in our minds, or in words standing for them, is not identical, but the negation of them one of another; viz. this proposition: "Extension or space is not body," is as true and evidently

certain as this maxim, It is impossible for the same thing to be and not to be, can make any proposition.

14. But they prove not the existence of things without us. But yet, though both these propositions (as you see) may be equally demonstrated, viz. that there may be a vacuum, and that there cannot be a vacuum, by these two certain principles, viz. what is, is, and the same thing cannot be and not be: yet neither of these principles will serve to prove to us, that any, or what bodies do exist: for that we are left to our senses to discover to us as far as they can. Those universal and self-evident principles being only our constant, clear, and distinct knowledge of our own ideas, more general or comprehensive, can assure us of nothing that passes without the mind: their certainty is founded only upon the knowledge we have of each idea by itself, and of its distinction from others, about which we cannot be mistaken whilst they are in our minds; though we may be and often are mistaken when we retain the names without the ideas; or use them confusedly, sometimes for one and sometimes for another idea. In which cases the

force of these axioms, reaching only to the sound, and not the signification of the words, serves only to lead us into confusion, mistake, and error. It is to show men that these maxims, however cried up for the great guards of truth, will not secure them from error in a careless loose use of their words, that I have made this remark. In all that is here suggested concerning their little use for the improvement of knowledge, or dangerous use in undetermined ideas, I have been far enough from saying or intending they should be laid aside; as some have been too forward to charge me. I affirm them to be truths, self-evident truths; and so cannot be laid aside. As far as their influence will reach, it is in vain to endeavour, nor will I attempt, to abridge it. But yet, without any injury to truth or knowledge, I may have reason to think their use is not answerable to the great stress which seems to be laid on them; and I may warn men not to make an ill use of them, for the confirming themselves in errors.

15. They cannot add to our knowledge of substances, and their application to complex ideas is dangerous. But

let them be of what use they will in verbal propositions, they cannot discover or prove to us the least knowledge of the nature of substances, as they are found and exist without us, any further than grounded on experience. And though the consequence of these two propositions, called principles, be very clear, and their use not dangerous or hurtful, in the probation of such things wherein there is no need at all of them for proof, but such as are clear by themselves without them, viz. where our ideas are [determined] and known by the names that stand for them: yet when these principles, viz. what is, is, and it is impossible for the same thing to be and not to be, are made use of in the probation of propositions wherein are words standing for complex ideas, v.g. man, horse, gold, virtue; there they are of infinite danger, and most commonly make men receive and retain falsehood for manifest truth, and uncertainty for demonstration: upon which follow error, obstinacy, and all the mischiefs that can happen from wrong reasoning. The reason whereof is not, that these principles are less true or of less force in proving propositions made of

terms standing for complex ideas, than where the propositions are about simple ideas. But because men mistake generally,—thinking that where the same terms are preserved, the propositions are about the same things, though the ideas they stand for are in truth different, therefore these maxims are made use of to support those which in sound and appearance are contradictory propositions; and is clear in the demonstrations above mentioned about a vacuum. So that whilst men take words for things, as usually they do, these maxims may and do commonly serve to prove contradictory propositions; as shall yet be further made manifest.

16. Instance in demonstrations about man, which can only be verbal. For instance: let man be that concerning which you would by these first principles demonstrate anything, and we shall see, that so far as demonstration is by these principles, it is only verbal, and gives us no certain, universal, true proposition, or knowledge, of any being existing without us. First, a child having framed the idea of a man, it is probable that his idea is just like that picture which the painter makes of the visible ap-

pearances joined together; and such a complication of ideas together in his understanding makes up the single complex idea which he calls man, whereof white or flesh-colour in England being one, the child can demonstrate to you that a negro is not a man, because white colour was one of the constant simple ideas of the complex idea he calls man; and therefore he can demonstrate, by the principle, It is impossible for the same thing to be and not to be, that a negro is not a man; the foundation of his certainty being not that universal proposition, which perhaps he never heard nor thought of, but the clear, distinct perception he hath of his own simple ideas of black and white, which he cannot be persuaded to take, nor can ever mistake one for another, whether he knows that maxim or no. And to this child, or any one who hath such an idea, which he calls man, can you never demonstrate that a man hath a soul, because his idea of man includes no such notion or idea in it. And therefore, to him, the principle of What is, is, proves not this matter; but it depends upon collection and observation, by which he is to make his complex idea called man.

17. Another instance. Secondly, Another that hath gone further in framing and collecting the idea he calls man, and to the outward shape adds laughter and rational discourse, may demonstrate that infants and changelings are no men, by this maxim, it is impossible for the same thing to be and not to be; and I have discoursed with very rational men, who have actually denied that they are men.

18. A third instance. Thirdly, Perhaps another makes up the complex idea which he calls man, only out of the ideas of body in general, and the powers of language and reason, and leaves out the shape wholly: this man is able to demonstrate that a man may have no hands, but be quadrupes, neither of those being included in his idea of man: and in whatever body or shape he found speech and reason joined, that was a man; because, having a clear knowledge of such a complex idea, it is certain that What is, is.

19. Little use of these maxims in proofs where we have clear and distinct ideas. So that, if rightly considered, I think we may say, That where our ideas are de-

termined in our minds, and have annexed to them by us known and steady names under those settled determinations, there is little need, or no use at all of these maxims, to prove the agreement or disagreement of any of them. He that cannot discern the truth or falsehood of such propositions, without the help of these and the like maxims, will not be helped by these maxims to do it: since he cannot be supposed to know the truth of these maxims themselves without proof, if he cannot know the truth of others without proof, which are as self-evident as these. Upon this ground it is that intuitive knowledge neither requires nor admits any proof, one part of it more than another. He that will suppose it does, takes away the foundation of all knowledge and certainty; and he that needs any proof to make him certain, and give his assent to this proposition, that two are equal to two, will also have need of a proof to make him admit, that what is, is. He that needs a probation to convince him that two are not three, that white is not black, that a triangle is not a circle, &c., or any other two [determined] distinct ideas are not one and

the same, will need also a demonstration to convince him that It is impossible for the same thing to be and not to be.

20. Their use dangerous, where our ideas are not determined. And as these maxims are of little use where we have determined ideas, so they are, as I have shown, of dangerous use where our ideas are not determined; and where we use words that are not annexed to determined ideas, but such as are of a loose and wandering signification, sometimes standing for one, and sometimes for another idea: from which follow mistake and error, which these maxims (brought as proofs to establish propositions, wherein the terms stand for undetermined ideas) do by their authority confirm and rivet.

Chapter VIII

Of Trifling Propositions

1. Some propositions bring no increase to our knowledge. Whether the maxims treated of in the foregoing chapter be of that use to real knowledge as is generally

supposed, I leave to be considered. This, I think, may confidently be affirmed, That there are universal propositions, which, though they be certainly true, yet they add no light to our understanding; bring no increase to our knowledge. Such are—

2. I. As identical propositions. First, All purely identical propositions. These obviously and at first blush appear to contain no instruction in them; for when we affirm the said term of itself, whether it be barely verbal, or whether it contains any clear and real idea, it shows us nothing but what we must certainly know before, whether such a proposition be either made by, or proposed to us. Indeed, that most general one, what is, is, may serve sometimes to show a man the absurdity he is guilty of, when, by circumlocution or equivocal terms, he would in particular instances deny the same thing of itself; because nobody will so openly bid defiance to common sense, as to affirm visible and direct contradictions in plain words; or, if he does, a man is excused if he breaks off any further discourse with him. But yet I think I may say, that neither that received

maxim, nor any other identical proposition, teaches us anything; and though in such kind of propositions this great and magnified maxim, boasted to be the foundation of demonstration, may be and often is made use of to confirm them, yet all it proves amounts to no more than this, That the same word may with great certainty be affirmed of itself, without any doubt of the truth of any such proposition; and let me add, also, without any real knowledge.

3. Examples. For, at this rate, any very ignorant person, who can but make a proposition, and knows what he means when he says ay or no, may make a million of propositions of whose truth he may be infallibly certain, and yet not know one thing in the world thereby; v.g. "what is a soul, is a soul,"; or, "a soul is a soul"; "a spirit is a spirit"; "a fetiche is a fetiche," &c. These all being equivalent to this proposition, viz. what is, is; i.e. what hath existence, hath existence; or, who hath a soul, hath a soul. What is this more than trifling with words? It is but like a monkey shifting his oyster from one hand to the other: and had he but words, might no

doubt have said, "Oyster in right hand is subject, and oyster in left hand is predicate": and so might have made a self-evident proposition of oyster, i.e. oyster is oyster; and yet, with all this, not have been one whit the wiser or more knowing: and that way of handling the matter would much at once have satisfied the monkey's hunger, or a man's understanding, and they would have improved in knowledge and bulk together.

How identical propositions are trifling. I know there are some who, because identical propositions are self-evident, show a great concern for them, and think they do great service to philosophy by crying them up; as if in them was contained all knowledge, and the understanding were led into all truth by them only. I grant as forwardly as any one, that they are all true and self-evident. I grant further, that the foundation of all our knowledge lies in the faculty we have of perceiving the same idea to be the same, and of discerning it from those that are different; as I have shown in the foregoing chapter. But how that vindicates the making use of identical propositions, for the improvement of knowl-

edge, from the imputation of trifling, I do not see. Let any one repeat, as often as he pleases, that "the will is the will," or lay what stress on it he thinks fit; of what use is this, and an infinite the like propositions, for the enlarging our knowledge? Let a man abound, as much as the plenty of words which he has will permit, in such propositions as these: "a law is a law," and "obligation is obligation"; "right is right," and "wrong is wrong":—will these and the like ever help him to an acquaintance with ethics, or instruct him or others in the knowledge of morality? Those who know not, nor perhaps ever will know, what is right and what is wrong, nor the measures of them, can with as much assurance make, and infallibly know, the truth of these and all such propositions, as he that is best instructed in morality can do. But what advance do such propositions give in the knowledge of anything necessary or useful for their conduct?

He would be thought to do little less than trifle, who, for the enlightening the understanding in any part of knowledge, should be busy with identical propositions and insist on such maxims as these: "substance is sub-

stance," and "body is body"; "a vacuum is a vacuum," and "a vortex is a vortex"; "a centaur is a centaur," and "a chimera is a chimera," &c. For these and all such are equally true, equally certain, and equally self-evident. But yet they cannot but be counted trifling, when made use of as principles of instruction, and stress laid on them as helps to knowledge; since they teach nothing but what every one who is capable of discourse knows without being told, viz. that the same term is the same term, and the same idea the same idea. And upon this account it was that I formerly did, and do still think, the offering and inculcating such propositions, in order to give the understanding any new light, or inlet into the knowledge of things, no better than trifling.

Instruction lies in something very different; and he that would enlarge his own or another's mind to truths he does not yet know, must find out intermediate ideas, and then lay them in such order one by another, that the understanding may see the agreement or disagreement of those in question. Propositions that do this are instructive; but they are far from such as affirm the

same term of itself; which is no way to advance one's self or others in any sort of knowledge. It no more helps to that than it would help any one in his learning to read, to have such propositions as these inculcated to him—"An A is an A," and "a B is a B"; which a man may know as well as any schoolmaster, and yet never be able to read a word as long as he lives. Nor do these, or any such identical propositions help him one jot forwards in the skill of reading, let him make what use of them he can.

If those who blame my calling them trifling propositions had but read and been at the pains to understand what I have above writ in very plain English, they could not but have seen that by identical propositions I mean only such wherein the same term, importing the same idea, is affirmed of itself: which I take to be the proper signification of identical propositions; and concerning all such, I think I may continue safely to say, that to propose them as instructive is no better than trifling. For no one who has the use of reason can miss them, where it is necessary they should be taken notice of;

nor doubt of their truth when he does take notice of them. But if men will call propositions identical, wherein the same term is not affirmed of itself, whether they speak more properly than I, others must judge; this is certain, all that they say of propositions that are not identical in my sense, concerns not me nor what I have said; all that I have said relating to those propositions wherein the same term is affirmed of itself. And I would fain see an instance wherein any such can be made use of, to the advantage and improvement of any one's knowledge. Instances of other kinds, whatever use may be made of them, concern not me, as not being such as I call identical.

4. II. Secondly, propositions in which a part of any complex idea is predicated of the whole. Another sort of trifling propositions is, when a part of the complex idea is predicated of the name of the whole; a part of the definition of the word defined. Such are all propositions wherein the genus is predicated of the species, or more comprehensive of less comprehensive terms. For what information, what knowledge, carries this proposition

in it, viz. "Lead is a metal" to a man who knows the complex idea the name lead stands for? All the simple ideas that go to the complex one signified by the term metal, being nothing but what he before comprehended and signified by the name lead. Indeed, to a man that knows the signification of the word metal, and not of the word lead, it is a shorter way to explain the signification of the word lead, by saying it is a metal, which at once expresses several of its simple ideas, than to enumerate them one by one, telling him it is a body very heavy, fusible, and malleable.

5. As part of the definition of the term defined. Alike trifling it is to predicate any other part of the definition of the term defined, or to affirm any one of the simple ideas of a complex one of the name of the whole complex idea; as, "All gold is fusible." For fusibility being one of the simple ideas that goes to the making up the complex one the sound gold stands for, what can it be but playing with sounds, to affirm that of the name gold, which is comprehended in its received signification? It would be thought little better than ridiculous

to affirm gravely, as a truth of moment, that gold is yellow; and I see not how it is any jot more material to say it is fusible, unless that quality be left out of the complex idea, of which the sound gold is the mark in ordinary speech. What instruction can it carry with it, to tell one that which he hath been told already, or he is supposed to know before? For I am supposed to know the signification of the word another uses to me, or else he is to tell me. And if I know that the name gold stands for this complex idea of body, yellow, heavy, fusible, malleable, it will not much instruct me to put it solemnly afterwards in a proposition, and gravely say, all gold is fusible. Such propositions can only serve to show the disingenuity of one who will go from the definition of his own terms, by reminding him sometimes of it; but carry no knowledge with them, but of the signification of words, however certain they be.

6. Instance, man and palfrey. "Every man is an animal, or living body," is as certain a proposition as can be; but no more conducing to the knowledge of things than to say, a palfrey is an ambling horse, or a neighing,

ambling animal, both being only about the signification of words, and make me know but this—That body, sense, and motion, or power of sensation and moving, are three of those ideas that I always comprehend and signify by the word man: and where they are not to be found together, the name man belongs not to that thing: and so of the other—That body, sense, and a certain way of going, with a certain kind of voice, are some of those ideas which I always comprehend and signify by the word palfrey; and when they are not to be found together, the name palfrey belongs not to that thing. It is just the same, and to the same purpose, when any term standing for any one or more of the simple ideas, that altogether make up that complex idea which is called man, is affirmed of the term man:—v.g. suppose a Roman signified by the word homo all these distinct ideas united in one subject, corporietas, sensibilitas, potentia se movendi rationalitas, risibilitas; he might, no doubt, with great certainty, universally affirm one, more, or all of these together of the word homo, but did no more than say that the word homo, in his country, compre-

hended in its signification all these ideas. Much like a romance knight, who by the word palfrey signified these ideas:—body of a certain figure, four-legged, with sense, motion, ambling, neighing, white, used to have a woman on his back—might with the same certainty universally affirm also any or all of these of the word palfrey: but did thereby teach no more, but that the word palfrey, in his or romance language, stood for all these, and was not to be applied to anything where any of these was wanting. But he that shall tell me, that in whatever thing sense, motion, reason, and laughter, were united, that thing had actually a notion of God, or would be cast into a sleep by opium, made indeed an instructive proposition: because neither having the notion of God, nor being cast into sleep by opium, being contained in the idea signified by the word man, we are by such propositions taught something more than barely what the word man stands for: and therefore the knowledge contained in it is more than verbal.

7. For this teaches but the signification of words. Before a man makes any proposition, he is supposed to

understand the terms he uses in it, or else he talks like a parrot, only making a noise by imitation, and framing certain sounds, which he has learnt of others; but not as a rational creature, using them for signs of ideas which he has in his mind. The hearer also is supposed to understand the terms as the speaker uses them, or else he talks jargon, and makes an unintelligible noise. And therefore he trifles with words who makes such a proposition, which, when it is made, contains no more than one of the terms does, and which a man was supposed to know before: v.g. a triangle hath three sides, or saffron is yellow. And this is no further tolerable than where a man goes to explain his terms to one who is supposed or declares himself not to understand him; and then it teaches only the signification of that word, and the use of that sign.

8. But adds no real knowledge. We can know then the truth of two sorts of propositions with perfect certainty. The one is, of those trifling propositions which have a certainty in them, but it is only a verbal certainty, but not instructive. And, secondly, we can know the truth,

and so may be certain in propositions, which affirm something of another, which is a necessary consequence of its precise complex idea, but not contained in it: as that the external angle of all triangles is bigger than either of the opposite internal angles. Which relation of the outward angle to either of the opposite internal angles, making no part of the complex idea signified by the name triangle, this is a real truth, and conveys with it instructive real knowledge.

9. General propositions concerning substances are often trifling. We having little or no knowledge of what combinations there be of simple ideas existing together in substances, but by our senses, we cannot make any universal certain propositions concerning them, any further than our nominal essences lead us. Which being to a very few and inconsiderable truths, in respect of those which depend on their real constitutions, the general propositions that are made about substances, if they are certain, are for the most part but trifling; and if they are instructive, are uncertain, and such as we can have no knowledge of their real truth, how much soever

constant observation and analogy may assist our judgment in guessing. Hence it comes to pass, that one may often meet with very clear and coherent discourses, that amount yet to nothing. For it is plain that names of substantial beings, as well as others, as far as they have relative significations affixed to them, may, with great truth, be joined negatively and affirmatively in propositions, as their relative definitions make them fit to be so joined; and propositions consisting of such terms, may, with the same clearness, be deduced one from another, as those that convey the most real truths: and all this without any knowledge of the nature or reality of things existing without us. By this method one may make demonstrations and undoubted propositions in words, and yet thereby advance not one jot in the knowledge of the truth of things: v.g. he that having learnt these following words, with their ordinary mutual relative acceptations annexed to them: v.g. substance, man, animal, form, soul, vegetative, sensitive, rational, may make several undoubted propositions about the soul, without knowing at all what the soul really is: and of this sort,

a man may find an infinite number of propositions, reasonings, and conclusions, in books of metaphysics, school-divinity, and some sort of natural philosophy: and, after all, know as little of God, spirits, or bodies, as he did before he set out.

10. And why. He that hath liberty to define, i.e. to determine the signification of his names of substances (as certainly every one does in effect, who makes them stand for his own ideas), and makes their significations at a venture, taking them from his own or other men's fancies, and not from an examination or inquiry into the nature of things themselves; may with little trouble demonstrate them one of another, according to those several respects and mutual relations he has given them one to another; wherein, however things agree or disagree in their own nature, he needs mind nothing but his own notions, with the names he hath bestowed upon them: but thereby no more increases in his own knowledge than he does his riches, who, taking a bag of counters, calls one in a certain place a pound, another in another place a shilling, and a third in a third place a

penny; and so proceeding, may undoubtedly reckon right, and cast up a great sum, according to his counters so placed, and standing for more or less as he pleases, without being one jot the richer, or without even knowing how much a pound, shilling, or penny is, but only that one is contained in the other twenty times, and contains the other twelve: which a man may also do in the signification of words, by making them, in respect of one another, more or less, or equally comprehensive.

11. Thirdly, using words variously is trifling with them. Though yet concerning most words used in discourses, equally argumentative and controversial, there is this more to be complained of, which is the worst sort of trifling, and which sets us yet further from the certainty of knowledge we hope to attain by them, or find in them; viz. that most writers are so far from instructing us in the nature and knowledge of things, that they use their words loosely and uncertainly, and do not. by using them constantly and steadily in the same significations, make plain and clear deductions of words one from another, and make their discourses coherent and

clear, (how little soever they were instructive); which were not difficult to do, did they not find it convenient to shelter their ignorance or obstinacy under the obscurity and perplexedness of their terms: to which, perhaps, inadvertency and ill custom do in many men much contribute.

12. Marks of verbal propositions. To conclude. Barely verbal propositions may be known by these following marks: Predication in abstract. First, All propositions wherein two abstract terms are affirmed one of another, are barely about the signification of sounds. For since no abstract idea can be the same with any other but itself, when its abstract name is affirmed of any other term, it can signify no more but this, that it may, or ought to be called by that name; or that these two names signify the same idea. Thus, should any one say that parsimony is frugality, that gratitude is justice, that this or that action is or is not temperate: however specious these and the like propositions may at first sight seem, yet when we come to press them, and examine nicely what they contain, we shall find that it all amounts

to nothing but the signification of those terms.

13. A part of the definition predicated of any term. Secondly, All propositions wherein a part of the complex idea which any term stands for is predicated of that term, are only verbal: v.g. to say that gold is a metal, or heavy. And thus all propositions wherein more comprehensive words, called genera, are affirmed of subordinate or less comprehensive, called species, or individuals, are barely verbal. When by these two rules we have examined the propositions that make up the discourses we ordinarily meet with, both in and out of books, we shall perhaps find that a greater part of them than is usually suspected are purely about the signification of words, and contain nothing in them but the use and application of these signs.

This I think I may lay down for an infallible rule, That, wherever the distinct idea any word stands for is not known and considered, and something not contained in the idea is not affirmed or denied of it, there our thoughts stick wholly in sounds, and are able to attain no real truth or falsehood. This, perhaps, if well heeded, might

save us a great deal of useless amusement and dispute; and very much shorten our trouble and wandering in the search of real and true knowledge.

Chapter IX

Of our Threefold Knowledge of Existence

1. General propositions that are certain concern not existence. Hitherto we have only considered the essences of things; which being only abstract ideas, and thereby removed in our thoughts from particular existence, (that being the proper operation of the mind, in abstraction, to consider an idea under no other existence but what it has in the understanding,) gives us no knowledge of real existence at all. Where, by the way, we may take notice, that universal propositions of whose truth or falsehood we can have certain knowledge concern not existence: and further, that all particular affirmations or negations that would not be certain if they were made general, are only concerning existence; they declaring only the accidental union or separation of ideas

in things existing, which, in their abstract natures, have no known necessary union or repugnancy.

2. A threefold knowledge of existence. But, leaving the nature of propositions, and different ways of predication to be considered more at large in another place, let us proceed now to inquire concerning our knowledge of the existence of things, and how we come by it. I say, then, that we have the knowledge of our own existence by intuition; of the existence of God by demonstration; and of other things by sensation.

3. Our knowledge of our own existence is intuitive. As for our own existence, we perceive it so plainly and so certainly, that it neither needs nor is capable of any proof. For nothing can be more evident to us than our own existence. I think, I reason, I feel pleasure and pain: can any of these be more evident to me than my own existence? If I doubt of all other things, that very doubt makes me perceive my own existence, and will not suffer me to doubt of that. For if I know I feel pain, it is evident I have as certain perception of my own existence, as of the existence of the pain I feel: or if I

know I doubt, I have as certain perception of the existence of the thing doubting, as of that thought which I call doubt. Experience then convinces us, that we have an intuitive knowledge of our own existence, and an internal infallible perception that we are. In every act of sensation, reasoning, or thinking, we are conscious to ourselves of our own being; and, in this matter, come not short of the highest degree of certainty.

Chapter X

Of our Knowledge of the Existence of a God

1. We are capable of knowing certainly that there is a God. Though God has given us no innate ideas of himself; though he has stamped no original characters on our minds, wherein we may read his being; yet having furnished us with those faculties our minds are endowed with, he hath not left himself without witness: since we have sense, perception, and reason, and cannot want a clear proof of him, as long as we carry ourselves about us. Nor can we justly complain of our ignorance in this

great point; since he has so plentifully provided us with the means to discover and know him; so far as is necessary to the end of our being, and the great concernment of our happiness. But, though this be the most obvious truth that reason discovers, and though its evidence be (if I mistake not) equal to mathematical certainty: yet it requires thought and attention; and the mind must apply itself to a regular deduction of it from some part of our intuitive knowledge, or else we shall be as uncertain and ignorant of this as of other propositions, which are in themselves capable of clear demonstration. To show, therefore, that we are capable of knowing, i.e. being certain that there is a God, and how we may come by this certainty, I think we need go no further than ourselves, and that undoubted knowledge we have of our own existence.

2. For man knows that he himself exists. I think it is beyond question, that man has a clear idea of his own being; he knows certainly he exists, and that he is something. He that can doubt whether he be anything or no, I speak not to; no more than I would argue with

pure nothing, or endeavour to convince nonentity that it were something. If any one pretends to be so sceptical as to deny his own existence, (for really to doubt of it is manifestly impossible,) let him for me enjoy his beloved happiness of being nothing, until hunger or some other pain convince him of the contrary. This, then, I think I may take for a truth, which every one's certain knowledge assures him of, beyond the liberty of doubting, viz. that he is something that actually exists.

3 He knows also that nothing cannot produce a being; therefore something must have existed from eternity. In the next place, man knows, by an intuitive certainty, that bare nothing can no more produce any real being, than it can be equal to two right angles. If a man knows not that nonentity, or the absence of all being, cannot be equal to two right angles, it is impossible he should know any demonstration in Euclid. If, therefore, we know there is some real being, and that nonentity cannot produce any real being, it is an evident demonstration, that from eternity there has been something; since what was not from eternity had a beginning; and what had a

beginning must be produced by something else.

4. And that eternal Being must be most powerful. Next, it is evident, that what had its being and beginning from another, must also have all that which is in and belongs to its being from another too. All the powers it has must be owing to and received from the same source. This eternal source, then, of all being must also be the source and original of all power; and so this eternal Being must be also the most powerful.

5. And most knowing. Again, a man finds in himself perception and knowledge. We have then got one step further; and we are certain now that there is not only some being, but some knowing, intelligent being in the world. There was a time, then, when there was no knowing being, and when knowledge began to be; or else there has been also a knowing being from eternity. If it be said, there was a time when no being had any knowledge, when that eternal being was void of all understanding; I reply, that then it was impossible there should ever have been any knowledge: it being as impossible that things wholly void of knowledge, and operating

blindly, and without any perception, should produce a knowing being, as it is impossible that a triangle should make itself three angles bigger than two right ones. For it is as repugnant to the idea of senseless matter, that it should put into itself sense, perception, and knowledge, as it is repugnant to the idea of a triangle, that it should put into itself greater angles than two right ones.

6. And therefore God. Thus, from the consideration of ourselves, and what we infallibly find in our own constitutions, our reason leads us to the knowledge of this certain and evident truth,—That there is an eternal, most powerful, and most knowing Being; which whether any one will please to call God, it matters not. The thing is evident; and from this idea duly considered, will easily be deduced all those other attributes, which we ought to ascribe to this eternal Being. If, nevertheless, any one should be found so senselessly arrogant, as to suppose man alone knowing and wise, but yet the product of mere ignorance and chance; and that all the rest of the universe acted only by that blind haphazard; I shall leave with him that very rational and emphatical rebuke

of Tully (I. ii. De Leg.), to be considered at his leisure: "What can be more sillily arrogant and misbecoming, than for a man to think that he has a mind and understanding in him, but yet in all the universe beside there is no such thing? Or that those things, which with the utmost stretch of his reason he can scarce comprehend, should be moved and managed without any reason at all?" *Quid est enim verius, quam neminem esse oportere tam stulte arrogantem, ut in se mentem et rationem putet inesse, in caelo mundoque non putet? Aut ea quae vix summa ingenii ratione comprehendat, nulla ratione moveri putet?*

From what has been said, it is plain to me we have a more certain knowledge of the existence of a God, than of anything our senses have not immediately discovered to us. Nay, I presume I may say, that we more certainly know that there is a God, than that there is anything else without us. When I say we know, I mean there is such a knowledge within our reach which we cannot miss, if we will but apply our minds to that, as we do to several other inquiries.

7. Our idea of a most perfect Being, not the sole proof of a God. How far the idea of a most perfect being, which a man may frame in his mind, does or does not prove the existence of a God, I will not here examine. For in the different make of men's tempers and application of their thoughts, some arguments prevail more on one, and some on another, for the confirmation of the same truth. But yet, I think, this I may say, that it is an ill way of establishing this truth, and silencing atheists, to lay the whole stress of so important a point as this upon that sole foundation: and take some men's having that idea of God in their minds, (for it is evident some men have none, and some worse than none, and the most very different,) for the only proof of a Deity; and out of an over fondness of that darling invention, cashier, or at least endeavour to invalidate all other arguments; and forbid us to hearken to those proofs, as being weak or fallacious, which our own existence, and the sensible parts of the universe offer so clearly and cogently to our thoughts, that I deem it impossible for a considering man to withstand them. For I judge it as

certain and clear a truth as can anywhere be delivered, that “the invisible things of God are clearly seen from the creation of the world, being understood by the things that are made, even his eternal power and Godhead.” Though our own being furnishes us, as I have shown, with an evident and incontestable proof of a Deity; and I believe nobody can avoid the cogency of it, who will but as carefully attend to it, as to any other demonstration of so many parts: yet this being so fundamental a truth, and of that consequence, that all religion and genuine morality depend thereon, I doubt not but I shall be forgiven by my reader if I go over some parts of this argument again, and enlarge a little more upon them.

8. Recapitulation—something from eternity. There is no truth more evident than that something must be from eternity. I never yet heard of any one so unreasonable, or that could suppose so manifest a contradiction, as a time wherein there was perfectly nothing. This being of all absurdities the greatest, to imagine that pure nothing, the perfect negation and absence of all beings, should ever produce any real existence.

It being, then, unavoidable for all rational creatures to conclude, that something has existed from eternity; let us next see what kind of thing that must be.

9. Two sorts of beings, cogitative and incogitative. There are but two sorts of beings in the world that man knows or conceives. First, such as are purely material, without sense, perception, or thought, as the clippings of our beards, and parings of our nails. Secondly, sensible, thinking, perceiving beings, such as we find ourselves to be. Which, if you please, we will hereafter call cogitative and incogitative beings; which to our present purpose, if for nothing else, are perhaps better terms than material and immaterial.

10. Incogitative being cannot produce a cogitative being. If, then, there must be something eternal, let us see what sort of being it must be. And to that it is very obvious to reason, that it must necessarily be a cogitative being. For it is as impossible to conceive that ever bare incogitative matter should produce a thinking intelligent being, as that nothing should of itself produce matter. Let us suppose any parcel of matter eternal,

great or small, we shall find it, in itself, able to produce nothing. For example: let us suppose the matter of the next pebble we meet with eternal, closely united, and the parts firmly at rest together; if there were no other being in the world, must it not eternally remain so, a dead inactive lump? Is it possible to conceive it can add motion to itself, being purely matter, or produce anything? Matter, then, by its own strength, cannot produce in itself so much as motion: the motion it has must also be from eternity, or else be produced, and added to matter by some other being more powerful than matter; matter, as is evident, having not power to produce motion in itself. But let us suppose motion eternal too: yet matter, incogitative matter and motion, whatever changes it might produce of figure and bulk, could never produce thought: knowledge will still be as far beyond the power of motion and matter to produce, as matter is beyond the power of nothing or nonentity to produce. And I appeal to every one's own thoughts, whether he cannot as easily conceive matter produced by nothing, as thought to be produced by pure matter,

when, before, there was no such thing as thought or an intelligent being existing? Divide matter into as many parts as you will, (which we are apt to imagine a sort of spiritualizing, or making a thinking thing of it,) vary the figure and motion of it as much as you please—a globe, cube, cone, prism, cylinder, &c., whose diameters are but 100,000th part of a gry, will operate no otherwise upon other bodies of proportionable bulk, than those of an inch or foot diameter; and you may as rationally expect to produce sense, thought, and knowledge, by putting together, in a certain figure and motion, gross particles of matter, as by those that are the very minutest that do anywhere exist. They knock, impel, and resist one another, just as the greater do; and that is all they can do. So that, if we will suppose nothing first or eternal, matter can never begin to be: if we suppose bare matter without motion, eternal, motion can never begin to be: if we suppose only matter and motion first, or eternal, thought can never begin to be. For it is impossible to conceive that matter, either with or without motion, could have, originally, in and from

itself, sense, perception, and knowledge; as is evident from hence, that then sense, perception, and knowledge, must be a property eternally inseparable from matter and every particle of it. Not to add, that, though our general or specific conception of matter makes us speak of it as one thing, yet really all matter is not one individual thing, neither is there any such thing existing as one material being, or one single body that we know or can conceive. And therefore, if matter were the eternal first cogitative being, there would not be one eternal, infinite, cogitative being, but an infinite number of eternal, finite, cogitative beings, independent one of another, of limited force, and distinct thoughts, which could never produce that order, harmony, and beauty which are to be found in nature. Since, therefore, whatsoever is the first eternal being must necessarily be cogitative; and whatsoever is first of all things must necessarily contain in it, and actually have, at least, all the perfections that can ever after exist; nor can it ever give to another any perfection that it hath not either actually in itself, or, at least, in a higher degree; it nec-

essarily follows, that the first eternal being cannot be matter.

11. Therefore, there has been an eternal cogitative Being. If, therefore, it be evident, that something necessarily must exist from eternity, it is also as evident, that that something must necessarily be a cogitative being: for it is as impossible that incogitative matter should produce a cogitative being, as that nothing, or the negation of all being, should produce a positive being or matter.

12. The attributes of the eternal cogitative Being. Though this discovery of the necessary existence of an eternal Mind does sufficiently lead us into the knowledge of God; since it will hence follow, that all other knowing beings that have a beginning must depend on him, and have no other ways of knowledge or extent of power than what he gives them; and therefore, if he made those, he made also the less excellent pieces of this universe,—all inanimate beings, whereby his omniscience, power, and providence will be established, and all his other attributes necessarily follow: yet, to clear

up this a little further, we will see what doubts can be raised against it.

13. Whether the eternal Mind may be also material or no. First, Perhaps it will be said, that, though it be as clear as demonstration can make it, that there must be an eternal Being, and that Being must also be knowing: yet it does not follow but that thinking Being may also be material. Let it be so, it equally still follows that there is a God. For if there be an eternal, omniscient, omnipotent Being, it is certain that there is a God, whether you imagine that Being to be material or no. But herein, I suppose, lies the danger and deceit of that supposition:—there being no way to avoid the demonstration, that there is an eternal knowing Being, men, devoted to matter, would willingly have it granted, that this knowing Being is material; and then, letting slide out of their minds, or the discourse, the demonstration whereby an eternal knowing Being was proved necessarily to exist, would argue all to be matter, and so deny a God, that is, an eternal cogitative Being: whereby they are so far from establishing, that they destroy their own

hypothesis. For, if there can be, in their opinion, eternal matter, without any eternal cogitative Being, they manifestly separate matter and thinking, and suppose no necessary connexion of the one with the other, and so establish the necessity of an eternal Spirit, but not of matter; since it has been proved already, that an eternal cogitative Being is unavoidably to be granted. Now, if thinking and matter may be separated, the eternal existence of matter will not follow from the eternal existence of a cogitative Being, and they suppose it to no purpose.

14. Not material: first, because each particle of matter is not cogitative. But now let us see how they can satisfy themselves, or others, that this eternal thinking Being is material.

I. I would ask them, whether they imagine that all matter, every particle of matter, thinks? This, I suppose, they will scarce say; since then there would be as many eternal thinking beings as there are particles of matter, and so an infinity of gods. And yet, if they will not allow matter as matter, that is, every particle of

matter, to be as well cogitative as extended, they will have as hard a task to make out to their own reasons a cogitative being out of incogitative particles, as an extended being out of unextended parts, if I may so speak.

15. II. Secondly, because one particle alone of matter cannot be cogitative. If all matter does not think, I next ask, Whether it be only one atom that does so? This has as many absurdities as the other; for then this atom of matter must be alone eternal or not. If this alone be eternal, then this alone, by its powerful thought or will, made all the rest of matter. And so we have the creation of matter by a powerful thought, which is that the materialists stick at; for if they suppose one single thinking atom to have produced all the rest of matter, they cannot ascribe that pre-eminency to it upon any other account than that of its thinking, the only supposed difference. But allow it to be by some other way which is above our conception, it must still be creation; and these men must give up their great maxim, *Ex nihilo nil fit*. If it be said, that all the rest of matter is equally eternal as that thinking atom, it will be to say anything

at pleasure, though ever so absurd. For to suppose all matter eternal, and yet one small particle in knowledge and power infinitely above all the rest, is without any the least appearance of reason to frame an hypothesis. Every particle of matter, as matter, is capable of all the same figures and motions of any other; and I challenge any one, in his thoughts, to add anything else to one above another.

16. III. Thirdly, because a system of incogitative matter cannot be cogitative. If then neither one peculiar atom alone can be this eternal thinking being; nor all matter, as matter, i.e. every particle of matter, can be it; it only remains, that it is some certain system of matter, duly put together, that is this thinking eternal Being. This is that which, I imagine, is that notion which men are aptest to have of God; who would have him a material being, as most readily suggested to them by the ordinary conceit they have of themselves and other men, which they take to be material thinking beings. But this imagination, however more natural, is no less absurd than the other: for to suppose the eternal think-

ing Being to be nothing else but a composition of particles of matter, each whereof is incogitative, is to ascribe all the wisdom and knowledge of that eternal Being only to the juxta-position of parts; than which nothing can be more absurd. For unthinking particles of matter, however put together, can have nothing thereby added to them, but a new relation of position, which it is impossible should give thought and knowledge to them.

17. And that whether this corporeal system is in motion or at rest. But further: this corporeal system either has all its parts at rest, or it is a certain motion of the parts wherein its thinking consists. If it be perfectly at rest, it is but one lump, and so can have no privileges above one atom.

If it be the motion of its parts on which its thinking depends, all the thoughts there must be unavoidably accidental and limited; since all the particles that by motion cause thought, being each of them in itself without any thought, cannot regulate its own motions, much less be regulated by the thought of the whole; since that thought is not the cause of motion, (for then it

must be antecedent to it, and so without it,) but the consequence of it; whereby freedom, power, choice, and all rational and wise thinking or acting, will be quite taken away: so that such a thinking being will be no better nor wiser than pure blind matter; since to resolve all into the accidental unguided motions of blind matter, or into thought depending on unguided motions of blind matter, is the same thing: not to mention the narrowness of such thoughts and knowledge that must depend on the motion of such parts. But there needs no enumeration of any more absurdities and impossibilities in this hypothesis (however full of them it be) than that before mentioned; since, let this thinking system be all or a part of the matter of the universe, it is impossible that any one particle should either know its own, or the motion of any other particle, or the whole know the motion of every particle; and so regulate its own thoughts or motions, or indeed have any thought resulting from such motion.

18. Matter not co-eternal with an eternal Mind. Secondly, Others would have Matter to be eternal, not-

withstanding that they allow an eternal, cogitative, immaterial Being. This, though it take not away the being of a God, yet, since it denies one and the first great piece of his workmanship, the creation, let us consider it a little. Matter must be allowed eternal: Why? because you cannot conceive how it can be made out of nothing: why do you not also think yourself eternal? You will answer, perhaps, Because, about twenty or forty years since, you began to be. But if I ask you, what that you is, which began then to be, you can scarce tell me. The matter whereof you are made began not then to be: for if it did, then it is not eternal: but it began to be put together in such a fashion and frame as makes up your body; but yet that frame of particles is not you, it makes not that thinking thing you are; (for I have now to do with one who allows an eternal, immaterial, thinking Being, but would have unthinking Matter eternal too;) therefore, when did that thinking thing begin to be? If it did never begin to be, then have you always been a thinking thing from eternity; the absurdity whereof I need not confute, till I meet with one who is so void of

understanding as to own it. If, therefore, you can allow a thinking thing to be made out of nothing, (as all things that are not eternal must be,) why also can you not allow it possible for a material being to be made out of nothing by an equal power, but that you have the experience of the one in view, and not of the other? Though, when well considered, creation of a spirit will be found to require no less power than the creation of matter. Nay, possibly, if we would emancipate ourselves from vulgar notions, and raise our thoughts, as far as they would reach, to a closer contemplation of things, we might be able to aim at some dim and seeming conception how matter might at first be made, and begin to exist, by the power of that eternal first Being: but to give beginning and being to a spirit would be found a more inconceivable effect of omnipotent power. But this being what would perhaps lead us too far from the notions on which the philosophy now in the world is built, it would not be pardonable to deviate so far from them; or to inquire, so far as grammar itself would authorize, if the common settled opinion opposes it: especially in

this place, where the received doctrine serves well enough to our present purpose, and leaves this past doubt, that the creation or beginning of any one *substance* out of nothing being once admitted, the creation of all other but the *Creator* himself, may, with the same ease, be supposed.

19. Objection: "Creation out of nothing." But you will say, Is it not impossible to admit of the making anything out of nothing, since we cannot possibly conceive it? I answer, No. Because it is not reasonable to deny the power of an infinite being, because we cannot comprehend its operations. We do not deny other effects upon this ground, because we cannot possibly conceive the manner of their production. We cannot conceive how anything but impulse of body can move body; and yet that is not a reason sufficient to make us deny it possible, against the constant experience we have of it in ourselves, in all our voluntary motions; which are produced in us only by the free action or thought of our own minds, and are not, nor can be, the effects of the impulse or determination of the motion of blind matter

in or upon our own bodies; for then it could not be in our power or choice to alter it. For example: my right hand writes, whilst my left hand is still: What causes rest in one, and motion in the other? Nothing but my will,—a thought of my mind; my thought only changing, the right hand rests, and the left hand moves. This is matter of fact, which cannot be denied: explain this and make it intelligible, and then the next step will be to understand creation. For the giving a new determination to the motion of the animal spirits (which some make use of to explain voluntary motion) clears not the difficulty one jot. To alter the determination of motion, being in this case no easier nor less, than to give motion itself: since the new determination given to the animal spirits must be either immediately by thought, or by some other body put in their way by thought which was not in their way before, and so must owe its motion to thought: either of which leaves voluntary motion as unintelligible as it was before. In the meantime, it is an overvaluing ourselves to reduce all to the narrow measure of our capacities, and to conclude all things impos-

sible to be done, whose manner of doing exceeds our comprehension. This is to make our comprehension infinite, or God finite, when what He can do is limited to what we can conceive of it. If you do not understand the operations of your own finite mind, that thinking thing within you, do not deem it strange that you cannot comprehend the operations of that eternal infinite Mind, who made and governs all things, and whom the heaven of heavens cannot contain.

Chapter XI

Of our Knowledge of the Existence of Other Things

1. Knowledge of the existence of other finite beings is to be had only by actual sensation. The knowledge of our own being we have by intuition. The existence of a God, reason clearly makes known to us, as has been shown.

The knowledge of the existence of any other thing we can have only by sensation: for there being no necessary connexion of real existence with any idea a man

hath in his memory; nor of any other existence but that of God with the existence of any particular man: no particular man can know the existence of any other being, but only when, by actual operating upon him, it makes itself perceived by him. For, the having the idea of anything in our mind, no more proves the existence of that thing, than the picture of a man evidences his being in the world, or the visions of a dream make thereby a true history.

2. Instance: whiteness of this paper. It is therefore the actual receiving of ideas from without that gives us notice of the existence of other things, and makes us know, that something doth exist at that time without us, which causes that idea in us; though perhaps we neither know nor consider how it does it. For it takes not from the certainty of our senses, and the ideas we receive by them, that we know not the manner wherein they are produced: v.g. whilst I write this, I have, by the paper affecting my eyes, that idea produced in my mind, which, whatever object causes, I call white; by which I know that that quality or accident (i.e. whose appearance

before my eyes always causes that idea) doth really exist, and hath a being without me. And of this, the greatest assurance I can possibly have, and to which my faculties can attain, is the testimony of my eyes, which are the proper and sole judges of this thing; whose testimony I have reason to rely on as so certain, that I can no more doubt, whilst I write this, that I see white and black, and that something really exists that causes that sensation in me, than that I write or move my hand; which is a certainty as great as human nature is capable of, concerning the existence of anything, but a man's self alone, and of God.

3. This notice by our senses, though not so certain as demonstration, yet may be called knowledge, and proves the existence of things without us. The notice we have by our senses of the existing of things without us, though it be not altogether so certain as our intuitive knowledge, or the deductions of our reason employed about the clear abstract ideas of our own minds; yet it is an assurance that deserves the name of knowledge. If we persuade ourselves that our faculties act and inform us

right concerning the existence of those objects that affect them, it cannot pass for an ill-grounded confidence: for I think nobody can, in earnest, be so sceptical as to be uncertain of the existence of those things which he sees and feels. At least, he that can doubt so far, (whatever he may have with his own thoughts,) will never have any controversy with me; since he can never be sure I say anything contrary to his own opinion. As to myself, I think God has given me assurance enough of the existence of things without me: since, by their different application, I can produce in myself both pleasure and pain, which is one great concernment of my present state. This is certain: the confidence that our faculties do not herein deceive us, is the greatest assurance we are capable of concerning the existence of material beings. For we cannot act anything but by our faculties; nor talk of knowledge itself, but by the help of those faculties which are fitted to apprehend even what knowledge is.

But besides the assurance we have from our senses themselves, that they do not err in the information they

give us of the existence of things without us, when they are affected by them, we are further confirmed in this assurance by other concurrent reasons:—

4. I. Confirmed by concurrent reasons:—First, because we cannot have ideas of sensation but by the inlet of the senses. It is plain those perceptions are produced in us by exterior causes affecting our senses: because those that want the organs of any sense, never can have the ideas belonging to that sense produced in their minds. This is too evident to be doubted: and therefore we cannot but be assured that they come in by the organs of that sense, and no other way. The organs themselves, it is plain, do not produce them: for then the eyes of a man in the dark would produce colours, and his nose smell roses in the winter: but we see nobody gets the relish of a pineapple, till he goes to the Indies, where it is, and tastes it.

5. II. Secondly, Because we find that an idea from actual sensation, and another from memory, are very distinct perceptions. Because sometimes I find that I cannot avoid the having those ideas produced in my mind.

For though, when my eyes are shut, or windows fast, I can at pleasure recall to my mind the ideas of light, or the sun, which former sensations had lodged in my memory; so I can at pleasure lay by that idea, and take into my view that of the smell of a rose, or taste of sugar. But, if I turn my eyes at noon towards the sun, I cannot avoid the ideas which the light or sun then produces in me. So that there is a manifest difference between the ideas laid up in my memory, (over which, if they were there only, I should have constantly the same power to dispose of them, and lay them by at pleasure,) and those which force themselves upon me, and I cannot avoid having. And therefore it must needs be some exterior cause, and the brisk acting of some objects without me, whose efficacy I cannot resist, that produces those ideas in my mind, whether I will or no. Besides, there is nobody who doth not perceive the difference in himself between contemplating the sun, as he hath the idea of it in his memory, and actually looking upon it: of which two, his perception is so distinct, that few of his ideas are more distinguishable one from another. And

therefore he hath certain knowledge that they are not both memory, or the actions of his mind, and fancies only within him; but that actual seeing hath a cause without.

6. III. Thirdly, because pleasure or pain, which accompanies actual sensation, accompanies not the returning of those ideas without the external objects. Add to this, that many of those ideas are produced in us with pain, which afterwards we remember without the least offence. Thus, the pain of heat or cold, when the idea of it is revived in our minds, gives us no disturbance; which, when felt, was very troublesome; and is again, when actually repeated: which is occasioned by the disorder the external object causes in our bodies when applied to them: and we remember the pains of hunger, thirst, or the headache, without any pain at all; which would either never disturb us, or else constantly do it, as often as we thought of it, were there nothing more but ideas floating in our minds, and appearances entertaining our fancies, without the real existence of things affecting us from abroad. The same may be said of pleasure, ac-

companying several actual sensations. And though mathematical demonstration depends not upon sense, yet the examining them by diagrams gives great credit to the evidence of our sight, and seems to give it a certainty approaching to that of demonstration itself. For, it would be very strange, that a man should allow it for an undeniable truth, that two angles of a figure, which he measures by lines and angles of a diagram, should be bigger one than the other, and yet doubt of the existence of those lines and angles, which by looking on he makes use of to measure that by.

7. IV. Fourthly, because our senses assist one another's testimony of the existence of outward things, and enable us to predict. Our senses in many cases bear witness to the truth of each other's report, concerning the existence of sensible things without us. He that sees a fire, may, if he doubt whether it be anything more than a bare fancy, feel it too; and be convinced, by putting his hand in it. Which certainly could never be put into such exquisite pain by a bare idea or phantom, unless that the pain be a fancy too: which yet he cannot,

when the burn is well, by raising the idea of it, bring upon himself again.

Thus I see, whilst I write this, I can change the appearance of the paper; and by designing the letters, tell beforehand what new idea it shall exhibit the very next moment, by barely drawing my pen over it: which will neither appear (let me fancy as much as I will) if my hands stand still; or though I move my pen, if my eyes be shut: nor, when those characters are once made on the paper, can I choose afterwards but see them as they are; that is, have the ideas of such letters as I have made. Whence it is manifest, that they are not barely the sport and play of my own imagination, when I find that the characters that were made at the pleasure of my own thoughts, do not obey them; nor yet cease to be, whenever I shall fancy it, but continue to affect my senses constantly and regularly, according to the figures I made them. To which if we will add, that the sight of those shall, from another man, draw such sounds as I beforehand design they shall stand for, there will be little reason left to doubt that those words I write do

really exist without me, when they cause a long series of regular sounds to affect my ears, which could not be the effect of my imagination, nor could my memory retain them in that order.

8. This certainty is as great as our condition needs. But yet, if after all this any one will be so sceptical as to distrust his senses, and to affirm that all we see and hear, feel and taste, think and do, during our whole being, is but the series and deluding appearances of a long dream, whereof there is no reality; and therefore will question the existence of all things, or our knowledge of anything: I must desire him to consider, that, if all be a dream, then he doth but dream that he makes the question, and so it is not much matter that a waking man should answer him. But yet, if he pleases, he may dream that I make him this answer, That the certainty of things existing in *rerum natura* when we have the testimony of our senses for it is not only as great as our frame can attain to, but as our condition needs. For, our faculties being suited not to the full extent of being, nor to a perfect, clear, comprehensive knowledge

of things free from all doubt and scruple; but to the preservation of us, in whom they are; and accommodated to the use of life: they serve to our purpose well enough, if they will but give us certain notice of those things, which are convenient or inconvenient to us. For he that sees a candle burning, and hath experimented the force of its flame by putting his finger in it, will little doubt that this is something existing without him, which does him harm, and puts him to great pain; which is assurance enough, when no man requires greater certainty to govern his actions by than what is as certain as his actions themselves. And if our dreamer pleases to try whether the glowing heat of a glass furnace be barely a wandering imagination in a drowsy man's fancy, by putting his hand into it, he may perhaps be wakened into a certainty greater than he could wish, that it is something more than bare imagination. So that this evidence is as great as we can desire, being as certain to us as our pleasure or pain, i.e. happiness or misery; beyond which we have no concernment, either of knowing or being. Such an assurance of the existence of things

without us is sufficient to direct us in the attaining the good and avoiding the evil which is caused by them, which is the important concernment we have of being made acquainted with them.

9. But reaches no further than actual sensation. In fine, then, when our senses do actually convey into our understandings any idea, we cannot but be satisfied that there doth something at that time really exist without us, which doth affect our senses, and by them give notice of itself to our apprehensive faculties, and actually produce that idea which we then perceive: and we cannot so far distrust their testimony, as to doubt that such collections of simple ideas as we have observed by our senses to be united together, do really exist together. But this knowledge extends as far as the present testimony of our senses, employed about particular objects that do then affect them, and no further. For if I saw such a collection of simple ideas as is wont to be called man, existing together one minute since, and am now alone, I cannot be certain that the same man exists now, since there is no necessary connexion of his exist-

ence a minute since with his existence now: by a thousand ways he may cease to be, since I had the testimony of my senses for his existence. And if I cannot be certain that the man I saw last to-day is now in being, I can less be certain that he is so who hath been longer removed from my senses, and I have not seen since yesterday, or since the last year: and much less can I be certain of the existence of men that I never saw. And, therefore, though it be highly probable that millions of men do now exist, yet, whilst I am alone, writing this, I have not that certainty of it which we strictly call knowledge; though the great likelihood of it puts me past doubt, and it be reasonable for me to do several things upon the confidence that there are men (and men also of my acquaintance, with whom I have to do) now in the world: but this is but probability, not knowledge.

10. Folly to expect demonstration in everything. Whereby yet we may observe how foolish and vain a thing it is for a man of a narrow knowledge, who having reason given him to judge of the different evidence and probability of things, and to be swayed accordingly; how

vain, I say, it is to expect demonstration and certainty in things not capable of it; and refuse assent to very rational propositions, and act contrary to very plain and clear truths, because they cannot be made out so evident, as to surmount every the least (I will not say reason, but) pretence of doubting. He that, in the ordinary affairs of life, would admit of nothing but direct plain demonstration, would be sure of nothing in this world, but of perishing quickly. The wholesomeness of his meat or drink would not give him reason to venture on it: and I would fain know what it is he could do upon such grounds as are capable of no doubt, no objection.

11. Past existence of other things is known by memory. As when our senses are actually employed about any object, we do know that it does exist; so by our memory we may be assured, that heretofore things that affected our senses have existed. And thus we have knowledge of the past existence of several things, whereof our senses having informed us, our memories still retain the ideas; and of this we are past all doubt, so long as we remember well. But this knowledge also reaches no further

than our senses have formerly assured us. Thus, seeing water at this instant, it is an unquestionable truth to me that water doth exist: and remembering that I saw it yesterday, it will also be always true, and as long as my memory retains it always an undoubted proposition to me, that water did exist the 10th of July, 1688; as it will also be equally true that a certain number of very fine colours did exist, which at the same time I saw upon a bubble of that water: but, being now quite out of sight both of the water and bubbles too, it is no more certainly known to me that the water doth now exist, than that the bubbles or colours therein do so: it being no more necessary that water should exist to-day, because it existed yesterday, than that the colours or bubbles exist to-day, because they existed yesterday, though it be exceedingly much more probable; because water hath been observed to continue long in existence, but bubbles, and the colours on them, quickly cease to be.

12. The existence of other finite spirits not knowable, and rests on faith. What ideas we have of spirits, and how we come by them, I have already shown. But though

we have those ideas in our minds, and know we have them there, the having the ideas of spirits does not make us know that any such things do exist without us, or that there are any finite spirits, or any other spiritual beings, but the Eternal God. We have ground from revelation, and several other reasons, to believe with assurance that there are such creatures: but our senses not being able to discover them, we want the means of knowing their particular existences. For we can no more know that there are finite spirits really existing, by the idea we have of such beings in our minds, than by the ideas any one has of fairies or centaurs, he can come to know that things answering those ideas do really exist.

And therefore concerning the existence of finite spirits, as well as several other things, we must content ourselves with the evidence of faith; but universal, certain propositions concerning this matter are beyond our reach. For however true it may be, v.g., that all the intelligent spirits that God ever created do still exist, yet it can never make a part of our certain knowledge. These and the like propositions we may assent to, as

highly probable, but are not, I fear, in this state capable of knowing. We are not, then, to put others upon demonstrating, nor ourselves upon search of universal certainty in all those matters; wherein we are not capable of any other knowledge, but what our senses give us in this or that particular.

13. Only particular propositions concerning concrete existences are knowable. By which it appears that there are two sorts of propositions:—(1) There is one sort of propositions concerning the existence of anything answerable to such an idea: as having the idea of an elephant, phoenix, motion, or an angel, in my mind, the first and natural inquiry is, Whether such a thing does anywhere exist? And this knowledge is only of particulars. No existence of anything without us, but only of God, can certainly be known further than our senses inform us. (2) There is another sort of propositions, wherein is expressed the agreement or disagreement of our abstract ideas, and their dependence on one another. Such propositions may be universal and certain. So, having the idea of God and myself, of fear and obe-

dience, I cannot but be sure that God is to be feared and obeyed by me: and this proposition will be certain, concerning man in general, if I have made an abstract idea of such a species, whereof I am one particular. But yet this proposition, how certain soever, that “men ought to fear and obey God” proves not to me the existence of men in the world; but will be true of all such creatures, whenever they do exist: which certainty of such general propositions depends on the agreement or disagreement to be discovered in those abstract ideas.

14. And all general propositions that are known to be true concern abstract ideas. In the former case, our knowledge is the consequence of the existence of things, producing ideas in our minds by our senses: in the latter, knowledge is the consequence of the ideas (be they what they will) that are in our minds, producing there general certain propositions. Many of these are called *aeternae veritates*, and all of them indeed are so; not from being written, all or any of them, in the minds of all men; or that they were any of them propositions in any one’s mind, till he, having got the abstract ideas, joined or

separated them by affirmation or negation. But wheresoever we can suppose such a creature as man is, endowed with such faculties, and thereby furnished with such ideas as we have, we must conclude, he must needs, when he applies his thoughts to the consideration of his ideas, know the truth of certain propositions that will arise from the agreement or disagreement which he will perceive in his own ideas. Such propositions are therefore called eternal truths, not because they are eternal propositions actually formed, and antecedent to the understanding that at any time makes them; nor because they are imprinted on the mind from any patterns that are anywhere out of the mind, and existed before: but because, being once made about abstract ideas, so as to be true, they will, whenever they can be supposed to be made again at any time, past or to come, by a mind having those ideas, always actually be true. For names being supposed to stand perpetually for the same ideas, and the same ideas having immutably the same habitudes one to another, propositions concerning any abstract ideas that are once true must needs be eternal verities.

Chapter XII Of the Improvement of our Knowledge

1. Knowledge is not got from maxims. It having been the common received opinion amongst men of letters, that maxims were the foundation of all knowledge; and that the sciences were each of them built upon certain *praecognita* from whence the understanding was to take its rise, and by which it was to conduct itself in its inquiries into the matters belonging to that science, the beaten road of the Schools has been, to lay down in the beginning one or more general propositions, as foundations whereon to build the knowledge that was to be had of that subject. These doctrines, thus laid down for foundations of any science, were called principles, as the beginnings from which we must set out, and look no further backwards in our inquiries, as we have already observed.

2. (The occasion of that opinion.) One thing which might probably give an occasion to this way of proceeding in other sciences, was (as I suppose) the good suc-

cess it seemed to have in mathematics, wherein men, being observed to attain a great certainty of knowledge, these sciences came by pre-eminence to be called Mathemata, and Mathesis, learning, or things learned, thoroughly learned, as having of all others the greatest certainty, clearness, and evidence in them.

3. But from comparing clear and distinct ideas. But if any one will consider, he will (I guess) find, that the great advancement and certainty of real knowledge which men arrived to in these sciences, was not owing to the influence of these principles, nor derived from any peculiar advantage they received from two or three general maxims, laid down in the beginning; but from the clear, distinct, complete ideas their thoughts were employed about, and the relation of equality and excess so clear between some of them, that they had an intuitive knowledge, and by that a way to discover it in others; and this without the help of those maxims. For I ask, Is it not possible for a young lad to know that his whole body is bigger than his little finger, but by virtue of this axiom, that the whole is bigger than a part; nor be

assured of it, till he has learned that maxim? Or cannot a country wench know that, having received a shilling from one that owes her three, and a shilling also from another that owes her three, the remaining debts in each of their hands are equal? Cannot she know this, I say, unless she fetch the certainty of it from this maxim, that if you take equals from equals, the remainder will be equals, a maxim which possibly she never heard or thought of? I desire any one to consider, from what has been elsewhere said, which is known first and clearest by most people, the particular instance, or the general rule; and which it is that gives life and birth to the other. These general rules are but the comparing our more general and abstract ideas, which are the workmanship of the mind, made, and names given to them for the easier dispatch in its reasonings, and drawing into comprehensive terms and short rules its various and multiplied observations. But knowledge began in the mind, and was founded on particulars; though afterwards, perhaps, no notice was taken thereof: it being natural for the mind (forward still to enlarge its knowl-

edge) most attentively to lay up those general notions, and make the proper use of them, which is to disburden the memory of the cumbersome load of particulars. For I desire it may be considered, what more certainty there is to a child, or any one, that his body, little finger, and all, is bigger than his little finger alone, after you have given to his body the name whole, and to his little finger the name part, than he could have had before; or what new knowledge concerning his body can these two relative terms give him, which he could not have without them? Could he not know that his body was bigger than his little finger, if his language were yet so imperfect that he had no such relative terms as whole and part? I ask, further, when he has got these names, how is he more certain that his body is a whole, and his little finger a part, than he was or might be certain before he learnt those terms, that his body was bigger than his little finger? Any one may as reasonably doubt or deny that his little finger is a part of his body, as that it is less than his body. And he that can doubt whether it be less, will as certainly doubt whether it be a part. So that

the maxim, the whole is bigger than a part, can never be made use of to prove the little finger less than the body, but when it is useless, by being brought to convince one of a truth which he knows already. For he that does not certainly know that any parcel of matter, with another parcel of matter joined to it, is bigger than either of them alone, will never be able to know it by the help of these two relative terms, whole and part, make of them what maxim you please.

4. Dangerous to build upon precarious principles. But be it in the mathematics as it will, whether it be clearer, that, taking an inch from a black line of two inches, and an inch from a red line of two inches, the remaining parts of the two lines will be equal, or that if you take equals from equals, the remainder will be equals: which, I say, of these two is the clearer and first known, I leave to any one to determine, it not being material to my present occasion. That which I have here to do, is to inquire, whether, if it be the readiest way to knowledge to begin with general maxims, and build upon them, it be yet a safe way to take the principles which are laid

down in any other science as unquestionable truths; and so receive them without examination, and adhere to them, without suffering them to be doubted of, because mathematicians have been so happy, or so fair, to use none but self-evident and undeniable. If this be so, I know not what may not pass for truth in morality, what may not be introduced and proved in natural philosophy.

Let that principle of some of the old philosophers, That all is Matter, and that there is nothing else, be received for certain and indubitable, and it will be easy to be seen by the writings of some that have revived it again in our days, what consequences it will lead us into. Let any one, with Polemo, take the world; or with the Stoics, the aether, or the sun; or with Anaximenes, the air, to be God; and what a divinity, religion, and worship must we needs have! Nothing can be so dangerous as principles thus taken up without questioning or examination; especially if they be such as concern morality, which influence men's lives, and give a bias to all their actions. Who might not justly expect another kind of life in Aristippus, who placed happiness in bodily plea-

sure; and in Antisthenes, who made virtue sufficient to felicity? And he who, with Plato, shall place beatitude in the knowledge of God, will have his thoughts raised to other contemplations than those who look not beyond this spot of earth, and those perishing things which are to be had in it. He that, with Archelaus, shall lay it down as a principle, that right and wrong, honest and dishonest, are defined only by laws, and not by nature, will have other measures of moral rectitude and pravity, than those who take it for granted that we are under obligations antecedent to all human constitutions.

5. To do so is no certain way to truth. If, therefore, those that pass for principles are not certain, (which we must have some way to know, that we may be able to distinguish them from those that are doubtful,) but are only made so to us by our blind assent, we are liable to be misled by them; and instead of being guided into truth, we shall, by principles, be only confirmed in mistake and error.

6. But to compare clear, complete ideas, under steady names. But since the knowledge of the certainty of prin-

ciples, as well as of all other truths, depends only upon the perception we have of the agreement or disagreement of our ideas, the way to improve our knowledge is not, I am sure, blindly, and with an implicit faith, to receive and swallow principles; but is, I think, to get and fix in our minds clear, distinct, and complete ideas, as far as they are to be had, and annex to them proper and constant names. And thus, perhaps, without any other principles, but barely considering those perfect ideas, and by comparing them one with another, finding their agreement and disagreement, and their several relations and habitudes; we shall get more true and clear knowledge by the conduct of this one rule than by taking up principles, and thereby putting our minds into the disposal of others.

7. The true method of advancing knowledge is by considering our abstract ideas. We must, therefore, if we will proceed as reason advises, adapt our methods of inquiry to the nature of the ideas we examine, and the truth we search after. General and certain truths are only founded in the habitudes and relations of abstract

ideas. A sagacious and methodical application of our thoughts, for the finding out these relations, is the only way to discover all that can be put with truth and certainty concerning them into general propositions. By what steps we are to proceed in these, is to be learned in the schools of the mathematicians, who, from very plain and easy beginnings, by gentle degrees, and a continued chain of reasonings, proceed to the discovery and demonstration of truths that appear at first sight beyond human capacity. The art of finding proofs, and the admirable methods they have invented for the singling out and laying in order those intermediate ideas that demonstratively show the equality or inequality of unapplicable quantities, is that which has carried them so far, and produced such wonderful and unexpected discoveries: but whether something like this, in respect of other ideas, as well as those of magnitude, may not in time be found out, I will not determine. This, I think, I may say, that if other ideas that are the real as well as nominal essences of their species, were pursued in the way familiar to mathematicians, they would carry our

thoughts further, and with greater evidence and clearness than possibly we are apt to imagine.

8. By which morality also may be made clearer. This gave me the confidence to advance that conjecture, which I suggest, (chap. iii.) viz. that morality is capable of demonstration as well as mathematics. For the ideas that ethics are conversant about, being all real essences, and such as I imagine have a discoverable connexion and agreement one with another; so far as we can find their habitudes and relations, so far we shall be possessed of certain, real, and general truths; and I doubt not but, if a right method were taken, a great part of morality might be made out with that clearness, that could leave, to a considering man, no more reason to doubt, than he could have to doubt of the truth of propositions in mathematics, which have been demonstrated to him.

9. Our knowledge of substances is to be improved, not by contemplation of abstract ideas, but only by experience. In our search after the knowledge of substances, our want of ideas that are suitable to such a way of

proceeding obliges us to a quite different method. We advance not here, as in the other, (where our abstract ideas are real as well as nominal essences,) by contemplating our ideas, and considering their relations and correspondences; that helps us very little, for the reasons, that in another place we have at large set down. By which I think it is evident, that substances afford matter of very little general knowledge; and the bare contemplation of their abstract ideas will carry us but a very little way in the search of truth and certainty. What, then, are we to do for the improvement of our knowledge in substantial beings? Here we are to take a quite contrary course: the want of ideas of their real essences sends us from our own thoughts to the things themselves as they exist. Experience here must teach me what reason cannot: and it is by trying alone, that I can certainly know, what other qualities co-exist with those of my complex idea, v.g. whether that yellow, heavy, fusible body I call gold, be malleable, or no; which experience (which way ever it prove in that particular body I examine) makes me not certain, that it is so in

all, or any other yellow, heavy, fusible bodies, but that which I have tried. Because it is no consequence one way or the other from my complex idea: the necessity or inconsistency of malleability hath no visible connexion with the combination of that colour, weight, and fusibility in any body. What I have said here of the nominal essence of gold, supposed to consist of a body of such a determinate colour, weight, and fusibility, will hold true, if malleableness, fixedness, and solubility in *aqua regia* be added to it. Our reasonings from these ideas will carry us but a little way in the certain discovery of the other properties in those masses of matter wherein all these are to be found. Because the other properties of such bodies, depending not on these, but on that unknown real essence on which these also depend, we cannot by them discover the rest; we can go no further than the simple ideas of our nominal essence will carry us, which is very little beyond themselves; and so afford us but very sparingly any certain, universal, and useful truths. For, upon trial, having found that particular piece (and all others of that colour, weight, and fusibility, that I ever tried) malleable,

that also makes now, perhaps, a part of my complex idea, part of my nominal essence of gold: whereby though I make my complex idea to which I affix the name gold, to consist of more simple ideas than before; yet still, it not containing the real essence of any species of bodies, it helps me not certainly to know (I say to know, perhaps it may be to conjecture) the other remaining properties of that body, further than they have a visible connexion with some or all of the simple ideas that make up my nominal essence. For example, I cannot be certain, from this complex idea, whether gold be fixed or no; because, as before, there is no necessary connexion or inconsistency to be discovered betwixt a complex idea of a body yellow, heavy, fusible, malleable; betwixt these, I say, and fixedness; so that I may certainly know, that in whatsoever body these are found, there fixedness is sure to be. Here, again, for assurance, I must apply myself to experience; as far as that reaches, I may have certain knowledge, but no further.

10. Experience may procure us convenience, not science. I deny not but a man, accustomed to rational and

regular experiments, shall be able to see further into the nature of bodies and guess righter at their yet unknown properties than one that is a stranger to them: but yet, as I have said, this is but judgment and opinion, not knowledge and certainty. This way of getting and improving our knowledge in substances only by experience and history, which is all that the weakness of our faculties in this state of mediocrity which we are in in this world can attain to, makes me suspect that natural philosophy is not capable of being made a science. We are able, I imagine, to reach very little general knowledge concerning the species of bodies and their several properties. Experiments and historical observations we may have, from which we may draw advantages of ease and health, and thereby increase our stock of conveniences for this life; but beyond this I fear our talents reach not, nor are our faculties, as I guess, able to advance.

11. We are fitted for moral science, but only for probable interpretations of external nature. From whence it is obvious to conclude that, since our faculties are not fitted to penetrate into the internal fabric and real es-

sences of bodies; but yet plainly discover to us the being of a God and the knowledge of ourselves, enough to lead us into a full and clear discovery of our duty and great concernment; it will become us, as rational creatures, to employ those faculties we have about what they are most adapted to, and follow the direction of nature, where it seems to point us out the way. For it is rational to conclude that our proper employment lies in those inquiries, and in that sort of knowledge which is most suited to our natural capacities, and carries in it our greatest interest, i.e. the condition of our eternal estate. Hence I think I may conclude that morality is the proper science and business of mankind in general, (who are both concerned and fitted to search out their summum bonum;) as several arts, conversant about several parts of nature, are the lot and private talent of particular men for the common use of human life and their own particular subsistence in this world. Of what consequence the discovery of one natural body and its properties may be to human life the whole great continent of America is a convincing instance: whose igno-

rance in useful arts, and want of the greatest part of the conveniences of life, in a country that abounded with all sorts of natural plenty, I think may be attributed to their ignorance of what was to be found in a very ordinary, despicable stone; I mean the mineral of iron. And whatever we think of our parts or improvements in this part of the world, where knowledge and plenty seem to vie with each other; yet to any one that will seriously reflect on it, I suppose it will appear past doubt, that, were the use of iron lost among us, we should in a few ages be unavoidably reduced to the wants and ignorance of the ancient savage Americans, whose natural endowments and provisions come no way short of those of the most flourishing and polite nations. So that he who first made known the use of that contemptible mineral, may be truly styled the father of arts, and author of plenty.

12. In the study of nature we must beware of hypotheses and wrong principles. I would not, therefore, be thought to disesteem or dissuade the study of nature. I readily agree the contemplation of his works gives us

occasion to admire, revere, and glorify their Author: and, if rightly directed, may be of greater benefit to mankind than the monuments of exemplary charity that have at so great charge been raised by the founders of hospitals and almshouses. He that first invented printing, discovered the use of the compass, or made public the virtue and right use of kin kina, did more for the propagation of knowledge, for the supply and increase of useful commodities, and saved more from the grave, than those who built colleges, workhouses, and hospitals. All that I would say is, that we should not be too forwardly possessed with the opinion or expectation of knowledge, where it is not to be had, or by ways that will not attain to it: that we should not take doubtful systems for complete sciences; nor unintelligible notions for scientific demonstrations. In the knowledge of bodies, we must be content to glean what we can from particular experiments: since we cannot, from a discovery of their real essences, grasp at a time whole sheaves, and in bundles comprehend the nature and properties of whole species together. Where our inquiry is con-

cerning co-existence, or repugnancy to co-exist, which by contemplation of our ideas we cannot discover; there experience, observation, and natural history, must give us, by our senses and by retail, an insight into corporeal substances. The knowledge of bodies we must get by our senses, warily employed in taking notice of their qualities and operations on one another: and what we hope to know of separate spirits in this world, we must, I think, expect only from revelation. He that shall consider how little general maxims, precarious principles, and hypotheses laid down at pleasure, have promoted true knowledge, or helped to satisfy the inquiries of rational men after real improvements; how little, I say, the setting out at that end has, for many ages together, advanced men's progress, towards the knowledge of natural philosophy, will think we have reason to thank those who in this latter age have taken another course, and have trod out to us, though not an easier way to learned ignorance, yet a surer way to profitable knowledge.

13. The true use of hypotheses. Not that we may not, to explain any phenomena of nature, make use of any

probable hypotheses whatsoever: hypotheses, if they are well made, are at least great helps to the memory, and often direct us to new discoveries. But my meaning is, that we should not take up any one too hastily (which the mind, that would always penetrate into the causes of things, and have principles to rest on, is very apt to do,) till we have very well examined particulars, and made several experiments, in that thing which we would explain by our hypothesis, and see whether it will agree to them all; whether our principles will carry us quite through, and not be as inconsistent with one phenomenon of nature, as they seem to accommodate and explain another. And at least that we take care that the name of principles deceive us not, nor impose on us, by making us receive that for an unquestionable truth, which is really at best but a very doubtful conjecture; such as are most (I had almost said all) of the hypotheses in natural philosophy.

14. Clear and distinct ideas with settled names, and the finding of those intermediate ideas which show their agreement or disagreement, are the ways to enlarge our

knowledge. But whether natural philosophy be capable of certainty or no, the ways to enlarge our knowledge, as far as we are capable, seem to me, in short, to be these two:—

First, The first is to get and settle in our minds determined ideas of those things whereof we have general or specific names; at least, so many of them as we would consider and improve our knowledge in, or reason about. And if they be specific ideas of substances, we should endeavour also to make them as complete as we can, whereby I mean, that we should put together as many simple ideas as, being constantly observed to co-exist, may perfectly determine the species; and each of those simple ideas which are the ingredients of our complex ones, should be clear and distinct in our minds. For it being evident that our knowledge cannot exceed our ideas; as far as they are either imperfect, confused, or obscure, we cannot expect to have certain, perfect, or clear knowledge.

Secondly, The other is the art of finding out those intermediate ideas, which may show us the agreement

or repugnancy of other ideas, which cannot be immediately compared.

15. Mathematics an instance of this. That these two (and not the relying on maxims, and drawing consequences from some general propositions) are the right methods of improving our knowledge in the ideas of other modes besides those of quantity, the consideration of mathematical knowledge will easily inform us. Where first we shall find that he that has not a perfect and clear idea of those angles or figures of which he desires to know anything, is utterly thereby incapable of any knowledge about them. Suppose but a man not to have a perfect exact idea of a right angle, a scalenum, or trapezium, and there is nothing more certain than that he will in vain seek any demonstration about them. Further, it is evident that it was not the influence of those maxims which are taken for principles in mathematics that hath led the masters of that science into those wonderful discoveries they have made. Let a man of good parts know all the maxims generally made use of in mathematics ever so perfectly, and contemplate their

extent and consequences as much as he pleases, he will, by their assistance, I suppose, scarce ever come to know that the square of the hypothenuse in a right-angled triangle is equal to the squares of the two other sides. The knowledge that “the whole is equal to all its parts,” and “if you take equals from equals, the remainder will be equal,” &c., helped him not, I presume, to this demonstration: and a man may, I think, pore long enough on those axioms without ever seeing one jot the more of mathematical truths. They have been discovered by the thoughts otherwise applied: the mind had other objects, other views before it, far different from those maxims, when it first got the knowledge of such truths in mathematics, which men, well enough acquainted with those received axioms, but ignorant of their method who first made these demonstrations, can never sufficiently admire. And who knows what methods to enlarge our knowledge in other parts of science may hereafter be invented, answering that of algebra in mathematics, which so readily finds out the ideas of quantities to measure others by; whose equality or proportion

we could otherwise very hardly, or, perhaps, never come to know?

Chapter XIII

Some Further Considerations Concerning our Knowledge

1. Our knowledge partly necessary, partly voluntary. Our knowledge, as in other things, so in this, has so great a conformity with our sight, that it is neither wholly necessary, nor wholly voluntary. If our knowledge were altogether necessary, all men’s knowledge would not only be alike, but every man would know all that is knowable; and if it were wholly voluntary, some men so little regard or value it that they would have extreme little, or none at all. Men that have senses cannot choose but receive some ideas by them; and if they have memory, they cannot but retain some of them; and if they have memory, they cannot but retain some of them; and if they have any distinguishing faculty, cannot but perceive the agreement or disagreement of

some of them one with another; as he that has eyes, if he will open them by day, cannot but see some objects and perceive a difference in them. But though a man with his eyes open in the light, cannot but see, yet there be certain objects which he may choose whether he will turn his eyes to; there may be in his reach a book containing pictures and discourses, capable to delight or instruct him, which yet he may never have the will to open, never take the pains to look into.

2. The application of our faculties voluntary; but, they being employed, we know as things are, not as we please. There is also another thing in a man's power, and that is, though he turns his eyes sometimes towards an object, yet he may choose whether he will curiously survey it, and with an intent application endeavour to observe accurately all that is visible in it. But yet, what he does see, he cannot see otherwise than he does. It depends not on his will to see that black which appears yellow; nor to persuade himself that what actually scalds him, feels cold. The earth will not appear painted with flowers, nor the fields covered with verdure, whenever

he has a mind to it: in the cold winter, he cannot help seeing it white and hoary, if he will look abroad. Just thus is it with our understanding: all that is voluntary in our knowledge is the employing or withholding any of our faculties from this or that sort of objects, and a more or less accurate survey of them: but, they being employed, our will hath no power to determine the knowledge of the mind one way or another; that is done only by the objects themselves, as far as they are clearly discovered. And therefore, as far as men's senses are conversant about external objects, the mind cannot but receive those ideas which are presented by them, and be informed of the existence of things without: and so far as men's thoughts converse with their own determined ideas, they cannot but in some measure observe the agreement or disagreement that is to be found amongst some of them, which is so far knowledge: and if they have names for those ideas which they have thus considered, they must needs be assured of the truth of those propositions which express that agreement or disagreement they perceive in them, and be undoubtedly con-

vinced of those truths. For what a man sees, he cannot but see; and what he perceives, he cannot but know that he perceives.

3. Instance in numbers. Thus he that has got the ideas of numbers, and hath taken the pains to compare one, two, and three, to six, cannot choose but know that they are equal: he that hath got the idea of a triangle, and found the ways to measure its angles and their magnitudes, is certain that its three angles are equal to two right ones; and can as little doubt of that, as of this truth, that it is impossible for the same thing to be, and not to be.

4. Instance in natural religion. He also that hath the idea of an intelligent, but frail and weak being, made by and depending on another, who is eternal, omnipotent, perfectly wise and good, will as certainly know that man is to honour, fear, and obey God, as that the sun shines when he sees it. For if he hath but the ideas of two such beings in his mind, and will turn his thoughts that way, and consider them, he will as certainly find that the inferior, finite, and dependent is under an ob-

ligation to obey the supreme and infinite, as he is certain to find that three, four, and seven are less than fifteen; if he will consider and compute those numbers: nor can he be surer in a clear morning that the sun is risen; if he will but open his eyes and turn them that way. But yet these truths, being ever so certain, ever so clear, he may be ignorant of either, or all of them, who will never take the pains to employ his faculties, as he should, to inform himself about them.

Chapter XIV Of Judgment

1. Our knowledge being short, we want something else. The understanding faculties being given to man, not barely for speculation, but also for the conduct of his life, man would be at a great loss if he had nothing to direct him but what has the certainty of true knowledge. For that being very short and scanty, as we have seen, he would be often utterly in the dark, and in most of the actions of his life, perfectly at a stand, had he

nothing to guide him in the absence of clear and certain knowledge. He that will not eat till he has demonstration that it will nourish him; he that will not stir till he infallibly knows the business he goes about will succeed, will have little else to do but to sit still and perish.

2. What use to be made of this twilight state. Therefore, as God has set some things in broad daylight; as he has given us some certain knowledge, though limited to a few things in comparison, probably as a taste of what intellectual creatures are capable of to excite in us a desire and endeavour after a better state: so, in the greatest part of our concernments, he has afforded us only the twilight, as I may so say, of probability; suitable, I presume, to that state of mediocrity and probationership he has been pleased to place us in here; wherein, to check our over-confidence and presumption, we might, by every day's experience, be made sensible of our short-sightedness and liableness to error; the sense whereof might be a constant admonition to us, to spend the days of this our pilgrimage with industry and care, in the search and following of that way

which might lead us to a state of greater perfection. It being highly rational to think, even were revelation silent in the case, that, as men employ those talents God has given them here, they shall accordingly receive their rewards at the close of the day, when their sun shall set and night shall put an end to their labours.

3. Judgment, or assent to probability, supplies our want of knowledge. The faculty which God has given man to supply the want of clear and certain knowledge, in cases where that cannot be had, is judgment: whereby the mind takes its ideas to agree or disagree; or, which is the same, any proposition to be true or false, without perceiving a demonstrative evidence in the proofs. The mind sometimes exercises this judgment out of necessity, where demonstrative proofs and certain knowledge are not to be had; and sometimes out of laziness, unskilfulness, or haste, even where demonstrative and certain proofs are to be had. Men often stay not warily to examine the agreement or disagreement of two ideas which they are desirous or concerned to know; but, either incapable of such attention as is requisite in a

long train of gradations, or impatient of delay, lightly cast their eyes on, or wholly pass by the proofs; and so, without making out the demonstration, determine of the agreement or disagreement of two ideas, as it were by a view of them as they are at a distance, and take it to be the one or the other, as seems most likely to them upon such a loose survey. This faculty of the mind, when it is exercised immediately about things, is called judgment; when about truths delivered in words, is most commonly called assent or dissent: which being the most usual way, wherein the mind has occasion to employ this faculty, I shall, under these terms, treat of it, as least liable in our language to equivocation.

4. Judgement is the presuming things to be so, without perceiving it. Thus the mind has two faculties conversant about truth and falsehood:—

First, *knowledge*, whereby it certainly perceives, and is undoubtedly satisfied of the agreement or disagreement of any ideas.

Secondly *judgment*, which is the putting ideas together, or separating them from one another in the mind,

when their certain agreement or disagreement is not perceived, but presumed to be so; which is, as the word imports, taken to be so before it certainly appears. And if it so unites or separates them as in reality things are, it is right judgment.

Chapter XV **Of Probability**

1. Probability is the appearance of agreement upon fallible proofs. As demonstration is the showing the agreement or disagreement of two ideas by the intervention of one or more proofs, which have a constant, immutable, and visible connexion one with another; so probability is nothing but the appearance of such an agreement or disagreement by the intervention of proofs, whose connexion is not constant and immutable, or at least is not perceived to be so, but is, or appears for the most part to be so, and is enough to induce the mind to judge the proposition to be true or false, rather than the contrary. For example: in the demonstration of it a

man perceives the certain, immutable connexion there is of equality between the three angles of a triangle, and those intermediate ones which are made use of to show their equality to two right ones; and so, by an intuitive knowledge of the agreement or disagreement of the intermediate ideas in each step of the progress, the whole series is continued with an evidence, which clearly shows the agreement or disagreement of those three angles in equality to two right ones: and thus he has certain knowledge that it is so. But another man, who never took the pains to observe the demonstration, hearing a mathematician, a man of credit, affirm the three angles of a triangle to be equal to two right ones, assents to it, i.e. receives it for true: in which case the foundation of his assent is the probability of the thing; the proof being such as for the most part carries truth with it: the man on whose testimony he receives it, not being wont to affirm anything contrary to or besides his knowledge, especially in matters of this kind: so that that which causes his assent to this proposition, that the three angles of a triangle are equal to two right ones, that

which makes him take these ideas to agree, without knowing them to do so, is the wonted veracity of the speaker in other cases, or his supposed veracity in this.

2. It is to supply our want of knowledge. Our knowledge, as has been shown, being very narrow, and we not happy enough to find certain truth in everything which we have occasion to consider; most of the propositions we think, reason, discourse—nay, act upon, are such as we cannot have undoubted knowledge of their truth: yet some of them border so near upon certainty, that we make no doubt at all about them; but assent to them as firmly, and act, according to that assent, as resolutely as if they were infallibly demonstrated, and that our knowledge of them was perfect and certain. But there being degrees herein, from the very neighbourhood of certainty and demonstration, quite down to improbability and unlikeness, even to the confines of impossibility; and also degrees of assent from full assurance and confidence, quite down to conjecture, doubt, and distrust: I shall come now, (having, as I think, found out the bounds of human knowledge and certainty,) in the

next place, to consider the several degrees and grounds of probability, and assent or faith.

3. Being that which makes us presume things to be true, before we know them to be so. Probability is likeliness to be true, the very notation of the word signifying such a proposition, for which there be arguments or proofs to make it pass, or be received for true. The entertainment the mind gives this sort of propositions is called belief, assent, or opinion, which is the admitting or receiving any proposition for true, upon arguments or proofs that are found to persuade us to receive it as true, without certain knowledge that it is so. And herein lies the difference between probability and certainty, faith, and knowledge, that in all the parts of knowledge there is intuition; each immediate idea, each step has its visible and certain connexion: in belief, not so. That which makes me believe, is something extraneous to the thing I believe; something not evidently joined on both sides to, and so not manifestly showing the agreement or disagreement of those ideas that are under consideration.

4. The grounds of probability are two: conformity with our own experience, or the testimony of others' experience. Probability then, being to supply the defect of our knowledge and to guide us where that fails, is always conversant about propositions whereof we have no certainty, but only some inducements to receive them for true. The grounds of it are, in short, these two following:—

First, The conformity of anything with our own knowledge, observation, and experience.

Secondly, The testimony of others, vouching their observation and experience. In the testimony of others is to be considered: 1. The number. 2. The integrity. 3. The skill of the witnesses. 4. The design of the author, where it is a testimony out of a book cited. 5. The consistency of the parts, and circumstances of the relation. 6. Contrary testimonies.

5. In this, all the arguments pro and con ought to be examined, before we come to a judgment. Probability wanting that intuitive evidence which infallibly determines the understanding and produces certain knowl-

edge, the mind, if it will proceed rationally, ought to examine all the grounds of probability, and see how they make more or less for or against any proposition, before it assents to or dissents from it; and, upon a due balancing the whole, reject or receive it, with a more or less firm assent, proportionably to the preponderancy of the greater grounds of probability on one side or the other. For example:—

If I myself see a man walk on the ice, it is past probability; it is knowledge. But if another tells me he saw a man in England, in the midst of a sharp winter, walk upon water hardened with cold, this has so great conformity with what is usually observed to happen that I am disposed by the nature of the thing itself to assent to it; unless some manifest suspicion attend the relation of that matter of fact. But if the same thing be told to one born between the tropics, who never saw nor heard of any such thing before, there the whole probability relies on testimony: and as the relators are more in number, and of more credit, and have no interest to speak contrary to the truth, so that matter of fact is like to

find more or less belief. Though to a man whose experience has always been quite contrary, and who has never heard of anything like it, the most untainted credit of a witness will scarce be able to find belief.

The king of Siam. As it happened to a Dutch ambassador, who entertaining the king of Siam with the particularities of Holland, which he was inquisitive after, amongst other things told him that the water in his country would sometimes, in cold weather, be so hard that men walked upon it, and that it would bear an elephant, if he were there. To which the king replied, Hitherto I have believed the strange things you have told me, because I look upon you as a sober fair man, but now I am sure you lie.

6. Probable arguments capable of great variety. Upon these grounds depends the probability of any proposition: and as the conformity of our knowledge, as the certainty of observations, as the frequency and constancy of experience and the number and credibility of testimonies do more or less agree or disagree with it, so is any proposition in itself more or less probable. There

is another, I confess, which, though by itself it be no true ground of probability, yet is often made use of for one, by which men most commonly regulate their assent, and upon which they pin their faith more than anything else, and that is, the opinion of others; though there cannot be a more dangerous thing to rely on, nor more likely to mislead one; since there is much more falsehood and error among men than truth and knowledge. And if the opinions and persuasions of others, whom we know and think well of, be a ground of assent, men have reason to be Heathens in Japan, Mahometans in Turkey, Papists in Spain, Protestants in England, and Lutherans in Sweden. But of this wrong ground of assent I shall have occasion to speak more at large in another place.

Chapter XVI

Of the Degrees of Assent

1. Our assent ought to be regulated by the grounds of probability. The grounds of probability we have laid down

in the foregoing chapter: as they are the foundations on which our assent is built, so are they also the measure whereby its several degrees are, or ought to be regulated: only we are to take notice that, whatever grounds of probability there may be, they yet operate no further on the mind which searches after truth, and endeavours to judge right, than they appear; at least, in the first judgment or search that the mind makes. I confess, in the opinions men have, and firmly stick to in the world, their assent is not always from an actual view of the reasons that at first prevailed with them: it being in many cases almost impossible, and in most, very hard, even for those who have very admirable memories, to retain all the proofs which, upon a due examination, made them embrace that side of the question. It suffices that they have once with care and fairness sifted the matter as far as they could; and that they have searched into all the particulars, that they could imagine to give any light to the question; and, with the best of their skill, cast up the account upon the whole evidence: and thus, having once found on which side the

probability appeared to them, after as full and exact an inquiry as they can make, they lay up the conclusion in their memories as a truth they have discovered; and for the future they remain satisfied with the testimony of their memories that this is the opinion that, by the proofs they have once seen of it, deserves such a degree of their assent as they afford it.

2. These cannot always be actually in view; and then we must content ourselves with the remembrance that we once saw ground for such a degree of assent. This is all that the greatest part of men are capable of doing, in regulating their opinions and judgments; unless a man will exact of them, either to retain distinctly in their memories all the proofs concerning any probable truth, and that too, in the same order, and regular deduction of consequences in which they have formerly placed or seen them; which sometimes is enough to fill a large volume on one single question: or else they must require a man, for every opinion that he embraces, every day to examine the proofs: both which are impossible. It is unavoidable, therefore, that the memory be relied on

in the case, and that men be persuaded of several opinions, whereof the proofs are not actually in their thoughts; nay, which perhaps they are not able actually to recall. Without this, the greatest part of men must be either very sceptic; or change every moment, and yield themselves up to whoever, having lately studied the question, offers them arguments, which, for want of memory, they are not able presently to answer.

3. The ill consequence of this, if our former judgments were not rightly made. I cannot but own, that men's sticking to their past judgment, and adhering firmly to conclusions formerly made, is often the cause of great obstinacy in error and mistake. But the fault is not that they rely on their memories for what they have before well judged, but because they judged before they had well examined. May we not find a great number (not to say the greatest part) of men that think they have formed right judgments of several matters; and that for no other reason, but because they never thought otherwise? that imagine themselves to have judged right, only because they never questioned, never examined, their own opin-

ions? Which is indeed to think they judged right, because they never judged at all. And yet these, of all men, hold their opinions with the greatest stiffness; those being generally the most fierce and firm in their tenets, who have least examined them. What we once know, we are certain is so: and we may be secure, that there are no latent proofs undiscovered, which may overturn our knowledge, or bring it in doubt. But, in matters of probability, it is not in every case we can be sure that we have all the particulars before us, that any way concern the question; and that there is no evidence behind, and yet unseen, which may cast the probability on the other side, and outweigh all that at present seems to preponderate with us. Who almost is there that hath the leisure, patience, and means to collect together all the proofs concerning most of the opinions he has, so as safely to conclude that he hath a clear and full view; and that there is no more to be alleged for his better information? And yet we are forced to determine ourselves on the one side or other. The conduct of our lives, and the management of our great concerns, will

not bear delay: for those depend, for the most part, on the determination of our judgment in points wherein we are not capable of certain and demonstrative knowledge, and wherein it is necessary for us to embrace the one side or the other.

4. The right use of it, mutual charity and forbearance, in a necessary diversity of opinions. Since, therefore, it is unavoidable to the greatest part of men, if not all, to have several opinions, without certain and indubitable proofs of their truth; and it carries too great an imputation of ignorance, lightness, or folly for men to quit and renounce their former tenets presently upon the offer of an argument which they cannot immediately answer, and show the insufficiency of: it would, methinks, become all men to maintain peace, and the common offices of humanity, and friendship, in the diversity of opinions; since we cannot reasonably expect that any one should readily and obsequiously quit his own opinion, and embrace ours, with a blind resignation to an authority which the understanding of man acknowledges not. For however it may often mistake, it can own

no other guide but reason, nor blindly submit to the will and dictates of another. If he you would bring over to your sentiments be one that examines before he assents, you must give him leave at his leisure to go over the account again, and, recalling what is out of his mind, examine all the particulars, to see on which side the advantage lies: and if he will not think our arguments of weight enough to engage him anew in so much pains, it is but what we often do ourselves in the like case; and we should take it amiss if others should prescribe to us what points we should study. And if he be one who takes his opinions upon trust, how can we imagine that he should renounce those tenets which time and custom have so settled in his mind, that he thinks them self-evident, and of an unquestionable certainty; or which he takes to be impressions he has received from God himself, or from men sent by him? How can we expect, I say, that opinions thus settled should be given up to the arguments or authority of a stranger or adversary, especially if there be any suspicion of interest or design, as there never fails to be, where men find

themselves ill treated? We should do well to commiserate our mutual ignorance, and endeavour to remove it in all the gentle and fair ways of information; and not instantly treat others ill, as obstinate and perverse, because they will not renounce their own, and receive our opinions, or at least those we would force upon them, when it is more than probable that we are no less obstinate in not embracing some of theirs. For where is the man that has incontestable evidence of the truth of all that he holds, or of the falsehood of all he condemns; or can say that he has examined to the bottom all his own, or other men's opinions? The necessity of believing without knowledge, nay often upon very slight grounds, in this fleeting state of action and blindness we are in, should make us more busy and careful to inform ourselves than constrain others. At least, those who have not thoroughly examined to the bottom all their own tenets, must confess they are unfit to prescribe to others; and are unreasonable in imposing that as truth on other men's belief, which they themselves have not searched into, nor weighed the arguments of probabil-

ity, on which they should receive or reject it. Those who have fairly and truly examined, and are thereby got past doubt in all the doctrines they profess and govern themselves by, would have a juster pretence to require others to follow them: but these are so few in number, and find so little reason to be magisterial in their opinions, that nothing insolent and imperious is to be expected from them: and there is reason to think, that, if men were better instructed themselves, they would be less imposing on others.

5. Probability is either of sensible matter of fact, capable of human testimony, or of what is beyond the evidence of our senses. But to return to the grounds of assent, and the several degrees of it, we are to take notice, that the propositions we receive upon inducements of probability are of two sorts: either concerning some particular existence, or, as it is usually termed, matter of fact, which, falling under observation, is capable of human testimony; or else concerning things, which, being beyond the discovery of our senses, are not capable of any such testimony.

6. The concurrent experience of all other men with ours, produces assurance approaching to knowledge. Concerning the first of these, viz. Particular matter of fact.

I. Where any particular thing, consonant to the constant observation of ourselves and others in the like case, comes attested by the concurrent reports of all that mention it, we receive it as easily, and build as firmly upon it, as if it were certain knowledge; and we reason and act thereupon with as little doubt as if it were perfect demonstration. Thus, if all Englishmen, who have occasion to mention it, should affirm that it froze in England the last winter, or that there were swallows seen there in the summer, I think a man could almost as little doubt of it as that seven and four are eleven. The first, therefore, and highest degree of probability, is, when the general consent of all men, in all ages, as far as it can be known, concurs with a man's constant and never-failing experience in like cases, to confirm the truth of any particular matter of fact attested by fair witnesses: such are all the stated constitutions and prop-

erties of bodies, and the regular proceedings of causes and effects in the ordinary course of nature. This we call an argument from the nature of things themselves. For what our own and other men's constant observation has found always to be after the same manner, that we with reason conclude to be the effect of steady and regular causes; though they come not within the reach of our knowledge. Thus, That fire warmed a man, made lead fluid, and changes the colour or consistency in wood or charcoal; that iron sunk in water, and swam in quicksilver: these and the like propositions about particular facts, being agreeable to our constant experience, as often as we have to do with these matters; and being generally spoke of (when mentioned by others) as things found constantly to be so, and therefore not so much as controverted by anybody—we are put past doubt that a relation affirming any such thing to have been, or any prediction that it will happen again in the same manner, is very true. These probabilities rise so near to certainty, that they govern our thoughts as absolutely, and influence all our actions as fully, as the most evi-

dent demonstration; and in what concerns us we make little or no difference between them and certain knowledge. Our belief, thus grounded, rises to assurance.

7. II. Unquestionable testimony, and our own experience that a thing is for the most part so, produce confidence. The next degree of probability is, when I find by my own experience, and the agreement of all others that mention it, a thing to be for the most part so, and that the particular instance of it is attested by many and undoubted witnesses: v.g. history giving us such an account of men in all ages, and my own experience, as far as I had an opportunity to observe, confirming it, that most men prefer their private advantage to the public: if all historians that write of Tiberius, say that Tiberius did so, it is extremely probable. And in this case, our assent has a sufficient foundation to raise itself to a degree which we may call confidence.

8. III. Fair testimony, and the nature of the thing indifferent, produce unavoidable assent. In things that happen indifferently, as that a bird should fly this or that way; that it should thunder on a man's right or left

hand, &c., when any particular matter of fact is vouched by the concurrent testimony of unsuspected witnesses, there our assent is also unavoidable. Thus: that there is such a city in Italy as Rome: that about one thousand seven hundred years ago, there lived in it a man, called Julius Caesar; that he was a general, and that he won a battle against another, called Pompey. This, though in the nature of the thing there be nothing for nor against it, yet being related by historians of credit, and contradicted by no one writer, a man cannot avoid believing it, and can as little doubt of it as he does of the being and actions of his own acquaintance, whereof he himself is a witness.

9. Experience and testimonies clashing infinitely vary the degrees of probability. Thus far the matter goes easy enough. Probability upon such grounds carries so much evidence with it, that it naturally determines the judgment, and leaves us as little liberty to believe or disbelieve, as a demonstration does, whether we will know, or be ignorant. The difficulty is, when testimonies contradict common experience, and the reports of history

and witnesses clash with the ordinary course of nature, or with one another; there it is, where diligence, attention, and exactness are required, to form a right judgment, and to proportion the assent to the different evidence and probability of the thing: which rises and falls, according as those two foundations of credibility, viz. common observation in like cases, and particular testimonies in that particular instance, favour or contradict it. These are liable to so great variety of contrary observations, circumstances, reports, different qualifications, tempers, designs, oversights, &c., of the reporters, that it is impossible to reduce to precise rules the various degrees wherein men give their assent. This only may be said in general, That as the arguments and proofs pro and con, upon due examination, nicely weighing every particular circumstance, shall to any one appear, upon the whole matter, in a greater or less degree to preponderate on either side; so they are fitted to produce in the mind such different entertainments, as we call belief, conjecture, guess, doubt, wavering, distrust, disbelief, &c.

10. Traditional testimonies, the further removed the less their proof becomes. This is what concerns assent in matters wherein testimony is made use of: concerning which, I think, it may not be amiss to take notice of a rule observed in the law of England; which is, That though the attested copy of a record be good proof, yet the copy of a copy, ever so well attested, and by ever so credible witnesses, will not be admitted as a proof in judicature. This is so generally approved as reasonable, and suited to the wisdom and caution to be used in our inquiry after material truths, that I never yet heard of any one that blamed it. This practice, if it be allowable in the decisions of right and wrong, carries this observation along with it, viz. That any testimony, the further off it is from the original truth, the less force and proof it has. The being and existence of the thing itself, is what I call the original truth. A credible man vouching his knowledge of it is a good proof; but if another equally credible do witness it from his report, the testimony is weaker: and a third that attests the hearsay of an hearsay is yet less considerable. So that in traditional truths, each

remove weakens the force of the proof: and the more hands the tradition has successively passed through, the less strength and evidence does it receive from them. This I thought necessary to be taken notice of: because I find amongst some men the quite contrary commonly practised, who look on opinions to gain force by growing older; and what a thousand years since would not, to a rational man contemporary with the first voucher, have appeared at all probable, is now urged as certain beyond all question, only because several have since, from him, said it one after another. Upon this ground propositions, evidently false or doubtful enough in their first beginning, come, by an inverted rule of probability, to pass for authentic truths; and those which found or deserved little credit from the mouths of their first authors, are thought to grow venerable by age, are urged as undeniable.

11. Yet history is of great use. I would not be thought here to lessen the credit and use of history: it is all the light we have in many cases, and we have in many cases, and we receive from it a great part of the useful truths we have, with a convincing evidence. I think nothing

more valuable than the records of antiquity: I wish we had more of them, and more uncorrupted. But this truth itself forces me to say, That no probability can rise higher than its first original. What has no other evidence than the single testimony of one only witness must stand or fall by his only testimony, whether good, bad, or indifferent; and though cited afterwards by hundreds of others, one after another, is so far from receiving any strength thereby, that it is only the weaker. Passion, interest, inadvertency, mistake of his meaning, and a thousand odd reasons, or capricios, men's minds are acted by, (impossible to be discovered,) may make one man quote another man's words or meaning wrong. He that has but ever so little examined the citations of writers, cannot doubt how little credit the quotations deserve, where the originals are wanting; and consequently how much less quotations of quotations can be relied on. This is certain, that what in one age was affirmed upon slight grounds, can never after come to be more valid in future ages by being often repeated. But the further still it is from the original, the less valid it is, and has

always less force in the mouth or writing of him that last made use of it than in his from whom he received it.

12. In things which sense cannot discover, analogy is the great rule of probability. [Secondly], The probabilities we have hitherto mentioned are only such as concern matter of fact, and such things as are capable of observation and testimony. There remains that other sort, concerning which men entertain opinions with variety of assent, though the things be such, that falling not under the reach of our senses, they are not capable of testimony. Such are, 1. The existence, nature and operations of finite immaterial beings without us; as spirits, angels, devils, &c. Or the existence of material beings which, either for their smallness in themselves or remoteness from us, our senses cannot take notice of—as, whether there be any plants, animals, and intelligent inhabitants in the planets, and other mansions of the vast universe. 2. Concerning the manner of operation in most parts of the works of nature: wherein, though we see the sensible effects, yet their causes are unknown, and we perceive not the ways and manner

how they are produced. We see animals are generated, nourished, and move; the loadstone draws iron; and the parts of a candle, successively melting, turn into flame, and give us both light and heat. These and the like effects we see and know: but the causes that operate, and the manner they are produced in, we can only guess and probably conjecture. For these and the like, coming not within the scrutiny of human senses, cannot be examined by them, or be attested by anybody; and therefore can appear more or less probable, only as they more or less agree to truths that are established in our minds, and as they hold proportion to other parts of our knowledge and observation. Analogy in these matters is the only help we have, and it is from that alone we draw all our grounds of probability. Thus, observing that the bare rubbing of two bodies violently one upon another, produces heat, and very often fire itself, we have reason to think, that what we call heat and fire consists in a violent agitation of the imperceptible minute parts of the burning matter. Observing likewise that the different refractions of pellucid bodies produce in our eyes the dif-

ferent appearances of several colours; and also, that the different ranging and laying the superficial parts of several bodies, as of velvet, watered silk, &c., does the like, we think it probable that the colour and shining of bodies is in them nothing but the different arrangement and refraction of their minute and insensible parts. Thus, finding in all parts of the creation, that fall under human observation, that there is a gradual connexion of one with another, without any great or discernible gaps between, in all that great variety of things we see in the world, which are so closely linked together, that, in the several ranks of beings, it is not easy to discover the bounds betwixt them; we have reason to be persuaded that, by such gentle steps, things ascend upwards in degrees of perfection. It is a hard matter to say where sensible and rational begin, and where insensible and irrational end: and who is there quick-sighted enough to determine precisely which is the lowest species of living things, and which the first of those which have no life? Things, as far as we can observe, lessen and augment, as the quantity does in a regular cone; where, though there be

a manifest odds betwixt the bigness of the diameter at a remote distance, yet the difference between the upper and under, where they touch one another, is hardly discernible. The difference is exceeding great between some men and some animals: but if we will compare the understanding and abilities of some men and some brutes, we shall find so little difference, that it will be hard to say, that that of the man is either clearer or larger. Observing, I say, such gradual and gentle descents downwards in those parts of the creation that are beneath man, the rule of analogy may make it probable, that it is so also in things above us and our observation; and that there are several ranks of intelligent beings, excelling us in several degrees of perfection, ascending upwards towards the infinite perfection of the Creator, by gentle steps and differences, that are every one at no great distance from the next to it. This sort of probability, which is the best conduct of rational experiments, and the rise of hypothesis, has also its use and influence; and a wary reasoning from analogy leads us often into the discovery of truths and useful productions, which would otherwise lie concealed.

13. One case where contrary experience lessens not the testimony. Though the common experience and the ordinary course of things have justly a mighty influence on the minds of men, to make them give or refuse credit to anything proposed to their belief; yet there is one case, wherein the strangeness of the fact lessens not the assent to a fair testimony given of it. For where such supernatural events are suitable to ends aimed at by Him who has the power to change the course of nature, there, under such circumstances, that may be the fitter to procure belief, by how much the more they are beyond or contrary to ordinary observation. This is the proper case of miracles, which, well attested, do not only find credit themselves, but give it also to other truths, which need such confirmation.

14. The bare testimony of divine revelation is the highest certainty. Besides those we have hitherto mentioned, there is one sort of propositions that challenge the highest degree of our assent, upon bare testimony, whether the thing proposed agree or disagree with common experience, and the ordinary course of things, or no. The

reason whereof is, because the testimony is of such an one as cannot deceive nor be deceived: and that is of God himself. This carries with it an assurance beyond doubt, evidence beyond exception. This is called by a peculiar name, revelation, and our assent to it, faith, which as absolutely determines our minds, and as perfectly excludes all wavering, as our knowledge itself; and we may as well doubt of our own being, as we can whether any revelation from God be true. So that faith is a settled and sure principle of assent and assurance, and leaves no manner of room for doubt or hesitation. Only we must be sure that it be a divine revelation, and that we understand it right: else we shall expose ourselves to all the extravagancy of enthusiasm, and all the error of wrong principles, if we have faith and assurance in what is not divine revelation. And therefore, in those cases, our assent can be rationally no higher than the evidence of its being a revelation, and that this is the meaning of the expressions it is delivered in. If the evidence of its being a revelation, or that this is its true sense, be only on probable proofs, our assent can reach

no higher than an assurance or diffidence, arising from the more or less apparent probability of the proofs. But of faith, and the precedency it ought to have before other arguments of persuasion, I shall speak more hereafter; where I treat of it as it is ordinarily placed, in contradistinction to reason; though in truth it be nothing else but an assent founded on the highest reason.

Chapter XVII

Of Reason

1. Various significations of the word “reason”. The word reason in the English language has different significations: sometimes it is taken for true and clear principles: sometimes for clear and fair deductions from those principles: and sometimes for the cause, and particularly the final cause. But the consideration I shall have of it here is in a signification different from all these; and that is, as it stands for a faculty in man, that faculty whereby man is supposed to be distinguished from beasts, and wherein it is evident he much surpasses them.

2. Wherein reasoning consists. If general knowledge, as has been shown, consists in a perception of the agreement or disagreement of our own ideas, and the knowledge of the existence of all things without us (except only of a God, whose existence every man may certainly know and demonstrate to himself from his own existence), be had only by our senses, what room is there for the exercise of any other faculty, but outward sense and inward perception? What need it there of reason? Very much: both for the enlargement of our knowledge, and regulating our assent. For it hath to do both in knowledge and opinion, and is necessary and assisting to all our other intellectual faculties, and indeed contains two of them, viz. sagacity and illation. By the one, it finds out; and by the other, it so orders the intermediate ideas as to discover what connexion there is in each link of the chain, whereby the extremes are held together; and thereby, as it were, to draw into view the truth sought for, which is that which we call illation or inference, and consists in nothing but the perception of the connexion there is between the ideas, in each step of

the deduction; whereby the mind comes to see, either the certain agreement or disagreement of any two ideas, as in demonstration, in which it arrives at knowledge; or their probable connexion, on which it gives or withholds its assent, as in opinion. Sense and intuition reach but a very little way. The greatest part of our knowledge depends upon deductions and intermediate ideas: and in those cases where we are fain to substitute assent instead of knowledge, and take propositions for true, without being certain they are so, we have need to find out, examine, and compare the grounds of their probability. In both these cases, the faculty which finds out the means, and rightly applies them, to discover certainty in the one, and probability in the other, is that which we call reason. For, as reason perceives the necessary and indubitable connexion of all the ideas or proofs one to another, in each step of any demonstration that produces knowledge; so it likewise perceives the probable connexion of all the ideas or proofs one to another, in every step of a discourse, to which it will think assent due. This is the lowest degree of that which can be

truly called reason. For where the mind does not perceive this probable connexion, where it does not discern whether there be any such connexion or no; there men's opinions are not the product of judgment, or the consequence of reason, but the effects of chance and hazard, of a mind floating at all adventures, without choice and without direction.

3. Reason in its four degrees. So that we may in reason consider these degrees: four the first and highest is the discovering and finding out of truths; the second, the regular and methodical disposition of them, and laying them in a clear and fit order, to make their connexion and force be plainly and easily perceived; the third is the perceiving their connexion; and the fourth, a making a right conclusion. These several degrees may be observed in any mathematical demonstration; it being one thing to perceive the connexion of each part, as the demonstration is made by another; another to perceive the dependence of the conclusion on all the parts; a third, to make out a demonstration clearly and neatly one's self; and something different from all these, to

have first found out these intermediate ideas or proofs by which it is made.

4. Whether syllogism is the great instrument of reason: first cause to doubt this. There is one thing more which I shall desire to be considered concerning reason; and that is, whether syllogism, as is generally thought, be the proper instrument of it, and the usefulest way of exercising this faculty. The causes I have to doubt are these:—

First, Because syllogism serves our reason but in one only of the forementioned parts of it; and that is, to show the connexion of the proofs in any one instance, and no more; but in this it is of no great use, since the mind can perceive such connexion, where it really is, as easily, nay, perhaps better, without it.

Men can reason well who cannot make a syllogism. If we will observe the actings of our own minds, we shall find that we reason best and clearest, when we only observe the connexion of the proof, without reducing our thoughts to any rule of syllogism. And therefore we may take notice, that there are many men that reason

exceeding clear and rightly, who know not how to make a syllogism. He that will look into many parts of Asia and America, will find men reason there perhaps as acutely as himself, who yet never heard of a syllogism, nor can reduce any one argument to those forms: and I believe scarce any one makes syllogisms in reasoning within himself. Indeed syllogism is made use of, on occasion, to discover a fallacy hid in a rhetorical flourish, or cunningly wrapt up in a smooth period; and, stripping an absurdity of the cover of wit and good language, show it in its naked deformity. But the weakness or fallacy of such a loose discourse it shows, by the artificial form it is put into, only to those who have thoroughly studied mode and figure, and have so examined the many ways that three propositions may be put together, as to know which of them does certainly conclude right, and which not, and upon what grounds it is that they do so. All who have so far considered syllogism, as to see the reason why in three propositions laid together in one form, the conclusion will be certainly right, but in another not certainly so, I grant are cer-

tain of the conclusion they draw from the premises in the allowed modes and figures. But they who have not so far looked into those forms, are not sure by virtue of syllogism, that the conclusion certainly follows from the premises; they only take it to be so by an implicit faith in their teachers and a confidence in those forms of argumentation; but this is still but believing, not being certain. Now, if, of all mankind those who can make syllogisms are extremely few in comparison of those who cannot; and if, of those few who have been taught logic, there is but a very small number who do any more than believe that syllogisms, in the allowed modes and figures do conclude right, without knowing certainly that they do so: if syllogisms must be taken for the only proper instrument of reason and means of knowledge, it will follow, that, before Aristotle, there was not one man that did or could know anything by reason; and that, since the invention of syllogisms, there is not one of ten thousand that doth.

Aristotle. But God has not been so sparing to men to make them barely two-legged creatures, and left it to

Aristotle to make them rational, i.e. those few of them that he could get so to examine the grounds of syllogisms, as to see that, in above three score ways that three propositions may be laid together, there are but about fourteen wherein one may be sure that the conclusion is right; and upon what grounds it is, that, in these few, the conclusion is certain, and in the other not. God has been more bountiful to mankind than so. He has given them a mind that can reason, without being instructed in methods of syllogizing: the understanding is not taught to reason by these rules; it has a native faculty to perceive the coherence or incoherence of its ideas, and can range them right, without any such perplexing repetitions. I say not this any way to lessen Aristotle, whom I look on as one of the greatest men amongst the ancients; whose large views, acuteness, and penetration of thought and strength of judgment, few have equalled; and who, in this very invention of forms of argumentation, wherein the conclusion may be shown to be rightly inferred, did great service against those who were not ashamed to deny anything. And I readily

own, that all right reasoning may be reduced to his forms of syllogism. But yet I think, without any diminution to him, I may truly say, that they are not the only nor the best way of reasoning, for the leading of those into truth who are willing to find it, and desire to make the best use they may of their reason, for the attainment of knowledge. And he himself, it is plain, found out some forms to be conclusive, and others not, not by the forms themselves, but by the original way of knowledge, i.e. by the visible agreement of ideas. Tell a country gentlewoman that the wind is south-west, and the weather lowering, and like to rain, and she will easily understand it is not safe for her to go abroad thin clad in such a day, after a fever: she clearly sees the probable connexion of all these, viz. south-west wind, and clouds, rain, wetting, taking cold, relapse, and danger of death, without tying them together in those artificial and cumbersome fetters of several syllogisms, that clog and hinder the mind, which proceeds from one part to another quicker and clearer without them: and the probability which she easily perceives in things thus in

their native state would be quite lost, if this argument were managed learnedly, and proposed in mode and figure. For it very often confounds the connexion; and, I think, every one will perceive in mathematical demonstrations, that the knowledge gained thereby comes shortest and clearest without syllogism.

Inference is looked on as the great act of the rational faculty, and so it is when it is rightly made: but the mind, either very desirous to enlarge its knowledge, or very apt to favour the sentiments it has once imbibed, is very forward to make inferences; and therefore often makes too much haste, before it perceives the connexion of the ideas that must hold the extremes together.

Syllogism does not discover ideas, or their connexions. To infer, is nothing but by virtue of one proposition laid down as true, to draw in another as true, i.e. to see or suppose such a connexion of the two ideas of the inferred proposition. V.g. Let this be the proposition laid down, "Men shall be punished in another world," and from thence be inferred this other, "Then men can determine themselves." The question now is, to know

whether the mind has made this inference right or no: if it has made it by finding out the intermediate ideas, and taking a view of the connexion of them, placed in a due order, it has proceeded rationally, and made a right inference: if it has done it without such a view, it has not so much made an inference that will hold, or an inference of right reason, as shown a willingness to have it be, or be taken for such. But in neither case is it syllogism that discovered those ideas, or showed the connexion of them; for they must be both found out, and the connexion everywhere perceived, before they can rationally be made use of in syllogism: unless it can be said, that any idea, without considering what connexion it hath with the two other, whose agreement should be shown by it, will do well enough in a syllogism, and may be taken at a venture for the medius terminus, to prove any conclusion. But this nobody will say; because it is by virtue of the perceived agreement of the intermediate idea with the extremes, that the extremes are concluded to agree; and therefore each intermediate idea must be such as in the whole chain hath

a visible connexion with those two it has been placed between, or else thereby the conclusion cannot be inferred or drawn in: for wherever any link of the chain is loose and without connexion, there the whole strength of it is lost, and it hath no force to infer or draw in anything. In the instance above mentioned, what is it shows the force of the inference, and consequently the reasonableness of it, but a view of the connexion of all the intermediate ideas that draw in the conclusion, or proposition inferred? V.g. "Men shall be punished"; "God the punisher"; "Just punishment"; "The punished guilty"; "Could have done otherwise"; "Freedom"; "Self-determination"; by which chain of ideas thus visibly linked together in train, i.e. each intermediate idea agreeing on each side with those two it is immediately placed between, the ideas of men and self-determination appear to be connected, i.e. this proposition "men can determine themselves" is drawn in or inferred from this, "that they shall be punished in the other world." For here the mind, seeing the connexion there is between the idea of men's punishment in the other world and

the idea of God punishing; between God punishing and the justice of the punishment; between justice of punishment and guilt; between guilt and a power to do otherwise; between a power to do otherwise and freedom; and between freedom and self-determination, sees the connexion between men and self-determination.

The connexion must be discovered before it can be put into syllogisms. Now I ask, whether the connexion of the extremes be not more clearly seen in this simple and natural disposition, than in the perplexed repetitions, and jumble of five or six syllogisms. I must beg pardon for calling it jumble, till somebody shall put these ideas into so many syllogisms, and then say that they are less jumbled, and their connexion more visible, when they are transposed and repeated, and spun out to a greater length in artificial forms, than in that short and natural plain order they are laid down in here, wherein everyone may see it, and wherein they must be seen before they can be put into a train of syllogisms. For the natural order of the connecting ideas must direct the order of the syllogisms, and a man must see the connexion

of each intermediate idea with those that it connects, before he can with reason make use of it in a syllogism. And when all those syllogisms are made, neither those that are nor those that are not logicians will see the force of the argumentation, i.e., the connexion of the extremes, one jot the better. [For those that are not men of art, not knowing the true forms of syllogism, nor the reasons of them, cannot know whether they are made in right and conclusive modes and figures or no, and so are not at all helped by the forms they are put into; though by them the natural order, wherein the mind could judge of their respective connexion, being disturbed, renders the illation much more uncertain than without them.] And as for the logicians themselves, they see the connexion of each intermediate idea with those it stands between, (on which the force of the inference depends,) as well before as after the syllogism is made, or else they do not see it at all. For a syllogism neither shows nor strengthens the connexion of any two ideas immediately put together, but only by the connexion seen in them shows what connexion the extremes have

one with another. But what connexion the intermediate has with either of the extremes in the syllogism, that no syllogism does or can show. That the mind only doth or can perceive as they stand there in that juxtaposition by its own view, to which the syllogistical form it happens to be in gives no help or light at all: it only shows that if the intermediate idea agrees with those it is on both sides immediately applied to; then those two remote ones, or, as they are called, extremes, do certainly agree; and therefore the immediate connexion of each idea to that which it is applied to on each side, on which the force of the reasoning depends, is as well seen before as after the syllogism is made, or else he that makes the syllogism could never see it at all. This, as has been already observed, is seen only by the eye, or the perceptive faculty, of the mind, taking a view of them laid together, in a juxtaposition; which view of any two it has equally, whenever they are laid together in any proposition, whether that proposition be placed as a major or a minor, in a syllogism or no.

Use of syllogism. Of what use, then are syllogisms? I

answer, their chief and main use is in the Schools, where men are allowed without shame to deny the agreement of ideas that do manifestly agree; or out of the Schools, to those who from thence have learned without shame to deny the connexion of ideas, which even to themselves is visible. But to an ingenuous searcher after truth, who has no other aim but to find it, there is no need of any such form to force the allowing of the inference: the truth and reasonableness of it is better seen in ranging of the ideas in a simple and plain order: and hence it is that men, in their own inquiries after truth, never use syllogisms to convince themselves or in teaching others to instruct willing learners. Because, before they can put them into a syllogism, they must see the connexion that is between the intermediate idea and the two other ideas it is set between and applied to, to show their agreement; and when they see that, they see whether the inference be good or no; and so syllogism comes too late to settle it. For to make use again of the former instance, I ask whether the mind, considering the idea of justice, placed as an intermediate idea be-

tween the punishment of men and the guilt of the punished, (and till it does so consider it, the mind cannot make use of it as a *medius terminus*,) does not as plainly see the force and strength of the inference as when it is formed into a syllogism. To show it in a very plain and easy example; let animal be the intermediate idea or *medius terminus* that the mind makes use of to show the connexion of *homo* and *vivens*; I ask whether the mind does not more readily and plainly see that connexion in the simple and proper position of the connecting idea in the middle thus:

Homo—Animal—Vivens,

than in this perplexed one,

Animal—Vivens—Homo—Animal:

which is the position these ideas have in a syllogism, to show the connexion between *homo* and *vivens* by the intervention of animal.

Not the only way to detect fallacies. Indeed syllogism is thought to be of necessary use, even to the lovers of truth, to show them the fallacies that are often concealed in florid, witty, or involved discourses. But that this is a mistake will appear, if we consider, that the reason why sometimes men who sincerely aim at truth are imposed upon by such loose, and, as they are called, rhetorical discourses, is, that their fancies being struck with some lively metaphorical representations, they neglect to observe, or do not easily perceive, what are the true ideas upon which the inference depends. Now, to show such men the weakness of such an argumentation, there needs no more but to strip it of the superfluous ideas, which, blended and confounded with those on which the inference depends, seem to show a connexion where there is none; or at least to hinder the discovery of the want of it; and then to lay the naked ideas on which the force of the argumentation depends in their due order; in which position the mind, taking a view of them, sees what connexion they have, and so is able to judge of the inference without any need of a syllogism at all.

I grant that mode and figure is commonly made use of in such cases, as if the detection of the incoherence of such loose discourses were wholly owing to the syllogistical form; and so I myself formerly thought, till, upon a stricter examination, I now find, that laying the intermediate ideas naked in their due order, shows the incoherence of the argumentation better than syllogism; not only as subjecting each link of the chain to the immediate view of the mind in its proper place, whereby its connexion is best observed; but also because syllogism shows the incoherence only to those (who are not one of ten thousand) who perfectly understand mode and figure, and the reason upon which those forms are established; whereas a due and orderly placing of the ideas upon which the inference is made, makes every one, whether logician or not logician, who understands the terms, and hath the faculty to perceive the agreement or disagreement of such ideas, (without which, in or out of syllogism, he cannot perceive the strength or weakness, coherence or incoherence of the discourse) see the want of connexion in the argumentation, and the absurdity of the inference.

And thus I have known a man unskilful in syllogism, who at first hearing could perceive the weakness and inconclusiveness of a long artificial and plausible discourse, wherewith others better skilled in syllogism have been misled: and I believe there are few of my readers who do not know such. And indeed, if it were not so, the debates of most princes' councils, and the business of assemblies, would be in danger to be mismanaged, since those who are relied upon, and have usually a great stroke in them, are not always such who have the good luck to be perfectly knowing in the forms of syllogism, or expert in mode and figure. And if syllogism were the only, or so much as the surest way to detect the fallacies of artificial discourses; I do not think that all mankind, even princes in matters that concern their crowns and dignities, are so much in love with falsehood and mistake, that they would everywhere have neglected to bring syllogism into the debates of moment; or thought it ridiculous so much as to offer them in affairs of consequence; a plain evidence to me, that men of parts and penetration, who were not idly to

dispute at their ease, but were to act according to the result of their debates, and often pay for their mistakes with their heads or fortunes, found those scholastic forms were of little use to discover truth or fallacy, whilst both the one and the other might be shown, and better shown without them, to those who would not refuse to see what was visibly shown them.

Another cause to doubt whether syllogism be the only proper instrument of reason, in the discovery of truth. Secondly, Another reason that makes me doubt whether syllogism be the only proper instrument of reason, in the discovery of truth, is, that of whatever use mode and figure is pretended to be in the laying open of fallacy, (which has been above considered,) those scholastic forms of discourse are not less liable to fallacies than the plainer ways of argumentation; and for this I appeal to common observation, which has always found these artificial methods of reasoning more adapted to catch and entangle the mind, than to instruct and inform the understanding. And hence it is that men, even when they are baffled and silenced in this scholastic way, are

seldom or never convinced, and so brought over to the conquering side: they perhaps acknowledge their adversary to be the more skilful disputant, but rest nevertheless persuaded of the truth on their side, and go away, worsted as they are, with the same opinion they brought with them: which they could not do if this way of argumentation carried light and conviction with it, and made men see where the truth lay; and therefore syllogism has been thought more proper for the attaining victory in dispute, than for the discovery or confirmation of truth in fair inquiries. And if it be certain, that fallacies can be couched in syllogism, as it cannot be denied; it must be something else, and not syllogism, that must discover them.

I have had experience how ready some men are, when all the use which they have been wont to ascribe to anything is not allowed, to cry out, that I am for laying it wholly aside. But to prevent such unjust and groundless imputations, I tell them, that I am not for taking away any helps to the understanding in the attainment of knowledge. And if men skilled in and used to syllo-

gisms, find them assisting to their reason in the discovery of truth, I think they ought to make use of them. All that I aim at, is, that they should not ascribe more to these forms than belongs to them, and think that men have no use, or not so full an use, of their reasoning faculties without them. Some eyes want spectacles to see things clearly and distinctly; but let not those that use them therefore say nobody can see clearly without them: those who do so will be thought, in favour of art (which, perhaps, they are beholden to,) a little too much to depress and discredit nature. Reason, by its own penetration, where it is strong and exercised, usually sees quicker and clearer without syllogism. If use of those spectacles has so dimmed its sight, that it cannot without them see consequences or inconsequences in argumentation, I am not so unreasonable as to be against the using them. Every one knows what best fits his own sight; but let him not thence conclude all in the dark, who use not just the same helps that he finds a need of.

5. Syllogism helps little in demonstration, less in probability. But however it be in knowledge, I think I may

truly say, it is of far less, or no use at all in probabilities. For the assent there being to be determined by the preponderancy, after due weighing of all the proofs, with all circumstances on both sides, nothing is so unfit to assist the mind in that as syllogism; which running away with one assumed probability, or one topical argument, pursues that till it has led the mind quite out of sight of the thing under consideration; and, forcing it upon some remote difficulty, holds it fast there; entangled perhaps, and, as it were, manacled, in the chain of syllogisms, without allowing it the liberty, much less affording it the helps, requisite to show on which side, all things considered, is the greater probability.

6. Serves not to increase our knowledge, but to fence with the knowledge we suppose we have. But let it help us (as perhaps may be said) in convincing men of their errors and mistakes: (and yet I would fain see the man that was forced out of his opinion by dint of syllogism,) yet still it fails our reason in that part, which, if not its highest perfection, is yet certainly its hardest task, and that which we most need its help in; and that is the

finding out of proofs, and making new discoveries. The rules of syllogism serve not to furnish the mind with those intermediate ideas that may show the connexion of remote ones. This way of reasoning discovers no new proofs, but is the art of marshalling and ranging the old ones we have already. The forty-seventh proposition of the first book of Euclid is very true; but the discovery of it, I think, not owing to any rules of common logic. A man knows first, and then he is able to prove syllogistically. So that syllogism comes after knowledge, and then a man has little or no need of it. But it is chiefly by the finding out those ideas that show the connexion of distant ones, that our stock of knowledge is increased, and that useful arts and sciences are advanced. Syllogism, at best, is but the art of fencing with the little knowledge we have, without making any addition to it. And if a man should employ his reason all this way, he will not do much otherwise than he who, having got some iron out of the bowels of the earth, should have it beaten up all into swords, and put it into his servants' hands to fence with and bang one another. Had the King of Spain

employed the hands of his people, and his Spanish iron so, he had brought to light but little of that treasure that lay so long hid in the dark entrails of America. And I am apt to think, that he who shall employ all the force of his reason only in brandishing of syllogisms, will discover very little of that mass of knowledge which lies yet concealed in the secret recesses of nature; and which, I am apt to think, native rustic reason (as it formerly has done) is likelier to open a way to, and add to the common stock of mankind, rather than any scholastic proceeding by the strict rules of mod, and figure.

7. Other helps to reason than syllogism should be sought. I doubt not, nevertheless, but there are ways to be found to assist our reason in this most useful part; and this the judicious Hooker encourages me to say, who in his Eccl. Pol. 1. i. SS 6, speaks thus: "If there might be added the right helps of true art and learning, (which helps, I must plainly confess, this age of the world, carrying the name of a learned age, doth neither much know nor generally regard,) there would undoubtedly be almost as much difference in maturity of judgment

between men therewith inured, and that which men now are, as between men that are now, and innocents." I do not pretend to have found or discovered here any of those "right helps of art," this great man of deep thought mentions: but that is plain, that syllogism, and the logic now in use, which were as well known in his days, can be none of those he means. It is sufficient for me, if by a Discourse, perhaps something out of the way, I am sure, as to me, wholly new and unborrowed, I shall have given occasion to others to cast about for new discoveries, and to seek in their own thoughts for those right helps of art, which will scarce be found, I fear, by those who servilely confine themselves to the rules and dictates of others. For beaten tracks lead this sort of cattle, (as an observing Roman calls them,) whose thoughts reach only to imitation, Non quo eundum est, sed quo itur. But I can be bold to say, that this age is adorned with some men of that strength of judgment and largeness of comprehension, that, if they would employ their thoughts on this subject, could open new and undiscovered ways to the advancement of knowledge.

8. We can reason about particulars; and the immediate object of all our reasonings is nothing but particular ideas. Having here had occasion to speak of syllogism in general, and the use of it in reasoning, and the improvement of our knowledge, it is fit, before I leave this subject, to take notice of one manifest mistake in the rules of syllogism: viz. that no syllogistical reasoning can be right and conclusive, but what has at least one general proposition in it. As if we could not reason, and have knowledge about particulars: whereas, in truth, the matter rightly considered, the immediate object of all our reasoning and knowledge, is nothing but particulars. Every man's reasoning and knowledge is only about the ideas existing in his own mind; which are truly, every one of them, particular existences: and our knowledge and reason about other things is only as they correspond with those particular ideas. So that the perception of the agreement or disagreement of our particular ideas is the whole and utmost of all our knowledge. Universality is but accidental to it, and consists only in this, that the particular ideas about which it is

are such as more than one particular thing can correspond with and be represented by. But the perception of the agreement or disagreement of any two ideas, and consequently our knowledge, is equally clear and certain, whether either, or both, or neither of those ideas, be capable of representing more real beings than one, or no. One thing more I crave leave to offer about syllogism, before I leave it, viz. May one not upon just ground inquire whether the form syllogism now has, is that which in reason it ought to have? For the medius terminus being to join the extremes, i.e. the intermediate ideas, by its intervention, to show the agreement or disagreement of the two in question, would not the position of the medius terminus be more natural, and show the agreement or disagreement of the extremes clearer and better, if it were placed in the middle between them? Which might be easily done by transposing the propositions, and making the medius terminus the predicate of the first, and the subject of the second. As thus:

Omnis homo est animal.

Omne animal est vivens.

Ergo, omnis homo est vivens.

Omne corpus est extensum et solidum.

Nullum extensum et solidum est pura extensio.

Ergo, corpus non est pura extensio.

I need not trouble my reader with instances in syllogisms whose conclusions are particular. The same reason hold for the same form in them, as well as in the general.

9. Our reason often fails us. Reason, though it penetrates into the depths of the sea and earth, elevates our thoughts as high as the stars, and leads us through the vast spaces and large rooms of this mighty fabric, yet it comes far short of the real extent of even corporeal being. And there are many instances wherein it fails us: as,

I. In cases when we have no ideas. It perfectly fails us where our ideas fail. It neither does nor can extend it-

self further than they do. And therefore, wherever we have no ideas, our reasoning stops, and we are at an end of our reckoning: and if at any time we reason about words which do not stand for any ideas, it is only about those sounds, and nothing else.

10. II. Because our ideas are often obscure or imperfect. Our reason is often puzzled and at a loss because of the obscurity, confusion, or imperfection of the ideas it is employed about; and there we are involved in difficulties and contradictions. Thus, not having any perfect idea of the least extension of matter, nor of infinity, we are at a loss about the divisibility of matter; but having perfect, clear, and distinct ideas of number, our reason meets with none of those inextricable difficulties in numbers, nor finds itself involved in any contradictions about them. Thus, we having but imperfect ideas of the operations of our minds, and of the beginning of motion, or thought how the mind produces either of them in us, and much imperfecter yet of the operation of God, run into great difficulties about free created agents, which reason cannot well extricate itself out of.

11. III. Because we perceive not intermediate ideas to show conclusions. Our reason is often at a stand because it perceives not those ideas, which could serve to show the certain or probable agreement or disagreement of any other two ideas: and in this some men's faculties far outgo others. Till algebra, that great instrument and instance of human sagacity, was discovered, men with amazement looked on several of the demonstrations of ancient mathematicians, and could scarce forbear to think the finding several of those proofs to be something more than human.

12. IV. Because we often proceed upon wrong principles. The mind, by proceeding upon false principles, is often engaged in absurdities and difficulties, brought into straits and contradictions, without knowing how to free itself: and in that case it is in vain to implore the help of reason, unless it be to discover the falsehood and reject the influence of those wrong principles. Reason is so far from clearing the difficulties which the building upon false foundations brings a man into, that if he will pursue it, it entangles him the more, and engages him deeper in perplexities.

13. V. Because we often employ doubtful terms. As obscure and imperfect ideas often involve our reason, so, upon the same ground, do dubious words and uncertain signs, often, in discourses and arguings, when not warily attended to, puzzle men's reason, and bring them to a nonplus. But these two latter are our fault, and not the fault of reason. But yet the consequences of them are nevertheless obvious; and the perplexities or errors they fill men's minds with are everywhere observable.

14. Our highest degree of knowledge is intuitive, without reasoning. Some of the ideas that are in the mind, are so there, that they can be by themselves immediately compared one with another: and in these the mind is able to perceive that they agree or disagree as clearly as that it has them. Thus the mind perceives, that an arch of a circle is less than the whole circle, as clearly as it does the idea of a circle: and this, therefore, as has been said, I call intuitive knowledge; which is certain, beyond all doubt, and needs no probation, nor can have any; this being the highest of all human certainty. In this consists the evidence of all those maxims which

nobody has any doubt about, but every man (does not, as is said, only assent to, but) knows to be true, as soon as ever they are proposed to his understanding. In the discovery of and assent to these truths, there is no use of the discursive faculty, no need of reasoning, but they are known by a superior and higher degree of evidence. And such, if I may guess at things unknown, I am apt to think that angels have now, and the spirits of just men made perfect shall have, in a future state, of thousands of things which now either wholly escape our apprehensions, or which our short-sighted reason having got some faint glimpse of, we, in the dark, grope after.

15. The next is got by reasoning. But though we have, here and there, a little of this clear light, some sparks of bright knowledge, yet the greatest part of our ideas are such, that we cannot discern their agreement or disagreement by an immediate comparing them. And in all these we have need of reasoning, and must, by discourse and inference, make our discoveries. Now of these there are two sorts, which I shall take the liberty to mention here again:—

Through reasonings that are demonstrative. First, Those whose agreement or disagreement, though it cannot be seen by an immediate putting them together, yet may be examined by the intervention of other ideas which can be compared with them. In this case, when the agreement or disagreement of the intermediate idea, on both sides, with those which we would compare, is plainly discerned: there it amounts to demonstration whereby knowledge is produced, which, though it be certain, yet it is not so easy, nor altogether so clear as intuitive knowledge. Because in that there is barely one simple intuition, wherein there is no room for any the least mistake or doubt: the truth is seen all perfectly at once. In demonstration, it is true, there is intuition too, but not altogether at once; for there must be a remembrance of the intuition of the agreement of the medium, or intermediate idea, with that we compared it with before, when we compare it with the other: and where there be many mediums, there the danger of the mistake is the greater. For each agreement or disagreement of the ideas must be observed and seen in each step of

the whole train, and retained in the memory, just as it is; and the mind must be sure that no part of what is necessary to make up the demonstration is omitted or overlooked. This makes some demonstrations long and perplexed, and too hard for those who have not strength of parts distinctly to perceive, and exactly carry so many particulars orderly in their heads. And even those who are able to master such intricate speculations, are fain sometimes to go over them again, and there is need of more than one review before they can arrive at certainty. But yet where the mind clearly retains the intuition it had of the agreement of any idea with another, and that with a third, and that with a fourth, &c., there the agreement of the first and the fourth is a demonstration, and produces certain knowledge; which may be called rational knowledge, as the other is intuitive.

16. To supply the narrowness of demonstrative and intuitive knowledge we have nothing but judgment upon probable reasoning. Secondly, There are other ideas, whose agreement or disagreement can no otherwise be

judged of but by the intervention of others which have not a certain agreement with the extremes, but an usual or likely one: and in these it is that the judgment is properly exercised; which is the acquiescing of the mind, that any ideas do agree, by comparing them with such probable mediums. This, though it never amounts to knowledge, no, not to that which is the lowest degree of it; yet sometimes the intermediate ideas tie the extremes so firmly together, and the probability is so clear and strong, that assent as necessarily follows it, as knowledge does demonstration. The great excellency and use of the judgment is to observe right, and take a true estimate of the force and weight of each probability; and then casting them up all right together, choose that side which has the overbalance.

17. Intuitive knowledge is the perception of the certain agreement or disagreement of two ideas immediately compared together. Rational knowledge is the perception of the certain agreement or disagreement of any two ideas, by the intervention of one or more other ideas.

Judgment is the thinking or taking two ideas to agree or disagree, by the intervention of one or more ideas, whose certain agreement or disagreement with them it does not perceive, but hath observed to be frequent and usual.

18. Consequences of words, and consequences of ideas. Though the deducing one proposition from another, or making inferences in words, be a great part of reason, and that which it is usually employed about; yet the principal act of ratiocination is the finding the agreement or disagreement of two ideas one with another, by the intervention of a third. As a man, by a yard, finds two houses to be of the same length, to measure their equality by juxta-position. Words have their consequences, as the signs of such ideas: and things agree or disagree, as really they are; but we observe it only by our ideas.

19. Four sorts of arguments. Before we quit this subject, it may be worth our while a little to reflect on four sorts of arguments, that men, in their reasonings with others, do ordinarily make use of to prevail on their assent; or at

least to awe them as to silence their opposition.

I. Argumentum ad verecundiam. The first is, to allege the opinions of men, whose parts, learning, eminency, power, or some other cause has gained a name, and settled their reputation in the common esteem with some kind of authority. When men are established in any kind of dignity, it is thought a breach of modesty for others to derogate any way from it, and question the authority of men who are in possession of it. This is apt to be censured, as carrying with it too much pride, when a man does not readily yield to the determination of approved authors, which is wont to be received with respect and submission by others: and it is looked upon as insolence, for a man to set up and adhere to his own opinion against the current stream of antiquity; or to put it in the balance against that of some learned doctor, or otherwise approved writer. Whoever backs his tenets with such authorities, thinks he ought thereby to carry the cause, and is ready to style it impudence in any one who shall stand out against them. This I think may be called argumentum ad verecundiam.

20. II. Argumentum ad ignorantiam. Secondly, Another way that men ordinarily use to drive others and force them to submit to their judgments, and receive their opinion in debate, is to require the adversary to admit what they allege as a proof, or to assign a better. And this I call argumentum ad ignorantiam.

21. III. Argumentum ad hominem. Thirdly, a third way is to press a man with consequences drawn from his own principles or concessions. This is already known under the name of argumentum ad hominem.

22. IV. Argumentum adjudicium. The fourth alone advances us in knowledge and judgment. The fourth is the using of proofs drawn from any of the foundations of knowledge or probability. This I call argumentum adjudicium. This alone, of all the four, brings true instruction with it, and advances us in our way to knowledge. For, 1. It argues not another man's opinion to be right, because I, out of respect, or any other consideration but that of conviction, will not contradict him. 2. It proves not another man to be in the right way, nor that I ought to take the same with him, because I know

not a better. 3. Nor does it follow that another man is in the right way because he has shown me that I am in the wrong. I may be modest, and therefore not oppose another man's persuasion: I may be ignorant, and not be able to produce a better: I may be in an error, and another may show me that I am so. This may dispose me, perhaps, for the reception of truth, but helps me not to it: that must come from proofs and arguments, and light arising from the nature of things themselves, and not from my shamefacedness, ignorance, or error.

23. Above, contrary, and according to reason. By what has been before said of reason, we may be able to make some guess at the distinction of things into those that are according to, above, and contrary to reason. 1. According to reason are such propositions whose truth we can discover by examining and tracing those ideas we have from sensation and reflection; and by natural deduction find to be true or probable. 2. Above reason are such propositions whose truth or probability we cannot by reason derive from those principles. 3. Contrary to reason are such propositions as are inconsistent with or

irreconcilable to our clear and distinct ideas. Thus the existence of one God is according to reason; the existence of more than one God, contrary to reason; the resurrection of the dead, above reason. Above reason also may be taken in a double sense, viz. either as signifying above probability, or above certainty: and in that large sense also, contrary to reason, is, I suppose, sometimes taken.

24. Reason and faith not opposite, for faith must be regulated by reason. There is another use of the word reason, wherein it is opposed to faith: which, though it be in itself a very improper way of speaking, yet common use has so authorized it, that it would be folly either to oppose or hope to remedy it. Only I think it may not be amiss to take notice that, however faith be opposed to reason, faith is nothing but a firm assent of the mind: which, if it be regulated, as is our duty, cannot be afforded to anything but upon good reason; and so cannot be opposite to it. He that believes without having any reason for believing, may be in love with his own fancies; but neither seeks truth as he ought, nor pays the obedi-

ence due to his Maker, who would have him use those discerning faculties he has given him, to keep him out of mistake and error. He that does not this to the best of his power, however he sometimes lights on truth, is in the right but by chance; and I know not whether the luckiness of the accident will excuse the irregularity of his proceeding. This at least is certain, that he must be accountable for whatever mistakes he runs into: whereas he that makes use of the light and faculties God has given him, and seeks sincerely to discover truth by those helps and abilities he has, may have this satisfaction in doing his duty as a rational creature, that, though he should miss truth, he will not miss the reward of it. For he governs his assent right, and places it as he should, who, in any case or matter whatsoever, believes or disbelieves according as reason directs him. He that doth otherwise, transgresses against his own light, and misuses those faculties which were given him to no other end, but to search and follow the clearer evidence and greater probability. But since reason and faith are by some men opposed, we will so consider them in the following chapter.

Chapter XVIII

Of Faith and Reason, and their Distinct Provinces

1. Necessary to know their boundaries. It has been above shown, 1. That we are of necessity ignorant, and want knowledge of all sorts, where we want ideas. 2. That we are ignorant, and want rational knowledge, where we want proofs. 3. That we want certain knowledge and certainty, as far as we want clear and determined specific ideas. 4. That we want probability to direct our assent in matters where we have neither knowledge of our own nor testimony of other men to bottom our reason upon.

From these things thus premised, I think we may come to lay down the measures and boundaries between faith and reason: the want whereof may possibly have been the cause, if not of great disorders, yet at least of great disputes, and perhaps mistakes in the world. For till it be resolved how far we are to be guided by reason, and how far by faith, we shall in vain dispute, and endeavour to convince one another in matters of religion.

2. Faith and reason, what, as contradistinguished. I find every sect, as far as reason will help them, make use of it gladly: and where it fails them, they cry out, It is matter of faith, and above reason. And I do not see how they can argue with any one, or ever convince a gainsayer who makes use of the same plea, without setting down strict boundaries between faith and reason; which ought to be the first point established in all questions where faith has anything to do.

Reason, therefore, here, as contradistinguished to faith, I take to be the discovery of the certainty or probability of such propositions or truths which the mind arrives at by deduction made from such ideas, which it has got by the use of its natural faculties; viz. by sensation or reflection.

Faith, on the other side, is the assent to any proposition, not thus made out by the deductions of reason, but upon the credit of the proposer, as coming from God, in some extraordinary way of communication. This way of discovering truths to men, we call revelation.

3. No new simple idea can be conveyed by traditional revelation. First, Then I say, that no man inspired by

God can by any revelation communicate to others any new simple ideas which they had not before from sensation or reflection. For, whatsoever impressions he himself may have from the immediate hand of God, this revelation, if it be of new simple ideas, cannot be conveyed to another, either by words or any other signs. Because words, by their immediate operation on us, cause no other ideas but of their natural sounds: and it is by the custom of using them for signs, that they excite and revive in our minds latent ideas; but yet only such ideas as were there before. For words, seen or heard, recall to our thoughts those ideas only which to us they have been wont to be signs of, but cannot introduce any perfectly new and formerly unknown simple ideas. The same holds in all other signs; which cannot signify to us things of which we have before never had any idea at all.

Thus whatever things were discovered to St. Paul, when he was rapt up into the third heaven; whatever new ideas his mind there received, all the description he can make to others of that place, is only this, That there are such things, “as eye hath not seen, nor ear heard, nor

hath it entered into the heart of man to conceive.” And supposing God should discover to any one, supernaturally, a species of creatures inhabiting, for example, Jupiter or Saturn, (for that it is possible there may be such, nobody can deny,) which had six senses; and imprint on his mind the ideas conveyed to theirs by that sixth sense: he could no more, by words, produce in the minds of other men those ideas imprinted by that sixth sense, than one of us could convey the idea of any colour, by the sound of words, into a man who, having the other four senses perfect, had always totally wanted the fifth, of seeing. For our simple ideas, then, which are the foundation, and sole matter of all our notions and knowledge, we must depend wholly on our reason; I mean our natural faculties; and can by no means receive them, or any of them, from traditional revelation. I say, traditional revelation, in distinction to original revelation. By the one, I mean that first impression which is made immediately by God on the mind of any man, to which we cannot set any bounds; and by the other, those impressions delivered over to others in words, and

the ordinary ways of conveying our conceptions one to another.

4. Traditional revelation may make us know propositions knowable also by reason, but not with the same certainty that reason doth. Secondly, I say that the same truths may be discovered, and conveyed down from revelation, which are discoverable to us by reason, and by those ideas we naturally may have. So God might, by revelation, discover the truth of any proposition in Euclid; as well as men, by the natural use of their faculties, come to make the discovery themselves. In all things of this kind there is little need or use of revelation, God having furnished us with natural and surer means to arrive at the knowledge of them. For whatsoever truth we come to the clear discovery of, from the knowledge and contemplation of our own ideas, will always be certainer to us than those which are conveyed to us by traditional revelation. For the knowledge we have that this revelation came at first from God can never be so sure as the knowledge we have from the clear and distinct perception of the agreement or disagreement of our own ideas: v.g. if it were

revealed some ages since, that the three angles of a triangle were equal to two right ones, I might assent to the truth of that proposition, upon the credit of that tradition, that it was revealed: but that would never amount to so great a certainty as the knowledge of it, upon the comparing and measuring my own ideas of two right angles, and the three angles of a triangle. The like holds in matter of fact knowable by our senses; v.g. the history of the deluge is conveyed to us by writings which had their original from revelation: and yet nobody, I think, will say he has as certain and clear a knowledge of the flood as Noah, that saw it; or that he himself would have had, had he then been alive and seen it. For he has no greater an assurance than that of his senses, that it is writ in the book supposed writ by Moses inspired: but he has not so great an assurance that Moses wrote that book as if he had seen Moses write it. So that the assurance of its being a revelation is less still than the assurance of his senses.

5. Even original revelation cannot be admitted against the clear evidence of reason. In propositions, then, whose

certainty is built upon the clear perception of the agreement or disagreement of our ideas, attained either by immediate intuition, as in self-evident propositions, or by evident deductions of reason in demonstrations we need not the assistance of revelation, as necessary to gain our assent, and introduce them into our minds. Because the natural ways of knowledge could settle them there, or had done it already; which is the greatest assurance we can possibly have of anything, unless where God immediately reveals it to us: and there too our assurance can be no greater than our knowledge is, that it is a revelation from God. But yet nothing, I think, can, under that title, shake or overrule plain knowledge; or rationally prevail with any man to admit it for true, in a direct contradiction to the clear evidence of his own understanding. For, since no evidence of our faculties, by which we receive such revelations, can exceed, if equal, the certainty of our intuitive knowledge, we can never receive for a truth anything that is directly contrary to our clear and distinct knowledge; v.g. the ideas of one body and one place do so clearly agree,

and the mind has so evident a perception of their agreement, that we can never assent to a proposition that affirms the same body to be in two distant places at once, however it should pretend to the authority of a divine revelation: since the evidence, first, that we deceive not ourselves, in ascribing it to God; secondly, that we understand it right; can never be so great as the evidence of our own intuitive knowledge, whereby we discern it impossible for the same body to be in two places at once. And therefore no proposition can be received for divine revelation, or obtain the assent due to all such, if it be contradictory to our clear intuitive knowledge. Because this would be to subvert the principles and foundations of all knowledge, evidence, and assent whatsoever: and there would be left no difference between truth and falsehood, no measures of credible and incredible in the world, if doubtful propositions shall take place before self-evident; and what we certainly know give way to what we may possibly be mistaken in. In propositions therefore contrary to the clear perception of the agreement or disagreement of any of

our ideas, it will be in vain to urge them as matters of faith. They cannot move our assent under that or any other title whatsoever. For faith can never convince us of anything that contradicts our knowledge. Because, though faith be founded on the testimony of God (who cannot lie) revealing any proposition to us: yet we cannot have an assurance of the truth of its being a divine revelation greater than our own knowledge. Since the whole strength of the certainty depends upon our knowledge that God revealed it; which, in this case, where the proposition supposed revealed contradicts our knowledge or reason, will always have this objection hanging to it, viz. that we cannot tell how to conceive that to come from God, the bountiful Author of our being, which, if received for true, must overturn all the principles and foundations of knowledge he has given us; render all our faculties useless; wholly destroy the most excellent part of his workmanship, our understandings; and put a man in a condition wherein he will have less light, less conduct than the beast that perisheth. For if the mind of man can never have a clearer (and perhaps not so clear) evidence of any-

thing to be a divine revelation, as it has of the principles of its own reason, it can never have a ground to quit the clear evidence of its reason, to give a place to a proposition, whose revelation has not a greater evidence than those principles have.

6. Traditional revelation much less. Thus far a man has use of reason, and ought to hearken to it, even in immediate and original revelation, where it is supposed to be made to himself. But to all those who pretend not to immediate revelation, but are required to pay obedience, and to receive the truths revealed to others, which, by the tradition of writings, or word of mouth, are conveyed down to them, reason has a great deal more to do, and is that only which can induce us to receive them. For matter of faith being only divine revelation, and nothing else, faith, as we use the word, (called commonly divine faith), has to do with no propositions, but those which are supposed to be divinely revealed. So that I do not see how those who make revelation alone the sole object of faith can say that it is a matter of faith, and not of reason, to believe that such or such a

proposition, to be found in such or such a book, is of divine inspiration; unless it be revealed that that proposition, or all in that book, was communicated by divine inspiration. Without such a revelation, the believing, or not believing, that proposition, or book, to be of divine authority, can never be matter of faith, but matter of reason; and such as I must come to an assent to only by the use of my reason, which can never require or enable me to believe that which is contrary to itself: it being impossible for reason ever to procure any assent to that which to itself appears unreasonable.

In all things, therefore, where we have clear evidence from our ideas, and those principles of knowledge I have above mentioned, reason is the proper judge; and revelation, though it may, in consenting with it, confirm its dictates, yet cannot in such cases invalidate its decrees: nor can we be obliged, where we have the clear and evident sentience of reason, to quit it for the contrary opinion, under a pretence that it is matter of faith: which can have no authority against the plain and clear dictates of reason.

7. Things above reason are, when revealed, the proper matter of faith. But, Thirdly, There being many things wherein we have very imperfect notions, or none at all; and other things, of whose past, present, or future existence, by the natural use of our faculties, we can have no knowledge at all; these, as being beyond the discovery of our natural faculties, and above reason, are, when revealed, the proper matter of faith. Thus, that part of the angels rebelled against God, and thereby lost their first happy state: and that the dead shall rise, and live again: these and the like, being beyond the discovery of reason, are purely matters of faith, with which reason has directly nothing to do.

8. Or not contrary to reason, if revealed, are matter of faith; and must carry it against probable conjectures of reason. But since God, in giving us the light of reason, has not thereby tied up his own hands from affording us, when he thinks fit, the light of revelation in any of those matters wherein our natural faculties are able to give a probable determination; revelation, where God has been pleased to give it, must carry it against the

probable conjectures of reason. Because the mind not being certain of the truth of that it does not evidently know, but only yielding to the probability that appears in it, is bound to give up its assent to such a testimony which, it is satisfied, comes from one who cannot err, and will not deceive. But yet, it still belongs to reason to judge of the truth of its being a revelation, and of the signification of the words wherein it is delivered. Indeed, if anything shall be thought revelation which is contrary to the plain principles of reason, and the evident knowledge the mind has of its own clear and distinct ideas; there reason must be hearkened to, as to a matter within its province. Since a man can never have so certain a knowledge that a proposition which contradicts the clear principles and evidence of his own knowledge was divinely revealed, or that he understands the words rightly wherein it is delivered, as he has that the contrary is true, and so is bound to consider and judge of it as a matter of reason, and not swallow it, without examination, as a matter of faith.

9. Revelation in matters where reason cannot judge, or

but probably, ought to be hearkened to. First, Whatever proposition is revealed, of whose truth our mind, by its natural faculties and notions, cannot judge, that is purely matter of faith, and above reason. Secondly, All propositions whereof the mind, by the use of its natural faculties, can come to determine and judge, from naturally acquired ideas, are matter of reason; with this difference still, that, in those concerning which it has but an uncertain evidence, and so is persuaded of their truth only upon probable grounds, which still admit a possibility of the contrary to be true, without doing violence to the certain evidence of its own knowledge, and overturning the principles of all reason; in such probable propositions, I say, an evident revelation ought to determine our assent, even against probability. For where the principles of reason have not evidenced a proposition to be certainly true or false, there clear revelation, as another principle of truth and ground of assent, may determine; and so it may be matter of faith, and be also above reason. Because reason, in that particular matter, being able to reach no higher than probability, faith

gave the determination where reason came short; and revelation discovered on which side the truth lay.

10. In matters where reason can afford certain knowledge, that is to be hearkened to. Thus far the dominion of faith reaches, and that without any violence or hindrance to reason; which is not injured or disturbed, but assisted and improved by new discoveries of truth, coming from the eternal fountain of all knowledge. Whatever God hath revealed is certainly true: no doubt can be made of it. This is the proper object of faith: but whether it be a divine revelation or no, reason must judge; which can never permit the mind to reject a greater evidence to embrace what is less evident, nor allow it to entertain probability in opposition to knowledge and certainty. There can be no evidence that any traditional revelation is of divine original, in the words we receive it, and in the sense we understand it, so clear and so certain as that of the principles of reason: and therefore Nothing that is contrary to, and inconsistent with, the clear and self-evident dictates of reason, has a right to be urged or assented to as a matter of faith, wherein

reason hath nothing to do. Whatsoever is divine revelation, ought to overrule all our opinions, prejudices, and interest, and hath a right to be received with full assent. Such a submission as this, of our reason to faith, takes not away the landmarks of knowledge: this shakes not the foundations of reason, but leaves us that use of our faculties for which they were given us.

11. If the boundaries be not set between faith and reason, no enthusiasm or extravagancy in religion can be contradicted. If the provinces of faith and reason are not kept distinct by these boundaries, there will, in matters of religion, be no room for reason at all; and those extravagant opinions and ceremonies that are to be found in the several religions of the world will not deserve to be blamed. For, to this crying up of faith in opposition to reason, we may, I think, in good measure ascribe those absurdities that fill almost all the religions which possess and divide mankind. For men having been principled with an opinion that they must not consult reason in the things of religion, however apparently contradictory to common sense and the very principles

of all their knowledge, have let loose their fancies and natural superstition; and have been by them led into so strange opinions, and extravagant practices in religion, that a considerate man cannot but stand amazed at their follies, and judge them so far from being acceptable to the great and wise God, that he cannot avoid thinking them ridiculous and offensive to a sober good man. So that, in effect, religion, which should most distinguish us from beasts, and ought most peculiarly to elevate us, as rational creatures, above brutes, is that wherein men often appear most irrational, and more senseless than beasts themselves. *Credo, quia impossibile est*: I believe, because it is impossible, might, in a good man, pass for a sally of zeal; but would prove a very ill rule for men to choose their opinions or religion by.

Chapter XIX Of Enthusiasm

1. Love of truth necessary. He that would seriously set upon the search of truth ought in the first place to

prepare his mind with a love of it. For he that loves it not will not take much pains to get it; nor be much concerned when he misses it. There is nobody in the commonwealth of learning who does not profess himself a lover of truth: and there is not a rational creature that would not take it amiss to be thought otherwise of. And yet, for all this, one may truly say, that there are very few lovers of truth, for truth's sake, even amongst those who persuade themselves that they are so. How a man may know whether he be so in earnest, is worth inquiry: and I think there is one unerring mark of it, viz. The not entertaining any proposition with greater assurance than the proofs it is built upon will warrant. Whoever goes beyond this measure of assent, it is plain, receives not the truth in the love of it; loves not truth for truth's sake, but for some other bye-end. For the evidence that any proposition is true (except such as are self-evident) lying only in the proofs a man has of it, whatsoever degrees of assent he affords it beyond the degrees of that evidence, it is plain that all the surplusage of assurance is owing to some other affec-

tion, and not to the love of truth: it being as impossible that the love of truth should carry my assent above the evidence there is to me that it is true, as that the love of truth should make me assent to any proposition for the sake of that evidence which it has not, that it is true: which is in effect to love it as a truth, because it is possible or probable that it may not be true. In any truth that gets not possession of our minds by the irresistible light of self-evidence, or by the force of demonstration, the arguments that gain it assent are the vouchers and gage of its probability to us; and we can receive it for no other than such as they deliver it to our understandings. Whatsoever credit or authority we give to any proposition more than it receives from the principles and proofs it supports itself upon, is owing to our inclinations that way, and is so far a derogation from the love of truth as such: which, as it can receive no evidence from our passions or interests, so it should receive no tincture from them.

2. A forwardness to dictate another's beliefs, from whence. The assuming an authority of dictating to others, and a

forwardness to prescribe to their opinions, is a constant concomitant of this bias and corruption of our judgments. For how almost can it be otherwise, but that he should be ready to impose on another's belief, who has already imposed on his own? Who can reasonably expect arguments and conviction from him in dealing with others, whose understanding is not accustomed to them in his dealing with himself? Who does violence to his own faculties, tyrannizes over his own mind, and usurps the prerogative that belongs to truth alone, which is to command assent by only its own authority, i.e. by and in proportion to that evidence which it carries with it.

3. Force of enthusiasm, in which reason is taken away. Upon this occasion I shall take the liberty to consider a third ground of assent, which with some men has the same authority, and is as confidently relied on as either faith or reason; I mean enthusiasm: which, laying by reason, would set up revelation without it. Whereby in effect it takes away both reason and revelation, and substitutes in the room of them the ungrounded fancies of a man's own brain, and assumes them for a founda-

tion both of opinion and conduct.

4. Reason and revelation. Reason is natural revelation, whereby the eternal Father of light and fountain of all knowledge, communicates to mankind that portion of truth which he has laid within the reach of their natural faculties: revelation is natural reason enlarged by a new set of discoveries communicated by God immediately; which reason vouches the truth of, by the testimony and proofs it gives that they come from God. So that he that takes away reason to make way for revelation, puts out the light of both, and does much what the same as if he would persuade a man to put out his eyes, the better to receive the remote light of an invisible star by a telescope.

5. Rise of enthusiasm. Immediate revelation being a much easier way for men to establish their opinions and regulate their conduct than the tedious and not always successful labour of strict reasoning, it is no wonder that some have been very apt to pretend to revelation, and to persuade themselves that they are under the peculiar guidance of heaven in their actions and opinions,

especially in those of them which they cannot account for by the ordinary methods of knowledge and principles of reason. Hence we see that, in all ages, men in whom melancholy has mixed with devotion, or whose conceit of themselves has raised them into an opinion of a greater familiarity with God, and a nearer admittance to his favour than is afforded to others, have often flattered themselves with a persuasion of an immediate intercourse with the Deity, and frequent communications from the Divine Spirit. God, I own, cannot be denied to be able to enlighten the understanding by a ray darted into the mind immediately from the fountain of light: this they understand he has promised to do, and who then has so good a title to expect it as those who are his peculiar people, chosen by him, and depending on him?

6. Enthusiastic impulse. Their minds being thus prepared, whatever groundless opinion comes to settle itself strongly upon their fancies is an illumination from the Spirit of God, and presently of divine authority: and whatsoever odd action they find in themselves a strong inclination to do, that impulse is concluded to be a call

or direction from heaven, and must be obeyed: it is a commission from above, and they cannot err in executing it.

7. What is meant by enthusiasm. This I take to be properly enthusiasm, which, though founded neither on reason nor divine revelation, but rising from the conceits of a warmed or overweening brain, works yet, where it once gets footing, more powerfully on the persuasions and actions of men than either of those two, or both together: men being most forwardly obedient to the impulses they receive from themselves; and the whole man is sure to act more vigorously where the whole man is carried by a natural motion. For strong conceit, like a new principle, carries all easily with it, when got above common sense, and freed from all restraint of reason and check of reflection, it is heightened into a divine authority, in concurrence with our own temper and inclination.

8. Enthusiasm accepts its supposed illumination without search and proof. Though the odd opinions and extravagant actions enthusiasm has run men into were

enough to warn them against this wrong principle, so apt to misguide them both in their belief and conduct: yet the love of something extraordinary, the ease and glory it is to be inspired, and be above the common and natural ways of knowledge, so flatters many men's laziness, ignorance, and vanity, that, when once they are got into this way of immediate revelation, of illumination without search, and of certainty without proof and without examination, it is a hard matter to get them out of it. Reason is lost upon them, they are above it: they see the light infused into their understandings, and cannot be mistaken; it is clear and visible there, like the light of bright sunshine; shows itself, and needs no other proof but its own evidence: they feel the hand of God moving them within, and the impulses of the Spirit, and cannot be mistaken in what they feel. Thus they support themselves, and are sure reasoning hath nothing to do with what they see and feel in themselves: what they have a sensible experience of admits no doubt, needs no probation. Would he not be ridiculous, who should require to have it proved to him that the light

shines, and that he sees it? It is its own proof, and can have no other. When the Spirit brings light into our minds, it dispels darkness. We see it as we do that of the sun at noon, and need not the twilight of reason to show it us. This light from heaven is strong, clear, and pure; carries its own demonstration with it: and we may as naturally take a glow-worm to assist us to discover the sun, as to examine the celestial ray by our dim candle, reason.

9. Enthusiasm how to be discovered. This is the way of talking of these men: they are sure, because they are sure: and their persuasions are right, because they are strong in them. For, when what they say is stripped of the metaphor of seeing and feeling, this is all it amounts to: and yet these similes so impose on them, that they serve them for certainty in themselves, and demonstration to others.

10. The supposed internal light examined. But to examine a little soberly this internal light, and this feeling on which they build so much. These men have, they say, clear light, and they see; they have awakened sense,

and they feel: this cannot, they are sure, be disputed them. For when a man says he sees or feels, nobody can deny him that he does so. But here let me ask: This seeing, is it the perception of the truth of the proposition, or of this, that it is a revelation from God? This feeling, is it a perception of an inclination or fancy to do something, or of the Spirit of God moving that inclination? These are two very different perceptions, and must be carefully distinguished, if we would not impose upon ourselves. I may perceive the truth of a proposition, and yet not perceive that it is an immediate revelation from God. I may perceive the truth of a proposition in Euclid, without its being, or my perceiving it to be, a revelation: nay, I may perceive I came not by this knowledge in a natural way, and so may conclude it revealed, without perceiving that it is a revelation of God. Because there be spirits which, without being divinely commissioned, may excite those ideas in me, and lay them in such order before my mind, that I may perceive their connexion. So that the knowledge of any proposition coming into my mind, I know not how, is

not a perception that it is from God. Much less is a strong persuasion that it is true, a perception that it is from God, or so much as true. But however it be called light and seeing, I suppose it is at most but belief and assurance: and the proposition taken for a revelation is not such as they know to be true, but take to be true. For where a proposition is known to be true, revelation is needless: and it is hard to conceive how there can be a revelation to any one of what he knows already. If therefore it be a proposition which they are persuaded, but do not know, to be true, whatever they may call it, it is not seeing, but believing. For these are two ways whereby truth comes into the mind, wholly distinct, so that one is not the other. What I see, I know to be so, by the evidence of the thing itself: what I believe, I take to be so upon the testimony of another. But this testimony I must know to be given, or else what ground have I of believing? I must see that it is God that reveals this to me, or else I see nothing. The question then here is: How do I know that God is the revealer of this to me; that this impression is made upon my mind by his Holy Spirit; and

that therefore I ought to obey it? If I know not this, how great soever the assurance is that I am possessed with, it is groundless; whatever light I pretend to, it is but enthusiasm. For, whether the proposition supposed to be revealed be in itself evidently true, or visibly probable, or, by the natural ways of knowledge, uncertain, the proposition that must be well grounded and manifested to be true, is this, That God is the revealer of it, and that what I take to be a revelation is certainly put into my mind by Him, and is not an illusion dropped in by some other spirit, or raised by my own fancy. For, if I mistake not, these men receive it for true, because they presume God revealed it. Does it not, then, stand them upon to examine upon what grounds they presume it to be a revelation from God? or else all their confidence is mere presumption: and this light they are so dazzled with is nothing but an ignis fatuus, that leads them constantly round in this circle; It is a revelation, because they firmly believe it; and they believe it, because it is a revelation.

11. Enthusiasm fails of evidence, that the proposition is from God. In all that is of divine revelation, there is

need of no other proof but that it is an inspiration from God: for he can neither deceive nor be deceived. But how shall it be known that any proposition in our minds is a truth infused by God; a truth that is revealed to us by him, which he declares to us, and therefore we ought to believe? Here it is that enthusiasm fails of the evidence it pretends to. For men thus possessed, boast of a light whereby they say they are enlightened, and brought into the knowledge of this or that truth. But if they know it to be a truth, they must know it to be so, either by its own self-evidence to natural reason, or by the rational proofs that make it out to be so. If they see and know it to be a truth, either of these two ways, they in vain suppose it to be a revelation. For they know it to be true the same way that any other man naturally may know that it is so, without the help of revelation. For thus, all the truths, of what kind soever, that men uninspired are enlightened with, came into their minds, and are established there. If they say they know it to be true, because it is a revelation from God, the reason is good: but then it will be demanded how they know it to

be a revelation from God. If they say, by the light it brings with it, which shines bright in their minds, and they cannot resist: I beseech them to consider whether this be any more than what we have taken notice of already, viz. that it is a revelation, because they strongly believe it to be true. For all the light they speak of is but a strong, though ungrounded persuasion of their own minds, that it is a truth. For rational grounds from proofs that it is a truth, they must acknowledge to have none; for then it is not received as a revelation, but upon the ordinary grounds that other truths are received: and if they believe it to be true because it is a revelation, and have no other reason for its being a revelation, but because they are fully persuaded, without any other reason, that it is true, then they believe it to be a revelation only because they strongly believe it to be a revelation; which is a very unsafe ground to proceed on, either in our tenets or actions. And what readier way can there be to run ourselves into the most extravagant errors and miscarriages, than thus to set up fancy for our supreme and sole guide, and to believe any

proposition to be true, any action to be right, only because we believe it to be so? The strength of our persuasions is no evidence at all of their own rectitude: crooked things may be as stiff and inflexible as straight: and men may be as positive and peremptory in error as in truth. How come else the untractable zealots in different and opposite parties? For if the light, which every one thinks he has in his mind, which in this case is nothing but the strength of his own persuasion, be an evidence that it is from God, contrary opinions have the same title to be inspirations; and God will be not only the Father of lights, but of opposite and contradictory lights, leading men contrary ways; and contradictory propositions will be divine truths, if an ungrounded strength of assurance be an evidence that any proposition is a Divine Revelation.

12. Firmness of persuasion no Proof that any proposition is from God. This cannot be otherwise, whilst firmness of persuasion is made the cause of believing, and confidence of being in the right is made an argument of truth. St. Paul himself believed he did well, and that he

had a call to it, when he persecuted the Christians, whom he confidently thought in the wrong: but yet it was he, and not they, who were mistaken. Good men are men still liable to mistakes, and are sometimes warmly engaged in errors, which they take for divine truths, shining in their minds with the clearest light.

13. Light in the mind, what. Light, true light, in the mind is, or can be, nothing else but the evidence of the truth of any proposition; and if it be not a self-evident proposition, all the light it has, or can have, is from the clearness and validity of those proofs upon which it is received. To talk of any other light in the understanding is to put ourselves in the dark, or in the power of the Prince of Darkness, and, by our own consent, to give ourselves up to delusion to believe a lie. For, if strength of persuasion be the light which must guide us; I ask how shall any one distinguish between the delusions of Satan, and the inspirations of the Holy Ghost? He can transform himself into an angel of light. And they who are led by this Son of the Morning are as fully satisfied of the illumination, i.e. are as strongly persuaded that they are en-

lightened by the Spirit of God as any one who is so: they acquiesce and rejoice in it, are actuated by it: and nobody can be more sure, nor more in the right (if their own belief may be judge) than they.

14. Revelation must be judged of by reason. He, therefore, that will not give himself up to all the extravagances of delusion and error must bring this guide of his light within to the trial. God when he makes the prophet does not unmake the man. He leaves all his faculties in the natural state, to enable him to judge of his inspirations, whether they be of divine original or no. When he illuminates the mind with supernatural light, he does not extinguish that which is natural. If he would have us assent to the truth of any proposition, he either evidences that truth by the usual methods of natural reason, or else makes it known to be a truth which he would have us assent to by his authority, and convinces us that it is from him, by some marks which reason cannot be mistaken in. Reason must be our last judge and guide in everything. I do not mean that we must consult reason, and examine whether a

proposition revealed from God can be made out by natural principles, and if it cannot, that then we may reject it: but consult it we must, and by it examine whether it be a revelation from God or no: and if reason finds it to be revealed from God, reason then declares for it as much as for any other truth, and makes it one of her dictates. Every conceit that thoroughly warms our fancies must pass for an inspiration, if there be nothing but the strength of our persuasions, whereby to judge of our persuasions: if reason must not examine their truth by something extrinsical to the persuasions themselves, inspirations and delusions, truth and falsehood, will have the same measure, and will not be possible to be distinguished.

15. Belief no proof of revelation. If this internal light, or any proposition which under that title we take for inspired, be conformable to the principles of reason, or to the word of God, which is attested revelation, reason warrants it, and we may safely receive it for true, and be guided by it in our belief and actions: if it receive no testimony nor evidence from either of these rules, we

cannot take it for a revelation, or so much as for true, till we have some other mark that it is a revelation, besides our believing that it is so. Thus we see the holy men of old, who had revelations from God, had something else besides that internal light of assurance in their own minds, to testify to them that it was from God. They were not left to their own persuasions alone, that those persuasions were from God, but had outward signs to convince them of the Author of those revelations. And when they were to convince others, they had a power given them to justify the truth of their commission from heaven, and by visible signs to assert the divine authority of a message they were sent with. Moses saw the bush burn without being consumed, and heard a voice out of it: this was something besides finding an impulse upon his mind to go to Pharaoh, that he might bring his brethren out of Egypt: and yet he thought not this enough to authorize him to go with that message, till God, by another miracle of his rod turned into a serpent, had assured him of a power to testify his mission, by the same miracle repeated before

them whom he was sent to. Gideon was sent by an angel to deliver Israel from the Midianites, and yet he desired a sign to convince him that this commission was from God. These, and several the like instances to be found among the prophets of old, are enough to show that they thought not an inward seeing or persuasion of their own minds, without any other proof, a sufficient evidence that it was from God; though the Scripture does not everywhere mention their demanding or having such proofs.

16. Criteria of a divine revelation. In what I have said I am far from denying, that God can, or doth sometimes enlighten men's minds in the apprehending of certain truths or excite them to good actions, by the immediate influence and assistance of the Holy Spirit, without any extraordinary signs accompanying it. But in such cases too we have reason and Scripture; unerring rules to know whether it be from God or no. Where the truth embraced is consonant to the revelation in the written word of God, or the action conformable to the dictates of right reason or holy writ, we may be assured that we

run no risk in entertaining it as such: because, though perhaps it be not an immediate revelation from God, extraordinarily operating on our minds, yet we are sure it is warranted by that revelation which he has given us of truth. But it is not the strength of our private persuasion within ourselves, that can warrant it to be a light or motion from heaven: nothing can do that but the written Word of God without us, or that standard of reason which is common to us with all men. Where reason or Scripture is express for any opinion or action, we may receive it as of divine authority: but it is not the strength of our own persuasions which can by itself give it that stamp. The bent of our own minds may favour it as much as we please: that may show it to be a fondling of our own, but will by no means prove it to be an offspring of heaven, and of divine original.

Chapter XX

Of Wrong Assent, or Error

1. Causes of error, or how men come to give assent

contrary to probability. Knowledge being to be had only of visible and certain truth, error is not a fault of our knowledge, but a mistake of our judgment giving assent to that which is not true.

But if assent be grounded on likelihood, if the proper object and motive of our assent be probability, and that probability consists in what is laid down in the foregoing chapters, it will be demanded how men come to give their assents contrary to probability. For there is nothing more common than contrariety of opinions; nothing more obvious than that one man wholly disbelieves what another only doubts of, and a third stedfastly believes and firmly adheres to. The reasons whereof, though they may be very various, yet, I suppose may all be reduced to these four:

- I. Want of proofs.
- II. Want of ability to use them.
- III. Want of will to see them.
- IV. Wrong measures of probability.

2. First cause of error, want of proofs. First, By want of proofs, I do not mean only the want of those proofs which are nowhere extant, and so are nowhere to be had; but the want even of those proofs which are in being, or might be procured. And thus men want proofs, who have not the convenience or opportunity to make experiments and observations themselves, tending to the proof of any proposition; nor likewise the convenience to inquire into and collect the testimonies of others: and in this state are the greatest part of mankind, who are given up to labour, and enslaved to the necessity of their mean condition, whose lives are worn out only in the provisions for living. These men's opportunities of knowledge and inquiry are commonly as narrow as their fortunes; and their understandings are but little instructed, when all their whole time and pains are laid out to still the croaking of their own bellies, or the cries of their children. It is not to be expected that a man who drudges on all his life in a laborious trade, should be more knowing in the variety of things done in the world than a packhorse, who is driven constantly for-

wards and backwards in a narrow lane and dirty road, only to market, should be skilled in the geography of the country. Nor is it at all more possible that he who wants leisure, books, and languages, and the opportunity of conversing with variety of men, should be in a condition to collect those testimonies and observations which are in being, and are necessary to make out many, nay most, of the propositions that, in the societies of men, are judged of the greatest moment; or to find out grounds of assurance so great as the belief of the points he would build on them is thought necessary. So that a great part of mankind are, by the natural and unalterable state of things in this world, and the constitution of human affairs, unavoidably given over to invincible ignorance of those proofs on which others build, and which are necessary to establish those opinions: the greatest part of men, having much to do to get the means of living, are not in a condition to look after those of learned and laborious inquiries.

3. Objection. "What shall become of those who want proofs?" Answered. What shall we say, then? Are the

greatest part of mankind, by the necessity of their condition, subjected to unavoidable ignorance in those things which are of greatest importance to them? (for of those it is obvious to inquire). Have the bulk of mankind no other guide but accident and blind chance to conduct them to their happiness or misery? Are the current opinions, and licensed guides of every country sufficient evidence and security to every man to venture his great concerns on; nay, his everlasting happiness or misery? Or can those be the certain and infallible oracles and standards of truth, which teach one thing in Christendom and another in Turkey? Or shall a poor countryman be eternally happy, for having the chance to be born in Italy; or a day-labourer be unavoidably lost, because he had the ill-luck to be born in England? How ready some men may be to say some of these things, I will not here examine: but this I am sure, that men must allow one or other of these to be true, (let them choose which they please,) or else grant that God has furnished men with faculties sufficient to direct them in the way they should take, if they will but seriously

employ them that way, when their ordinary vocations allow them the leisure. No man is so wholly taken up with the attendance on the means of living, as to have no spare time at all to think of his soul, and inform himself in matters of religion. Were men as intent upon this as they are on things of lower concernment, there are none so enslaved to the necessities of life who might not find many vacancies that might be husbanded to this advantage of their knowledge.

4. People hindered from inquiry. Besides those whose improvements and informations are straitened by the narrowness of their fortunes, there are others whose largeness of fortune would plentifully enough supply books, and other requisites for clearing of doubts, and discovering of truth: but they are cooped in close, by the laws of their countries, and the strict guards of those whose interest it is to keep them ignorant, lest, knowing more, they should believe the less in them. These are as far, nay further, from the liberty and opportunities of a fair inquiry, than these poor and wretched labourers we before spoke of: and however

they may seem high and great, are confined to narrowness of thought, and enslaved in that which should be the freest part of man, their understandings. This is generally the case of all those who live in places where care is taken to propagate truth without knowledge; where men are forced, at a venture, to be of the religion of the country; and must therefore swallow down opinions, as silly people do empiric's pills, without knowing what they are made of, or how they will work, and having nothing to do but believe that they will do the cure: but in this are much more miserable than they, in that they are not at liberty to refuse swallowing what perhaps they had rather let alone; or to choose the physician, to whose conduct they would trust themselves.

5. Second cause of error, want of skill to use proofs. Secondly, Those who want skill to use those evidences they have of probabilities; who cannot carry a train of consequences in their heads; nor weigh exactly the preponderancy of contrary proofs and testimonies, making every circumstance its due allowance; may be easily misled to assent to positions that are not probable. There are

some men of one, some but of two syllogisms, and no more; and others that can but advance one step further. These cannot always discern that side on which the strongest proofs lie; cannot constantly follow that which in itself is the more probable opinion. Now that there is such a difference between men, in respect of their understandings, I think nobody, who has had any conversation with his neighbours, will question: though he never was at Westminster-Hall or the Exchange on the one hand, nor at Alms-houses or Bedlam on the other. Which great difference in men's intellectuals, whether it rises from any defect in the organs of the body particularly adapted to thinking; or in the dullness or untractableness of those faculties for want of use; or, as some think, in the natural differences of men's souls themselves; or some, or all of these together; it matters not here to examine: only this is evident, that there is a difference of degrees in men's understandings, apprehensions, and reasonings, to so great a latitude, that one may, without doing injury to mankind, affirm that there is a greater distance between

some men and others in this respect than between some men and some beasts. But how this comes about is a speculation, though of great consequence, yet not necessary to our present purpose.

6. Third cause of error, want of will to use them. Thirdly, There are another sort of people that want proofs, not because they are out of their reach, but because they will not use them: who though they have riches and leisure enough and want neither parts nor other helps, are yet never the better for them. Their hot pursuit of pleasure, or constant drudgery in business, engages some men's thoughts elsewhere: laziness and oscitancy in general, or a particular aversion for books, study, and meditation, keep others from any serious thoughts at all; and some out of fear that an impartial inquiry would not favour those opinions which best suit their prejudices, lives, and designs, content themselves, without examination, to take upon trust what they find convenient and in fashion. Thus, most men, even of those that might do otherwise, pass their lives without an acquaintance with, much less a rational assent to, prob-

abilities they are concerned to know, though they lie so much within their view that, to be convinced of them, they need but turn their eyes that way. We know some men will not read a letter which is supposed to bring ill news; and many men forbear to cast up their accounts, or so much as think upon their estates, who have reason to fear their affairs are in no very good posture. How men, whose plentiful fortunes allow them leisure to improve their understandings, can satisfy themselves with a lazy ignorance, I cannot tell: but methinks they have a low opinion of their souls, who lay out all their incomes in provisions for the body, and employ none of it to procure the means and helps of knowledge; who take great care to appear always in a neat and splendid outside, and would think themselves miserable in coarse clothes, or a patched coat, and yet contentedly suffer their minds to appear abroad in a piebald livery of coarse patches and borrowed shreds, such as it has pleased chance, or their country tailor (I mean the common opinion of those they have conversed with) to clothe them in. I will not here mention how unreasonable this

is for men that ever think of a future state, and their concernment in it, which no rational man can avoid to do sometimes: nor shall I take notice what a shame and confusion it is to the greatest contemners of knowledge, to be found ignorant in things they are concerned to know. But this at least is worth the consideration of those who call themselves gentlemen, That, however they may think credit, respect, power, and authority the concomitants of their birth and fortune, yet they will find all these still carried away from them by men of lower condition, who surpass them in knowledge. They who are blind will always be led by those that see, or else fall into the ditch: and he is certainly the most subjected, the most enslaved, who is so in his understanding. In the foregoing instances some of the causes have been shown of wrong assent, and how it comes to pass that probable doctrines are not always received with an assent proportionable to the reasons which are to be had for their probability: but hitherto we have considered only such probabilities whose proofs do exist, but do not appear to him who embraces the error.

7. Fourth cause of error, wrong measures of Probability. Fourthly, There remains yet the last sort, who, even where the real probabilities appear, and are plainly laid before them, do not admit of the conviction, nor yield unto manifest reasons, but do either epechein, suspend their assent, or give it to the less probable opinion. And to this danger are those exposed who have taken up wrong measures of probability, which are:

I. Propositions that are not in themselves certain and evident, but doubtful and false, taken up for principles.

II. Received hypotheses.

III. Predominant passions or inclinations.

IV. Authority.

8. I. Doubtful propositions taken for principles. The first and firmest ground of probability is the conformity anything has to our own knowledge; especially that part of our knowledge which we have embraced, and continue to look on as principles. These have so great an influence upon our opinions, that it is usually by them we judge of truth, and measure probability; to that degree, that what is inconsistent with our principles, is so

far from passing for probable with us, that it will not be allowed possible. The reverence borne to these principles is so great, and their authority so paramount to all other, that the testimony, not only of other men, but the evidence of our own senses are often rejected, when they offer to vouch anything contrary to these established rules. How much the doctrine of innate principles, and that principles are not to be proved or questioned, has contributed to this, I will not here examine. This I readily grant, that one truth cannot contradict another: but withal I take leave also to say, that every one ought very carefully to beware what he admits for a principle, to examine it strictly, and see whether he certainly knows it to be true of itself, by its own evidence, or whether he does only with assurance believe it to be so upon the authority of others. For he hath a strong bias put into his understanding, which will unavoidably misguide his assent, who hath imbibed wrong principles, and has blindly given himself up to the authority of any opinion in itself not evidently true.

9. Instilled in childhood. There is nothing more ordi-

nary than children's receiving into their minds propositions (especially about matters of religion) from their parents, nurses, or those about them: which being insinuated into their unwary as well as unbiassed understandings, and fastened by degrees, are at last (equally whether true or false) riveted there by long custom and education, beyond all possibility of being pulled out again. For men, when they are grown up, reflecting upon their opinions, and finding those of this sort to be as ancient in their minds as their very memories, not having observed their early insinuation, nor by what means they got them, they are apt to reverence them as sacred things, and not to suffer them to be profaned, touched, or questioned: they look on them as the Urim and Thummim set up in their minds immediately by God himself, to be the great and unerring deciders of truth and falsehood, and the judges to which they are to appeal in all manner of controversies.

10. Of irresistible efficacy. This opinion of his principles (let them be what they will) being once established in any one's mind, it is easy to be imagined what reception

any proposition shall find, how clearly soever proved, that shall invalidate their authority, or at all thwart these internal oracles; whereas the grossest absurdities and improbabilities, being but agreeable to such principles, go down glibly, and are easily digested. The great obstinacy that is to be found in men firmly believing quite contrary opinions, though many times equally absurd, in the various religions of mankind, are as evident a proof as they are an unavoidable consequence of this way of reasoning from received traditional principles. So that men will disbelieve their own eyes, renounce the evidence of their senses, and give their own experience the lie, rather than admit of anything disagreeing with these sacred tenets. Take an intelligent Romanist that, from the first dawning of any notions in his understanding, hath had this principle constantly inculcated, viz. that he must believe as the church (i.e. those of his communion) believes, or that the pope is infallible, and this he never so much as heard questioned, till at forty or fifty years old he met with one of other principles: how is he prepared easily to swallow,

not only against all probability, but even the clear evidence of his senses, the doctrine of transubstantiation? This principle has such an influence on his mind, that he will believe that to be flesh which he sees to be bread. And what way will you take to convince a man of any improbable opinion he holds, who, with some philosophers, hath laid down this as a foundation of reasoning, That he must believe his reason (for so men improperly call arguments drawn from their principles) against his senses? Let an enthusiast be principled that he or his teacher is inspired, and acted by an immediate communication of the Divine Spirit, and you in vain bring the evidence of clear reasons against his doctrine. Whoever, therefore, have imbibed wrong principles, are not, in things inconsistent with these principles, to be moved by the most apparent and convincing probabilities, till they are so candid and ingenuous to themselves, as to be persuaded to examine even those very principles, which many never suffer themselves to do.

11. II. Received hypotheses. Next to these are men whose understandings are cast into a mould, and fashioned

just to the size of a received hypothesis. The difference between these and the former, is, that they will admit of matter of fact, and agree with dissenters in that; but differ only in assigning of reasons and explaining the manner of operation. These are not at that open defiance with their senses, with the former: they can endure to hearken to their information a little more patiently; but will by no means admit of their reports in the explanation of things; nor be prevailed on by probabilities, which would convince them that things are not brought about just after the same manner that they have decreed within themselves that they are. Would it not be an insufferable thing for a learned professor, and that which his scarlet would blush at, to have his authority of forty years, standing, wrought out of hard rock, Greek and Latin, with no small expense of time and candle, and confirmed by general tradition and a reverend beard, in an instant overturned by an upstart novelist? Can any one expect that he should be made to confess, that what he taught his scholars thirty years ago was all error and mistake; and that he sold them

hard words and ignorance at a very dear rate. What probabilities, I say, are sufficient to prevail in such a case? And who ever, by the most cogent arguments, will be prevailed with to disrobe himself at once of all his old opinions, and pretences to knowledge and learning, which with hard study he hath all this time been labouring for; and turn himself out stark naked, in quest afresh of new notions? All the arguments that can be used will be as little able to prevail, as the wind did with the traveller to part with his cloak, which he held only the faster. To this of wrong hypothesis may be reduced the errors that may be occasioned by a true hypothesis, or right principles, but not rightly understood. There is nothing more familiar than this. The instances of men contending for different opinions, which they all derive from the infallible truth of the Scripture, are an undeniable proof of it. All that call themselves Christians, allow the text that says, metanoëite, to carry in it the obligation to a very weighty duty. But yet how very erroneous will one of their practices be, who, understanding nothing but the French, take this rule with

one translation to be, Repentez-vous, repent; or with the other, Fatiez penitence, do penance.

12. III. Predominant passions. Probabilities which cross men's appetites and prevailing passions run the same fate. Let ever so much probability hang on one side of a covetous man's reasoning, and money on the other; it is easy to foresee which will outweigh. Earthly minds, like mud walls, resist the strongest batteries: and though, perhaps, sometimes the force of a clear argument may make some impression, yet they nevertheless stand firm, and keep out the enemy, truth, that would captivate or disturb them. Tell a man passionately in love that he is jilted; bring a score of witnesses of the falsehood of his mistress, it is ten to one but three kind words of hers shall invalidate all their testimonies. Quod volumus, facile credimus; what suits our wishes, is forwardly believed, is, I suppose, what every one hath more than once experimented: and though men cannot always openly gain-say or resist the force of manifest probabilities that make against them, yet yield they not to the argument. Not but that it is the nature of the understanding constantly

to close with the more probable side; but yet a man hath a power to suspend and restrain its inquiries, and not permit a full and satisfactory examination, as far as the matter in question is capable, and will bear it to be made. Until that be done, there will be always these two ways left of evading the most apparent probabilities:

13. Two means of evading probabilities: I. Supposed fallacy latent in the words employed. First, That the arguments being (as for the most part they are) brought in words, there may be a fallacy latent in them: and the consequences being, perhaps, many in train, they may be some of them incoherent. There are very few discourses so short, clear, and consistent, to which most men may not, with satisfaction enough to themselves, raise this doubt; and from whose conviction they may not, without reproach of disingenuity or unreasonableness, set themselves free with the old reply, *Non persuadebis, etiamsi persuaseris*; though I cannot answer, I will not yield.

14. Supposed unknown arguments for the contrary. Secondly, Manifest probabilities may be evaded, and the

assent withheld, upon this suggestion, That I know not yet all that may be said on the contrary side. And therefore, though I be beaten, it is not necessary I should yield, not knowing what forces there are in reserve behind. This is a refuge against conviction so open and so wide, that it is hard to determine when a man is quite out of the verge of it.

15. What probabilities naturally determine the assent. But yet there is some end of it; and a man having carefully inquired into all the grounds of probability and unlikeliness; done his utmost to inform himself in all particulars fairly, and cast up the sum total on both sides; may, in most cases, come to acknowledge, upon the whole matter, on which side the probability rests: wherein some proofs in matter of reason, being suppositions upon universal experience, are so cogent and clear, and some testimonies in matter of fact so universal, that he cannot refuse his assent. So that I think we may conclude, that, in propositions, where though the proofs in view are of most moment, yet there are sufficient grounds to suspect that there is either fallacy in

words, or certain proofs as considerable to be produced on the contrary side; there assent, suspense, or dissent, are often voluntary actions. But where the proofs are such as make it highly probable, and there is not sufficient ground to suspect that there is either fallacy of words (which sober and serious consideration may discover) nor equally valid proofs yet undiscovered, latent on the other side (which also the nature of the thing may, in some cases, make plain to a considerate man); there, I think, a man who has weighed them can scarce refuse his assent to the side on which the greater probability appears. Whether it be probable that a promiscuous jumble of printing letters should often fall into a method and order, which should stamp on paper a coherent discourse; or that a blind fortuitous concourse of atoms, not guided by an understanding agent, should frequently constitute the bodies of any species of animals: in these and the like cases, I think, nobody that considers them can be one jot at a stand which side to take, nor at all waver in his assent. Lastly, when there can be no supposition (the thing in its own nature in-

different, and wholly depending upon the testimony of witnesses) that there is as fair testimony against, as for the matter of fact attested; which by inquiry is to be learned, v.g. whether there was one thousand seven hundred years ago such a man at Rome as Julius Caesar: in all such cases, I say, I think it is not in any rational man's power to refuse his assent; but that it necessarily follows, and closes with such probabilities. In other less clear cases, I think it is in man's power to suspend his assent; and perhaps content himself with the proofs he has, if they favour the opinion that suits with his inclination or interest, and so stop from further search. But that a man should afford his assent to that side on which the less probability appears to him, seems to me utterly impracticable, and as impossible as it is to believe the same thing probable and improbable at the same time.

16. Where it is in our power to suspend our judgment. As knowledge is no more arbitrary than perception; so, I think, assent is no more in our power than knowledge. When the agreement of any two ideas appears to our minds, whether immediately or by the assistance of rea-

son, I can no more refuse to perceive, no more avoid knowing it, than I can avoid seeing those objects which I turn my eyes to, and look on in daylight; and what upon full examination I find the most probable, I cannot deny my assent to. But, though we cannot hinder our knowledge, where the agreement is once perceived; nor our assent, where the probability manifestly appears upon due consideration of all the measures of it: yet we can hinder both knowledge and assent, by stopping our inquiry, and not employing our faculties in the search of any truth. If it were not so, ignorance, error, or infidelity, could not in any case be a fault. Thus, in some cases we can prevent or suspend our assent: but can a man versed in modern or ancient history doubt whether there is such a place as Rome, or whether there was such a man as Julius Caesar? Indeed, there are millions of truths that a man is not, or may not think himself concerned to know; as whether our king Richard the Third was crooked or no; or whether Roger Bacon was a mathematician or a magician. In these and such like cases, where the assent one way or other is of

no importance to the interest of any one; no action, no concernment of his following or depending thereon, there it is not strange that the mind should give itself up to the common opinion, or render itself to the first comer. These and the like opinions are of so little weight and moment, that, like motes in the sun, their tendencies are very rarely taken notice of. They are there, as it were, by chance, and the mind lets them float at liberty. But where the mind judges that the proposition has concernment in it: where the assent or not assenting is thought to draw consequences of moment after it, and good and evil to depend on choosing or refusing the right side, and the mind sets itself seriously to inquire and examine the probability: there I think it is not in our choice to take which side we please, if manifest odds appear on either. The greater probability, I think, in that case will determine the assent: and a man can no more avoid assenting, or taking it to be true, where he perceives the greater probability, than he can avoid knowing it to be true, where he perceives the agreement or disagreement of any two ideas.

If this be so, the foundation of error will lie in wrong measures of probability; as the foundation of vice in wrong measures of good.

17. IV. Authority. The fourth and last wrong measure of probability I shall take notice of, and which keeps in ignorance or error more people than all the other together, is that which I have mentioned in the foregoing chapter: I mean the giving up our assent to the common received opinions, either of our friends or party, neighbourhood or country. How many men have no other ground for their tenets, than the supposed honesty, or learning, or number of those of the same profession? As if honest or bookish men could not err; or truth were to be established by the vote of the multitude: yet this with most men serves the turn. The tenet has had the attestation of reverend antiquity; it comes to me with the passport of former ages, and therefore I am secure in the reception I give it: other men have been and are of the same opinion, (for that is all is said,) and therefore it is reasonable for me to embrace it. A man may more justifiably throw up cross and pile for his opin-

ions, than take them up by such measures. All men are liable to error, and most men are in many points, by passion or interest, under temptation to it. If we could but see the secret motives that influenced the men of name and learning in the world, and the leaders of parties, we should not always find that it was the embracing of truth for its own sake, that made them espouse the doctrines they owned and maintained. This at least is certain, there is not an opinion so absurd, which a man may not receive upon this ground. There is no error to be named, which has not had its professors: and a man shall never want crooked paths to walk in, if he thinks that he is in the right way, wherever he has the footsteps of others to follow.

18. Not so many men in errors as is commonly supposed. But, notwithstanding the great noise is made in the world about errors and opinions, I must do mankind that right as to say, There are not so many men in errors and wrong opinions as is commonly supposed. Not that I think they embrace the truth; but indeed, because concerning those doctrines they keep such a

stir about, they have no thought, no opinion at all. For if any one should a little catechise the greatest part of the partizans of most of the sects in the world, he would not find, concerning those matters they are so zealous for, that they have any opinions of their own: much less would he have reason to think that they took them upon the examination of arguments and appearance of probability. They are resolved to stick to a party that education or interest has engaged them in; and there, like the common soldiers of an army, show their courage and warmth as their leaders direct, without ever examining, or so much as knowing, the cause they contend for. If a man's life shows that he has no serious regard for religion; for what reason should we think that he beats his head about the opinions of his church, and troubles himself to examine the grounds of this or that doctrine? It is enough for him to obey his leaders, to have his hand and his tongue ready for the support of the common cause, and thereby approve himself to those who can give him credit, preferment, or protection in that society. Thus men become professors of, and com-

batants for, those opinions they were never convinced of nor proselytes to; no, nor ever had so much as floating in their heads: and though one cannot say there are fewer improbable or erroneous opinions in the world than there are, yet this is certain; there are fewer that actually assent to them, and mistake them for truths, than is imagined.

Chapter XXI Of the Division of the Sciences

1. Science may be divided into three sorts. All that can fall within the compass of human understanding, being either, First, the nature of things, as they are in themselves, their relations, and their manner of operation: or, Secondly, that which man himself ought to do, as a rational and voluntary agent, for the attainment of any end, especially happiness: or, Thirdly, the ways and means whereby the knowledge of both the one and the other of these is attained and communicated; I think science may be divided properly into these three sorts:—

2. Physica. First, The knowledge of things, as they are in their own proper beings, their constitution, properties, and operations; whereby I mean not only matter and body, but spirits also, which have their proper natures, constitutions, and operations, as well as bodies. This, in a little more enlarged sense of the word, I call Phusike, or natural philosophy. The end of this is bare speculative truth: and whatsoever can afford the mind of man any such, falls under this branch, whether it be God himself, angels, spirits, bodies; or any of their affections, as number, and figure, &c.

3. Practica. Secondly, Praktike, The skill of right applying our own powers and actions, for the attainment of things good and useful. The most considerable under this head is ethics, which is the seeking out those rules and measures of human actions, which lead to happiness, and the means to practise them. The end of this is not bare speculation and the knowledge of truth; but right, and a conduct suitable to it.

4. Semeiotike. Thirdly, the third branch may be called Semeiotike, or the doctrine of signs; the most usual

whereof being words, it is aptly enough termed also Logike, logic: the business whereof is to consider the nature of signs, the mind makes use of for the understanding of things, or conveying its knowledge to others. For, since the things the mind contemplates are none of them, besides itself, present to the understanding, it is necessary that something else, as a sign or representation of the thing it considers, should be present to it: and these are ideas. And because the scene of ideas that makes one man's thoughts cannot be laid open to the immediate view of another, nor laid up anywhere but in the memory, a no very sure repository: therefore to communicate our thoughts to one another, as well as record them for our own use, signs of our ideas are also necessary: those which men have found most convenient, and therefore generally make use of, are articulate sounds. The consideration, then, of ideas and words as the great instruments of knowledge, makes no despicable part of their contemplation who would take a view of human knowledge in the whole extent of it. And perhaps if they were distinctly weighed, and duly con-

sidered, they would afford us another sort of logic and critic, than what we have been hitherto acquainted with.

5. This is the first and most general division of the objects of our understanding. This seems to me the first and most general, as well as natural division of the objects of our understanding. For a man can employ his thoughts about nothing, but either, the contemplation of things themselves, for the discovery of truth; or about the things in his own power, which are his own actions, for the attainment of his own ends; or the signs the mind makes use of both in the one and the other, and the right ordering of them, for its clearer information. All which three, viz, things, as they are in themselves knowable; actions as they depend on us, in order to happiness; and the right use of signs in order to knowledge, being toto coelo different, they seemed to me to be the three great provinces of the intellectual world, wholly separate and distinct one from another.

THE END

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